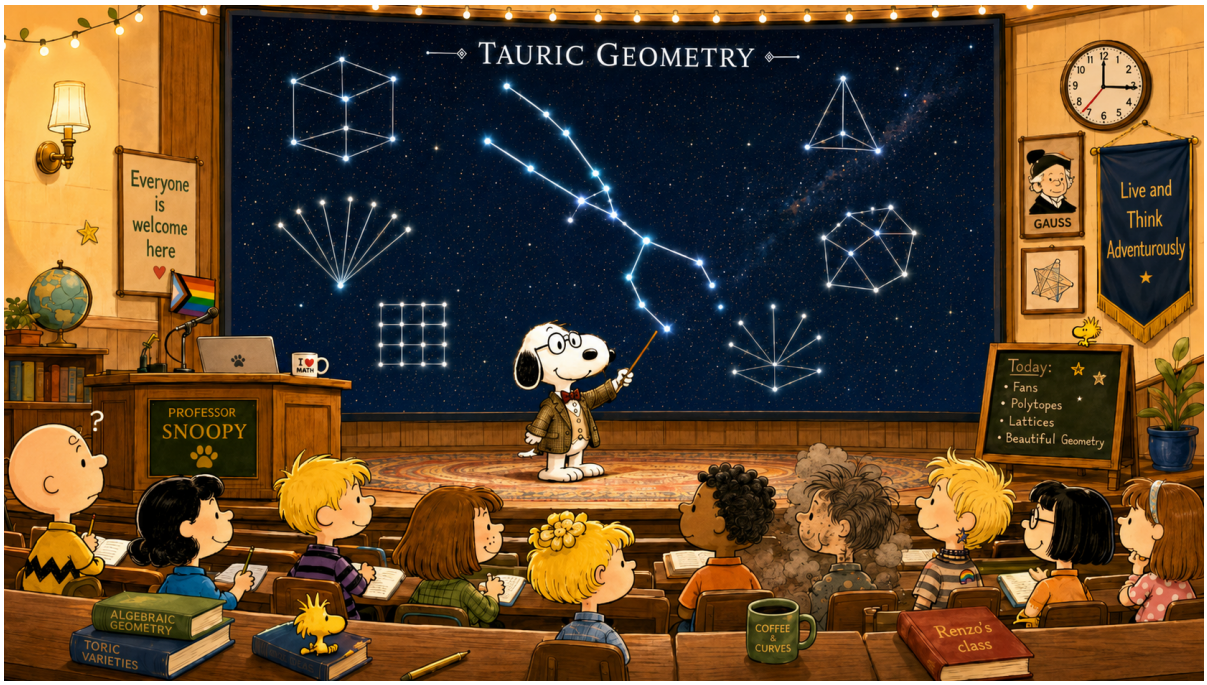
The Three Dictionaries of Toric Geometry

Notes for Renzo's Fall 2026 course



Colorado State University

Contents

Preface xv

Conventions and notation xix

I Basic Concepts 1

0 $\mathbb{P}^2$ Reloaded 3

0.1 The affine model: $\mathbb{A}^2$ . . . 3
0.1.1 The coordinate strata . . . 3
0.1.2 Monomial functions and lattice points . . . 5
0.1.3 Approaching the boundary along monomial curves . . . 6
0.2 What we already know about $\mathbb{P}^2$ . . . 8
0.2.1 Lines through the origin . . . 8
0.2.2 The standard affine charts . . . 8
0.2.3 Changing affine coordinates . . . 10
0.2.4 Homogeneous equations . . . 10
0.2.5 The coordinate triangle . . . 11
0.3 The dense torus in $\mathbb{P}^2$ . . . 12
0.3.1 Removing the coordinate triangle . . . 12
0.3.2 Multiplication inside the open set . . . 14
0.3.3 Letting the torus move the whole projective plane . . . 15
0.3.4 The affine torus reappears in every chart . . . 16
0.4 Monomials, triangles, and global sections . . . 16
0.4.1 Why homogeneous polynomials are not ordinary functions . . . 17
0.4.2 The monomial basis . . . 18
0.4.3 The triangle of degree-d monomials . . . 19
0.4.4 Counting sections by counting lattice points . . . 21
0.5 Orbits and coordinate strata . . . 22
0.5.1 The open orbit . . . 22
0.5.2 The three one-dimensional orbits . . . 22
0.5.3 The three fixed points . . . 23
0.5.4 Orbit closures . . . 23
0.5.5 Two triangles, two different meanings . . . 26
0.6 The triangle and the fan . . . 27
0.6.1 Three sides, three inward directions . . . 28
0.6.2 A direction asks a question of the triangle . . . 28
0.6.3 Faces of the triangle and cones of the fan . . . 30
0.6.4 The fan and one-parameter limits . . . 30

0.6.5 The three maximal cones recover the three affine charts . . . 31
0.6.6 What the fan remembers . . . 32
0.7 Blowing up a torus-fixed point . . . 33
0.7.1 Replacing a point by its directions . . . 34
0.7.2 The two affine charts of the blowup . . . 34
0.7.3 The star subdivision of the fan . . . 36
0.7.4 The exceptional divisor in the orbit picture . . . 37
0.7.5 The exceptional curve has self-intersection -1 . . . 37
What this chapter has accomplished . . . 39
Comprehension exercises . . . 43
Challenge exercise . . . 44
Notes and references . . . 45

1 Algebraic Tori and Their Representation Theory 47
1.1 Why "torus"? . . . 47
1.1.1 From the circle to the punctured complex line . . . 48
1.1.2 The algebraic definition . . . 49
1.1.3 Examples close to home . . . 49
1.1.4 Why monomials are waiting for us . . . 50
1.2 The algebraic torus $\mathbb{G}_m^n$ . . . 51
1.2.1 Removing the coordinate hyperplanes . . . 51
1.2.2 Laurent polynomials are the regular functions . . . 52
1.2.3 The group law is algebraic . . . 53
1.2.4 Maps into $\mathbb{G}_m$ and invertible functions . . . 54
1.2.5 The dense torus of $\mathbb{P}^2$ revisited . . . 55
1.3 Monomials and characters . . . 56
1.3.1 Multiplicative regular functions . . . 56
1.3.2 There are no other characters . . . 56
1.3.3 What the exponents record . . . 58
1.3.4 Multiplying characters adds exponents . . . 59
1.3.5 Characters as one-dimensional representations . . . 60
1.4 The character lattice . . . 60
1.4.1 The intrinsic definition . . . 61
1.4.2 Coordinates are bases of the character lattice . . . 61
1.4.3 The coordinate ring as a group algebra . . . 62
1.4.4 Morphisms reverse the direction on character lattices . . . 64
1.4.5 Passing to real vector spaces . . . 64
1.5 One-parameter subgroups . . . 65
1.5.1 Turning the arrow around . . . 65
1.5.2 The one-parameter subgroup lattice . . . 66
1.5.3 One-parameter subgroups as motions . . . 67
1.5.4 Limits in affine space . . . 68
1.5.5 The same motion in a different compactification . . . 69
1.5.6 Primitive one-parameter subgroups . . . 70
1.5.7 Functoriality . . . 71

1.5.8 A first meeting between M and N . . . 72
1.6 Dual lattices and the pairing . . . 73
1.6.1 The pairing by composition . . . 73
1.6.2 Why the pairing is bilinear . . . 74
1.6.3 Perfect duality . . . 75
1.6.4 Real vector spaces and linear functions . . . 76
1.6.5 Orders along one-parameter motions . . . 77
1.6.6 Compatibility with torus morphisms . . . 78
1.7 Changing coordinates on a torus . . . 79
1.7.1 From a new basis to new torus coordinates . . . 80
1.7.2 Monomial automorphisms of a split torus . . . 81
1.7.3 What happens to the dual basis? . . . 82
1.7.4 Coordinates of vectors versus bases . . . 83
1.7.5 Matrices for morphisms of tori . . . 83
1.7.6 Isomorphisms, finite maps, and dimensions . . . 85
1.8 Representation theory in one page . . . 86
1.8.1 What is a representation? . . . 86
1.8.2 Weight vectors and weight spaces . . . 87
1.8.3 The weight-space decomposition . . . 88
1.8.4 Why the theorem is plausible . . . 89
1.8.5 Invariant vectors . . . 89
1.8.6 Equivariant linear maps preserve weights . . . 90
1.8.7 Subrepresentations and quotients . . . 91
1.8.8 Weights as a finite configuration in a lattice . . . 91
1.8.9 Operations on representations . . . 91
1.8.10 The basic classification . . . 93
What this chapter has accomplished . . . 93
Comprehension exercises . . . 96
Challenge exercise . . . 99
Notes and references . . . 101

2 Affine Toric Varieties 103
2.1 From the affine plane to affine toric varieties . . . 103
2.1.1 Which characters extend? . . . 104
2.1.2 The torus action on the affine plane . . . 105
2.1.3 The intrinsic definition . . . 106
2.1.4 The coordinate-ring viewpoint . . . 107
2.2 Affine semigroups, monomial maps, and equations . . . 108
2.2.1 Affine semigroups . . . 108
2.2.2 The semigroup algebra . . . 109
2.2.3 From generators to a monomial map . . . 110
2.2.4 Relations among generators become equations . . . 112
2.2.5 The cusp: the simplest missing exponent . . . 113
2.2.6 A quadratic cone and the correct lattice . . . 114
2.2.7 Four equivalent descriptions . . . 115

2.3 Cones, saturation, and normality . . . 116
2.3.1 From a semigroup to a cone . . . 117
2.3.2 A small amount of cone geometry . . . 118
2.3.3 Gordan's lemma . . . 118
2.3.4 The Hilbert basis . . . 119
2.3.5 Saturation fills the holes . . . 120
2.3.6 Normality is saturation in algebraic form . . . 121
2.3.7 Normalization of an affine semigroup variety . . . 123
2.4 Dual cones and one-parameter subgroup limits . . . 125
2.4.1 Dual cones . . . 125
2.4.2 From a cone in $N_{\mathbb{R}}$ to an affine variety . . . 127
2.4.3 What does it mean for a limit to exist? . . . 128
2.4.4 The one-parameter subgroup limit criterion . . . 129
2.4.5 The limit point . . . 130
2.5 Properties of affine toric varieties . . . 132
2.5.1 Points as semigroup homomorphisms . . . 132
2.5.2 Invertible characters and torus factors . . . 134
2.5.3 Fixed points . . . 136
2.5.4 Smoothness . . . 137
2.5.5 Toric morphisms . . . 139
2.5.6 Faces give affine open subsets . . . 141
2.6 The classification dictionary . . . 143
What this chapter has accomplished . . . 145
Comprehension exercises . . . 146
Challenge exercise . . . 148
Notes and references . . . 151

3 Fans and Toric Varieties 153
3.1 Why one cone is not enough . . . 154
3.1.1 The affine construction is a local model . . . 154
3.1.2 $\mathbb{P}^1$ from two affine lines . . . 154
3.1.3 Common faces are the transition charts . . . 156
3.1.4 $\mathbb{P}^2$ from three affine planes . . . 157
3.2 Fans and their basic language . . . 158
3.2.1 The definition . . . 158
3.2.2 What can go wrong? . . . 159
3.2.3 Support and cones of each dimension . . . 160
3.2.4 The dimension of a fan . . . 161
3.2.5 Subfans . . . 161
3.2.6 Complete fans . . . 162
3.3 Gluing affine toric varieties . . . 163
3.3.1 The canonical overlap of two charts . . . 163
3.3.2 The separation lemma . . . 164
3.3.3 The cocycle condition is automatic . . . 165
3.3.4 Constructing X_Σ . . . 166

3.3.5 The toric variety of a fan . . . 167
3.4 The orbit–cone correspondence . . . 169
3.4.1 Distinguished points . . . 169
3.4.2 What kind of torus is $O(\sigma)$? . . . 170
3.4.3 The theorem . . . 171
3.4.4 Reading the blowup fan . . . 172
3.4.5 Reading the fan of $\mathbb{P}^2$ . . . 173
3.4.6 Orbit closures are toric varieties . . . 174
3.5 Properties read from the fan . . . 175
3.5.1 Smoothness is checked cone by cone . . . 175
3.5.2 Completeness and one-parameter limits . . . 176
3.5.3 Product fans . . . 177
3.5.4 Toric morphisms from lattice maps . . . 178
3.6 A catalogue of fans and toric surfaces . . . 180
3.6.1 Four recurring complete fans . . . 180
3.6.2 A weighted projective plane . . . 181
3.6.3 Hirzebruch surfaces . . . 182
3.6.4 Blowing up a torus-fixed point . . . 182
3.6.5 A compact recognition table . . . 184
What this chapter has accomplished . . . 185
Comprehension exercises . . . 185
Challenge exercise . . . 188
Notes and references . . . 192

4 Divisors from Fans 193
4.1 Rays become prime divisors . . . 194
4.1.1 A brief divisor dictionary . . . 194
4.1.2 One divisor for every ray . . . 195
4.1.3 The toric boundary . . . 196
4.1.4 The fan remembers incidence . . . 197
4.1.5 The coordinate triangle in $\mathbb{P}^2$ . . . 198
4.1.6 A new ray creates the exceptional divisor . . . 199
4.2 Characters and principal divisors . . . 200
4.2.1 The divisor of a character . . . 200
4.2.2 First examples . . . 202
4.2.3 The coordinate characters on $\mathbb{P}^2$ . . . 203
4.2.4 Principal divisors and linear equivalence . . . 204
4.2.5 The blowup remembers total transforms . . . 204
4.3 Weil divisors and the divisor class group . . . 205
4.3.1 Invariant Weil divisors . . . 206
4.3.2 A matrix for the class group . . . 208
4.3.3 Three familiar complete surfaces . . . 209
4.3.4 The blowup of $\mathbb{P}^2$ . . . 210
4.3.5 A singular comparison: torsion . . . 211
4.4 Cartier divisors and local equations . . . 212

4.4.1 From multiplicities to local equations . . . 212
4.4.2 Cartier data on a fan . . . 213
4.4.3 A coordinate line on $\mathbb{P}^2$ . . . 215
4.4.4 A Weil divisor that is not Cartier . . . 216
4.4.5 The exceptional divisor on the blowup . . . 217
4.4.6 Cartier classes . . . 218
4.5 Support functions . . . 218
4.5.1 One function, linear on every cone . . . 218
4.5.2 Cartier divisors and integral support functions . . . 219
4.5.3 The support function of a line in $\mathbb{P}^2$ . . . 221
4.5.4 Linear equivalence means changing the global tilt . . . 222
4.5.5 Surface laboratory: the exceptional divisor . . . 223
4.6 The canonical divisor and singularity criteria . . . 224
4.6.1 The canonical divisor is minus the boundary . . . 224
4.6.2 Canonical classes of familiar surfaces . . . 226
4.6.3 Factorial and $\mathbb{Q}$-factorial toric varieties . . . 226
4.6.4 Gorenstein and $\mathbb{Q}$-Gorenstein cones . . . 227
4.6.5 Two singular comparisons . . . 228
4.6.6 A recognition table . . . 230
What this chapter has accomplished . . . 231
Comprehension exercises . . . 232
Challenge exercise . . . 235
Notes and references . . . 240

5 Projective Toric Varieties and Polytopes 241
5.1 Divisor polyhedra and global sections . . . 242
5.1.1 The monomial basis theorem . . . 243
5.1.2 Polyhedra need not be bounded . . . 245
5.1.3 Blowing up cuts off a corner . . . 246
5.2 Projective maps from lattice points . . . 248
5.2.1 From sections to a projective map . . . 249
5.2.2 Any finite set of lattice points gives a projective toric variety . . . 250
5.2.3 The first examples . . . 250
5.2.4 Very ample versus ample . . . 253
5.3 Convexity and positivity . . . 253
5.3.1 The distinguished section on an affine chart . . . 254
5.3.2 Convex support functions and global generation . . . 256
5.3.3 Strict convexity and ampleness . . . 258
5.3.4 Surface laboratory: the positivity interval on a blowup . . . 259
5.4 Polytopes and normal fans . . . 261
5.4.1 Which face does a direction see? . . . 261
5.4.2 The normal fan . . . 262
5.4.3 Recovering a variety from its vertex charts . . . 262
5.4.4 When the vertex coordinates give an embedding . . . 265
5.4.5 What changes the normal fan? . . . 267

5.4.6 The normal fan of a divisor polytope . . . 267
5.4.7 The morphism encoded by the normal fan . . . 268
5.5 Toric surfaces through their polygons . . . 270
5.5.1 A four-polygon catalogue . . . 270
5.5.2 Smoothness is visible at the corners . . . 271
5.5.3 Edge lengths are intersection numbers . . . 272
5.5.4 Supplementary example: the classroom trapezoid . . . 274
5.5.5 Area is self-intersection . . . 277
5.5.6 Blowing up cuts a corner . . . 278
5.5.7 A surface-recognition workflow . . . 279
What this chapter has accomplished . . . 280
Comprehension exercises . . . 281
Challenge exercise . . . 283
Notes and references . . . 286

6 Cox Coordinates and Quotient Constructions 287
6.1 Why homogeneous coordinates need a grading . . . 288
6.1.1 One scaling direction: projective space . . . 288
6.1.2 Two scaling directions: $\mathbb{P}^1 \times \mathbb{P}^1$ . . . 289
6.1.3 One variable for every boundary divisor . . . 291
6.1.4 Surface laboratory: the blowup grading . . . 292
6.2 The Cox ring and its homogeneous pieces . . . 294
6.2.1 One polynomial ring for all divisor classes . . . 294
6.2.2 A degree fiber is a section polytope . . . 295
6.2.3 Homogeneous pieces are spaces of sections . . . 296
6.2.4 Multiplication of sections becomes multiplication of monomials . . . 297
6.2.5 Surface laboratory: solving the blowup prediction . . . 298
6.3 The irrelevant ideal and the exceptional set . . . 300
6.3.1 The monomial attached to a cone . . . 300
6.3.2 Which coordinates are allowed to vanish together? . . . 301
6.3.3 Minimal forbidden collections . . . 302
6.3.4 Projective space: delete only the origin . . . 302
6.3.5 Two independent projective factors . . . 303
6.3.6 Surface laboratory: the blowup of $\mathbb{P}^2$ . . . 303
6.4 The quotient construction . . . 305
6.4.1 The rescaling group . . . 305
6.4.2 Recovering one affine chart . . . 306
6.4.3 Gluing the local quotients . . . 308
6.4.4 Surface laboratory: one chart of the blowup . . . 309
6.5 Four quotient constructions . . . 310
6.5.1 Weight matrices . . . 311
6.5.2 Projective space . . . 311
6.5.3 A product: $\mathbb{P}^1 \times \mathbb{P}^1$ . . . 312
6.5.4 A twist: the Hirzebruch surface $\mathbb{F}_r$ . . . 313
6.5.5 The blowup of $\mathbb{P}^2$ . . . 314

6.6 Singular and non-simplicial fans . . . 316
6.6.1 Which rescalings fix a Cox point? . . . 316
6.6.2 The smooth–simplicial–nonsimplicial trichotomy . . . 317
6.6.3 Finite stabilizers: a cyclic quotient surface . . . 318
6.6.4 Positive-dimensional stabilizers: the cone over a square . . . 319
What this chapter has accomplished . . . 322
Comprehension exercises . . . 323
Challenge exercise . . . 326
Notes and references . . . 330

II (More) Advanced Topics 331

7 Canonical Geometry and Reflexive Polytopes 333
7.1 Logarithmic forms and the toric boundary . . . 334
7.1.1 Logarithmically differentiating a character . . . 335
7.1.2 What happens at the boundary . . . 336
7.1.3 The logarithmic cotangent bundle is trivial . . . 338
7.1.4 Taking the top exterior power . . . 339
7.1.5 Two compactifications of a torus . . . 340
7.2 The anticanonical divisor and Fano varieties . . . 341
7.2.1 The anticanonical polytope . . . 342
7.2.2 The toric Fano criterion . . . 343
7.2.3 The first anticanonical polygons . . . 344
7.2.4 The Hirzebruch surfaces: ample, borderline, and negative . . . 344
7.3 Reflexive polytopes and polar duality . . . 346
7.3.1 The polar remembers supporting hyperplanes . . . 347
7.3.2 The anticanonical triangle and its polar . . . 348
7.3.3 Reflexive polytopes . . . 348
7.3.4 Faces reverse dimension . . . 350
7.3.5 The normal fan is already inside the polar . . . 351
7.4 Gorenstein Fano toric varieties . . . 352
7.4.1 Three levels of anticanonical positivity . . . 352
7.4.2 Local height-one data become polytope vertices . . . 352
7.4.3 Gorenstein Fano varieties and reflexive polytopes . . . 353
7.4.4 Weighted projective spaces . . . 354
7.4.5 A singular comparison: $\mathbb{P}(1,1,2)$ and $\mathbb{P}(1,1,3)$ . . . 355
7.4.6 Reflexive does not mean smooth . . . 356
7.5 Toric del Pezzo surfaces through their polygons . . . 357
7.5.1 The five smooth toric del Pezzo surfaces . . . 357
7.5.2 Degree, area, and boundary lattice points . . . 358
7.5.3 Toric blowups are anticanonical corner cuts . . . 359
What this chapter has accomplished . . . 361
Comprehension exercises . . . 362
Challenge exercise . . . 365

Notes and references . . . 369

8 Moment Polytopes and Topology 371
8.1 Compact torus actions and the moment map . . . 372
8.1.1 The compact torus inside the algebraic torus . . . 372
8.1.2 A projective toric variety with weighted coordinates . . . 373
8.1.3 The interval inside $\mathbb{P}^1$ . . . 375
8.1.4 Triangles, rectangles, and corner cuts . . . 376
8.1.5 What information has the map forgotten? . . . 378
8.2 The polytope as an orbit space . . . 378
8.2.1 Faces and torus strata . . . 378
8.2.2 The compact stabilizer of a face . . . 379
8.2.3 Every fiber is one compact-torus orbit . . . 380
8.2.4 A toric surface over its polygon . . . 381
8.2.5 Reconstructing the space from collapsing tori . . . 382
8.2.6 What changes in the singular case? . . . 383
8.3 One-parameter flows, cells, and Betti numbers . . . 384
8.3.1 One vector, two kinds of motion . . . 384
8.3.2 Attracting sets at the fixed points . . . 385
8.3.3 Reading the cells from a polygon . . . 386
8.3.4 Odd cohomology vanishes . . . 387
8.3.5 The h-vector packages the count . . . 387
8.3.6 Toric surfaces and blowups . . . 388
8.4 The cohomology ring from the fan . . . 390
8.4.1 Invariant divisors as degree-two classes . . . 390
8.4.2 Cox variables and cohomology variables . . . 392
8.4.3 A four-step computation . . . 392
8.4.4 The projective plane . . . 392
8.4.5 The product of two projective lines . . . 393
8.4.6 The blowup of the projective plane . . . 394
8.4.7 From ring products to intersection numbers . . . 395
What this chapter has accomplished . . . 397
Comprehension exercises . . . 398
Challenge exercise . . . 400
Notes and references . . . 404

9 Localization and Toric Riemann–Roch 405
9.1 Chern characters, Todd classes, and Riemann–Roch . . . 406
9.1.1 The tangent bundle from the boundary divisors . . . 406
9.1.2 The Chern character is an exponential . . . 407
9.1.3 The Todd class is the correction factor . . . 408
9.1.4 Hirzebruch–Riemann–Roch . . . 409
9.1.5 The first calculation: line bundles on $\mathbb{P}^1$ . . . 410
9.1.6 Surfaces: Riemann–Roch becomes intersection theory . . . 410
9.1.7 A triangle already knows Riemann–Roch . . . 411

9.2 Equivariant cohomology and piecewise polynomials . . . 412
9.2.1 The Borel construction: remembering the action . . . 412
9.2.2 What a toric equivariant class looks like . . . 415
9.2.3 The first picture: $\mathbb{P}^1$ . . . 417
9.2.4 Degree two: our old support functions return . . . 418
9.2.5 Forgetting the action . . . 419
9.2.6 A full two-dimensional example: $\mathbb{P}^2$ . . . 420
9.3 Fixed points and Atiyah–Bott localization . . . 422
9.3.1 The tangent weights at a fixed point . . . 422
9.3.2 Equivariant integration . . . 423
9.3.3 The first localization calculation: $\mathbb{P}^1$ . . . 424
9.3.4 A two-dimensional calculation: $\mathbb{P}^2$ . . . 425
9.3.5 Why this is exactly the tool Riemann–Roch needs . . . 426
9.4 Toric Riemann–Roch and lattice points . . . 427
9.4.1 The equivariant Euler characteristic remembers the weights . . . 427
9.4.2 The one-dimensional identity in full . . . 429
9.4.3 The toric Riemann–Roch counting formula . . . 429
9.4.4 A corner cut: two computations on $\mathrm{Bl}_p(\mathbb{P}^2)$ . . . 430
9.4.5 Dilating the polytope . . . 432
What this chapter has accomplished . . . 434
Comprehension exercises . . . 435
Challenge exercise . . . 437
Notes and references . . . 440

10 VGIT and Toric Birational Geometry 441
10.1 One Cox quotient, many linearizations . . . 442
10.1.1 Why the affine quotient is not enough . . . 442
10.1.2 Semistable points and the GIT quotient . . . 443
10.1.3 An ample class recovers the Cox quotient . . . 445
10.1.4 Surface laboratory: the same action gives $\mathbb{F}_1$ and $\mathbb{P}^2$ . . . 445
10.1.5 What lives on the wall? . . . 447
10.2 The secondary fan and the secondary polytope . . . 448
10.2.1 From weight cones to chambers . . . 448
10.2.2 The $\mathbb{F}_1$ picture really is a fan . . . 450
10.2.3 Where triangulations enter: Gale duality . . . 451
10.2.4 The secondary polytope . . . 452
10.2.5 A square: two triangulations and one wall . . . 453
10.3 Crossing a wall: contractions, flips, and fibrations . . . 454
10.3.1 The three geometric possibilities . . . 455
10.3.2 Surface laboratory: the two boundary rays of $\mathrm{Nef}(\mathbb{F}_1)$ . . . 456
10.3.3 Why a genuine flip first appears in dimension three . . . 457
10.3.4 Reading a wall crossing in four steps . . . 459
10.4 Extremal rays and contractions . . . 459
10.4.1 Divisors and curves modulo numerical equivalence . . . 460
10.4.2 The toric cone theorem . . . 461

10.4.3 Wall relations compute curve classes . . . 462
10.4.4 A nef facet produces an extremal contraction . . . 463
10.4.5 Surface laboratory: the dual cones of $\mathbb{F}_1$ . . . 464
10.4.6 The conifold relation revisited . . . 465
10.4.7 Nef chambers inside the secondary fan . . . 466
10.5 The toric minimal model program . . . 466
10.5.1 The algorithm: contract, flip, or stop . . . 467
10.5.2 The MMP as a directed path through the secondary fan . . . 468
10.5.3 A complete run on $\mathbb{F}_1$ . . . 469
10.5.4 Flips are directed; flops are not MMP steps . . . 470
10.5.5 The chapter's final dictionary . . . 471
What this chapter has accomplished . . . 473
Comprehension exercises . . . 474
Challenge exercise . . . 476
A final perspective . . . 479
Notes and references . . . 480

Index **481**

Preface

These notes were prepared for Renzo's Fall 2026 course at Colorado State University. Their organizing premise is that toric geometry is best learned by moving repeatedly among three languages:

$$\boxed{\text{algebra} \quad \longleftrightarrow \quad \text{geometry} \quad \longleftrightarrow \quad \text{combinatorics.}}$$

Characters become monomials, cones become affine charts, rays become divisors, and lattice points become sections. None of these translations is merely decorative: each language makes visible features that are difficult to see in the other two.

The book is divided into two parts.

Part I, Basic Concepts (Chapters 0–6), builds the three dictionaries themselves. Chapter 0 is a panoramic return to $\mathbb{P}^2$: it introduces most of the recurring constructions in one familiar example before the later chapters separate them into definitions and theorems. Chapters 1–6 then pass from tori and affine toric varieties to fans, divisors, polytopes, and Cox coordinates. By the end of Part I, the reader can translate a basic question among the algebraic, geometric, and combinatorial languages and choose the one in which it is easiest to solve.

Part II, (More) Advanced Topics (Chapters 7–10), puts those dictionaries to work. Its chapters study canonical and Fano geometry, moment maps and topology, localization and Riemann–Roch, and finally variation of quotients and the toric minimal model program. The word "more" is intentional: these topics use additional machinery, but the calculations remain grounded in the same cones, fans, divisors, lattice points, and homogeneous coordinates developed in Part I. The division is therefore a change of emphasis rather than a hard boundary. Part I builds the translation tools; Part II uses them to organize deeper geometry.

The notes are deliberately conversational. Chapter overviews state the central question; dictionary boxes keep the three languages in view; surface laboratories make the general theory calculable; and pause, prediction, and guiding-question boxes invite the reader to anticipate the next step. Every chapter closes with a summary, comprehension exercises, a cumulative challenge, and notes and references. The intended rhythm is to compute first, translate second, and only then compress the pattern into a theorem.

The notation below records the conventions used throughout. When a chapter introduces temporary coordinates or labels for a particular example, those local choices are stated explicitly. The index at the end of the book collects the main constructions, theorems, and recurring examples; entries under a broad term often point to more specific subentries.

These notes were developed with substantial expository, editorial, and typesetting assistance from OpenAI's ChatGPT.

How this class is going to run

I don't want this course to be a standard "I lecture you listen" kind of class; at the same time I am wanting to cover quite a bit of material in this one semester that we have, so I have thought about the following shared-responsibility structures, one for the basic concepts, one for the advanced topics.

Basic concepts

For each of the basic concepts chapters, I will first lecture for a few hours, aiming to cover all the important concepts covered in the chapter. For each chapter, the last two hours will be the responsibility of a group of students. These two hours should not be a repetition of my previous lectures - but rather the opportunity to fill in gaps I might have left, or to go into more depth into the material. Some of the things you can do during this time are:

- present examples of concepts;
- work out significant exercises - both in the comprehension and the challange category;
- ask me questions about things are still not clear (but be ready to reason about the questions you ask);
- describe to the class modifications to the book you have made (more on this later).

But really, you can do anything which is meaningful for the goal of the class, which is understanding the relevant concepts in toric geometry. If you are in doubt if your plan is legit, just check with me. And the assignment of the chapters is - drum roll...

Chapter 1: Brendan, Kandin, Vivian;

Chapter 2: Ayodeji, Eamon, Kobe, Ryan;

Chapter 3: Derek, Makenna, Ross;

Chapter 4: Derek, Eamon, Kobe;

Chapter 5: Ayodeji, Brendan, Makenna, Ryan;

Chapter 6: Kandin, Ross, Vivian.

Advanced topics

The four chapters of advanced topics, you will be completely in charge. We will devote one week (= three hours) to each chapter, and you will be leading the class the whole time. You shouldn't think of just covering all the material in the chapter, but make a selection that makes sense for three hours. Of course, I do encourage you to work with me throughout the semester to prepare for your advanced topics week. Do not leave it last minute. While the assignment of topics to people in the basic concepts part was more or less random, here I am trying to match topics with people's interest. If I am mistaken, and some of you would like to trade topics with someone else, no problem, just let me know.

Chapter 7: Derek, Kandin, Kobe;

Chapter 8: Makenna, Ryan, Vivian;

Chapter 9: Ayodeji, (Ignacio), Ross;

Chapter 10: Brendan, Eamon, (Jacob).

The book

Over the last month I had quite a lot of conversations with ChatGPT, and came up with this book. It is a bit chatty and not super efficient, but I think it overall does a good job of charting an interesting initial path through all the material covered in Cox-Little-Schenck, with a lot of examples and pictures and emphasizing the conceptual rather than the computational aspects of the theory.

While I did put a bit of effort trying to catch missteps, I make no claim of having produced an error free product, and also - more simply - this is the book that **I** came up with. I would like by the end of the semester for this book to be **our** book. So you are encouraged - especially for the chapters you are responsible for - but not necessarily only for those, to proofread and make modification to the textbook so that it serves the best possible purpose for your learning.

AI usage

AI is here and we can't put our head in the sand about it, but rather we need to learn to use it as a tool for better learning (and not as a substitute for learning). You are totally welcome to use AI to work on the book. You are totally welcome to have conversations with AI as starting points for your understanding. But there are two things I ask of you:

1. don't let the conversations with AI completely replace conversations with your peers and with me; use your AI conversations as a starting point, but always use another human as the endpoint of your understanding - even if it is just to confirm what you have understdood chatting with your favorite AI;
2. with the exercises, especially the comprehension exercises, spend a little time on your own (or with human friends) before you turn for help to AI. There is value and good training even in being confused and making unsuccessful attempts at solving problems. If and when you then turn to AI for help, try to have a strategy, and to ask questions that lead towards following that strategy, or understanding why it doesn't work and devising a better one, rather than just asking for a solution.

Why does all of this matter anyway?

Evaluation

Grades - ugh - don't we all hate them? I do. Let me premise that I want to give an A to everyone in this class...but (there is always a but) I do want to make you work for it, with the payoff being that you learn toric geometry. So here is what the evaluation will be based on:

1. the two basic concept chapters you are responsible for;
2. the advanced topic presentation you are responsible for;
3. one meeting (say 30sh minutes) towards the end of the semester, in which we will have a conversation about some of the comprehension exercises in the book. I don't need you to be able to cleanly solve the random exercise I choose, but I do need you to have an idea for how to tackle it, and to be able to describe it to me. The idea with this format is that throughout the semester you should have a look at each of the comprehension exercises, and get to the point of knowing what they are asking and what the first couple steps towards a solution would be.

Conventions and notation

Unless a chapter says otherwise, $\Bbbk$ is an algebraically closed field. Chapters 8–10 work over $\mathbb{C}$ whenever classical topology, equivariant cohomology, or symplectic language is used.

Notation	**Meaning**
$T = T_N$	an algebraic torus
$M = X^*(T)$	the character lattice
$N = X_*(T) = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$	the lattice of one-parameter subgroups
χ^m, λ^u	the character associated to $m \in M$, and the one-parameter subgroup associated to $u \in N$
$\langle m, u\rangle$	the perfect pairing $M \times N \to \mathbb{Z}$, characterized by $\chi^m(\lambda^u(t)) = t^{\langle m,u\rangle}$
$M_{\mathbb{R}}, N_{\mathbb{R}}$	the real scalar extensions of the two lattices
$\sigma^\vee$	the dual cone of $\sigma \subseteq N_{\mathbb{R}}$
U_σ	the affine toric variety $\operatorname{Spec} \Bbbk[\sigma^\vee \cap M]$
Σ, X_Σ	a fan in $N_{\mathbb{R}}$, and its toric variety
$\Sigma(1)$	the set of rays of Σ
u_ρ	the primitive lattice generator of a ray $\rho \in \Sigma(1)$
$O(\sigma)$, $V(\sigma)$	the orbit associated to σ, and its closure
D_ρ	the invariant prime divisor corresponding to ρ
$\mathrm{Cl}(X)$, $\mathrm{Pic}(X)$	the divisor class group and Picard group
$P_D \subseteq M_{\mathbb{R}}$	the divisor polyhedron $\{m : \langle m, u_\rho\rangle \geq -a_\rho\}$ for $D = \sum_\rho a_\rho D_\rho$
ψ_D	the support function normalized by $\psi_D(u_\rho) = -a_\rho$
$\overline{\mathrm{NE}}(X)$, $\mathrm{Nef}(X)$	the Mori cone of curves and the nef cone of divisors

We write $\mathrm{Cone}(u_1, \ldots, u_r)$ for the convex polyhedral cone generated by the displayed vectors and $\mathrm{Conv}\{m_1, \ldots, m_s\}$ for their convex hull. The symbol u_ρ is reserved for a primitive ray generator; letters such as u_i, v_i, w_i may be used locally when an example comes with an ordered list of rays or weights.

Part I

Basic Concepts

Chapter 0

$\mathbb{P}^2$ Reloaded

Chapter overview

In this chapter we revisit the projective plane from three perspectives: its affine charts and homogeneous coordinate ring, its torus action, and the combinatorics of a lattice triangle and its normal fan. We also see for the first time that monomial sections of the line bundles $\mathcal{O}_{\mathbb{P}^2}(d)$ are indexed by the lattice points of a dilated triangle. The goal is not yet to develop the general theory, but to encounter its central correspondences in a familiar example.

0.1 The affine model: $\mathbb{A}^2$

Our first example is the affine plane. Fix a field $\Bbbk$, and write

$$\mathbb{A}^2 = \operatorname{Spec} \Bbbk[x, y].$$

The symbols x and y are coordinate functions. Once they have been chosen, every $\Bbbk$-point p determines a pair

$$(x(p), y(p)) \in \Bbbk^2,$$

and this gives the familiar identification

$$\mathbb{A}^2(\Bbbk) \simeq \Bbbk^2.$$

The notation $\mathbb{A}^2$, however, remembers more than the set $\Bbbk^2$: it remembers which functions are polynomial and which subsets are cut out by polynomial equations. This extra structure is the algebraic geometry of the affine plane.

Guiding question

When $\Bbbk = \mathbb{C}$, what is the difference between $\mathbb{A}^2_{\mathbb{C}}$ and $\mathbb{C}^2$?

Their sets of complex points are naturally identified. The notation $\mathbb{C}^2$, however, often asks us to remember its structure as a complex vector space or a complex manifold, together with the Euclidean topology and holomorphic functions. The notation $\mathbb{A}^2_{\mathbb{C}}$ asks us instead to remember the Zariski topology and the polynomial functions. The points are the same; the chosen geometric structure is different.

0.1.1 The coordinate strata

The coordinates x and y single out the two coordinate axes,

$$V(x) = \{x = 0\}, \qquad V(y) = \{y = 0\},$$

and their intersection, the origin

$$0 = V(x, y).$$

According to which coordinates vanish, the affine plane breaks into four disjoint pieces:

$$\mathbb{A}^2 = T \sqcup (V(x) \setminus \{0\}) \sqcup (V(y) \setminus \{0\}) \sqcup \{0\},$$

where

$$T = \mathbb{A}^2 \setminus V(xy) = \{(a, b) : a \neq 0,\ b \neq 0\}.$$

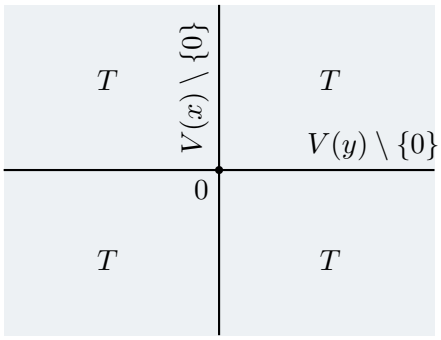


Figure 1: A schematic real-coordinate picture of the four strata of $\mathbb{A}^2$. The shading represents the open set T; over a general field, the picture should be read combinatorially rather than literally.

On T, both coordinate functions are nonzero and hence invertible. Its coordinate ring is therefore

$$\Bbbk[T] = \Bbbk[x^{\pm 1}, y^{\pm 1}],$$

so

$$T \simeq \mathbb{G}_m^2.$$

We call T the *dense torus* of $\mathbb{A}^2$. It is open because it is the complement of $V(xy)$, and it is dense because $\mathbb{A}^2$ is irreducible and T is nonempty.

The name "torus" reflects the coordinatewise multiplication law

$$(t_1, t_2)(s_1, s_2) = (t_1 s_1, t_2 s_2).$$

Using this law, T acts on the whole affine plane by rescaling:

$$(t_1, t_2) \cdot (a, b) = (t_1 a, t_2 b).$$

The four pieces above are exactly the four orbits of this action. The open set T is the largest orbit; each punctured coordinate axis is a one-dimensional orbit; and the origin is fixed. Thus the boundary of the torus already comes naturally organized into strata.

0.1.2 Monomial functions and lattice points

The polynomial functions most naturally adapted to coordinatewise rescaling are the monomials

$$x^a y^b, \qquad a, b \in \mathbb{Z}_{\geq 0}.$$

Every such monomial is regular on $\mathbb{A}^2$, and none vanishes at a point of T. Its two exponents provide a compact combinatorial label, so we record the monomial $x^a y^b$ by the lattice point

$$(a, b) \in M := \mathbb{Z}^2.$$

The monomials in $\Bbbk[x, y]$ occupy precisely the lattice points in the nonnegative quadrant:

$$\mathbb{Z}^2_{\geq 0} \subseteq M_{\mathbb{R}} \simeq \mathbb{R}^2.$$

The corresponding real cone is

$$\sigma^\vee = \mathbb{R}^2_{\geq 0}.$$

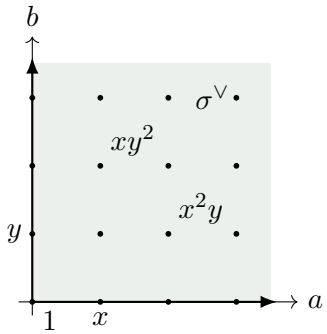


Figure 2: The lattice point (a, b) labels the monomial $x^a y^b$. The lattice points of $\sigma^\vee$ are exactly the monomials regular on $\mathbb{A}^2$.

The significance of the quadrant becomes clearer if we begin on the dense torus. Since x and y are invertible on T, every Laurent monomial

$$x^a y^b, \qquad (a, b) \in \mathbb{Z}^2,$$

is regular there. Such a Laurent monomial extends across the boundary to a regular function on all of $\mathbb{A}^2$ precisely when

$$a \geq 0 \qquad \text{and} \qquad b \geq 0.$$

Indeed, a negative exponent of x produces a pole along $V(x)$, and a negative exponent of y produces a pole along $V(y)$.

This gives a useful way to recover the affine plane from its dense torus: retain exactly those Laurent monomials whose exponent vectors lie in $\sigma^\vee$. In symbols,

$$\Bbbk[x, y] = \bigoplus_{(a,b) \in \sigma^\vee \cap M} \Bbbk\, x^a y^b.$$

The cone is therefore not merely a picture attached to the coordinate ring; it records exactly which monomial functions survive when the boundary is added.

Observation 0.1. In the representation theory of algebraic tori, a Laurent monomial $x^a y^b$ is called a *character* of T, and M is called the *character lattice*. No representation theory is required for the discussion here. For now, M may simply be regarded as the lattice indexing Laurent monomials on T.

0.1.3 Approaching the boundary along monomial curves

There is another first quadrant, obtained by looking at simple curves in the torus rather than functions on it. For an integer pair $(u, v) \in \mathbb{Z}^2$, consider

$$\mathbb{G}_m \longrightarrow T, \qquad t \longmapsto (t^u, t^v).$$

We ask whether this curve acquires a value at $t = 0$. Algebraically, this means asking whether the map extends to a morphism

$$\mathbb{A}^1 \longrightarrow \mathbb{A}^2.$$

The extension exists exactly when

$$u \geq 0 \qquad \text{and} \qquad v \geq 0.$$

If both exponents are nonnegative, the two coordinates t^u and t^v are polynomial functions of t. If either exponent is negative, the corresponding coordinate has a pole at $t = 0$.

The position of (u, v) in the quadrant determines where the extended curve meets the boundary:

position of (u, v)	value at $t = 0$
$u > 0,\ v > 0$	$(0, 0)$
$u > 0,\ v = 0$	$(0, 1) \in V(x) \setminus \{0\}$
$u = 0,\ v > 0$	$(1, 0) \in V(y) \setminus \{0\}$
$u = v = 0$	$(1, 1) \in T$.

Thus the relative interior of the two-dimensional quadrant points toward the origin, the relative interior of each ray points toward one of the punctured coordinate axes, and the zero-dimensional face points toward the torus itself.

The particular points $(0, 1)$, $(1, 0)$, and $(1, 1)$ arise because our curve begins at the identity of the torus. More generally, for $(c_1, c_2) \in T$, consider the translated curve

$$t \longmapsto (c_1 t^u, c_2 t^v).$$

If $u > 0$ and $v = 0$, its value at 0 is $(0, c_2)$, and every point of $V(x) \setminus \{0\}$ occurs in this way. Similarly, the other ray produces all of $V(y) \setminus \{0\}$. The interior always produces the origin. In this precise sense, the faces of the quadrant encode the four strata of $\mathbb{A}^2$.

The exponent vectors of the extendable curves form the cone

$$\sigma = \mathbb{R}^2_{\geq 0} \subseteq N_{\mathbb{R}}, \qquad N := \mathbb{Z}^2.$$

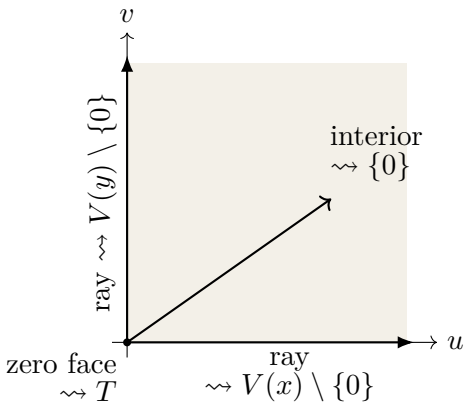


Figure 3: Faces of the cone of extendable monomial curves correspond to the torus-orbit strata of $\mathbb{A}^2$. The arrows indicate the exponent vectors (u, v), not paths in the affine plane.

Although σ and $\sigma^\vee$ are both drawn as first quadrants in our chosen coordinates, they answer different questions: $\sigma^\vee$ records monomial functions that extend to $\mathbb{A}^2$, while σ records monomial curves that extend to $\mathbb{A}^2$.

The connection between the two is visible in one calculation. Evaluate the monomial $x^a y^b$ along the curve $t \mapsto (t^u, t^v)$:

$$x^a y^b(t^u, t^v) = t^{au+bv}.$$

The integer $au + bv$ records the behavior of the function along the curve. A positive value means that it vanishes at $t = 0$, a zero value means that it remains nonzero, and a negative value means that it has a pole.

Observation 0.2. The curves $t \mapsto (t^u, t^v)$ are called *one-parameter subgroups* of T, and N is called the *lattice of one-parameter subgroups*. The integer $au + bv$ is written

$$\langle (a, b), (u, v) \rangle .$$

This is the natural pairing between the two exponent lattices M and N. The terminology is useful, but the essential content is the elementary identity

$$x^a y^b(t^u, t^v) = t^{au+bv}.$$

We can now state the duality precisely:

$$\sigma^\vee = \{m \in M_{\mathbb{R}} : \langle m, n \rangle \geq 0 \text{ for every } n \in \sigma\} .$$

Indeed, a monomial indexed by m extends across every monomial curve indexed by σ exactly when all of these pairings are nonnegative. The two cones look alike in the chosen coordinates, but they live in paired vector spaces and encode complementary aspects of the affine plane.

The three dictionaries

For the affine plane, our first dictionary is

Algebra	Geometry	Combinatorics
$x^a y^b \in \Bbbk[x,y]$	a monomial extending across the boundary	$(a,b) \in \sigma^\vee \cap M$
$t \mapsto (t^u, t^v)$	a curve acquiring a boundary value	$(u,v) \in \sigma \cap N$
$x^a y^b(t^u, t^v) = t^{au+bv}$	order along the curve	$\langle (a,b),(u,v)\rangle = au+bv.$

0.2 What we already know about $\mathbb{P}^2$

We now pass from the affine plane to the projective plane. The point of this review is not merely to repeat definitions we already know. Instead, we isolate a few familiar features of $\mathbb{P}^2$ that will soon acquire a toric interpretation: its three affine charts, its homogeneous coordinates, and the triangle formed by the coordinate lines.

0.2.1 Lines through the origin

For the moment, think of the $\Bbbk$-points of $\mathbb{P}^2$ as one-dimensional linear subspaces of $\Bbbk^3$. A nonzero vector

$$(x_0, x_1, x_2) \in \Bbbk^3$$

determines the line that it spans, and multiplying the vector by a nonzero scalar does not change that line. Thus

$$\mathbb{P}^2(\Bbbk) = \frac{\Bbbk^3 \setminus \{0\}}{\Bbbk^\times}.$$

We write the equivalence class of (x_0, x_1, x_2) as

$$[x_0 : x_1 : x_2].$$

These are the *homogeneous coordinates* on $\mathbb{P}^2$. By definition,

$$[x_0 : x_1 : x_2] = [\lambda x_0 : \lambda x_1 : \lambda x_2] \qquad \text{for every } \lambda \in \Bbbk^\times,$$

and the triple $(0,0,0)$ is not permitted.

A projective point therefore records a direction in $\Bbbk^3$, rather than a vector with a specified size. One may choose a convenient representative of the direction, but there is usually no preferred representative valid for every point at once.

0.2.2 The standard affine charts

Suppose first that $x_0 \neq 0$. We may rescale the homogeneous coordinates by x_0^{-1} and write the point uniquely in the form

$$[1 : z_1 : z_2], \qquad z_1 = \frac{x_1}{x_0}, \quad z_2 = \frac{x_2}{x_0}.$$

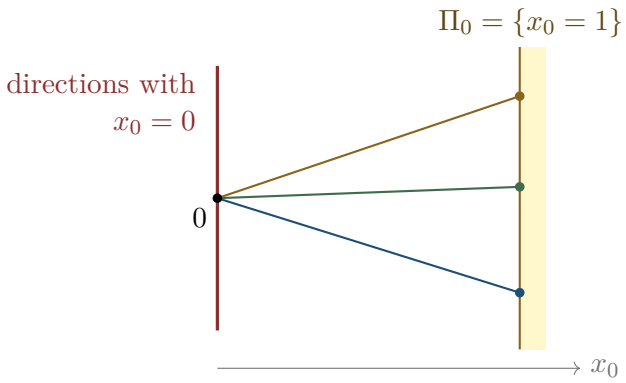


Figure 4: Projectivization in a two-dimensional schematic. A line through the origin with $x_0 \neq 0$ meets the normalization slice $\Pi_0 = \{x_0 = 1\}$ exactly once. A direction contained in $\{x_0 = 0\}$ never meets that slice.

This identifies the open subset

$$U_0 = \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 : x_0 \neq 0\}$$

with the affine plane:

$$\phi_0 : U_0 \longrightarrow \mathbb{A}^2,$$
$$[x_0 : x_1 : x_2] \longmapsto \left(\frac{x_1}{x_0}, \frac{x_2}{x_0}\right).$$

The inverse map is

$$(a, b) \longmapsto [1 : a : b].$$

There is nothing special about x_0. Repeating the construction with the other two coordinates gives

$$U_i = \{x_i \neq 0\} \simeq \mathbb{A}^2, \qquad i = 0, 1, 2.$$

Every projective point has at least one nonzero coordinate, so

$$\mathbb{P}^2 = U_0 \cup U_1 \cup U_2.$$

Thus $\mathbb{P}^2$ is covered by three copies of the affine plane.

The complement of U_0 is the projective line

$$D_0 = \{x_0 = 0\} \simeq \mathbb{P}^1.$$

Relative to the chart U_0, this is the *line at infinity*. Hence one may think schematically of

$$\mathbb{P}^2 = \mathbb{A}^2 \sqcup \mathbb{P}^1,$$

where the affine plane is U_0 and the projective line is D_0. This description depends on the choice of x_0: choosing another chart produces another affine plane and another line at infinity.

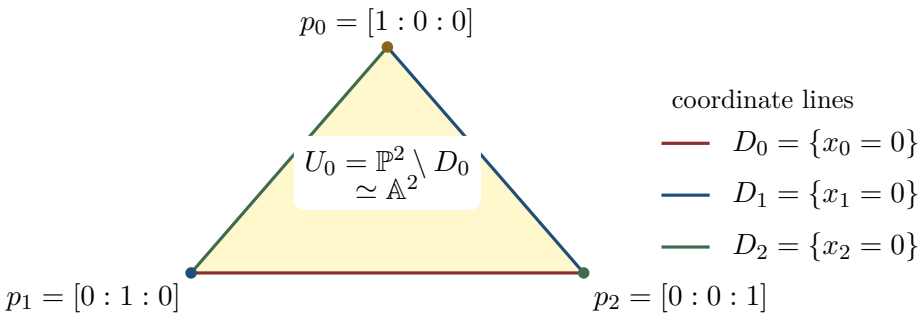


Figure 5: The coordinate triangle. The three colors distinguish the coordinate lines. In this schematic, the pale yellow region indicates the chart U_0, obtained by removing the red side D_0.

0.2.3 Changing affine coordinates

The affine charts overlap, and their coordinates are related by explicit formulas. On U_0, write

$$[x_0 : x_1 : x_2] = [1 : z_1 : z_2].$$

On U_1, write the same point as

$$[x_0 : x_1 : x_2] = [w_0 : 1 : w_2].$$

A point lies in $U_0 \cap U_1$ exactly when $z_1 \neq 0$. To pass from the U_0-coordinates to the U_1-coordinates, we simply renormalize so that the second homogeneous coordinate becomes 1:

$$[1 : z_1 : z_2] = \left[\frac{1}{z_1} : 1 : \frac{z_2}{z_1}\right].$$

Thus the transition map is

$$(z_1, z_2) \longmapsto \left(\frac{1}{z_1}, \frac{z_2}{z_1}\right).$$

The inverse is obtained by the same kind of renormalization:

$$(w_0, w_2) \longmapsto \left(\frac{1}{w_0}, \frac{w_2}{w_0}\right).$$

This simple calculation contains an important lesson about projective space. Each chart has two ordinary coordinates, but no single pair of affine coordinates works everywhere. On overlaps, the coordinates are related by rational expressions whose denominators are nonzero precisely where the two charts meet.

0.2.4 Homogeneous equations

Because homogeneous coordinates are defined only up to a common nonzero scalar, not every polynomial expression gives a well-defined condition on projective points. The appropriate equations are homogeneous polynomials.

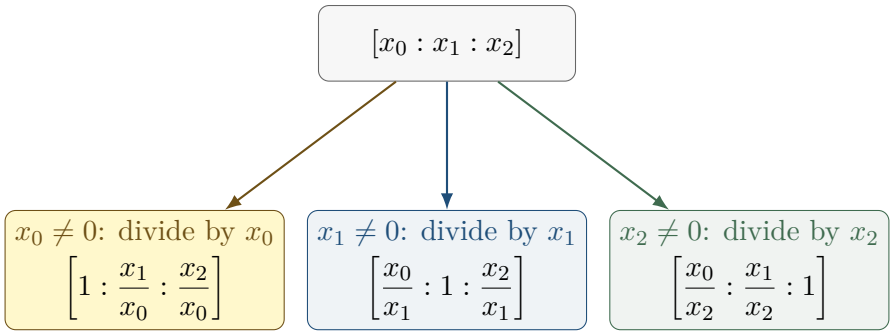


Figure 6: The three affine charts arise from the same operation: choose a nonzero homogeneous coordinate and normalize it to 1. Transition maps are obtained by normalizing the same point in a different way.

A polynomial $F(x_0, x_1, x_2)$ is homogeneous of degree d if

$$F(\lambda x_0, \lambda x_1, \lambda x_2) = \lambda^d F(x_0, x_1, x_2)$$

for every scalar λ. It follows that

$$F(x_0, x_1, x_2) = 0$$

is independent of the chosen representative of $[x_0 : x_1 : x_2]$. For example, a homogeneous linear equation cuts out a projective line, while a homogeneous quadratic equation cuts out a projective conic.

The polynomial ring

$$S = \Bbbk[x_0, x_1, x_2]$$

is naturally graded by total degree:

$$S = \bigoplus_{d \geq 0} S_d,$$

where S_d is the vector space of homogeneous polynomials of degree d. This is the *homogeneous coordinate ring* of $\mathbb{P}^2$. The grading is essential: it records how an expression changes when the homogeneous coordinates are simultaneously rescaled.

Observation 0.3. The phrase "coordinate ring" is being used differently here than it was for $\mathbb{A}^2$. The ring $\Bbbk[x, y]$ is the ring of global regular functions on $\mathbb{A}^2$. By contrast, $\Bbbk[x_0, x_1, x_2]$ is not the ring of global regular functions on $\mathbb{P}^2$; its grading encodes projective equations and, later, sections of line bundles. This distinction is one reason homogeneous coordinates are richer than ordinary affine coordinates.

0.2.5 The coordinate triangle

The three homogeneous coordinate functions define three projective lines

$$D_0 = V(x_0), \qquad D_1 = V(x_1), \qquad D_2 = V(x_2).$$

We call their union

$$D_0 \cup D_1 \cup D_2 = V(x_0 x_1 x_2)$$

the *coordinate triangle*. Its vertices are the three coordinate points

$$p_0 = [1 : 0 : 0], \qquad p_1 = [0 : 1 : 0], \qquad p_2 = [0 : 0 : 1].$$

More precisely,

$$p_0 = D_1 \cap D_2, \qquad p_1 = D_0 \cap D_2, \qquad p_2 = D_0 \cap D_1.$$

There is no common intersection of all three coordinate lines, because $[0 : 0 : 0]$ is not a projective point.

The coordinate triangle organizes $\mathbb{P}^2$ according to which homogeneous coordinates vanish:

vanishing pattern	resulting piece
$x_0x_1x_2 \neq 0$	one open piece
exactly one $x_i = 0$	three punctured coordinate lines
exactly two $x_i = 0$	the three coordinate points.

Thus the coordinate decomposition of $\mathbb{P}^2$ has seven pieces. In the next section, we will see that these are not merely convenient subsets: they are the natural strata for a two-dimensional torus action.

Guiding question

What remains after the coordinate triangle $V(x_0x_1x_2)$ is removed from $\mathbb{P}^2$, and how is it related to the dense torus in $\mathbb{A}^2$?

0.3 The dense torus in $\mathbb{P}^2$

At the end of the previous section, we decomposed $\mathbb{P}^2$ according to which homogeneous coordinates vanish. The largest piece is the one on which no coordinate vanishes:

$$T = \mathbb{P}^2 \setminus V(x_0x_1x_2) = \{[x_0 : x_1 : x_2] : x_0x_1x_2 \neq 0\}.$$

We now take a closer look at this open set. It will play the same role in $\mathbb{P}^2$ that $\mathbb{G}_m^2$ played inside $\mathbb{A}^2$.

Throughout this chapter, the phrase *coordinate triangle* means the union

$$D_0 \cup D_1 \cup D_2 = V(x_0x_1x_2)$$

of the three coordinate lines. We are not asserting that this union is a triangle in any Euclidean metric on $\mathbb{P}^2$. The terminology is combinatorial: the three lines meet pairwise in three points, producing the incidence pattern of three sides and three vertices.

0.3.1 Removing the coordinate triangle

The condition $x_0x_1x_2 \neq 0$ means that all three homogeneous coordinates are nonzero. In particular, every point of T lies in the chart U_0. After dividing by x_0, it has a unique representative

$$[1 : z_1 : z_2] \qquad \text{with} \qquad z_1, z_2 \in \Bbbk^\times.$$

Therefore the map

$$T \longrightarrow \mathbb{G}_m^2,$$
$$[x_0 : x_1 : x_2] \longmapsto \left(\frac{x_1}{x_0}, \frac{x_2}{x_0}\right)$$

is an isomorphism, with inverse

$$(z_1, z_2) \longmapsto [1 : z_1 : z_2].$$

Thus

$$T \simeq \mathbb{G}_m^2.$$

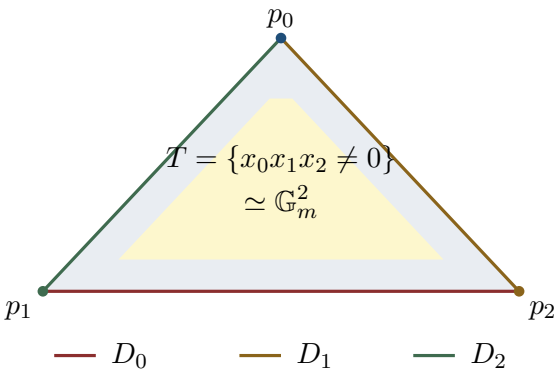


Figure 7: The dense torus is what remains after the three coordinate lines are removed. The drawing is only a combinatorial model: the "triangle" means the union $D_0 \cup D_1 \cup D_2$, not a Euclidean triangle sitting inside $\mathbb{P}^2$. The yellow region represents the single open set on which all three coordinates are nonzero.

This description also explains why the word *dense* is appropriate. The complement of T is the union of the three coordinate lines—what we are calling the coordinate triangle—and this union is a proper closed subset of $\mathbb{P}^2$. Since $\mathbb{P}^2$ is irreducible, every nonempty open subset is dense.

Guiding question

Why do three nonzero homogeneous coordinates produce only a two-dimensional torus?

The answer is that homogeneous coordinates contain one redundant degree of freedom. A triple

$$(x_0, x_1, x_2) \in (\Bbbk^\times)^3$$

and its scalar multiple

$$(\lambda x_0, \lambda x_1, \lambda x_2)$$

represent the same projective point. Dividing by the common scalar leaves two independent ratios. On U_0, those ratios are

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}.$$

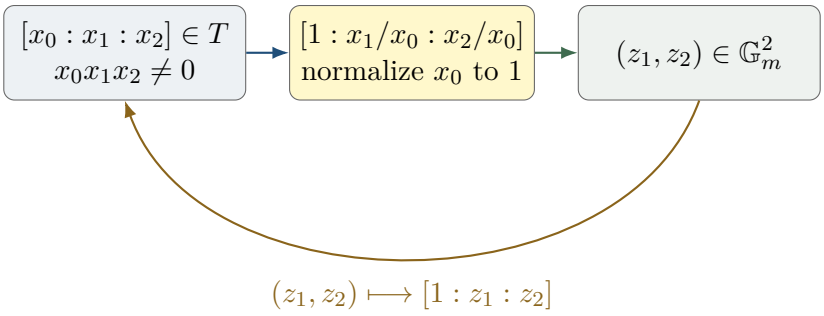


Figure 8: The identification $T \simeq \mathbb{G}_m^2$. Moving from left to right means normalizing x_0 to 1 and retaining the two independent ratios.

0.3.2 Multiplication inside the open set

The identification $T \simeq \mathbb{G}_m^2$ transports the usual coordinatewise multiplication on $\mathbb{G}_m^2$ to T. In normalized coordinates,

$$[1 : z_1 : z_2] \cdot [1 : w_1 : w_2] = [1 : z_1w_1 : z_2w_2].$$

The identity element is

$$e = [1 : 1 : 1],$$

and

$$[1 : z_1 : z_2]^{-1} = [1 : z_1^{-1} : z_2^{-1}].$$

So the open subset T is not merely isomorphic to a torus as a variety: it carries a natural algebraic group structure.

There is also a symmetric way to write this multiplication without choosing x_0 as the preferred coordinate. If

$$p = [x_0 : x_1 : x_2], \qquad q = [y_0 : y_1 : y_2]$$

belong to T, define

$$p \cdot q = [x_0y_0 : x_1y_1 : x_2y_2].$$

This is well defined. Indeed, rescaling p or q only rescales all three products by a common nonzero scalar. The identity for this description is again $[1 : 1 : 1]$.

Observation 0.4. A more representation-theoretic description is

$$T \simeq \mathbb{G}_m^3/\mathbb{G}_m,$$

where the copy of $\mathbb{G}_m$ in the denominator is embedded diagonally:

$$\lambda \longmapsto (\lambda, \lambda, \lambda).$$

This says exactly that three nonzero coordinates are being considered up to simultaneous rescaling. Nothing in this chapter requires the language of quotient groups; the two affine ratios x_1/x_0 and x_2/x_0 contain the same information.

0.3.3 Letting the torus move the whole projective plane

The multiplication on T extends to an action on all of $\mathbb{P}^2$. Using the coordinates $T \simeq \mathbb{G}_m^2$, define

$$(t_1, t_2) \cdot [x_0 : x_1 : x_2] = [x_0 : t_1 x_1 : t_2 x_2].$$

This formula makes sense even when some of the x_i vanish. It is also compatible with projective scaling: replacing $[x_0 : x_1 : x_2]$ by $[\lambda x_0 : \lambda x_1 : \lambda x_2]$ changes the output by the same common scalar λ.

On the open set T, this action is simply multiplication. Indeed,

$$(t_1, t_2) \cdot [1 : z_1 : z_2] = [1 : t_1 z_1 : t_2 z_2].$$

In particular, every point of T can be moved to every other point of T. Given

$$[1 : z_1 : z_2], \qquad [1 : w_1 : w_2] \in T,$$

the unique torus element carrying the first point to the second is

$$\left(\frac{w_1}{z_1}, \frac{w_2}{z_2}\right).$$

Thus T is a single orbit for the action.

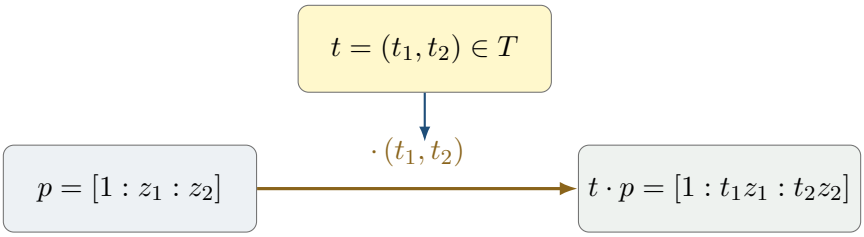


Figure 9: The torus action on its open orbit is coordinatewise multiplication. The same formula continues to make sense on the coordinate lines and coordinate points.

The action preserves each coordinate line. For example, if $x_1 = 0$, then

$$(t_1, t_2) \cdot [x_0 : 0 : x_2] = [x_0 : 0 : t_2 x_2],$$

so the second coordinate remains zero. The three coordinate points are fixed:

$$(t_1, t_2) \cdot p_i = p_i \qquad (i = 0, 1, 2).$$

The complete orbit decomposition appears in Section 0.5. For now, the essential picture is that the torus moves freely through the large open set, while the coordinate triangle records the ways in which the orbit can degenerate.

0.3.4 The affine torus reappears in every chart

Although we used U_0 to identify T with $\mathbb{G}_m^2$, the same open torus appears in every standard affine chart. In U_1, it is described by

$$\frac{x_0}{x_1} \neq 0, \qquad \frac{x_2}{x_1} \neq 0,$$

and in U_2 by

$$\frac{x_0}{x_2} \neq 0, \qquad \frac{x_1}{x_2} \neq 0.$$

Thus each chart contains the same open torus, but uses a different pair of coordinates on it.

On $U_0 \cap T$, write

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}.$$

On $U_1 \cap T$, write

$$w_0 = \frac{x_0}{x_1}, \qquad w_2 = \frac{x_2}{x_1}.$$

Then

$$w_0 = z_1^{-1}, \qquad w_2 = z_1^{-1} z_2.$$

Notice that the change of coordinates is expressed entirely using Laurent monomials. This is our first hint that monomials are not an artifact of one chosen affine chart: they are the natural coordinate changes on the torus itself.

The three dictionaries

The dense open set of $\mathbb{P}^2$ may be read in three parallel ways.

Algebra	Geometry	Combinatorics
$\Bbbk[z_1^{\pm 1}, z_2^{\pm 1}]$	$T = \mathbb{P}^2 \setminus V(x_0x_1x_2)$	two independent exponent directions
Laurent monomial coordinate changes	the same torus inside all three affine charts	integral changes of lattice coordinates
coordinatewise multiplication	a torus action on $\mathbb{P}^2$	rescaling in two independent directions.

Guiding question

The action fixes the three coordinate points and preserves each coordinate line. What are the remaining orbits on the coordinate triangle?

0.4 Monomials, triangles, and global sections

We have now met two kinds of coordinate-like objects on $\mathbb{P}^2$. Homogeneous coordinates

$$[x_0 : x_1 : x_2]$$

treat the three coordinate directions symmetrically, while affine coordinates such as

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}$$

describe one chart at a time.

The qualifier "coordinate-like" matters. The symbols $[x_0 : x_1 : x_2]$ are not coordinates in the ordinary affine sense, because the triple is not uniquely determined by the point:

$$[x_0 : x_1 : x_2] = [\lambda x_0 : \lambda x_1 : \lambda x_2] \qquad (\lambda \in \Bbbk^\times).$$

Homogeneous coordinates are therefore redundant labels for projective points rather than genuine globally defined coordinate functions. By contrast, once a chart U_i has been chosen and x_i normalized to 1, the resulting ratios are honest affine coordinates on that chart.

In this section, monomials will provide the bridge between these two descriptions.

The reward will be our first compact polyhedral picture of a genuinely global object: the global sections of $\mathcal{O}_{\mathbb{P}^2}(d)$ will be indexed by the lattice points in a triangle of side length d.

0.4.1 Why homogeneous polynomials are not ordinary functions

Let

$$F(x_0, x_1, x_2)$$

be a homogeneous polynomial of degree d. If we replace a representative (x_0, x_1, x_2) of a projective point by the scalar multiple

$$(\lambda x_0, \lambda x_1, \lambda x_2),$$

then

$$F(\lambda x_0, \lambda x_1, \lambda x_2) = \lambda^d F(x_0, x_1, x_2).$$

Unless $d = 0$, the value of F therefore depends on the chosen representative. A homogeneous polynomial of positive degree is not an ordinary function on $\mathbb{P}^2$.

The failure is not arbitrary: changing the representative always changes the value by the same power λ^d. Rather than discarding homogeneous polynomials, we package this predictable scaling into a geometric object called the line bundle $\mathcal{O}_{\mathbb{P}^2}(d)$.

There is an intuitive way to see what this means. On the chart $U_0 = \{x_0 \neq 0\}$, divide by x_0^d:

$$\frac{F(x_0, x_1, x_2)}{x_0^d} = F\left(1, \frac{x_1}{x_0}, \frac{x_2}{x_0}\right).$$

This is an honest polynomial function in the affine coordinates

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}.$$

Likewise, on U_1 and U_2, the expressions

$$\frac{F}{x_1^d} \qquad \text{and} \qquad \frac{F}{x_2^d}$$

are regular functions.

These three local functions do not agree literally on the overlaps. Instead, they differ by the predictable factors obtained by changing which homogeneous coordinate has been normalized to 1. For example, on $U_0 \cap U_1$,

$$\frac{F}{x_0^d} = \left(\frac{x_1}{x_0}\right)^d \frac{F}{x_1^d}.$$

This controlled compatibility is precisely what it means for F to define a global section of $\mathcal{O}_{\mathbb{P}^2}(d)$.

U_0
$$F_0 = \frac{F}{x_0^d} = F(1, z_1, z_2)$$

U_1
$$F_1 = \frac{F}{x_1^d}$$

U_2
$$F_2 = \frac{F}{x_2^d}$$

$$F_0 = \left(\frac{x_1}{x_0}\right)^d F_1 \quad \text{on } U_0 \cap U_1, \qquad F_1 = \left(\frac{x_2}{x_1}\right)^d F_2 \quad \text{on } U_1 \cap U_2.$$

Figure 10: A homogeneous polynomial of degree d becomes an ordinary function after one coordinate is normalized. The lower box records how the local functions agree on overlaps.

Observation 0.5. The line bundle $\mathcal{O}_{\mathbb{P}^2}(d)$ is the device that remembers these transition factors. No general theory of line bundles is needed at this stage. For this chapter, one may safely think of a global section of $\mathcal{O}_{\mathbb{P}^2}(d)$ as a homogeneous polynomial of degree d, viewed through its compatible local expressions on the three affine charts.

For $d \geq 0$, we therefore have

$$H^0\Big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)\Big) \simeq \Bbbk[x_0, x_1, x_2]_d,$$

where the right-hand side denotes the vector space of homogeneous polynomials of degree d.

0.4.2 The monomial basis

A homogeneous monomial of degree d has the form

$$x_0^{a_0} x_1^{a_1} x_2^{a_2},$$

where

$$a_0, a_1, a_2 \in \mathbb{Z}_{\geq 0}, \qquad a_0 + a_1 + a_2 = d.$$

Thus the monomial basis of $H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d))$ is indexed by the nonnegative integer solutions of one linear equation.

There are three exponents but only two independent choices. Set

$$a = a_1, \qquad b = a_2.$$

Then

$$a_0 = d - a - b,$$

and the conditions $a_0, a_1, a_2 \geq 0$ become

$$a \geq 0, \qquad b \geq 0, \qquad a + b \leq d.$$

The monomial may therefore be written as

$$x_0^{d-a-b} x_1^a x_2^b.$$

On the chart U_0, after dividing by x_0^d, this section becomes

$$\frac{x_0^{d-a-b} x_1^a x_2^b}{x_0^d} = \left(\frac{x_1}{x_0}\right)^a \left(\frac{x_2}{x_0}\right)^b = z_1^a z_2^b.$$

This formula is the key point: a homogeneous monomial on projective space becomes an ordinary monomial on the affine chart.

Guiding question

Which monomials $z_1^a z_2^b$ can arise from degree-d homogeneous monomials?

The answer is encoded by the three inequalities

$$a \geq 0, \qquad b \geq 0, \qquad a + b \leq d.$$

These are exactly the inequalities cutting out a triangle.

0.4.3 The triangle of degree-d monomials

Let

$$\Delta = \mathrm{Conv}\{(0,0), (1,0), (0,1)\} \subseteq M_{\mathbb{R}}.$$

Its d-fold dilation is

$$d\Delta = \mathrm{Conv}\{(0,0), (d,0), (0,d)\}.$$

The lattice points in $d\Delta$ are precisely the pairs

$$(a, b) \in \mathbb{Z}^2$$

satisfying

$$a \geq 0, \qquad b \geq 0, \qquad a + b \leq d.$$

Hence we obtain the correspondence

$$x_0^{d-a-b} x_1^a x_2^b \longleftrightarrow (a, b) \in d\Delta \cap M.$$

For $d = 2$, the six lattice points correspond to the six monomials

$$x_0^2, \quad x_0 x_1, \quad x_1^2, \quad x_0 x_2, \quad x_1 x_2, \quad x_2^2.$$

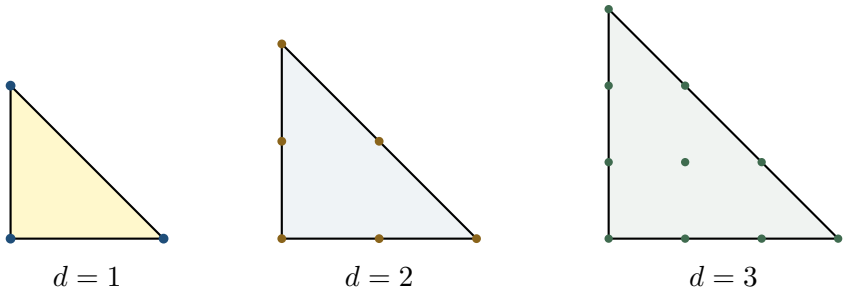


Figure 11: The lattice points of $d\Delta$ index the degree-d monomials. Increasing d dilates the triangle and creates more global sections.

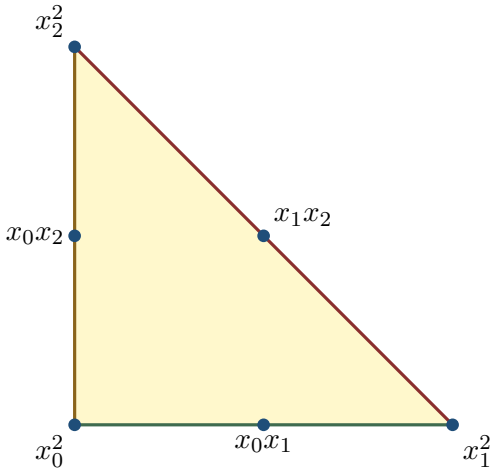


Figure 12: The degree-two triangle. The six lattice points label the six quadratic monomials. The three edge colors match the three coordinate lines D_0, D_1, D_2.

The colors on the three edges match the coordinate triangle from the previous sections:

edge of $d\Delta$	missing factor	coordinate line
$a + b = d$	x_0	$D_0 = \{x_0 = 0\}$
$a = 0$	x_1	$D_1 = \{x_1 = 0\}$
$b = 0$	x_2	$D_2 = \{x_2 = 0\}$.

For example, a lattice point on the edge $a + b = d$ labels a monomial with exponent $a_0 = 0$, so that monomial is not divisible by x_0. A lattice point in the interior has

$$a_0 > 0, \qquad a_1 > 0, \qquad a_2 > 0,$$

so the corresponding monomial is divisible by all three homogeneous coordinates.

This is our first glimpse of a general principle: the faces of a polytope record how sections interact with the boundary of a toric variety.

0.4.4 Counting sections by counting lattice points

The triangle also computes the dimension of the space of global sections. In the horizontal row with fixed exponent b, the possible values of a are

$$0, 1, \dots, d-b,$$

so that row contains $d-b+1$ lattice points. Summing over $b = 0, \dots, d$, we find

$$\#(d\Delta \cap M) = 1 + 2 + \cdots + (d+1) = \frac{(d+1)(d+2)}{2}.$$

Therefore

$$\dim_{\Bbbk} H^0\Big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)\Big) = \frac{(d+1)(d+2)}{2} = \binom{d+2}{2}.$$

This familiar dimension formula is now visible in a picture: it is simply the number of lattice points in a dilated triangle.

Observation 0.6. Under the torus action

$$(t_1, t_2) \cdot [x_0 : x_1 : x_2] = [x_0 : t_1 x_1 : t_2 x_2],$$

the monomial section

$$x_0^{d-a-b} x_1^a x_2^b$$

is rescaled by

$$t_1^a t_2^b.$$

In representation-theoretic language, it is a torus *eigensection* of weight (a, b). No representation theory is needed to use the correspondence: the lattice point (a, b) simply records how the monomial changes under coordinatewise rescaling.

The three dictionaries

For the line bundle $\mathcal{O}_{\mathbb{P}^2}(d)$, the three viewpoints meet in one picture.

Algebra	Geometry	Combinatorics
$\Bbbk[x_0, x_1, x_2]_d$	$H^0\big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)\big)$	$d\Delta \cap M$
$x_0^{d-a-b} x_1^a x_2^b$	a monomial section	(a, b)
absence of the factor x_i	restriction not identically zero on D_i	a lattice point on the corresponding edge.

Guiding question

The vertices, edges, and interior of the triangle distinguish different kinds of monomial sections. Do the vertices, edges, and interior also know something about the torus orbits in $\mathbb{P}^2$?

0.5 Orbits and coordinate strata

The torus action

$$(t_1, t_2) \cdot [x_0 : x_1 : x_2] = [x_0 : t_1 x_1 : t_2 x_2]$$

has a simple but powerful feature: it can rescale a nonzero coordinate, but it can never turn a zero coordinate into a nonzero one, or vice versa. Thus the pattern of vanishing coordinates is constant along every orbit.

For $\mathbb{P}^2$, there are only seven possible nonempty vanishing patterns:

no coordinate vanishes,
exactly one coordinate vanishes,
exactly two coordinates vanish.

In fact, each such pattern determines exactly one torus orbit. This gives a decomposition of $\mathbb{P}^2$ into seven pieces.

Guiding question

Why should the vanishing pattern determine the entire orbit, rather than merely a union of several orbits?

The reason is that the torus has enough freedom to change all the nonzero coordinates relative to one another. Let us work through the three cases.

0.5.1 The open orbit

If no coordinate vanishes, the point belongs to

$$T = \{[x_0 : x_1 : x_2] : x_0 x_1 x_2 \neq 0\}.$$

We already know that T is a single orbit. Indeed, after normalizing $x_0 = 1$, any point has the form

$$[1 : z_1 : z_2], \qquad z_1, z_2 \in \Bbbk^\times,$$

and

$$(z_1^{-1}, z_2^{-1}) \cdot [1 : z_1 : z_2] = [1 : 1 : 1].$$

Thus every point of T can be moved to the identity $[1 : 1 : 1]$.

This orbit has dimension 2, the same as $\mathbb{P}^2$, and it is open and dense.

0.5.2 The three one-dimensional orbits

Suppose exactly one homogeneous coordinate vanishes. For example, consider the coordinate line

$$D_0 = \{x_0 = 0\}.$$

Removing its two coordinate points leaves

$$O_0 = D_0 \setminus \{p_1, p_2\} = \{[0 : x_1 : x_2] : x_1 x_2 \neq 0\}.$$

Every point of O_0 may be written uniquely as

$$[0 : 1 : z], \qquad z \in \Bbbk^\times.$$

The torus action becomes

$$(t_1, t_2) \cdot [0 : 1 : z] = [0 : t_1 : t_2 z] = \left[0 : 1 : \frac{t_2}{t_1} z\right].$$

By choosing the ratio t_2/t_1, we may move any point of O_0 to any other. Hence O_0 is a single orbit.

Exactly the same argument applies to the other two coordinate lines:

$$O_1 = D_1 \setminus \{p_0, p_2\} = \{[x_0 : 0 : x_2] : x_0 x_2 \neq 0\},$$
$$O_2 = D_2 \setminus \{p_0, p_1\} = \{[x_0 : x_1 : 0] : x_0 x_1 \neq 0\}.$$

Each O_i is isomorphic to $\mathbb{G}_m$, and each is a one-dimensional torus orbit.

Observation 0.7. The torus $T \simeq \mathbb{G}_m^2$ acts transitively on $O_i \simeq \mathbb{G}_m$, but not freely: one direction of the two-dimensional torus acts trivially. For instance, on O_0, only the ratio t_2/t_1 matters. The diagonal subgroup

$$\{(t, t) : t \in \mathbb{G}_m\}$$

fixes every point of O_0. This explains geometrically why the orbit has dimension 1 rather than 2.

0.5.3 The three fixed points

If exactly two coordinates vanish, the point must be one of

$$p_0 = [1 : 0 : 0], \qquad p_1 = [0 : 1 : 0], \qquad p_2 = [0 : 0 : 1].$$

Each is fixed by the entire torus. For example,

$$(t_1, t_2) \cdot [1 : 0 : 0] = [1 : 0 : 0].$$

Thus each coordinate point is a zero-dimensional orbit.

We have proved the orbit decomposition

$$\mathbb{P}^2 = T \sqcup O_0 \sqcup O_1 \sqcup O_2 \sqcup \{p_0\} \sqcup \{p_1\} \sqcup \{p_2\}.$$

0.5.4 Orbit closures

The orbits do not sit independently inside $\mathbb{P}^2$. Their closures remember how one stratum can degenerate into another.

The closure of the open orbit is all of $\mathbb{P}^2$:

$$\overline{T} = \mathbb{P}^2.$$

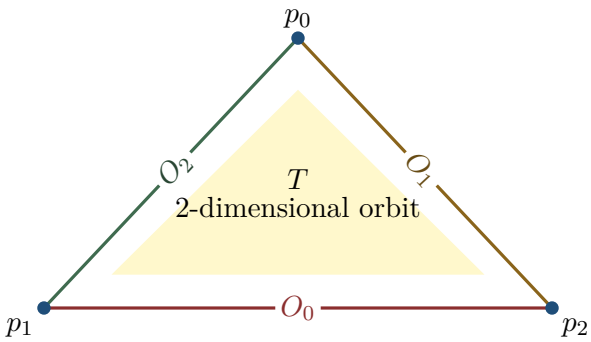


Figure 13: The seven torus orbits of $\mathbb{P}^2$: one open two-dimensional orbit, three one-dimensional orbits, and three fixed points. As before, the triangular drawing is a combinatorial model of the coordinate-line incidence pattern.

The closure of a one-dimensional orbit is the corresponding coordinate line:

$$\overline{O_i} = D_i.$$

For example,

$$\overline{O_0} = D_0 = O_0 \sqcup \{p_1, p_2\}.$$

Finally, each fixed point is already closed:

$$\overline{\{p_i\}} = \{p_i\}.$$

These relations may be displayed in a small incidence diagram.

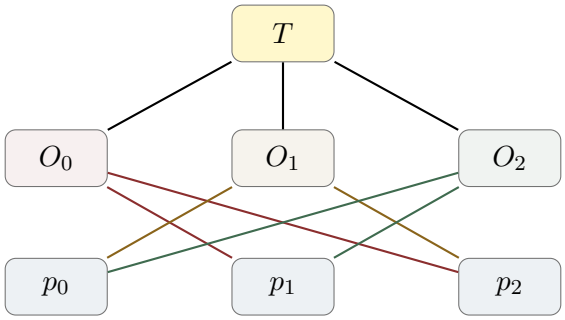


Figure 14: The orbit-closure relations. A downward path means that the lower orbit lies in the closure of the upper one.

There is also a direct way to see these degenerations, but in projective space one must remember that a limit may become visible only after changing the homogeneous representative.

For $(u, v) \in \mathbb{Z}^2$, consider the one-parameter motion

$$\lambda^{(u,v)}(t) = (t^u, t^v) \in T.$$

Acting on the identity point $e = [1 : 1 : 1]$ gives

$$\gamma_{u,v}(t) = \lambda^{(u,v)}(t) \cdot e = [1 : t^u : t^v].$$

The exponents u and v are allowed to be negative. Although an individual coordinate may then have a pole at $t = 0$, the *projective point* can still have a limit, because we may rescale all three homogeneous coordinates by the same power of t.

Set

$$m = \min\{0, u, v\}.$$

For $t \neq 0$, we may rewrite the same projective point as

$$\gamma_{u,v}(t) = [t^{-m} : t^{u-m} : t^{v-m}].$$

All three exponents are now nonnegative, and at least one is zero. Consequently, the limit is obtained by keeping precisely those coordinates whose original exponent attains the minimum of $\{0, u, v\}$.

This rule reaches strata that are invisible in the affine chart U_0. For example,

$$\begin{aligned}
\lambda^{(1,1)}(t) \cdot e &= [1 : t : t] \longrightarrow [1 : 0 : 0] = p_0, \\
\lambda^{(-1,0)}(t) \cdot e &= [1 : t^{-1} : 1] = [t : 1 : t] \longrightarrow [0 : 1 : 0] = p_1, \\
\lambda^{(0,-1)}(t) \cdot e &= [1 : 1 : t^{-1}] = [t : t : 1] \longrightarrow [0 : 0 : 1] = p_2.
\end{aligned}$$

The second and third limits land naturally in the charts U_1 and U_2, respectively.

The same idea reaches distinguished points on the one-dimensional orbits:

$$\begin{aligned}
\lambda^{(-1,-1)}(t) \cdot e &= [1 : t^{-1} : t^{-1}] = [t : 1 : 1] \longrightarrow [0 : 1 : 1] \in O_0, \\
\lambda^{(1,0)}(t) \cdot e &= [1 : t : 1] \longrightarrow [1 : 0 : 1] \in O_1, \\
\lambda^{(0,1)}(t) \cdot e &= [1 : 1 : t] \longrightarrow [1 : 1 : 0] \in O_2.
\end{aligned}$$

More generally, begin at an arbitrary point

$$q = [1 : c_1 : c_2] \in T.$$

If $u = v < 0$, then

$$\lambda^{(u,u)}(t) \cdot q = [1 : c_1 t^u : c_2 t^u] = [t^{-u} : c_1 : c_2] \longrightarrow [0 : c_1 : c_2] \in O_0.$$

As c_1 and c_2 vary, every point of O_0 occurs in this way. Analogous choices produce every point of O_1 and O_2.

Thus the limit is governed not by whether u and v are nonnegative in one chosen chart, but by which of the three numbers

$$0, \qquad u, \qquad v$$

are minimal. A unique minimum produces a fixed point; a minimum attained twice produces a one-dimensional orbit; and equality of all three exponents leaves us in the open orbit. This is the projective counterpart of the limit behavior we saw in $\mathbb{A}^2$.

0.5.5 Two triangles, two different meanings

At this point, two triangular objects have appeared:

1. the *coordinate triangle*
$$D_0 \cup D_1 \cup D_2 \subseteq \mathbb{P}^2,$$
which is the union of three projective lines;
2. the *lattice triangle*
$$\Delta = \mathrm{Conv}\{(0,0),(1,0),(0,1)\} \subseteq M_{\mathbb{R}},$$
which is an actual Euclidean triangle in a real vector space.

These should not be confused. The first is a reducible algebraic curve inside $\mathbb{P}^2$; the second is a convex polygon in $M_{\mathbb{R}}$. Nevertheless, they encode the same incidence pattern, and this is one of the central miracles of toric geometry.

The face structure of Δ matches the orbit structure of $\mathbb{P}^2$:

face of Δ	torus orbit in $\mathbb{P}^2$	dimension
Δ	T	2
an edge	O_i	1
a vertex	p_i	0.

Here the whole triangle Δ, regarded as its unique two-dimensional face, corresponds to the open orbit. The edges correspond to the punctured coordinate lines, and the vertices correspond to the fixed points.

To make the bijection completely explicit, regard Δ as a cell complex whose cells are the relative interiors of its faces.

Cell of Δ	**Closed face**	**Orbit**	**Representative**
$\mathrm{Relint}(\Delta)$	Δ	T	$[1:1:1]$
$\{a,b>0,\ a+b=1\}$	$F_0=\{a+b=1\}$	O_0	$[0:1:1]$
$\{a=0,\ 0<b<1\}$	$F_1=\{a=0\}$	O_1	$[1:0:1]$
$\{b=0,\ 0<a<1\}$	$F_2=\{b=0\}$	O_2	$[1:1:0]$
$\{(0,0)\}$	$\{(0,0)\}$	$\{p_0\}$	$[1:0:0]$
$\{(1,0)\}$	$\{(1,0)\}$	$\{p_1\}$	$[0:1:0]$
$\{(0,1)\}$	$\{(0,1)\}$	$\{p_2\}$	$[0:0:1]$

Passing from a cell to its closed face corresponds geometrically to passing from an orbit to its orbit closure:
$$F_0 \longleftrightarrow D_0, \qquad F_1 \longleftrightarrow D_1, \qquad F_2 \longleftrightarrow D_2.$$

Warning

A lattice point in the interior of $d\Delta$ does not by itself represent a torus orbit. Individual lattice points index monomial sections. Torus orbits correspond to faces of the polytope: the whole polytope, its edges, and its vertices.

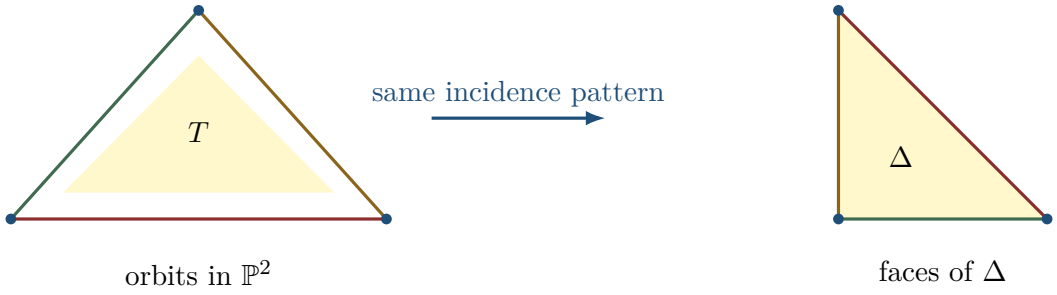


Figure 15: The coordinate triangle in $\mathbb{P}^2$ and the lattice triangle Δ are different mathematical objects, but their strata have the same incidence pattern. Matching colors indicate the corresponding one-dimensional strata.

The three dictionaries

The orbit decomposition of $\mathbb{P}^2$ can be read in three parallel languages.

Algebra	**Geometry**	**Combinatorics**
Record which of $x_0,\ x_1,\ x_2$ vanish. The vanishing set is constant along an orbit.	The vanishing pattern determines one torus orbit: $T,\quad O_i,\quad$ or $\quad p_i.$ Allowing additional coordinates to vanish describes orbit closure.	The orbit corresponds to the relative interior of a face of Δ. Orbit closure corresponds to replacing that cell by its closed face.

vanishing pattern $\longleftrightarrow$ torus orbit $\longleftrightarrow$ cell of Δ

Guiding question

The triangle Δ explains the orbit structure beautifully, but it lives in the space of monomial exponents. Is there another combinatorial object that records how the three affine charts of $\mathbb{P}^2$ fit together?

0.6 The triangle and the fan

The lattice triangle

$$\Delta = \mathrm{Conv}\{(0,0),(1,0),(0,1)\} \subseteq M_{\mathbb{R}}$$

has already encoded two global features of $\mathbb{P}^2$: its monomial sections and its torus orbits. We now extract a second combinatorial object from the same triangle.

There is one important change of viewpoint. The triangle lives in $M_{\mathbb{R}}$, the space of monomial exponents, whereas the new object lives in the paired vector space $N_{\mathbb{R}}$, where we record one-parameter motions. Its job is different as well: rather than recording sections, it remembers the three affine charts of $\mathbb{P}^2$ and how they fit together.

0.6.1 Three sides, three inward directions

Write a point of $M_{\mathbb{R}}$ as $m = (a, b)$. The triangle Δ is cut out by the inequalities

$$a \geq 0, \qquad b \geq 0, \qquad a + b \leq 1.$$

Its three edges are

$$F_0 = \{a + b = 1\}, \qquad F_1 = \{a = 0\}, \qquad F_2 = \{b = 0\}.$$

Each edge has a distinguished primitive direction pointing into the triangle:

$$v_0 = (-1, -1), \qquad v_1 = (1, 0), \qquad v_2 = (0, 1).$$

Indeed, moving in the direction v_1 increases a, moving in the direction v_2 increases b, and moving in the direction v_0 decreases $a + b$. Thus v_i points inward from the edge F_i.

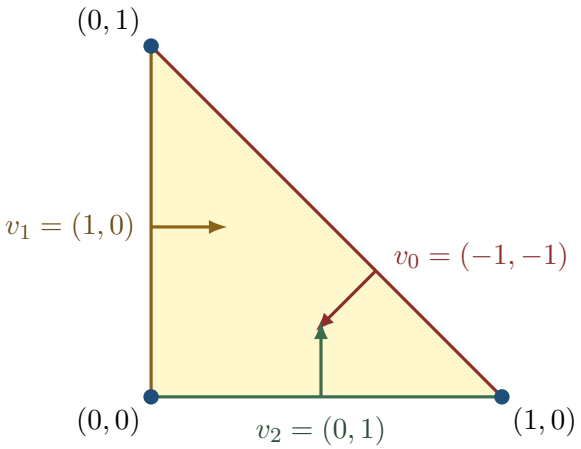


Figure 16: The three primitive inward normals of the standard triangle. The colors match the coordinate lines D_i and the corresponding edges F_i.

There is a mild abuse hidden in this picture. The triangle lies in $M_{\mathbb{R}}$, while the normal vectors lie in $N_{\mathbb{R}}$. After choosing dual bases, both spaces look like $\mathbb{R}^2$, so it is useful to draw the vectors next to the triangle. Conceptually, however, they belong to paired spaces.

0.6.2 A direction asks a question of the triangle

Let $n = (u, v) \in N_{\mathbb{R}}$. Pairing with n produces a linear function on the triangle:

$$m = (a, b) \longmapsto \langle m, n \rangle = au + bv.$$

We ask:

On which part of Δ is this linear function smallest?

The answer depends only on the direction of n.

For example, if $n = v_1 = (1, 0)$, then

$$\langle (a, b), v_1 \rangle = a.$$

Its minimum is attained on the entire edge $F_1 = \{a = 0\}$. Similarly, v_2 singles out F_2, while v_0 singles out F_0.

A more general direction can single out a vertex. If $u > 0$ and $v > 0$, then $au + bv$ is smallest uniquely at $(0, 0)$. The set of all such directions, together with its boundary, is the cone

$$\sigma_0 = \text{Cone}(v_1, v_2).$$

Likewise,

$$\sigma_1 = \text{Cone}(v_2, v_0)$$

is the set of directions whose minimum is attained at $(1, 0)$, and

$$\sigma_2 = \text{Cone}(v_0, v_1)$$

is the set of directions whose minimum is attained at $(0, 1)$.

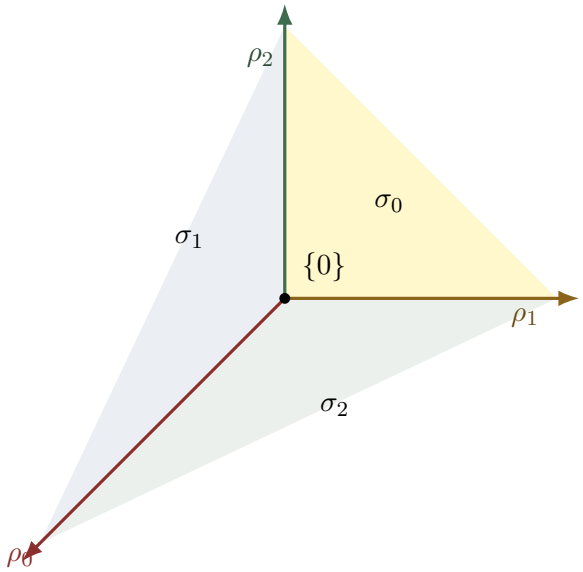


Figure 17: The normal fan of Δ. The rays are $\rho_i = \mathbb{R}_{\geq 0} v_i$, and the three two-dimensional cones are $\sigma_0, \sigma_1, \sigma_2$. Together they cover $N_{\mathbb{R}}$.

The collection

$$\Sigma_\Delta = \{\{0\}, \rho_0, \rho_1, \rho_2, \sigma_0, \sigma_1, \sigma_2\}$$

is called the *normal fan* of Δ. Here

$$\rho_i = \mathbb{R}_{\geq 0} v_i.$$

The word *fan* refers to a collection of cones meeting cleanly along common faces. In this example, the three two-dimensional cones fit around the origin like the sectors of a fan and cover the whole plane $N_{\mathbb{R}}$.

There is an intuitive reason they cover everything. Every linear function on the compact triangle Δ attains a minimum somewhere: at a vertex, along an edge, or on all of Δ if the function is constant. The normal fan records exactly where that minimum occurs.

0.6.3 Faces of the triangle and cones of the fan

The passage from a face of Δ to its normal cone reverses dimensions:

face of Δ	normal cone in Σ_Δ
Δ	$\{0\}$
F_i	ρ_i
a vertex	a two-dimensional cone.

More explicitly,

face of Δ	normal cone
Δ	$\{0\}$
$F_0 = \{a+b=1\}$	$\rho_0 = \mathbb{R}_{\geq 0} v_0$
$F_1 = \{a=0\}$	$\rho_1 = \mathbb{R}_{\geq 0} v_1$
$F_2 = \{b=0\}$	$\rho_2 = \mathbb{R}_{\geq 0} v_2$
$(0,0)$	$\sigma_0 = \mathrm{Cone}(v_1, v_2)$
$(1,0)$	$\sigma_1 = \mathrm{Cone}(v_2, v_0)$
$(0,1)$	$\sigma_2 = \mathrm{Cone}(v_0, v_1)$.

If one face is contained in another, the corresponding normal cones are contained in the opposite order. For example,

$$(0,0) \in F_1 \qquad \Longleftrightarrow \qquad \rho_1 \subseteq \sigma_0.$$

This reversal is not an accident. Requiring a linear function to be minimal along an entire edge is more restrictive than requiring it to be minimal at one of the edge's vertices. The larger face therefore has the smaller normal cone.

0.6.4 The fan and one-parameter limits

The fan packages the limit calculations of the previous section.

A direction in the relative interior of a ray ρ_i leads to the one-dimensional orbit O_i. For the primitive ray generators,

$$\begin{aligned} v_0 = (-1,-1): &\quad [1 : t^{-1} : t^{-1}] = [t : 1 : 1] \longrightarrow [0 : 1 : 1] \in O_0, \\ v_1 = (1,0): &\quad [1 : t : 1] \longrightarrow [1 : 0 : 1] \in O_1, \\ v_2 = (0,1): &\quad [1 : 1 : t] \longrightarrow [1 : 1 : 0] \in O_2. \end{aligned}$$

A direction in the relative interior of a two-dimensional cone leads to the corresponding fixed point:

cone containing (u, v)	$\lim_{t\to 0}[1 : t^u : t^v]$
$\mathrm{Relint}(\sigma_0)$	$p_0 = [1 : 0 : 0]$
$\mathrm{Relint}(\sigma_1)$	$p_1 = [0 : 1 : 0]$
$\mathrm{Relint}(\sigma_2)$	$p_2 = [0 : 0 : 1]$.

Finally, the zero vector gives the constant point $[1 : 1 : 1]$ in the open orbit.

Thus cones of dimension k correspond to orbits of dimension $2 - k$:

$$\dim O(\sigma) = 2 - \dim \sigma.$$

At this stage, the formula is a summary of the seven calculations above. Later it will become part of the general orbit–cone correspondence.

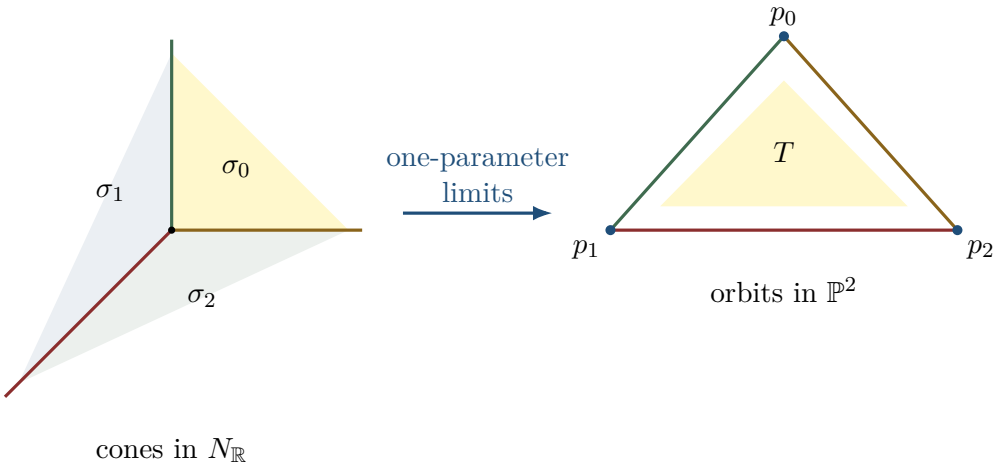


Figure 18: The cone containing a one-parameter direction determines the orbit in which the limit lies. Two-dimensional cones lead to fixed points, rays lead to one-dimensional orbits, and the zero cone corresponds to the open orbit.

0.6.5 The three maximal cones recover the three affine charts

The fan remembers more than the orbit decomposition. Each two-dimensional cone recovers one of the standard affine charts.

Start with

$$\sigma_0 = \mathrm{Cone}(v_1, v_2),$$

the first quadrant in $N_{\mathbb{R}}$. A monomial $z_1^a z_2^b$ extends regularly along every one-parameter motion in σ_0 precisely when

$$a \geq 0, \qquad b \geq 0.$$

These monomials form the polynomial ring

$$\Bbbk[z_1, z_2],$$

whose spectrum is the chart

$$U_0 \simeq \mathbb{A}^2.$$

For the second cone,

$$\sigma_1 = \mathrm{Cone}(v_2, v_0),$$

the extendable Laurent monomials are generated by

$$z_1^{-1} \qquad \text{and} \qquad z_1^{-1} z_2.$$

But these are exactly the affine coordinates on U_1:

$$w_0 = \frac{x_0}{x_1} = z_1^{-1}, \qquad w_2 = \frac{x_2}{x_1} = z_1^{-1} z_2.$$

Hence

$$\Bbbk[\sigma_1^\vee \cap M] = \Bbbk[w_0, w_2], \qquad U_1 \simeq \mathbb{A}^2.$$

Similarly, for

$$\sigma_2 = \mathrm{Cone}(v_0, v_1),$$

the extendable Laurent monomials are generated by

$$z_2^{-1} \qquad \text{and} \qquad z_1 z_2^{-1},$$

which are the coordinates

$$r_0 = \frac{x_0}{x_2}, \qquad r_1 = \frac{x_1}{x_2}$$

on U_2. Thus

$$\Bbbk[\sigma_2^\vee \cap M] = \Bbbk[r_0, r_1].$$

Vertex of Δ	**Maximal cone**	**Affine coordinates**	**Chart and fixed point**
$(0,0)$	$\sigma_0 = \mathrm{Cone}(v_1, v_2)$	$z_1 = x_1/x_0,\ z_2 = x_2/x_0$	$U_0,\ p_0$
$(1,0)$	$\sigma_1 = \mathrm{Cone}(v_2, v_0)$	$w_0 = x_0/x_1,\ w_2 = x_2/x_1$	$U_1,\ p_1$
$(0,1)$	$\sigma_2 = \mathrm{Cone}(v_0, v_1)$	$r_0 = x_0/x_2,\ r_1 = x_1/x_2$	$U_2,\ p_2$

This table is the first complete appearance of our three dictionaries: a vertex of the polytope determines a cone; the cone determines a semigroup of monomials; and those monomials form the coordinate ring of an affine chart.

0.6.6 What the fan remembers

For $\mathbb{P}^2$, the normal fan packages the following correspondences:

$$\begin{aligned} \{0\} &\longleftrightarrow \text{the dense torus } T, \\ \rho_i &\longleftrightarrow \text{the coordinate line } D_i, \\ \sigma_i &\longleftrightarrow \text{the affine chart } U_i \text{ and its fixed point } p_i. \end{aligned}$$

Containment of cones reverses containment of orbit closures. For example,

$$\rho_1, \rho_2 \subseteq \sigma_0$$

records that

$$p_0 = D_1 \cap D_2.$$

The three dictionaries

The normal fan translates between local algebra, projective geometry, and convex combinatorics.

Algebra	Geometry	Combinatorics
A cone σ_i selects the Laurent monomials that extend regularly. They form the coordinate ring $\Bbbk[\sigma_i^\vee \cap M]$.	That ring describes the affine chart U_i. Faces of the cone describe the torus orbits meeting the chart.	Vertices, edges, and the whole polytope have normal cones of dimensions $2, 1, 0$, respectively.

Surface laboratory

Starting from

$$\Delta = \mathrm{Conv}\{(0,0), (1,0), (0,1)\},$$

perform the following reconstruction of $\mathbb{P}^2$.

1. Find the primitive inward normal to each edge of Δ.
2. Form the three two-dimensional cones generated by adjacent normal vectors.
3. For each maximal cone, determine the Laurent monomials that pair nonnegatively with it.
4. Recognize the resulting three polynomial rings as the coordinate rings of U_0, U_1, U_2.

Guiding question

If changing the fan changes the way affine charts fit together, what geometric operation should correspond to inserting a new ray into one of its two-dimensional cones?

0.7 Blowing up a torus-fixed point

The fan of $\mathbb{P}^2$ contains three two-dimensional cones, one for each torus-fixed point. We now modify one of these cones and watch the projective plane change.

Our chosen point is

$$p_0 = [1 : 0 : 0].$$

It lies in the affine chart

$$U_0 = \{x_0 \neq 0\} \simeq \mathbb{A}^2$$

and becomes the origin in the affine coordinates

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}.$$

Thus blowing up p_0 in $\mathbb{P}^2$ is, near p_0, the familiar operation of blowing up the origin in the affine plane.

0.7.1 Replacing a point by its directions

The origin of $\mathbb{A}^2$ has many possible directions of approach. A line through the origin is determined by a projective point

$$[s : t] \in \mathbb{P}^1.$$

The blowup remembers this direction while approaching the origin.

Concretely, define

$$\widetilde{\mathbb{A}^2} = \left\{ ((z_1, z_2), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 : z_1 t = z_2 s \right\}.$$

The equation says that the vector (z_1, z_2) lies on the line with direction $[s : t]$. Projection onto the first factor gives a map

$$\pi : \widetilde{\mathbb{A}^2} \longrightarrow \mathbb{A}^2.$$

If $(z_1, z_2) \neq (0, 0)$, there is a unique compatible direction:

$$[s : t] = [z_1 : z_2].$$

Hence π is an isomorphism away from the origin. Over the origin, however, the equation becomes

$$0 = 0,$$

so every direction is allowed:

$$\pi^{-1}(0, 0) \simeq \mathbb{P}^1.$$

The blowup therefore replaces the origin by a projective line parameterizing directions through it. This projective line is called the *exceptional divisor* and will be denoted by E.

Observation 0.8. It is tempting to say that a blowup simply removes a point and inserts a copy of $\mathbb{P}^1$. This is a useful picture, but the morphism π carries more information: it explains how nearby points approach the exceptional divisor and which limiting direction they select.

0.7.2 The two affine charts of the blowup

The projective line of directions has two standard charts, and these give two affine charts on the blowup.

On the chart $s \neq 0$, write

$$q = \frac{t}{s}.$$

The equation $z_1 t = z_2 s$ becomes

$$z_2 = z_1 q.$$

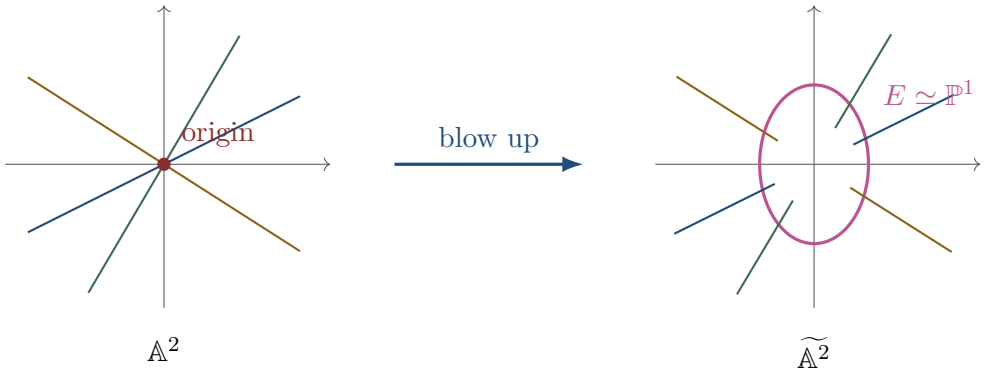


Figure 19: The blowup replaces the origin by the projective line of directions through it. The drawing is schematic: different directions meet the exceptional divisor at different points.

Thus this part of the blowup has coordinates (z_1, q), and the map to $\mathbb{A}^2$ is

$$(z_1, q) \longmapsto (z_1, z_1 q).$$

The exceptional divisor is given by

$$z_1 = 0.$$

On the chart $t \neq 0$, write

$$p = \frac{s}{t}.$$

The defining equation becomes

$$z_1 = p z_2.$$

This chart has coordinates (p, z_2), the blowup map is

$$(p, z_2) \longmapsto (p z_2, z_2),$$

and the exceptional divisor is given by

$$z_2 = 0.$$

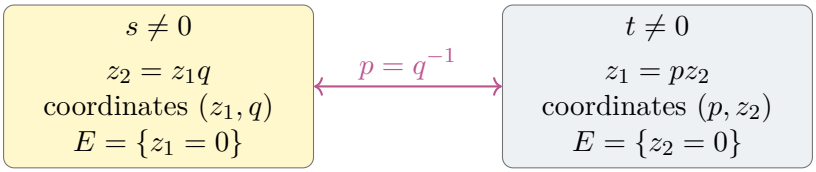


Figure 20: The two affine charts of the blowup. On their overlap, the two coordinates p and q are reciprocal coordinates on the exceptional projective line.

The two coordinate axes through the origin are separated by the blowup. In U_0, the coordinate lines through p_0 are

$$D_1 \cap U_0 = \{z_1 = 0\}, \qquad D_2 \cap U_0 = \{z_2 = 0\}.$$

Their strict transforms no longer meet one another. Instead, each meets E at the point corresponding to its own direction:

$$\widetilde{D}_1 \cap E = \{[s:t] = [0:1]\}, \qquad \widetilde{D}_2 \cap E = \{[s:t] = [1:0]\}.$$

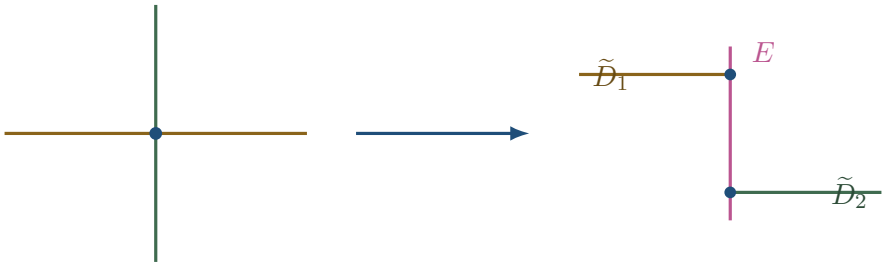


Figure 21: Before the blowup, D_1 and D_2 meet at p_0. Afterward, their strict transforms meet the exceptional divisor at two different points.

0.7.3 The star subdivision of the fan

The fixed point p_0 corresponds to the cone

$$\sigma_0 = \mathrm{Cone}(v_1, v_2), \qquad v_1 = (1,0), \quad v_2 = (0,1).$$

Insert the new primitive vector

$$v_E = v_1 + v_2 = (1,1)$$

and the ray

$$\rho_E = \mathbb{R}_{\geq 0} v_E.$$

The old cone σ_0 is replaced by two cones:

$$\mathrm{Cone}(v_1, v_E) \qquad \text{and} \qquad \mathrm{Cone}(v_E, v_2).$$

All other cones of the fan are left unchanged.

This operation is called the *star subdivision* of σ_0 by the ray ρ_E. The resulting fan describes the blowup

$$\pi : \mathrm{Bl}_{p_0}(\mathbb{P}^2) \longrightarrow \mathbb{P}^2.$$

Why should the new vector be $v_1 + v_2$? In the local chart $U_0 \simeq \mathbb{A}^2$, the rays generated by v_1 and v_2 represent the two coordinate directions. The vector $(1,1)$ records approaching the origin while both coordinates vanish at the same rate:

$$t \longmapsto (t,t).$$

It is the simplest integral direction lying strictly between the two old rays.

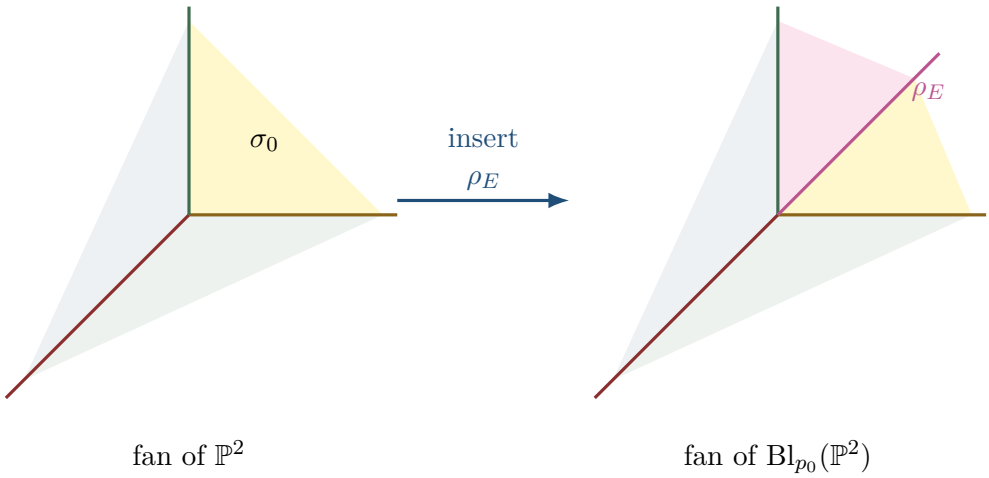


Figure 22: The star subdivision replaces the cone for p_0 by two cones separated by the new ray ρ_E.

0.7.4 The exceptional divisor in the orbit picture

The new ray ρ_E corresponds to a new one-dimensional torus orbit and its closure, the exceptional divisor E. The two new maximal cones correspond to the two torus-fixed points on E:

$$\mathrm{Cone}(v_1, v_E) \longleftrightarrow \widetilde{D}_1 \cap E,$$

and

$$\mathrm{Cone}(v_E, v_2) \longleftrightarrow E \cap \widetilde{D}_2.$$

The old fixed point p_0 has disappeared. In its place we find the one-dimensional orbit inside E together with two new fixed points. The orbit count changes from

$$1 + 3 + 3 = 7$$

to

$$1 + 4 + 4 = 9 :$$

one open orbit, four one-dimensional orbits, and four fixed points.

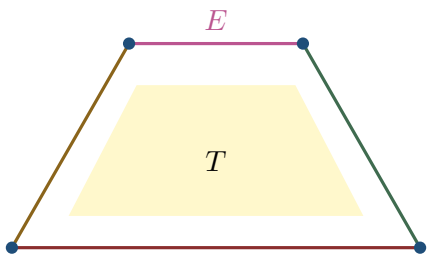
Observation 0.9. The surface $\mathrm{Bl}_{p_0}(\mathbb{P}^2)$ is isomorphic to the first Hirzebruch surface $\mathbb{F}_1$. Hirzebruch surfaces form the next natural family of smooth complete toric surfaces after $\mathbb{P}^2$, and they will give us a useful supply of examples.

0.7.5 The exceptional curve has self-intersection -1

One numerical feature of the exceptional divisor is worth previewing:

$$E^2 = -1.$$

Intersection theory comes later, so this formula is a preview rather than a completed calculation.



boundary incidence after blowing up p_0

Figure 23: The boundary incidence pattern after the blowup is a quadrilateral. The new side represents the exceptional divisor, and its two endpoints represent the two new fixed points.

Geometrically, the negative self-intersection records that the exceptional curve can be contracted back to a smooth point. The blowup map

$$\pi : \mathrm{Bl}_{p_0}(\mathbb{P}^2) \longrightarrow \mathbb{P}^2$$

does exactly this: it sends the whole curve E to p_0 and is an isomorphism elsewhere. In the fan, contraction means deleting the ray ρ_E and restoring the original cone $\mathrm{Cone}(v_1, v_2)$.

The three dictionaries

The blowup of p_0 is one operation described in three languages.

Algebra	**Geometry**	**Combinatorics**
Locally, blow up the ideal (z_1, z_2) at the origin of $\mathrm{Spec}\,\Bbbk[z_1, z_2]$. The two charts satisfy $z_2 = z_1 q, \quad z_1 = p z_2$.	Replace p_0 by the exceptional curve $E \simeq \mathbb{P}^1$, whose points record directions through p_0.	Insert the ray $\rho_E = \mathbb{R}_{\geq 0}(v_1 + v_2)$ and subdivide $\mathrm{Cone}(v_1, v_2)$ into two cones.

Surface laboratory

Track the blowup of p_0 through the following parallel changes.

Before	**Blowup operation**	**After**
the ideal (z_1, z_2)	record the ratio $[z_1 : z_2]$	two affine blowup charts
the point $p_0 = D_1 \cap D_2$	replace the point by its directions	$E \simeq \mathbb{P}^1$, meeting $\widetilde{D}_1, \widetilde{D}_2$
the cone $\mathrm{Cone}(v_1, v_2)$	insert $v_E = v_1 + v_2$	two cones and one new ray.

What this chapter has accomplished

We have spent an entire chapter revisiting a single familiar surface. The point was not to make $\mathbb{P}^2$ look more complicated. It was to notice that several kinds of information we usually study separately are shadows of the same combinatorial structure.

In $\mathbb{P}^2$, we have been able to check everything directly. The next chapters turn these observations into definitions and theorems for arbitrary toric varieties.

The affine dictionary

For the affine plane, the first quadrant

$$\sigma = \mathbb{R}^2_{\geq 0} \subseteq N_{\mathbb{R}}$$

controlled two complementary questions:

- which monomial curves acquire limits in $\mathbb{A}^2$;
- which Laurent monomials extend to regular functions on $\mathbb{A}^2$.

The corresponding algebra was

$$\Bbbk[\sigma^\vee \cap M] = \Bbbk[x, y],$$

and its spectrum was the affine plane.

This suggests the general construction

$$\sigma \quad \rightsquigarrow \quad \sigma^\vee \cap M \quad \rightsquigarrow \quad \Bbbk[\sigma^\vee \cap M] \quad \rightsquigarrow \quad \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].$$

The general construction developed in the next chapters will show that rational polyhedral cones produce affine toric varieties in precisely this way.

Fans as gluing instructions

The fan of $\mathbb{P}^2$ consisted of three maximal cones. Each maximal cone produced one affine chart, and shared faces produced the overlap data. This suggests that a fan should be understood as a set of instructions for gluing affine toric varieties.

The orbit decomposition was visible in the same fan:

$$\{0\} \longleftrightarrow T, \qquad \rho_i \longleftrightarrow O_i, \qquad \sigma_i \longleftrightarrow p_i.$$

The dimensions added to 2:

$$\dim \sigma + \dim O(\sigma) = 2.$$

In higher dimensions, the same pattern will become the orbit–cone correspondence.

Polytopes, sections, and projective embeddings

The dilated triangle $d\Delta$ indexed the monomial sections of $\mathcal{O}_{\mathbb{P}^2}(d)$:

$$x_0^{d-a-b} x_1^a x_2^b \longleftrightarrow (a, b) \in d\Delta \cap M.$$

Counting global sections became counting lattice points.

This suggests that lattice polytopes should encode projective geometry. Their lattice points should give monomial maps to projective space, their faces should record boundary behavior, and their normal fans should construct projective toric varieties.

Warning

Fans are more general than normal fans of polytopes. Every lattice polytope has a normal fan, but not every fan comes from a polytope. Consequently, not every toric variety is projective.

Rays, divisors, and birational modifications

In the fan of $\mathbb{P}^2$, each ray corresponded to one coordinate line:

$$\rho_i \longleftrightarrow D_i.$$

After blowing up p_0, the new ray corresponded to the exceptional divisor:

$$\rho_E \longleftrightarrow E.$$

This predicts a general correspondence between rays of a fan and torus-invariant prime divisors.

The star subdivision also suggests that combinatorial refinements should produce geometric modifications:

$$\text{subdivide a fan} \quad \rightsquigarrow \quad \text{modify a toric variety birationally.}$$

For a smooth toric surface, inserting the ray generated by the sum of two adjacent primitive ray generators describes the blowup of the corresponding fixed point. When the reverse operation is allowed, deleting that ray contracts the exceptional curve.

What we have seen, and what remains to prove

The following table separates the evidence supplied by $\mathbb{P}^2$ from the general theory still ahead.

Seen directly in $\mathbb{P}^2$	**General principle to be developed**
The first-quadrant cone gives $\mathbb{A}^2$.	A rational polyhedral cone gives an affine toric variety.
The three maximal cones give the three affine charts.	A fan glues affine toric varieties into a global toric variety.
Cones index the seven torus orbits.	Cones of a fan correspond to torus orbits, with complementary dimensions.
The three rays give the coordinate lines.	Rays correspond to torus-invariant prime divisors.
Lattice points in $d\Delta$ index sections of $\mathcal{O}_{\mathbb{P}^2}(d)$.	Lattice polytopes and polyhedra encode sections of toric line bundles.
The normal fan of Δ recovers $\mathbb{P}^2$.	Normal fans connect lattice polytopes with projective toric varieties.
Adding the ray $(1,1)$ blows up p_0.	Under suitable hypotheses, star subdivisions describe toric blowups of invariant strata.
Deleting the exceptional ray contracts E.	Fan refinements and contractions organize toric birational geometry.

The chapter's organizing principle may now be displayed in one picture. Algebra, geometry, and combinatorics are not three separate topics: they are three ways of reading the same constructions.

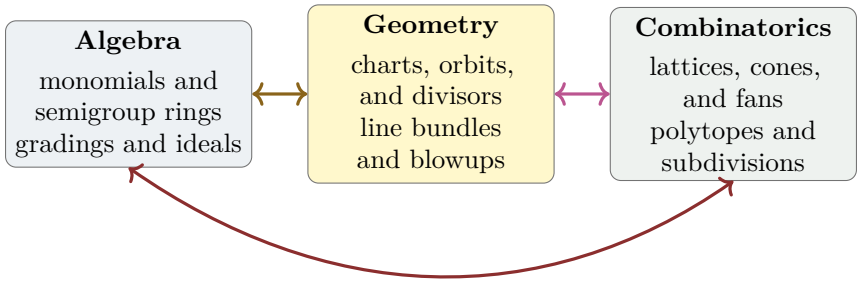


Figure 24: The organizing principle of the course. None of the three languages is merely decorative: each makes some constructions transparent that are difficult to see in the other two.

The three dictionaries

The chapter may be compressed into four recurring translations.

monomial functions	$\longleftrightarrow$	regular functions on torus charts	$\longleftrightarrow$	lattice points in dual cones
coordinate rings	$\longleftrightarrow$	affine toric charts	$\longleftrightarrow$	cones of a fan
graded monomials	$\longleftrightarrow$	sections of line bundles	$\longleftrightarrow$	lattice points in polytopes
blowup ideals	$\longleftrightarrow$	exceptional divisors	$\longleftrightarrow$	star subdivisions.

A provisional picture of a toric variety

The examples suggest the following working picture. A toric variety is an algebraic variety containing an algebraic torus as a dense open subset, together with an action of that torus extending its multiplication on itself.

This is not yet our formal starting point. We still need to understand algebraic tori, their monomial functions, and their one-parameter motions in a coordinate-free way. That is the task of the next chapter.

Guiding question

Which features of $\mathbb{G}_m^n$ are intrinsic, and which arose only because we chose coordinates $x_1, \dots, x_n$?

Comprehension exercises

0.1. Determine which characters of $T = \mathbb{G}_m^2$ extend to regular functions on $\mathbb{A}^2$.

0.2. Let $\lambda^{(u,v)}(t) = (t^u, t^v)$. Determine when this map extends to a morphism $\mathbb{A}^1 \to \mathbb{A}^2$.

0.3. Write the transition maps between the three standard affine charts of $\mathbb{P}^2$.

0.4. List all torus orbits in $\mathbb{P}^2$, compute their dimensions, and describe their closures.

0.5. For $d = 2$ and $d = 3$, draw $d\Delta \cap M$ and label each lattice point by its corresponding monomial section of $\mathcal{O}_{\mathbb{P}^2}(d)$.

0.6. Compute

$$\dim_{\Bbbk} H^0\Big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)\Big)$$

by counting lattice points in $d\Delta$.

0.7. Compute the inward-pointing primitive normal vector to each edge of Δ, and use these vectors to draw the normal fan.

0.8. Draw the star subdivision corresponding to the blowup of each of the three torus-fixed points of $\mathbb{P}^2$.

0.9. Explain why blowing up any one of the three torus-fixed points produces isomorphic toric surfaces.

Challenge exercise

Compute the topological Euler characteristic of $\mathbb{P}^2(\mathbb{C})$ using the torus action.

The goal is to discover a map from $\mathbb{P}^2(\mathbb{C})$ to the triangle

$$\Delta = \mathrm{Conv}\{(0,0),(1,0),(0,1)\}$$

whose fibers reflect the orbit decomposition. Do not use the usual cell decomposition $\mathbb{P}^2 = \mathbb{C}^2 \sqcup \mathbb{C} \sqcup \{\mathrm{pt}\}$.

Hint 1. Consider the map

$$\mu : \mathbb{P}^2(\mathbb{C}) \longrightarrow \mathbb{R}^2, \qquad \mu([z_0 : z_1 : z_2]) = \left(\frac{|z_1|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2}, \frac{|z_2|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2} \right).$$

Check that it is well defined on projective points and that its image is exactly Δ. This is the moment map for the compact torus $(S^1)^2 \subset (\mathbb{C}^\times)^2$, up to the usual harmless choices of normalization.

Hint 2. Let (a,b) lie in the interior of Δ. Fixing $\mu([z_0 : z_1 : z_2]) = (a,b)$ fixes the absolute values of the three homogeneous coordinates, up to common scaling. Determine which phase parameters remain. Show that the fiber is a real two-torus:

$$\mu^{-1}(a,b) \simeq (S^1)^2.$$

Hint 3. Now let (a,b) lie in the relative interior of an edge of Δ. One homogeneous coordinate must vanish. Show that the fiber is a circle:

$$\mu^{-1}(a,b) \simeq S^1.$$

Hint 4. Determine the fibers over the three vertices of Δ. Show that they are the three torus-fixed points

$$p_0, \qquad p_1, \qquad p_2.$$

Hint 5. Stratify Δ into the relative interiors of its face, its three edges, and its three vertices. Over each stratum, μ is a locally trivial fibration of the type found above. Use additivity of Euler characteristic over a finite stratification and multiplicativity for locally trivial fibrations. On the noncompact strata, one may use compactly supported Euler characteristic; since $\mathbb{P}^2(\mathbb{C})$ is compact, the final result agrees with its ordinary Euler characteristic.

Hint 6. Recall that

$$\chi((S^1)^2) = 0, \qquad \chi(S^1) = 0, \qquad \chi(\{\mathrm{pt}\}) = 1.$$

Conclude that only the fibers above the three vertices contribute. What is

$$\chi(\mathbb{P}^2(\mathbb{C}))?$$

Hint 7. Now forget the special geometry of $\mathbb{P}^2$ for a moment and look only at its decomposition into torus orbits. Every positive-dimensional orbit is a copy of $(\mathbb{C}^\times)^r$ for some $r > 0$, and hence

has Euler characteristic zero. What conjecture does this suggest for the Euler characteristic of an arbitrary complex toric variety? For a noncomplete variety, you may formulate the conjecture first using compactly supported Euler characteristic. Can you express your answer in terms of torus-fixed points, or equivalently in terms of the full-dimensional cones of the fan?

Notes and references

This chapter is intentionally panoramic. It uses $\mathbb{P}^2$ to preview the constructions that will be defined and proved in the chapters that follow: affine toric varieties from cones, global varieties from fans, invariant divisors from rays, monomial sections from lattice points, and toric blowups from star subdivisions.

For systematic introductions to these correspondences, see Fulton, *Introduction to Toric Varieties*, Chapters 1–3, and Cox–Little–Schenck, *Toric Varieties*, Chapters 1–4 and 6. The moment-map viewpoint used in the challenge is developed from first principles in Chapter 8 of these notes.

Chapter 1

Algebraic Tori and Their Representation Theory

Chapter overview

Why are monomials so central in toric geometry?
The answer begins with representation theory. Algebraic tori are among the simplest algebraic groups, and their finite-dimensional representations are completely controlled by monomial functions. In this chapter we develop the basic language of characters, one-parameter subgroups, weights, and dual lattices. The examples of $\mathbb{G}_m$, $\mathbb{G}_m^2$, and the dense torus of $\mathbb{P}^2$ remain in the foreground throughout.
Most structural results about representations are stated without proof. References and proofs may be supplied from standard texts on linear algebraic groups; our goal here is to establish the language and geometric intuition needed for toric varieties.

The three dictionaries

The central objects of the chapter already come in parallel families.

Algebra	Geometry	Combinatorics
the group algebra $\Bbbk[M]$	the algebraic torus T	the character lattice M
characters $T \to \mathbb{G}_m$	monomial functions on T	lattice points $m \in M$
one-parameter subgroups $\mathbb{G}_m \to T$	one-parameter motions and limits	lattice points $u \in N$
the composition $\chi^m \circ \lambda^u$	the order of vanishing or pole at $t = 0$	the pairing $\langle m, u \rangle$

1.1 Why "torus"?

The word *torus* comes to algebraic geometry from topology. A topologist begins with the circle

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}$$

and calls a product

$$(S^1)^n$$

an n-dimensional torus. When $n = 2$, this is the familiar surface of a doughnut; in higher dimensions, the product description is more useful than any attempted drawing.

What does this have to do with the algebraic group

$$\mathbb{G}_m = \operatorname{Spec} \Bbbk[x, x^{-1}]?$$

The connection is easiest to see over $\mathbb{C}$.

1.1.1 From the circle to the punctured complex line

The complex points of $\mathbb{G}_m$ are

$$\mathbb{G}_m(\mathbb{C}) = \mathbb{C}^\times = \mathbb{C} \setminus \{0\}.$$

Every nonzero complex number has a polar decomposition

$$z = r\, e^{i\theta}, \qquad r > 0.$$

Thus, as a topological space,

$$\mathbb{C}^\times \simeq S^1 \times \mathbb{R}_{>0}.$$

The second factor is contractible. We may continuously push every point radially toward the unit circle by

$$H(z, s) = \left((1 - s) + \frac{s}{|z|}\right) z, \qquad 0 \le s \le 1.$$

At $s = 0$, this is the identity; at $s = 1$, it sends z to $z/|z| \in S^1$. In topological language, $\mathbb{C}^\times$ deformation retracts onto S^1.

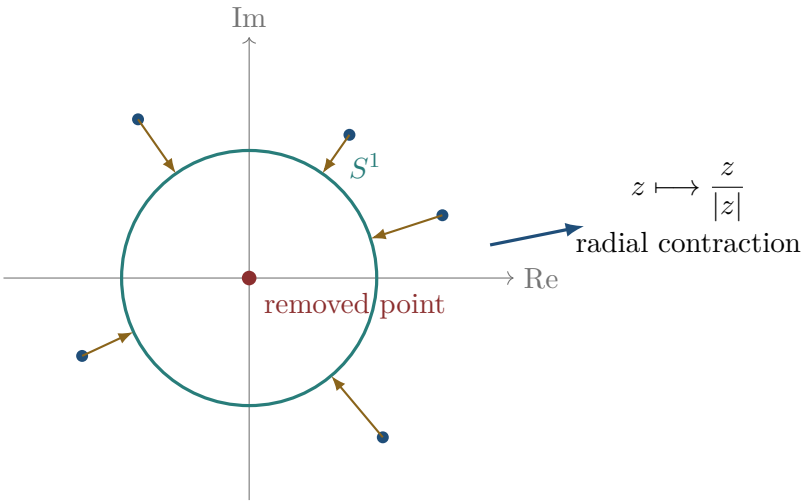


Figure 1.1: The punctured complex line retracts radially onto the unit circle. This topological circle is the compact core behind the name $\mathbb{G}_m$.

Taking products gives

$$(\mathbb{C}^\times)^n \simeq (S^1)^n \times (\mathbb{R}_{>0})^n,$$

and therefore

$$(\mathbb{C}^\times)^n \quad \text{deformation retracts onto} \quad (S^1)^n.$$

This is why $\mathbb{G}_m^n$ is called an algebraic torus.

Warning

The complex algebraic torus $(\mathbb{C}^\times)^n$ is not itself compact. It contains the compact topological torus $(S^1)^n$ as a deformation retract. The word “torus” records this topological relationship; it does not mean that the algebraic variety is literally a compact doughnut.

1.1.2 The algebraic definition

For most of this course, the base field $\Bbbk$ is algebraically closed.

Definition 1.1. An n-dimensional algebraic torus over $\Bbbk$ is an algebraic group isomorphic to

$$\mathbb{G}_m^n.$$

The word *algebraic group* means that multiplication and inversion are given by morphisms of algebraic varieties. On points, the group law of $\mathbb{G}_m^n$ is coordinatewise multiplication:

$$(t_1, \dots, t_n)(s_1, \dots, s_n) = (t_1 s_1, \dots, t_n s_n),$$

the identity is

$$(1, \dots, 1),$$

and inversion is

$$(t_1, \dots, t_n)^{-1} = (t_1^{-1}, \dots, t_n^{-1}).$$

The coordinate ring

$$\Bbbk[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$$

is exactly what allows these inverse coordinates to remain regular.

Observation 1.2. Over a field that is not algebraically closed, one distinguishes between *split* tori, which are already isomorphic to $\mathbb{G}_m^n$, and tori that become isomorphic to $\mathbb{G}_m^n$ only after extending the base field. The distinction is important in arithmetic geometry, but it will not play a role in the main development of these notes.

1.1.3 Examples close to home

The first examples are already familiar.

Example 1.3. The one-dimensional torus is

$$\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}.$$

As an affine variety,

$$\mathbb{G}_m = \operatorname{Spec} \Bbbk[x, x^{-1}].$$

Example 1.4. The dense open subset of $\mathbb{P}^2$ studied in Section 0.3 is

$$T = \{[x_0 : x_1 : x_2] : x_0 x_1 x_2 \neq 0\} \simeq \mathbb{G}_m^2.$$

After normalizing $x_0 = 1$, its group coordinates are

$$\left(\frac{x_1}{x_0}, \frac{x_2}{x_0}\right).$$

Thus Chapter 0 did not merely exhibit an open subset isomorphic to $\mathbb{G}_m^2$: it exhibited the algebraic torus that acts on all of $\mathbb{P}^2$.

Example 1.5. The group of invertible diagonal $n \times n$ matrices is an n-dimensional torus:

$$\left\{ \begin{pmatrix} t_1 & & 0 \\ & \ddots & \\ 0 & & t_n \end{pmatrix} : t_i \in \Bbbk^\times \right\} \simeq \mathbb{G}_m^n.$$

The determinant-one diagonal matrices form an $(n-1)$-dimensional torus:

$$\{(t_1, \dots, t_n) \in \mathbb{G}_m^n : t_1 \cdots t_n = 1\} \simeq \mathbb{G}_m^{n-1}.$$

For example, one may solve for t_n as

$$t_n = (t_1 \cdots t_{n-1})^{-1}.$$

Pause and think

The affine line with addition,

$$\mathbb{G}_a = (\mathbb{A}^1, +),$$

is also a one-dimensional algebraic group. Could it be isomorphic to an algebraic torus? Try to approach the question both algebraically and, over $\mathbb{C}$, topologically.

The algebraic answer is that $\mathbb{G}_a$ is not isomorphic to $\mathbb{G}_m$: its coordinate ring is $\Bbbk[x]$, not $\Bbbk[x, x^{-1}]$, and its group law is addition rather than multiplication. Over $\mathbb{C}$, the spaces are topologically different as well:

$$\mathbb{C} \quad \text{is contractible,} \qquad \mathbb{C}^\times \quad \text{retracts onto } S^1.$$

1.1.4 Why monomials are waiting for us

The multiplication law on a torus strongly favors monomials. On $\mathbb{G}_m$, the functions

$$t \longmapsto t^m, \qquad m \in \mathbb{Z},$$

satisfy

$$(ts)^m = t^m s^m.$$

On $\mathbb{G}_m^n$, the Laurent monomials

$$t_1^{m_1} \cdots t_n^{m_n}$$

have the same property:

$$(t_1 s_1)^{m_1} \cdots (t_n s_n)^{m_n} = (t_1^{m_1} \cdots t_n^{m_n})(s_1^{m_1} \cdots s_n^{m_n}).$$

Prediction

Suppose

$$f : \mathbb{G}_m^n \longrightarrow \mathbb{G}_m$$

is both a regular map and a group homomorphism. What forms can such maps take? Explore the cases $n = 1$ and $n = 2$ first, and try to organize the examples you find.

The answer is yes. Such maps are called *characters*, and their exponent vectors form a lattice. This simple fact is the entrance to the representation theory of algebraic tori and the explanation for the persistent appearance of monomials in Chapter 0.

1.2 The algebraic torus $\mathbb{G}_m^n$

The notation

$$\mathbb{G}_m^n$$

is compact, but it hides several compatible structures. It is at once

an affine variety, an algebraic group, and a space carrying many monomial functions.

In this section we put these structures on the table. The point is not to introduce a large amount of machinery. It is to understand exactly why Laurent polynomials are the natural algebraic language of a torus.

1.2.1 Removing the coordinate hyperplanes

Begin with affine space

$$\mathbb{A}^n = \operatorname{Spec} \Bbbk[x_1, \dots, x_n].$$

The algebraic torus is the open subset on which no coordinate vanishes:

$$\mathbb{G}_m^n = \{(t_1, \dots, t_n) \in \mathbb{A}^n : t_1 \cdots t_n \neq 0\}.$$

Equivalently,

$$\mathbb{G}_m^n = D(x_1 \cdots x_n) = \mathbb{A}^n \setminus V(x_1 \cdots x_n).$$

Since

$$V(x_1 \cdots x_n) = V(x_1) \cup \cdots \cup V(x_n),$$

we obtain $\mathbb{G}_m^n$ by removing the n coordinate hyperplanes.

For $n = 2$, this is the familiar picture

$$\mathbb{G}_m^2 = \mathbb{A}^2 \setminus (V(x) \cup V(y)).$$

This description immediately explains why the coordinates x_i become invertible on the torus. None of them vanishes there, so each reciprocal

$$x_i^{-1}$$

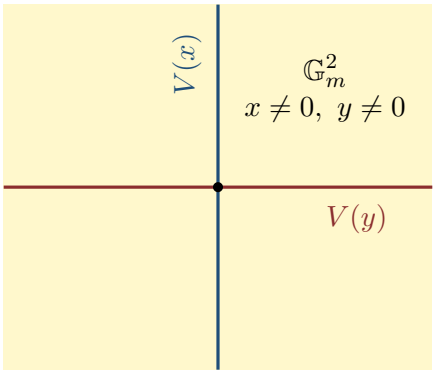


Figure 1.2: The torus $\mathbb{G}_m^2$ is obtained from $\mathbb{A}^2$ by removing the two coordinate axes. The colored lines are not part of the torus.

is a regular function on $\mathbb{G}_m^n$.

1.2.2 Laurent polynomials are the regular functions

The coordinate ring of $\mathbb{G}_m^n$ is obtained by inverting the product $x_1 \cdots x_n$:

$$\Bbbk[\mathbb{G}_m^n] = \Bbbk[x_1, \ldots, x_n]_{x_1 \cdots x_n}.$$

Since inverting the product makes every factor invertible, this ring is more familiarly written as

$$\Bbbk[\mathbb{G}_m^n] = \Bbbk[x_1^{\pm 1}, \ldots, x_n^{\pm 1}].$$

Its elements are Laurent polynomials:

$$f = \sum_{m \in F} c_m x_1^{m_1} \cdots x_n^{m_n},$$

where

$$F \subseteq \mathbb{Z}^n$$

is finite and the exponents m_i may be positive, zero, or negative.

Example 1.6. The function

$$f(x, y) = x^{-2} y^3 + 4xy^{-1} - 7$$

is regular on $\mathbb{G}_m^2$. It is not regular on all of $\mathbb{A}^2$, because its negative exponents produce poles along the coordinate axes.

Thus passing from affine space to the torus has one simple algebraic effect:

$$\begin{array}{ccc} \mathbb{A}^n & \rightsquigarrow & \mathbb{G}_m^n \\ \Bbbk[x_1, \ldots, x_n] & \rightsquigarrow & \Bbbk[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]. \end{array}$$

Removing the coordinate hyperplanes allows negative exponents.

Pause and think

Consider the Laurent monomial

$$x_1^{-3}x_2^2.$$

On which open subset of $\mathbb{A}^2$ is it regular? What is the smallest standard principal open subset on which it becomes a regular function?

Only the negative exponent creates a denominator. Hence the monomial is regular on

$$D(x_1) = \{x_1 \neq 0\}.$$

There is no need to remove $V(x_2)$, because the positive power x_2^2 is regular everywhere.

1.2.3 The group law is algebraic

On points, multiplication is coordinatewise:

$$\mu : \mathbb{G}_m^n \times \mathbb{G}_m^n \longrightarrow \mathbb{G}_m^n,$$

$$\mu\big((t_1,\dots,t_n),(s_1,\dots,s_n)\big) = (t_1s_1,\dots,t_ns_n).$$

The inverse map is

$$\iota : \mathbb{G}_m^n \longrightarrow \mathbb{G}_m^n, \qquad (t_1,\dots,t_n) \longmapsto (t_1^{-1},\dots,t_n^{-1}),$$

and the identity element is

$$e = (1,\dots,1).$$

Pause and think

Why are multiplication and inversion regular maps on $\mathbb{G}_m^n$? Write each output coordinate as a regular function on the source, and identify where the fact that the coordinates t_i are invertible is being used.

Multiplication and inversion are regular because their coordinate functions are Laurent monomials.

Multiplication	**Inversion**	**Identity**
$(t_i),(s_i) \longmapsto (t_is_i)$	$(t_i) \longmapsto (t_i^{-1})$	$e = (1,\dots,1)$

all formulas are built from Laurent monomials

Figure 1.3: The algebraic group structure of $\mathbb{G}_m^n$. Laurent monomials make both multiplication and inversion regular.

There is an equivalent description on coordinate rings. A regular map of affine varieties produces a homomorphism of coordinate rings in the opposite direction. For multiplication, inversion, and the identity, the coordinate functions satisfy

$$\begin{aligned}\mu^*(x_i) &= x_i \otimes x_i,\\ \iota^*(x_i) &= x_i^{-1},\\ e^*(x_i) &= 1.\end{aligned}$$

The first formula says that the i-th coordinate of a product is the product of the two i-th coordinates.

Observation 1.7. The three displayed formulas are the beginning of the Hopf algebra description of an algebraic group. We will not need that language in the main development. What matters here is the concrete principle: the group law on points becomes a rule for pulling back regular functions.

1.2.4 Maps into $\mathbb{G}_m$ and invertible functions

There is a useful general principle. Let X be an affine variety. A morphism

$$f : X \longrightarrow \mathbb{G}_m$$

is the same thing as an invertible regular function on X.

Indeed, the coordinate z on $\mathbb{G}_m$ is invertible, so its pullback

$$f^*(z)$$

must be a unit in $\Bbbk[X]$. Conversely, a unit

$$g \in \Bbbk[X]^\times$$

defines a morphism

$$X \longrightarrow \mathbb{G}_m, \qquad p \longmapsto g(p).$$

Specializing to $X = \mathbb{G}_m^n$, maps

$$\mathbb{G}_m^n \longrightarrow \mathbb{G}_m$$

are therefore controlled by the units of the Laurent polynomial ring

$$\Bbbk[x_1^{\pm 1}, \dots, x_n^{\pm 1}].$$

In the next section, we impose one additional condition: the map should respect multiplication. This will isolate the characters of the torus.

Prediction

Among all invertible Laurent polynomials, which ones should define group homomorphisms

$$\mathbb{G}_m^n \longrightarrow \mathbb{G}_m?$$

Try testing sums of monomials, scalar multiples of monomials, and pure Laurent monomials against the identity

$$f(ts) = f(t)f(s).$$

1.2.5 The dense torus of $\mathbb{P}^2$ revisited

The dense torus of $\mathbb{P}^2$ is

$$T = \{[x_0 : x_1 : x_2] : x_0x_1x_2 \neq 0\}.$$

On the chart U_0, it has coordinates

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0},$$

and hence

$$\Bbbk[T] = \Bbbk[z_1^{\pm 1}, z_2^{\pm 1}].$$

The other projective charts give different Laurent coordinates on the same torus. On U_1, for instance,

$$w_0 = \frac{x_0}{x_1} = z_1^{-1}, \qquad w_2 = \frac{x_2}{x_1} = z_1^{-1}z_2.$$

Thus the change of coordinates is monomial:

$$(z_1, z_2) \longmapsto (z_1^{-1}, z_1^{-1}z_2).$$

This is our first indication that the torus itself does not come with a preferred coordinate system. Different choices of coordinates are related by invertible monomial transformations. Later we will encode such changes by matrices in

$$\mathrm{GL}(n, \mathbb{Z}).$$

The three dictionaries

The algebraic torus may be read in three compatible ways.

Algebra	**Geometry**	**Combinatorics**
Its coordinate ring is $\Bbbk[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$. Regular functions are finite sums of Laurent monomials.	It is the complement of the coordinate hyperplanes: $\mathbb{G}_m^n = \mathbb{A}^n \setminus V(x_1 \cdots x_n)$.	Laurent monomials are indexed by exponent vectors $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$.

Guiding question

Regular functions on a torus are sums of Laurent monomials. Which of these functions are compatible with the group law?

1.3 Monomials and characters

At the end of the previous section, we asked which regular maps

$$f : \mathbb{G}_m^n \longrightarrow \mathbb{G}_m$$

are compatible with multiplication. There are many regular functions on a torus, but the group law is highly restrictive. It singles out the Laurent monomials.

1.3.1 Multiplicative regular functions

Definition 1.8. A *character* of an algebraic torus T is a morphism of algebraic groups

$$\chi : T \longrightarrow \mathbb{G}_m.$$

Thus χ is a regular map satisfying

$$\chi(ts) = \chi(t)\chi(s) \qquad \text{and} \qquad \chi(e) = 1.$$

For $T = \mathbb{G}_m^n$, every exponent vector

$$m = (m_1, \ldots, m_n) \in \mathbb{Z}^n$$

defines a character

$$\chi^m(t_1, \ldots, t_n) = t_1^{m_1} \cdots t_n^{m_n}.$$

It will often be convenient to abbreviate this Laurent monomial as

$$t^m.$$

Negative exponents cause no problem: every coordinate t_i is invertible on the torus. Moreover,

$$\chi^m(ts) = (t_1 s_1)^{m_1} \cdots (t_n s_n)^{m_n} = \chi^m(t)\chi^m(s),$$

so χ^m is indeed a group homomorphism.

Example 1.9. On $\mathbb{G}_m^2$, the exponent vector

$$m = (2, -3)$$

defines the character

$$\chi^{(2,-3)}(t_1, t_2) = t_1^2 t_2^{-3}.$$

The vector $m = (0, 0)$ gives the constant character

$$\chi^0 = 1.$$

1.3.2 There are no other characters

The examples above give every character of $\mathbb{G}_m^n$.

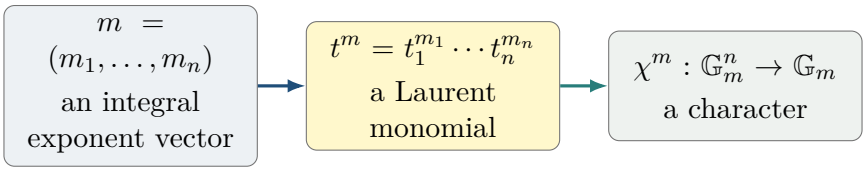


Figure 1.4: For a split torus, an integral exponent vector, a Laurent monomial, and a character are three descriptions of the same object.

Theorem 1.10 (Characters of a split torus)**.** *Every character*

$$\chi : \mathbb{G}_m^n \longrightarrow \mathbb{G}_m$$

has the form

$$\chi(t_1, \dots, t_n) = t_1^{m_1} \cdots t_n^{m_n}$$

for a unique vector

$$m = (m_1, \dots, m_n) \in \mathbb{Z}^n.$$

The statement follows from two elementary facts.

First, a morphism

$$\mathbb{G}_m^n \longrightarrow \mathbb{G}_m$$

is an invertible regular function on $\mathbb{G}_m^n$. The units in the Laurent polynomial ring are precisely

$$c\, t_1^{m_1} \cdots t_n^{m_n}, \qquad c \in \Bbbk^\times, \quad m_i \in \mathbb{Z}.$$

Second, a group homomorphism must send the identity to the identity. Evaluating at

$$e = (1, \dots, 1)$$

forces

$$c = 1.$$

Thus the multiplicative regular functions are exactly the pure Laurent monomials.

Observation 1.11. The description of the units in a Laurent polynomial ring is standard. One may prove it using the $\mathbb{Z}^n$-grading, or by comparing the extreme exponent vectors in a product. We will use the result freely; the Notes and references section lists standard sources for this background.

Pause and think

Why does a scalar multiple

$$c\, t^m \qquad (c \in \Bbbk^\times)$$

fail to be a character unless $c = 1$? Can you see the obstruction both by evaluating at the identity and by testing the equation

$$f(ts) = f(t)f(s)?$$

Not every regular function is a character. For instance,

$$t_1 + t_2$$

is a regular function on $\mathbb{G}_m^2$, but it can vanish and therefore does not even define a map to $\mathbb{G}_m$. A sum of several Laurent monomials usually loses the multiplicative behavior that characters require.

1.3.3 What the exponents record

Once coordinates have been chosen, the torus sits naturally inside affine space:

$$\mathbb{G}_m^n = \mathbb{A}^n \setminus (H_1 \cup \cdots \cup H_n), \qquad H_i = V(x_i).$$

A character

$$\chi^m = x_1^{m_1} \cdots x_n^{m_n}$$

is a regular function on $\mathbb{G}_m^n$, but it may have zeros or poles when regarded as a rational function on the surrounding affine space $\mathbb{A}^n$.

The exponent m_i records exactly what happens along the coordinate hyperplane H_i:

$$\operatorname{ord}_{H_i}(\chi^m) = m_i.$$

Thus

$m_i > 0$	χ^m vanishes to order m_i along H_i,
$m_i = 0$	χ^m has neither a zero nor a pole along H_i,
$m_i < 0$	χ^m has a pole of order $-m_i$ along H_i.

Equivalently, as a rational function on $\mathbb{A}^n$,

$$\operatorname{div}_{\mathbb{A}^n}(\chi^m) = m_1 H_1 + \cdots + m_n H_n.$$

This interpretation depends on the chosen coordinates: the affine space $\mathbb{A}^n$ and its coordinate hyperplanes arise from the chosen identification $T \simeq \mathbb{G}_m^n$. Later, when we work intrinsically, orders obtained by composing characters with one-parameter subgroups will be encoded by the pairing between the character and one-parameter subgroup lattices.

Example 1.12. Let

$$\chi(t_1, t_2, t_3) = t_1^2 t_2^{-1} t_3^4.$$

As a rational function on $\mathbb{A}^3$,

$$\operatorname{div}_{\mathbb{A}^3}(\chi) = 2H_1 - H_2 + 4H_3.$$

Thus χ vanishes to order 2 along H_1, has a simple pole along H_2, and vanishes to order 4 along H_3.

Observation 1.13. For the coordinate one-parameter map

$$\epsilon_i(s) = (1, \ldots, 1, s, 1, \ldots, 1),$$

one also has

$$\chi^m(\epsilon_i(s)) = s^{m_i}.$$

The same integer m_i therefore appears as an order at $s = 0$. This is a special coordinate case of the character–one-parameter subgroup pairing developed later in the chapter.

Prediction

Suppose no coordinates have been chosen on an abstract torus T. How might one record all of its characters without first identifying T with $\mathbb{G}_m^n$? What algebraic structure should the collection of characters carry?

1.3.4 Multiplying characters adds exponents

Characters can be multiplied pointwise:

$$(\chi\psi)(t) = \chi(t)\psi(t).$$

The product is again a character. For monomial characters,

$$\chi^m\chi^{m'} = \chi^{m+m'}.$$

Indeed,

$$t^m t^{m'} = t^{m+m'}.$$

Similarly,

$$(\chi^m)^{-1} = \chi^{-m},$$

and the identity element is the constant character

$$\chi^0 = 1.$$

Consequently, the characters of $\mathbb{G}_m^n$ form an abelian group under multiplication, and the exponent correspondence gives an isomorphism

$$\mathbb{Z}^n \longrightarrow \mathrm{Hom}_{\mathrm{alg.grp}}(\mathbb{G}_m^n, \mathbb{G}_m), \qquad m \longmapsto \chi^m.$$

Addition on the left corresponds to multiplication on the right.

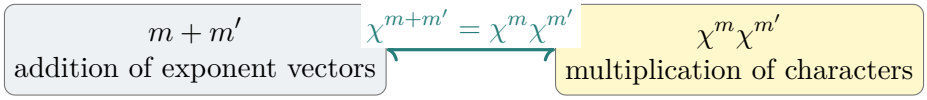


Figure 1.5: The group law on characters becomes ordinary addition of exponent vectors.

This group is traditionally denoted

$$X^*(T) = \mathrm{Hom}_{\mathrm{alg.grp}}(T, \mathbb{G}_m).$$

For $T = \mathbb{G}_m^n$, the chosen coordinates identify it with $\mathbb{Z}^n$. The next section studies this group intrinsically and explains why it should be regarded as a lattice.

1.3.5 Characters as one-dimensional representations

A character also determines an action of T on a one-dimensional vector space. If $V = \Bbbk v$, define

$$t \cdot v = \chi(t)v.$$

The group law for the action is exactly the identity

$$\chi(ts) = \chi(t)\chi(s).$$

Conversely, every algebraic action of T on a one-dimensional vector space arises from a character.

Observation 1.14. This is the representation-theoretic reason for the word *character*: characters are precisely the one-dimensional algebraic representations of T. This use of the word differs from the character of a finite-dimensional representation defined as its trace. Both notions will coexist comfortably once weight-space decompositions enter the picture.

Example 1.15. Let $T = \mathbb{G}_m^2$ act on the one-dimensional vector space $V = \Bbbk v$ by

$$(t_1, t_2) \cdot v = t_1^3 t_2^{-2} v.$$

Then V is a one-dimensional representation with character

$$\chi^{(3,-2)}.$$

The three dictionaries

A character is one object seen from three viewpoints.

Algebra	Geometry	Combinatorics
It is a group homomorphism $\chi : T \to \mathbb{G}_m$ or, equivalently, a one-dimensional representation of T.	It is a nowhere-vanishing regular function compatible with multiplication on the torus.	After choosing torus coordinates, it is indexed by an exponent vector $m \in \mathbb{Z}^n$.

Guiding question

The identification $X^*(\mathbb{G}_m^n) \simeq \mathbb{Z}^n$ used a chosen coordinate system. What remains of this description when the torus is presented without coordinates?

1.4 The character lattice

For the coordinatized torus $\mathbb{G}_m^n$, characters were indexed by vectors in $\mathbb{Z}^n$. The coordinates made this description easy, but they also hid what was intrinsic. An abstract torus does not come with preferred functions

$$t_1, \dots, t_n.$$

What it does come with is the collection of all its characters.

1.4.1 The intrinsic definition

Definition 1.16. Let T be an algebraic torus. Its *character group* is

$$X^*(T) = \mathrm{Hom}_{\mathrm{alg.grp}}(T, \mathbb{G}_m).$$

The group law is pointwise multiplication:

$$(\chi\psi)(t) = \chi(t)\psi(t).$$

The identity element is the constant character 1, and the inverse of χ is the character

$$\chi^{-1} : t \longmapsto \chi(t)^{-1}.$$

Thus $X^*(T)$ is an abelian group.

For $T = \mathbb{G}_m^n$, the classification of characters from theorem 1.10 gives

$$X^*(\mathbb{G}_m^n) \simeq \mathbb{Z}^n.$$

Under this identification,

$$m + m' \quad \longleftrightarrow \quad \chi^m \chi^{m'}.$$

Definition 1.17. A *lattice* is a free abelian group of finite rank.

Theorem 1.18. *If T is an n-dimensional algebraic torus, then*

$$M := X^*(T)$$

is a lattice of rank n.

No preferred isomorphism

$$M \simeq \mathbb{Z}^n$$

is built into the torus. Such an isomorphism appears only after coordinates have been chosen.

Observation 1.19. Because our tori are split by definition, the theorem follows by choosing an isomorphism $T \simeq \mathbb{G}_m^n$. The important point is not the existence of such an isomorphism, but the fact that the resulting lattice $X^*(T)$ is intrinsic even though a basis is not.

1.4.2 Coordinates are bases of the character lattice

Suppose

$$\varphi : T \xrightarrow{\sim} \mathbb{G}_m^n$$

is a choice of torus coordinates. Pulling back the coordinate characters on $\mathbb{G}_m^n$ gives characters

$$\chi_1, \ldots, \chi_n \in X^*(T).$$

Every character of T can then be written uniquely as

$$\chi_1^{m_1} \cdots \chi_n^{m_n}, \qquad (m_1, \ldots, m_n) \in \mathbb{Z}^n.$$

Thus $\chi_1, \ldots, \chi_n$ form a basis of the lattice M.

Conversely, a basis

$$\chi_1, \ldots, \chi_n$$

of M defines a map

$$T \longrightarrow \mathbb{G}_m^n, \qquad t \longmapsto (\chi_1(t), \ldots, \chi_n(t)),$$

and this map is an isomorphism of algebraic tori.

Therefore:

choosing coordinates on T $\iff$ choosing a basis of $X^*(T)$.

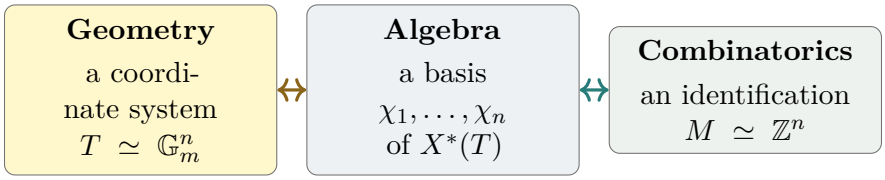


Figure 1.6: Choosing coordinates on a torus is the same as choosing a basis of its character lattice. The lattice is intrinsic; the basis is not.

Pause and think

A vector space does not come with a preferred basis. In what sense is the same true of a torus? Which formulas from the previous sections depended on a chosen basis of $X^*(T)$, and which statements did not?

1.4.3 The coordinate ring as a group algebra

Let M be a lattice. Its *group algebra* over $\Bbbk$ is

$$\Bbbk[M] = \bigoplus_{m \in M} \Bbbk\, \chi^m,$$

where the symbols χ^m multiply according to

$$\chi^m \chi^{m'} = \chi^{m+m'}.$$

An element of $\Bbbk[M]$ is a finite sum

$$\sum_{m \in F} c_m \chi^m, \qquad F \subseteq M \text{ finite}.$$

For $M = \mathbb{Z}^n$, choosing the standard basis identifies

$$\Bbbk[M] \simeq \Bbbk[x_1^{\pm 1}, \dots, x_n^{\pm 1}].$$

The basis element $\chi^{(m_1,\dots,m_n)}$ becomes the Laurent monomial

$$x_1^{m_1} \cdots x_n^{m_n}.$$

For an arbitrary torus T, evaluation of characters gives a canonical isomorphism

$$\Bbbk[M] \xrightarrow{\sim} \Bbbk[T], \qquad \chi^m \longmapsto \big(t \mapsto \chi^m(t)\big).$$

Thus the coordinate ring of a torus is canonically the group algebra of its character lattice:

$$\boxed{\Bbbk[T] \simeq \Bbbk[X^*(T)].}$$

The word *canonically* matters. The Laurent polynomial presentation

$$\Bbbk[T] \simeq \Bbbk[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$$

depends on a basis of M, but the group algebra

$$\Bbbk[M]$$

does not.

Example 1.20. Let $T \subseteq \mathbb{P}^2$ be the dense torus. On the chart U_0, the coordinates

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}$$

give a basis m_1, m_2 of $M = X^*(T)$.

On U_1, the torus coordinates are

$$w_0 = \frac{x_0}{x_1} = z_1^{-1}, \qquad w_2 = \frac{x_2}{x_1} = z_1^{-1} z_2.$$

In the basis m_1, m_2, these new coordinate characters have exponent vectors

$$(-1, 0) \qquad \text{and} \qquad (-1, 1).$$

They form another basis of the same lattice M.

Likewise, the coordinates on U_2,

$$r_0 = \frac{x_0}{x_2} = z_2^{-1}, \qquad r_1 = \frac{x_1}{x_2} = z_1 z_2^{-1},$$

correspond to the basis vectors

$$(0, -1) \qquad \text{and} \qquad (1, -1).$$

The torus has not changed; only the chosen basis of its character lattice has changed.

1.4.4 Morphisms reverse the direction on character lattices

Let

$$\phi : T_1 \longrightarrow T_2$$

be a morphism of algebraic tori. Every character

$$\chi : T_2 \longrightarrow \mathbb{G}_m$$

can be pulled back along ϕ:

$$\phi^*(\chi) = \chi \circ \phi : T_1 \longrightarrow \mathbb{G}_m.$$

Thus ϕ induces a homomorphism of lattices

$$\phi^* : X^*(T_2) \longrightarrow X^*(T_1).$$

Notice the reversal of direction:

$$T_1 \longrightarrow T_2 \qquad \text{produces} \qquad X^*(T_2) \longrightarrow X^*(T_1).$$

The character lattice is therefore a contravariant construction.

Example 1.21. Consider the diagonal embedding

$$\delta : \mathbb{G}_m \longrightarrow \mathbb{G}_m^2, \qquad t \longmapsto (t, t).$$

A character of $\mathbb{G}_m^2$ has the form

$$\chi^{(a,b)}(t_1, t_2) = t_1^a t_2^b.$$

Pulling it back gives

$$\delta^*(\chi^{(a,b)})(t) = t^{a+b}.$$

On lattices, the map is therefore

$$\delta^* : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (a, b) \longmapsto a + b.$$

Pause and think

Why must the direction reverse? Start with a character on the target torus and ask where it can be evaluated after a point of the source torus has been mapped forward.

1.4.5 Passing to real vector spaces

Once the lattice M has been identified, we form the associated real vector space

$$M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}.$$

If $\operatorname{rank} M = n$, then

$$M_{\mathbb{R}} \simeq \mathbb{R}^n,$$

but again only after choosing a basis.

The lattice points

$$M \subseteq M_{\mathbb{R}}$$

index characters. Convex combinations of these points do not usually correspond to characters, but they allow us to build polytopes and polyhedra. This is why the character lattice is the natural home for the triangles and polytopes that appeared in Chapter 0.

Guiding question

The character lattice records maps $T \to \mathbb{G}_m$. What should record maps in the opposite direction,

$$\mathbb{G}_m \longrightarrow T?$$

How might such maps interact with characters?

The three dictionaries

The character lattice packages the intrinsic algebra of a torus.

Algebra	**Geometry**	**Combinatorics**
$\Bbbk[T] \simeq \Bbbk[M]$.	A basis of M is a coordinate system $T \simeq \mathbb{G}_m^n$.	M is a lattice, and $M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}$
Characters form a multiplicative basis of the coordinate ring.	Different coordinates describe the same torus.	is the space in which exponent polytopes live.

1.5 One-parameter subgroups

Characters are morphisms

$$T \longrightarrow \mathbb{G}_m.$$

We now reverse the arrow and study morphisms

$$\mathbb{G}_m \longrightarrow T.$$

Geometrically, these maps are parameterized algebraic curves inside the torus: the parameter t ranges through $\mathbb{G}_m$. Because the curve respects the group law, it is natural to think of it as an algebraic motion. When the torus sits inside a larger variety, following such a motion toward $t = 0$ often reveals a boundary point.

1.5.1 Turning the arrow around

Definition 1.22. A *one-parameter subgroup* of an algebraic torus T is a morphism of algebraic groups

$$\lambda : \mathbb{G}_m \longrightarrow T.$$

The collection of all one-parameter subgroups is denoted

$$X_*(T) = \mathrm{Hom}_{\mathrm{alg.grp}}(\mathbb{G}_m, T).$$

It is also called the *cocharacter group* of T.

A one-parameter subgroup is therefore a very special parameterized curve in T. It passes through the identity at $t = 1$, and its parameterization is compatible with multiplication:

$$\lambda(tt') = \lambda(t)\lambda(t').$$

Not every algebraic curve in T has this property.

For the coordinatized torus $T = \mathbb{G}_m^n$, every vector

$$u = (u_1, \ldots, u_n) \in \mathbb{Z}^n$$

defines a one-parameter subgroup

$$\lambda^u(t) = (t^{u_1}, \ldots, t^{u_n}).$$

Negative exponents are allowed because $t \neq 0$ on $\mathbb{G}_m$.

Example 1.23. The vector

$$u = (2, -1, 3)$$

defines

$$\lambda^u(t) = (t^2, t^{-1}, t^3)$$

in $\mathbb{G}_m^3$.

The converse also holds.

Theorem 1.24 (One-parameter subgroups of a split torus)**.** *Every one-parameter subgroup*

$$\lambda : \mathbb{G}_m \longrightarrow \mathbb{G}_m^n$$

has the form

$$\lambda(t) = (t^{u_1}, \ldots, t^{u_n})$$

for a unique vector

$$u = (u_1, \ldots, u_n) \in \mathbb{Z}^n.$$

Indeed, composing λ with each coordinate projection

$$\mathbb{G}_m^n \longrightarrow \mathbb{G}_m$$

produces a character of $\mathbb{G}_m$, and every such character is a power $t \mapsto t^{u_i}$.

1.5.2 The one-parameter subgroup lattice

One-parameter subgroups can be multiplied pointwise:

$$(\lambda\lambda')(t) = \lambda(t)\lambda'(t).$$

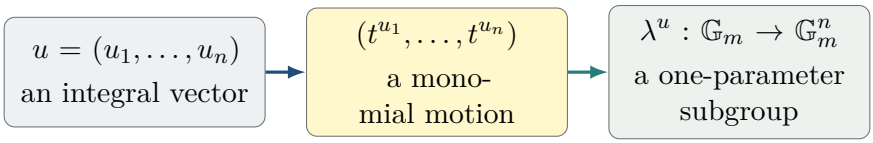


Figure 1.7: After choosing coordinates, an integral vector and a one-parameter subgroup are two descriptions of the same object.

For a coordinatized torus,

$$\lambda^u(t)\lambda^{u'}(t) = \lambda^{u+u'}(t).$$

The inverse of λ^u is λ^{-u}, and the identity is the constant one-parameter subgroup

$$\lambda^0(t) = e.$$

Thus $X_*(T)$ is an abelian group.

Theorem 1.25. *If T is an n-dimensional algebraic torus, then*

$$N := X_*(T)$$

is a lattice of rank n.

For $T = \mathbb{G}_m^n$, chosen coordinates give an identification

$$N \simeq \mathbb{Z}^n.$$

As with the character lattice, this identification depends on the choice of coordinates, while the lattice N itself is intrinsic.

We will usually call N the *one-parameter subgroup lattice*. The notation is chosen to match the standard conventions of toric geometry:

$$M = X^*(T), \qquad N = X_*(T).$$

Observation 1.26. The word *cocharacter* is common in the theory of algebraic groups. In toric geometry, the phrase *one-parameter subgroup* often carries more geometric intuition, because the map $\mathbb{G}_m \to T$ can be followed toward the boundary of a compactification.

1.5.3 One-parameter subgroups as motions

A one-parameter subgroup always passes through the identity:

$$\lambda(1) = e.$$

To obtain a motion through an arbitrary point $p \in T$, translate the curve:

$$t \longmapsto p\lambda(t).$$

For $T = \mathbb{G}_m^n$, if

$$p = (a_1, \dots, a_n)$$

and

$$\lambda = \lambda^u,$$

then the translated motion is

$$t \longmapsto (a_1 t^{u_1}, \dots, a_n s^{u_n}).$$

The image of λ remains inside T as long as $t \neq 0$. The interesting question is what happens as $t \to 0$. Since 0 is not a point of $\mathbb{G}_m$, this is really a question about whether the map extends across $t = 0$ after T has been placed inside a larger variety.

Pause and think

Why is the phrase "take the limit as $t \to 0$" not intrinsic to the torus T alone? What extra geometric object must be chosen before such a limit can be discussed?

A torus contains no boundary point at which the motion could arrive. Limits become meaningful only after choosing an embedding

$$T \subseteq X$$

into a larger variety.

1.5.4 Limits in affine space

Consider the standard inclusion

$$\mathbb{G}_m^n \subseteq \mathbb{A}^n.$$

The one-parameter subgroup

$$\lambda^u(t) = (t^{u_1}, \dots, t^{u_n})$$

extends to a morphism

$$\mathbb{A}^1 \longrightarrow \mathbb{A}^n$$

if and only if

$$u_i \geq 0 \qquad \text{for every } i.$$

Indeed, a negative exponent produces a pole at $t = 0$.

When all $u_i \geq 0$, the limit is obtained coordinate by coordinate:

$$\lim_{t \to 0} t^{u_i} = \begin{cases} 1, & u_i = 0, \\ 0, & u_i > 0. \end{cases}$$

Thus the vanishing pattern of the exponents determines the coordinate stratum reached by the motion.

Example 1.27. Inside $\mathbb{G}_m^2 \subseteq \mathbb{A}^2$, consider:

$$\lambda^{(2,1)}(t) = (t^2, t),$$
$$\lambda^{(0,3)}(t) = (1, t^3),$$
$$\lambda^{(-1,2)}(t) = (t^{-1}, t^2).$$

The first motion extends to $t = 0$ and reaches the origin. The second extends and reaches $(1, 0)$. The third does not extend to $\mathbb{A}^1 \to \mathbb{A}^2$, because its first coordinate has a pole.

For a translated motion

$$t \longmapsto (a_1 t^{u_1}, \dots, a_n s^{u_n}),$$

the same criterion applies. When $u_i = 0$, the i-th coordinate survives as a_i; when $u_i > 0$, it tends to zero.

Example 1.28. Let $(a, b) \in \mathbb{G}_m^2$. The translated motion

$$t \longmapsto (a, bs^3)$$

approaches

$$(a, 0).$$

As a varies, these motions reach every point of

$$V(y) \setminus \{(0, 0)\}.$$

Thus one exponent vector describes a whole family of limits after translation.

Pause and think

For $u = (u_1, u_2)$, classify the limit of

$$(a, b)\lambda^u(t) = (as^{u_1}, bs^{u_2})$$

inside $\mathbb{A}^2$, whenever the limit exists. Which choices of u reach the open torus orbit, an open coordinate axis, or the origin?

The answer reproduces the orbit stratification of $\mathbb{A}^2$:

conditions on u	limit of $(a, b)\lambda^u(t)$
$u_1 = u_2 = 0$	$(a, b) \in \mathbb{G}_m^2$
$u_1 > 0,\ u_2 = 0$	$(0, b) \in V(x) \setminus \{0\}$
$u_1 = 0,\ u_2 > 0$	$(a, 0) \in V(y) \setminus \{0\}$
$u_1 > 0,\ u_2 > 0$	$(0, 0)$.

1.5.5 The same motion in a different compactification

A one-parameter subgroup that has no limit in $\mathbb{A}^n$ may acquire a limit in another variety containing the same torus.

For the dense torus of $\mathbb{P}^2$, write

$$T = \{[1 : t_1 : t_2] : t_1, t_2 \in \Bbbk^\times\}.$$

The one-parameter subgroup determined by $u = (u_1, u_2)$ traces

$$t \longmapsto [1 : t^{u_1} : t^{u_2}].$$

Even when one of the exponents is negative, projective rescaling may produce a limit.

Set

$$c = \min\{0, u_1, u_2\}.$$

Then

$$[1 : t^{u_1} : t^{u_2}] = [t^{-c} : t^{u_1-c} : t^{u_2-c}],$$

and all three exponents on the right are nonnegative. As $t \to 0$, the coordinates whose original exponents attain the minimum survive.

Example 1.29. We obtain:

$$\begin{aligned}
u = (1,2) &\implies [1 : t : t^2] \longrightarrow [1 : 0 : 0] = p_0,\\
u = (-1,0) &\implies [1 : t^{-1} : 1] = [t : 1 : t] \longrightarrow [0 : 1 : 0] = p_1,\\
u = (0,-1) &\implies [1 : 1 : t^{-1}] = [t : t : 1] \longrightarrow [0 : 0 : 1] = p_2,\\
u = (-1,-1) &\implies [1 : t^{-1} : t^{-1}] = [t : 1 : 1] \longrightarrow [0 : 1 : 1].
\end{aligned}$$

The last limit lies in the one-dimensional orbit O_0, rather than at a fixed point.

This is a first indication that different regions of

$$N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}$$

should correspond to different boundary limits. Cones in $N_{\mathbb{R}}$ will eventually organize precisely this behavior.

Prediction

For the compactification $T \subseteq \mathbb{P}^2$, which vectors $u \in N_{\mathbb{R}}$ lead to the same limiting orbit? Try to divide the plane into regions according to the answer.

1.5.6 Primitive one-parameter subgroups

A nonzero lattice vector $u \in N$ is called *primitive* if it is not a positive integer multiple of another lattice vector. Thus

$$u = ku_0, \qquad k \geq 1,$$

with u_0 primitive.

The one-parameter subgroups

$$\lambda^u \qquad \text{and} \qquad \lambda^{u_0}$$

have the same image subgroup, but they parametrize it differently:

$$\lambda^u(t) = \lambda^{u_0}(t^k).$$

The primitive vector is therefore the most economical lattice generator of that one-dimensional direction.

This distinction becomes important when rays appear. A ray contains many lattice points, but it has a unique primitive lattice generator.

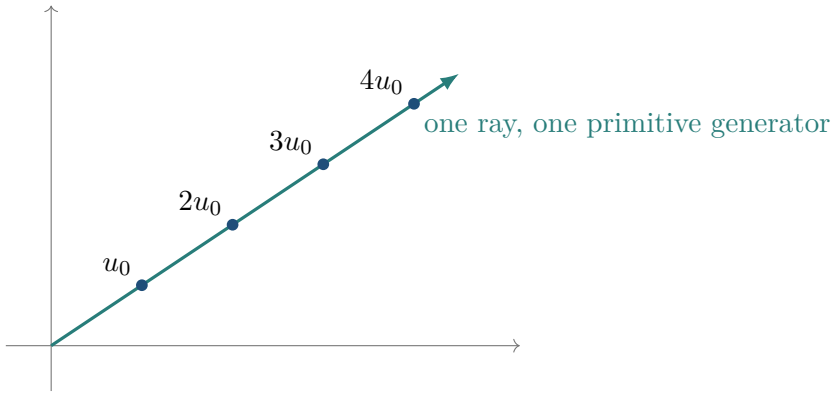


Figure 1.8: All positive multiples of u_0 lie on the same ray, but u_0 is its unique primitive lattice generator.

1.5.7 Functoriality

Let

$$\phi : T_1 \longrightarrow T_2$$

be a morphism of algebraic tori. A one-parameter subgroup of T_1 can be followed by ϕ:

$$\mathbb{G}_m \xrightarrow{\lambda} T_1 \xrightarrow{\phi} T_2.$$

Hence ϕ induces a homomorphism

$$\phi_* : X_*(T_1) \longrightarrow X_*(T_2), \qquad \lambda \longmapsto \phi \circ \lambda.$$

This time the direction is preserved:

$$T_1 \longrightarrow T_2 \qquad \text{produces} \qquad X_*(T_1) \longrightarrow X_*(T_2).$$

Example 1.30. Consider the multiplication map

$$\mu : \mathbb{G}_m^2 \longrightarrow \mathbb{G}_m, \qquad (t_1, t_2) \longmapsto t_1 t_2.$$

For $u = (a, b) \in X_*(\mathbb{G}_m^2) \simeq \mathbb{Z}^2$,

$$(\mu \circ \lambda^u)(t) = t^a s^b = t^{a+b}.$$

Thus

$$\mu_* : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (a, b) \longmapsto a + b.$$

Characters and one-parameter subgroups therefore behave in opposite ways under a torus morphism:

characters	**one-parameter subgroups**
$\phi^* : X^*(T_2) \to X^*(T_1)$	$\phi_* : X_*(T_1) \to X_*(T_2)$
pull back	push forward.

1.5.8 A first meeting between M and N

Take a character

$$\chi : T \longrightarrow \mathbb{G}_m$$

and a one-parameter subgroup

$$\lambda : \mathbb{G}_m \longrightarrow T.$$

Their composition is a character of $\mathbb{G}_m$:

$$\chi \circ \lambda : \mathbb{G}_m \longrightarrow \mathbb{G}_m.$$

Therefore there is a unique integer r such that

$$(\chi \circ \lambda)(t) = t^r.$$

This integer has two equivalent interpretations:

$$r = \text{the exponent of the character } \chi \circ \lambda = \operatorname{ord}_{t=0}(\chi \circ \lambda),$$

where the second expression regards t^r as a rational function on $\mathbb{A}^1$. A positive value gives a zero, a negative value gives a pole, and zero gives neither.

For coordinatized tori, if

$$\chi = \chi^m, \qquad m = (m_1, \ldots, m_n),$$

and

$$\lambda = \lambda^u, \qquad u = (u_1, \ldots, u_n),$$

then

$$(\chi^m \circ \lambda^u)(t) = t^{m_1 u_1 + \cdots + m_n u_n}.$$

The next section will isolate this integer as the natural pairing between the two lattices M and N.

The three dictionaries

A one-parameter subgroup has three compatible interpretations.

Algebra	Geometry	Combinatorics
It is a homomorphism $\lambda : \mathbb{G}_m \to T$. Such maps form the lattice $N = X_*(T)$.	It is an algebraic motion inside T. In a compactification, its closure may meet a boundary orbit.	After choosing coordinates, it is an integral vector $u \in \mathbb{Z}^n$. Rays remember primitive directions.

Guiding question

Why do characters and one-parameter subgroups form dual lattices, and what geometric information is measured by their pairing?

1.6 Dual lattices and the pairing

We now have two lattices attached to a torus:

$$M = X^*(T) = \mathrm{Hom}_{\mathrm{alg.grp}}(T, \mathbb{G}_m)$$

and

$$N = X_*(T) = \mathrm{Hom}_{\mathrm{alg.grp}}(\mathbb{G}_m, T).$$

The arrows point in opposite directions, so the two kinds of maps can be composed:

$$\mathbb{G}_m \xrightarrow{\lambda} T \xrightarrow{\chi} \mathbb{G}_m.$$

This composition is again a character of $\mathbb{G}_m$, and therefore it is an integral power map. The exponent of that power is the basic numerical invariant connecting M and N.

1.6.1 The pairing by composition

Let

$$m \in M \qquad \text{and} \qquad u \in N.$$

We write the corresponding character and one-parameter subgroup as

$$\chi^m : T \longrightarrow \mathbb{G}_m, \qquad \lambda^u : \mathbb{G}_m \longrightarrow T.$$

Their composition is a character of $\mathbb{G}_m$, so there is a unique integer $\langle m, u \rangle$ such that

$$\chi^m \circ \lambda^u(t) = t^{\langle m,u \rangle}.$$

Definition 1.31. The integer

$$\langle m, u \rangle \in \mathbb{Z}$$

is the *pairing* of the character m with the one-parameter subgroup u.

The pairing may be read in several equivalent ways:

$$\langle m, u \rangle = \text{the exponent of } \chi^m \circ \lambda^u = \mathrm{ord}_{t=0}(\chi^m \circ \lambda^u).$$

Thus

$$\begin{array}{l|l} \langle m,u \rangle > 0 & \chi^m \circ \lambda^u \text{ has a zero of order } \langle m,u \rangle \text{ at } 0, \\ \langle m,u \rangle = 0 & \chi^m \circ \lambda^u \text{ has neither a zero nor a pole at } 0, \\ \langle m,u \rangle < 0 & \chi^m \circ \lambda^u \text{ has a pole of order } -\langle m,u \rangle \text{ at } 0. \end{array}$$

This is the intrinsic version of the coordinate calculation previewed in the previous section.

Example 1.32. Let $T = \mathbb{G}_m^3$, and take

$$m = (2, -1, 4) \in M \simeq \mathbb{Z}^3, \qquad u = (-1, 3, 2) \in N \simeq \mathbb{Z}^3.$$

Then

$$\chi^m(t_1, t_2, t_3) = t_1^2 t_2^{-1} t_3^4$$

and

$$\lambda^u(t) = (t^{-1}, t^3, t^2).$$

Therefore

$$\begin{aligned} \chi^m(\lambda^u(t)) &= (t^{-1})^2 (t^3)^{-1} (t^2)^4 \\ &= t^{-2-3+8} \\ &= t^3. \end{aligned}$$

Hence

$$\langle m, u \rangle = 3.$$

Pause and think

Compute $\langle m, u \rangle$ in the previous example without composing the maps explicitly. What familiar operation on the two coordinate vectors appears?

After choosing dual coordinate bases, the pairing becomes the usual dot product:

$$\boxed{\langle m, u \rangle = m_1 u_1 + \cdots + m_n u_n.}$$

The formula is convenient, but the pairing itself was defined without coordinates.

1.6.2 Why the pairing is bilinear

The pairing is additive in each variable:

$$\langle m + m', u \rangle = \langle m, u \rangle + \langle m', u \rangle$$

and

$$\langle m, u + u' \rangle = \langle m, u \rangle + \langle m, u' \rangle .$$

The first identity follows because multiplication of characters corresponds to addition in M:

$$\begin{aligned} \chi^{m+m'} \circ \lambda^u &= (\chi^m \chi^{m'}) \circ \lambda^u \\ &= (\chi^m \circ \lambda^u)(\chi^{m'} \circ \lambda^u) \\ &= t^{\langle m, u \rangle} t^{\langle m', u \rangle}. \end{aligned}$$

The exponents add. The second identity follows similarly from

$$\lambda^{u+u'}(t) = \lambda^u(t) \lambda^{u'}(t).$$

Thus we have a bilinear map of lattices

$$\langle \quad , \quad \rangle : M \times N \longrightarrow \mathbb{Z}.$$

Observation 1.33. The word *bilinear* here means additive in each variable because M and N are abelian groups. After tensoring with $\mathbb{R}$, the same pairing becomes bilinear in the usual real-vector-space sense.

1.6.3 Perfect duality

Fix $m \in M$. The rule

$$u \longmapsto \langle m, u \rangle$$

is a homomorphism

$$N \longrightarrow \mathbb{Z}.$$

Therefore every character determines an element of the dual lattice

$$N^\vee = \mathrm{Hom}_{\mathbb{Z}}(N, \mathbb{Z}).$$

This gives a homomorphism

$$M \longrightarrow N^\vee.$$

Likewise, fixing $u \in N$ gives a homomorphism

$$M \longrightarrow \mathbb{Z}, \qquad m \longmapsto \langle m, u \rangle,$$

and therefore a map

$$N \longrightarrow M^\vee.$$

Theorem 1.34 (Duality of characters and one-parameter subgroups)**.** *For an algebraic torus T, the pairing is perfect. Equivalently, the natural maps*

$$M \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$$

and

$$N \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$$

are isomorphisms.

Thus M and N are dual lattices:

$$\boxed{M = N^\vee, \qquad N = M^\vee.}$$

For $T = \mathbb{G}_m^n$, this statement is visible in coordinates. Let

$$e_1, \ldots, e_n$$

be the coordinate character basis of M, and let

$$e_1^\vee, \ldots, e_n^\vee$$

be the coordinate one-parameter subgroup basis of N. Then

$$\left\langle e_i, e_j^\vee \right\rangle = \delta_{ij}.$$

These are dual bases.

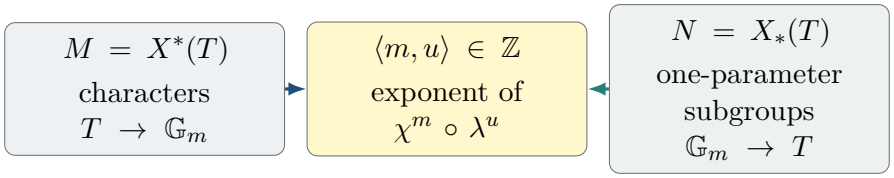


Figure 1.9: A character and a one-parameter subgroup compose to an integral power map on $\mathbb{G}_m$. This composition produces the perfect pairing between M and N.

Pause and think

Suppose $m \in M$ satisfies

$$\langle m, u \rangle = 0 \qquad \text{for every } u \in N.$$

Why should $m = 0$? Interpret this first in coordinates, and then in terms of the perfect-pairing theorem.

1.6.4 Real vector spaces and linear functions

Tensoring with $\mathbb{R}$ gives dual real vector spaces

$$M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}, \qquad N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}.$$

The integral pairing extends uniquely to a real bilinear pairing

$$\langle \quad , \quad \rangle : M_{\mathbb{R}} \times N_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

Consequently,

$$M_{\mathbb{R}} \simeq N_{\mathbb{R}}^\vee, \qquad N_{\mathbb{R}} \simeq M_{\mathbb{R}}^\vee.$$

A point $m \in M_{\mathbb{R}}$ may therefore be viewed as a linear function on $N_{\mathbb{R}}$:

$$u \longmapsto \langle m, u \rangle .$$

Similarly, a point $u \in N_{\mathbb{R}}$ defines a linear function on $M_{\mathbb{R}}$.

This is the reason inequalities in toric geometry naturally take the form

$$\langle m, u \rangle \geq c.$$

The vector u determines a linear functional on $M_{\mathbb{R}}$, and the inequality cuts out a half-space.

Observation 1.35. In dimension two, the zero set of a nonzero linear functional is a line through the origin. In higher dimensions it is a hyperplane. This linear-algebraic picture will organize cones, dual cones, and polyhedra throughout the course.

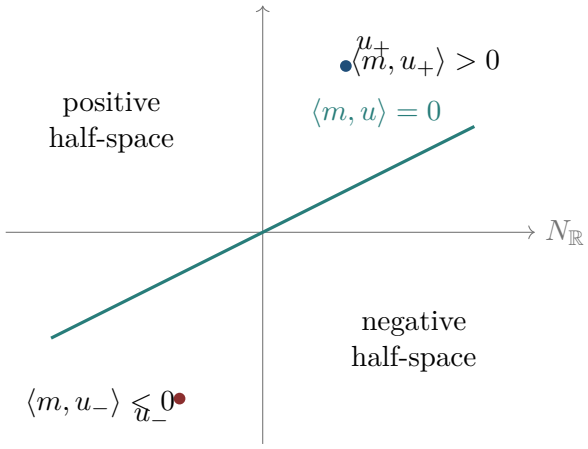


Figure 1.10: A character $m \in M_{\mathbb{R}}$ acts as a linear function on $N_{\mathbb{R}}$. Its zero set is a hyperplane separating positive and negative regions.

1.6.5 Orders along one-parameter motions

The pairing has a direct geometric meaning. Suppose T is embedded as a dense open subset of a variety X, and suppose the one-parameter subgroup λ^u extends to a map

$$\overline{\lambda^u} : \mathbb{A}^1 \longrightarrow X.$$

Then a character χ^m, viewed as a rational function on X, pulls back to

$$\overline{\lambda^u}^*(\chi^m) = t^{\langle m,u\rangle}.$$

Hence

$$\langle m, u\rangle = \operatorname{ord}_{t=0}\big(\overline{\lambda^u}^*(\chi^m)\big).$$

This is different from the coordinate-hyperplane interpretation of an individual exponent m_i. There, after choosing a specific compactification

$$\mathbb{G}_m^n \subseteq \mathbb{A}^n,$$

the integer m_i measures the order of χ^m along the coordinate divisor $V(x_i)$. Here, for an arbitrary one-parameter subgroup u, the integer $\langle m, u\rangle$ measures the order seen along the motion determined by u.

The two viewpoints agree in the coordinate case. If

$$u = e_i^\vee,$$

then

$$\langle m, e_i^\vee\rangle = m_i.$$

Example 1.36. In $T = \mathbb{G}_m^2$, let

$$\chi^m(x, y) = x^2 y^{-3}$$

and let

$$\lambda^u(t) = (t^4, t).$$

Then

$$\langle m, u\rangle = 2 \cdot 4 - 3 \cdot 1 = 5.$$

Along this motion,

$$\chi^m(t^4, t) = t^5,$$

so the character vanishes to order 5 at the limiting parameter $t = 0$.

Prediction

Suppose a collection of characters

$$\chi^{m_1}, \ldots, \chi^{m_r}$$

is evaluated along the same one-parameter subgroup u. How might the numbers

$$\langle m_1, u\rangle, \ldots, \langle m_r, u\rangle$$

help predict which monomials dominate, vanish, or survive as $t \to 0$?

1.6.6 Compatibility with torus morphisms

Let

$$\phi : T_1 \longrightarrow T_2$$

be a morphism of algebraic tori. Recall that it induces

$$\phi^* : M_2 \longrightarrow M_1$$

on character lattices and

$$\phi_* : N_1 \longrightarrow N_2$$

on one-parameter subgroup lattices.

These two maps are adjoint with respect to the pairings:

$$\boxed{\langle \phi^*(m), u\rangle = \langle m, \phi_*(u)\rangle\,.}$$

Indeed, both sides are the exponent of the same composition:

$$\mathbb{G}_m \xrightarrow{\lambda^u} T_1 \xrightarrow{\phi} T_2 \xrightarrow{\chi^m} \mathbb{G}_m.$$

Example 1.37. Let

$$\phi : \mathbb{G}_m^2 \longrightarrow \mathbb{G}_m^2, \qquad (t_1, t_2) \longmapsto (t_1^2 t_2, t_1 t_2^3).$$

On character lattices,

$$\phi^*(a, b) = (2a + b, a + 3b).$$

On one-parameter subgroup lattices,

$$\phi_*(p, q) = (2p + q, p + 3q).$$

For

$$m = (a, b), \qquad u = (p, q),$$

one checks that

$$\langle \phi^*(m), u \rangle = (2a + b)p + (a + 3b)q$$

and

$$\langle m, \phi_*(u) \rangle = a(2p + q) + b(p + 3q),$$

which are equal.

In this example, the numerical matrices representing ϕ^* and ϕ_* happen to agree because the matrix is symmetric. In general, the two matrices are transposes of one another once dual bases are fixed. We will return to this carefully in the next section.

The three dictionaries

The pairing is the bridge between the two lattices.

Algebra	**Geometry**	**Combinatorics**
Composition gives $\chi^m \circ \lambda^u = t^{\langle m,u \rangle}$. The pairing is bilinear and perfect.	$\langle m, u \rangle$ is the order of vanishing or pole of a character along a one-parameter motion.	$M = N^\vee, \qquad N = M^\vee.$ Elements of one lattice are integral linear functions on the other.

Guiding question

How do the lattices, the pairing, and the formulas for torus morphisms change when we replace one basis by another?

1.7 Changing coordinates on a torus

A torus does not come with preferred coordinates. Choosing coordinates means choosing a basis of its character lattice, and changing coordinates means replacing that basis by another one. The allowable changes are therefore controlled by

$$\mathrm{GL}(n, \mathbb{Z}),$$

the group of invertible $n \times n$ integer matrices.

This is an important difference between affine space and a torus. Coordinate changes on $\mathbb{A}^n$ may involve many kinds of polynomials. Coordinate changes that preserve the algebraic group structure of $\mathbb{G}_m^n$ are monomial.

1.7.1 From a new basis to new torus coordinates

Let T be an n-dimensional torus, and choose a basis

$$m_1, \dots, m_n$$

of its character lattice M. The corresponding torus coordinates are

$$x_i = \chi^{m_i}, \qquad i = 1, \dots, n.$$

Now choose another basis

$$m'_1, \dots, m'_n.$$

Write the new basis vectors in terms of the old basis:

$$m'_j = \sum_{i=1}^{n} a_{ij} m_i.$$

The coefficients form an integer matrix

$$A = (a_{ij}).$$

Because both collections are bases of the same lattice,

$$A \in \mathrm{GL}(n, \mathbb{Z}).$$

The new coordinate characters are

$$x'_j = \chi^{m'_j} = \prod_{i=1}^{n} x_i^{a_{ij}}.$$

Thus every change of basis gives an invertible monomial change of coordinates:

$$(x_1, \dots, x_n) \longmapsto (x'_1, \dots, x'_n).$$

Example 1.38. Let $T = \mathbb{G}_m^2$, and begin with the standard character basis

$$m_1 = (1, 0), \qquad m_2 = (0, 1).$$

Choose the new basis

$$m'_1 = -m_1, \qquad m'_2 = -m_1 + m_2.$$

Its change-of-basis matrix is

$$A = \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix},$$

whose columns record m'_1 and m'_2 in the old basis.

If

$$x = \chi^{m_1}, \qquad y = \chi^{m_2},$$

then

$$x' = x^{-1}, \qquad y' = x^{-1}y.$$

These are exactly the coordinate changes that appeared between the charts U_0 and U_1 on the dense torus of $\mathbb{P}^2$.

Pause and think

Why must the matrix A have determinant ± 1, rather than merely a nonzero determinant? What would go wrong if the columns generated a proper finite-index sublattice of M?

If $|\det A| > 1$, the new vectors do not form a lattice basis. The associated monomial map may still be a finite morphism of tori, but it is not an isomorphism.

1.7.2 Monomial automorphisms of a split torus

Let

$$A = (a_{ij}) \in \mathrm{GL}(n, \mathbb{Z}).$$

It defines a map

$$\Phi_A : \mathbb{G}_m^n \longrightarrow \mathbb{G}_m^n$$

by

$$\Phi_A(t_1, \ldots, t_n) = \left(\prod_{i=1}^{n} t_i^{a_{i1}}, \ldots, \prod_{i=1}^{n} t_i^{a_{in}} \right).$$

The columns of A contain the exponent vectors of the output coordinates.

Because A^{-1} also has integer entries, the map

$$\Phi_{A^{-1}}$$

is again monomial and is inverse to Φ_A. Hence Φ_A is an automorphism of algebraic tori.

Conversely, every algebraic group automorphism of $\mathbb{G}_m^n$ arises in this way. Therefore

$$\mathrm{Aut}_{\mathrm{alg.grp}}(\mathbb{G}_m^n) \simeq \mathrm{GL}(n, \mathbb{Z}).$$

Observation 1.39. Translations

$$t \longmapsto p\,t, \qquad p \in T,$$

are automorphisms of the underlying variety, but they do not preserve the identity unless $p = e$. They are therefore not automorphisms of the algebraic group. The group automorphisms are precisely the monomial coordinate changes coming from lattice automorphisms.

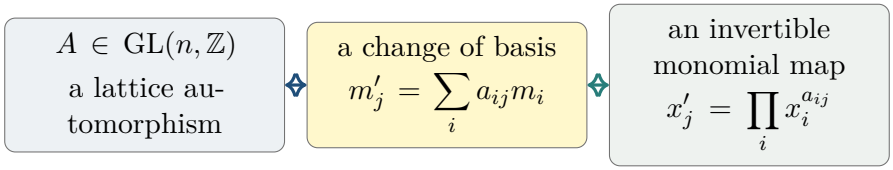


Figure 1.11: Integer change-of-basis matrices, lattice automorphisms, and monomial automorphisms of a torus are three descriptions of the same operation.

1.7.3 What happens to the dual basis?

Let

$$m_1, \dots, m_n$$

be a basis of M, and let

$$u_1, \dots, u_n$$

be the dual basis of N:

$$\langle m_i, u_j \rangle = \delta_{ij}.$$

Suppose the new character basis is given by

$$m'_j = \sum_i a_{ij} m_i.$$

Let

$$u'_1, \dots, u'_n$$

be the basis of N dual to the m'_j.

In matrix notation, the new character basis is obtained from the old one by A. The new one-parameter subgroup basis is obtained by

$$A^{-T} = (A^{-1})^T.$$

Indeed, if the columns of a matrix B record the u'_j in the old basis, the duality condition says

$$A^T B = I,$$

and therefore

$$B = A^{-T}.$$

Thus:

characters change by A, one-parameter subgroups change by A^{-T}.

Example 1.40. For

$$A = \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix},$$

one has

$$A^{-1} = A \qquad \text{and} \qquad A^{-T} = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Hence the basis of N dual to

$$m'_1 = -m_1, \qquad m'_2 = -m_1 + m_2$$

is

$$u'_1 = -u_1 - u_2, \qquad u'_2 = u_2.$$

A direct computation verifies

$$\left\langle m'_i, u'_j \right\rangle = \delta_{ij}.$$

Pause and think

Why does the inverse transpose appear? Start with the requirement

$$\left\langle m'_i, u'_j \right\rangle = \delta_{ij}$$

and translate it into a matrix equation.

1.7.4 Coordinates of vectors versus bases

There are two related matrix operations that are easy to confuse.

First, the basis vectors themselves change:

$$(m'_1, \dots, m'_n) = (m_1, \dots, m_n)A.$$

Second, the coordinate column of a fixed element changes in the opposite direction.

If

$$m = \sum_i c_i m_i = \sum_j c'_j m'_j,$$

then

$$c = Ac', \qquad c' = A^{-1}c.$$

Thus the basis changes by A, while the coordinate column of a fixed character changes by A^{-1}.

Similarly, if

$$u = \sum_i d_i u_i = \sum_j d'_j u'_j,$$

then

$$d = A^{-T}d', \qquad d' = A^T d.$$

The pairing is unchanged:

$$c^T d = (c')^T d'.$$

Indeed,

$$(Ac')^T (A^{-T}d') = (c')^T d'.$$

Warning

A matrix may describe either a change of basis or the coordinates of a linear map. Transposes and inverses depend on which convention is being used. In these notes, columns record images of basis vectors. Whenever a matrix formula matters, we will state the convention explicitly.

1.7.5 Matrices for morphisms of tori

Now let

$$\phi : T_1 \longrightarrow T_2$$

be a morphism of tori. Choose bases

$$m_1^{(1)}, \ldots, m_r^{(1)}$$

of $M_1 = X^*(T_1)$ and

$$m_1^{(2)}, \ldots, m_s^{(2)}$$

of $M_2 = X^*(T_2)$.

The pullback

$$\phi^* : M_2 \longrightarrow M_1$$

is represented by an integer $r \times s$ matrix

$$B = (b_{ij}),$$

defined by

$$\phi^*(m_j^{(2)}) = \sum_{i=1}^{r} b_{ij} m_i^{(1)}.$$

Equivalently, if

$$x_i = \chi^{m_i^{(1)}} \qquad \text{and} \qquad y_j = \chi^{m_j^{(2)}},$$

then

$$\phi^*(y_j) = \prod_{i=1}^{r} x_i^{b_{ij}}.$$

On points, the torus morphism is therefore

$$\phi(t_1, \ldots, t_r) = \left(\prod_i t_i^{b_{i1}}, \ldots, \prod_i t_i^{b_{is}} \right).$$

The induced map on one-parameter subgroup lattices,

$$\phi_* : N_1 \longrightarrow N_2,$$

is represented in the dual bases by the transpose matrix

$$B^T.$$

This follows from the adjointness identity

$$\langle \phi^*(m), u \rangle = \langle m, \phi_*(u) \rangle .$$

Example 1.41. Consider

$$\phi : \mathbb{G}_m^2 \longrightarrow \mathbb{G}_m^2, \qquad (t_1, t_2) \longmapsto (t_1^2 t_2,\ t_1 t_2^3).$$

The pullback of the two target coordinate characters is

$$\phi^*(y_1) = x_1^2 x_2, \qquad \phi^*(y_2) = x_1 x_2^3.$$

Thus

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

In this example $B = B^T$, so the same numerical matrix represents ϕ^* and ϕ_*. This is accidental: the matrix happens to be symmetric.

Example 1.42. Consider instead

$$\psi : \mathbb{G}_m^2 \longrightarrow \mathbb{G}_m^2, \qquad (t_1, t_2) \longmapsto (t_1^2 t_2^3,\ t_1^{-1} t_2^4).$$

Then

$$B = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$$

represents

$$\psi^* : M_2 \longrightarrow M_1,$$

while

$$B^T = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}$$

represents

$$\psi_* : N_1 \longrightarrow N_2.$$

For $u = (p, q) \in N_1$,

$$\psi(\lambda^u(t)) = (t^{2p+3q}, t^{-p+4q}),$$

which indeed has exponent vector

$$B^T \begin{pmatrix} p \\ q \end{pmatrix}.$$

1.7.6 Isomorphisms, finite maps, and dimensions

The matrix description makes several basic properties visible.

If B is square and

$$\det B = \pm 1,$$

then $B \in \mathrm{GL}(n, \mathbb{Z})$, and the corresponding torus morphism is an isomorphism.

If B is square and

$$\det B \neq 0$$

but $|\det B| > 1$, then the map is finite and surjective but not an isomorphism. For example,

$$\mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad t \longmapsto t^d$$

corresponds to multiplication by d on the character lattice.

Over an algebraically closed field whose characteristic does not divide d, the kernel consists of the d-th roots of unity. In positive characteristic dividing d, the scheme-theoretic kernel requires more care; we will not need that refinement here.

When the source and target have different dimensions, the ranks of the lattice maps help describe kernels and images. We will use these ideas later, but the main lesson for now is simpler:

morphisms of tori are encoded by homomorphisms of lattices.

Guiding question

Suppose a torus acts linearly on a finite-dimensional vector space. After choosing coordinates on the torus, its elements are tuples of nonzero scalars. What kind of basis of the vector space might make the action as simple as possible?

The three dictionaries

Changing coordinates on a torus is an integral operation.

Algebra	Geometry	Combinatorics
Torus morphisms pull back characters: $\phi^* : M_2 \to M_1$.	Automorphisms of $\mathbb{G}_m^n$ preserving the identity are invertible monomial coordinate changes.	Changes of lattice basis lie in $\mathrm{GL}(n, \mathbb{Z})$.
After choosing bases, this is an integer matrix.		The dual lattice changes by the inverse transpose.

1.8 Representation theory in one page

Characters are the one-dimensional representations of a torus. The basic representation theory of tori says that every finite-dimensional representation is assembled from these one-dimensional pieces.

This is the central structural fact of the chapter. We will state it without proof and focus on learning how to use it.

1.8.1 What is a representation?

Let V be a finite-dimensional vector space over $\Bbbk$.

Definition 1.43. A *finite-dimensional algebraic representation* of a torus T on V is a morphism of algebraic groups

$$\rho : T \longrightarrow \mathrm{GL}(V).$$

Equivalently, it is an algebraic action

$$T \times V \longrightarrow V$$

such that, for each $t \in T$, the map

$$v \longmapsto t \cdot v$$

is linear.

The adjective *algebraic* matters. It requires the matrix entries of $\rho(t)$ to be regular functions of t, after bases are chosen.

Example 1.44. Let $T = \mathbb{G}_m$ act on $V = \Bbbk^2$ by

$$t \cdot (x, y) = (t^2 x, t^{-3} y).$$

In the standard basis,

$$\rho(t) = \begin{pmatrix} t^2 & 0 \\ 0 & t^{-3} \end{pmatrix}.$$

The two coordinate lines are one-dimensional representations with characters $t \mapsto t^2$ and $t \mapsto t^{-3}$.

Example 1.45. Let $T = \mathbb{G}_m^2$ act on $\Bbbk^3$ by

$$(t_1, t_2) \cdot (x_0, x_1, x_2) = (x_0, t_1 x_1, t_2 x_2).$$

The three coordinate lines transform by the characters

$$1, \qquad t_1, \qquad t_2.$$

Projectivizing this representation gives the torus action on $\mathbb{P}^2$ used throughout Chapter 0:

$$(t_1, t_2) \cdot [x_0 : x_1 : x_2] = [x_0 : t_1 x_1 : t_2 x_2].$$

1.8.2 Weight vectors and weight spaces

A character tells us how a torus may scale a single vector.

Definition 1.46. Let V be a representation of T, and let $m \in M = X^*(T)$. A nonzero vector $v \in V$ is a *weight vector of weight* m if

$$t \cdot v = \chi^m(t) v \qquad \text{for every } t \in T.$$

The corresponding *weight space* is

$$V_m = \{v \in V : t \cdot v = \chi^m(t) v \text{ for every } t \in T\}.$$

The zero vector lies in every V_m, but a nonzero vector cannot have two distinct weights.

Example 1.47. For the action

$$t \cdot (x, y) = (t^2 x, t^{-3} y)$$

on $\Bbbk^2$, let e_1, e_2 be the standard basis. Then

$$e_1 \in V_2, \qquad e_2 \in V_{-3},$$

and

$$V = V_2 \oplus V_{-3}.$$

The vector $e_1 + e_2$ is not a weight vector: its two components scale by different characters.

Pause and think

For the action of $\mathbb{G}_m$ on $\Bbbk^3$ given by

$$t \cdot (x, y, z) = (t^2 x, t^2 y, t^{-1} z),$$

find all weight spaces. Which vectors are weight vectors, and which are not?

Here

$$V_2 = \operatorname{span}\{e_1, e_2\}, \qquad V_{-1} = \operatorname{span}\{e_3\}.$$

Every nonzero vector in either of these subspaces is a weight vector. A vector having nonzero components in both subspaces is not.

This example is worth remembering: a weight space need not be one-dimensional. A weight records how the torus acts, not the dimension of the subspace on which it acts that way.

1.8.3 The weight-space decomposition

The representation theory of a torus is governed by the following theorem.

Theorem 1.48 (Weight-space decomposition)**.** *Let T be an algebraic torus, and let V be a finite-dimensional algebraic representation of T. Then*

$$V = \bigoplus_{m \in M} V_m.$$

Only finitely many of the weight spaces V_m are nonzero.

In other words, every vector can be written uniquely as a finite sum of weight vectors belonging to distinct weights.

The finite set

$$\operatorname{Wt}(V) = \{m \in M : V_m \neq 0\}$$

is called the *set of weights* of the representation. The dimension

$$\dim V_m$$

is the *multiplicity* of the weight m.

Observation 1.49. After choosing a basis of each nonzero weight space, every matrix $\rho(t)$ becomes diagonal:

$$\rho(t) = \operatorname{diag}(\chi^{m_1}(t), \dots, \chi^{m_r}(t)),$$

where weights may be repeated according to their multiplicities.

Thus a torus representation may be simultaneously diagonalized: one basis diagonalizes the action of every element of T at once.

Warning

The theorem does not say that every element of V is a weight vector. It says that every element is a sum of weight vectors. A vector is itself a weight vector only when all of its nonzero components lie in a single weight space.

$$V = \bigoplus_{m \in \mathrm{Wt}(V)} V_m$$

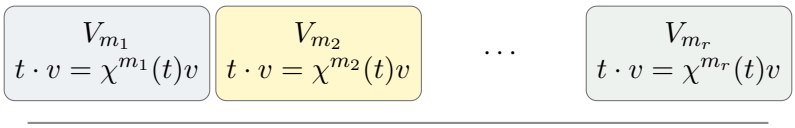


Figure 1.12: A finite-dimensional representation of a torus decomposes into weight spaces indexed by characters.

1.8.4 Why the theorem is plausible

We will not prove the weight-space decomposition here, but it is useful to understand why one should expect it.

For $T = \mathbb{G}_m^n$, the operators $\rho(t)$ commute with one another because T is abelian. More importantly, the algebraic nature of the action forces the matrix coefficients to be built from characters. The representation therefore separates into subspaces on which the torus acts through individual Laurent monomials.

Observation 1.50. Another viewpoint uses the coordinate ring

$$\Bbbk[T] = \Bbbk[M].$$

The action gives V the structure of a comodule over $\Bbbk[M]$, and the basis elements χ^m of the group algebra separate the representation into the spaces V_m. This is the standard algebraic proof, but we will not need the comodule language later.

Observation 1.51. The weight-space decomposition holds for algebraic tori over an algebraically closed field in arbitrary characteristic. Standard references prove it as part of the representation theory of diagonalizable algebraic groups.

1.8.5 Invariant vectors

The zero element

$$0 \in M$$

corresponds to the trivial character

$$\chi^0 = 1.$$

Its weight space is

$$V_0 = \{v \in V : t \cdot v = v \text{ for every } t \in T\}.$$

This is precisely the subspace of T-invariant vectors, often written

$$V^T.$$

Thus

$$V^T = V_0.$$

Example 1.52. For

$$(t_1, t_2) \cdot (x_0, x_1, x_2) = (x_0, t_1 x_1, t_2 x_2),$$

the invariant subspace is

$$V^T = \Bbbk e_0.$$

The other two coordinate lines have nonzero weights.

Pause and think

Let $\mathbb{G}_m$ act on $\Bbbk[x, y]_d$ by

$$(t \cdot f)(x, y) = f(t^{-1}x, ty).$$

Determine the weight of each monomial

$$x^{d-i}y^i.$$

For which values of d does the representation contain a nonzero invariant vector?

We have

$$t \cdot x^{d-i}y^i = t^{-(d-i)}t^i x^{d-i}y^i = t^{2i-d}x^{d-i}y^i.$$

Thus the weight is

$$2i - d.$$

A nonzero invariant monomial exists exactly when $2i = d$, hence when d is even.

1.8.6 Equivariant linear maps preserve weights

Let V and W be representations of T.

Definition 1.53. A linear map

$$F : V \longrightarrow W$$

is *T-equivariant* if

$$F(t \cdot v) = t \cdot F(v)$$

for every $t \in T$ and $v \in V$.

If $v \in V_m$, then

$$\begin{aligned} t \cdot F(v) &= F(t \cdot v) \\ &= F(\chi^m(t)v) \\ &= \chi^m(t)F(v). \end{aligned}$$

Therefore

$$F(V_m) \subseteq W_m.$$

So an equivariant map acts weight by weight. In particular, if $W_m = 0$, then every equivariant map must vanish on V_m.

Example 1.54. Suppose

$$V = V_1 \oplus V_3 \qquad \text{and} \qquad W = W_1 \oplus W_2$$

are representations of $\mathbb{G}_m$. A $\mathbb{G}_m$-equivariant map

$$F : V \longrightarrow W$$

may map V_1 into W_1, but it must vanish on V_3, since $W_3 = 0$.

This simple observation is extremely useful. It turns many questions about equivariant maps into separate linear-algebra problems on the individual weight spaces.

1.8.7 Subrepresentations and quotients

A subspace $U \subseteq V$ is a *subrepresentation* if it is preserved by T. The weight decomposition restricts to

$$U = \bigoplus_{m \in M} (U \cap V_m).$$

Likewise, the quotient V/U inherits a representation, and its weights are obtained from the corresponding quotients of weight spaces.

Thus the representation theory of a torus has no hidden interaction between distinct weights. Each weight may be studied independently.

Observation 1.55. This is one reason tori are among the simplest algebraic groups from the viewpoint of representation theory. Their finite-dimensional representations are controlled by:

$$\text{a finite set of lattice points } m \in M \quad \text{and} \quad \text{their multiplicities } \dim V_m.$$

1.8.8 Weights as a finite configuration in a lattice

The set of weights

$$\mathrm{Wt}(V) \subseteq M$$

may be drawn as a finite configuration of lattice points in $M_{\mathbb{R}}$. Multiplicity may be recorded by labels, repeated points, or the dimensions of the associated weight spaces.

Example 1.56. Consider the representation of $\mathbb{G}_m^2$ on $\Bbbk^5$ with weights

$$(0,0), \quad (1,0), \quad (1,0), \quad (0,1), \quad (2,-1).$$

Then

$$\mathrm{Wt}(V) = \{(0,0), (1,0), (0,1), (2,-1)\},$$

with multiplicity 2 at $(1,0)$ and multiplicity 1 at each of the other weights.

This turns a representation into combinatorial data. Later, when a torus acts on spaces of sections, the weights will often be the lattice points of a polytope.

1.8.9 Operations on representations

The weight description makes standard constructions transparent.

If

$$V = \bigoplus_m V_m \qquad \text{and} \qquad W = \bigoplus_{m'} W_{m'},$$

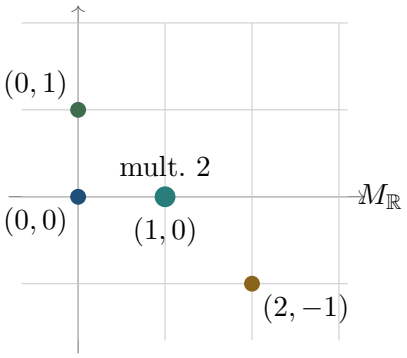


Figure 1.13: A torus representation may be encoded by a finite set of weights in M, together with their multiplicities.

then

$$(V \oplus W)_m = V_m \oplus W_m.$$

For tensor products,

$$v \in V_m, \quad w \in W_{m'} \quad \Longrightarrow \quad v \otimes w \in (V \otimes W)_{m+m'}.$$

Thus weights add under tensor product.

For dual representations,

$$(V^\vee)_{-m} \simeq (V_m)^\vee.$$

Thus weights change sign under duality.

Example 1.57. Let V and W be one-dimensional representations of weights m and m', respectively. Then

$$V \otimes W$$

has weight

$$m + m',$$

while $V^\vee$ has weight

$$-m.$$

This is the representation-theoretic version of

$$\chi^m \chi^{m'} = \chi^{m+m'} \qquad \text{and} \qquad (\chi^m)^{-1} = \chi^{-m}.$$

Prediction

A torus acts on the space

$$H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d))$$

of homogeneous degree-d polynomials. What should its weight vectors look like? How might the triangle $d\Delta \cap M$ from Chapter 0 reappear in this representation?

1.8.10 The basic classification

The weight-space theorem may be summarized as a classification statement.

Theorem 1.58 (Classification of finite-dimensional torus representations)**.** *Up to isomorphism, a finite-dimensional representation of T is determined by the multiplicities*

$$a_m = \dim V_m, \qquad m \in M,$$

where a_m is a nonnegative integer and only finitely many a_m are nonzero.

Equivalently,

$$V \simeq \bigoplus_{m \in M} (\Bbbk_m)^{\oplus a_m},$$

where $\Bbbk_m$ denotes the one-dimensional representation of weight m.

No extension data are needed: the representation is the direct sum of its weight spaces.

Observation 1.59. Proofs of the weight-space theorem and this classification may be found in standard treatments of diagonalizable algebraic groups or linear algebraic groups; the Notes and references section lists the sources used here.

The three dictionaries

The representation theory of a torus is controlled by its character lattice.

Algebra	Geometry	Combinatorics
A representation decomposes as $V = \bigoplus_{m \in M} V_m$. Equivariant maps preserve each weight.	After choosing a weight basis, the torus acts by diagonal matrices: each coordinate direction is scaled independently.	A representation is encoded by finitely many lattice points $m \in M$ and their multiplicities.

Guiding question

How do the monomials, exponent vectors, one-parameter limits, and lattice triangles of Chapter 0 fit into the character and weight language developed in this chapter?

What this chapter has accomplished

Chapter 0 already contained nearly every object introduced in this chapter. We first met them concretely in $\mathbb{P}^2$; we can now name them and place them in the two lattices attached to its dense torus.

Pointers back to Chapter 0

1. Torus coordinates are bases of M. On the chart U_0, the functions

$$z_1 = \frac{x_1}{x_0}, \qquad z_2 = \frac{x_2}{x_0}$$

are characters of the dense torus $T \subseteq \mathbb{P}^2$. Their lattice points form a basis of

$$M = X^*(T).$$

The Laurent coordinate changes between U_0, U_1, U_2 are changes of basis in M, hence are governed by $\mathrm{GL}(2, \mathbb{Z})$.

2. Exponent vectors are character lattice points. A Laurent monomial

$$z_1^a z_2^b$$

is the character χ^m corresponding, in the chosen basis, to

$$m = (a, b) \in M.$$

Thus exponent vectors are coordinates for intrinsic characters.

3. Monomial sections form weight bases. The degree-d monomials

$$x_0^{d-a-b} x_1^a x_2^b, \qquad (a, b) \in d\Delta \cap M,$$

form a weight basis of

$$H^0\big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)\big).$$

Depending on the convention for the induced action on sections, the weights may be the exponent points or their negatives. The essential statement is unchanged: the lattice points of $d\Delta$ index the monomial weight spaces.

4. Monomial curves are one-parameter subgroups. The parameterized curves

$$t \longmapsto (t^{u_1}, t^{u_2})$$

are the one-parameter subgroups

$$\lambda^u : \mathbb{G}_m \longrightarrow T, \qquad u = (u_1, u_2) \in N = X_*(T).$$

Characters live in M; algebraic motions live in the dual lattice N.

5. The pairing records orders along motions. For $m \in M$ and $u \in N$,

$$\chi^m\big(\lambda^u(t)\big) = t^{\langle m, u \rangle}.$$

Hence $\langle m, u \rangle$ is the order of vanishing or pole of the character along the one-parameter motion. This is the bridge between functions in M and directions in N.

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tem **Fans organize boundary behavior.** For

$$t \longmapsto [1 : t^{u_1} : t^{u_2}]$$

in $\mathbb{P}^2$, vectors u lying in the same cone of the fan produce compatible boundary limits. The rays

$$(-1, -1), \qquad (1, 0), \qquad (0, 1)$$

record the three torus-invariant boundary divisors, while the maximal cones correspond to the three torus-fixed points.
The blowup of a fixed point inserts a new primitive ray. Thus the star subdivision from Chapter 0 may now be read as adding a new one-parameter direction, and geometrically as introducing the exceptional divisor.

The three dictionaries

The chapter may be summarized by one compact dictionary.

Algebra	Geometry	Combinatorics
$\Bbbk[T] = \Bbbk[M]$	the torus T	the character lattice M
a character χ^m	a monomial function or weight	a lattice point $m \in M$
a one-parameter subgroup λ^u	a parameterized motion toward the boundary	a lattice point $u \in N$
$\chi^m \circ \lambda^u = t^{\langle m,u\rangle}$	order of vanishing or pole along the motion	the pairing $M \times N \to \mathbb{Z}$

The next construction

The next chapter begins with the observation that a cone

$$\sigma \subseteq N_{\mathbb{R}}$$

specifies a collection of allowed one-parameter directions. The characters having no pole along any of them are precisely those $m \in M$ satisfying

$$\langle m, u\rangle \geq 0 \qquad \text{for every } u \in \sigma.$$

They form the semigroup

$$\sigma^\vee \cap M,$$

and its semigroup algebra

$$\Bbbk[\sigma^\vee \cap M]$$

will produce an affine variety containing T.

For the first quadrant in $N_{\mathbb{R}}$, this construction gives

$$\sigma^\vee \cap M = \mathbb{Z}^2_{\geq 0}, \qquad \Bbbk[\sigma^\vee \cap M] = \Bbbk[x, y],$$

and therefore recovers $\mathbb{A}^2$.

Guiding question

Given a cone of one-parameter directions in $N_{\mathbb{R}}$, what geometry is encoded by the algebra $\Bbbk[\sigma^\vee \cap M]$?

Comprehension exercises

1.1. Determine all characters

$$\chi : \mathbb{G}_m \longrightarrow \mathbb{G}_m.$$

Then determine which regular maps of the form

$$f(t) = c\,t^a \qquad (c \in \Bbbk^\times,\ a \in \mathbb{Z})$$

are characters. Explain the role of the identity element.

1.2. Let $T = \mathbb{G}_m^2$, with coordinate characters x and y. For each of the following pairs, decide whether the two characters form a basis of $M = X^*(T)$. When they do, write the corresponding invertible monomial change of coordinates.

(a) $x^2y,\ xy$, (b) $x^{-1},\ x^{-1}y$,
(c) $x^2,\ y$, (d) $xy,\ x^{-1}y$.

1.3. Consider the one-parameter subgroup

$$\lambda^u(t) = (t^{u_1}, t^{u_2}, t^{u_3})$$

of $\mathbb{G}_m^3$.

(a) Determine when it extends to a morphism

$$\mathbb{A}^1 \longrightarrow \mathbb{A}^3.$$

(b) When it extends, describe its value at $t = 0$ in terms of the vanishing pattern of u_1, u_2, u_3.

(c) Repeat the question for a translated curve

$$t \longmapsto (a_1t^{u_1}, a_2t^{u_2}, a_3t^{u_3}), \qquad a_i \in \Bbbk^\times.$$

1.4. Let

$$m = (3, -2, 1) \in M \simeq \mathbb{Z}^3, \qquad u = (-1, 4, 2) \in N \simeq \mathbb{Z}^3.$$

Compute $\langle m, u\rangle$, and interpret the answer as the order of vanishing or pole of

$$\chi^m \circ \lambda^u$$

at $t = 0$. Find a nonzero vector $u' \in N$ satisfying

$$\langle m, u'\rangle = 0.$$

1.5. Let m_1, m_2 be a basis of M, with dual basis u_1, u_2 of N. Define

$$m_1' = 2m_1 + m_2, \qquad m_2' = m_1 + m_2.$$

(a) Verify that m_1', m_2' form a basis of M.

(b) Compute the basis u_1', u_2' of N dual to m_1', m_2'.

(c) Check directly that

$$\left\langle m_i', u_j' \right\rangle = \delta_{ij}.$$

1.6. Consider the torus morphism

$$\phi : \mathbb{G}_m^2 \longrightarrow \mathbb{G}_m^3, \qquad (t_1, t_2) \longmapsto (t_1^2 t_2,\ t_1^{-1} t_2^3,\ t_1 t_2^2).$$

(a) Write the matrix of

$$\phi^* : M_{\mathbb{G}_m^3} \longrightarrow M_{\mathbb{G}_m^2}$$

in the standard bases.

(b) Write the matrix of

$$\phi_* : N_{\mathbb{G}_m^2} \longrightarrow N_{\mathbb{G}_m^3}.$$

(c) Verify

$$\langle \phi^*(m), u \rangle = \langle m, \phi_*(u) \rangle$$

for a general $m \in \mathbb{Z}^3$ and $u \in \mathbb{Z}^2$.

1.7. Let $T = \mathbb{G}_m^2$ act on $V = \Bbbk^5$ by

$$(t_1, t_2) \cdot (v_1, v_2, v_3, v_4, v_5) = (t_1 v_1,\ t_2 v_2,\ t_1 t_2 v_3,\ t_1 v_4,\ v_5).$$

Determine every weight space, its weight, and its multiplicity. Describe the invariant subspace V^T.

1.8. Let

$$V = V_{(1,0)} \oplus V_{(0,1)} \oplus V_{(1,1)}$$

and

$$W = W_{(1,0)} \oplus W_{(1,1)}$$

be representations of $T = \mathbb{G}_m^2$, with

$$\dim V_{(1,0)} = 2, \quad \dim V_{(0,1)} = 1, \quad \dim V_{(1,1)} = 3,$$

and

$$\dim W_{(1,0)} = 1, \quad \dim W_{(1,1)} = 2.$$

Compute the dimension of the vector space

$$\mathrm{Hom}_T(V, W)$$

of T-equivariant linear maps.

1.9. For the standard torus action on $\mathbb{P}^2$, list the monomial basis of

$$H^0\big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3)\big)$$

and label each monomial by its exponent point in $3\Delta \cap M$. Using the standard pullback convention for the induced action on sections, also record its weight. Which section spans the invariant subspace?

1.10. Let

$$\sigma = \mathrm{Cone}((1,0),(1,2)) \subseteq N_{\mathbb{R}} \simeq \mathbb{R}^2.$$

Determine all

$$m = (a,b) \in M \simeq \mathbb{Z}^2$$

satisfying

$$\langle m, u \rangle \geq 0 \qquad \text{for every } u \in \sigma.$$

Describe the semigroup $\sigma^\vee \cap M$ by inequalities, and find a finite collection of lattice points that generates it additively.

Challenge exercise

Projective limits, weights, and exposed faces.

Let T be an algebraic torus, and let

$$V = \bigoplus_{m \in M} V_m$$

be a finite-dimensional representation. Choose a nonzero vector

$$v = \sum_{m \in S} v_m, \qquad 0 \neq v_m \in V_m,$$

where $S \subseteq M$ is the finite set of weights occurring in v. The torus acts on the projective space $\mathbb{P}(V)$.

Fix a one-parameter subgroup $u \in N$. The goal is to determine

$$\lim_{t \to 0} \lambda^u(t) \cdot [v]$$

directly from the pairings

$$\langle m, u \rangle, \qquad m \in S.$$

Hint 1. Show that

$$\lambda^u(t) \cdot v = \sum_{m \in S} t^{\langle m,u \rangle} v_m.$$

Hint 2. Let

$$a = \min_{m \in S} \langle m, u \rangle .$$

Use projective rescaling by t^{-a} to show that the limit exists and equals

$$\left[\sum_{\substack{m \in S \\ \langle m,u \rangle = a}} v_m \right].$$

Explain why this vector is nonzero.

Hint 3. Let

$$P(v) = \mathrm{Conv}(S) \subseteq M_{\mathbb{R}}$$

be the convex hull of the weights occurring in v. Show that the weights contributing to the limit are precisely the points of S lying on the face

$$F_u = \left\{ m \in P(v) : \langle m, u \rangle = \min_{m' \in P(v)} \langle m', u \rangle \right\}.$$

This is the face of $P(v)$ exposed by the linear functional $m \mapsto \langle m, u \rangle$.

Hint 4. Explain why replacing u by another vector u' that exposes the same face produces a limit lying in the same projective weight subspace

$$\mathbb{P}\left(\bigoplus_{m\in F_u\cap S} V_m\right).$$

Hint 5. Specialize to

$$V=\Bbbk^3$$

with the $\mathbb{G}_m^2$-action

$$(t_1,t_2)\cdot(x_0,x_1,x_2)=(x_0,t_1x_1,t_2x_2).$$

The weights are

$$0,\qquad m_1,\qquad m_2.$$

For a point

$$[x_0:x_1:x_2]\in\mathbb{P}^2,$$

use the formula above to recover the one-parameter limits described in Chapter 0. Pay attention to the possibility that one or more coordinates vanish, since the set S depends on the point.

Hint 6. Draw the weight triangle

$$\mathrm{Conv}\{0,m_1,m_2\}\subseteq M_{\mathbb{R}}.$$

Partition $N_{\mathbb{R}}$ according to which face it exposes by taking the minimum of $\langle m,u\rangle$. Compare the resulting decomposition with the fan of $\mathbb{P}^2$.

Hint 7. Formulate a general conjecture: given a finite weight polytope $P\subseteq M_{\mathbb{R}}$, how should the regions of vectors $u\in N_{\mathbb{R}}$ exposing a fixed face of P fit together? What familiar fan should they form?

Notes and references

This chapter is intended as a working introduction to the language of algebraic tori rather than a self-contained treatment of linear algebraic groups. The classification of characters and one-parameter subgroups, the perfect pairing between their lattices, and the weight-space decomposition of a finite-dimensional torus representation were stated with only brief indications of proof.

For the toric-geometric viewpoint on lattices, characters, one-parameter subgroups, and group algebras, see the opening chapter of Cox–Little–Schenck, *Toric Varieties*. Standard references for algebraic groups and the representation theory of tori include Humphreys, *Linear Algebraic Groups*; Borel, *Linear Algebraic Groups*; and Springer, *Linear Algebraic Groups*. In those texts, tori are treated as examples of diagonalizable algebraic groups, and the weight-space decomposition appears as part of the general representation theory of diagonalizable groups.

Over a non-algebraically closed field, one must distinguish split tori from tori that become split only after extending scalars. That arithmetic direction lies outside the main path of these notes.

Chapter 2

Affine Toric Varieties: Semigroups, Cones, and Limits

Chapter overview

Which monomial functions on a torus remain regular after we add a boundary?
For the affine plane, the answer is familiar: the Laurent monomial $x^a y^b$ extends precisely when $a, b \geq 0$. This chapter turns that example into a classification. Additive collections of characters produce semigroup algebras and affine varieties; normality turns those collections into the lattice points of rational polyhedral cones.
The same variety then admits two dual descriptions:

a cone in $M_{\mathbb{R}}$ records the characters that extend,
while a cone in $N_{\mathbb{R}}$ records the one-parameter subgroups whose limits exist.

Our main goal is to understand both readings of

$$U_\sigma = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].$$

When a result has a lengthy or technical proof, we will give a precise reference to Cox–Little–Schenck and a conceptual proof sketch that isolates the driving idea and the logical flow.

The three dictionaries

Affine toric geometry begins with one object viewed in three ways.

Algebra	Geometry	Combinatorics
a semigroup algebra $\Bbbk[S]$	an affine toric variety $\operatorname{Spec} \Bbbk[S]$	an additive semigroup $S \subseteq M$
regular characters	monomial functions extending to the boundary	lattice points in a cone in $M_{\mathbb{R}}$
pullback along λ^u	limits of one-parameter motions	a dual cone in $N_{\mathbb{R}}$

2.1 From the affine plane to affine toric varieties

We begin where Chapter 0 began:

$$\mathbb{A}^2 = \operatorname{Spec} \Bbbk[x, y].$$

The affine plane contains the torus

$$T = \mathbb{G}_m^2 = \{(x, y) \in \mathbb{A}^2 : xy \neq 0\}$$

as a dense open subset. On T, both x and y are invertible, so every Laurent monomial

$$x^a y^b, \qquad (a, b) \in \mathbb{Z}^2,$$

is a regular function. In the language of Chapter 1, these Laurent monomials are precisely the characters of T:

$$x^a y^b = \chi^{(a,b)}.$$

But most of these characters do not remain regular on all of $\mathbb{A}^2$. The function x^{-1}, for instance, has a pole along the coordinate line $V(x)$. Likewise, y^{-1} has a pole along $V(y)$. The characters that do extend regularly are exactly

$$x^a y^b, \qquad a, b \geq 0.$$

Thus

$$\Bbbk[\mathbb{A}^2] = \Bbbk[x, y] = \bigoplus_{(a,b) \in \mathbb{Z}^2_{\geq 0}} \Bbbk\, \chi^{(a,b)}.$$

This gives us the question that drives the entire chapter:

Guiding question

Given a torus T, which collections of characters can occur as the regular functions on an affine variety containing T?

2.1.1 Which characters extend?

For the affine plane, the answer is visible in the character lattice

$$M \simeq \mathbb{Z}^2.$$

The lattice point (a, b) corresponds to the character $x^a y^b$, and this character extends to $\mathbb{A}^2$ exactly when

$$(a, b) \in \mathbb{Z}^2_{\geq 0}.$$

The allowed exponent vectors form the lattice points of the first quadrant in $M_{\mathbb{R}}$.

Two nearby examples help separate the roles of affine and invertible coordinates.

X	$\Bbbk[X]$	characters that extend
$\mathbb{G}_m^2$	$\Bbbk[x^{\pm 1}, y^{\pm 1}]$	$(a, b) \in \mathbb{Z}^2$
$\mathbb{A}^1 \times \mathbb{G}_m$	$\Bbbk[x, y^{\pm 1}]$	$a \geq 0, \quad b \in \mathbb{Z}$
$\mathbb{A}^2$	$\Bbbk[x, y]$	$a \geq 0, \quad b \geq 0.$

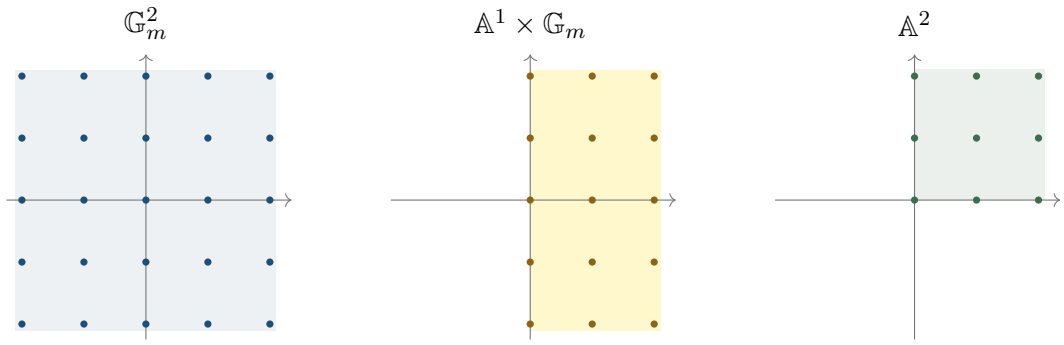


Figure 2.1: The lattice points indexing regular characters depend on which boundary directions have been added to the torus.

Pause and think

Consider

$$X = \mathbb{A}^2 \times \mathbb{G}_m$$

with dense torus $\mathbb{G}_m^3$. Which characters

$$\chi^{(a,b,c)} = x^a y^b z^c$$

extend regularly to X? Draw the corresponding lattice points in $M_{\mathbb{R}} \simeq \mathbb{R}^3$.

The pattern is already suggestive. Whenever two characters extend, their product extends as well:

$$\chi^m \chi^{m'} = \chi^{m+m'}.$$

The constant character $\chi^0 = 1$ also extends. Therefore the set of extending characters is closed under addition and contains 0. In the next section, we will call such a collection a *semigroup*.

2.1.2 The torus action on the affine plane

The dense torus does more than sit inside $\mathbb{A}^2$. It acts on the whole affine plane by coordinatewise multiplication:

$$(t_1, t_2) \cdot (x, y) = (t_1 x, t_2 y).$$

On the open subset where $xy \neq 0$, this is simply the group law of $T = \mathbb{G}_m^2$. Thus the multiplication action of the torus on itself extends across the coordinate axes.

The coordinate axes record what has been added to the torus:

$$\mathbb{A}^2 = T \sqcup (V(x) \setminus \{(0,0)\}) \sqcup (V(y) \setminus \{(0,0)\}) \sqcup \{(0,0)\}.$$

Each piece is a torus orbit. The two one-dimensional orbits are obtained by allowing one coordinate to vanish, and the origin is the unique fixed point.

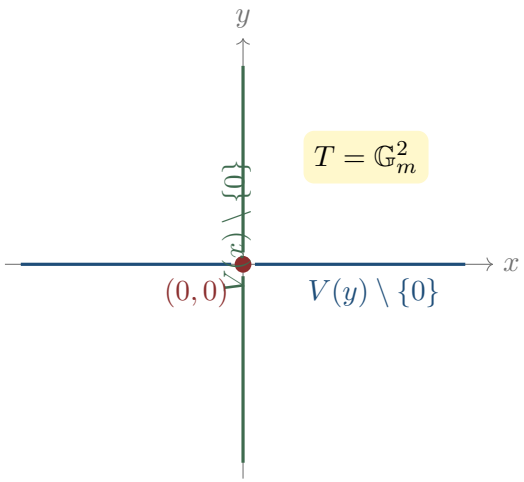


Figure 2.2: The affine plane consists of its dense torus, two one-dimensional boundary orbits, and one fixed point.

The orbit decomposition will eventually be read directly from cones. For now, it simply illustrates the two features that matter:

$\mathbb{A}^2$ contains T densely, and the action of T extends to all of $\mathbb{A}^2$.

2.1.3 The intrinsic definition

The affine plane suggests the general definition.

Definition 2.1 (Affine toric variety)**.** An *affine toric variety* is an irreducible affine variety X containing an algebraic torus T as a dense open subset such that the multiplication action

$$T \times T \longrightarrow T$$

extends to an algebraic action

$$T \times X \longrightarrow X.$$

The torus T is called the *dense torus* or *big torus* of X.

Reference. This is Cox–Little–Schenck, Definition 1.1.3, with $\mathbb{C}$ replaced by our algebraically closed base field $\Bbbk$.

Each condition has a role.

- **Affine** means that X is determined by its coordinate ring:

$$X = \operatorname{Spec} \Bbbk[X].$$

- **Irreducible** ensures that the dense torus lies in one geometric piece and that regular functions on X are determined by their restriction to T.
- **Dense and open** means that X is obtained from T by adding boundary points, while retaining the torus as the generic part of the variety.

- **The action extends** means that the added boundary is compatible with the group structure of T. This condition is what forces monomials and lattice combinatorics to control the geometry.

Example 2.2. The torus T itself is an affine toric variety. Its boundary is empty, and

$$\Bbbk[T] = \Bbbk[M].$$

Example 2.3. Affine space $\mathbb{A}^n$ is an affine toric variety with dense torus

$$\mathbb{G}_m^n = \mathbb{A}^n \setminus V(x_1 \cdots x_n).$$

The action is coordinatewise multiplication, and

$$\Bbbk[\mathbb{A}^n] = \bigoplus_{m \in \mathbb{Z}^n_{\geq 0}} \Bbbk\, \chi^m.$$

Example 2.4. More generally,

$$\mathbb{A}^r \times \mathbb{G}_m^{n-r}$$

is an affine toric variety with dense torus $\mathbb{G}_m^n$. Its coordinate ring is

$$\Bbbk[x_1, \ldots, x_r, x_{r+1}^{\pm 1}, \ldots, x_n^{\pm 1}],$$

and its extending characters are indexed by

$$\mathbb{Z}^r_{\geq 0} \times \mathbb{Z}^{n-r} \subseteq M.$$

Warning

Normality is not part of theorem 2.1. In these notes, as in CLS Chapter 1, affine toric varieties may be nonnormal. The distinction is essential: later in this chapter, normality will be recognized combinatorially as a condition that fills in missing lattice points.
Some authors build normality into the definition of a toric variety. Whenever conventions could matter, we will state them explicitly.

2.1.4 The coordinate-ring viewpoint

Let X be an affine toric variety with dense torus T and character lattice M. Restricting regular functions from X to the dense open subset T gives an injective homomorphism

$$\Bbbk[X] \hookrightarrow \Bbbk[T] = \Bbbk[M].$$

It is injective because a regular function on an irreducible variety that vanishes on a dense open subset vanishes everywhere.

Thus we may regard $\Bbbk[X]$ as a subalgebra of the Laurent polynomial algebra $\Bbbk[M]$. In concrete terms:

$$\boxed{\Bbbk[X] = \{\text{regular functions on } T \text{ that extend to } X\}.}$$

The torus action places a strong restriction on this subalgebra. The representation theory of Chapter 1 suggests that it should decompose into character spaces. Consequently, the coordinate ring should be determined by the set

$$S_X = \{m \in M : \chi^m \text{ extends regularly to } X\}.$$

For $\mathbb{A}^2$, this set is $\mathbb{Z}^2_{\geq 0}$.

The precise theorem—that every affine toric coordinate ring is a semigroup algebra $\Bbbk[S]$—is Cox–Little–Schenck, Theorem 1.1.17. We will build and explain that equivalence in the next section rather than use it as a black box here.

The three dictionaries

The affine-plane example already contains the basic dictionary.

Algebra	Geometry	Combinatorics
$\Bbbk[X] \subseteq \Bbbk[M]$	an affine variety X containing T	a collection of lattice points in M
characters that extend regularly	monomial functions without poles on the boundary	allowed exponent vectors
multiplication of characters	multiplication of regular functions	addition of lattice points
the M-grading of the coordinate ring	the action of the dense torus	weights indexed by M

Guiding question

How does an additive collection

$$S \subseteq M$$

produce an affine toric variety, a monomial parametrization, and a system of binomial equations?

2.2 Affine semigroups, monomial maps, and equations

Section 2.1 suggested that an affine toric variety should be determined by the characters that extend regularly from its dense torus. Because products of extending characters still extend, their lattice points form an additive collection in the character lattice.

We now make that idea precise.

2.2.1 Affine semigroups

Throughout this chapter,

$$\mathbb{N} = \{0, 1, 2, \ldots\}.$$

Definition 2.5. Let M be a lattice. An *affine semigroup* is a finitely generated additive subsemigroup

$$S \subseteq M$$

containing 0.

Equivalently, there are lattice points

$$m_1, \dots, m_r \in M$$

such that

$$S = \mathbb{N}m_1 + \cdots + \mathbb{N}m_r.$$

We often write

$$S = \mathbb{N}\mathcal{A} \qquad \text{for} \qquad \mathcal{A} = \{m_1, \dots, m_r\}.$$

The subgroup of M generated by S will be denoted

$$\mathbb{Z}S = \{a_1 m_1 + \cdots + a_r m_r : a_i \in \mathbb{Z}\}.$$

Although S remembers which nonnegative combinations are allowed, $\mathbb{Z}S$ remembers the lattice of all differences between elements of S.

Example 2.6. Inside $M = \mathbb{Z}$, consider

$$S = \mathbb{N}\{2, 3\}.$$

Then

$$S = \{0, 2, 3, 4, 5, \dots\}.$$

The integer 1 does not lie in S, even though

$$1 = 3 - 2 \in \mathbb{Z}S.$$

This missing lattice point will become important when we discuss normality.

Pause and think

Find finite generating sets for the following affine semigroups:

$$\mathbb{Z}^2_{\geq 0}, \qquad 2\mathbb{Z}_{\geq 0}, \qquad \{(a, b) \in \mathbb{Z}^2 : a \geq 0,\ a + b \geq 0\}.$$

A generating set need not be unique. Can one of your generating sets be made smaller?

2.2.2 The semigroup algebra

An affine semigroup can be turned directly into an algebra.

Definition 2.7. Let $S \subseteq M$ be an affine semigroup. Its *semigroup algebra* is

$$\Bbbk[S] = \bigoplus_{m \in S} \Bbbk\, \chi^m,$$

with multiplication determined by

$$\chi^m \chi^{m'} = \chi^{m+m'}.$$

This notation treats the elements χ^m as formal basis vectors. Since $S \subseteq M$, we may also regard them as actual characters of the torus with character lattice M. In that realization,

$$\Bbbk[S] \subseteq \Bbbk[M].$$

If

$$S = \mathbb{N}m_1 + \cdots + \mathbb{N}m_r,$$

then

$$\Bbbk[S] = \Bbbk[\chi^{m_1}, \ldots, \chi^{m_r}].$$

Thus finite generation of the semigroup gives finite generation of the algebra. The basic examples from the previous section become

S	$\Bbbk[S]$	$\operatorname{Spec}\Bbbk[S]$
$\mathbb{N}^n$	$\Bbbk[x_1, \ldots, x_n]$	$\mathbb{A}^n$
$\mathbb{Z}_{\geq 0} \times \mathbb{Z}$	$\Bbbk[x, y^{\pm 1}]$	$\mathbb{A}^1 \times \mathbb{G}_m$
$\mathbb{Z}^n$	$\Bbbk[t_1^{\pm 1}, \ldots, t_n^{\pm 1}]$	$\mathbb{G}_m^n$.

The notation $\Bbbk[S]$ encodes both the basis and the multiplication. Adding lattice points corresponds to multiplying monomials:

$$m + m' \qquad \longleftrightarrow \qquad \chi^m \chi^{m'}.$$

2.2.3 From generators to a monomial map

Let

$$S = \mathbb{N}\mathcal{A}, \qquad \mathcal{A} = \{m_1, \ldots, m_r\} \subseteq M.$$

The generators determine a monomial map

$$\Phi_{\mathcal{A}} : T \longrightarrow \mathbb{A}^r$$

defined by

$$\Phi_{\mathcal{A}}(t) = \big(\chi^{m_1}(t), \ldots, \chi^{m_r}(t)\big).$$

Every coordinate of the map is a character. We therefore obtain a parameterized subset of affine space using monomials alone.

Definition 2.8. The *affine semigroup variety* associated to $\mathcal{A}$ is the Zariski closure

$$Y_{\mathcal{A}} = \overline{\Phi_{\mathcal{A}}(T)} \subseteq \mathbb{A}^r.$$

The closure is necessary: the image of the torus has all coordinates nonzero, while the closure adds the boundary points obtained when some monomials approach zero.

Example 2.9. Take

$$\mathcal{A} = \{(1,0), (1,1), (0,1)\} \subseteq \mathbb{Z}^2.$$

The monomial map is

$$\Phi_{\mathcal{A}}(s,t) = (s, st, t).$$

Writing the coordinates of $\mathbb{A}^3$ as x, z, y, the image satisfies

$$z = xy.$$

Its closure is the surface

$$Y_{\mathcal{A}} = V(z - xy) \subseteq \mathbb{A}^3,$$

which is isomorphic to $\mathbb{A}^2$. The extra generator $(1, 1)$ did not change the abstract variety; it produced a redundant affine embedding.

Proposition 2.10. *Let*

$$S = \mathbb{N}\mathcal{A} \subseteq M$$

be an affine semigroup. Then:

(a) $\Bbbk[S]$ *is a finitely generated integral domain;*

(b) $\operatorname{Spec} \Bbbk[S]$ *is an affine toric variety whose dense torus has character lattice* $\mathbb{Z}S$*;*

(c) there is a natural isomorphism

$$\operatorname{Spec} \Bbbk[S] \simeq Y_{\mathcal{A}}.$$

In particular,

$$\dim \operatorname{Spec} \Bbbk[S] = \operatorname{rank}(\mathbb{Z}S).$$

Reference. Over $\mathbb{C}$, this combines Cox–Little–Schenck, Propositions 1.1.8 and 1.1.14. The same argument works over our algebraically closed base field $\Bbbk$.

Conceptual proof sketch. The generators $m_1, \dots, m_r$ give a homomorphism

$$\pi : \Bbbk[x_1, \dots, x_r] \longrightarrow \Bbbk[M], \qquad x_i \longmapsto \chi^{m_i}.$$

Its image is exactly

$$\Bbbk[\chi^{m_1}, \dots, \chi^{m_r}] = \Bbbk[S].$$

On the geometric side, π is the pullback of the monomial map $\Phi_{\mathcal{A}}$. Therefore the coordinate ring of the closure of its image is the image of π, namely $\Bbbk[S]$.

The image of the torus is again a torus. Its character lattice is the subgroup generated by the characters used in the map, which is $\mathbb{Z}S$. The closure is irreducible because it is the closure of an irreducible torus, and the torus action preserves the closure.

To complete every detail, one checks that the ideal of the closure is $\ker \pi$, verifies that the image torus is open in the closure, and identifies its character lattice through pullback. □

Warning

The dense torus of $\operatorname{Spec} \Bbbk[S]$ has character lattice

$$\mathbb{Z}S,$$

which may be smaller than the ambient lattice M in which S was first written.
Thus the intrinsic lattice of the variety is not automatically the largest lattice visible in the notation. Working in the correct lattice will be essential when we discuss saturation and normality.

2.2.4 Relations among generators become equations

The monomial map gives a parametrization. To find equations for its closure, we look for additive relations among the generators

$$m_1, \dots, m_r.$$

Define the lattice homomorphism

$$\beta : \mathbb{Z}^r \longrightarrow M, \qquad e_i \longmapsto m_i,$$

and let

$$L = \ker \beta.$$

An element

$$\ell = (\ell_1, \dots, \ell_r) \in L$$

records an integral relation

$$\ell_1 m_1 + \cdots + \ell_r m_r = 0.$$

Write

$$\ell = \ell^+ - \ell^-,$$

where $\ell^+, \ell^- \in \mathbb{N}^r$ have disjoint supports. The relation $\beta(\ell) = 0$ becomes

$$\sum_i \ell_i^+ m_i = \sum_i \ell_i^- m_i.$$

Consequently,

$$x^{\ell^+} - x^{\ell^-}$$

vanishes after the substitution

$$x_i = \chi^{m_i}.$$

Example 2.11. For

$$\mathcal{A} = \{(1,0), (1,1), (0,1)\},$$

the relation

$$(1,0) + (0,1) = (1,1)$$

gives

$$x_1 x_3 - x_2.$$

This is the equation $xy - z = 0$ found above.

Definition 2.12. The kernel

$$I_{\mathcal{A}} = \ker \left(\Bbbk[x_1, \dots, x_r] \longrightarrow \Bbbk[S] \right), \qquad x_i \longmapsto \chi^{m_i},$$

is the *toric ideal* associated to $\mathcal{A}$.

Proposition 2.13 (Equations of a monomial image)**.** *With notation as above,*

$$I_{\mathcal{A}} = \left\langle x^{\ell^+} - x^{\ell^-} : \ell \in L \right\rangle.$$

Equivalently, $I_{\mathcal{A}}$ is generated by all binomials

$$x^{\alpha} - x^{\beta}$$

for which

$$\sum_i \alpha_i m_i = \sum_i \beta_i m_i.$$

Reference. This is Cox–Little–Schenck, Proposition 1.1.9.

Conceptual proof sketch. Every displayed binomial lies in the kernel because its two monomials are sent to the same character.

For the reverse inclusion, take a polynomial relation among the monomials χ^{m_i}. After substitution, different terms can cancel only when they become the same character of the torus. Group together all terms with the same resulting exponent in M. Within each group, subtract one chosen monomial from the others. Every such difference is one of the binomials above.

A complete proof organizes this reduction using a monomial order, or equivalently decomposes the polynomial according to its M-degree. The driving idea is simply that distinct characters are linearly independent in $\Bbbk[M]$. □

Thus additive relations in the lattice become binomial equations in affine space:

same sum of exponent vectors $\iff$ same monomial after parametrization.

2.2.5 The cusp: the simplest missing exponent

Let

$$S = \mathbb{N}\{2, 3\} \subseteq \mathbb{Z}.$$

Then

$$\Bbbk[S] = \Bbbk[t^2, t^3].$$

The monomial map is

$$\Phi_{\mathcal{A}} : \mathbb{G}_m \longrightarrow \mathbb{A}^2, \qquad t \longmapsto (t^2, t^3).$$

Let the coordinates of $\mathbb{A}^2$ be x, y, corresponding to the generators $2, 3$. The relation

$$3 \cdot 2 = 2 \cdot 3$$

gives

$$x^3 - y^2.$$

Hence

$$\Bbbk[S] \simeq \Bbbk[x, y]/(x^3 - y^2)$$

and

$$\operatorname{Spec} \Bbbk[S] \simeq V(x^3 - y^2) \subseteq \mathbb{A}^2.$$

This is the cuspidal cubic. It is an affine toric variety, even though it is not normal. The character $t = \chi^1$ is regular on the normalization $\mathbb{A}^1$, but it does not belong to $\Bbbk[t^2, t^3]$.

Algebraic geometry cheatsheet

For an irreducible affine variety $X = \operatorname{Spec} A$, being *normal* means that A is integrally closed in its fraction field. Concretely, if $f \in \operatorname{Frac}(A)$ satisfies a monic equation

$$f^r + a_{r-1} f^{r-1} + \cdots + a_0 = 0, \qquad a_i \in A,$$

then normality requires $f \in A$.
Normal varieties may still be singular, but they are regular in codimension one; consequently, their singularities can occur only in codimension at least two. Every irreducible variety X has a canonical normalization

$$\nu : X^\nu \longrightarrow X,$$

a finite birational morphism from a normal variety, unique up to unique isomorphism.
For further reading, see Hartshorne, *Algebraic Geometry*, Exercise I.3.17; the Stacks Project, Tags 031O and 0BXQ; and Cox–Little–Schenck, §1.3.

Pause and think

Why does

$$t \notin \Bbbk[t^2, t^3]$$

even though

$$t = \frac{t^3}{t^2}$$

belongs to the fraction field? Why does

$$t^2 \in \Bbbk[t^2, t^3]$$

suggest that the missing exponent 1 should be relevant to normality?

2.2.6 A quadratic cone and the correct lattice

Consider

$$\mathcal{A} = \{(2,0), (1,1), (0,2)\} \subseteq \mathbb{Z}^2$$

and

$$S = \mathbb{N}\mathcal{A}.$$

The monomial map

$$\mathbb{G}_m^2 \longrightarrow \mathbb{A}^3$$

is

$$(s,t) \longmapsto (s^2, st, t^2).$$

Writing the coordinates of $\mathbb{A}^3$ as a, b, c, the relation

$$(2,0) + (0,2) = 2(1,1)$$

gives

$$ac - b^2 = 0.$$

Therefore

$$\operatorname{Spec} \Bbbk[S] \simeq V(ac - b^2) \subseteq \mathbb{A}^3.$$

At first glance, S appears to omit many lattice points of the first quadrant, such as $(1, 0)$. But the character lattice of the dense torus is not all of $\mathbb{Z}^2$. It is

$$\mathbb{Z}S = \{(a, b) \in \mathbb{Z}^2 : a + b \text{ is even}\}.$$

Inside this intrinsic lattice, the apparent gaps look very different.

This example anticipates an important rule:

Questions about missing lattice points must be asked in the character lattice of the dense torus.

2.2.7 Four equivalent descriptions

We can now state the basic classification of arbitrary affine toric varieties, before imposing normality.

Theorem 2.14 (Equivalent constructions of affine toric varieties)**.** *Let X be an affine variety. The following are equivalent:*

(a) X is an affine toric variety;

(b) $X \simeq Y_{\mathcal{A}}$ for a finite set $\mathcal{A}$ in a lattice;

(c) X is isomorphic to an affine variety defined by a toric ideal;

(d)

$$X \simeq \operatorname{Spec} \Bbbk[S]$$

for an affine semigroup S.

Reference. This is Cox–Little–Schenck, Theorem 1.1.17, adapted from $\mathbb{C}$ to our algebraically closed field $\Bbbk$.

Conceptual proof sketch. Starting with a finite set of lattice points, the monomial map, its binomial equations, and the semigroup algebra are three presentations of the same object:

$$Y_{\mathcal{A}} \simeq \operatorname{Spec} \Bbbk[\mathbb{N}\mathcal{A}].$$

The preceding propositions establish these implications.

The only genuinely new direction starts with an intrinsic affine toric variety X. Since the dense torus T is open and dense, restriction embeds

$$\Bbbk[X] \hookrightarrow \Bbbk[T] = \Bbbk[M].$$

The extended torus action makes $\Bbbk[X]$ stable under T. The weight-space decomposition then says that this subalgebra is spanned by the characters it contains:

$$\Bbbk[X] = \bigoplus_{\chi^m \in \Bbbk[X]} \Bbbk\, \chi^m.$$

The set

$$S_X = \{m \in M : \chi^m \in \Bbbk[X]\}$$

is closed under addition because $\Bbbk[X]$ is an algebra. Finally, finite generation of the coordinate ring implies that S_X is finitely generated. Hence

$$\Bbbk[X] = \Bbbk[S_X].$$

To complete the proof in detail, one verifies that a finite collection of algebra generators can be replaced by the finitely many characters appearing in their weight decompositions. □

The theorem says that we are free to move between four viewpoints:

an intrinsic torus action,
a monomial parametrization,
binomial equations,
a semigroup algebra.

Guiding question

An affine semigroup S generates a real cone

$$\mathrm{Cone}(S) \subseteq M_{\mathbb{R}}.$$

When can S be recovered as all lattice points of that cone, and what geometric property does this recovery encode?

2.3 Cones, saturation, and normality

An affine semigroup

$$S = \mathbb{N}m_1 + \cdots + \mathbb{N}m_r \subseteq M$$

contains discrete information: it records exactly which lattice points are allowed. If we permit nonnegative real coefficients instead, we obtain a continuous geometric object:

$$\mathrm{Cone}(S) = \mathbb{R}_{\geq 0}m_1 + \cdots + \mathbb{R}_{\geq 0}m_r \subseteq M_{\mathbb{R}}.$$

Passing from S to $\mathrm{Cone}(S)$ remembers the directions in which the semigroup grows, but it may forget holes. The main question of this section is whether the original semigroup can be recovered from its cone:

$$S \stackrel{?}{=} \mathrm{Cone}(S) \cap M.$$

The answer is controlled simultaneously by a combinatorial condition on S and an algebraic condition on $\Bbbk[S]$:

$$\boxed{S \text{ has no hidden lattice-point holes} \iff \Bbbk[S] \text{ is normal.}}$$

2.3.1 From a semigroup to a cone

Definition 2.15. Let V be a finite-dimensional real vector space. A *polyhedral cone* is a subset of the form

$$C = \mathbb{R}_{\geq 0} v_1 + \cdots + \mathbb{R}_{\geq 0} v_r \subseteq V$$

for finitely many vectors $v_1, \ldots, v_r \in V$.

When $V = M_{\mathbb{R}}$ for a lattice M, the cone is *rational* if the generators may be chosen in M.

Thus the cone of every affine semigroup is a rational polyhedral cone.

Example 2.16. Let

$$S = \mathbb{N}(2,0) + \mathbb{N}(0,1) + \mathbb{N}(1,1) \subseteq \mathbb{Z}^2.$$

Its cone is again the first quadrant:

$$\operatorname{Cone}(S) = \mathbb{R}^2_{\geq 0}.$$

However,

$$(1,0) \notin S, \qquad 2(1,0) = (2,0) \in S.$$

Hence

$$S \subsetneq \operatorname{Cone}(S) \cap \mathbb{Z}^2.$$

The cone remembers the first quadrant but does not remember the missing point $(1,0)$.

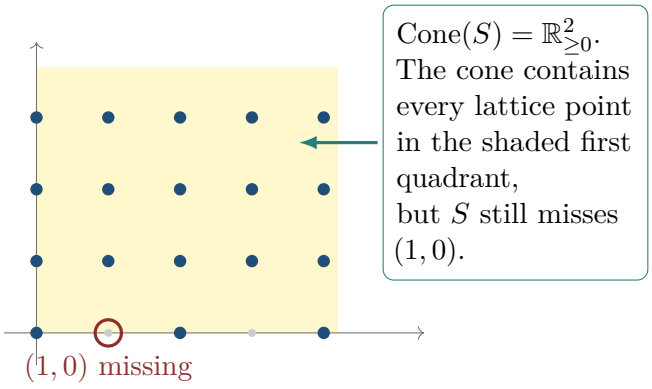


Figure 2.3: The semigroup generated by $(2,0), (0,1), (1,1)$ has the whole first quadrant as its cone, but it misses $(1,0)$.

Pause and think

For

$$S = \mathbb{N}\{2, 3\} \subseteq \mathbb{Z},$$

determine $\mathrm{Cone}(S)$, $\mathrm{Cone}(S) \cap \mathbb{Z}$, and the lattice points missing from S. Which of the missing points remain missing after multiplication by every positive integer?

2.3.2 A small amount of cone geometry

We record the pieces of cone language that will be used throughout the course.

The *span* of a cone C is the smallest linear subspace containing it:

$$\mathrm{span}_{\mathbb{R}}(C) = C - C.$$

Its dimension is the dimension of this span.

A cone is *strongly convex*, or *pointed*, if it contains no nonzero linear subspace:

$$C \cap (-C) = \{0\}.$$

A cone that contains a line has directions in which one may move both forward and backward while remaining inside the cone.

Definition 2.17. Let $C \subseteq M_{\mathbb{R}}$ be a polyhedral cone. A *face* of C is a subset

$$F = C \cap H$$

where H is a supporting hyperplane: C lies in one of the two closed half-spaces determined by H.

A one-dimensional face of a strongly convex cone is called a *ray*. A codimension-one face is called a *facet*.

For a rational ray ρ, the lattice points

$$\rho \cap M$$

are generated by a unique primitive lattice point. We call it the *primitive ray generator*.

Observation 2.18. Faces will acquire algebraic and geometric meanings later:

$$\text{faces of a cone} \quad \longleftrightarrow \quad \text{localizations and torus orbits.}$$

For the moment, faces simply describe the boundary structure of a cone.

Reference. The basic cone terminology appears in Cox–Little–Schenck, Definitions 1.2.2, 1.2.5, and 1.2.14. Primitive ray generators are discussed immediately before Lemma 1.2.15.

2.3.3 Gordan's lemma

Starting with a finite set of lattice points always produces a finitely generated semigroup. The converse direction is less obvious: if we start with a rational cone and take *all* of its lattice points, do we still obtain a finitely generated semigroup?

Theorem 2.19 (Gordan's lemma)**.** *Let $C \subseteq M_{\mathbb{R}}$ be a rational polyhedral cone. Then*

$$C \cap M$$

is a finitely generated semigroup.

Reference. This is Cox–Little–Schenck, Proposition 1.2.17, applied to a rational polyhedral cone in the character space $M_{\mathbb{R}}$.

Conceptual proof sketch. Choose lattice generators

$$v_1, \dots, v_r \in M$$

for C. Every point of C is a nonnegative real combination

$$a_1 v_1 + \cdots + a_r v_r.$$

Split each coefficient into its integer and fractional parts:

$$a_i = \lfloor a_i \rfloor + \delta_i, \qquad 0 \le \delta_i < 1.$$

The integer parts produce a nonnegative integral combination of the v_i. The fractional remainder lies in the bounded region

$$K = \{\delta_1 v_1 + \cdots + \delta_r v_r : 0 \le \delta_i < 1\}.$$

Since a bounded region contains only finitely many lattice points, $K \cap M$ is finite. Every lattice point of C is therefore the sum of one of these finitely many remainder points and a nonnegative integral combination of the v_i.

To complete the proof, retain from $K \cap M$ only the points that actually lie in C, and add them to the original generators. □

The proof explains the finite generation geometrically:

Integer parts move along the cone generators, while only finitely many fractional remainders must be remembered.

2.3.4 The Hilbert basis

When C is strongly convex, the semigroup $C \cap M$ has a canonical minimal generating set.

Definition 2.20. Let $S = C \cap M$, where C is strongly convex. A nonzero element $m \in S$ is *irreducible* if

$$m = m' + m'', \qquad m', m'' \in S,$$

implies $m' = 0$ or $m'' = 0$.

The set of irreducible elements of S is called the *Hilbert basis* of $C \cap M$.

Proposition 2.21. *The Hilbert basis of $C \cap M$ is finite, generates $C \cap M$, and is its unique minimal generating set.*

Reference. This is the character-space version of Cox–Little–Schenck, Proposition 1.2.23.

Conceptual proof sketch. Gordan's lemma supplies some finite generating set. Any irreducible element must occur in every generating set, because it cannot be written as a sum of smaller nonzero semigroup elements. Conversely, repeatedly decompose any reducible element. Strong convexity provides a linear function that is strictly positive away from the origin, so this process cannot continue indefinitely. The resulting pieces are irreducible.

Thus the irreducible elements generate and are forced to occur in every generating set. □

Warning

The primitive ray generators of C need not generate the whole semigroup $C \cap M$. A cone may have very few rays but require additional lattice points between them in its Hilbert basis.
Cone generators and semigroup generators answer different questions.

2.3.5 Saturation fills the holes

We now give a name to the condition suggested by the cusp and by Figure 2.3.

Definition 2.22. Let S be an affine semigroup, and let

$$M_S = \mathbb{Z}S$$

be its group of differences. The semigroup S is *saturated* if

$$m \in M_S, \quad k > 0, \quad km \in S \qquad \Longrightarrow \qquad m \in S.$$

The use of the intrinsic lattice $M_S = \mathbb{Z}S$ is essential. A point outside M_S is not a character of the dense torus of $\operatorname{Spec} \Bbbk[S]$, so it should not count as a missing character.

Example 2.23. The semigroup

$$S = \mathbb{N}\{2, 3\} \subseteq \mathbb{Z}$$

is not saturated because

$$1 \notin S, \qquad 2 \cdot 1 = 2 \in S.$$

Example 2.24. For

$$S = \mathbb{N}(2, 0) + \mathbb{N}(0, 1) + \mathbb{N}(1, 1) \subseteq \mathbb{Z}^2,$$

we have

$$\mathbb{Z}S = \mathbb{Z}^2.$$

The point $(1, 0)$ witnesses the failure of saturation:

$$(1, 0) \notin S, \qquad 2(1, 0) \in S.$$

Proposition 2.25 (Saturation and lattice points)**.** *Let S be an affine semigroup and set*

$$M_S = \mathbb{Z}S, \qquad C = \operatorname{Cone}(S) \subseteq (M_S)_{\mathbb{R}}.$$

Then

$$S \text{ is saturated} \quad \Longleftrightarrow \quad S = C \cap M_S.$$

Reference. This is Cox–Little–Schenck, Exercise 1.3.4(a).

Proof. The inclusion

$$S \subseteq C \cap M_S$$

always holds.

Suppose first that S is saturated, and let $m \in C \cap M_S$. Because C is a rational cone generated by elements of S, the point m can be expressed using nonnegative rational coefficients:

$$m = q_1 m_1 + \cdots + q_r m_r, \qquad q_i \in \mathbb{Q}_{\geq 0}.$$

Choose a positive integer k clearing all denominators. Then

$$km \in S.$$

Saturation gives $m \in S$.

Conversely, suppose

$$S = C \cap M_S.$$

If $km \in S$ for $m \in M_S$ and $k > 0$, then

$$m = \frac{1}{k}(km) \in C.$$

Hence

$$m \in C \cap M_S = S.$$

□

This proposition is the precise version of the slogan:

saturation fills every lattice-point hole inside the cone.

2.3.6 Normality is saturation in algebraic form

Recall from the algebraic geometry cheatsheet in Section 2.2 that an affine domain A is normal when it is integrally closed in its fraction field.

For semigroup algebras, integral closure has a direct combinatorial meaning.

Theorem 2.26 (Normality criterion for affine toric varieties)**.** *Let S be an affine semigroup, and let*

$$M_S = \mathbb{Z}S$$

be its intrinsic lattice. Set

$$C = \mathrm{Cone}(S) \subseteq (M_S)_{\mathbb{R}}.$$

The following are equivalent:

(a) $\operatorname{Spec} \Bbbk[S]$ *is normal;*

(b) S is saturated in M_S;
(c)

$$S = C \cap M_S.$$

Consequently, every normal affine toric variety arising from S has coordinate ring

$$\Bbbk[C \cap M_S],$$

where $C = \mathrm{Cone}(S) \subseteq (M_S)_{\mathbb{R}}$. More intrinsically, if an affine toric variety has dense-torus character lattice M_T, then it has the form

$$X_C = \operatorname{Spec} \Bbbk[C \cap M_T]$$

for a full-dimensional rational polyhedral cone $C \subseteq (M_T)_{\mathbb{R}}$.

Reference. This is the character-space formulation of Cox–Little–Schenck, Theorem 1.3.5.

Conceptual proof sketch. The implication

$$\text{normal} \Longrightarrow \text{saturated}$$

contains the key algebraic idea. Suppose

$$km \in S$$

for some $m \in M_S$ and $k > 0$. Since $m \in \mathbb{Z}S$, the character χ^m belongs to the fraction field of $\Bbbk[S]$. Moreover, it satisfies the monic equation

$$Z^k - \chi^{km} = 0$$

with coefficient $\chi^{km} \in \Bbbk[S]$. Thus χ^m is integral over $\Bbbk[S]$. Normality forces $\chi^m \in \Bbbk[S]$, so $m \in S$.

The equivalence between saturation and

$$S = C \cap M_S$$

is theorem 2.25.

For the remaining direction, let

$$N_S = \mathrm{Hom}_{\mathbb{Z}}(M_S, \mathbb{Z})$$

be the lattice dual to M_S, and write

$$\sigma = C^\vee \subseteq (N_S)_{\mathbb{R}}.$$

Since C spans $(M_S)_{\mathbb{R}}$, the cone σ is strongly convex. The algebra $\Bbbk[C \cap M_S]$ may be recovered as an intersection of the algebras associated to the rays of σ. After choosing a basis adapted

to a primitive ray generator, each ray algebra becomes a polynomial ring in one variable with Laurent variables in the other directions:

$$\Bbbk[x_1, x_2^{\pm 1}, \dots, x_n^{\pm 1}].$$

These rings are normal, and an intersection of integrally closed domains inside a common fraction field is integrally closed.

The technical part of the complete proof verifies the intersection description and the reduction to primitive ray generators. □

Pause and think

Return to

$$S = \mathbb{N}(2,0) + \mathbb{N}(0,1) + \mathbb{N}(1,1).$$

Determine its saturation. What is the normalization of

$$\operatorname{Spec} \Bbbk[S]?$$

Using

$$a = x^2, \qquad b = y, \qquad c = xy,$$

find an equation for the original affine semigroup variety and write the normalization map explicitly.

Here

$$\overline{S} = \mathbb{Z}^2_{\geq 0}.$$

The original algebra is

$$\Bbbk[S] = \Bbbk[x^2, y, xy] \simeq \Bbbk[a,b,c]/(c^2 - ab^2),$$

while the saturated algebra is

$$\Bbbk[\overline{S}] = \Bbbk[x,y].$$

The normalization map is

$$\mathbb{A}^2 \longrightarrow V(c^2 - ab^2) \subseteq \mathbb{A}^3, \qquad (x,y) \longmapsto (x^2, y, xy).$$

2.3.7 Normalization of an affine semigroup variety

The previous theorem identifies the ring that should normalize $\Bbbk[S]$.

Proposition 2.27 (Normalization by filling the cone)**.** *Let S be an affine semigroup, set*

$$M_S = \mathbb{Z}S, \qquad C = \operatorname{Cone}(S) \subseteq (M_S)_{\mathbb{R}},$$

and write

$$\overline{S} = C \cap M_S.$$

Then the inclusion

$$\Bbbk[S] \subseteq \Bbbk[\overline{S}]$$

induces the normalization morphism

$$\nu : \operatorname{Spec} \Bbbk[\overline{S}] \longrightarrow \operatorname{Spec} \Bbbk[S].$$

Reference. This is the character-space form of Cox–Little–Schenck, Proposition 1.3.8.

Conceptual proof sketch. By Gordan's lemma, $\overline{S} = C \cap M_S$ is finitely generated. It is saturated, so $\Bbbk[\overline{S}]$ is normal.

For every $m \in \overline{S}$, rationality of the cone implies that some positive multiple km lies in S. Hence

$$\chi^m$$

satisfies

$$Z^k - \chi^{km} = 0$$

and is integral over $\Bbbk[S]$. Thus $\Bbbk[\overline{S}]$ is integral, and therefore finite, over $\Bbbk[S]$.

Finally, S and $\overline{S}$ generate the same group M_S, so their semigroup algebras have the same fraction field. The induced morphism is therefore finite and birational, with normal source, which characterizes the normalization. □

For the cusp this recovers the normalization already anticipated:

$$\Bbbk[t^2, t^3] \subseteq \Bbbk[t], \qquad t \longmapsto (t^2, t^3).$$

Different semigroups can generate the same cone, so cones do not classify arbitrary affine toric varieties. If S_1 and S_2 generate the same cone and the same intrinsic lattice, then they have the same saturation and hence the same normalization.

All intersections with the cone must be taken in

$$M_S = \mathbb{Z}S,$$

the character lattice of the dense torus determined by S. Using a larger ambient lattice can create apparent holes that are not intrinsic to the variety.

The three dictionaries

Normality is the same phenomenon in three languages.

Algebra	Geometry	Combinatorics
$\Bbbk[S]$	$\operatorname{Spec} \Bbbk[S]$	an affine semigroup $S \subseteq M$, with $M_S = \mathbb{Z}S$
integral closure	normalization	fill lattice-point holes
$\Bbbk[S]$ integrally closed	the affine toric variety is normal	S is saturated
$\Bbbk[C \cap M_S]$	a normal affine toric variety	all lattice points of a rational cone C

Guiding question

If a cone

$$C \subseteq M_{\mathbb{R}}$$

records the characters that extend regularly, what does the dual cone in $N_{\mathbb{R}}$ record about one-parameter subgroups and their limits?

2.4 Dual cones and one-parameter subgroup limits

The previous section described a normal affine toric variety by a cone

$$C \subseteq M_{\mathbb{R}}$$

whose lattice points index the characters that extend regularly. We now turn this picture around.

Fix a torus T, with character and one-parameter subgroup lattices

$$M = X^*(T), \qquad N = X_*(T),$$

and perfect pairing

$$\langle \quad , \quad \rangle : M \times N \longrightarrow \mathbb{Z}.$$

When a variety is first presented by an affine semigroup S, the lattices in this section should be understood intrinsically:

$$M = M_S = \mathbb{Z}S, \qquad N = N_S = \mathrm{Hom}_{\mathbb{Z}}(M_S, \mathbb{Z}).$$

An ambient lattice containing S plays no role in the geometry of the dense torus unless it equals M_S.

2.4.1 Dual cones

A vector $u \in N_{\mathbb{R}}$ defines a linear function on $M_{\mathbb{R}}$:

$$m \longmapsto \langle m, u \rangle .$$

A cone in one of the two spaces therefore determines a cone in the other by requiring all these values to be nonnegative.

Definition 2.28. Let

$$\sigma \subseteq N_{\mathbb{R}}$$

be a polyhedral cone. Its *dual cone* is

$$\sigma^\vee = \{m \in M_{\mathbb{R}} : \langle m, u \rangle \geq 0 \text{ for every } u \in \sigma\} .$$

Likewise, for a cone $C \subseteq M_{\mathbb{R}}$, its dual is

$$C^\vee = \{u \in N_{\mathbb{R}} : \langle m, u \rangle \geq 0 \text{ for every } m \in C\} .$$

Thus $\sigma^\vee$ consists of the linear functions on $N_\mathbb{R}$ that are nonnegative throughout σ.

Reference. This is Cox–Little–Schenck, Definition 1.2.3.

If

$$\sigma = \mathrm{Cone}(u_1, \dots, u_r),$$

then it is enough to test the generators:

$$\sigma^\vee = \{m \in M_\mathbb{R} : \langle m, u_i \rangle \geq 0 \text{ for } i = 1, \dots, r\}.$$

Indeed, nonnegativity on the u_i implies nonnegativity on every nonnegative linear combination of them.

Example 2.29. Let

$$\sigma = \mathrm{Cone}((1,0),(1,2)) \subseteq N_\mathbb{R}.$$

Then

$$\sigma^\vee = \{(a,b) \in M_\mathbb{R} : a \geq 0,\ a + 2b \geq 0\}.$$

Its two rays are generated by

$$(0,1) \qquad \text{and} \qquad (2,-1).$$

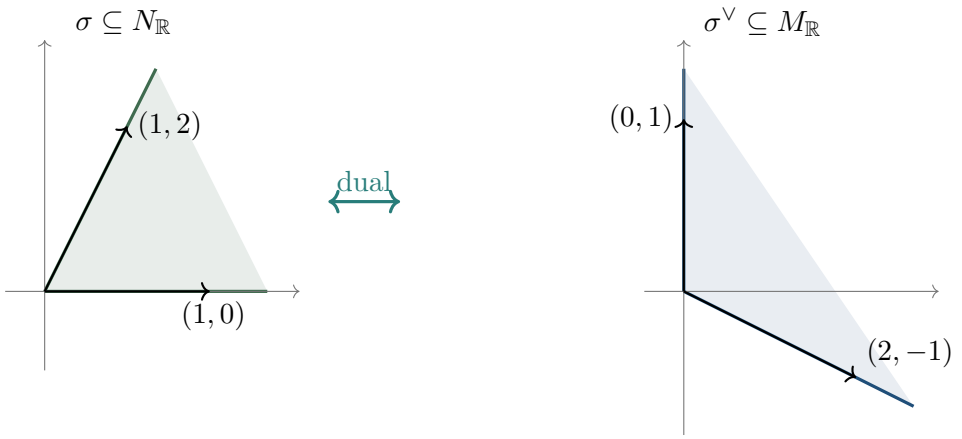


Figure 2.4: The cone $\sigma = \mathrm{Cone}((1,0),(1,2))$ and its dual $\sigma^\vee = \mathrm{Cone}((0,1),(2,-1))$.

Proposition 2.30 (Biduality)**.** *For every polyhedral cone,*

$$(\sigma^\vee)^\vee = \sigma.$$

Moreover, the dual of a rational polyhedral cone is again rational polyhedral.

Reference. The biduality statement is Cox–Little–Schenck, Proposition 1.2.4. Rationality of dual cones is recorded in the discussion following Definition 1.2.14.

Conceptual proof sketch. Every $u \in \sigma$ pairs nonnegatively with every $m \in \sigma^\vee$, so

$$\sigma \subseteq (\sigma^\vee)^\vee.$$

For the reverse inclusion, a point outside a closed polyhedral cone can be separated from it by a linear functional. That functional is an element $m \in \sigma^\vee$ that takes a negative value on the outside point. Hence that point cannot lie in the bidual.

For a rational cone, the required supporting functionals may be chosen rational and then cleared of denominators to become lattice vectors. □

Pause and think

Compute the duals of the following cones in $N_\mathbb{R} \simeq \mathbb{R}^2$:

$$\{0\}, \qquad \mathrm{Cone}((1,0)), \qquad \mathrm{Cone}((1,0),(0,1)), \qquad \mathrm{Cone}((1,0),(-1,0)).$$

What happens to the dimension of the dual when the original cone contains a line?

2.4.2 From a cone in $N_\mathbb{R}$ to an affine variety

Let

$$\sigma \subseteq N_\mathbb{R}$$

be a strongly convex rational polyhedral cone. Define

$$S_\sigma = \sigma^\vee \cap M$$

and

$$U_\sigma = \operatorname{Spec} \Bbbk[S_\sigma] = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].$$

Gordan's lemma makes S_σ finitely generated. Because it contains every lattice point of the rational cone $\sigma^\vee$, it is saturated. Therefore U_σ is normal.

Strong convexity of σ ensures that

$$\sigma^\vee$$

spans $M_\mathbb{R}$. Consequently,

$$\mathbb{Z}(\sigma^\vee \cap M) = M,$$

so the dense torus of U_σ is the original torus T.

Theorem 2.31. *Let $\sigma \subseteq N_\mathbb{R}$ be a strongly convex rational polyhedral cone. Then*

$$U_\sigma = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M]$$

is a normal affine toric variety with dense torus T.

Reference. The construction and dense-torus statement are Cox–Little–Schenck, Theorem 1.2.18 and Remark 1.2.19. Normality follows from Theorem 1.3.5.

Conceptual proof sketch. Gordan's lemma supplies finite generation, so U_σ is an affine semigroup variety. The semigroup $\sigma^\vee \cap M$ is saturated, hence its algebra is normal. Finally, strong convexity is dual to the condition that $\sigma^\vee$ be full-dimensional, which guarantees that its lattice points generate the full character lattice M. □

This construction translates the result of the previous section. If

$$C \subseteq M_{\mathbb{R}}$$

is a full-dimensional rational polyhedral cone and

$$X_C = \operatorname{Spec} \Bbbk[C \cap M],$$

then

$$\sigma = C^\vee \subseteq N_{\mathbb{R}}$$

is strongly convex and

$$C = \sigma^\vee.$$

Hence

$$X_C = U_\sigma.$$

The standard cones recover the basic affine toric varieties:

σ	$\sigma^\vee \cap M$	U_σ
$\{0\}$	M	T
$\operatorname{Cone}((1,0)) \subseteq \mathbb{R}^2$	$\{(a,b) \in \mathbb{Z}^2 : a \geq 0\}$	$\mathbb{A}^1 \times \mathbb{G}_m$
$\operatorname{Cone}((1,0),(0,1))$	$\mathbb{Z}^2_{\geq 0}$	$\mathbb{A}^2$.

The nonsmooth cone

$$\operatorname{Cone}((1,0),(1,2))$$

produces the normal quadratic cone

$$V(AC - B^2) \subseteq \mathbb{A}^3,$$

as computed in Section 2.2 and used again in the smoothness discussion of Section 2.5.

2.4.3 What does it mean for a limit to exist?

For $u \in N$, let

$$\lambda^u : \mathbb{G}_m \longrightarrow T$$

be the corresponding one-parameter subgroup.

When we write

$$\lim_{t \to 0} \lambda^u(t) \quad \text{in } U_\sigma,$$

we mean algebraically that the morphism λ^u extends across the missing parameter value:

$$\begin{array}{ccc} \mathbb{G}_m & \xrightarrow{\lambda^u} & T \\ \big\downarrow & & \big\downarrow \\ \mathbb{A}^1 & \xrightarrow[\overline{\lambda^u}]{} & U_\sigma. \end{array}$$

The limit is then the point

$$\overline{\lambda^u}(0).$$

Because U_σ is affine, the extension can be tested on its coordinate ring. For every

$$m \in \sigma^\vee \cap M,$$

we have

$$(\lambda^u)^*(\chi^m) = t^{\langle m,u\rangle}.$$

This Laurent monomial extends regularly to $\mathbb{A}^1$ precisely when

$$\langle m, u\rangle \geq 0.$$

2.4.4 The one-parameter subgroup limit criterion

We have reached the geometric reason for placing the defining cone in the one-parameter subgroup space.

Theorem 2.32 (One-parameter subgroup limit criterion)**.** *Let*

$$\sigma \subseteq N_{\mathbb{R}}$$

be a strongly convex rational polyhedral cone, and let $u \in N$. Then

$$u \in \sigma \quad \Longleftrightarrow \quad \lambda^u : \mathbb{G}_m \longrightarrow T \text{ extends to a morphism } \mathbb{A}^1 \longrightarrow U_\sigma.$$

Equivalently,

$$u \in \sigma \quad \Longleftrightarrow \quad \lim_{t\to 0} \lambda^u(t) \text{ exists in } U_\sigma.$$

Reference. This is Cox–Little–Schenck, Proposition 3.2.2. Although it appears there in the discussion of the orbit–cone correspondence, its proof uses only the affine coordinate ring $\Bbbk[\sigma^\vee \cap M]$. The same algebraic proof works over our base field $\Bbbk$.

Proof. A morphism

$$\mathbb{A}^1 \longrightarrow U_\sigma$$

is equivalent to a $\Bbbk$-algebra homomorphism

$$\Bbbk[\sigma^\vee \cap M] \longrightarrow \Bbbk[t].$$

The pullback along λ^u is already defined with values in $\Bbbk[t, t^{-1}]$, and sends

$$\chi^m \longmapsto t^{\langle m,u\rangle}.$$

It takes values in $\Bbbk[t]$ exactly when

$$\langle m, u\rangle \geq 0 \qquad \text{for every } m \in \sigma^\vee \cap M.$$

Since the lattice points generate the rational cone $\sigma^\vee$, this is equivalent to

$$u \in (\sigma^\vee)^\vee.$$

By biduality,

$$(\sigma^\vee)^\vee = \sigma.$$

□

The theorem gives the promised second description of U_σ:

$$\boxed{\sigma \cap N = \{u \in N : \lambda^u \text{ has a limit in } U_\sigma \text{ as } t \to 0\}.}$$

On the character side,

$$\sigma^\vee \cap M = \{m \in M : \chi^m \text{ extends regularly to } U_\sigma\}.$$

Thus the two cones encode two complementary extension problems:

$\sigma^\vee \subseteq M_\mathbb{R}$	$\sigma \subseteq N_\mathbb{R}$
which functions extend	which motions acquire limits.

2.4.5 The limit point

Suppose $u \in \sigma \cap N$. Evaluating the extended map at $t = 0$ gives the semigroup homomorphism

$$\gamma_u : \sigma^\vee \cap M \longrightarrow \Bbbk$$

defined by

$$\gamma_u(m) = \begin{cases} 1, & \langle m, u\rangle = 0, \\ 0, & \langle m, u\rangle > 0. \end{cases}$$

This is the point

$$\lim_{t\to 0} \lambda^u(t) \in U_\sigma.$$

If u lies in the relative interior of σ, then

$$\langle m, u\rangle = 0 \quad \Longleftrightarrow \quad m \in \sigma^\perp \cap M.$$

The resulting point is called the *distinguished point* γ_σ of U_σ.

More generally, let $\tau \preceq \sigma$ be the smallest face containing u. Then

$$u \in \operatorname{Relint}(\tau),$$

and the limit depends only on τ:

$$\gamma_u(m) = \begin{cases} 1, & m \in \tau^\perp \cap M, \\ 0, & m \notin \tau^\perp. \end{cases}$$

We will later interpret this point as the distinguished point associated to the face τ.

Example 2.33. Let

$$U_\sigma = \mathbb{A}^2, \qquad \sigma = \mathbb{R}^2_{\geq 0}.$$

For

$$u = (a, b) \in \sigma \cap \mathbb{Z}^2,$$

the one-parameter subgroup is

$$\lambda^u(t) = (t^a, t^b).$$

Its limit is

(a, b)	$\lim_{t\to 0}(t^a, t^b)$
$a > 0,\ b > 0$	$(0, 0)$,
$a > 0,\ b = 0$	$(0, 1)$,
$a = 0,\ b > 0$	$(1, 0)$,
$a = b = 0$	$(1, 1)$.

The four possibilities correspond to the relative interiors of the four faces of the first quadrant: the two-dimensional cone, its two rays, and the zero face.

Surface laboratory

For $\mathbb{A}^2$, a one-parameter subgroup by itself reaches one distinguished point on each orbit. Translating the motion by a torus point,

$$t \longmapsto p\,\lambda^u(t),$$

moves that distinguished limit through the corresponding orbit.
For example, if $u = (1, 0)$, then

$$(a, b)\,(t, 1) = (at, b) \longrightarrow (0, b).$$

As b varies through $\Bbbk^\times$, these limits fill the one-dimensional orbit

$$V(x) \setminus \{(0, 0)\}.$$

Pause and think

For

$$\sigma = \mathrm{Cone}((1, 0), (1, 2))$$

and $u = (p, q) \in N = \mathbb{Z}^2$, use the inequalities defining σ to determine exactly when

$$\lim_{t\to 0} \lambda^u(t)$$

exists in

$$U_\sigma \simeq V(AC - B^2).$$

Then compute the pullbacks of A, B, C and determine the limit point when u lies:

$$\text{(a) in } \operatorname{Relint}(\sigma), \qquad \text{(b) on each ray of } \sigma.$$

For a single affine variety, the cones

$$\sigma \subseteq N_{\mathbb{R}} \qquad \text{and} \qquad \sigma^\vee \subseteq M_{\mathbb{R}}$$

contain the same information. The advantage of σ is geometric: its lattice points are one-parameter motions, and its faces organize their limit points. This is why cones in $N_{\mathbb{R}}$ become the natural objects for gluing affine toric varieties.

Guiding question

Which geometric properties of U_σ—its units, torus factors, fixed points, smoothness, morphisms, and affine open subsets—can be read directly from the cone σ?

2.5 Properties of affine toric varieties

We now know how to construct a normal affine toric variety from a strongly convex rational polyhedral cone:

$$U_\sigma = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].$$

The purpose of this section is to learn how much geometry can be read directly from σ.

Normality has already provided the first instance of this principle:

$$U_\sigma \text{ is normal} \iff \sigma^\vee \cap M \text{ contains every lattice point of } \sigma^\vee.$$

We now add several more entries to the dictionary.

2.5.1 Points as semigroup homomorphisms

Let

$$S \subseteq M$$

be an affine semigroup. We regard $\Bbbk$ as a multiplicative semigroup, including the absorbing element 0.

Proposition 2.34 (Points of an affine semigroup variety)**.** *For*

$$X_S = \operatorname{Spec} \Bbbk[S],$$

there are natural correspondences among:

(a) points $p \in X_S$;

(b) maximal ideals of $\Bbbk[S]$;

(c) semigroup homomorphisms

$$\gamma : S \longrightarrow \Bbbk.$$

The homomorphism associated to p is

$$\gamma_p(m) = \chi^m(p).$$

Reference. This is Cox–Little–Schenck, Proposition 1.3.1.

Proof. A point p gives the evaluation homomorphism

$$\mathrm{ev}_p : \Bbbk[S] \longrightarrow \Bbbk.$$

Restricting it to the monomial basis gives

$$\gamma_p(m) = \mathrm{ev}_p(\chi^m).$$

Since

$$\chi^{m+m'} = \chi^m \chi^{m'},$$

the map γ_p is multiplicative.

Conversely, a semigroup homomorphism $\gamma : S \to \Bbbk$ extends linearly to

$$\Bbbk[S] \longrightarrow \Bbbk, \qquad \sum_m c_m \chi^m \longmapsto \sum_m c_m \gamma(m).$$

Multiplicativity of γ makes this a $\Bbbk$-algebra homomorphism, hence a point of $\operatorname{Spec} \Bbbk[S]$. □

This description is especially effective because zeros are allowed. A value

$$\gamma(m) = 0$$

records that the character χ^m vanishes at the point.

Example 2.35. For

$$\mathbb{A}^2 = \operatorname{Spec} \Bbbk[\mathbb{N}^2],$$

a semigroup homomorphism

$$\gamma : \mathbb{N}^2 \longrightarrow \Bbbk$$

is determined by

$$a = \gamma(1,0), \qquad b = \gamma(0,1).$$

Indeed,

$$\gamma(i,j) = a^i b^j.$$

Thus semigroup homomorphisms $\mathbb{N}^2 \to \Bbbk$ are exactly the points

$$(a,b) \in \mathbb{A}^2.$$

The torus action also has an intrinsic semigroup description. If $t \in T$ and $p \leftrightarrow \gamma_p$, then

$$t \cdot p \quad \longleftrightarrow \quad (m \longmapsto \chi^m(t)\gamma_p(m)).$$

The nonzero factors $\chi^m(t)$ rescale the values of γ_p, but they do not change which values vanish.

Pause and think

A point $p \in X_S$ belongs to the dense torus precisely when

$$\gamma_p(m) \neq 0 \qquad \text{for every } m \in S.$$

Explain why. Then identify the dense torus inside $\mathbb{A}^2$ using semigroup homomorphisms $\mathbb{N}^2 \to \Bbbk$.

2.5.2 Invertible characters and torus factors

A character χ^m is invertible in

$$\Bbbk[\sigma^\vee \cap M]$$

exactly when both m and $-m$ lie in $\sigma^\vee$. This means

$$\langle m, u\rangle = 0 \qquad \text{for every } u \in \sigma.$$

Define

$$\sigma^\perp = \{m \in M_{\mathbb{R}} : \langle m, u\rangle = 0 \text{ for every } u \in \sigma\}.$$

Proposition 2.36 (Units)**.** *The units of the coordinate ring of U_σ are*

$$\Bbbk[U_\sigma]^\times = \left\{c\,\chi^m : c \in \Bbbk^\times,\ m \in \sigma^\perp \cap M\right\}.$$

Conceptual proof sketch. Every displayed element is visibly invertible, with inverse

$$c^{-1}\chi^{-m}.$$

Conversely, suppose $fg = 1$. Pass from M to the quotient

$$M/(\sigma^\perp \cap M).$$

The image of $\sigma^\vee \cap M$ in this quotient is pointed. Choose a linear functional that is strictly positive on every nonzero degree. Looking at the largest degree occurring in fg shows that both f and g must have only one degree in the quotient. Their remaining factors lie in the group algebra $\Bbbk[\sigma^\perp \cap M]$, whose units are scalar multiples of characters. □

The orthogonal space $\sigma^\perp$ therefore measures the directions in which the affine toric variety still behaves like a torus.

Let

$$N_\sigma = N \cap \operatorname{span}_{\mathbb{R}}(\sigma).$$

This is a saturated sublattice of N of rank

$$r = \dim \sigma.$$

Choose a complementary lattice N' such that

$$N = N_\sigma \oplus N'.$$

The choice is generally not canonical.

Proposition 2.37 (Splitting off the torus factor)**.** *With notation as above,*

$$U_{\sigma,N} \simeq U_{\sigma,N_\sigma} \times T_{N'}.$$

In particular,

$$U_{\sigma,N} \simeq U_{\sigma,N_\sigma} \times \mathbb{G}_m^{\operatorname{rank} N - \dim \sigma}.$$

Reference. This is the decomposition in Cox–Little–Schenck, Equation (1.3.2), in the discussion following Theorem 1.3.12.

Conceptual proof sketch. Dualizing the decomposition

$$N = N_\sigma \oplus N'$$

gives

$$M = M_\sigma \oplus M'.$$

Since σ lies entirely in $(N_\sigma)_{\mathbb{R}}$, its dual semigroup splits as

$$\sigma^\vee \cap M = (\sigma^\vee \cap M_\sigma) \oplus M'.$$

The corresponding semigroup algebra is therefore a tensor product:

$$\Bbbk[\sigma^\vee \cap M] \simeq \Bbbk[\sigma^\vee \cap M_\sigma] \otimes_\Bbbk \Bbbk[M'].$$

Taking spectra turns the tensor product into the displayed product of varieties. □

Example 2.38. For the ray

$$\sigma = \operatorname{Cone}((1,0)) \subseteq \mathbb{R}^2,$$

we have

$$U_\sigma \simeq \mathbb{A}^1 \times \mathbb{G}_m.$$

The character y lies in $\sigma^\perp \cap M = \mathbb{Z}(0,1)$, so y is invertible. The $\mathbb{G}_m$-factor is visible both algebraically and geometrically.

The three dictionaries

The directions perpendicular to σ describe the same feature in three languages.

Algebra	**Geometry**	**Combinatorics**
invertible characters χ^m	torus directions that never meet a boundary	$m \in \sigma^\perp \cap M$
a Laurent polynomial factor $\Bbbk[M']$	a factor $T_{N'}$	a complementary lattice to N_σ

2.5.3 Fixed points

A torus-fixed point must remain unchanged under every rescaling by the torus. In semigroup language this leaves only one possible candidate.

Definition 2.39. An affine semigroup S is *pointed* if

$$S \cap (-S) = \{0\}.$$

Equivalently, 0 is the only invertible element of S.

Proposition 2.40 (Fixed points)**.** *Let*

$$X_S = \operatorname{Spec} \Bbbk[S].$$

The torus action on X_S has a fixed point if and only if S is pointed. When it exists, the fixed point is unique and is represented by

$$\gamma_0(m) = \begin{cases} 1, & m = 0, \\ 0, & m \neq 0. \end{cases}$$

Reference. This is Cox–Little–Schenck, Proposition 1.3.2.

Proof. If $p \leftrightarrow \gamma$ is fixed, then

$$\chi^m(t)\gamma(m) = \gamma(m)$$

for every $t \in T$. For $m \neq 0$, one can choose t with $\chi^m(t) \neq 1$, so $\gamma(m) = 0$. Hence a fixed point, if it exists, must be γ_0.

The map γ_0 is multiplicative exactly when two nonzero elements of S cannot add to 0. This is equivalent to

$$S \cap (-S) = \{0\}.$$

□

For the cone construction, pointedness has a simple geometric translation.

Corollary 2.41. *Let*

$$\sigma \subseteq N_{\mathbb{R}}$$

be strongly convex. The torus action on U_σ has a fixed point if and only if

$$\dim \sigma = \dim N_{\mathbb{R}}.$$

When it exists, it is the distinguished point γ_σ obtained as the limit of every one-parameter subgroup in $\operatorname{Relint}(\sigma)$*.*

Reference. The fixed-point criterion is Cox–Little–Schenck, Corollary 1.3.3. The interpretation by one-parameter limits follows from theorem 2.32.

Proof. The semigroup

$$\sigma^\vee \cap M$$

is pointed precisely when

$$\sigma^\perp = \{0\}.$$

This is equivalent to σ spanning $N_\mathbb{R}$, or

$$\dim \sigma = \dim N_\mathbb{R}.$$

□

Example 2.42. The affine plane corresponds to the full-dimensional first quadrant, so it has one fixed point: the origin.

The variety

$$\mathbb{A}^1 \times \mathbb{G}_m$$

corresponds to a ray in $\mathbb{R}^2$. It has no fixed point because the $\mathbb{G}_m$-factor acts freely in its own direction.

2.5.4 Smoothness

The singularities of U_σ are controlled by how the primitive ray generators of σ sit in the lattice.

Definition 2.43. A rational polyhedral cone

$$\sigma \subseteq N_\mathbb{R}$$

is *smooth* if its primitive ray generators form part of a basis of N.

For a full-dimensional cone, smoothness means that its primitive ray generators themselves form a basis of N.

Theorem 2.44 (Smoothness criterion)**.** *Let*

$$\sigma \subseteq N_\mathbb{R}$$

be a strongly convex rational polyhedral cone. Then

$$U_\sigma \text{ is smooth} \iff \sigma \text{ is smooth.}$$

Consequently, every smooth affine toric variety is isomorphic to

$$\mathbb{A}^r \times \mathbb{G}_m^{n-r}$$

for suitable $r \leq n$.

Reference. This is Cox–Little–Schenck, Theorem 1.3.12, together with the product decomposition in Equation (1.3.2).

Conceptual proof sketch. If the primitive ray generators extend to a basis of N, choose that basis. In the dual basis,

$$\sigma^\vee \cap M \simeq \mathbb{N}^r \oplus \mathbb{Z}^{n-r}.$$

Hence

$$U_\sigma \simeq \mathbb{A}^r \times \mathbb{G}_m^{n-r},$$

which is smooth.

For the converse, first split off the torus factor and reduce to a full-dimensional cone. Its distinguished fixed point has maximal ideal generated by the monomials corresponding to the Hilbert basis of $\sigma^\vee \cap M$. Modulo the square of that ideal, those generators form a basis of the Zariski cotangent space. Smoothness forces the tangent-space dimension to equal $\dim U_\sigma = n$.

But every ray of $\sigma^\vee$ contributes at least one Hilbert basis element, and a full-dimensional cone has at least n rays. Equality of dimensions forces exactly n rays and no additional Hilbert basis elements. Those primitive ray generators must then form a lattice basis. Dualizing shows that σ is smooth.

The complete proof carefully identifies the tangent space and handles the reduction from a non-full-dimensional cone. □

Example 2.45. The first quadrant

$$\mathrm{Cone}((1,0),(0,1))$$

is smooth, and its affine toric variety is $\mathbb{A}^2$.

Example 2.46. The cone

$$\sigma = \mathrm{Cone}((1,0),(1,2))$$

is not smooth because

$$\det \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = 2.$$

Its primitive ray generators span a sublattice of index 2, rather than a basis of $\mathbb{Z}^2$. Correspondingly,

$$U_\sigma \simeq V(AC - B^2)$$

is singular at the origin.

Surface laboratory

For a two-dimensional full-dimensional cone

$$\sigma = \mathrm{Cone}(u_1, u_2),$$

smoothness is equivalent to

$$|\det(u_1, u_2)| = 1.$$

Thus the determinant measures how far the two primitive ray generators are from being a lattice basis.
The value 1 gives a smooth affine plane chart. A value larger than 1 gives a toric surface singularity.

Pause and think

Determine which of the following cones in $\mathbb{Z}^2$ are smooth:

$$\mathrm{Cone}((1,0),(0,1)), \qquad \mathrm{Cone}((1,0),(1,3)), \qquad \mathrm{Cone}((1,1),(-1,0)).$$

For each smooth cone, write the associated affine toric variety as $\mathbb{A}^r \times \mathbb{G}_m^{2-r}$.

2.5.5 Toric morphisms

The morphisms compatible with torus actions are also controlled by lattice maps.

Definition 2.47. Let

$$X_{S_1} = \operatorname{Spec} \Bbbk[S_1], \qquad X_{S_2} = \operatorname{Spec} \Bbbk[S_2].$$

A morphism

$$\phi : X_{S_1} \longrightarrow X_{S_2}$$

is *toric* if its pullback

$$\phi^* : \Bbbk[S_2] \longrightarrow \Bbbk[S_1]$$

is induced by a semigroup homomorphism

$$\widehat{\phi} : S_2 \longrightarrow S_1.$$

Reference. This is Cox–Little–Schenck, Definition 1.3.13.

Such a morphism restricts to a group homomorphism between the dense tori and is equivariant:

$$\phi(t \cdot p) = \phi(t) \cdot \phi(p).$$

Conversely, a morphism that maps the dense torus into the dense torus, restricts there to a group homomorphism, and respects the torus actions is toric.

Reference. These statements are Cox–Little–Schenck, Proposition 1.3.14.

For normal affine toric varieties, the criterion becomes particularly simple.

Proposition 2.48 (Cone criterion for toric morphisms)**.** *Let*

$$\sigma_i \subseteq (N_i)_{\mathbb{R}}, \qquad i = 1, 2,$$

be strongly convex rational polyhedral cones, and let

$$F : N_1 \longrightarrow N_2$$

be a lattice homomorphism. The induced torus homomorphism

$$T_{N_1} \longrightarrow T_{N_2}$$

extends to a toric morphism

$$U_{\sigma_1} \longrightarrow U_{\sigma_2}$$

if and only if

$$F_{\mathbb{R}}(\sigma_1) \subseteq \sigma_2.$$

Reference. This is Cox–Little–Schenck, Proposition 1.3.15.

Proof. Let

$$F^* : M_2 \longrightarrow M_1$$

be the dual map. The torus homomorphism extends precisely when the pullback of every regular character on U_{σ_2} remains regular on U_{σ_1}:

$$F^*(\sigma_2^\vee \cap M_2) \subseteq \sigma_1^\vee \cap M_1.$$

Suppose first that

$$F_{\mathbb{R}}(\sigma_1) \subseteq \sigma_2.$$

For $m \in \sigma_2^\vee$ and $u \in \sigma_1$,

$$\langle F^*(m), u\rangle = \langle m, F(u)\rangle \geq 0.$$

Thus

$$F^*(m) \in \sigma_1^\vee.$$

Conversely, suppose the pullback condition holds. By rationality, the lattice points of $\sigma_2^\vee$ generate the cone. Hence for every $u \in \sigma_1$ and every $m \in \sigma_2^\vee$,

$$\langle m, F(u)\rangle = \langle F^*(m), u\rangle \geq 0.$$

Biduality gives

$$F(u) \in (\sigma_2^\vee)^\vee = \sigma_2.$$

□

Example 2.49. Let

$$F : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (a, b) \longmapsto a + b.$$

The first quadrant maps into the ray

$$\mathbb{R}_{\geq 0} \subseteq \mathbb{R}.$$

Therefore F induces the toric morphism

$$\mathbb{A}^2 \longrightarrow \mathbb{A}^1, \qquad (x, y) \longmapsto xy.$$

Indeed, the target character z pulls back to xy.

Prediction

Suppose two cones share a common face. The identity lattice map sends that face into each cone. What morphisms should this produce between the corresponding affine toric varieties, and what kind of subsets should their images be?

2.5.6 Faces give affine open subsets

Let

$$\tau \preceq \sigma$$

be a face. There is a lattice point

$$m \in \sigma^\vee \cap M$$

such that

$$\tau = \sigma \cap m^\perp = \{u \in \sigma : \langle m, u\rangle = 0\}.$$

The character χ^m is nonnegative along all directions of σ, and it vanishes in order precisely zero along the directions of τ.

Proposition 2.50 (Face localization)**.** *With notation as above,*

$$\Bbbk[\tau^\vee \cap M] = \Bbbk[\sigma^\vee \cap M]_{\chi^m}.$$

Consequently,

$$U_\tau \simeq (U_\sigma)_{\chi^m} = D(\chi^m) \subseteq U_\sigma.$$

Thus every face determines a torus-invariant affine open subset.

Reference. This is Cox–Little–Schenck, Proposition 1.3.16 and Equation (1.3.4).

Conceptual proof sketch. Localizing at χ^m allows arbitrary negative powers of that character. On exponent vectors, this replaces the semigroup

$$\sigma^\vee \cap M$$

by

$$(\sigma^\vee \cap M) + \mathbb{Z}(-m).$$

The defining property

$$\tau = \sigma \cap m^\perp$$

implies that this enlarged semigroup is exactly

$$\tau^\vee \cap M.$$

Taking spectra identifies the localization with the principal open subset where $\chi^m \neq 0$.

The only substantive verification is the equality of the two semigroups; geometrically, subtracting multiples of m removes the inequality transverse to the face while retaining all inequalities along τ. □

Example 2.51. Let

$$\sigma = \mathrm{Cone}(e_1, e_2) \subseteq \mathbb{R}^2, \qquad U_\sigma = \mathbb{A}^2.$$

For the face

$$\tau = \mathrm{Cone}(e_1),$$

take

$$m = e_2^\vee.$$

Then

$$\chi^m = y,$$

and

$$U_\tau = D(y) = \operatorname{Spec} \Bbbk[x, y^{\pm 1}] \simeq \mathbb{A}^1 \times \mathbb{G}_m.$$

Example 2.52. For the zero face

$$\tau = \{0\},$$

we may take

$$m = e_1^\vee + e_2^\vee.$$

Then

$$D(\chi^m) = D(xy) = \mathbb{G}_m^2,$$

the dense torus of $\mathbb{A}^2$.

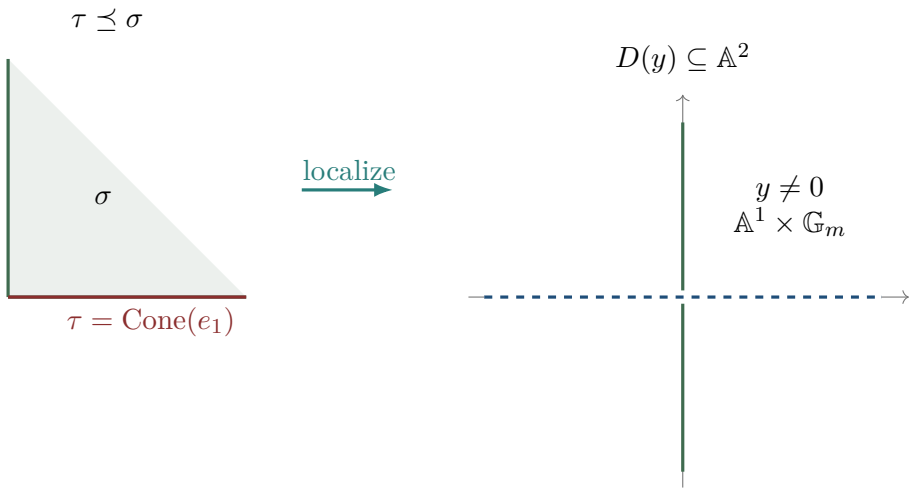


Figure 2.5: The face $\tau = \mathrm{Cone}(e_1)$ gives the affine open subset $U_\tau = D(y) \subseteq U_\sigma = \mathbb{A}^2$.

Faces are therefore the combinatorial instructions for passing to torus-invariant affine open subsets. This is exactly the mechanism needed to glue affine toric varieties: when two cones meet in a common face, the corresponding affine varieties contain a common affine open subset.

The three dictionaries

The main properties of U_σ can now be read in three languages.

Algebra	Geometry	Combinatorics
semigroup maps $\sigma^\vee \cap M \to \Bbbk$	points of U_σ	multiplicative assignments to lattice points
units $c\chi^m$	torus factors	$m \in \sigma^\perp \cap M$
the maximal monomial ideal	the unique fixed point	σ is full-dimensional
a polynomial–Laurent coordinate ring	smoothness	primitive ray generators extend to a basis
pullback of characters	toric morphisms	$F_{\mathbb{R}}(\sigma_1) \subseteq \sigma_2$
localization at χ^m	a torus-invariant affine open subset	a face $\tau = \sigma \cap m^\perp$

Guiding question

How can we assemble the constructions and properties of this chapter into one classification dictionary, and what do the standard examples look like in all three languages?

2.6 The classification dictionary

We can now assemble the chapter into one statement. Fix an algebraic torus T, with lattices

$$M = X^*(T), \qquad N = X_*(T).$$

Theorem 2.53 (Classification of normal affine toric varieties)**.** *The following data determine one another:*

(a) a normal affine toric variety X with dense torus T;

(b) a saturated affine semigroup $S \subseteq M$ satisfying

$$\mathbb{Z}S = M;$$

(c) a full-dimensional rational polyhedral cone

$$C \subseteq M_{\mathbb{R}};$$

(d) a strongly convex rational polyhedral cone

$$\sigma \subseteq N_{\mathbb{R}}.$$

The correspondences are

$$S = C \cap M, \qquad C = \sigma^\vee, \qquad \sigma = C^\vee,$$

and

$$X \simeq \operatorname{Spec} \Bbbk[S] = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].$$

Reference. This combines Cox–Little–Schenck, Theorems 1.2.18 and 1.3.5.

Conceptual proof sketch. Starting with σ, Gordan's lemma makes

$$S = \sigma^\vee \cap M$$

finitely generated. It is saturated because it contains every lattice point of its cone, so $\Bbbk[S]$ is normal. Strong convexity ensures that S generates M, and therefore

$$U_\sigma = \operatorname{Spec} \Bbbk[S]$$

has dense torus T.

Conversely, the regular characters on a normal affine toric variety form a saturated affine semigroup $S \subseteq M$. Hence

$$S = \operatorname{Cone}(S) \cap M.$$

Taking

$$C = \operatorname{Cone}(S), \qquad \sigma = C^\vee$$

recovers the variety. Biduality shows that no information is lost. □

Warning

Without normality, the cone does not determine the variety: it forgets the holes of the semigroup. It determines only the normalization

$$\operatorname{Spec} \Bbbk[\operatorname{Cone}(S) \cap M_S].$$

What this chapter has accomplished

Starting with the affine plane, we followed the chain

$$\text{extending characters} \longrightarrow S \subseteq M \longrightarrow \Bbbk[S] \longrightarrow \operatorname{Spec} \Bbbk[S].$$

For a general affine toric variety, the semigroup S is the discrete data that remembers every monomial allowed in the coordinate ring.

Normality removes the holes:

$$S \text{ saturated} \iff S = \operatorname{Cone}(S) \cap M_S \iff \Bbbk[S] \text{ normal.}$$

At that point the cone contains all the information. Passing to the dual lattice gives the standard formula

$$\boxed{U_\sigma = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].}$$

The formula has two complementary readings:

$$\boxed{\begin{array}{c} m \in \sigma^\vee \cap M \iff \chi^m \text{ extends regularly to } U_\sigma, \\ u \in \sigma \cap N \iff \lambda^u \text{ extends across } t = 0. \end{array}}$$

Equivalently, the dual cone records regular functions, while the original cone records one-parameter motions whose limits exist.

The cone also reads the basic geometry of U_σ: orthogonal directions give torus factors, full dimension gives a fixed point, lattice bases give smoothness, linear maps give toric morphisms, and faces give affine open subsets.

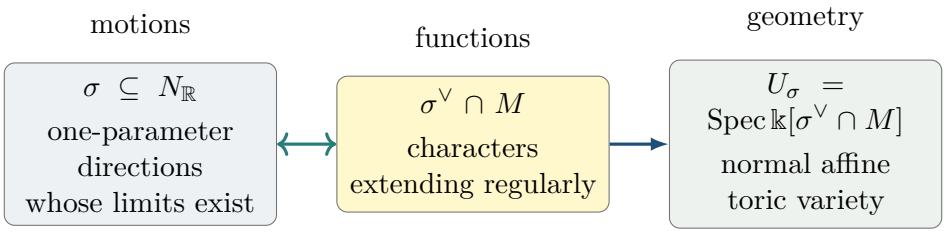


Figure 2.6: The classification dictionary for normal affine toric varieties.

Prediction

A single cone produces one affine toric variety. Suppose several cones meet along common faces. Since faces give common affine open subsets, how might the varieties U_σ be glued to produce nonaffine toric varieties?

Guiding question

How can compatible cones be glued along their common faces to produce general normal toric varieties?

Comprehension exercises

2.1. Let

$$S = \mathbb{N}(2,0) + \mathbb{N}(0,1) + \mathbb{N}(1,1) \subseteq \mathbb{Z}^2.$$

List all points of S of total degree at most 4, where total degree means the sum of the coordinates. Compute $\mathbb{Z}S$ and $\mathrm{Cone}(S)$.

2.2. For each of the following affine semigroups, write down the semigroup algebra $\Bbbk[S]$ in familiar coordinates and identify $\operatorname{Spec}\Bbbk[S]$:

$$\mathbb{N}^2, \qquad \mathbb{Z}_{\geq 0} \times \mathbb{Z}, \qquad \mathbb{Z}^2.$$

2.3. Let

$$\mathcal{A} = \{(1,0),(1,1),(0,1)\} \subseteq \mathbb{Z}^2.$$

Write the monomial map

$$\Phi_{\mathcal{A}} : \mathbb{G}_m^2 \longrightarrow \mathbb{A}^3.$$

Find a binomial equation for the closure of its image and identify the resulting affine toric variety.

2.4. Let

$$S = \mathbb{N}\{2,3\} \subseteq \mathbb{Z}.$$

Show that S is not saturated. Compute $\mathrm{Cone}(S) \cap \mathbb{Z}$, the normalization of $\Bbbk[S]$, and the normalization map of $\operatorname{Spec}\Bbbk[S]$.

2.5. Let

$$S = \mathbb{N}(2,0) + \mathbb{N}(0,1) + \mathbb{N}(1,1) \subseteq \mathbb{Z}^2.$$

Show that

$$S \subsetneq \mathrm{Cone}(S) \cap \mathbb{Z}^2.$$

Find a point witnessing the failure of saturation, and compute the normalization of $\Bbbk[S]$.

2.6. Compute the dual cones of the following cones in $\mathbb{R}^2$:

$$\mathrm{Cone}((1,0)), \qquad \mathrm{Cone}((1,0),(0,1)), \qquad \mathrm{Cone}((1,0),(1,2)).$$

For each cone σ, describe the semigroup

$$\sigma^\vee \cap M.$$

2.7. Let

$$\sigma = \mathrm{Cone}((1,0),(0,1)) \subseteq N_{\mathbb{R}} \simeq \mathbb{R}^2.$$

For each lattice point

$$u = (a,b) \in N,$$

determine whether the limit

$$\lim_{t \to 0} \lambda^u(t)$$

exists in $U_\sigma = \mathbb{A}^2$. When it exists, determine the limit point.

2.8. Let

$$\sigma = \mathrm{Cone}((1,0),(1,2)).$$

Using the Hilbert basis of $\sigma^\vee \cap M$, write the coordinate ring of U_σ as a quotient of a polynomial ring. Show that U_σ is normal but singular.

2.9. For each of the following cones in $\mathbb{Z}^2$, determine whether the associated affine toric variety is smooth:

$$\mathrm{Cone}((1,0),(0,1)), \qquad \mathrm{Cone}((1,0),(1,3)), \qquad \mathrm{Cone}((1,1),(-1,0)).$$

In the smooth cases, identify the variety as

$$\mathbb{A}^2, \qquad \mathbb{A}^1 \times \mathbb{G}_m, \qquad \text{or } \mathbb{G}_m^2.$$

2.10. Let

$$F : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (a,b) \longmapsto a+b.$$

Show that

$$F_{\mathbb{R}}\big(\mathrm{Cone}((1,0),(0,1))\big) \subseteq \mathbb{R}_{\geq 0},$$

and use the cone criterion to produce a toric morphism

$$\mathbb{A}^2 \longrightarrow \mathbb{A}^1.$$

Write the induced map on coordinate rings and the corresponding map of varieties.

2.11. Let

$$\sigma = \mathrm{Cone}(e_1, e_2) \subseteq \mathbb{R}^2, \qquad \tau = \mathrm{Cone}(e_1).$$

Choose a lattice point $m \in \sigma^\vee \cap M$ such that

$$\tau = \sigma \cap m^\perp.$$

Then verify directly that

$$U_\tau \simeq D(\chi^m) \subseteq U_\sigma.$$

Challenge exercise

Cyclic quotient orbifolds and their coarse toric surfaces.

Assume that the characteristic of $\Bbbk$ does not divide n. Choose an integer q satisfying

$$1 \leq q < n, \qquad \gcd(q, n) = 1,$$

and let μ_n act on $\mathbb{A}^2$ by

$$\zeta \cdot (x, y) = (\zeta x, \zeta^q y).$$

We write

$$\left[\mathbb{A}^2/\mu_n\right]$$

for the resulting smooth toric orbifold, and

$$X = \mathbb{A}^2/\mu_n = \operatorname{Spec} \Bbbk[x, y]^{\mu_n}$$

for its coarse moduli space.

No general theory of stacks is needed for this exercise. Think of $[\mathbb{A}^2/\mu_n]$ as the affine plane together with its finite symmetry, and of X as the affine variety whose regular functions are precisely the invariant polynomials.

The goal is to see that the orbifold and its coarse space use the same real cone but different lattices. The ordinary coordinates x, y will become *orbifold pseudo-coordinates*: they are honest coordinates upstairs, but usually not regular functions on the coarse space.

Let

$$\widetilde{T} = \mathbb{G}_m^2, \qquad \widetilde{M} = X^*(\widetilde{T}) \simeq \mathbb{Z}^2, \qquad \widetilde{N} = X_*(\widetilde{T}) \simeq \mathbb{Z}^2,$$

and let

$$\sigma = \operatorname{Cone}(e_1, e_2) \subseteq \widetilde{N}_{\mathbb{R}}.$$

Thus

$$\mathbb{A}^2 = U_{\sigma, \widetilde{N}}.$$

Hint 1. Show that the inclusion

$$\mu_n \hookrightarrow \widetilde{T}, \qquad \zeta \longmapsto (\zeta, \zeta^q),$$

recovers the given action on $\mathbb{A}^2$. Explain why the coordinate axes and the dense torus are preserved.

Hint 2. A character

$$\chi^{(a,b)} = x^a y^b$$

of $\widetilde{T}$ restricts to the character

$$\zeta \longmapsto \zeta^{a+qb}$$

of μ_n. Show that the characters descending to the quotient torus

$$T = \widetilde{T}/\mu_n$$

form the sublattice

$$M = \left\{(a,b) \in \widetilde{M} : a + qb \equiv 0 \pmod{n}\right\}.$$

Show that

$$[\widetilde{M} : M] = n.$$

Hint 3. Prove that

$$\Bbbk[x,y]^{\mu_n} = \bigoplus_{\substack{a,b\geq 0 \\ a+qb\equiv 0 \pmod{n}}} \Bbbk\, x^a y^b.$$

Deduce that

$$\Bbbk[x,y]^{\mu_n} = \Bbbk[\sigma^\vee \cap M].$$

Hint 4. Let

$$N = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z}).$$

Dualizing the finite-index inclusion

$$M \subseteq \widetilde{M}$$

produces a finite-index inclusion

$$\widetilde{N} \subseteq N.$$

Show that

$$N = \widetilde{N} + \mathbb{Z}\left(\frac{1}{n}(1,q)\right) \subseteq \widetilde{N}_{\mathbb{Q}}.$$

In particular,

$$[N : \widetilde{N}] = n.$$

Hint 5. Regard the same real cone

$$\sigma = \mathrm{Cone}(e_1, e_2)$$

inside

$$N_{\mathbb{R}} = \widetilde{N}_{\mathbb{R}}.$$

Show that the coarse moduli space is the affine toric variety

$$X = U_{\sigma,N} = \mathrm{Spec}\, \Bbbk[\sigma^\vee \cap M].$$

Explain why the inclusion of lattices

$$\widetilde{N} \hookrightarrow N$$

induces the quotient morphism

$$\mathbb{A}^2 = U_{\sigma,\widetilde{N}} \longrightarrow U_{\sigma,N} = X.$$

Hint 6. The coordinate functions x and y correspond to the characters

$$(1,0), (0,1) \in \widetilde{M}.$$

Determine when either of these characters belongs to M. Explain why x and y are generally not functions on the coarse space, even though invariant monomials in x and y are.
Interpret x, y as orbifold pseudo-coordinates.

Hint 7. Show that e_1, e_2 are primitive lattice vectors in N, but that they span the proper sublattice

$$\widetilde{N} \subseteq N$$

of index n. Conclude from the smoothness criterion that, for $n > 1$,

$$U_{\sigma,N}$$

is singular at its torus-fixed point.
Contrast this with the smooth orbifold

$$[\mathbb{A}^2/\mu_n].$$

The same real cone is smooth in the lattice $\widetilde{N}$, but not in the larger lattice N.

Hint 8. Specialize to

$$q = n - 1,$$

so that

$$\zeta \cdot (x, y) = (\zeta x, \zeta^{-1} y).$$

Show that every invariant monomial is generated by

$$u = x^n, \qquad v = y^n, \qquad w = xy.$$

Find the relation among u, v, w, and conclude that

$$X \simeq V(uv - w^n) \subseteq \mathbb{A}^3.$$

This is the A_{n-1} surface singularity.

Hint 9. Summarize the two toric descriptions:

$$\mathbb{A}^2 = U_{\sigma,\widetilde{N}}, \qquad \mathbb{A}^2/\mu_n = U_{\sigma,N}.$$

Explain in your own words how changing the lattice while keeping the same real cone turns genuine coordinates upstairs into orbifold pseudo-coordinates and creates a singular coarse moduli space.

Notes and references

This chapter follows the first half of Cox–Little–Schenck, Chapters 1 and 3, with the emphasis shifted toward the three dictionaries:

$$\text{algebra}, \qquad \text{geometry}, \qquad \text{combinatorics}.$$

For the general theory of affine semigroups, Hilbert bases, and normality, standard complementary references include Bruns–Gubeladze, *Polytopes, Rings, and K-Theory*, and Miller–Sturmfels, *Combinatorial Commutative Algebra*.

From the toric viewpoint, the central result is that normal affine toric varieties are classified by strongly convex rational polyhedral cones. The nonnormal examples in this chapter are included to show exactly where cones stop being enough and semigroups must take over.

The next chapter globalizes this picture. Faces already provide common affine open subsets, so compatible cones can be glued. This leads from single cones to fans, and from affine toric varieties to general normal toric varieties.

Chapter 3

Fans and Toric Varieties: Gluing Affine Pieces

Chapter overview

A single strongly convex rational cone

$$\sigma \subseteq N_{\mathbb{R}}$$

produces one normal affine toric variety

$$U_\sigma = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M].$$

But many of the toric varieties we care about most—projective spaces, Hirzebruch surfaces, and blowups—are not affine. They cannot be described by one cone.
The solution is local-to-global. Several cones will provide several affine toric charts, and common faces will provide the affine open subsets along which those charts are glued. A compatible collection of cones is called a *fan*.
This chapter develops the passage

$$\begin{array}{c} \text{cones and common faces} \\ \downarrow \\ \text{affine toric charts and common open subsets} \\ \downarrow \\ \text{a global toric variety } X_\Sigma. \end{array}$$

The running examples will be toric curves and surfaces, especially $\mathbb{P}^1$, $\mathbb{P}^2$, products, and blowups.

The three dictionaries

The global theory adds one new operation to each dictionary.

Algebra	**Geometry**	**Combinatorics**
semigroup algebras $\Bbbk[\sigma^\vee \cap M]$	affine charts U_σ	cones $\sigma \subseteq N_{\mathbb{R}}$
localization at characters	common affine open subsets	common faces
compatible restriction maps	gluing affine charts	a fan Σ

3.1 Why one cone is not enough

Every cone gives an affine variety. This is a strength of the construction

$$\sigma \longmapsto U_\sigma = \operatorname{Spec} \Bbbk[\sigma^\vee \cap M],$$

but it is also a limitation. Neither $\mathbb{P}^1$ nor $\mathbb{P}^2$ is affine, so neither can come from a single cone.

They are, however, covered by affine toric varieties. The question is therefore not how to replace cones, but how to use several of them at once.

Guiding question

How can several cones encode affine toric charts together with the instructions for gluing those charts?

3.1.1 The affine construction is a local model

Let

$$\tau \preceq \sigma$$

be a face. Chapter 2 showed that

$$U_\tau \subseteq U_\sigma$$

is a torus-invariant affine open subset. More precisely, if

$$\tau = \sigma \cap m^\perp \qquad \text{for } m \in \sigma^\vee \cap M,$$

then

$$U_\tau = D(\chi^m) \subseteq U_\sigma.$$

Thus a face already carries gluing information:

a face of a cone $\longleftrightarrow$ an affine open subset of the corresponding chart.

Suppose now that two cones $\sigma_1, \sigma_2 \subseteq N_\mathbb{R}$ meet in a cone

$$\tau = \sigma_1 \cap \sigma_2$$

that is a face of both. Then the same affine toric variety U_τ appears as an open subset of both:

$$U_{\sigma_1} \supseteq U_\tau \subseteq U_{\sigma_2}.$$

The identity on the common dense torus identifies these two copies of U_τ. This gives exactly the kind of overlap needed to glue the two affine charts.

We first watch this happen in dimension one.

3.1.2 $\mathbb{P}^1$ from two affine lines

Take

$$N = \mathbb{Z}, \qquad M = \mathbb{Z},$$

and let $t = \chi^1$ be the coordinate on the torus

$$T = \mathbb{G}_m.$$

Consider the two rays

$$\sigma_+ = \mathbb{R}_{\geq 0}, \qquad \sigma_- = \mathbb{R}_{\leq 0},$$

whose intersection is the zero cone

$$\tau = \{0\}.$$

Their dual semigroups are

$$\sigma_+^\vee \cap M = \mathbb{N}, \qquad \sigma_-^\vee \cap M = -\mathbb{N}, \qquad \tau^\vee \cap M = \mathbb{Z}.$$

Consequently,

$$U_{\sigma_+} = \operatorname{Spec} \Bbbk[t] \simeq \mathbb{A}^1,$$
$$U_{\sigma_-} = \operatorname{Spec} \Bbbk[t^{-1}] \simeq \mathbb{A}^1,$$

and

$$U_\tau = \operatorname{Spec} \Bbbk[t, t^{-1}] = \mathbb{G}_m.$$

The face inclusions identify U_τ with

$$D(t) \subseteq U_{\sigma_+}$$

and

$$D(t^{-1}) \subseteq U_{\sigma_-}.$$

If we write

$$x = t \qquad \text{on } U_{\sigma_+},$$

and

$$y = t^{-1} \qquad \text{on } U_{\sigma_-},$$

then the transition function on the overlap is

$$y = x^{-1}.$$

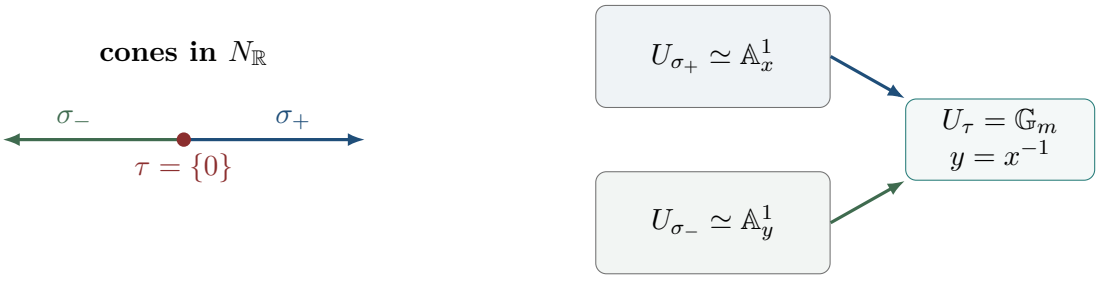


Figure 3.1: The two rays give affine-line charts, while their common zero face gives the torus on which the charts overlap.

Gluing these two affine lines along $\mathbb{G}_m$ using

$$y = x^{-1}$$

produces $\mathbb{P}^1$. Indeed, the maps

$$\mathbb{A}^1_x \longrightarrow \mathbb{P}^1, \qquad x \longmapsto [1 : x],$$

and

$$\mathbb{A}^1_y \longrightarrow \mathbb{P}^1, \qquad y \longmapsto [y : 1],$$

agree on the overlap precisely because $y = x^{-1}$.

Reference. This is the one-dimensional case of Cox–Little–Schenck, Examples 3.1.10 and 3.1.11. The general affine gluing model for projective space is developed in §3.0.

Pause and think

The point $0 \in \mathbb{A}^1_x$ becomes $[1 : 0]$, while $0 \in \mathbb{A}^1_y$ becomes $[0 : 1]$. Why are these two points not part of the overlap? Explain how the two origins supply the two boundary points of

$$\mathbb{P}^1 \setminus \mathbb{G}_m.$$

This first example contains the global construction in miniature:

$$\begin{array}{ccc} \sigma_+, \sigma_- & \longleftrightarrow & \mathbb{A}^1_x, \mathbb{A}^1_y \\ \sigma_+ \cap \sigma_- = \{0\} & \longleftrightarrow & \mathbb{A}^1_x \cap \mathbb{A}^1_y = \mathbb{G}_m. \end{array}$$

3.1.3 Common faces are the transition charts

The gluing in the previous example was not chosen independently of the cones. It was forced by the shared face.

For a face

$$\tau \preceq \sigma,$$

the inclusion

$$U_\tau \hookrightarrow U_\sigma$$

restricts to the identity on the dense torus T. Hence whenever

$$\tau = \sigma_1 \cap \sigma_2$$

is a face of both cones, the diagram

$$\begin{array}{ccc} & U_\tau & \\ \swarrow & & \searrow \\ U_{\sigma_1} & & U_{\sigma_2} \end{array}$$

provides canonical gluing data.

On coordinate rings, these open embeddings come from localizations. If

$$\tau = \sigma_1 \cap m^\perp = \sigma_2 \cap (-m)^\perp$$

for a suitable $m \in M$, then

$$\Bbbk[U_\tau] = \Bbbk[U_{\sigma_1}]_{\chi^m} = \Bbbk[U_{\sigma_2}]_{\chi^{-m}}.$$

The transition map is the identity inside the common Laurent polynomial ring $\Bbbk[M]$.

Prediction

Suppose a finite collection of cones is meant to describe affine charts of one global toric variety.
What conditions should be imposed so that every boundary piece needed for an overlap is present, and so that the intersection of any two charts is represented by one common face rather than by incompatible pieces?

3.1.4 $\mathbb{P}^2$ from three affine planes

Let

$$N = \mathbb{Z}^2, \qquad e_0 = -e_1 - e_2,$$

and consider

$$\sigma_0 = \mathrm{Cone}(e_1, e_2), \qquad \sigma_1 = \mathrm{Cone}(e_0, e_2), \qquad \sigma_2 = \mathrm{Cone}(e_1, e_0).$$

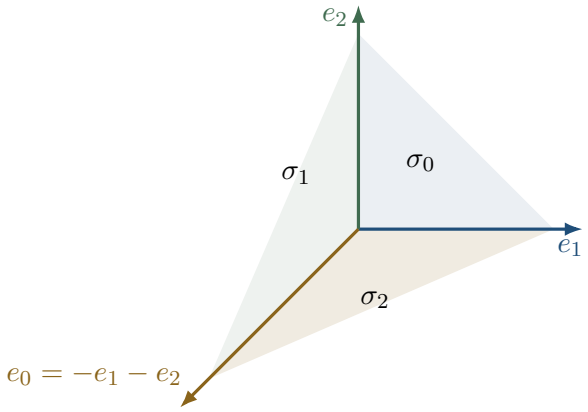


Figure 3.2: The three maximal cones for $\mathbb{P}^2$. Their rays and the zero cone encode all chart intersections.

Let m_1, m_2 be dual to e_1, e_2, and write

$$x = \chi^{m_1}, \qquad y = \chi^{m_2}.$$

The complete affine atlas is contained in the following table.

Cone or common face	**Affine toric piece**	**Projective interpretation**
$\sigma_0 = \mathrm{Cone}(e_1, e_2)$	$\Bbbk[x, y]$, hence $U_{\sigma_0} \simeq \mathbb{A}^2$	$V_0 = \{z_0 \neq 0\}$; $(x, y) = (z_1/z_0, z_2/z_0)$
$\sigma_1 = \mathrm{Cone}(e_0, e_2)$	$\Bbbk[x^{-1}, x^{-1}y]$, hence $U_{\sigma_1} \simeq \mathbb{A}^2$	$V_1 = \{z_1 \neq 0\}$; $(x^{-1}, x^{-1}y) = (z_0/z_1, z_2/z_1)$
$\sigma_2 = \mathrm{Cone}(e_1, e_0)$	$\Bbbk[y^{-1}, xy^{-1}]$, hence $U_{\sigma_2} \simeq \mathbb{A}^2$	$V_2 = \{z_2 \neq 0\}$; $(y^{-1}, xy^{-1}) = (z_0/z_2, z_1/z_2)$
$\sigma_0 \cap \sigma_1 = \mathrm{Cone}(e_2)$	$\Bbbk[x^{\pm 1}, y] \simeq \Bbbk[x^{-1}, x^{-1}y]_{x^{-1}}$	$V_0 \cap V_1$; $(x, y) \mapsto (x^{-1}, y/x)$
$\sigma_0 \cap \sigma_2 = \mathrm{Cone}(e_1)$	$\Bbbk[x, y^{\pm 1}]$	$V_0 \cap V_2$; $(x, y) \mapsto (y^{-1}, x/y)$
$\sigma_1 \cap \sigma_2 = \mathrm{Cone}(e_0)$	$\Bbbk[a, b^{\pm 1}]$, with $a = x^{-1}$, $b = x^{-1}y$	$V_1 \cap V_2$; $(a, b) \mapsto (a/b, 1/b)$
$\sigma_0 \cap \sigma_1 \cap \sigma_2 = \{0\}$	$\Bbbk[x^{\pm 1}, y^{\pm 1}]$, so $U_{\{0\}} = \mathbb{G}_m^2$	$V_0 \cap V_1 \cap V_2$, the dense torus

Reference. This is Cox–Little–Schenck, Example 3.1.9. Its n-dimensional generalization is Example 3.1.10.

Surface laboratory

Read the atlas directly from the table.

(a) Verify that each maximal-cone ring is a polynomial ring in two variables.

(b) For each pair of charts, identify which character is inverted and derive the displayed transition map from the two coordinate systems.

(c) Explain why every pairwise overlap is $\mathbb{A}^1 \times \mathbb{G}_m$, while the triple overlap is $\mathbb{G}_m^2$.

(d) What two compatibility rules on a collection of cones are suggested by this atlas? Your rules should ensure that all needed overlap charts are present and that the intersection of any two maximal charts is represented by one common cone.

The examples of $\mathbb{P}^1$ and $\mathbb{P}^2$ therefore lead directly to the next definition: a compatible collection of cones should contain all faces and should have the property that the intersection of any two cones is a face of each.

Guiding question

How should we define such a collection precisely, and which features of a global toric variety can already be read from the arrangement of its cones?

3.2 Fans and their basic language

The examples of $\mathbb{P}^1$ and $\mathbb{P}^2$ forced two compatibility conditions: faces must be retained, and intersections must be common faces. We now make those conditions precise.

3.2.1 The definition

Definition 3.1 (Fan)**.** A *fan* Σ in $N_{\mathbb{R}}$ is a finite collection of cones such that:

(a) every $\sigma \in \Sigma$ is a strongly convex rational polyhedral cone;

(b) if $\sigma \in \Sigma$ and $\tau \preceq \sigma$, then $\tau \in \Sigma$;

(c) if $\sigma_1, \sigma_2 \in \Sigma$, then

$$\sigma_1 \cap \sigma_2$$

is a face of both σ_1 and σ_2.

Reference. This is Cox–Little–Schenck, Definition 3.1.2.

The zero cone belongs to every nonempty fan, because it is a face of every strongly convex cone.

The first condition ensures that every cone produces a normal affine toric variety. The second ensures that every torus-invariant affine open subset needed for gluing is represented. The third ensures that two affine charts have one canonical common chart:

$$\sigma_1 \cap \sigma_2 = \tau \quad \rightsquigarrow \quad U_{\sigma_1} \supseteq U_\tau \subseteq U_{\sigma_2}.$$

Example 3.2 (The affine fan of a cone)**.** Let $\sigma \subseteq N_{\mathbb{R}}$ be a strongly convex rational polyhedral cone. Then

$$\Sigma_\sigma = \{\tau : \tau \preceq \sigma\}$$

is a fan. It consists of σ together with all of its faces.

The toric variety associated to this fan will be the affine variety U_σ.

3.2.2 What can go wrong?

Each part of theorem 3.1 is necessary.

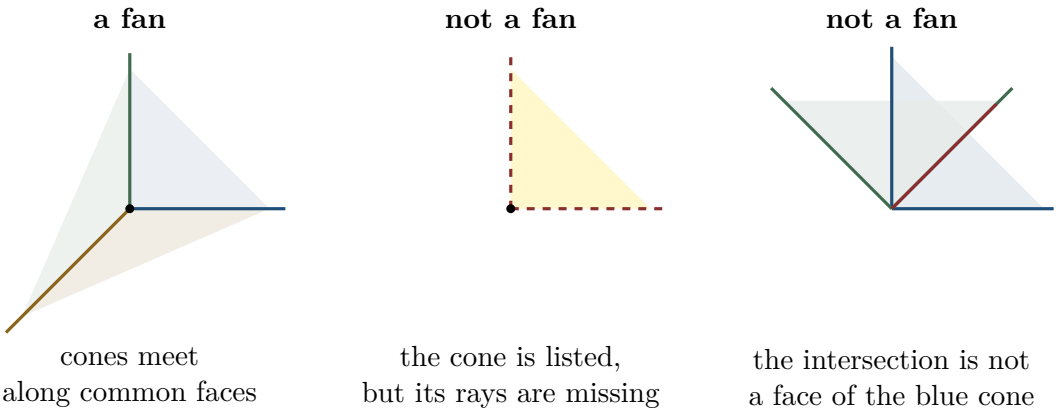


Figure 3.3: The face and intersection conditions distinguish fans from arbitrary collections of cones.

A cone containing a line also fails the definition: it is not strongly convex, and the associated affine construction does not contain the chosen torus as its dense torus in the required way.

Pause and think

Let

$$\sigma_1 = \mathrm{Cone}((1,0),(0,1))$$

and

$$\sigma_2 = \mathrm{Cone}((1,1),(-1,1)).$$

Compute $\sigma_1 \cap \sigma_2$. Why is it not a face of σ_1? Which new ray would have to be inserted to subdivide σ_1 so that the two regions could become part of one fan?

3.2.3 Support and cones of each dimension

Definition 3.3. Let Σ be a fan in $N_{\mathbb{R}}$.

Its *support* is

$$|\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma \subseteq N_{\mathbb{R}}.$$

For $r \geq 0$, we write

$$\Sigma(r) = \{\sigma \in \Sigma : \dim \sigma = r\}.$$

Reference. See Cox–Little–Schenck, immediately after Definition 3.1.2.

The set

$$\Sigma(1)$$

is the set of rays of the fan. For each ray $\rho \in \Sigma(1)$, we write

$$u_\rho \in N$$

for its primitive lattice generator.

A cone $\sigma \in \Sigma$ is *maximal* if it is not properly contained in another cone of Σ. We denote the set of maximal cones by

$$\Sigma_{\max}.$$

Because the fan is finite, every cone lies in a maximal cone. Since all faces of every maximal cone must be present, a fan is completely determined by its maximal cones.

Proposition 3.4 (Relative interiors partition the support)**.** *For every fan Σ,*

$$|\Sigma| = \bigsqcup_{\sigma \in \Sigma} \operatorname{Relint}(\sigma).$$

Equivalently, every point of the support lies in the relative interior of a unique cone of the fan.

Proof. Choose $u \in |\Sigma|$, and let $\sigma \in \Sigma$ contain u. Every point of a polyhedral cone lies in the relative interior of a unique face of that cone. Hence

$$u \in \operatorname{Relint}(\tau)$$

for a unique face $\tau \preceq \sigma$, and $\tau \in \Sigma$.

Suppose also that

$$u \in \operatorname{Relint}(\tau')$$

for another cone $\tau' \in \Sigma$. The intersection

$$\tau \cap \tau'$$

is a face of both. Since a relative-interior point cannot lie in a proper face, we must have

$$\tau = \tau \cap \tau' = \tau'.$$

□

Thus a fan is not merely a union of cones. It decomposes its support into nonoverlapping pieces:

$$\operatorname{Relint}(\sigma), \qquad \sigma \in \Sigma.$$

Observation 3.5. This partition is the combinatorial shadow of the orbit decomposition that will appear later:

$$\text{one cone} \quad \longleftrightarrow \quad \text{one torus orbit.}$$

For now, it simply tells us that a direction $u \in |\Sigma|$ belongs unambiguously to one cone at the level of relative interiors.

3.2.4 The dimension of a fan

We define

$$\dim \Sigma = \max_{\sigma \in \Sigma} \dim \sigma = \dim |\Sigma|.$$

The second equality holds because Σ is finite.

A fan is *pure of dimension* r if every maximal cone has dimension r. The fans of $\mathbb{P}^1$ and $\mathbb{P}^2$ are pure of dimensions 1 and 2, respectively.

Warning

The dimension of the fan need not equal the dimension of its toric variety.
The fan

$$\Sigma_{\mathrm{Cone}((1,0))} \subseteq \mathbb{R}^2$$

has dimension 1, but its affine toric variety is

$$\mathbb{A}^1 \times \mathbb{G}_m,$$

which has dimension 2. The variety always has dimension $\operatorname{rank} N$; a lower-dimensional fan signals a torus factor.

3.2.5 Subfans

Definition 3.6. A *subfan* of Σ is a subset

$$\Sigma' \subseteq \Sigma$$

that is itself a fan.

The phrase “itself a fan” matters. One may not remove a face while retaining a cone that contains it.

A convenient way to produce a subfan is to choose some cones and then include all of their faces. The intersection condition is inherited from Σ.

Example 3.7. Let $\Sigma_{\mathbb{P}^2}$ be the fan of $\mathbb{P}^2$, with maximal cones

$$\sigma_0, \sigma_1, \sigma_2.$$

The collection consisting of

$$\sigma_0, \sigma_1$$

and all of their faces is a subfan. Geometrically, it will correspond to the union of the two affine charts

$$V_0 \cup V_1 = \mathbb{P}^2 \setminus \{[0:0:1]\}.$$

This example previews a general rule:

$$\text{subfan} \quad \rightsquigarrow \quad \text{torus-invariant open subvariety.}$$

We will prove it after constructing X_Σ.

Pause and think

Starting with the fan of $\mathbb{P}^2$, decide whether each of the following collections is a subfan:

(a) one maximal cone and all of its faces;
(b) the three rays and the zero cone;
(c) the three maximal cones and the zero cone, but no rays;
(d) two maximal cones together with only their common ray and the zero cone.

3.2.6 Complete fans

Definition 3.8. A fan Σ in $N_\mathbb{R}$ is *complete* if

$$|\Sigma| = N_\mathbb{R}.$$

Reference. This is Cox–Little–Schenck, Definition 3.1.18(c).

The fans of $\mathbb{P}^1$ and $\mathbb{P}^2$ are complete: their cones cover the entire one-parameter subgroup space.

By contrast, the affine fan

$$\Sigma_\sigma = \{\text{faces of } \sigma\}$$

has support σ, so in positive dimension it is not complete.

Completeness has a direct interpretation through one-parameter subgroups. If Σ is complete, every vector

$$u \in N$$

lies in some cone $\sigma \in \Sigma$. Therefore

$$\lambda^u$$

has a limit in the affine chart U_σ. No one-parameter direction can escape all of the charts.

Prediction

For $\mathbb{P}^1$ and $\mathbb{P}^2$, completeness of the fan accompanies completeness of the variety. Formulate the general relationship you expect between

$$|\Sigma| = N_{\mathbb{R}}$$

and the geometric property that X_Σ has no points "at infinity" still missing.
Why might the existence of limits for every one-parameter subgroup be relevant?

The three dictionaries

The basic fan terminology anticipates global geometry.

Algebra	Geometry	Combinatorics
semigroup algebras for maximal cones	an affine open cover	$\Sigma_{\max}$
localizations of those algebras	chart intersections	common faces
restriction to selected charts	a torus-invariant open subvariety	a subfan
every one-parameter subgroup acquires a limit somewhere	no direction at infinity is missing	$\lvert\Sigma\rvert = N_{\mathbb{R}}$

Guiding question

Given a fan Σ, how do the affine varieties U_σ glue to form a normal separated toric variety X_Σ?

3.3 Gluing affine toric varieties

Let Σ be a fan in $N_{\mathbb{R}}$. Every cone

$$\sigma \in \Sigma$$

gives a normal affine toric variety

$$U_\sigma = \operatorname{Spec} \Bbbk[S_\sigma], \qquad S_\sigma = \sigma^\vee \cap M.$$

We now glue these affine pieces into one global variety.

The construction is canonical: the fan has already specified every overlap and every transition map.

3.3.1 The canonical overlap of two charts

Take

$$\sigma_1, \sigma_2 \in \Sigma, \qquad \tau = \sigma_1 \cap \sigma_2.$$

By the definition of a fan, τ is a face of both cones. Hence

$$U_{\sigma_1} \supseteq U_\tau \subseteq U_{\sigma_2}$$

are affine open embeddings.

More explicitly, one can choose

$$m \in \sigma_1^\vee \cap (-\sigma_2)^\vee \cap M$$

such that

$$\tau = \sigma_1 \cap m^\perp = \sigma_2 \cap m^\perp.$$

Then

$$U_\tau = (U_{\sigma_1})_{\chi^m} = (U_{\sigma_2})_{\chi^{-m}}.$$

Both copies of U_τ have the same coordinate ring

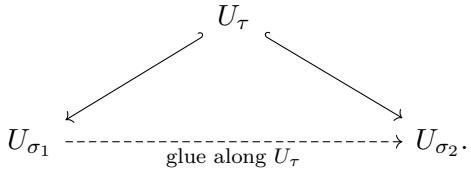
$$\Bbbk[S_\tau] \subseteq \Bbbk[M].$$

The transition map

$$g_{\sigma_2,\sigma_1} : (U_{\sigma_1})_{\chi^m} \longrightarrow (U_{\sigma_2})_{\chi^{-m}}$$

is therefore induced by the identity map on $\Bbbk[S_\tau]$. In particular, it restricts to the identity on the common dense torus

$$T = \operatorname{Spec} \Bbbk[M].$$



No extra coordinate change has to be invented. It is already encoded by viewing the same characters in two different affine coordinate rings.

3.3.2 The separation lemma

The following equality is stronger than merely saying that U_τ is open in both charts.

Proposition 3.9 (Separation lemma)**.** *Let*

$$\sigma_1, \sigma_2 \in \Sigma, \qquad \tau = \sigma_1 \cap \sigma_2.$$

Then

$$S_\tau = S_{\sigma_1} + S_{\sigma_2},$$

where the right-hand side is the Minkowski sum of semigroups.

Reference. This is Cox–Little–Schenck, Proposition 3.1.3.

Proof. Since

$$\sigma_i^\vee \subseteq \tau^\vee,$$

we immediately have

$$S_{\sigma_1} + S_{\sigma_2} \subseteq S_\tau.$$

For the reverse inclusion, choose

$$m \in \sigma_1^\vee \cap (-\sigma_2)^\vee \cap M$$

cutting out the common face τ. Let $p \in S_\tau$. Because $\tau \preceq \sigma_1$, the face-localization equality gives

$$S_\tau = S_{\sigma_1} + \mathbb{Z}(-m).$$

Thus

$$p = q + \ell(-m)$$

for some $q \in S_{\sigma_1}$ and some $\ell \in \mathbb{Z}$.

If $\ell \geq 0$, then

$$-m \in \sigma_2^\vee \cap M = S_{\sigma_2}$$

implies

$$\ell(-m) \in S_{\sigma_2}.$$

If $\ell < 0$, then

$$p = q + (-\ell)m \in S_{\sigma_1},$$

because $m \in S_{\sigma_1}$. In either case,

$$p \in S_{\sigma_1} + S_{\sigma_2}.$$

□

The lemma says that every character regular on the overlap can be written as a product of one character regular on the first chart and one regular on the second:

$$\chi^p = \chi^q \chi^r, \qquad q \in S_{\sigma_1}, \quad r \in S_{\sigma_2}.$$

This algebraic statement will be the key to separatedness.

3.3.3 The cocycle condition is automatic

Suppose

$$\sigma_1, \sigma_2, \sigma_3 \in \Sigma.$$

Their triple intersection

$$\omega = \sigma_1 \cap \sigma_2 \cap \sigma_3$$

is a common face, and

$$U_\omega$$

is an affine open subset of every pairwise overlap.

On U_ω, all transition maps are induced by the identity on

$$\Bbbk[S_\omega] \subseteq \Bbbk[M].$$

Consequently,

$$g_{\sigma_3,\sigma_2} \circ g_{\sigma_2,\sigma_1} = g_{\sigma_3,\sigma_1}$$

wherever all three maps are defined.

Thus the gluing maps satisfy the cocycle condition without an additional calculation: they are all restrictions of the identity on the dense torus and its Laurent coordinate ring.

3.3.4 Constructing X_Σ

Form the disjoint union

$$\bigsqcup_{\sigma\in\Sigma} U_\sigma.$$

Whenever

$$p \in U_{\sigma_1} \qquad \text{and} \qquad q \in U_{\sigma_2}$$

come from the same point of

$$U_{\sigma_1\cap\sigma_2},$$

identify p and q. The cocycle condition makes this an equivalence relation.

Definition 3.10. The variety obtained by gluing the affine toric varieties

$$\{U_\sigma : \sigma \in \Sigma\}$$

along the canonical common opens

$$U_{\sigma_1\cap\sigma_2}$$

is denoted

$$X_\Sigma.$$

It is called the *toric variety of the fan* Σ.

Every U_σ maps isomorphically onto an affine open subset of X_Σ, and

$$U_{\sigma_1} \cap U_{\sigma_2} = U_{\sigma_1\cap\sigma_2}.$$

Only maximal cones are needed to cover the result:

$$X_\Sigma = \bigcup_{\sigma\in\Sigma_{\max}} U_\sigma.$$

Indeed, every cone $\tau \in \Sigma$ is a face of some maximal cone σ, and then

$$U_\tau \subseteq U_\sigma.$$

Algebraic geometry cheatsheet

A variety is *separated* when its diagonal

$$\Delta_X : X \longrightarrow X \times X, \qquad p \longmapsto (p,p),$$

has closed image.

This is the algebraic-geometric replacement for the Hausdorff condition. Affine varieties are separated, but an arbitrary gluing of affine varieties need not be.
To prove that a variety assembled from affine charts is separated, it is enough to check that the diagonal is closed on every product of two charts.

3.3.5 The toric variety of a fan

We use the global version of the definition from Chapter 2.

Definition 3.11. A *toric variety* with torus T is an irreducible variety X containing T as a dense open subset such that the multiplication action of T on itself extends to an algebraic action

$$T \times X \longrightarrow X.$$

Reference. This is Cox–Little–Schenck, Definition 3.1.1.

Theorem 3.12 (The toric variety of a fan)**.** *Let Σ be a fan in $N_{\mathbb{R}}$. Then X_Σ is a normal separated toric variety with dense torus*

$$T = \operatorname{Spec} \Bbbk[M].$$

Reference. This is Cox–Little–Schenck, Theorem 3.1.5. The proof there is written over $\mathbb{C}$; the same algebraic argument works over our base field $\Bbbk$.

Conceptual proof sketch. The proof has four parts.

The torus and its action. The zero cone is a face of every cone in Σ, so every chart contains the same open torus

$$U_{\{0\}} = T.$$

The gluing maps identify these copies by the identity. The torus actions on the affine charts agree on every overlap and therefore glue to an action on X_Σ.

Irreducibility. Every chart U_σ is irreducible and contains the same dense open torus. Their union is therefore irreducible, with T dense.

Normality. Each U_σ is normal because

$$S_\sigma = \sigma^\vee \cap M$$

is saturated. Normality is local, so X_Σ is normal.

Separatedness. For

$$\tau = \sigma_1 \cap \sigma_2,$$

consider the diagonal on the two charts:

$$\Delta : U_\tau \longrightarrow U_{\sigma_1} \times U_{\sigma_2}.$$

On coordinate rings it is induced by

$$\Delta^* : \Bbbk[S_{\sigma_1}] \otimes_{\Bbbk} \Bbbk[S_{\sigma_2}] \longrightarrow \Bbbk[S_\tau],$$

$$\chi^m \otimes \chi^n \longmapsto \chi^{m+n}.$$

The separation lemma says

$$S_\tau = S_{\sigma_1} + S_{\sigma_2},$$

so Δ^* is surjective. Hence $\Delta(U_\tau)$ is closed in

$$U_{\sigma_1} \times U_{\sigma_2}.$$

Checking this for every pair of affine charts proves that X_Σ is separated. □

Warning

The affine pieces alone do not guarantee separatedness.
If two copies of $\mathbb{A}^1$ are glued along

$$\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$$

by the identity, the result has two distinct origins that cannot be separated by disjoint open sets. This is the *affine line with doubled origin*, and it is not separated.
The fan construction rules out this pathology. Its canonical overlaps satisfy the separation lemma, which forces the diagonal to be closed.

Example 3.13 (The toric blowup of $\mathbb{A}^2$ at the origin)**.** Start with the affine fan of the first quadrant

$$\Sigma_{\mathbb{A}^2} = \{\{0\}, \rho_1, \rho_2, \sigma\}, \qquad \rho_1 = \mathrm{Cone}(e_1),\ \rho_2 = \mathrm{Cone}(e_2),\ \sigma = \mathrm{Cone}(e_1, e_2).$$

Insert the new ray

$$\rho = \mathrm{Cone}(e_1 + e_2).$$

This subdivides σ into two smooth cones

$$\sigma_1 = \mathrm{Cone}(e_1, e_1 + e_2), \qquad \sigma_2 = \mathrm{Cone}(e_1 + e_2, e_2).$$

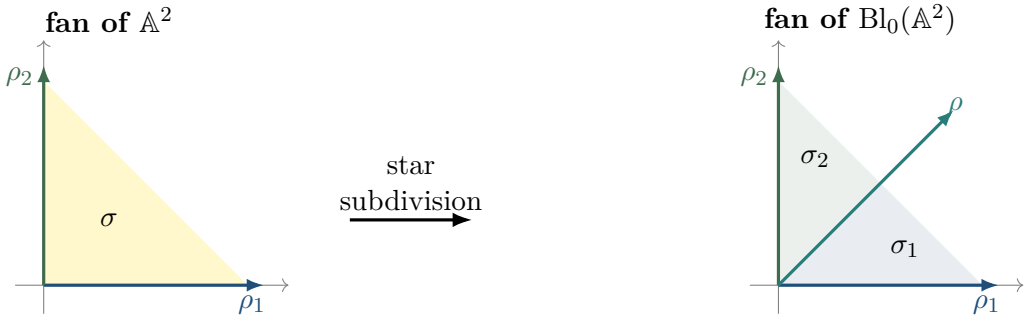


Figure 3.4: Subdividing the first quadrant by the ray through $e_1 + e_2$ produces the fan of the toric blowup of $\mathbb{A}^2$ at the origin.

Each maximal cone is smooth, so each affine chart is isomorphic to $\mathbb{A}^2$. Indeed,

$$U_{\sigma_1} \simeq \operatorname{Spec} \Bbbk[x, xy^{-1}], \qquad U_{\sigma_2} \simeq \operatorname{Spec} \Bbbk[y, x^{-1}y].$$

The identity map of lattices

$$N \longrightarrow N$$

is compatible from the subdivided fan to the single-cone fan of $\mathbb{A}^2$, so it induces a toric morphism

$$\pi : X_{\Sigma'} \longrightarrow \mathbb{A}^2.$$

Geometrically, π is the blowup of the origin. The new ray ρ contributes a new one-dimensional orbit; its closure is the exceptional divisor

$$V(\rho) \simeq \mathbb{P}^1.$$

The old rays still correspond to the strict transforms of the two coordinate axes.

The three dictionaries

The gluing construction is the global form of the three dictionaries.

Algebra	Geometry	Combinatorics
$\Bbbk[S_\sigma]$	an affine chart U_σ	a cone σ
localization to $\Bbbk[S_{\sigma_1 \cap \sigma_2}]$	the overlap $U_{\sigma_1} \cap U_{\sigma_2}$	the common face $\sigma_1 \cap \sigma_2$
identity maps inside $\Bbbk[M]$	transition maps satisfying the cocycle condition	fan compatibility
surjectivity from $S_{\sigma_1 \cap \sigma_2} = S_{\sigma_1} + S_{\sigma_2}$	separatedness	the separation lemma

Guiding question

Once the affine pieces have been glued, how are the torus orbits of X_Σ indexed by the cones of the fan, and how do face relations control orbit closures?

3.4 The orbit–cone correspondence

A fan is more than an atlas of affine charts. Its individual cones also index the torus orbits of the glued variety.

The starting point is the distinguished limit attached to a cone.

3.4.1 Distinguished points

Let

$$\sigma \in \Sigma, \qquad S_\sigma = \sigma^\vee \cap M.$$

Define a point

$$\gamma_\sigma \in U_\sigma$$

by the semigroup homomorphism

$$\gamma_\sigma(m) = \begin{cases} 1, & m \in \sigma^\perp \cap M, \\ 0, & m \notin \sigma^\perp. \end{cases}$$

This is the *distinguished point* corresponding to σ.

If

$$u \in \operatorname{Relint}(\sigma) \cap N,$$

then the one-parameter subgroup criterion from Chapter 2 gives

$$\lim_{t \to 0} \lambda^u(t) = \gamma_\sigma.$$

Thus every lattice direction in the relative interior of σ selects the same distinguished point.

Reference. The distinguished point and its interpretation by one-parameter limits are Cox–Little–Schenck, Proposition 3.2.2 and the discussion preceding it.

Now let the torus move this point.

Definition 3.14. For $\sigma \in \Sigma$, define

$$O(\sigma) = T \cdot \gamma_\sigma \subseteq X_\Sigma.$$

3.4.2 What kind of torus is $O(\sigma)$?

Let

$$N_\sigma = N \cap \operatorname{span}_{\mathbb{R}}(\sigma)$$

and set

$$N(\sigma) = N / N_\sigma.$$

The lattice dual to $N(\sigma)$ is

$$\sigma^\perp \cap M.$$

Proposition 3.15 (The orbit attached to a cone)**.** *There are natural isomorphisms*

$$O(\sigma) \simeq \operatorname{Hom}_{\mathbb{Z}}\big(\sigma^\perp \cap M, \Bbbk^\times\big) \simeq T_{N(\sigma)}.$$

Consequently,

$$\dim O(\sigma) = \operatorname{rank} N - \dim \sigma.$$

Reference. This is Cox–Little–Schenck, Lemmas 3.2.4 and 3.2.5.

Conceptual proof sketch. A point of U_σ is a semigroup homomorphism

$$\gamma : S_\sigma \longrightarrow \Bbbk.$$

The orbit of γ_σ consists exactly of those γ satisfying

$$\gamma(m) \neq 0 \quad \Longleftrightarrow \quad m \in \sigma^\perp \cap M.$$

Restricting such a γ to $\sigma^\perp \cap M$ gives a group homomorphism to $\Bbbk^\times$, and every such group homomorphism extends by zero outside $\sigma^\perp$.

The perfect pairing

$$(\sigma^\perp \cap M) \times N(\sigma) \longrightarrow \mathbb{Z}$$

identifies this group of homomorphisms with the torus $T_{N(\sigma)}$. Its dimension is

$$\operatorname{rank} N(\sigma) = \operatorname{rank} N - \operatorname{rank} N_\sigma = \operatorname{rank} N - \dim \sigma.$$

□

Two extreme cases are already familiar:

$$O(\{0\}) = T,$$

while a full-dimensional cone gives a zero-dimensional orbit, hence a torus-fixed point.

Pause and think

Let Σ be a fan in a rank-n lattice.
What orbit dimensions correspond to:

the zero cone, a ray, a maximal n-dimensional cone?

What geometric strata should the rays represent?

3.4.3 The theorem

Theorem 3.16 (Orbit–cone correspondence)**.** *Let X_Σ be the toric variety of a fan $\Sigma \subseteq N_\mathbb{R}$. Then:*

(a) The assignment

$$\sigma \longmapsto O(\sigma)$$

is a bijection

$$\{\text{cones of } \Sigma\} \longleftrightarrow \{T\text{-orbits in } X_\Sigma\}.$$

(b) For every cone,

$$\dim O(\sigma) = \operatorname{rank} N - \dim \sigma.$$

(c) Each affine chart decomposes as

$$U_\sigma = \bigsqcup_{\tau \preceq \sigma} O(\tau).$$

(d) Face relations reverse under orbit closure:

$$\tau \preceq \sigma \quad \Longleftrightarrow \quad O(\sigma) \subseteq \overline{O(\tau)}.$$

In particular,

$$\overline{O(\tau)} = \bigsqcup_{\tau \preceq \sigma} O(\sigma).$$

Reference. This is Cox–Little–Schenck, Theorem 3.2.6.

Conceptual proof sketch. The previous proposition constructs one orbit $O(\sigma)$ for every cone.

Conversely, let O be any orbit. Among the cones whose affine charts meet O, choose a minimal one, say σ. Represent a point of $O \cap U_\sigma$ by a semigroup homomorphism

$$\gamma : S_\sigma \to \Bbbk.$$

The set of exponents on which γ is nonzero is the lattice part of a face of $\sigma^\vee$. Duality identifies that face with a face $\tau \preceq \sigma$, and then the point lies in U_τ. Minimality forces

$$\tau = \sigma,$$

so the orbit is $O(\sigma)$. This proves the bijection.

The dimension formula follows from

$$O(\sigma) \simeq T_{N(\sigma)}.$$

The description of U_σ follows because its orbits are exactly those indexed by faces of σ.

Finally, if $\tau \preceq \sigma$, move the distinguished point γ_τ by a one-parameter subgroup whose lattice vector lies in $\operatorname{Relint}(\sigma)$. Its limit lies in $O(\sigma)$, showing

$$O(\sigma) \subseteq \overline{O(\tau)}.$$

The converse follows from the affine-chart decomposition. The technical part of the complete proof verifies that the displayed union is Zariski closed. □

The theorem reverses both dimension and incidence:

$$\begin{array}{ccc} \text{larger cone} & \longleftrightarrow & \text{smaller orbit} \\ \tau \preceq \sigma & \longleftrightarrow & O(\sigma) \subseteq \overline{O(\tau)}. \end{array}$$

Corollary 3.17. *For every cone $\sigma \in \Sigma$, the affine chart U_σ contains a unique closed torus orbit, namely $O(\sigma)$.*

Proof. The chart contains precisely the orbits $O(\tau)$ with $\tau \preceq \sigma$. The closure of every such orbit contains $O(\sigma)$, while no strictly larger cone belongs to the affine fan of σ. □

3.4.4 Reading the blowup fan

Return to the blowup fan from theorem 3.13. It has three rays

$$\rho_1 = \operatorname{Cone}(e_1), \qquad \rho = \operatorname{Cone}(e_1 + e_2), \qquad \rho_2 = \operatorname{Cone}(e_2),$$

and two maximal cones

$$\sigma_1 = \operatorname{Cone}(e_1, e_1 + e_2), \qquad \sigma_2 = \operatorname{Cone}(e_1 + e_2, e_2).$$

The orbit–cone correspondence can be read directly from the picture.

The figure makes three features visible at once.

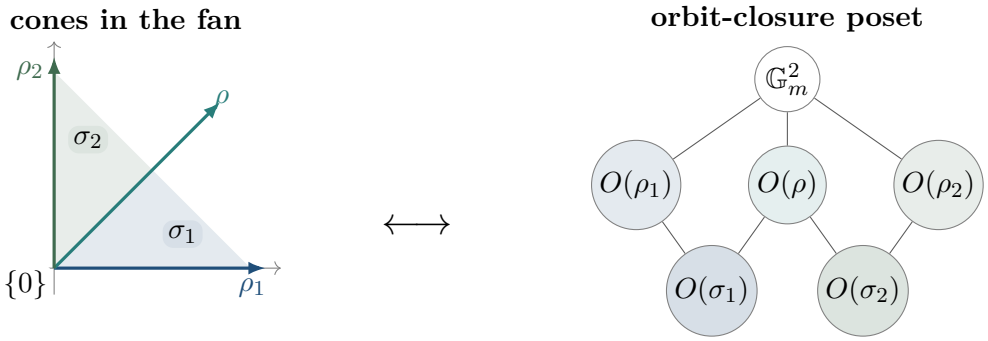


Figure 3.5: Color-coded orbit–cone correspondence for the fan of $\mathrm{Bl}_0(\mathbb{A}^2)$. The colors and labels establish the bijection between cones and orbits, while the Hasse diagram records the orbit-closure order. Ordering cones by inclusion and orbit closures by reverse inclusion, the correspondence is an isomorphism of posets.

- The zero cone corresponds to the dense torus.
- The three rays correspond to the three boundary curves: the two strict transforms of the coordinate axes and the exceptional divisor.
- The two maximal cones correspond to the two torus-fixed points, which lie on the exceptional divisor.

This is a good first reminder that the orbit decomposition is not only a list of strata: it also remembers how those strata fit together.

3.4.5 Reading the fan of $\mathbb{P}^2$

Let

$$\rho_i = \mathrm{Cone}(e_i), \qquad i = 0, 1, 2,$$

and let $\sigma_0, \sigma_1, \sigma_2$ be the three maximal cones from Section 3.1. The complete orbit decomposition is:

Cones	**Orbit dimension**	**Orbit in $\mathbb{P}^2$**	**Orbit closure**
$\{0\}$	2	$\{[z_0 : z_1 : z_2] : z_0 z_1 z_2 \neq 0\} \simeq \mathbb{G}_m^2$	$\mathbb{P}^2$
ρ_0, ρ_1, ρ_2	1	a coordinate line with its two coordinate points removed	a coordinate line $\simeq \mathbb{P}^1$
$\sigma_0, \sigma_1, \sigma_2$	0	the three coordinate points	the point itself

Surface laboratory

Use the fan and the table to answer the following.

(a) Why does each ray belong to exactly two maximal cones, and how does this predict that the closure of the corresponding one-dimensional orbit contains exactly two fixed points?

(b) Choose one maximal cone σ_i. List its faces and recover the three orbit types appearing in its affine-plane chart.

(c) Starting from a coordinate line in $\mathbb{P}^2$, identify its open torus orbit and the two lower-dimensional orbits in its boundary.

(d) Count the orbits of each dimension directly from

$$\Sigma(0), \qquad \Sigma(1), \qquad \Sigma(2).$$

3.4.6 Orbit closures are toric varieties

For $\tau \in \Sigma$, write

$$V(\tau) = \overline{O(\tau)}.$$

The theorem gives

$$V(\tau) = \bigsqcup_{\tau \preceq \sigma} O(\sigma).$$

To describe $V(\tau)$ by a fan, project every cone containing τ to

$$N(\tau)_{\mathbb{R}} = N_{\mathbb{R}} / \operatorname{span}_{\mathbb{R}}(\tau).$$

If $\overline{\sigma}$ denotes the image of σ, define

$$\operatorname{Star}(\tau) = \{\overline{\sigma} \subseteq N(\tau)_{\mathbb{R}} : \tau \preceq \sigma \in \Sigma\}.$$

Proposition 3.18 (The fan of an orbit closure)**.** *The collection* $\operatorname{Star}(\tau)$ *is a fan, and*

$$V(\tau) \simeq X_{\operatorname{Star}(\tau)}.$$

Reference. This is Cox–Little–Schenck, Proposition 3.2.7.

Conceptual proof sketch. The orbit $O(\tau)$ is the torus with one-parameter subgroup lattice

$$N(\tau) = N/N_\tau.$$

The orbits contained in its closure are indexed exactly by the cones σ containing τ. After quotienting by the directions of τ, these cones form the fan $\operatorname{Star}(\tau)$. The orbit–cone correspondence for that quotient fan reproduces the same orbits and the same incidence relations as $V(\tau)$. □

For a ray ρ in a surface fan,

$$N(\rho)$$

has rank one. Hence $V(\rho)$ is a toric curve. In the fan of $\mathbb{P}^2$, the two maximal cones containing ρ project to the two opposite rays in $N(\rho)_{\mathbb{R}}$, so

$$V(\rho) \simeq \mathbb{P}^1.$$

The three dictionaries

The orbit–cone correspondence translates the face structure of the fan into the stratification of the variety.

Algebra	Geometry	Combinatorics
characters in $\sigma^\perp \cap M$	the torus $O(\sigma)$	a cone σ
rank of $\sigma^\perp \cap M$	$\dim O(\sigma)$	$\operatorname{rank} N - \dim \sigma$
vanishing of additional characters	movement toward the boundary	passing to a larger cone
quotient character and one-parameter lattices	the orbit closure $V(\tau)$	the fan $\operatorname{Star}(\tau)$

Guiding question

Which global geometric properties of X_Σ—smoothness, completeness, products, and morphisms—can be recognized directly from the fan?

3.5 Properties read from the fan

The construction

$$\Sigma \longmapsto X_\Sigma$$

turns combinatorial conditions into geometric properties. The most basic entries are:

Geometry of X_Σ	Combinatorics of Σ
smooth	every cone is smooth
finite quotient singularities	every cone is simplicial
complete	$\lvert\Sigma\rvert = N_{\mathbb{R}}$
product	product fan
toric morphism	a compatible lattice map

We now explain these correspondences, concentrating on the criterion and the idea behind it.

3.5.1 Smoothness is checked cone by cone

Definition 3.19. A fan Σ is *smooth* if every cone $\sigma \in \Sigma$ is smooth: the primitive generators of its rays form part of a basis of N.

A fan is *simplicial* if every cone is generated by linearly independent vectors.

In dimension two, every strongly convex cone is simplicial, but it need not be smooth. If

$$\sigma = \operatorname{Cone}(u_1, u_2)$$

has primitive ray generators, then

$$\sigma \text{ is smooth} \quad \Longleftrightarrow \quad |\det(u_1, u_2)| = 1.$$

Theorem 3.20 (Global smoothness criterion)**.** *The toric variety X_Σ is smooth if and only if the fan Σ is smooth.*

Reference. This is Cox–Little–Schenck, Theorem 3.1.19(a).

Proof. The affine charts U_σ, for $\sigma \in \Sigma_{\max}$, cover X_Σ. Smoothness is local, and the affine smoothness criterion gives

$$U_\sigma \text{ smooth} \quad \Longleftrightarrow \quad \sigma \text{ smooth}.$$

Thus every point of X_Σ is smooth exactly when every maximal cone is smooth. Since every cone is a face of a maximal cone, this is equivalent to smoothness of the entire fan. □

Observation 3.21. Over a field of characteristic zero, one also has

$$X_\Sigma \text{ has only finite quotient singularities} \quad \Longleftrightarrow \quad \Sigma \text{ is simplicial}.$$

This is Cox–Little–Schenck, Theorem 3.1.19(b). Its proof requires more singularity theory than we need here and will be revisited later.

The cyclic quotient surface from the challenge exercise of Chapter 2 is the local model: the cone is simplicial, but its primitive generators need not form a basis of the chosen lattice.

Pause and think

Let Σ be a two-dimensional fan. Explain why its singular points can occur only at the zero-dimensional orbits associated to nonsmooth maximal cones.
Why can the one-dimensional orbits not be singular?

3.5.2 Completeness and one-parameter limits

Recall that a fan is complete when

$$|\Sigma| = N_{\mathbb{R}}.$$

This condition has an exact geometric meaning.

Algebraic geometry cheatsheet

A variety X is *complete* if its structure morphism

$$X \longrightarrow \operatorname{Spec} \Bbbk$$

is proper.
Over $\mathbb{C}$, complete varieties are precisely those whose complex points are compact in the classical topology. Every projective variety is complete, although a complete variety need not be projective.

Theorem 3.22 (Completeness criterion)**.** *For a toric variety X_Σ, the following are equivalent:*

(a) X_Σ is complete;

(b) every one-parameter subgroup

$$\lambda^u : \mathbb{G}_m \longrightarrow T, \qquad u \in N,$$

has a limit in X_Σ as $t \to 0$;

(c) the fan is complete:

$$|\Sigma| = N_{\mathbb{R}}.$$

Reference. This is Cox–Little–Schenck, Theorem 3.4.6.

Conceptual proof sketch. The equivalence of (b) and (c) is the one-parameter subgroup criterion applied chart by chart:

$$\lim_{t\to 0} \lambda^u(t) \text{ exists in } U_\sigma \quad \Longleftrightarrow \quad u \in \sigma.$$

Therefore the limit exists somewhere in X_Σ exactly when

$$u \in |\Sigma|.$$

If the support is not all of $N_{\mathbb{R}}$, rationality and finiteness of the fan allow us to choose a lattice vector

$$u \notin |\Sigma|.$$

The corresponding map λ^u has no extension across $t = 0$, so the valuative criterion rules out completeness.

Conversely, if the support is complete, every one-parameter subgroup has a limit in some affine chart. The substantive part of the proof uses the valuative criterion to show that, for toric varieties, these torus limits suffice to establish properness. □

This criterion makes the pictures of $\mathbb{A}^2$ and $\mathbb{P}^2$ immediately different. The affine-plane fan occupies only one quadrant, while the three maximal cones of the projective-plane fan cover all of $N_{\mathbb{R}}$.

Prediction

A complete fan in $\mathbb{R}^2$ must surround the origin. What is the smallest possible number of rays in a complete two-dimensional fan? What toric surface should the smallest smooth example produce?

3.5.3 Product fans

Let Σ_1 and Σ_2 be fans in $(N_1)_{\mathbb{R}}$ and $(N_2)_{\mathbb{R}}$.

Definition 3.23. Their *product fan* is

$$\Sigma_1 \times \Sigma_2 = \{\sigma_1 \times \sigma_2 : \sigma_i \in \Sigma_i\}$$

in

$$(N_1 \oplus N_2)_{\mathbb{R}}.$$

Faces and intersections are computed factor by factor, so this is indeed a fan.

Proposition 3.24 (Products)**.** *There is a canonical isomorphism*

$$X_{\Sigma_1 \times \Sigma_2} \simeq X_{\Sigma_1} \times X_{\Sigma_2}.$$

Reference. This is Cox–Little–Schenck, Proposition 3.1.14.

Proof. For a product cone,

$$(\sigma_1 \times \sigma_2)^\vee = \sigma_1^\vee \times \sigma_2^\vee,$$

and hence

$$S_{\sigma_1 \times \sigma_2} = S_{\sigma_1} \oplus S_{\sigma_2}.$$

Therefore

$$\Bbbk[S_{\sigma_1 \times \sigma_2}] \simeq \Bbbk[S_{\sigma_1}] \otimes_{\Bbbk} \Bbbk[S_{\sigma_2}],$$

which gives

$$U_{\sigma_1 \times \sigma_2} \simeq U_{\sigma_1} \times U_{\sigma_2}.$$

These affine isomorphisms agree on common face charts and therefore glue. □

Smoothness and completeness behave as expected:

$$\Sigma_1 \times \Sigma_2 \text{ is smooth} \quad \Longleftrightarrow \quad \Sigma_1, \Sigma_2 \text{ are smooth},$$

and

$$|\Sigma_1 \times \Sigma_2| = |\Sigma_1| \times |\Sigma_2|.$$

3.5.4 Toric morphisms from lattice maps

Let Σ_i be a fan in $(N_i)_{\mathbb{R}}$, and let

$$F : N_1 \longrightarrow N_2$$

be a lattice homomorphism.

Definition 3.25. The map F is *compatible with the fans* Σ_1, Σ_2 if, for every cone $\sigma \in \Sigma_1$, there is a cone $\sigma' \in \Sigma_2$ such that

$$F_{\mathbb{R}}(\sigma) \subseteq \sigma'.$$

This is the global version of the affine cone criterion.

Theorem 3.26 (Toric morphisms and maps of fans)**.** *Compatible lattice maps*

$$F : (N_1, \Sigma_1) \longrightarrow (N_2, \Sigma_2)$$

are in natural correspondence with toric morphisms

$$\phi_F : X_{\Sigma_1} \longrightarrow X_{\Sigma_2}.$$

Reference. The definition of toric morphism is Cox–Little–Schenck, Definition 3.3.3, and the correspondence is Theorem 3.3.4.

Conceptual proof sketch. Suppose F is compatible. For every $\sigma \in \Sigma_1$, choose $\sigma' \in \Sigma_2$ with

$$F_{\mathbb{R}}(\sigma) \subseteq \sigma'.$$

The affine cone criterion produces a toric morphism

$$U_\sigma \longrightarrow U_{\sigma'}.$$

All of these maps restrict to the same torus homomorphism, so they agree on common face charts and glue to a global morphism.

Conversely, a toric morphism restricts on the dense tori to a group homomorphism, hence induces a lattice map

$$F : N_1 \to N_2.$$

The image of each affine chart U_σ meets a suitable affine chart of the target. Applying the affine cone criterion shows that $F_{\mathbb{R}}(\sigma)$ lies in a cone of Σ_2.

The bookkeeping needed to choose target charts and verify gluing is the only global step. □

The product projections are the basic examples. The lattice map

$$\mathrm{pr}_1 : N_1 \oplus N_2 \longrightarrow N_1$$

sends

$$\sigma_1 \times \sigma_2 \longmapsto \sigma_1,$$

so it induces

$$X_{\Sigma_1} \times X_{\Sigma_2} \longrightarrow X_{\Sigma_1}.$$

Observation 3.27. Finite-index changes of lattice fit the same theorem. If

$$N' \subseteq N$$

has finite index and the same real fan Σ is viewed in both lattices, then the inclusion is compatible and induces

$$X_{\Sigma,N'} \longrightarrow X_{\Sigma,N}.$$

This is the global version of the lattice change behind the cyclic quotient orbifolds in the Chapter 2 challenge.

Surface laboratory

Let $\Sigma_\square$ be the fan in $\mathbb{R}^2$ whose rays are

$$\mathbb{R}_{\geq 0}e_1, \quad \mathbb{R}_{\geq 0}e_2, \quad \mathbb{R}_{\geq 0}(-e_1), \quad \mathbb{R}_{\geq 0}(-e_2),$$

and whose maximal cones are the four quadrants.

(a) Verify directly that $\Sigma_\square$ is smooth and complete.

(b) Show that it is the product of two copies of the fan of $\mathbb{P}^1$.

(c) Deduce that

$$X_{\Sigma_\square} \simeq \mathbb{P}^1 \times \mathbb{P}^1.$$

(d) Use the coordinate projections

$$\mathbb{Z}^2 \longrightarrow \mathbb{Z}$$

to recover the two morphisms

$$\mathbb{P}^1 \times \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

(e) Count the two-dimensional, one-dimensional, and zero-dimensional torus orbits directly from the fan.

Guiding question

How do the standard fans of projective spaces, products, weighted projective surfaces, and blowups display these properties in concrete pictures?

3.6 A catalogue of fans and toric surfaces

We finish the chapter by collecting a small number of surface fans that will reappear throughout the course. The goal is not to memorize the pictures, but to learn which features should be read immediately:

support, primitive ray generators, determinants of adjacent rays, subdivisions.

3.6.1 Four recurring complete fans

The following four fans form a useful visual vocabulary.

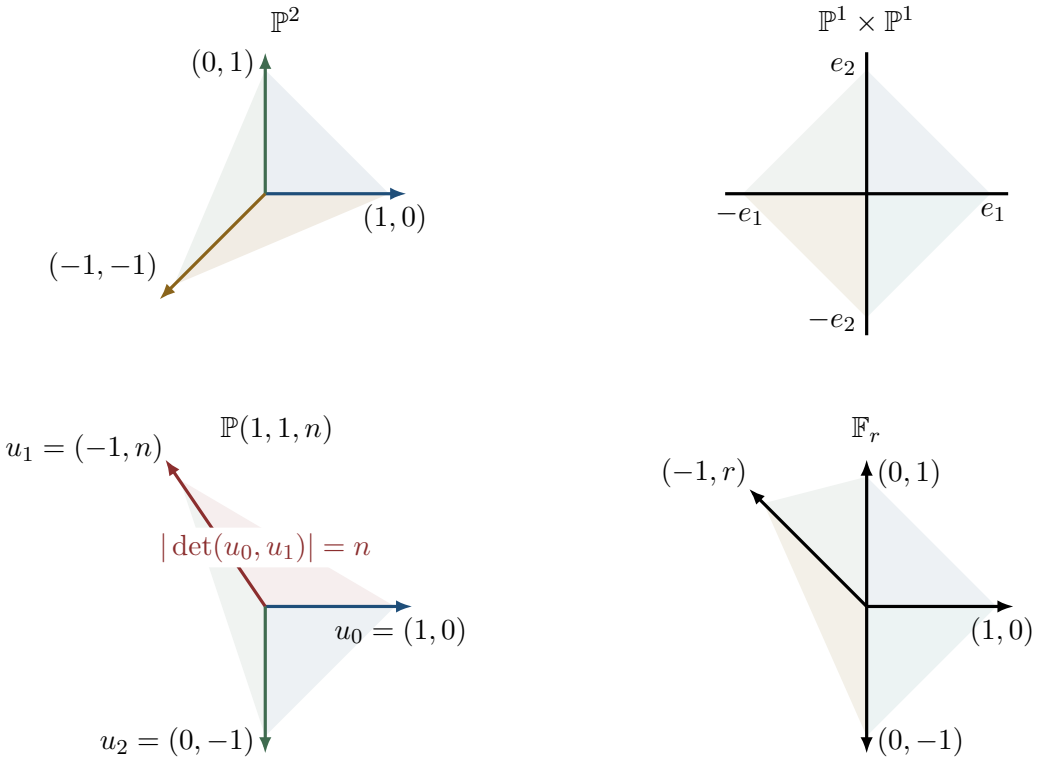


Figure 3.6: Four standard complete surface fans. The first, second, and fourth are smooth. The weighted projective fan is simplicial but has one nonsmooth maximal cone when $n > 1$.

The first two fans are already familiar. Their primitive rays are

$$\begin{array}{c|c} \mathbb{P}^2 & (1,0),\ (0,1),\ (-1,-1) \\ \mathbb{P}^1 \times \mathbb{P}^1 & (1,0),\ (0,1),\ (-1,0),\ (0,-1). \end{array}$$

Every adjacent determinant has absolute value 1, and both supports fill $\mathbb{R}^2$. Thus both surfaces are smooth and complete.

Reference. These are Cox–Little–Schenck, Examples 3.1.9 and 3.1.12.

3.6.2 A weighted projective plane

Fix $n \geq 2$, and consider the three primitive rays generated by

$$u_0 = (1,0), \qquad u_1 = (-1,n), \qquad u_2 = (0,-1).$$

They satisfy the weighted relation

$$u_0 + u_1 + nu_2 = 0.$$

Taking the cones generated by each proper subset gives the standard fan of

$$\mathbb{P}(1,1,n).$$

The fan is complete and simplicial. Two maximal cones are smooth:

$$|\det(u_1,u_2)| = 1, \qquad |\det(u_2,u_0)| = 1.$$

The third has

$$|\det(u_0,u_1)| = n,$$

so the corresponding torus-fixed point is singular when $n > 1$.

This is the global version of the cyclic quotient phenomenon from Chapter 2: the real cone is simple, but the primitive generators span a sublattice of index n.

Example 3.28. For $n = 2$, the singular affine chart is the quadratic cone

$$V(ac - b^2) \subseteq \mathbb{A}^3.$$

Thus $\mathbb{P}(1,1,2)$ is complete and has exactly one singular torus-fixed point.

Reference. Weighted projective fans are Cox–Little–Schenck, Example 3.1.17. The special case $\mathbb{P}(1,1,2)$ and its quadratic-cone chart are worked out there explicitly.

Pause and think

Use the orbit–cone correspondence to count the torus orbits of $\mathbb{P}(1,1,n)$. Which orbit contains the singular point, and which cone detects it?

3.6.3 Hirzebruch surfaces

For $r \geq 0$, let Σ_r be the complete fan whose rays, in cyclic order, are

$$v_1 = (1,0), \qquad v_2 = (0,1), \qquad v_3 = (-1,r), \qquad v_4 = (0,-1).$$

The associated toric surface is the *r-th Hirzebruch surface*

$$\mathbb{F}_r = X_{\Sigma_r}.$$

Every adjacent determinant has absolute value 1:

$$\det(v_1,v_2) = \det(v_2,v_3) = \det(v_3,v_4) = \det(v_4,v_1) = 1.$$

Hence every $\mathbb{F}_r$ is smooth. The fan is complete, so every $\mathbb{F}_r$ is complete.

The lattice projection

$$p : N = \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad p(a,b) = a,$$

is compatible with the fan of $\mathbb{P}^1$. It therefore induces a toric morphism

$$\mathbb{F}_r \longrightarrow \mathbb{P}^1.$$

This is the ruling of the Hirzebruch surface.

When $r = 0$, the four rays are the coordinate rays, so

$$\mathbb{F}_0 \simeq \mathbb{P}^1 \times \mathbb{P}^1.$$

Reference. This family is Cox–Little–Schenck, Example 3.1.16.

Surface laboratory

For the fan of $\mathbb{F}_r$:

(a) Verify the four determinant calculations.

(b) Check compatibility of

$$p(a,b) = a$$

with the two rays in the fan of $\mathbb{P}^1$.

(c) Count the torus orbits of dimensions $2, 1, 0$.

(d) Compare the fans of $\mathbb{F}_0$ and $\mathbb{P}^1 \times \mathbb{P}^1$. Why are they literally the same fan?

3.6.4 Blowing up a torus-fixed point

The blowup of $\mathbb{A}^2$ from Section 3.3 is the local model for a general operation.

Let

$$\sigma = \mathrm{Cone}(u_1, u_2)$$

be a smooth two-dimensional cone in a fan. Insert the ray

$$\rho = \mathrm{Cone}(u_1 + u_2)$$

and replace σ by

$$\mathrm{Cone}(u_1, u_1 + u_2), \qquad \mathrm{Cone}(u_1 + u_2, u_2).$$

This operation is the *star subdivision* of the fan at σ.

Proposition 3.29 (Star subdivision and blowup)**.** *The toric morphism from the subdivided fan to the original fan is the blowup of the torus-fixed point $O(\sigma)$.*

Reference. This is the two-dimensional case of Cox–Little–Schenck, Proposition 3.3.15.

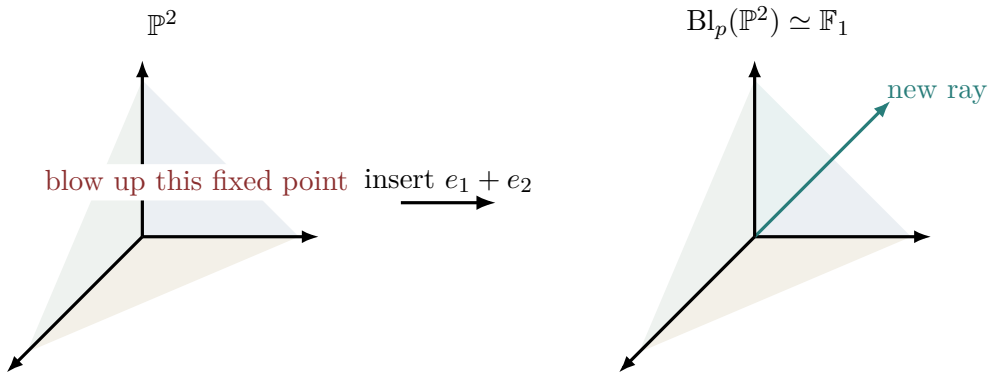


Figure 3.7: Blowing up a torus-fixed point of $\mathbb{P}^2$ inserts one ray and replaces one maximal cone by two. The resulting fan is equivalent by a lattice change to the fan of $\mathbb{F}_1$.

For the fixed point corresponding to

$$\mathrm{Cone}(e_1, e_2),$$

the new fan has rays

$$e_1, \qquad e_1 + e_2, \qquad e_2, \qquad -e_1 - e_2.$$

The matrix

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}(2, \mathbb{Z})$$

takes these rays to

$$(1, 0), \qquad (0, 1), \qquad (-1, 1), \qquad (0, -1),$$

the fan of $\mathbb{F}_1$. Hence

$$\mathrm{Bl}_p(\mathbb{P}^2) \simeq \mathbb{F}_1$$

for a torus-fixed point p.

Prediction

Each blowup at a torus-fixed point inserts one new ray and one new torus-invariant boundary curve. What combinatorial patterns might arise after repeatedly blowing up fixed points of a smooth complete toric surface?

3.6.5 A compact recognition table

Surface	Rays	Smooth	Complete	Feature to recognize
$\mathbb{P}^2$	$e_1, e_2, -e_1 - e_2$	yes	yes	three rays and three fixed points
$\mathbb{P}^1 \times \mathbb{P}^1$	$\pm e_1, \pm e_2$	yes	yes	product fan
$\mathbb{P}(1,1,n)$	$(1,0), (-1,n), (0,-1)$	only if $n = 1$	yes	one cone of determinant n
$\mathbb{F}_r$	$(1,0), (0,1), (-1,r), (0,-1)$	yes	yes	a toric ruling to $\mathbb{P}^1$

Guiding question

How do divisor classes, torus-invariant divisors, and line bundles translate the rays of these fans into intersection theory and projective embeddings?

What this chapter has accomplished

A single cone gives one affine toric chart. A fan records how many such charts fit together and how they overlap. The chapter has developed the global dictionary

$$\boxed{\text{fan } \Sigma \quad \longleftrightarrow \quad \text{normal separated toric variety } X_\Sigma.}$$

More precisely, we have seen that:

$$\begin{array}{ccc} \sigma \in \Sigma & \longleftrightarrow & U_\sigma \subseteq X_\Sigma \\ \sigma \cap \tau & \longleftrightarrow & U_\sigma \cap U_\tau \\ \sigma & \longleftrightarrow & O(\sigma) \\ \tau \preceq \sigma & \longleftrightarrow & O(\sigma) \subseteq \overline{O(\tau)} \\ |\Sigma| = N_{\mathbb{R}} & \longleftrightarrow & X_\Sigma \text{ complete} \\ \Sigma \text{ smooth} & \longleftrightarrow & X_\Sigma \text{ smooth.} \end{array}$$

The surface laboratory supplied a second thread:

$$\mathbb{P}^2, \qquad \mathbb{P}^1 \times \mathbb{P}^1, \qquad \mathbb{P}(1,1,n), \qquad \mathbb{F}_r, \qquad \mathrm{Bl}_p(\mathbb{P}^2).$$

In these examples, rays became boundary curves, maximal cones became fixed points, and subdivisions became blowups.

The next chapter will attach divisor classes and equations to this combinatorial skeleton.

Comprehension exercises

3.1. Let

$$\sigma_1 = \mathrm{Cone}((1,0),(0,1)), \qquad \sigma_2 = \mathrm{Cone}((1,1),(-1,2)).$$

Compute $\sigma_1 \cap \sigma_2$. Decide whether the two cones and all of their faces form a fan. If not, add the smallest possible collection of rays and cones that repairs the problem.

3.2. Let $\sigma \subseteq N_{\mathbb{R}}$ be a strongly convex rational polyhedral cone, and let

$$\Sigma_\sigma = \{\tau : \tau \preceq \sigma\}.$$

Show directly from the gluing construction that

$$X_{\Sigma_\sigma} \simeq U_\sigma.$$

In other words, explain why gluing all face charts adds no new points.

3.3. Let $\Sigma' \subseteq \Sigma$ be a subfan. Prove that the canonical map

$$X_{\Sigma'} \longrightarrow X_\Sigma$$

identifies $X_{\Sigma'}$ with the torus-invariant open subset

$$\bigcup_{\sigma\in\Sigma'} U_\sigma.$$

3.4. Consider the fan of $\mathbb{P}^2$. Remove one maximal cone and every face that belongs only to that cone.

(a) Verify that the remaining collection is a subfan.

(b) Identify the corresponding open subvariety of $\mathbb{P}^2$.

(c) List its torus orbits and their closure relations.

3.5. Let Σ' be the fan of $\mathrm{Bl}_0(\mathbb{A}^2)$, with rays generated by

$$e_1, \qquad e_1+e_2, \qquad e_2.$$

(a) Compute the coordinate rings of the two maximal-cone charts.

(b) Compute their common open chart.

(c) Identify the three one-dimensional orbits and the two zero-dimensional orbits.

(d) Use the orbit–cone correspondence to explain why the closure of the orbit corresponding to e_1+e_2 is a copy of $\mathbb{P}^1$.

3.6. Let Σ be a fan in a rank-3 lattice with

$$|\Sigma| = N_{\mathbb{R}}.$$

Suppose it has 6 rays, 10 two-dimensional cones, and 6 maximal three-dimensional cones. Determine the number of torus orbits of each dimension. Which orbits are closed?

3.7. For each of the following fans, decide whether it is smooth and whether it is complete:

(a) $\mathrm{Cone}((1,0),(1,2))$ and its faces;

(b) the complete fan with rays $(1,0),(0,1),(-1,0),(0,-1)$;

(c) the complete fan with rays $(1,0),(-1,3),(0,-1)$.

For every nonsmooth maximal cone, compute the absolute determinant of its primitive ray generators.

3.8. Let Σ_r be the fan of the Hirzebruch surface $\mathbb{F}_r$.

(a) Verify that the lattice map

$$p(a,b) = a$$

is compatible with the fan of $\mathbb{P}^1$.

(b) Determine the image under p of every ray and every maximal cone.

(c) Describe the induced toric morphism

$$\mathbb{F}_r \longrightarrow \mathbb{P}^1$$

on the dense tori.

3.9. Let Σ_1, Σ_2 be fans. Prove directly from the orbit–cone correspondence that

$$O(\sigma_1 \times \sigma_2) \simeq O(\sigma_1) \times O(\sigma_2).$$

Use this to recover the orbit decomposition of $\mathbb{P}^1 \times \mathbb{P}^1$.

3.10. Let

$$F : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad F(a, b) = a.$$

Determine whether F is compatible with:

(a) the fan of $\mathbb{A}^2$ and the fan of $\mathbb{A}^1$;

(b) the fan of $\mathbb{P}^1 \times \mathbb{P}^1$ and the fan of $\mathbb{P}^1$;

(c) the fan of $\mathbb{P}^2$ and the fan of $\mathbb{P}^1$.

Whenever it is compatible, describe the induced map on the dense tori.

3.11. Start with the fan of $\mathbb{P}^2$, and subdivide the cone

$$\mathrm{Cone}(e_1, e_2)$$

by the ray through $e_1 + e_2$.

(a) Check that the resulting fan is smooth and complete.

(b) Count its torus orbits.

(c) Identify its toric surface in two different ways: as a blowup of $\mathbb{P}^2$, and as a Hirzebruch surface.

Challenge exercise

Weighted toric blowups and the resolution of an A_{n-1} singularity.

Assume throughout that $\Bbbk = \mathbb{C}$. Consider the Gorenstein cyclic quotient

$$X_n = \mathbb{A}^2/\mu_n, \qquad \zeta \cdot (x, y) = (\zeta x, \zeta^{-1} y).$$

As in the Chapter 2 challenge,

$$X_n \simeq V(uv - w^n) \subseteq \mathbb{A}^3,$$

the A_{n-1} surface singularity. Torically, it is the affine variety of the first-quadrant cone

$$\sigma = \mathrm{Cone}(e_1, e_2)$$

in the enlarged lattice

$$N = \mathbb{Z}^2 + \mathbb{Z}\left(\frac{1}{n}(1, n-1)\right).$$

The goal is to resolve the singularity by inserting the primitive lattice rays lying between e_2 and e_1. Each insertion is a toric weighted blowup, and the new rays will produce a chain of rational exceptional curves.

Warning

In this exercise, a *toric weighted blowup* means the proper birational toric morphism produced by inserting one primitive interior ray into a two-dimensional cone.
You may use:

(i) a refinement with the same support induces a proper birational toric morphism;

(ii) the toric variety of the complete one-dimensional fan is $\mathbb{P}^1$;

(iii) a two-dimensional cone is smooth exactly when its primitive ray generators form a basis of the lattice.

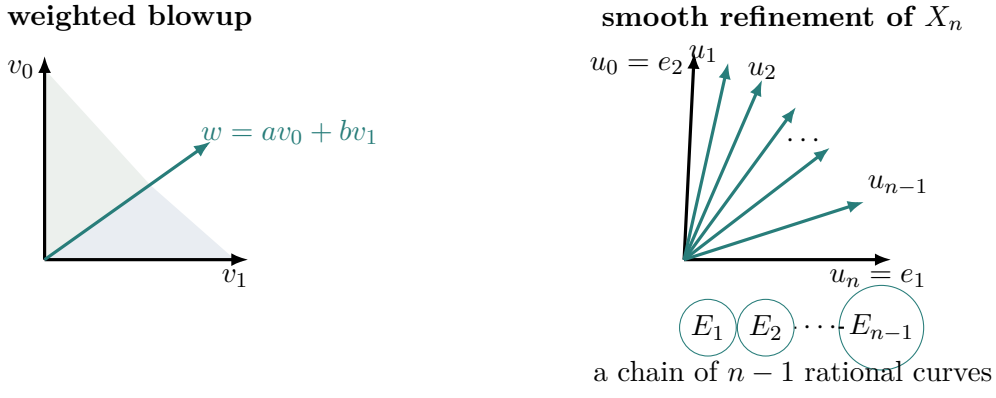


Figure 3.8: A primitive interior ray gives a toric weighted blowup. For the A_{n-1} singularity, inserting all intermediate primitive rays produces a smooth fan and an exceptional chain.

Hint 1. Let

$$\tau = \mathrm{Cone}(v_0, v_1)$$

be a smooth two-dimensional cone, and let

$$w = av_0 + bv_1$$

be a primitive interior lattice vector, where

$$a, b > 0, \qquad \gcd(a, b) = 1.$$

Insert the ray $\rho_w = \mathrm{Cone}(w)$.
Show that the two new maximal cones are

$$\mathrm{Cone}(v_0, w), \qquad \mathrm{Cone}(w, v_1).$$

In the basis v_0, v_1, compute

$$|\det(v_0, w)| = b, \qquad |\det(w, v_1)| = a.$$

Explain why the ordinary blowup corresponds to $a = b = 1$, while general weights may leave quotient singularities in the two new charts.

Hint 2. Let

$$\pi_w : Y_w \longrightarrow U_\tau$$

be the morphism defined by this subdivision. Use the orbit–cone correspondence to show that:

(a) π_w is an isomorphism away from the fixed point $O(\tau)$;

(b) the fiber over $O(\tau)$ is

$$E = V(\rho_w);$$

(c) the star of ρ_w is the complete one-dimensional fan.

Deduce that

$$E \simeq \mathbb{P}^1.$$

Hint 3. Return to X_n. Set

$$f_1 = \frac{1}{n}(1, n-1), \qquad f_2 = e_2.$$

Show that f_1, f_2 form a basis of N, and that

$$e_1 = nf_1 - (n-1)f_2.$$

For $0 \le i \le n$, define

$$u_i = if_1 - (i-1)f_2.$$

Verify the simpler formula

$$u_i = \left(\frac{i}{n}, \frac{n-i}{n}\right).$$

In particular,

$$u_0 = e_2, \qquad u_n = e_1.$$

Hint 4. Show that the vectors

$$u_1, \ldots, u_{n-1}$$

are the primitive lattice points on the line segment joining e_2 to e_1, ordered from left to right. Now compute, in the basis f_1, f_2,

$$\det(u_{i-1}, u_i) = -1 \qquad (1 \le i \le n).$$

Conclude that every cone

$$\mathrm{Cone}(u_{i-1}, u_i)$$

is smooth.

Hint 5. Let Σ' be the fan whose rays are

$$\rho_i = \mathrm{Cone}(u_i), \qquad 0 \le i \le n,$$

and whose maximal cones are

$$\mathrm{Cone}(u_{i-1}, u_i), \qquad 1 \le i \le n.$$

Show that

$$|\Sigma'| = \sigma.$$

Therefore the identity map of N induces a proper birational toric morphism

$$\pi : Y = X_{\Sigma'} \longrightarrow X_n.$$

Why is Y smooth? Why does this make π a resolution of singularities?
Explain how the refinement can be built by inserting

$$\rho_1, \rho_2, \ldots, \rho_{n-1}$$

one at a time. Interpret π as a sequence of toric weighted blowups of torus-fixed points.

Hint 6. For $1 \le i \le n-1$, let

$$E_i = V(\rho_i).$$

Show that the star fan of ρ_i has two opposite rays. Deduce

$$E_i \simeq \mathbb{P}^1.$$

Use the orbit–cone correspondence to prove

$$E_i \cap E_j \neq \varnothing \quad \Longleftrightarrow \quad |i - j| = 1.$$

When $|i - j| = 1$, identify the unique fixed point in the intersection.

Conclude that the exceptional fiber is the chain

$$E_1 - E_2 - \cdots - E_{n-1}.$$

Hint 7. Verify the relation

$$u_{i-1} + u_{i+1} = 2u_i \qquad (1 \le i \le n-1).$$

A formula from the intersection theory of smooth toric surfaces, proved later in the course, says that this relation implies

$$E_i^2 = -2.$$

Record the intersection matrix

$$E_i \cdot E_j = \begin{cases} -2, & i = j, \\ 1, & |i-j| = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the resolution of

$$V(uv - w^n)$$

has an exceptional fiber consisting of $n-1$ rational (-2)-curves arranged in a chain.

Hint 8. Work out the case $n = 5$. List

$$u_0, u_1, \ldots, u_5,$$

draw the refined fan, and draw the corresponding chain

$$E_1 - E_2 - E_3 - E_4.$$

Mark the five torus-fixed points of the resolution on your picture.

Beyond the Gorenstein case

For a general quotient $\mathbb{A}^2/\frac{1}{n}(1,q)$, the same strategy gives a chain of rational curves, but the rays need not be equally spaced. The Hirzebruch–Jung expansion $\frac{n}{q} = [\![b_1, \ldots, b_r]\!]$ determines relations $u_{i-1} + u_{i+1} = b_i u_i$, and hence $E_i^2 = -b_i$. In the Gorenstein case $q = n-1$, all the b_i are 2.

Reference. See Cox–Little–Schenck, Theorem 3.4.11 and Theorems 10.2.3, 10.2.5, and 10.4.4.

Notes and references

The construction of X_Σ, the orbit–cone correspondence, toric morphisms, and completeness follow Cox–Little–Schenck, Chapter 3. The surface examples in the final catalogue are drawn from Examples 3.1.9, 3.1.12, 3.1.16, and 3.1.17.

The blowup examples illustrate the general principle that refinements of fans give proper birational toric morphisms. The challenge exercise looks ahead to Cox–Little–Schenck, Chapter 10, where cyclic quotient surface singularities, Hirzebruch–Jung continued fractions, and their minimal toric resolutions are developed systematically.

The next chapter turns from the orbit stratification to divisors. Every ray

$$\rho \in \Sigma(1)$$

will determine a torus-invariant prime divisor

$$D_\rho = V(\rho),$$

and linear relations among the primitive ray generators will begin to control divisor classes and intersection numbers.

Chapter 4

Divisors from Fans: Rays, Characters, and Piecewise-Linear Functions

Chapter overview

The orbit–cone correspondence tells us that a ray

$$\rho \in \Sigma(1)$$

corresponds to a codimension-one torus orbit. Its closure is therefore an irreducible, torus-invariant prime divisor

$$D_\rho = \overline{O(\rho)}.$$

This is the first principle of the chapter:

Rays record the irreducible components of the toric boundary.

We will use the primitive ray generators u_ρ to measure zeros and poles of characters:

$$\operatorname{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho.$$

These principal divisors provide the relations among the invariant prime divisors and lead to a combinatorial description of the divisor class group.
Cartier divisors will then be described by one local character on each cone. Compatibility of those local equations is equivalent to piecewise linearity on the fan. Thus the chapter develops the passage

rays and integral linear functions
↓
prime divisors and local equations
↓
divisor classes and support functions.

The three dictionaries

Divisors add codimension-one information to the three dictionaries.

Algebra	Geometry	Combinatorics
a valuation along D_ρ	order of zero or pole	the pairing with u_ρ
a principal divisor	relations among boundary components	an element $m \in M$
local monomial equations	a Cartier divisor	linear functions on cones
compatible local equations	a global divisor class	an integral piecewise-linear function

Surface laboratory

For a toric surface, every ray gives a torus-invariant curve and every two-dimensional cone gives a torus-fixed point.
Throughout the chapter, we will repeatedly ask:

- Which boundary curve corresponds to each ray?
- Which curves meet?
- Which integer relations among the ray generators become linear equivalences among divisors?
- Which divisors admit compatible local equations?

4.1 Rays become prime divisors

The previous chapter indexed every torus orbit by a cone:

$$\sigma \in \Sigma \quad \longleftrightarrow \quad O(\sigma) \subseteq X_\Sigma,$$

with

$$\dim O(\sigma) = \operatorname{rank} N - \dim \sigma.$$

We now focus on the cones of dimension one.

4.1.1 A brief divisor dictionary

Algebraic geometry cheatsheet

Let X be an irreducible normal variety.
A *prime divisor* is an irreducible subvariety

$$D \subseteq X$$

of codimension one.
A *Weil divisor* is a finite formal sum

$$D = \sum_i a_i D_i, \qquad a_i \in \mathbb{Z},$$

of prime divisors. The group of Weil divisors is denoted

$$\mathrm{Div}(X).$$

The divisor is *effective*, written $D \geq 0$, if every $a_i \geq 0$. Its support is

$$\mathrm{Supp}(D) = \bigcup_{a_i \neq 0} D_i.$$

At this stage, a Weil divisor should be viewed as codimension-one geometry with integer multiplicities. Zeros, poles, and linear equivalence will enter in the next section.

Normality is important because codimension-one subvarieties carry well-behaved orders of vanishing. Toric varieties constructed from fans are normal, so Weil divisors are available throughout this chapter.

4.1.2 One divisor for every ray

Let

$$\rho \in \Sigma(1).$$

The orbit–cone correspondence gives

$$\dim O(\rho) = \operatorname{rank} N - 1.$$

Its closure is therefore irreducible and has codimension one.

Definition 4.1. The *torus-invariant prime divisor associated to* ρ is

$$D_\rho = V(\rho) = \overline{O(\rho)} \subseteq X_\Sigma.$$

The notation $V(\rho)$ emphasizes orbit closure; the notation D_ρ emphasizes codimension one. We will use D_ρ from now on.

Theorem 4.2 (Rays and invariant prime divisors)**.** *The assignment*

$$\rho \longmapsto D_\rho$$

is a bijection

$$\Sigma(1) \longleftrightarrow \left\{ \begin{array}{c} \textit{torus-invariant} \\ \textit{prime divisors of } X_\Sigma \end{array} \right\}.$$

Reference. The notation $D_\rho = \overline{O(\rho)}$ is introduced at the beginning of Cox–Little–Schenck, §4.1. The characterization of all torus-invariant Weil divisors is Exercise 4.1.1 there.

Proof. Every D_ρ is irreducible because it is the closure of one orbit, and it has codimension one by the orbit-dimension formula. It is torus-invariant because the closure of a torus orbit is preserved by the torus action.

Conversely, let $D \subseteq X_\Sigma$ be a torus-invariant prime divisor. Since the torus acts transitively on itself, the intersection

$$D \cap T$$

is either empty or all of T. It cannot be all of T, because D is a proper closed subset. Hence

$$D \subseteq X_\Sigma \setminus T.$$

Every nonzero cone contains a ray. If

$$\sigma \neq \{0\} \qquad \text{and} \qquad \rho \preceq \sigma,$$

then the orbit–cone correspondence gives

$$O(\sigma) \subseteq \overline{O(\rho)} = D_\rho.$$

Therefore

$$X_\Sigma \setminus T = \bigcup_{\rho \in \Sigma(1)} D_\rho.$$

The irreducible divisor D must lie in one of these finitely many closed sets. Since D and D_ρ are both irreducible of codimension one, they are equal. □

Thus the fan lists every invariant codimension-one subvariety before we write a single equation.

4.1.3 The toric boundary

Definition 4.3. The *toric boundary* is the complement of the dense torus:

$$\partial X_\Sigma = X_\Sigma \setminus T.$$

As a reduced Weil divisor, it is

$$\partial X_\Sigma = \sum_{\rho \in \Sigma(1)} D_\rho.$$

The same symbol is traditionally used for the closed subset and for its associated reduced divisor; the intended meaning will be clear from context.

Example 4.4 (The boundary of $\mathbb{A}^2$)**.** Let

$$\mathbb{A}^2 = \operatorname{Spec} \Bbbk[x, y],$$

and let the rays of its first-quadrant fan be

$$\rho_1 = \operatorname{Cone}(e_1), \qquad \rho_2 = \operatorname{Cone}(e_2).$$

The coordinate axes are precisely the corresponding invariant prime divisors:

$$V(x) = D_{\rho_1}, \qquad V(y) = D_{\rho_2}.$$

Therefore, as a closed subset,

$$\partial\mathbb{A}^2 = V(x) \cup V(y),$$

while as a reduced Weil divisor,

$$\partial\mathbb{A}^2 = D_{\rho_1} + D_{\rho_2} = V(x) + V(y).$$

For a complete toric variety, every direction has been filled in by the fan, but the toric boundary remains: completeness does not mean that the dense torus has no boundary. It means that every one-parameter direction approaches some boundary orbit.

Pause and think

The fan of $\mathbb{P}^1$ has two rays. Identify the corresponding prime divisors, remembering that prime divisors on a curve are points.
What is the toric boundary of $\mathbb{P}^1$?

4.1.4 The fan remembers incidence

The fan records not only the boundary components but also how they meet.

Proposition 4.5 (Boundary incidence)**.** *Let $\rho_1, \dots, \rho_s \in \Sigma(1)$. Then*

$$D_{\rho_1} \cap \cdots \cap D_{\rho_s} \neq \varnothing$$

if and only if the rays $\rho_1, \dots, \rho_s$ are all contained in a cone of Σ.

For a toric surface, two distinct boundary curves meet if and only if their rays are adjacent. When they meet, their intersection is the torus-fixed point corresponding to the two-dimensional cone that they span.

Proof. Suppose the rays lie in a cone σ. Then

$$\rho_i \preceq \sigma$$

for every i, so

$$O(\sigma) \subseteq \overline{O(\rho_i)} = D_{\rho_i}.$$

Thus the intersection is nonempty.

Conversely, let p lie in the intersection. The point p lies in a unique orbit $O(\sigma)$. Since

$$O(\sigma) \subseteq D_{\rho_i},$$

the orbit-closure relation implies

$$\rho_i \preceq \sigma$$

for every i.

On a surface, adjacent rays span a two-dimensional cone, whose orbit is one point. No larger cone exists, so this fixed point is the whole intersection. □

This proposition turns the cyclic ordering of rays in a surface fan into the incidence graph of its boundary curves.

4.1.5 The coordinate triangle in $\mathbb{P}^2$

Let the rays of the fan of $\mathbb{P}^2$ be

$$\rho_0 = \mathrm{Cone}(-e_1 - e_2), \qquad \rho_1 = \mathrm{Cone}(e_1), \qquad \rho_2 = \mathrm{Cone}(e_2).$$

The associated invariant divisors are the coordinate lines

$$D_{\rho_i} = \{z_i = 0\}.$$

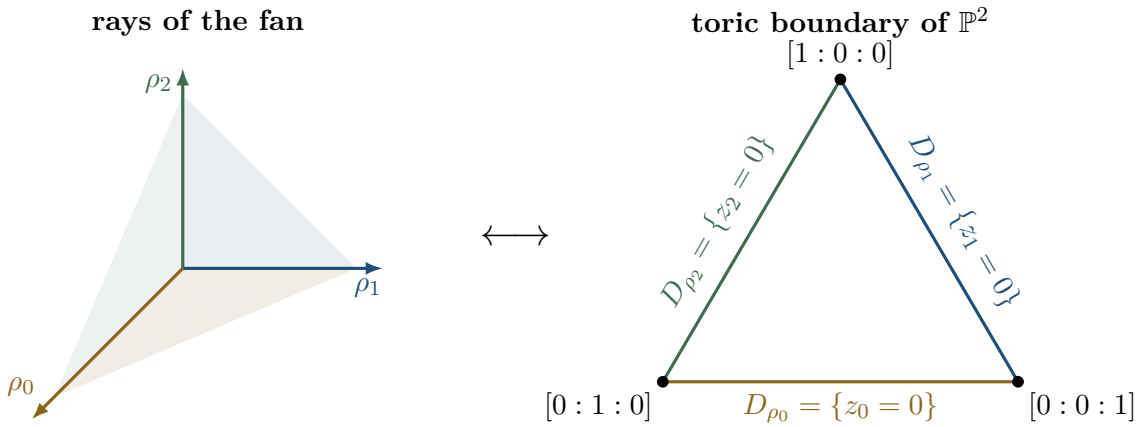


Figure 4.1: The three rays of the fan correspond to the three coordinate lines in $\mathbb{P}^2$. Adjacent rays span maximal cones, just as pairs of coordinate lines meet at coordinate points. The triangle on the right is an incidence picture of the boundary.

Thus

$$\partial\mathbb{P}^2 = D_{\rho_0} + D_{\rho_1} + D_{\rho_2}.$$

Every pair of rays spans a maximal cone, so every pair of coordinate lines meets. The three lines have no common point, matching the fact that no cone contains all three rays.

Surface laboratory

Read the boundary of $\mathbb{P}^2$ from Figure 4.1.

(a) Match each maximal cone with the coordinate point at which its two boundary divisors meet.

(b) Explain why

$$D_{\rho_0} \cap D_{\rho_1} \cap D_{\rho_2} = \varnothing$$

without using homogeneous coordinates.

(c) Remove the ray ρ_0 and every cone containing it. Which toric open subset remains, and which coordinate boundary components does it contain?

4.1.6 A new ray creates the exceptional divisor

Consider again the fan of the blowup

$$\pi : \mathrm{Bl}_0(\mathbb{A}^2) \longrightarrow \mathbb{A}^2.$$

It is obtained by inserting

$$\rho_E = \mathrm{Cone}(e_1 + e_2)$$

between

$$\rho_1 = \mathrm{Cone}(e_1) \qquad \text{and} \qquad \rho_2 = \mathrm{Cone}(e_2).$$

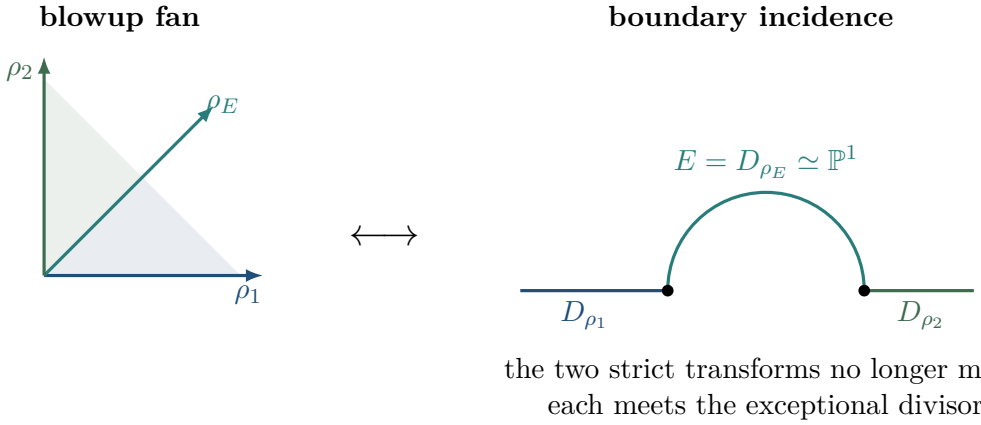


Figure 4.2: The new ray of the blowup fan creates the exceptional divisor. The subdivision also changes incidence: the strict transforms of the coordinate axes are separated by E.

The old rays correspond to the strict transforms of the coordinate axes. The new ray corresponds to the exceptional curve

$$E = D_{\rho_E}.$$

Because ρ_E lies in two maximal cones, its star is the complete one-dimensional fan, so

$$E \simeq \mathbb{P}^1.$$

The original rays ρ_1, ρ_2 no longer span a cone. Hence

$$D_{\rho_1} \cap D_{\rho_2} = \varnothing.$$

Instead,

$$D_{\rho_1} \cap E \neq \varnothing, \qquad E \cap D_{\rho_2} \neq \varnothing.$$

Pause and think

Compare the fans before and after the blowup.
Which single combinatorial change simultaneously explains:

(a) the appearance of a new boundary curve;

(b) the appearance of one additional torus-fixed point;
(c) the separation of the strict transforms of the coordinate axes?

The fan tells us which prime divisors exist, but not yet how to write equations for them or compare them.

Guiding question

How can the integer $\langle m, u_\rho \rangle$ measure the order of vanishing or pole of χ^m along D_ρ?

4.2 Characters and principal divisors

A character

$$\chi^m : T \longrightarrow \mathbb{G}_m, \qquad m \in M,$$

is regular and nowhere zero on the dense torus. Since

$$T \subseteq X_\Sigma$$

is dense, χ^m is a rational function on X_Σ.

It can acquire zeros and poles only when we approach the toric boundary. The primitive ray generators tell us exactly what happens.

4.2.1 The divisor of a character

Algebraic geometry cheatsheet

Let X be a normal irreducible variety, let $D \subseteq X$ be a prime divisor, and let

$$f \in \Bbbk(X)^\times$$

be a nonzero rational function.
The local ring at the generic point of D is a discrete valuation ring. Consequently, near the generic point of D, one can write

$$f = \pi^\ell u,$$

where π is a local equation for D, u is a unit, and $\ell \in \mathbb{Z}$. We define

$$\operatorname{ord}_D(f) = \ell.$$

Thus

$\operatorname{ord}_D(f) > 0$	f has a zero along D,
$\operatorname{ord}_D(f) = 0$	f is a unit at the generic point of D,
$\operatorname{ord}_D(f) < 0$	f has a pole along D.

The *principal divisor* of f is

$$\operatorname{div}(f) = \sum_D \operatorname{ord}_D(f) D,$$

where the sum runs over all prime divisors of X. Only finitely many coefficients are nonzero.

The order is measured at the generic point of the divisor. A function may vanish at isolated points of D without vanishing *along* D.

For a toric divisor D_ρ, we abbreviate

$$\mathrm{ord}_{D_\rho}(f)$$

by

$$\mathrm{ord}_\rho(f).$$

Proposition 4.6 (The order of a character)**.** *Let $\rho \in \Sigma(1)$ have primitive generator u_ρ. Then, for every $m \in M$,*

$$\mathrm{ord}_{D_\rho}(\chi^m) = \langle m, u_\rho \rangle.$$

Reference. This is Cox–Little–Schenck, Proposition 4.1.1.

Conceptual proof sketch. Because u_ρ is primitive, it can be completed to a basis of N. In the resulting toric coordinates,

$$U_\rho \simeq \mathbb{A}^1 \times \mathbb{G}_m^{n-1}, \qquad D_\rho \cap U_\rho = V(x_1).$$

The character χ^m is then $x_1^{\langle m, u_\rho \rangle}$ times a unit along D_ρ. Its order is therefore $\langle m, u_\rho \rangle$. □

Remark 4.7 (Why the primitive generator matters)*.* The valuation ord_{D_ρ} is normalized so that a local parameter has order one. If we replaced the primitive generator by

$$v = k u_\rho, \qquad k > 1,$$

then

$$\langle m, v \rangle = k \langle m, u_\rho \rangle,$$

which would multiply every order by k. The ray determines the divisor, but its *primitive* lattice generator gives the correctly normalized valuation.

The formula has a useful one-parameter interpretation. Along λ^{u_ρ},

$$\chi^m(\lambda^{u_\rho}(t)) = t^{\langle m, u_\rho \rangle}.$$

The same integer therefore controls both the limit along the ray and the order along the corresponding divisor.

Theorem 4.8 (Principal divisor formula)**.** *For every $m \in M$,*

$$\boxed{\mathrm{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho.}$$

Reference. This is Cox–Little–Schenck, Proposition 4.1.2.

Pause and think

Fill in the proof. There are only two steps:

(a) Explain why $\operatorname{div}(\chi^m)$ is supported on the toric boundary.

(b) Use theorem 4.6 to compute the coefficient of every D_ρ.

The signs have a direct geometric meaning:

$$\begin{aligned} \langle m, u_\rho \rangle > 0 &\qquad D_\rho \text{ occurs in the zero divisor,} \\ \langle m, u_\rho \rangle = 0 &\qquad \chi^m \text{ is generically nonzero along } D_\rho, \\ \langle m, u_\rho \rangle < 0 &\qquad D_\rho \text{ occurs in the pole divisor.} \end{aligned}$$

Thus a linear function on $N_\mathbb{R}$ becomes a signed collection of multiplicities on the boundary.

4.2.2 First examples

Example 4.9 (Monomials on $\mathbb{A}^2$)**.** Let

$$\rho_1 = \operatorname{Cone}(e_1), \qquad \rho_2 = \operatorname{Cone}(e_2),$$

so that

$$D_{\rho_1} = V(x), \qquad D_{\rho_2} = V(y).$$

For

$$m = am_1 + bm_2,$$

we have

$$\chi^m = x^a y^b$$

as a rational function, and

$$\operatorname{div}(x^a y^b) = aD_{\rho_1} + bD_{\rho_2}.$$

In particular,

$$\operatorname{div}(x) = D_{\rho_1}, \qquad \operatorname{div}(y) = D_{\rho_2},$$

while

$$\operatorname{div}(x/y) = D_{\rho_1} - D_{\rho_2}.$$

Example 4.10 (Zeros and poles on $\mathbb{P}^1$)**.** Let

$$\rho_+ = \mathbb{R}_{\geq 0}, \qquad \rho_- = \mathbb{R}_{\leq 0},$$

with primitive generators 1 and -1. The character

$$t = \chi^1$$

satisfies

$$\operatorname{div}(t) = D_{\rho_+} - D_{\rho_-}.$$

Under the usual identification,

$$D_{\rho_+} = \{0\}, \qquad D_{\rho_-} = \{\infty\},$$

so this is the familiar formula

$$\operatorname{div}(t) = [0] - [\infty].$$

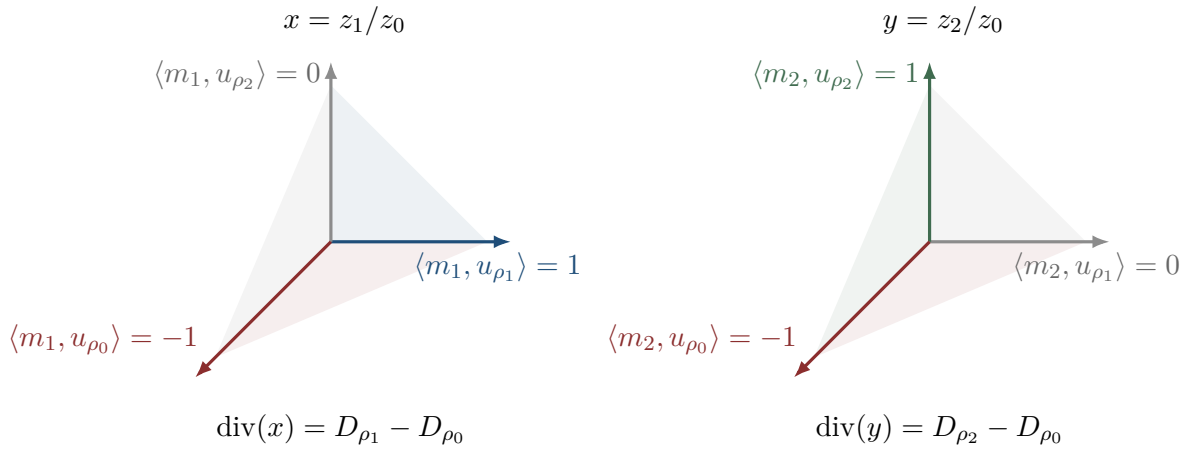
4.2.3 The coordinate characters on $\mathbb{P}^2$

Let

$$u_{\rho_0} = -e_1 - e_2, \qquad u_{\rho_1} = e_1, \qquad u_{\rho_2} = e_2.$$

Write

$$x = \chi^{m_1} = \frac{z_1}{z_0}, \qquad y = \chi^{m_2} = \frac{z_2}{z_0}.$$



$$\operatorname{div}(x) = D_{\rho_1} - D_{\rho_0} \qquad \operatorname{div}(y) = D_{\rho_2} - D_{\rho_0}$$

Figure 4.3: The coefficients in the divisor of a character are obtained by evaluating the corresponding linear function on the primitive ray generators. Positive coefficients record zeros, red negative coefficients record poles, and gray zero coefficients contribute nothing.

The figure gives

$$\operatorname{div}(x) = D_{\rho_1} - D_{\rho_0}, \qquad \operatorname{div}(y) = D_{\rho_2} - D_{\rho_0}.$$

More generally,

$$\operatorname{div}(x^a y^b) = aD_{\rho_1} + bD_{\rho_2} - (a+b)D_{\rho_0}.$$

This agrees with homogeneous coordinates:

$$x^a y^b = \frac{z_1^a z_2^b}{z_0^{a+b}}.$$

The numerator records the zeros, and the denominator records the poles.

4.2.4 Principal divisors and linear equivalence

Definition 4.11. Two Weil divisors D, D' are *linearly equivalent*, written

$$D \sim D',$$

if

$$D - D' = \operatorname{div}(f)$$

for some $f \in \Bbbk(X)^\times$.

A divisor of the form $\operatorname{div}(f)$ is called *principal*. Thus every principal divisor is linearly equivalent to zero.

The character calculations on $\mathbb{P}^2$ imply

$$D_{\rho_1} \sim D_{\rho_0}, \qquad D_{\rho_2} \sim D_{\rho_0}.$$

Hence the three coordinate lines are linearly equivalent.

This is the first appearance of a general principle:

$$\text{characters} \quad \rightsquigarrow \quad \text{relations among invariant divisors.}$$

Pause and think

Let

$$m = 2m_1 - m_2.$$

Use Figure 4.3, or pair directly with the three ray generators, to compute

$$\operatorname{div}(x^2/y).$$

Which coordinate lines occur as zeros, and which occur as poles?

4.2.5 The blowup remembers total transforms

Let

$$\pi : \mathrm{Bl}_0(\mathbb{A}^2) \longrightarrow \mathbb{A}^2$$

be the toric blowup. Its rays have primitive generators

$$e_1, \qquad e_1 + e_2, \qquad e_2.$$

Write

$$D_1 = D_{\mathrm{Cone}(e_1)}, \qquad E = D_{\mathrm{Cone}(e_1+e_2)}, \qquad D_2 = D_{\mathrm{Cone}(e_2)}.$$

The pairing table is

	e_1	$e_1 + e_2$	e_2
m_1	1	1	0
m_2	0	1	1.

Therefore

$$\operatorname{div}(x) = D_1 + E, \qquad \operatorname{div}(y) = D_2 + E.$$

Surface laboratory

Interpret these formulas geometrically.

(a) Explain why D_1 and D_2 are the strict transforms of the coordinate axes.

(b) Explain why E appears in both

$$\operatorname{div}(x) \qquad \text{and} \qquad \operatorname{div}(y).$$

Relate this to the fact that both coordinate axes pass through the blown-up point.

(c) Compute

$$\operatorname{div}(x/y).$$

Why does the exceptional divisor cancel?

(d) Use the formulas to obtain the linear equivalences

$$D_1 \sim -E, \qquad D_2 \sim -E$$

on the noncomplete surface $\mathrm{Bl}_0(\mathbb{A}^2)$.

Warning

The equality

$$\operatorname{div}(x) = D_1 + E$$

is an equality of divisors on the blowup. It should not be confused with the zero set of the function x on the original affine plane.
The divisor on the blowup records the total transform: the strict transform D_1 and the exceptional contribution E.

We now know the torus-invariant prime divisors and the relations among them produced by rational functions. The next step is to decide when two divisors should count as representing the same geometry.

Guiding question

Why might it be useful to regard two divisors as equivalent whenever their difference is the divisor of a rational function?
What geometric information would you expect the quotient

$$\operatorname{Div}(X_\Sigma)/\operatorname{Prin}(X_\Sigma)$$

to retain, and what information would it deliberately forget?

4.3 Weil divisors and the divisor class group

The preceding sections produced two pieces of information:

$$\rho \in \Sigma(1) \quad \longmapsto \quad D_\rho,$$

and

$$m \in M \quad \longmapsto \quad \operatorname{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho.$$

We now organize the invariant prime divisors modulo the relations coming from characters.

4.3.1 Invariant Weil divisors

Definition 4.12. The group of *torus-invariant Weil divisors* is

$$\operatorname{Div}_T(X_\Sigma) = \bigoplus_{\rho \in \Sigma(1)} \mathbb{Z} D_\rho.$$

Thus an invariant Weil divisor has the form

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho, \qquad a_\rho \in \mathbb{Z}.$$

Since the divisors D_ρ are distinct prime divisors, they form a basis of this free abelian group. After ordering the rays, we may identify

$$\operatorname{Div}_T(X_\Sigma) \simeq \mathbb{Z}^{\Sigma(1)}.$$

Definition 4.13. The *divisor class group* is

$$\operatorname{Cl}(X) = \operatorname{Div}(X) / \operatorname{Prin}(X),$$

the group of Weil divisors modulo principal divisors.

The class of a divisor D is denoted

$$[D] \in \operatorname{Cl}(X).$$

Thus

$$[D] = [D'] \quad \Longleftrightarrow \quad D \sim D'.$$

At first sight, computing $\operatorname{Cl}(X_\Sigma)$ seems to require all prime divisors of X_Σ, not merely the invariant ones. The torus action makes the answer much simpler.

Theorem 4.14 (The class-group sequence)**.** *There is an exact sequence*

$$M \xrightarrow{\operatorname{div}} \operatorname{Div}_T(X_\Sigma) \longrightarrow \operatorname{Cl}(X_\Sigma) \longrightarrow 0,$$

where

$$m \longmapsto \operatorname{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho,$$

and the second map sends an invariant divisor to its divisor class.

The first map is injective if and only if the primitive ray generators span $N_\mathbb{R}$. Equivalently, the sequence is short exact precisely when X_Σ has no torus factor.

Reference. This is Cox–Little–Schenck, Theorem 4.1.3.

Conceptual proof sketch. The complement of the dense torus is the union of the invariant prime divisors:

$$X_\Sigma \setminus T = \bigcup_{\rho \in \Sigma(1)} D_\rho.$$

Since the coordinate ring

$$\Bbbk[M]$$

of the torus is a Laurent polynomial ring, it is a unique factorization domain and

$$\mathrm{Cl}(T) = 0.$$

The localization sequence for divisor class groups therefore says that every divisor class on X_Σ has a representative supported on the boundary. This proves surjectivity:

$$\mathrm{Div}_T(X_\Sigma) \twoheadrightarrow \mathrm{Cl}(X_\Sigma).$$

Now suppose an invariant divisor is principal:

$$D = \mathrm{div}(f).$$

Its support misses the torus, so the restriction of f to T has neither zeros nor poles. The units of $\Bbbk[M]$ are

$$c\chi^m, \qquad c \in \Bbbk^\times, \quad m \in M.$$

Hence

$$D = \mathrm{div}(\chi^m),$$

which identifies the kernel.

Finally,

$$\mathrm{div}(\chi^m) = 0$$

exactly when

$$\langle m, u_\rho \rangle = 0 \qquad \text{for every } \rho.$$

Thus the first map is injective exactly when the u_ρ span $N_\mathbb{R}$. □

The most striking part of the theorem is the surjectivity statement:

Every Weil divisor class has a torus-invariant representative.

A general divisor need not itself be invariant. It is only its class that can always be represented by a sum of boundary divisors.

Pause and think

Suppose $\Sigma = \{\{0\}\}$ in a nonzero lattice N. Then

$$X_\Sigma = T_N.$$

What are

$$\mathrm{Div}_T(X_\Sigma) \qquad \text{and} \qquad \mathrm{Cl}(X_\Sigma)?$$

Why is the map

$$M \longrightarrow \mathrm{Div}_T(X_\Sigma)$$

not injective?

4.3.2 A matrix for the class group

Choose a basis

$$m_1, \dots, m_n$$

of M, and order the rays

$$\Sigma(1) = \{\rho_1, \dots, \rho_r\}.$$

Write the primitive generators as column vectors

$$u_i = u_{\rho_i} \in \mathbb{Z}^n.$$

Under the identifications

$$M \simeq \mathbb{Z}^n, \qquad \mathrm{Div}_T(X_\Sigma) \simeq \mathbb{Z}^r,$$

the divisor map is represented by the $r \times n$ matrix

$$A = \begin{pmatrix} - - - u_1^T - - - \\ - - - u_2^T - - - \\ \vdots \\ - - - u_r^T - - - \end{pmatrix}.$$

Indeed, its i-th entry is

$$\langle m, u_i \rangle.$$

Consequently,

$$\mathrm{Cl}(X_\Sigma) \simeq \mathrm{coker}(A).$$

Observation 4.15. The Smith normal form of A separates the free and torsion parts of the class group. When the rays span $N_\mathbb{R}$,

$$\operatorname{rank} \mathrm{Cl}(X_\Sigma) = |\Sigma(1)| - \operatorname{rank} N.$$

Torsion can still occur: it measures a finite-index lattice phenomenon rather than a failure of the rays to span the real vector space.

For hand calculations, it is often faster to write down the relations

$$\sum_\rho \langle m_i, u_\rho \rangle [D_\rho] = 0$$

coming from a basis $m_1, \dots, m_n$ of M.

4.3.3 Three familiar complete surfaces

Example 4.16 (The projective plane)**.** For $\mathbb{P}^2$, take

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_2 = e_2.$$

The characters m_1, m_2 give the relations

$$-D_0 + D_1 \sim 0, \qquad -D_0 + D_2 \sim 0.$$

Thus

$$[D_0] = [D_1] = [D_2].$$

Writing their common class as H, we obtain

$$\mathrm{Cl}(\mathbb{P}^2) \simeq \mathbb{Z}H.$$

The three coordinate lines are different divisors but represent the same divisor class.

Example 4.17 (The product $\mathbb{P}^1 \times \mathbb{P}^1$)**.** Order the rays

$$e_1, \quad e_2, \quad -e_1, \quad -e_2,$$

and denote the corresponding divisors by

$$D_1, D_2, D_3, D_4.$$

The two character relations are

$$D_1 - D_3 \sim 0, \qquad D_2 - D_4 \sim 0.$$

Hence

$$\mathrm{Cl}(\mathbb{P}^1 \times \mathbb{P}^1) \simeq \mathbb{Z}[D_1] \oplus \mathbb{Z}[D_2].$$

The two generators have a geometric interpretation:

$$[D_1] = [D_3]$$

is the class of a fiber of one projection, while

$$[D_2] = [D_4]$$

is the class of a fiber of the other.

These two examples illustrate the rank count:

$$3 - 2 = 1 \quad \text{for } \mathbb{P}^2, \qquad 4 - 2 = 2 \quad \text{for } \mathbb{P}^1 \times \mathbb{P}^1.$$

4.3.4 The blowup of $\mathbb{P}^2$

Blow up the fixed point corresponding to

$$\mathrm{Cone}(e_1, e_2).$$

The resulting fan has primitive ray generators

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2.$$

Let the corresponding divisors be

$$D_0, \qquad D_1, \qquad E, \qquad D_2.$$

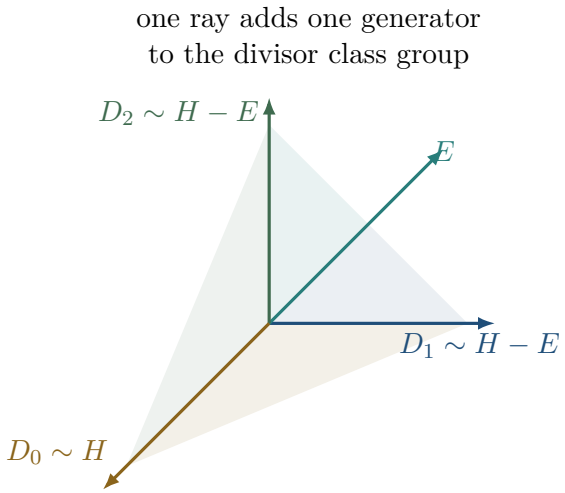


Figure 4.4: Divisor classes on $\mathrm{Bl}_p(\mathbb{P}^2)$. The new ray contributes the exceptional class E; the two boundary lines through the blown-up point have class $H - E$.

The character relations are

$$-D_0 + D_1 + E \sim 0, \qquad -D_0 + E + D_2 \sim 0.$$

Therefore

$$[D_1] = [D_0] - [E], \qquad [D_2] = [D_0] - [E].$$

If

$$H = [D_0],$$

then

$$\mathrm{Cl}(\mathrm{Bl}_p(\mathbb{P}^2)) \simeq \mathbb{Z}H \oplus \mathbb{Z}E.$$

Geometrically, D_0 is the strict transform of the coordinate line that misses the blown-up point, so its class is the pullback of a line. The divisors D_1, D_2 are strict transforms of lines through the point, hence their class is

$$H - E.$$

Surface laboratory

Use the fan and the principal-divisor formula to recover the class group of $\mathrm{Bl}_p(\mathbb{P}^2)$.

(a) Write the 4×2 matrix representing

$$M \longrightarrow \mathrm{Div}_T(X).$$

(b) Verify directly that its cokernel is free of rank two.

(c) Explain why adding one ray to the fan of $\mathbb{P}^2$ increases the rank of the class group by one.

(d) Compare the relation

$$[D_1] = H - E$$

with the total-transform formula

$$\pi^* L = \widetilde{L} + E$$

for a line L through the blown-up point.

4.3.5 A singular comparison: torsion

Let

$$X_n = \mathbb{A}^2/\tfrac{1}{n}(1, n-1)$$

be the A_{n-1} cyclic quotient from the Chapter 3 challenge. Its fan consists of the first-quadrant cone in

$$N = \mathbb{Z}^2 + \mathbb{Z}\left(\frac{1}{n}(1, n-1)\right).$$

The two primitive ray generators are still

$$e_1, e_2,$$

but they generate a sublattice

$$\mathbb{Z}e_1 + \mathbb{Z}e_2 \subseteq N$$

of index n.

The class-group matrix has full real rank, but its image is not saturated. Its Smith normal form gives

$$\mathrm{Cl}(X_n) \simeq \mathbb{Z}/n\mathbb{Z}.$$

Thus the class group detects the finite quotient hidden in the lattice. The contrast is worth remembering:

extra rays	free divisor classes
finite-index lattice change	torsion divisor classes.

Pause and think

Why does the resolution fan of X_n, which inserts $n-1$ new rays and is smooth, have a torsion-free class group?
Use only the facts that adjacent primitive generators form lattice bases and that the full collection of ray generators therefore spans N as an abelian group.

The class group treats Weil divisors modulo rational equations. The next question is more local: when can a divisor itself be described by one equation on each affine chart?

Guiding question

For an invariant divisor

$$D = \sum_{\rho} a_\rho D_\rho,$$

when can we choose, on every U_σ, a character whose divisor agrees with D along all rays of σ?

4.4 Cartier divisors and local equations

A Weil divisor records integer multiplicities along codimension-one subvarieties. A Cartier divisor asks for more: those multiplicities must arise from compatible local equations.

4.4.1 From multiplicities to local equations

Algebraic geometry cheatsheet

Let X be an irreducible variety.
A *Cartier divisor* is described by an open cover

$$X = \bigcup_i U_i$$

and nonzero rational functions

$$f_i \in \Bbbk(X)^\times$$

such that

$$f_i/f_j$$

is regular and nowhere zero on every overlap $U_i \cap U_j$.
The function f_i is a *local equation* for the divisor on U_i. On a normal variety, the local equations determine a Weil divisor by

$$D|_{U_i} = \operatorname{div}(f_i)|_{U_i}.$$

Thus a Cartier divisor is a Weil divisor that is locally principal. A global principal divisor is the special case in which one rational function works on all of X.

We will identify a Cartier divisor with its associated Weil divisor. In particular, an invariant Weil divisor

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

is Cartier precisely when it can be cut out locally by one rational function on each affine chart.

The toric setting is especially rigid: invariant local equations can be chosen to be characters.

Proposition 4.18 (Invariant Cartier divisors on an affine chart)**.** *Let $\sigma \subseteq N_{\mathbb{R}}$ be a strongly convex rational polyhedral cone. Every torus-invariant Cartier divisor on*

$$U_\sigma$$

is the divisor of a character. Consequently,

$$\mathrm{Pic}(U_\sigma) = 0.$$

Reference. This is Cox–Little–Schenck, Proposition 4.2.2.

Conceptual proof sketch. The affine chart U_σ contains a unique closed torus orbit. Begin with a local equation near that orbit. Because the divisor is torus-invariant, translating the equation by the torus changes it only by a unit.

The key step is to replace the local equation by one of its torus eigencomponents. Eigenfunctions for the torus action are characters, so one obtains a character having the same divisor near the closed orbit. Every invariant boundary divisor of U_σ meets that orbit, forcing equality of the two divisors on the entire affine chart.

For a general, not necessarily effective, Cartier divisor, add a sufficiently positive principal character divisor, apply the effective case, and subtract it again. □

The technical point in this proposition is not that every Cartier divisor on an affine variety is principal—that statement is false in general. It is the combination of affineness and torus invariance that forces a character equation.

4.4.2 Cartier data on a fan

Let

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho.$$

On the affine chart U_σ, only the boundary divisors corresponding to rays of σ appear:

$$D|_{U_\sigma} = \sum_{\rho \in \sigma(1)} a_\rho D_\rho.$$

If this restriction is the divisor of χ^{-m_σ}, then the principal-divisor formula gives

$$a_\rho = \langle -m_\sigma, u_\rho \rangle = -\langle m_\sigma, u_\rho \rangle.$$

This explains the signs in the following criterion.

Theorem 4.19 (Toric Cartier criterion)**.** *Let*

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be a torus-invariant Weil divisor. The following are equivalent:

(a) D is Cartier.
(b) $D|_{U_\sigma}$ is principal for every $\sigma \in \Sigma$.
(c) For every $\sigma \in \Sigma$, there exists

$$m_\sigma \in M$$

such that

$$\langle m_\sigma, u_\rho \rangle = -a_\rho \qquad \text{for every } \rho \in \sigma(1).$$

(d) The condition in (c) *holds for every maximal cone of Σ.*

When these conditions hold, the local equation on U_σ is

$$\chi^{-m_\sigma}.$$

Moreover:

$$m_\sigma \text{ is unique modulo } M(\sigma) := \sigma^\perp \cap M,$$

and whenever $\tau \preceq \sigma$,

$$m_\sigma \equiv m_\tau \pmod{M(\tau)}.$$

The collection

$$\{m_\sigma\}_{\sigma \in \Sigma}$$

is called the Cartier data *of D.*

Reference. This is Cox–Little–Schenck, Theorem 4.2.8.

Proof. A Cartier divisor restricts to an invariant Cartier divisor on every affine chart. By theorem 4.18, that restriction is the divisor of a character. The principal-divisor formula shows that the character can be written as χ^{-m_σ} precisely when

$$\langle m_\sigma, u_\rho \rangle = -a_\rho$$

for all rays of σ. This proves the equivalence of (a), (b), and (c).

It is enough to work on maximal cones because their affine charts cover X_Σ, and any character equation for a maximal cone restricts to every face.

If m_σ and m'_σ give the same coefficients, then

$$\langle m_\sigma - m'_\sigma, u_\rho \rangle = 0$$

on every ray of σ, hence on all of σ. Thus

$$m_\sigma - m'_\sigma \in \sigma^\perp \cap M.$$

The compatibility along faces follows by applying this uniqueness statement to the restricted divisor on U_τ. □

Warning

The coefficient convention is

$$D = \sum_\rho a_\rho D_\rho, \qquad \langle m_\sigma, u_\rho \rangle = -a_\rho,$$

because the local equation is

$$\chi^{-m_\sigma}.$$

It is easy to lose this minus sign when passing between a divisor and its Cartier data.

If σ is full-dimensional, then

$$M(\sigma) = 0,$$

so m_σ is uniquely determined. For a lower-dimensional cone, one may add any character that is trivial on the subtorus directions seen by σ.

Pause and think

Suppose τ is a ray. Why can two different elements of M give the same local equation data along D_τ?
Describe the ambiguity geometrically using

$$M(\tau) = \tau^\perp \cap M.$$

4.4.3 A coordinate line on $\mathbb{P}^2$

Let the primitive ray generators be

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_2 = e_2,$$

and let

$$D_0 = \{z_0 = 0\}.$$

The maximal cones are

$$\sigma_0 = \mathrm{Cone}(u_1, u_2), \qquad \sigma_1 = \mathrm{Cone}(u_0, u_2), \qquad \sigma_2 = \mathrm{Cone}(u_0, u_1).$$

For

$$D_0 = 1D_0 + 0D_1 + 0D_2,$$

the Cartier data and local equations are:

Chart	Rays in the cone	Cartier datum	Local equation
U_{σ_0}	u_1, u_2	$m_{\sigma_0} = 0$	1
U_{σ_1}	u_0, u_2	$m_{\sigma_1} = m_1$	$\chi^{-m_1} = z_0/z_1$
U_{σ_2}	u_0, u_1	$m_{\sigma_2} = m_2$	$\chi^{-m_2} = z_0/z_2$

On U_{σ_0}, the divisor D_0 is absent, so the local equation is a unit. On the other two charts, the displayed ratios are the familiar equations of the coordinate line.

On an overlap, the quotient of two local equations is a character whose pairing with every ray of the common face is zero. It is therefore a unit on that overlap.

Surface laboratory

Repeat the calculation for

$$D_1 = \{z_1 = 0\}.$$

(a) Find the three elements

$$m_{\sigma_0}, m_{\sigma_1}, m_{\sigma_2}.$$

(b) Write the local equations in homogeneous coordinates.

(c) Verify explicitly that the ratios of the equations are units on the pairwise overlaps.

4.4.4 A Weil divisor that is not Cartier

Return to the cyclic quotient

$$X_n = \mathbb{A}^2/\tfrac{1}{n}(1, n-1).$$

Its cone is the first quadrant, but its character lattice is

$$M = \left\{(a,b) \in \mathbb{Z}^2 : a - b \equiv 0 \pmod{n}\right\}.$$

Let D_1, D_2 be the invariant prime divisors corresponding to the two coordinate rays.

Since X_n is affine, an invariant Cartier divisor must be the divisor of a character. To have

$$\operatorname{div}(\chi^m) = D_1,$$

we would need

$$m = (1, 0).$$

But

$$(1,0) \notin M \qquad (n > 1),$$

so D_1 is not Cartier.

On the other hand,

$$(n, 0) \in M,$$

and

$$\operatorname{div}(\chi^{(n,0)}) = nD_1.$$

Thus D_1 is not Cartier, but a positive multiple of it is.

Definition 4.20. A Weil divisor D is $\mathbb{Q}$-*Cartier* if some positive integer multiple of D is Cartier.

In this example, D_1 is $\mathbb{Q}$-Cartier of Cartier index n. The same is true of D_2.

This gives a sharp distinction:

$$\operatorname{Pic}(X_n) = 0, \qquad \operatorname{Cl}(X_n) \simeq \mathbb{Z}/n\mathbb{Z}.$$

The class group sees Weil divisors that do not admit local equations; the Picard group sees only Cartier classes.

4.4.5 The exceptional divisor on the blowup

Let

$$X = \mathrm{Bl}_p(\mathbb{P}^2)$$

with ray generators

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2.$$

We compute local equations for the exceptional divisor $E = D_{\rho_E}$.

local equations for the exceptional divisor E

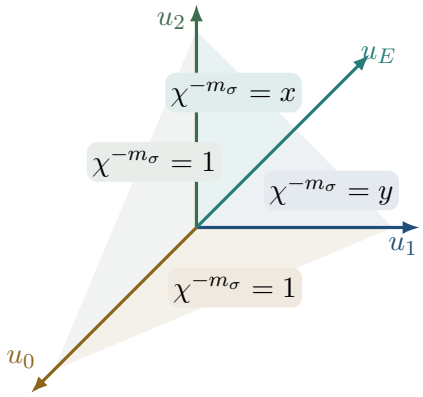


Figure 4.5: Cartier data for the exceptional divisor. On the two charts meeting E, its local equations are y and x; on the other two charts, E is absent and the equation is 1.

For the four maximal cones, we may take

maximal cone	m_σ	χ^{-m_σ}
$\mathrm{Cone}(u_0, u_1)$	0	1
$\mathrm{Cone}(u_1, u_E)$	$-m_2$	y
$\mathrm{Cone}(u_E, u_2)$	$-m_1$	x
$\mathrm{Cone}(u_2, u_0)$	0	1.

Indeed, on the cone $\mathrm{Cone}(u_1, u_E)$,

$$\langle -m_2, u_1 \rangle = 0, \qquad \langle -m_2, u_E \rangle = -1,$$

and similarly on $\mathrm{Cone}(u_E, u_2)$.

The two nontrivial equations agree up to a unit on their overlap:

$$\frac{x}{y}$$

is invertible there. This is the local-equation version of the gluing that produced the blowup.

Observation 4.21. The calculation succeeds because the primitive generators of each maximal cone form a lattice basis. Given arbitrary integer coefficients on the two rays, there is a unique integral linear function taking the required values.

This is the local reason smooth toric varieties have no distinction between Weil and Cartier divisors. We will return to the precise criterion later in this chapter.

4.4.6 Cartier classes

Cartier divisors modulo principal divisors form the Picard group:

$$\operatorname{Pic}(X) = \operatorname{CDiv}(X)/\operatorname{Prin}(X).$$

Equivalently, this is the group of line bundles on X; that interpretation will become more prominent in the next chapter.

Every Cartier class on a toric variety has an invariant representative. Consequently, there is an exact sequence

$$M \longrightarrow \operatorname{CDiv}_T(X_\Sigma) \longrightarrow \operatorname{Pic}(X_\Sigma) \longrightarrow 0.$$

Reference. This is Cox–Little–Schenck, Theorem 4.2.1.

The difference between

$$\operatorname{Pic}(X_\Sigma) \qquad \text{and} \qquad \operatorname{Cl}(X_\Sigma)$$

measures the failure of Weil divisors to admit compatible local equations.

The Cartier data are a collection of integral linear functions, one on each cone, compatible on common faces. There should be a more economical way to package them.

Guiding question

Can the collection

$$\{m_\sigma\}_{\sigma\in\Sigma}$$

be assembled into one function on $|\Sigma|$ that is linear on each cone and remembers the coefficients of the divisor on every ray?

4.5 Support functions

The Cartier data

$$\{m_\sigma\}_{\sigma\in\Sigma}$$

assign one integral linear function to every cone. Compatibility says that these linear functions agree wherever the cones meet.

This collection can therefore be viewed as one function on the entire support of the fan.

4.5.1 One function, linear on every cone

Definition 4.22. Let Σ be a fan in $N_\mathbb{R}$.

A *support function* is a function

$$\psi : |\Sigma| \longrightarrow \mathbb{R}$$

whose restriction to every cone $\sigma \in \Sigma$ is linear.

The support function is *integral* if

$$\psi(|\Sigma| \cap N) \subseteq \mathbb{Z}.$$

We write

$$\mathrm{SF}(\Sigma)$$

for the group of support functions and

$$\mathrm{SF}(\Sigma, N)$$

for the subgroup of integral support functions.

Because adjacent linear pieces agree on their common face, a support function is automatically continuous. It is also homogeneous:

$$\psi(tu) = t\psi(u) \qquad (t \geq 0).$$

Thus "piecewise linear" here means linear, rather than affine linear, on each cone.

For a rational cone σ, integrality is equivalent to the existence of some $m_\sigma \in M$ such that

$$\psi(u) = \langle m_\sigma, u \rangle \qquad (u \in \sigma).$$

The element m_σ is determined only modulo

$$M(\sigma) = \sigma^\perp \cap M,$$

exactly as in the Cartier criterion.

Reference. This is Cox–Little–Schenck, Definition 4.2.11.

4.5.2 Cartier divisors and integral support functions

Let

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be a torus-invariant Cartier divisor with Cartier data

$$\{m_\sigma\}_{\sigma \in \Sigma}.$$

Define

$$\psi_D(u) = \langle m_\sigma, u \rangle \qquad \text{when } u \in \sigma.$$

Theorem 4.23 (The support-function dictionary)**.** *The function ψ_D is a well-defined integral support function. Its values on the primitive ray generators recover the divisor:*

$$\psi_D(u_\rho) = -a_\rho,$$

and hence

$$\boxed{D = -\sum_{\rho \in \Sigma(1)} \psi_D(u_\rho) D_\rho.}$$

Moreover, the assignment

$$D \longmapsto \psi_D$$

is an isomorphism of abelian groups

$$\mathrm{CDiv}_T(X_\Sigma) \simeq \mathrm{SF}(\Sigma, N).$$

Reference. This is Cox–Little–Schenck, Theorem 4.2.12.

Proof. If $u \in \sigma \cap \sigma'$, compatibility of the Cartier data gives

$$m_\sigma - m_{\sigma'} \in (\sigma \cap \sigma')^\perp \cap M.$$

Therefore

$$\langle m_\sigma, u \rangle = \langle m_{\sigma'}, u \rangle,$$

so ψ_D is well defined. It is integral and linear on each cone because every m_σ lies in M.

For a ray $\rho \preceq \sigma$, the Cartier criterion gives

$$\psi_D(u_\rho) = \langle m_\sigma, u_\rho \rangle = -a_\rho.$$

Thus the ray values recover D, proving injectivity.

Conversely, let

$$\psi \in \mathrm{SF}(\Sigma, N).$$

On each cone, choose $m_\sigma \in M$ with

$$\psi|_\sigma = \langle m_\sigma, - \rangle.$$

The pieces agree on common faces, so these m_σ are compatible Cartier data. The divisor

$$D_\psi = -\sum_{\rho \in \Sigma(1)} \psi(u_\rho) D_\rho$$

has support function ψ. This proves surjectivity. □

The theorem compresses the Cartier criterion into a picture:

local characters m_σ	$\longleftrightarrow$	linear pieces of ψ_D
$\langle m_\sigma, u_\rho \rangle = -a_\rho$	$\longleftrightarrow$	$\psi_D(u_\rho) = -a_\rho$
compatibility on overlaps	$\longleftrightarrow$	one well-defined function on $\lvert\Sigma\rvert$.

Warning

An arbitrary assignment of integers to the rays need not extend to a support function. On every cone, the assigned values must come from one linear function.

This is exactly the obstruction to the corresponding invariant Weil divisor being Cartier.

4.5.3 The support function of a line in $\mathbb{P}^2$

Consider

$$D_0 = \{z_0 = 0\} \subseteq \mathbb{P}^2.$$

With

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_2 = e_2,$$

the ray values are

$$\psi_{D_0}(u_0) = -1, \qquad \psi_{D_0}(u_1) = 0, \qquad \psi_{D_0}(u_2) = 0.$$

The Cartier data from the previous section give

$$\psi_{D_0}(x, y) = \begin{cases} 0, & (x, y) \in \mathrm{Cone}(e_1, e_2), \\ x, & (x, y) \in \mathrm{Cone}(-e_1 - e_2, e_2), \\ y, & (x, y) \in \mathrm{Cone}(e_1, -e_1 - e_2). \end{cases}$$

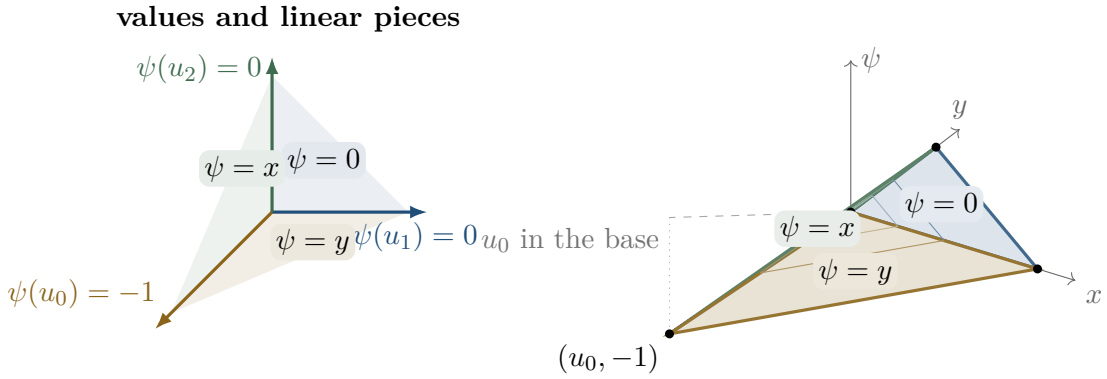


Figure 4.6: The support function of the coordinate line D_0. The right-hand panel shows a truncated portion of its graph in $N_{\mathbb{R}} \times \mathbb{R}$. Its three facets lie in the planes $\psi = 0$, $\psi = x$, and $\psi = y$.

The right-hand picture is the graph

$$\Gamma_{\psi_{D_0}} = \{(u, \psi_{D_0}(u)) : u \in N_{\mathbb{R}}\} \subseteq N_{\mathbb{R}} \times \mathbb{R}.$$

Only a bounded triangular portion of each unbounded planar piece is drawn. Over $\mathrm{Cone}(u_1, u_2)$ the graph is horizontal; over $\mathrm{Cone}(u_0, u_2)$ it lies in the plane $\psi = x$; and over $\mathrm{Cone}(u_1, u_0)$ it lies in the plane $\psi = y$. The three planes meet along the lifted common rays.

The bending of the graph records that D_0 is not principal. A principal divisor would produce one global linear function, with no change of slope from cone to cone.

Pause and think

Use the ray values in Figure 4.6 to recover the divisor D_0 from

$$D = -\sum_{\rho} \psi_D(u_\rho) D_\rho.$$

Then verify directly that the three displayed formulas agree on every common ray.

4.5.4 Linear equivalence means changing the global tilt

Let D be an invariant Cartier divisor and let $m \in M$. Since

$$\operatorname{div}(\chi^m) = \sum_{\rho} \langle m, u_\rho \rangle D_\rho,$$

we obtain

$$\psi_{D+\operatorname{div}(\chi^m)}(u) = \psi_D(u) - \langle m, u \rangle.$$

Thus adding a principal divisor changes the support function by one global integral linear function. It does not alter how the linear pieces bend relative to one another.

Consequently, the Picard group has the presentation

$$M \longrightarrow \mathrm{SF}(\Sigma, N) \longrightarrow \mathrm{Pic}(X_\Sigma) \longrightarrow 0,$$

where

$$m \longmapsto (u \longmapsto -\langle m, u \rangle).$$

Equivalently,

$$\mathrm{Pic}(X_\Sigma) \simeq \frac{\mathrm{SF}(\Sigma, N)}{\{\text{restrictions of global integral linear functions}\}}.$$

Reference. This is the support-function form of Cox–Little–Schenck, Theorem 4.2.1, following Theorem 4.2.12.

Example 4.24 (Why the coordinate lines of $\mathbb{P}^2$ are equivalent)**.** Let ψ_0 and ψ_1 be the support functions of D_0 and D_1. Since

$$D_1 - D_0 = \operatorname{div}(z_1/z_0) = \operatorname{div}(\chi^{m_1}),$$

we have

$$\psi_1(u) = \psi_0(u) - \langle m_1, u \rangle.$$

Their graphs differ by a global tilt, so they represent the same Picard class.

This gives a useful geometric interpretation:

a Cartier class remembers the bending of a support function, not its overall tilt.

4.5.5 Surface laboratory: the exceptional divisor

Let

$$X = \mathrm{Bl}_p(\mathbb{P}^2)$$

and use the rays

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2.$$

For the exceptional divisor E, the support function satisfies

$$\psi_E(u_E) = -1, \qquad \psi_E(u_0) = \psi_E(u_1) = \psi_E(u_2) = 0.$$

Surface laboratory

(a) Use the Cartier data from Figure 4.5 to show that

$$\psi_E(x,y) = \begin{cases} 0, & \mathrm{Cone}(u_0,u_1) \text{ and } \mathrm{Cone}(u_2,u_0), \\ -y, & \mathrm{Cone}(u_1,u_E), \\ -x, & \mathrm{Cone}(u_E,u_2). \end{cases}$$

(b) Check that the four pieces agree on their common rays.

(c) Sketch the graph of ψ_E. Where does it bend?

(d) Explain why the newly inserted ray is exactly where the support function acquires a new linear piece.

(e) Compute the support function of the invariant divisor

$$D_0 - E$$

by subtracting support functions. Which Picard class does this divisor represent?

Observation 4.25. Support functions will return in the next chapter with additional shape conditions. Convexity and strict convexity will translate into global generation and ampleness, respectively, while the supporting inequalities

$$\langle m, u_\rho \rangle \geq -a_\rho$$

will cut out a polyhedron in $M_{\mathbb{R}}$.

For now, the essential point is only the Cartier dictionary: integral piecewise-linear functions encode compatible local equations.

A particularly important invariant divisor assigns the same coefficient to every boundary component. Its Cartier condition asks for all primitive ray generators of each cone to lie on a common integral affine hyperplane.

Guiding question

What geometric divisor has coefficient -1 on every D_ρ, and how does the existence of a support function with

$$\psi(u_\rho) = 1$$

detect whether that divisor is Cartier?

4.6 The canonical divisor and singularity criteria

The toric boundary is the reduced divisor

$$\partial X_\Sigma = \sum_{\rho \in \Sigma(1)} D_\rho.$$

The canonical divisor is its negative:

$$K_{X_\Sigma} = -\partial X_\Sigma.$$

This simple formula brings together every dictionary developed in the chapter. It also lets us recognize several singularity properties directly from the primitive ray generators.

4.6.1 The canonical divisor is minus the boundary

Algebraic geometry cheatsheet

Let X be a smooth irreducible variety of dimension n. A nonzero rational top differential form

$$\omega$$

has an order of vanishing along every prime divisor. Its divisor

$$\operatorname{div}(\omega)$$

is called a *canonical divisor*.
Different rational top forms differ by a nonzero rational function, so their divisors are linearly equivalent. Thus the canonical divisor class

$$[K_X]$$

is intrinsic even though a particular representative is not.
For a normal variety, the smooth locus contains every codimension-one point. One may therefore compute the canonical divisor on the smooth locus and take the closures of its codimension-one components.

Choose a basis

$$m_1, \dots, m_n$$

of M, and write

$$x_i = \chi^{m_i}.$$

On the dense torus, consider the invariant logarithmic volume form

$$\Omega_T = \frac{dx_1}{x_1} \wedge \cdots \wedge \frac{dx_n}{x_n}.$$

Theorem 4.26 (The canonical divisor of a toric variety)**.** *For every normal toric variety X_Σ,*

$$\boxed{K_{X_\Sigma} = -\sum_{\rho \in \Sigma(1)} D_\rho}$$

is a torus-invariant canonical divisor. Equivalently,

$$-K_{X_\Sigma} = \partial X_\Sigma.$$

Reference. This is Cox–Little–Schenck, Theorem 8.2.3.

Proof. The form Ω_T is regular and nowhere zero on the dense torus, so any zeros or poles occur along the boundary.

Fix a ray ρ. Because its generator u_ρ is primitive, it extends to a basis

$$u_\rho = e_1, e_2, \ldots, e_n$$

of N. Let $m_1, \ldots, m_n$ now denote the dual basis. On the ray chart,

$$U_\rho \simeq \mathbb{A}^1 \times \mathbb{G}_m^{n-1},$$

with

$$D_\rho \cap U_\rho = V(x_1).$$

Up to a nonzero constant, the invariant form becomes

$$\Omega_T = \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge \cdots \wedge \frac{dx_n}{x_n}.$$

The factors $x_2, \ldots, x_n$ are units along D_ρ, while dx_1/x_1 has a simple pole. Therefore

$$\operatorname{ord}_{D_\rho}(\Omega_T) = -1.$$

This holds for every ray, proving the formula. □

Observation 4.27. The displayed divisor is one convenient canonical representative, not necessarily the divisor of the most familiar volume form.

For example, on $\mathbb{A}^2$,

$$dx \wedge dy$$

has divisor 0, while

$$\frac{dx}{x} \wedge \frac{dy}{y} = \frac{1}{xy}\, dx \wedge dy$$

has divisor

$$-D_{\rho_1} - D_{\rho_2}.$$

The two canonical divisors differ by

$$\operatorname{div}(1/xy).$$

4.6.2 Canonical classes of familiar surfaces

The formula becomes especially transparent once the boundary classes are known.

Surface	Boundary class	Canonical class
$\mathbb{P}^2$	$D_0 + D_1 + D_2 \sim 3H$	$K_{\mathbb{P}^2} \sim -3H$
$\mathbb{P}^1 \times \mathbb{P}^1$	$2F_1 + 2F_2$	$K \sim -2F_1 - 2F_2$
$\mathrm{Bl}_p(\mathbb{P}^2)$	$D_0 + D_1 + E + D_2 \sim 3H - E$	$K \sim -3H + E$

For the blowup, recall

$$[D_0] = H, \qquad [D_1] = [D_2] = H - E.$$

Hence

$$-K = D_0 + D_1 + E + D_2 \sim H + (H - E) + E + (H - E) = 3H - E.$$

Pause and think

Compute the canonical class of the Hirzebruch surface $\mathbb{F}_r$ by summing its four invariant boundary divisors.
At this stage, it is enough to express the answer using any two boundary divisor classes that generate $\mathrm{Cl}(\mathbb{F}_r)$.

4.6.3 Factorial and $\mathbb{Q}$-factorial toric varieties

Definition 4.28. A normal variety is *factorial* if every Weil divisor is Cartier.

It is $\mathbb{Q}$-*factorial* if every Weil divisor is $\mathbb{Q}$-Cartier; that is, every Weil divisor has a positive multiple that is Cartier.

These conditions ask whether arbitrary codimension-one multiplicities admit integral or rational local equations.

Theorem 4.29 (Cartier properties and the fan)**.** *Let X_Σ be a normal toric variety.*

(a) The following are equivalent:

$$X_\Sigma \text{ is smooth}, \qquad X_\Sigma \text{ is factorial}, \qquad \mathrm{Pic}(X_\Sigma) = \mathrm{Cl}(X_\Sigma).$$

(b) The following are equivalent:

$$\Sigma \text{ is simplicial}, \qquad X_\Sigma \text{ is } \mathbb{Q}\text{-factorial}, \qquad \mathrm{Pic}(X_\Sigma) \text{ has finite index in } \mathrm{Cl}(X_\Sigma).$$

Reference. These are Cox–Little–Schenck, Propositions 4.2.6 and 4.2.7.

Conceptual proof sketch. Work on an affine chart

$$U_\sigma$$

with ray generators

$$u_1, \dots, u_s.$$

Prescribing a boundary divisor means prescribing arbitrary integers

$$a_1, \dots, a_s.$$

The Cartier criterion asks whether the linear system

$$\langle m, u_i \rangle = -a_i$$

has a solution $m \in M$.

It has an integral solution for every integer right-hand side exactly when the u_i extend to a lattice basis. This is the smoothness criterion.

It has a rational solution for every rational right-hand side exactly when the u_i are linearly independent over $\mathbb{R}$. This says that σ is simplicial.

Applying the local statement to every cone gives the theorem. The statements involving Pic and Cl merely rephrase whether all Weil classes are Cartier or become Cartier after multiplication. □

Observation 4.30. Every strongly convex two-dimensional cone is simplicial. Therefore every normal toric surface is $\mathbb{Q}$-factorial, even though it may be singular.

Thus, for toric surfaces,

$$\text{smooth} \iff \text{factorial},$$

but $\mathbb{Q}$-factorial is automatic.

The cyclic quotient

$$X_n = \mathbb{A}^2/\tfrac{1}{n}(1, n-1)$$

is a good test case. For $n > 1$, it is singular and therefore not factorial. Its cone is simplicial, so it is nevertheless $\mathbb{Q}$-factorial.

4.6.4 Gorenstein and $\mathbb{Q}$-Gorenstein cones

Definition 4.31. A normal variety is *Gorenstein* if its canonical divisor is Cartier. It is $\mathbb{Q}$-*Gorenstein* if its canonical divisor is $\mathbb{Q}$-Cartier.

Because

$$K_{X_\Sigma} = -\sum_\rho D_\rho,$$

its coefficient along every boundary divisor is -1. The Cartier criterion therefore becomes a height-one condition.

Proposition 4.32 (The height-one criterion)**.** *Let X_Σ be a normal toric variety.*

(a) X_Σ is Gorenstein if and only if, for every maximal cone σ, there exists $m_\sigma \in M$ such that

$$\langle m_\sigma, u_\rho \rangle = 1 \qquad \text{for every } \rho \in \sigma(1).$$

(b) X_Σ is $\mathbb{Q}$-Gorenstein if and only if the same condition holds with

$$m_\sigma \in M_{\mathbb{Q}}.$$

Reference. This is Cox–Little–Schenck, Proposition 8.2.12; the Gorenstein terminology is Definition 8.2.14.

The condition says that the primitive ray generators of each maximal cone lie on an affine hyperplane

$$\{u \in N_{\mathbb{R}} : \langle m_\sigma, u \rangle = 1\}.$$

The normal vector must be integral for Gorensteinness and merely rational for $\mathbb{Q}$-Gorenstein.

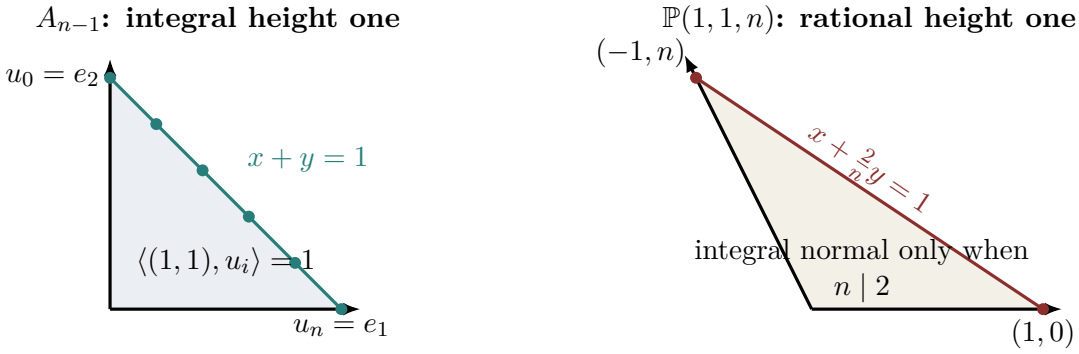


Figure 4.7: Gorensteinness is an integral height-one condition. For the A_{n-1} cone, the character $(1,1)$ is integral in the intrinsic character lattice. For the singular cone of $\mathbb{P}(1,1,n)$, the height-one functional is $(1, 2/n)$, which is integral only for $n = 1, 2$.

4.6.5 Two singular comparisons

Example 4.33 (The A_{n-1} singularity is Gorenstein)**.** For

$$X_n = \mathbb{A}^2/\tfrac{1}{n}(1, n-1),$$

the character lattice is

$$M = \{(a,b) \in \mathbb{Z}^2 : a - b \equiv 0 \pmod{n}\}.$$

The element

$$m = (1,1)$$

belongs to M, and

$$\langle m, e_1 \rangle = \langle m, e_2 \rangle = 1.$$

Hence X_n is Gorenstein.

This recovers geometrically the special role of the equation

$$uv - w^n = 0.$$

The singularity is not smooth for $n > 1$, but its canonical divisor is Cartier.

Moreover, the rays in the resolution from the Chapter 3 challenge,

$$u_i = \left(\frac{i}{n}, \frac{n-i}{n}\right),$$

all satisfy

$$\langle (1,1), u_i \rangle = 1.$$

The new rays remain on the same height-one line.

Observation 4.34. For the general quotient

$$\mathbb{A}^2/\tfrac{1}{n}(1,q),$$

the unique rational functional taking value 1 on e_1, e_2 is

$$m = (1,1).$$

It lies in the intrinsic character lattice exactly when

$$1 + q \equiv 0 \pmod{n}.$$

For $1 \le q < n$, this means

$$q = n - 1.$$

Thus the Gorenstein quotients in this family are precisely the A_{n-1} singularities. Every member of the family is $\mathbb{Q}$-Gorenstein, and its Gorenstein index is

$$\frac{n}{\gcd(n, q+1)}.$$

Example 4.35 (The weighted projective plane)**.** The fan of $\mathbb{P}(1,1,n)$ has a singular cone generated by

$$u_0 = (1,0), \qquad u_1 = (-1,n).$$

The height-one equations force

$$m = \left(1, \frac{2}{n}\right).$$

Therefore

$$\mathbb{P}(1,1,n) \text{ is Gorenstein} \quad \Longleftrightarrow \quad n \mid 2.$$

Hence the singular surface

$$\mathbb{P}(1,1,2)$$

is Gorenstein, while $\mathbb{P}(1,1,n)$ is only $\mathbb{Q}$-Gorenstein for $n > 2$.

The fan is simplicial for every n, so all these weighted projective planes are $\mathbb{Q}$-factorial.

4.6.6 A recognition table

Property	Divisor formulation	Fan formulation
smooth / factorial	every Weil divisor is Cartier	every cone is smooth
$\mathbb{Q}$-factorial	every Weil divisor is $\mathbb{Q}$-Cartier	every cone is simplicial
Gorenstein	K_X is Cartier	each maximal cone has an integral height-one functional
$\mathbb{Q}$-Gorenstein	K_X is $\mathbb{Q}$-Cartier	each maximal cone has a rational height-one functional

The implications should not be confused:

$$\text{smooth} \Longrightarrow \begin{cases} \text{factorial}, \\ \text{Gorenstein}, \\ \mathbb{Q}\text{-factorial}. \end{cases}$$

But a singular toric variety may still be Gorenstein, $\mathbb{Q}$-factorial, or both.

The chapter has translated divisors into coefficients on rays, relations from characters, local linear equations, and support functions. The next step is to ask which characters become global sections.

Guiding question

Given

$$D = \sum_{\rho} a_\rho D_\rho,$$

why should the inequalities

$$\langle m, u_\rho \rangle \geq -a_\rho \qquad \text{for every } \rho$$

select exactly the characters χ^m for which

$$\operatorname{div}(\chi^m) + D \geq 0?$$

What geometric object in $M_{\mathbb{R}}$ is cut out by these inequalities?

What this chapter has accomplished

The rays of a fan record the irreducible components of the toric boundary:

$$\rho \in \Sigma(1) \quad \longleftrightarrow \quad D_\rho = \overline{O(\rho)}.$$

Characters then measure zeros and poles along those components:

$$\operatorname{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho.$$

From this formula, the chapter developed the sequence

$$\begin{array}{|r|} \hline \text{rays} \longrightarrow \text{invariant prime divisors}, \\ \text{characters} \longrightarrow \text{principal relations}, \\ \text{Cartier data} \longrightarrow \text{compatible local equations}, \\ \text{support functions} \longrightarrow \text{one piecewise-linear encoding}. \\ \hline \end{array}$$

The two basic divisor groups are organized by

$$M \longrightarrow \operatorname{Div}_T(X_\Sigma) \longrightarrow \operatorname{Cl}(X_\Sigma) \longrightarrow 0$$

and

$$M \longrightarrow \operatorname{SF}(\Sigma, N) \longrightarrow \operatorname{Pic}(X_\Sigma) \longrightarrow 0.$$

The difference between Cl and Pic measures the failure of Weil divisors to admit compatible local equations.

Finally,

$$K_{X_\Sigma} = -\sum_{\rho \in \Sigma(1)} D_\rho$$

turned singularity properties into fan conditions:

$$\begin{array}{rcc} \text{factorial} & \longleftrightarrow & \text{smooth fan}, \\ \mathbb{Q}\text{-factorial} & \longleftrightarrow & \text{simplicial fan}, \\ \text{Gorenstein} & \longleftrightarrow & \text{integral height-one data}, \\ \mathbb{Q}\text{-Gorenstein} & \longleftrightarrow & \text{rational height-one data}. \end{array}$$

The next chapter begins with a divisor

$$D = \sum_\rho a_\rho D_\rho$$

and asks which characters satisfy

$$\operatorname{div}(\chi^m) + D \geq 0.$$

The answer will be encoded by lattice points in the polyhedron

$$P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho\}.$$

Comprehension exercises

4.1. Let $X = \mathbb{P}^1$, with boundary divisors

$$D_0 = \{0\}, \qquad D_\infty = \{\infty\}.$$

(a) Use the character t to compute

$$\mathrm{Cl}(\mathbb{P}^1).$$

(b) Write Cartier data for D_0 on the two affine charts.

(c) Determine the support function of D_0.

(d) Recover

$$K_{\mathbb{P}^1} \sim -2[D_0].$$

4.2. Let

$$X = \mathbb{A}^1 \times \mathbb{G}_m$$

with its usual toric structure.

(a) Describe its fan and its invariant prime divisors.

(b) Compute the map

$$M \longrightarrow \mathrm{Div}_T(X).$$

(c) Determine $\mathrm{Cl}(X)$.

(d) Find the kernel of the divisor map and explain it using the torus factor.

4.3. For the fan of $\mathbb{P}^2$, let

$$D = a_0 D_0 + a_1 D_1 + a_2 D_2.$$

(a) Show directly that every such invariant Weil divisor is Cartier.

(b) Find Cartier data on the three maximal cones.

(c) Prove

$$[D] = (a_0 + a_1 + a_2)H.$$

(d) Characterize the principal invariant divisors among these triples.

4.4. For $\mathbb{P}^1 \times \mathbb{P}^1$, order the invariant boundary divisors cyclically as

$$D_1, D_2, D_3, D_4.$$

(a) Recover the two relations

$$D_1 \sim D_3, \qquad D_2 \sim D_4.$$

(b) Compute the support function of $D_1 + D_2$.

(c) Express the canonical class in the basis

$$[D_1], [D_2].$$

4.5. Let $\mathbb{F}_r$ have rays

$$(1,0), \quad (0,1), \quad (-1,r), \quad (0,-1),$$

with corresponding divisors

$$D_1, D_2, D_3, D_4.$$

(a) Compute the two principal relations among the D_i.

(b) Determine $\mathrm{Cl}(\mathbb{F}_r)$.

(c) Express $K_{\mathbb{F}_r}$ using two chosen generators.

(d) Check that every invariant Weil divisor is Cartier.

4.6. Consider the affine cone

$$\sigma = \mathrm{Cone}((1,0),(1,d)) \subseteq \mathbb{R}^2, \qquad d \geq 2,$$

in the lattice $\mathbb{Z}^2$.

(a) Compute $\mathrm{Cl}(U_\sigma)$.

(b) Show that each invariant prime divisor is $\mathbb{Q}$-Cartier.

(c) Determine the smallest positive multiple of the divisor corresponding to $(1,0)$ that is Cartier.

(d) Decide when U_σ is Gorenstein.

4.7. Assign integers

$$c_0, c_1, c_2$$

to the three rays of the fan of $\mathbb{P}^2$.

(a) Show that there is a unique support function taking those values on the rays.

(b) Write its three linear pieces explicitly.

(c) Which choice of c_i gives a global linear function?

(d) Translate your answer into a statement about principal divisors.

4.8. Let Σ consist of the cone

$$\mathrm{Cone}(e_1, e_2, e_3, e_1 + e_2 - e_3) \subseteq \mathbb{R}^3$$

and all its faces.

(a) Explain why the maximal cone is not simplicial.

(b) Construct a Weil divisor whose coefficients on the four rays cannot arise from a rational linear function.

(c) Deduce that X_Σ is not $\mathbb{Q}$-factorial.

4.9. Let $X = \mathbb{P}(1,1,n)$.

(a) Use the fan to verify that X is $\mathbb{Q}$-factorial.

(b) Compute the rational height-one functional on its singular cone.

(c) Determine the Cartier index of K_X.

(d) Verify that X is Gorenstein precisely when $n \mid 2$.

4.10. Let Σ' be a refinement of a fan Σ, with the same support, and let

$$\pi : X_{\Sigma'} \longrightarrow X_\Sigma$$

be the induced proper birational toric morphism. Suppose D is a torus-invariant Cartier divisor on X_Σ with support function ψ_D.

(a) Explain why the same function is a support function on Σ'.

(b) Show that the corresponding divisor on $X_{\Sigma'}$ is $\pi^* D$.

(c) If ρ' is a new ray, determine its coefficient in $\pi^* D$ from $\psi_D(u_{\rho'})$.

Challenge exercise

Divisors and the crepant resolution of an A_{n-1} singularity.

Assume $\Bbbk = \mathbb{C}$, and consider

$$X_n = \mathbb{A}^2/\tfrac{1}{n}(1, n-1) \simeq V(uv - w^n) \subseteq \mathbb{A}^3.$$

Its fan consists of the first-quadrant cone

$$\sigma = \mathrm{Cone}(e_2, e_1)$$

in

$$N = \mathbb{Z}^2 + \mathbb{Z}\left(\frac{1}{n}(1, n-1)\right).$$

The intrinsic character lattice is

$$M = \left\{(a, b) \in \mathbb{Z}^2 : a - b \equiv 0 \pmod{n}\right\}.$$

As in the Chapter 3 challenge, define

$$u_i = \left(\frac{i}{n}, \frac{n-i}{n}\right), \qquad 0 \le i \le n.$$

Thus

$$u_0 = e_2, \qquad u_n = e_1.$$

Let Σ' be the smooth fan with maximal cones

$$\sigma_i = \mathrm{Cone}(u_{i-1}, u_i), \qquad 1 \le i \le n,$$

and let

$$\pi : Y = X_{\Sigma'} \longrightarrow X_n$$

be the resulting resolution.

Write

$$D_0 = V(\mathrm{Cone}(u_0)), \qquad D_n = V(\mathrm{Cone}(u_n)),$$

for the strict transforms of the two original boundary divisors, and

$$E_i = V(\mathrm{Cone}(u_i)), \qquad 1 \le i \le n-1,$$

for the exceptional curves.

The goal is to understand how divisor classes and canonical divisors behave under the resolution.

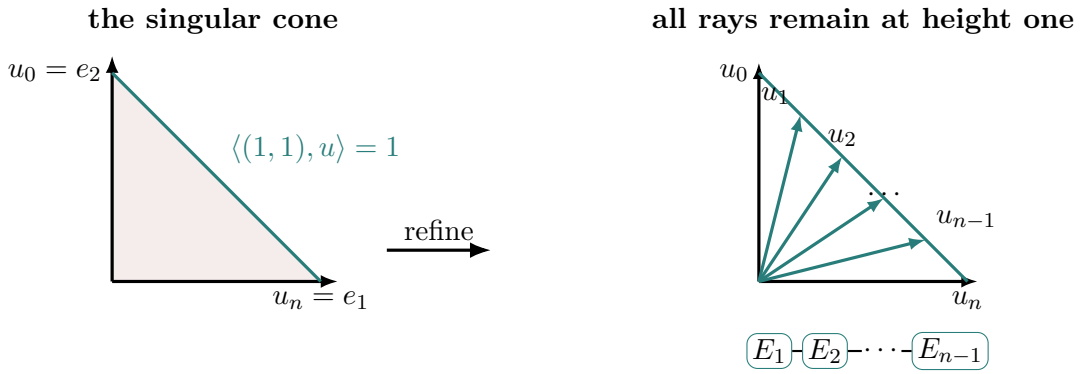


Figure 4.8: The resolution inserts the intermediate lattice rays. Every ray lies on the same integral height-one line, which will force the resolution to preserve the canonical divisor.

Hint 1. The class group of the singularity.
Let

$$B_0 = V(\mathrm{Cone}(u_0)), \qquad B_n = V(\mathrm{Cone}(u_n))$$

be the invariant boundary divisors on X_n. Show that

$$m_0 = (1,1), \qquad m_1 = (n,0)$$

form a basis of M, and compute

$$\mathrm{div}_{X_n}(\chi^{m_0}) = B_0 + B_n,$$

$$\mathrm{div}_{X_n}(\chi^{m_1}) = nB_n.$$

Deduce that

$$\mathrm{Cl}(X_n) \simeq \mathbb{Z}/n\mathbb{Z},$$

generated by $[B_n]$, with

$$[B_0] = -[B_n].$$

Hint 2. The class group upstairs.
On Y, write

$$D_i = \begin{cases} D_0, & i = 0, \\ E_i, & 1 \le i \le n-1, \\ D_n, & i = n. \end{cases}$$

Verify

$$\langle m_0, u_i \rangle = 1, \qquad \langle m_1, u_i \rangle = i.$$

Hence

$$\mathrm{div}_Y(\chi^{m_0}) = \sum_{i=0}^{n} D_i,$$

and

$$\operatorname{div}_Y(\chi^{m_1}) = \sum_{i=0}^{n} iD_i.$$

Use these two relations to show that

$$\operatorname{Cl}(Y) \simeq \mathbb{Z}^{n-1}.$$

Why is the answer torsion-free without performing a Smith normal form calculation?

Hint 3. What the exceptional curves generate.
Let $L \subseteq \operatorname{Cl}(Y)$ be the subgroup generated by

$$[E_1], \dots, [E_{n-1}].$$

Show that

$$\operatorname{Cl}(Y)/L \simeq \mathbb{Z}/n\mathbb{Z}.$$

Compare this quotient with $\operatorname{Cl}(X_n)$.
Explain informally why contracting the exceptional chain should recover the torsion class group of the singular surface.

Hint 4. Cartier data for one exceptional curve.
Fix $1 \le i \le n-1$. Since Y is smooth, E_i is Cartier. Show that the only maximal cones on which a nontrivial local equation is needed are

$$\sigma_i = \operatorname{Cone}(u_{i-1}, u_i), \qquad \sigma_{i+1} = \operatorname{Cone}(u_i, u_{i+1}).$$

Define

$$m_i^- = (i-1-n, i-1), \qquad m_i^+ = (n-i-1, -i-1).$$

Verify that both belong to M and that

$$\begin{array}{c|cc} & u_{i-1} & u_i \\ \hline m_i^- & 0 & -1 \end{array}, \qquad \begin{array}{c|cc} & u_i & u_{i+1} \\ \hline m_i^+ & -1 & 0. \end{array}$$

Conclude that Cartier data for E_i are

$$m_{\sigma_i} = m_i^-, \qquad m_{\sigma_{i+1}} = m_i^+, \qquad m_\sigma = 0$$

on every other maximal cone.
Write the corresponding local character equations and check their compatibility on overlaps.

Hint 5. Total transforms of the boundary.
Since

$$nB_n = \operatorname{div}_{X_n}(\chi^{m_1}),$$

pulling back this principal divisor gives

$$\pi^*(nB_n) = \operatorname{div}_Y(\chi^{m_1}).$$

Deduce

$$\boxed{\pi^*(nB_n) = nD_n + \sum_{i=1}^{n-1} iE_i.}$$

Similarly, use

$$(0, n) = nm_0 - m_1$$

to show

$$\boxed{\pi^*(nB_0) = nD_0 + \sum_{i=1}^{n-1}(n-i)E_i.}$$

How do the increasing and decreasing coefficients reflect the positions of the rays in the fan?

Hint 6. The canonical divisor downstairs.
Use the toric canonical-divisor formula to write

$$K_{X_n} = -B_0 - B_n.$$

Show that

$$K_{X_n} = \mathrm{div}_{X_n}(\chi^{-m_0}).$$

Thus the canonical divisor is not merely Cartier: this chosen representative is principal.
Relate this calculation to the fact that the two primitive ray generators lie on the integral height-one line

$$\langle m_0, u\rangle = 1.$$

Hint 7. The canonical divisor upstairs.
Since every ray of Σ' satisfies

$$\langle m_0, u_i\rangle = 1,$$

show that

$$K_Y = -\sum_{i=0}^{n} D_i = \mathrm{div}_Y(\chi^{-m_0}).$$

Now use pullback of principal divisors to conclude

$$\boxed{K_Y = \pi^* K_{X_n}.}$$

A resolution satisfying this equality is called *crepant*. Explain, using only the fan picture, why no exceptional divisor appears with a nonzero correction coefficient.

Hint 8. The support-function interpretation.
The support function of K_{X_n} is the global linear function

$$\psi_K(u) = \langle m_0, u\rangle.$$

Regard the same function on the refined fan Σ'.

(a) What coefficient does the pullback divisor assign to each new ray u_i?
(b) Why does this agree with the coefficient of E_i in K_Y?

(c) Explain why the equality

$$K_Y = \pi^* K_{X_n}$$

is equivalent to saying that every inserted ray remains at height one.

Hint 9. A concrete case.
Work out the case $n = 5$.

(a) Draw the six rays $u_0, \dots, u_5$.
(b) Write the two principal relations in $\mathrm{Div}_T(Y)$.
(c) Compute Cartier data for E_2.
(d) Write the total transforms of $5B_0$ and $5B_5$.
(e) Verify directly that

$$K_Y = \pi^* K_{X_5}.$$

Beyond the A_{n-1} case

For a general cyclic quotient $\mathbb{A}^2/\frac{1}{n}(1,q)$, the canonical support function is rational and takes value 1 on the two original rays. A new primitive ray u contributes

$$\langle m_K, u\rangle - 1$$

to the coefficient of its exceptional curve in $K_Y - \pi^* K_X$.
For A_{n-1}, every inserted ray remains at height 1, so all these corrections vanish.

Reference. The divisor class sequence is Cox–Little–Schenck, Theorem 4.1.3, and support functions are developed in §4.2. Canonical divisors and the height-one criterion are treated in §8.2. The toric resolution of cyclic quotient surface singularities and its exceptional chain are developed in Chapter 10, especially §§10.2 and 10.4.

Notes and references

The invariant divisor sequence, the Cartier criterion, and support functions follow Cox–Little–Schenck, Chapter 4. The formula

$$K_{X_\Sigma} = -\sum_{\rho \in \Sigma(1)} D_\rho$$

and the Gorenstein height-one criterion are developed in Cox–Little–Schenck, §8.2.

The chapter deliberately stops before global sections and positivity. Those topics require one further translation:

$$\operatorname{div}(\chi^m) + D \geq 0 \quad \Longleftrightarrow \quad \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho.$$

Chapter 5 will turn these inequalities into polyhedra, interpret their lattice points as sections, and connect convex support functions to projective embeddings.

The challenge exercise returns to the A_{n-1} resolution from Chapter 3. It illustrates the principle that a toric resolution is crepant precisely when the new rays remain on the canonical height-one hyperplane.

Chapter 5

Projective Toric Varieties and Polytopes

Chapter overview

Chapter 4 translated codimension-one geometry into data on the rays of a fan. We now reverse the direction: the inequalities attached to a divisor carve out a region in the character space $M_{\mathbb{R}}$, and the lattice points in that region become global sections.
The central construction is

$$D = \sum_{\rho} a_\rho D_\rho \quad \longmapsto \quad P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho\}.$$

It begins a new chain of translations:

divisors $\longrightarrow$ polyhedra and lattice points $\longrightarrow$ linear systems and projective maps.

Later in the chapter, convexity and strict convexity of support functions will detect basepoint freeness and ampleness. Normal fans will then explain how polytopes build projective toric varieties.

The three dictionaries

The three dictionaries now meet in the theory of sections.

Algebra	Geometry	Combinatorics
characters and monomial sections	line bundles and linear systems	lattice points in P_D
section multiplication	projective maps	Minkowski addition
graded section rings	projective embeddings	polytopes and normal fans

Surface laboratory

For a toric surface, every inequality cuts out a half-plane in $M_{\mathbb{R}}$. The polyhedron P_D can therefore be drawn explicitly.
Under a toric blowup, the new ray contributes one new inequality. On the polytope side, this often cuts off a corner. We will repeatedly use this visible operation to interpret vanishing conditions, positivity, and projective maps on blown-up surfaces.

5.1 Divisor polyhedra and global sections

Algebraic geometry cheatsheet

Let X be a normal irreducible variety and let D be a Weil divisor. The sheaf $\mathcal{O}_X(D)$ is defined inside the constant sheaf of rational functions by

$$\Gamma(U, \mathcal{O}_X(D)) = \{f \in \Bbbk(X)^\times : (\operatorname{div}(f) + D)|_U \geq 0\} \cup \{0\}.$$

Thus

$$H^0(X, \mathcal{O}_X(D)) = \{f \in \Bbbk(X)^\times : \operatorname{div}(f) + D \geq 0\} \cup \{0\}.$$

The inequality has a concrete meaning. The rational function f may have poles, but along every prime divisor its pole order must be no larger than the multiplicity supplied by D. Equivalently, after adding the zeros and poles of f to D, the result must be effective.
When D is Cartier, $\mathcal{O}_X(D)$ is a line bundle and has a canonical rational section s_D with

$$\operatorname{div}(s_D) = D.$$

Multiplication by this section identifies rational functions with rational sections:

$$f \longmapsto f s_D.$$

Under this identification, $f s_D$ is regular exactly when

$$\operatorname{div}(f) + D \geq 0.$$

Hence the inequality is precisely the regularity condition for the corresponding section of the line bundle.
For a non-Cartier Weil divisor on a normal variety, the same formula defines the associated rank-one reflexive sheaf.

Now let

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be torus invariant. By the principal-divisor formula from theorem 4.8,

$$\operatorname{div}(\chi^m) + D = \sum_{\rho \in \Sigma(1)} (\langle m, u_\rho \rangle + a_\rho) D_\rho.$$

Therefore

$$\chi^m \in H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))$$

precisely when

$$\langle m, u_\rho \rangle \geq -a_\rho \qquad \text{for every } \rho \in \Sigma(1).$$

When D is Cartier, this says equivalently that the rational section $\chi^m s_D$ is regular. These inequalities cut out a polyhedron in $M_\mathbb{R}$.

5.1.1 The monomial basis theorem

Proposition 5.1 (Global sections decompose into characters)**.** *Let*

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be a torus-invariant Weil divisor on X_Σ. Then

$$H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) = \bigoplus_{\operatorname{div}(\chi^m)+D \geq 0} \Bbbk\, \chi^m.$$

Reference. This is Cox–Little–Schenck, Proposition 4.3.2.

Proof. A global section f restricts to a regular function on the dense torus, because D has no support there. Hence

$$f|_T \in \Bbbk[M].$$

The space of global sections is stable under the torus action, while $\Bbbk[M]$ has the character basis

$$\{\chi^m : m \in M\}.$$

Therefore the space of sections is the direct sum of the character weight spaces that it contains. Finally,

$$\chi^m \in H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))$$

if and only if

$$\operatorname{div}(\chi^m) + D \geq 0.$$

□

Observation 5.2. The short representation-theoretic phrase "weight-space decomposition" is optional language for a simple fact: Laurent monomials form a basis of $\Bbbk[M]$, and a torus-stable subspace is spanned by the monomials it contains.

Definition 5.3 (The polyhedron of a divisor)**.** For a torus-invariant Weil divisor

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho,$$

define

$$\boxed{P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho \in \Sigma(1)\}.}$$

Each ray contributes one closed half-space. The coefficient a_ρ controls the position of its boundary hyperplane:

$$\langle m, u_\rho \rangle = -a_\rho.$$

Increasing a_ρ moves that wall outward and weakens the pole restriction along D_ρ.

Theorem 5.4 (Lattice points are monomial sections)**.** *For every torus-invariant Weil divisor D,*

$$\boxed{H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) = \bigoplus_{m \in P_D \cap M} \Bbbk\, \chi^m.}$$

In particular, whenever $P_D \cap M$ is finite,

$$h^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) = |P_D \cap M|.$$

Reference. This is Cox–Little–Schenck, Proposition 4.3.3.

Proof. By theorem 4.8,

$$\operatorname{div}(\chi^m) + D \geq 0$$

if and only if

$$\langle m, u_\rho \rangle + a_\rho \geq 0$$

for every ray ρ. This is exactly the condition

$$m \in P_D \cap M.$$

Now apply theorem 5.1. □

The theorem is the first major bridge of this chapter:

global sections $\longleftrightarrow$ lattice points in a polyhedron.

Example 5.5 (The simplex of sections on $\mathbb{P}^2$)**.** Let

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_2 = e_2,$$

and consider

$$D = dD_0, \qquad d \geq 0.$$

For $m = (a, b) \in M \simeq \mathbb{Z}^2$, the inequalities defining P_D are

$$a \geq 0, \qquad b \geq 0, \qquad a + b \leq d.$$

Hence

$$P_D = d\Delta_2 = \operatorname{Conv}\big((0,0), (d,0), (0,d)\big).$$

The lattice point (a, b) corresponds to

$$\chi^{(a,b)} = \left(\frac{z_1}{z_0}\right)^a \left(\frac{z_2}{z_0}\right)^b.$$

As a section of $\mathcal{O}_{\mathbb{P}^2}(dD_0)$, this becomes the homogeneous monomial

$$z_0^{d-a-b} z_1^a z_2^b.$$

Thus

$$H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d))$$

has the familiar basis of homogeneous degree-d monomials.

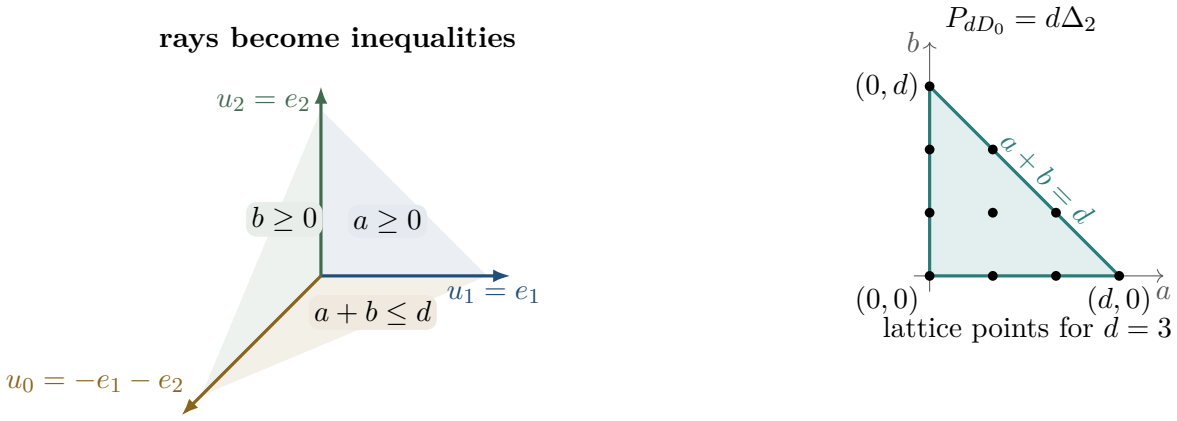


Figure 5.1: For $D = dD_0$ on $\mathbb{P}^2$, each ray contributes one inequality. Their intersection is the dilated standard simplex, and its lattice points index the degree-d monomial sections.

Pause and think

Count the lattice points in $d\Delta_2$ by grouping them according to the value of $a + b$. Recover

$$h^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)) = \binom{d+2}{2}.$$

5.1.2 Polyhedra need not be bounded

Example 5.6 (The structure sheaf of $\mathbb{A}^2$)**.** For the first-quadrant cone

$$\sigma = \mathrm{Cone}(e_1, e_2),$$

take $D = 0$. Then

$$P_0 = \left\{(a, b) \in \mathbb{R}^2 : a \geq 0,\ b \geq 0\right\} = \sigma^\vee.$$

This polyhedron is unbounded, and

$$P_0 \cap M = \mathbb{Z}^2_{\geq 0}.$$

The lattice-point theorem gives

$$H^0(\mathbb{A}^2, \mathcal{O}_{\mathbb{A}^2}) = \bigoplus_{a,b \geq 0} \Bbbk x^a y^b = \Bbbk[x, y].$$

The infinitely many lattice points reflect the infinitely many regular functions on the affine plane.

For a general divisor, the recession cone of P_D is independent of the coefficients of D:

$$\mathrm{rec}(P_D) = \{v \in M_{\mathbb{R}} : \langle v, u_\rho \rangle \geq 0 \text{ for every } \rho\}.$$

It records the directions in which the polyhedron can continue forever.

Proposition 5.7 (Complete fans give bounded section polyhedra)**.** *If Σ is complete, then for every torus-invariant Weil divisor D:*

(a) P_D is a polytope, possibly empty;
(b) $H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))$ is finite dimensional;
(c) $H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}) = \Bbbk$.

Reference. This is Cox–Little–Schenck, Proposition 4.3.8.

Conceptual proof sketch. A direction v lies in the recession cone of P_D precisely when

$$\langle v, u_\rho \rangle \geq 0$$

for every ray. If the fan is complete, its cones fill all of $N_\mathbb{R}$, so the only linear functional nonnegative in every direction is $v = 0$. Hence P_D has no nonzero unbounded direction and is therefore bounded.

A bounded polyhedron has finitely many lattice points, giving part (b). For $D = 0$, completeness forces

$$P_0 = \{0\},$$

so the only global regular functions are constants. □

5.1.3 Blowing up cuts off a corner

Let

$$\pi : X = \mathrm{Bl}_p(\mathbb{P}^2) \longrightarrow \mathbb{P}^2, \qquad p = [1 : 0 : 0],$$

be the toric blowup. The fan has rays

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2.$$

Recall that

$$[D_0] = H, \qquad [E] = E.$$

For integers $0 \leq r \leq d$, choose the invariant representative

$$D = dD_0 - rE$$

of the class $dH - rE$. Its polytope is cut out by

$$a \geq 0, \qquad b \geq 0, \qquad r \leq a + b \leq d.$$

The new ray $u_E = e_1 + e_2$ contributes the new inequality

$$a + b \geq r.$$

Prediction

Start with the simplex $d\Delta_2$, whose lattice points represent all degree-d monomials in z_0, z_1, z_2.

Before looking at the next figure, predict which corner should be removed when we require the corresponding polynomial to vanish at

$$p = [1 : 0 : 0]$$

to order at least r. Which exponent controls that order of vanishing?

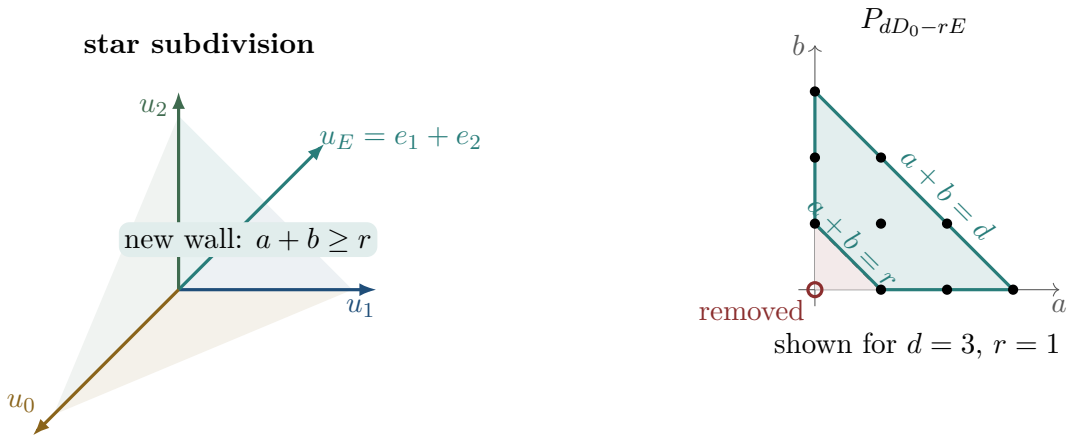


Figure 5.2: Blowing up the fixed point inserts the ray $e_1 + e_2$, and the coefficient $-r$ of the exceptional divisor inserts the inequality $a + b \geq r$. In the section polytope, this cuts off the corner corresponding to monomials that do not vanish sufficiently at the blown-up point.

The lattice point (a, b) corresponds to

$$z_0^{d-a-b} z_1^a z_2^b.$$

At the point $p = [1 : 0 : 0]$, its order of vanishing is

$$a + b.$$

Consequently,

$$H^0(X, \mathcal{O}_X(dH - rE))$$

has the monomial basis indexed by

$$\left\{(a, b) \in \mathbb{Z}^2_{\geq 0} : r \leq a + b \leq d\right\}.$$

The exceptional inequality is therefore the algebraic condition of vanishing to order at least r at the blown-up point.

Surface laboratory

For $0 \leq r \leq d$, use the truncated simplex to answer the following.

(a) Compute

$$h^0(X, \mathcal{O}_X(dH - rE))$$

by counting lattice points.

(b) Check that your answer equals

$$\binom{d+2}{2} - \binom{r+1}{2}.$$

(c) Interpret the subtracted term as the number of independent conditions imposed by vanishing to order at least r at one point.

(d) Draw the polytopes for $d = 4$ and $r = 0, 1, 2, 3, 4$. What happens when $r = d$?

Guiding question

The polyhedron P_D tells us all global sections of $\mathcal{O}_X(D)$. How can its shape tell whether, at every point of X_Σ, at least one of those sections is nonzero?
Equivalently, what condition on the support function ψ_D makes $\mathcal{O}_X(D)$ generated by its global sections?

5.2 Projective maps from lattice points

Algebraic geometry cheatsheet

Let $\mathcal{L}$ be a line bundle on a complete variety X, and let

$$V \subseteq H^0(X, \mathcal{L})$$

be a finite-dimensional space of global sections.
If the sections in V have no common zero, then they define a morphism

$$\phi_V : X \longrightarrow \mathbb{P}(V^\vee).$$

After choosing a basis $s_0, \dots, s_r$ of V, this map is given in a local trivialization of $\mathcal{L}$ by

$$x \longmapsto [s_0(x) : \cdots : s_r(x)].$$

Changing the trivialization multiplies all coordinates by the same nonzero scalar, so the point of projective space is well defined.
When

$$V = H^0(X, \mathcal{L}),$$

this is the map associated to the *complete linear system*.
The line bundle $\mathcal{L}$ is *very ample* if its complete linear system gives a closed embedding into projective space. It is *ample* if some positive tensor power

$$\mathcal{L}^{\otimes k}$$

is very ample.

Observation 5.8 (Evaluation in a one-dimensional fiber). Intrinsically, evaluation at x is a linear map

$$\mathrm{ev}_x : V \longrightarrow \mathcal{L}|_x.$$

Its target is the fiber, rather than a canonically chosen copy of $\Bbbk$. When evaluation is nonzero, identifying that fiber with $\Bbbk$ gives a nonzero element of $V^\vee$, well defined up to scalar. This explains the point of $\mathbb{P}(V^\vee)$ without choosing coordinates on the fiber.

On the open set where $s_j \neq 0$, the affine coordinates of the projective map are the regular ratios s_i/s_j. Thus division requires a nonvanishing denominator. The basepoints of a chosen subspace V concern that subspace; a line bundle is basepoint free when its entire space of global sections has no common zero.

Let X_Σ be a complete toric variety, and let

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be a torus-invariant Cartier divisor. By theorem 5.4, its monomial sections are indexed by the lattice points of

$$P_D = \{m \in M_\mathbb{R} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho\}.$$

Write

$$P_D \cap M = \{m_0, \ldots, m_r\}.$$

Each lattice point determines a global section

$$s_i = \chi^{m_i} s_D,$$

where s_D is the distinguished rational section from Section 5.1.

5.2.1 From sections to a projective map

If the sections $s_0, \ldots, s_r$ have no common zero, then they give a morphism

$$\phi_D : X_\Sigma \longrightarrow \mathbb{P}^r.$$

On the dense torus, the section s_D is nowhere zero, so it can be cancelled from all homogeneous coordinates. Therefore

$$\phi_D|_T(t) = [\chi^{m_0}(t) : \cdots : \chi^{m_r}(t)].$$

This is the most concrete bridge from lattice points to projective geometry: *the exponents become homogeneous coordinates on the image.*

Observation 5.9. If the whole polytope is translated by one lattice vector $m \in M$, then every exponent becomes $m_i + m$. On the torus, this multiplies every projective coordinate by the common factor $\chi^m(t)$, so the projective map does not change.

Thus the projective map depends only on the affine shape of the set of lattice points, not on its absolute position in M.

Definition 5.10. When D is basepoint free, we write

$$\phi_D : X_\Sigma \longrightarrow \mathbb{P}^{|P_D \cap M|-1}$$

for the morphism defined by the complete linear system $|D|$.

At this stage, the word "basepoint free" is just a condition saying that ϕ_D is everywhere defined. The next section will explain how to detect that condition directly from the support function.

5.2.2 Any finite set of lattice points gives a projective toric variety

The same construction makes sense before a divisor enters the story. Let

$$A = \{m_0, \dots, m_r\} \subseteq M$$

be any finite set of lattice points. Define a map on the torus by

$$\phi_A : T \longrightarrow \mathbb{P}^r, \qquad \phi_A(t) = [\chi^{m_0}(t) : \cdots : \chi^{m_r}(t)].$$

Proposition 5.11 (Projective toric varieties from lattice points)**.** *Let*

$$Y_A = \overline{\phi_A(T)} \subseteq \mathbb{P}^r.$$

Then Y_A is a projective toric variety.

Reference. This is the construction in Cox–Little–Schenck, §7.1.

Conceptual proof sketch. Let the torus act on $\mathbb{P}^r$ diagonally by

$$t \cdot [z_0 : \cdots : z_r] = [\chi^{m_0}(t) z_0 : \cdots : \chi^{m_r}(t) z_r].$$

Then $\phi_A(T)$ is precisely the orbit of the point $[1 : \cdots : 1]$. Its closure is therefore a projective variety containing a dense torus orbit, so it is a projective toric variety. □

This proposition says that *projective toric varieties are born from finite sets of lattice points.* When $A = P_D \cap M$, the image of ϕ_D is exactly one such variety.

5.2.3 The first examples

Example 5.12 ($\mathbb{P}^2$ and the Veronese picture)**.** Let $H = D_0$ on $\mathbb{P}^2$. From the $\mathbb{P}^2$ example in the previous section, the polytope of dH is

$$P_{dH} = d\Delta_2,$$

whose lattice points are the pairs (a, b) with

$$a, b \geq 0, \qquad a + b \leq d.$$

The corresponding sections are the degree-d monomials

$$x_1^a x_2^b x_0^{d-a-b},$$

more conveniently written as

$$x_0^{d-a-b} x_1^a x_2^b.$$

For $d = 1$, the three lattice points of Δ_2 give the three homogeneous coordinates, and

$$\phi_H : \mathbb{P}^2 \longrightarrow \mathbb{P}^2$$

is the identity map.

For general d, the lattice points of $d\Delta_2$ give the d-th Veronese map

$$\mathbb{P}^2 \longrightarrow \mathbb{P}^{\binom{d+2}{2}-1}.$$

Thus dilating the triangle produces the familiar projective embedding by degree-d plane curves.

Example 5.13 (The unit square and the Segre quadric)**.** On $\mathbb{P}^1 \times \mathbb{P}^1$, take the rays $e_1, e_2, -e_1, -e_2$, with divisor coefficients $0, 0, 1, 1$, respectively. The associated line bundle is $\mathcal{O}(1, 1)$, and

$$P_D = [0, 1]^2.$$

The four lattice points give the torus map

$$(s, t) \longmapsto [1 : s : t : st].$$

In homogeneous coordinates this extends to

$$\begin{aligned} \mathbb{P}^1 \times \mathbb{P}^1 &\longrightarrow \mathbb{P}^3, \\ ([u_0 : u_1], [v_0 : v_1]) &\longmapsto [u_0v_0 : u_1v_0 : u_0v_1 : u_1v_1]. \end{aligned}$$

The image is the quadric

$$z_0 z_3 = z_1 z_2.$$

Indeed, its coordinates form a nonzero rank-one matrix $\left(\begin{smallmatrix} z_0 & z_2 \\ z_1 & z_3 \end{smallmatrix}\right)$. Such a matrix recovers its two projective factors uniquely, and this recovery is regular on the standard coordinate charts. Thus the map is the Segre embedding. The square therefore gives both a familiar surface and a concrete equation for its projective image.

Example 5.14 (A blowup map from a truncated triangle)**.** On

$$X = \mathrm{Bl}_p(\mathbb{P}^2),$$

consider

$$D = 2H - E.$$

From Figure 5.2,

$$P_D = \left\{ (a, b) \in \mathbb{R}^2 : a, b \geq 0, 1 \leq a + b \leq 2 \right\}.$$

Its lattice points are

$$(1,0),\ (2,0),\ (1,1),\ (0,2),\ (0,1).$$

Hence the complete linear system of $2H - E$ gives a map

$$\phi_{2H-E} : X \longrightarrow \mathbb{P}^4.$$

On the torus, it is the monomial map

$$(t_1, t_2) \longmapsto [t_1 : t_1^2 : t_1 t_2 : t_2^2 : t_2].$$

Compared with the full triangle for $2H$, the missing lattice point $(0,0)$ reflects the vanishing condition imposed at the blown-up point.

Example 5.15 (Resolving projection from a point)**.** Keep the chapter's coordinates $p = [1:0:0]$, and consider the two-dimensional subsystem

$$V = \langle z_1, z_2 \rangle \subset H^0(\mathbb{P}^2, \mathcal{O}(1)).$$

It defines the rational projection

$$\phi_V : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1, \qquad [z_0 : z_1 : z_2] \longmapsto [z_1 : z_2],$$

whose basepoint is p. Geometrically, it records the line through p and the chosen point. The segment with endpoints e_1, e_2 indexes this subsystem of $|H|$, rather than its complete set of sections on $\mathbb{P}^2$.

On the blowup $X = \mathrm{Bl}_p(\mathbb{P}^2)$, removing the common exceptional factor from the pullbacks of z_1, z_2 gives two generating sections of $H - E$. Here the complete section polytope is exactly

$$P_{H-E} = \{(a,b) : a, b \geq 0,\ a + b = 1\}.$$

The rational projection has become a morphism:

$$\begin{array}{ccc} X & & \\ {\scriptstyle \pi}\downarrow & \searrow^{\phi_{H-E}} & \\ \mathbb{P}^2 & \xrightarrow[\phi_V]{} & \mathbb{P}^1. \end{array}$$

An explicit model makes the two arrows easy to distinguish:

$$X \cong \{([z_0 : z_1 : z_2], [u : v]) \in \mathbb{P}^2 \times \mathbb{P}^1 : z_1 v = z_2 u\}.$$

The first projection is π, and the second is ϕ_{H-E}. Its fiber over $[u:v]$ is the strict transform of the line $z_1 v = z_2 u$ through p, hence a copy of $\mathbb{P}^1$. The exceptional curve is $E = \{p\} \times \mathbb{P}^1$. The map π contracts it to p, whereas $\phi_{H-E}|_E$ is an isomorphism onto the target $\mathbb{P}^1$: a point of E records a tangent direction at p.

Thus resolving the basepoint makes the map regular, but does not make it injective. The whole strict transform of each line remains one fiber. We return to this boundary case in the positivity discussion.

Pause and think

What happens if one starts with a finite subset

$$A \subseteq M$$

that is *not* the whole set of lattice points of its convex hull? Do you expect the orbit closure Y_A to remember only the convex hull, or the actual chosen set A?

5.2.4 Very ample versus ample

The map ϕ_D packages all sections of $\mathcal{O}(D)$ at once. There are then three increasingly strong possibilities:

- The sections may fail to define a morphism everywhere. Then D is not basepoint free.
- The sections may define a morphism

$$\phi_D : X_\Sigma \to \mathbb{P}^r,$$

  but different points or tangent directions may still map together.
- The sections may give a closed embedding. Then D is very ample.

Ampleness means that this last outcome occurs after replacing D by a sufficiently large positive multiple.

So the natural questions become:

> Can we tell from the support function whether ϕ_D is defined everywhere? Can we tell when it is an embedding, or becomes one after scaling?

Those are precisely the positivity questions answered by convexity and strict convexity.

Guiding question

In theorem 5.16, each maximal cone σ had a distinguished exponent m_σ. Why should the condition

$$m_\sigma \in P_D$$

mean that the complete linear system has no basepoint at the fixed point x_σ?
And if two neighboring maximal cones have the same distinguished exponent, what failure of separation should you expect for the map ϕ_D?

5.3 Convexity and positivity

The previous section attached to a basepoint-free divisor a projective morphism

$$\phi_D : X_\Sigma \longrightarrow \mathbb{P}(H^0(X_\Sigma, \mathcal{O}(D))^\vee).$$

We now ask how to recognize this situation directly from the fan. The answer is encoded by the support function. Convexity will tell us when the map is defined everywhere, and strict convexity will tell us when the associated projective geometry is positive enough to produce an embedding after taking multiples.

Algebraic geometry cheatsheet

Let $\mathcal{L}$ be a line bundle on a complete variety X.
A point $p \in X$ is a *basepoint* of $\mathcal{L}$ if every section in

$$H^0(X, \mathcal{L})$$

vanishes at p. Equivalently, the evaluation map

$$H^0(X, \mathcal{L}) \longrightarrow \mathcal{L}|_p$$

is zero.
The line bundle is *generated by global sections*, or *basepoint free*, if it has no basepoints. This means that near every point one can find a global section that serves as a local frame.
It is *very ample* if the complete linear system gives a closed embedding into projective space.
It is *ample* if some positive tensor power is very ample.

Throughout this section, Σ is a complete fan and

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

is a torus-invariant Cartier divisor. Write

$$\{m_\sigma\}_{\sigma \in \Sigma(n)}$$

for its Cartier data and

$$\psi_D : |\Sigma| \longrightarrow \mathbb{R}$$

for its support function.

5.3.1 The distinguished section on an affine chart

Let $\sigma \in \Sigma(n)$. Since

$$\langle m_\sigma, u_\rho \rangle = -a_\rho \qquad (\rho \in \sigma(1)),$$

the rational function χ^{m_σ} is a local frame of $\mathcal{O}_{X_\Sigma}(D)$ on U_σ. Indeed,

$$\operatorname{div}(\chi^{m_\sigma}) + D$$

has coefficient zero along every boundary divisor that meets the closed orbit of U_σ.

For any lattice point $m \in P_D \cap M$, the corresponding section can be written on this chart as

$$\chi^m = \chi^{m - m_\sigma} \chi^{m_\sigma}.$$

The first factor is regular on U_σ, because

$$m - m_\sigma \in \sigma^\vee.$$

Lemma 5.16 (The section that does not vanish at a fixed point)**.** *Let $x_\sigma = O(\sigma)$ be the torus fixed point corresponding to a maximal cone σ. A monomial section χ^m, with $m \in P_D \cap M$, is nonzero at x_σ if and only if*

$$m = m_\sigma.$$

Consequently, x_σ is not a basepoint if and only if

$$m_\sigma \in P_D.$$

Proof. Relative to the local frame χ^{m_σ}, the section is the regular monomial

$$\chi^{m-m_\sigma} \in \Bbbk[\sigma^\vee \cap M].$$

At the distinguished point x_σ, a monomial is nonzero only when its exponent lies in

$$\sigma^\perp \cap M.$$

Since σ is full dimensional, this group is zero. Hence the section is nonzero precisely when

$$m - m_\sigma = 0.$$

□

Observation 5.17 (The monomial hypothesis matters)*.* The preceding fixed-point argument is specific to monomial sections. A general regular section nonzero at x_σ need only be a frame near that point: $1 + x$ on $\mathbb{A}^1$ is nonzero at 0 but vanishes at -1. Nor do boundary orders alone determine the value of an arbitrary section at a fixed point: $x + y$ vanishes at the origin of $\mathbb{A}^2$, although it has order zero along both coordinate axes. Passing to a nonvanishing weight component is what makes the monomial criterion applicable to global generation.

This lemma gives a useful interpretation of the Cartier datum:

m_σ is the exponent of the monomial section that should trivialize $\mathcal{O}(D)$ on U_σ.

The only question is whether that exponent actually lies in P_D, so that the local frame extends to a global section.

Example 5.18 (A local equation and its reciprocal frame)**.** Suppose $U_\sigma \simeq \mathbb{A}^2$, with boundary divisors $V(x), V(y)$, and

$$D|_{U_\sigma} = 2V(x) - V(y) = \operatorname{div}\left(\frac{x^2}{y}\right).$$

The local defining equation has a zero of order two along $V(x)$ and a pole of order one along $V(y)$. Its reciprocal is a frame of $\mathcal{O}(D)$: in the rational-function realization used here,

$$m_\sigma = (-2, 1), \qquad e_\sigma = \chi^{m_\sigma} = \frac{y}{x^2}.$$

Every regular section on this chart is

$$f = p(x,y)e_\sigma = p(x,y)\frac{y}{x^2}, \qquad p \in \Bbbk[x,y].$$

Its value in the line-bundle fiber at $x_\sigma = (0,0)$ is nonzero exactly when $p(0,0) \neq 0$. In that case,

$$\operatorname{ord}_{V(x)}(f) = -2, \qquad \operatorname{ord}_{V(y)}(f) = 1.$$

The rational function f can therefore have a pole while still representing a regular section of $\mathcal{O}(D)$. The coefficient p, not the numerical value of f, is evaluated in this frame. The monomial frame extends globally exactly when $(-2,1) \in P_D$; rays outside σ supply the remaining inequalities for that extension.

5.3.2 Convex support functions and global generation

Definition 5.19 (Convexity in the toric convention)**.** The support function ψ_D is *convex* if, for every maximal cone σ,

$$\psi_D(u) \leq \langle m_\sigma, u\rangle \qquad \text{for every } u \in N_{\mathbb{R}}.$$

Equivalently,

$$\psi_D(u) = \min_{\sigma\in\Sigma(n)} \langle m_\sigma, u\rangle.$$

Each linear piece agrees with ψ_D on its own cone. Convexity says that when this piece is extended across all of $N_{\mathbb{R}}$, the graph of ψ_D never rises above it.

Warning

Because ψ_D is written as a *minimum* of linear functions, some books on convex analysis would call it concave or upper convex. We follow the standard toric convention used by Cox–Little–Schenck: this condition is called convex.

Observation 5.20 (Translating the classroom maximum formula). If the local defining equation is written as χ^{q_σ}, then $q_\sigma = -m_\sigma$. The opposite support function $\eta_D = -\psi_D$ has $\eta_D(u_\rho) = a_\rho$, and

$$\psi_D(u) = \min_\sigma \langle m_\sigma, u\rangle \quad \Longleftrightarrow \quad \eta_D(u) = \max_\sigma \langle q_\sigma, u\rangle.$$

This is the maximum formula used in class, with the local-equation exponents in place of the frame exponents. Throughout this chapter, ψ_D retains the minimum convention. Both formulas express the same global-generation condition.

Theorem 5.21 (Basepoint freeness equals convexity)**.** *The following conditions are equivalent:*

(a) D is basepoint free.

(b)

$$m_\sigma \in P_D \qquad \textit{for every } \sigma \in \Sigma(n).$$

(c)

$$\psi_D(u) = \min_{\sigma\in\Sigma(n)} \langle m_\sigma, u\rangle \qquad \textit{for every } u \in N_{\mathbb{R}}.$$

(d) The support function ψ_D is convex.

Reference. This is Cox–Little–Schenck, Theorem 6.1.7.

Proof. Suppose first that every m_σ lies in P_D. By theorem 5.4, χ^{m_σ} is a global section. It is a local frame on U_σ, so these sections generate $\mathcal{O}(D)$ on the affine cover

$$X_\Sigma = \bigcup_{\sigma \in \Sigma(n)} U_\sigma.$$

Thus (b) implies (a).

Conversely, suppose D is basepoint free. At the fixed point x_σ, some global section is nonzero. Decomposing that section into its monomial weight components, at least one monomial section is nonzero there. By theorem 5.16, its exponent must be m_σ. Hence $m_\sigma \in P_D$, proving (a) implies (b).

Now fix maximal cones σ, τ, and let $u \in \tau$. If $m_\sigma \in P_D$, then the inequalities defining P_D imply

$$\langle m_\sigma, u \rangle \geq \langle m_\tau, u \rangle = \psi_D(u).$$

Since equality holds when $u \in \sigma$, this is equivalent to

$$\psi_D(u) = \min_\sigma \langle m_\sigma, u \rangle.$$

Thus (b), (c), and (d) are equivalent. □

The theorem turns a local geometric question into the shape of one piecewise-linear function:

$\mathcal{O}(D)$ is globally generated $\iff$ ψ_D bends convexly across every wall.

Example 5.22 (The support function of a line on $\mathbb{P}^2$)**.** For the ray generators $u_0 = -e_1 - e_2, u_1 = e_1, u_2 = e_2$, take $D = D_0$. Its polytope is Δ_2, and its Cartier data and linear pieces are

σ	m_σ	$\psi_D(x, y)$ on σ
$\mathrm{Cone}(e_1, e_2)$	$(0, 0)$	0
$\mathrm{Cone}(e_2, -e_1 - e_2)$	$(1, 0)$	x
$\mathrm{Cone}(-e_1 - e_2, e_1)$	$(0, 1)$	y

All three Cartier data lie in Δ_2. On each cone, the listed linear piece is the smallest of $0, x, y$, so

$$\psi_D(x, y) = \min(0, x, y).$$

The classroom expression $\max(0, -x, -y)$ is its negative, as in theorem 5.20. Along a common ray the neighboring formulas agree, making continuity and the convexity criterion visible in this example.

Observation 5.23. There is also a numerical name for this condition. A Cartier divisor is *nef* if

$$D \cdot C \geq 0$$

for every complete curve C. On a toric variety, a Cartier divisor is nef exactly when it is basepoint free; it is enough to test the torus-invariant complete curves.

This is Cox–Little–Schenck, Theorem 6.3.12. No intersection theory is needed for the arguments in this section, so this terminology may be viewed as an optional preview.

5.3.3 Strict convexity and ampleness

Convexity allows two adjacent maximal cones to carry the same linear piece. Geometrically, the support function then fails to notice the wall between them, and the projective map from the previous section cannot separate the geometry on the two sides. Ampleness requires a genuine bend at every wall.

Definition 5.24 (Strict convexity)**.** A convex support function ψ_D is *strictly convex* if, for every maximal cone σ,

$$\psi_D(u) = \langle m_\sigma, u \rangle \quad \Longleftrightarrow \quad u \in \sigma.$$

Equivalently, neighboring maximal cones have different Cartier data:

$$m_\sigma \neq m_{\sigma'}$$

whenever σ and σ' meet along a wall.

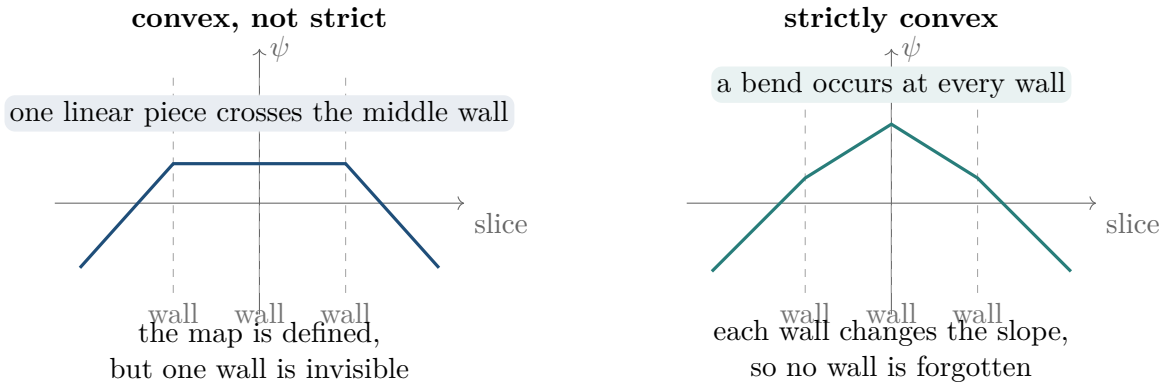


Figure 5.3: A transverse slice through the graph of a support function. Both panels use the same three walls. On the left, the middle wall is invisible to the support function because one linear piece continues straight across it. On the right, the slope changes at every wall.

Theorem 5.25 (Ampleness equals strict convexity)**.** *Let D be a torus-invariant Cartier divisor on the complete toric variety X_Σ. Then*

$$\boxed{D \text{ is ample} \quad \Longleftrightarrow \quad \psi_D \text{ is strictly convex.}}$$

Reference. This is Cox–Little–Schenck, Theorem 6.1.14.

Conceptual proof sketch. If a positive multiple of D gives an embedding, then the resulting projective map must separate the torus-invariant curve attached to any wall. If two neighboring maximal cones had the same Cartier datum, the corresponding support function would fail to bend at that wall, and the linear system would not detect that invariant curve. Thus ampleness forces a genuine change of slope everywhere.

Conversely, strict convexity ensures that different maximal cones have different Cartier data, so the polytope has enough distinct vertices to encode the fan faithfully. The projective map attached to a large multiple of D then separates the toric strata, and one obtains a closed embedding. The complete proof is given in the cited theorem. □

Corollary 5.26 (Projectivity criterion)**.** *A complete toric variety X_Σ is projective if and only if its fan admits an integral strictly convex support function.*

This criterion is easy to state but powerful: completeness is visible from whether the cones cover $N_\mathbb{R}$, while projectivity requires a compatible strictly convex shape on those cones.

5.3.4 Surface laboratory: the positivity interval on a blowup

Let

$$X = \mathrm{Bl}_p(\mathbb{P}^2)$$

and consider

$$D = dD_0 - rE$$

with $d > 0$ and $0 \le r \le d$. As in Figure 5.2,

$$P_D = \Big\{(a,b) \in \mathbb{R}^2 : a, b \ge 0, \quad r \le a + b \le d\Big\}.$$

For the four maximal cones, the Cartier data are

$$(0,d), \qquad (0,r), \qquad (r,0), \qquad (d,0).$$

Each lies in P_D, so D is basepoint free throughout the range

$$0 \le r \le d.$$

They are distinct exactly when

$$0 < r < d.$$

Therefore

$$\boxed{dH - rE \text{ is basepoint free for } 0 \le r \le d, \qquad dH - rE \text{ is ample for } 0 < r < d.}$$

The two boundary cases fail strict convexity for different visible reasons:

- When $r = 0$, the inequality from the exceptional ray is redundant. The support function does not bend along the newly inserted wall.
- When $r = d$, the polytope loses dimension. The linear system remembers only the pencil of lines through the blown-up point.

The classroom projection in theorem 5.15 also makes both boundary cases explicit. For every $k \ge 1$,

$$\phi_{kH} = \nu_k \circ \pi, \qquad \phi_{k(H-E)} = \nu_k \circ \phi_{H-E},$$

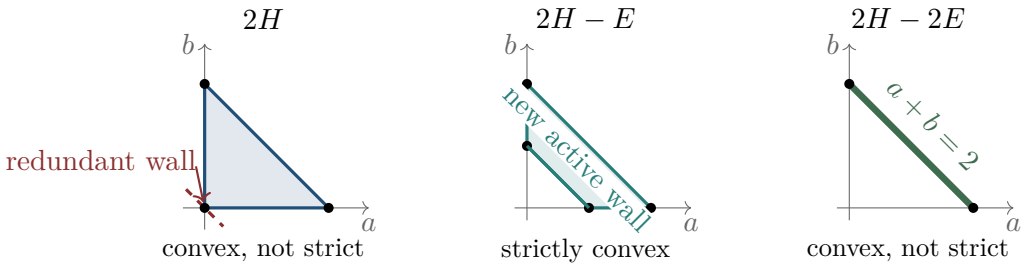


Figure 5.4: Positivity for divisors $dH - rE$ on the blowup. For $2H$, the exceptional wall is redundant. For $2H - E$, the new wall is active and all four Cartier data are distinct. For $2H - 2E$, the polytope collapses to a segment.

where the Veronese maps are those of $\mathbb{P}^2$ and $\mathbb{P}^1$, respectively. Thus every positive multiple of H still contracts E, and every positive multiple of $H - E$ still contracts the strict transforms of the lines through p. These divisors are basepoint free without becoming very ample after scaling.

Pause and think

On $\mathbb{P}^1 \times \mathbb{P}^1$, show that the polytope of $aF_1 + bF_2$ is a rectangle with side lengths a and b. Use its shape to determine when the divisor is basepoint free and when it is ample. What collapses when one coefficient is zero?

For a basepoint-free divisor, the Cartier data lie in P_D. When different maximal cones have the same Cartier datum, the polytope forgets the wall between them. Strict convexity is exactly the condition that no wall is forgotten.

The three dictionaries

At this point, the three descriptions of positivity line up.

Algebra	Geometry	Combinatorics
χ^{m_σ} extends to a global section	x_σ is not a basepoint	$m_\sigma \in P_D \cap M$
global sections generate $\mathcal{O}(D)$	ϕ_D is defined everywhere	ψ_D is convex
sections of kD separate toric strata for $k \gg 0$	D is ample	ψ_D is strictly convex

convexity makes the map exist; strict convexity makes it eventually embed.

Guiding question

Given a polytope $P \subseteq M_{\mathbb{R}}$, how can one construct a fan in $N_{\mathbb{R}}$ from the directions in which each face of P is minimized?
For a basepoint-free divisor D, how does that fan compare with the original fan Σ, and why does strict convexity force the two fans to agree?

5.4 Polytopes and normal fans

A polytope does more than provide lattice points for projective coordinates. It also remembers a fan.

The guiding idea is simple. Choose a direction

$$u \in N_{\mathbb{R}}.$$

The linear function

$$m \longmapsto \langle m, u\rangle$$

reaches its minimum on a face of the polytope. Directions that select the same face form a cone. As the face changes, these cones fit together into the *normal fan*.

5.4.1 Which face does a direction see?

Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional lattice polytope. For $u \in N_{\mathbb{R}}$, define the minimizing face

$$\mathrm{face}_u(P) = \left\{ m \in P : \langle m, u\rangle = \min_{p\in P} \langle p, u\rangle \right\}.$$

The vector u should be thought of as an inward-pointing normal: it pushes a supporting hyperplane toward P until that hyperplane first meets the polytope.

Definition 5.27 (Normal cone)**.** Let $Q \preceq P$ be a face. The *normal cone* of P along Q is

$$\sigma_Q = \{u \in N_{\mathbb{R}} : Q \subseteq \mathrm{face}_u(P)\}.$$

Equivalently, $u \in \sigma_Q$ precisely when

$$\langle q, u\rangle \le \langle p, u\rangle \qquad \text{for every } q \in Q, p \in P.$$

The relative interior has the sharper interpretation

$$u \in \mathrm{Relint}(\sigma_Q) \quad \Longleftrightarrow \quad \mathrm{face}_u(P) = Q.$$

Thus the closed cone allows a larger minimizing face, while its relative interior remembers exactly Q.

Suppose the facets of P have equations

$$P = \{m \in M_{\mathbb{R}} : \langle m, u_F\rangle \ge -a_F \text{ for every facet } F\},$$

where $u_F \in N$ is the primitive inward normal to F. Then

$$\boxed{\sigma_Q = \mathrm{Cone}(u_F : Q \subseteq F).}$$

In particular:

$$\begin{array}{ccc} \text{facet } F & \longleftrightarrow & \text{ray } \mathrm{Cone}(u_F), \\ \text{vertex } v & \longleftrightarrow & \text{maximal cone generated by the normals at } v. \end{array}$$

5.4.2 The normal fan

Theorem 5.28 (Faces assemble into a fan)**.** *Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional lattice polytope. Then*

$$\Sigma_P = \{\sigma_Q : Q \preceq P\}$$

is a complete fan in $N_{\mathbb{R}}$, called the normal fan *of P.*

Reference. This is Cox–Little–Schenck, Theorem 2.3.2; see also Proposition 6.2.3.

Conceptual proof sketch. If $Q \subseteq Q'$, then every supporting direction that minimizes on Q' also minimizes on Q. Hence inclusions reverse:

$$Q \subseteq Q' \quad \Longrightarrow \quad \sigma_{Q'} \preceq \sigma_Q.$$

The faces of a normal cone therefore come from larger faces of the polytope.

For two faces Q, Q', the intersection

$$\sigma_Q \cap \sigma_{Q'}$$

consists of directions minimizing on both. This is the normal cone of the smallest face containing Q and Q', so the intersection is a face of each cone.

Finally, every $u \in N_{\mathbb{R}}$ minimizes somewhere on the compact polytope P. Therefore u lies in the normal cone of its minimizing face, proving completeness. □

The face–cone correspondence reverses dimensions and inclusions:

$$\boxed{\dim Q + \dim \sigma_Q = \dim P.}$$

For a polygon, this gives the compact dictionary

face of P	cone of Σ_P
vertex	two-dimensional cone
edge	ray
P	$\{0\}$.

Prediction

Before looking at the assembled fan, focus on the vertex $(1, 0)$. Which two inward facet normals should generate its normal cone?
Choose a vector in the interior of your predicted cone and determine where its linear function is minimized on P. Then compare your answer with Figure 5.5.

5.4.3 Recovering a variety from its vertex charts

Example 5.29 (The five-monomial trapezoid)**.** The classroom question "which variety does a polytope produce?" can be answered directly in affine coordinates. Consider

$$P = \mathrm{Conv}\{(0,0), (2,0), (1,1), (0,1)\}$$

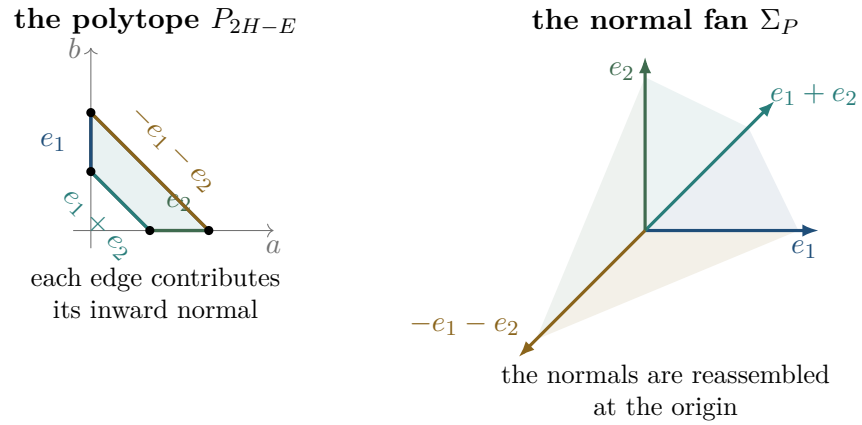


Figure 5.5: The truncated triangle for $2H - E$ and its normal fan. Each edge contributes its primitive inward normal. At a vertex, the normals of the two incident edges generate the corresponding maximal cone.

and its five lattice points, ordered as

$$m_0 = (0,0), \quad m_1 = (1,0), \quad m_2 = (0,1), \quad m_3 = (2,0), \quad m_4 = (1,1).$$

Writing (s,t) for coordinates on the torus, set

$$\phi_P(s,t) = [1 : s : t : s^2 : st], \qquad Y = \overline{\phi_P(T)} \subseteq \mathbb{P}^4.$$

Let $Y_i = Y \cap \{x_i \neq 0\}$. On the chart $i = 0$, the image has coordinates (s, t, s^2, st). Its closure is the graph of the polynomial map $(s,t) \mapsto (s^2, st)$, so

$$Y_0 = V(z_3 - z_1^2, z_4 - z_1 z_2) \cong \mathbb{A}^2, \qquad z_j = x_j/x_0.$$

The remaining vertex charts work in the same way. In each tuple below, the coordinates x_j/x_i appear in increasing order of $j \neq i$.

i	m_i	parameters (u,v)	normalized image
0	$(0,0)$	(s,t)	(u, v, u^2, uv)
2	$(0,1)$	(s,t^{-1})	(v, uv, u^2v, u)
3	$(2,0)$	$(s^{-1}, t/s)$	(u^2, u, uv, v)
4	$(1,1)$	$(s^{-1}, s/t)$	(u^2v, uv, u, v)

Each tuple contains u, v themselves, with the other entries polynomial in them. Thus each of these four charts is an affine plane.

Why do normal cones appear? Dividing by the coordinate of a vertex m_i replaces each character χ^m by χ^{m-m_i}. Their exponents generate the tangent cone

$$\sigma_i^\vee = \mathrm{Cone}(P - m_i) \subseteq M_{\mathbb{R}}.$$

Its dual is precisely the inward normal cone at m_i:

$$\sigma_i = \{u \in N_{\mathbb{R}} : \langle m - m_i, u\rangle \geq 0 \text{ for every } m \in P\}.$$

The tangent cone lies in the character space, whereas the normal cone lies in the dual space. In the present example the normal cones are

$$\begin{aligned} \sigma_0 &= \mathrm{Cone}(e_1, e_2), & \sigma_2 &= \mathrm{Cone}(e_1, -e_2), \\ \sigma_3 &= \mathrm{Cone}(e_2, -e_1 - e_2), & \sigma_4 &= \mathrm{Cone}(-e_1 - e_2, -e_2). \end{aligned}$$

At each vertex the two parameter exponents form a lattice basis, so the displayed coordinate ring is exactly $\Bbbk[\sigma_i^\vee \cap M]$.

The lattice point $(1, 0)$ is not a vertex. Its chart has coordinates

$$(s^{-1}, t/s, s, t) = (u^{-1}, v, u, uv),$$

and therefore

$$Y_1 \cong \operatorname{Spec} \Bbbk[u, u^{-1}, v] \cong \mathbb{G}_m \times \mathbb{A}^1.$$

The relation $x_1^2 = x_0 x_3$ on Y gives $Y_1 = Y_0 \cap Y_3$. Consequently the four vertex charts cover Y. They glue through their common torus characters; for example, on $Y_0 \cap Y_3$ the coordinate change is $(s, t) \mapsto (s^{-1}, t/s)$. These are the affine charts and gluing maps of the normal fan, so

$$Y \cong X_{\Sigma_P}.$$

This is also the earlier $2H - E$ example in different lattice coordinates: the affine lattice transformation

$$(r, q) \longmapsto (a, b) = (2 - r - q, r)$$

sends P to $\{a, b \geq 0 : 1 \leq a + b \leq 2\} = P_{2H-E}$. The linear part of this transformation has determinant one. The chart calculation therefore explains concretely why the five sections of $2H - E$ embed the blowup into $\mathbb{P}^4$.

Observation 5.30 (The semigroup check). For a finite set A affinely spanning $M_{\mathbb{R}}$ and a vertex v of its convex hull, the corresponding affine chart of the orbit closure has coordinate ring

$$\Bbbk[S_v], \qquad S_v = \mathbb{N}\{m - v : m \in A\}.$$

By theorem 2.26, its cone alone need not determine this ring: the semigroup may have missing lattice points. When the differences in A generate M, saturating these semigroups gives the affine charts of $X_{\Sigma_{\mathrm{Conv}(A)}}$, hence the normalization of the orbit closure. If they generate a proper sublattice, that sublattice is the character lattice of the image torus and must be used in the normalization statement. In the trapezoid example the parameter calculations verify saturation in M at every vertex, so no normalization is needed.

Observation 5.31 (Local functions versus normalized global sections). In the trapezoid example, let D be the invariant Cartier divisor with $P_D = P$, and take $\sigma = \mathrm{Cone}(e_2, -e_1 - e_2)$, whose Cartier datum is $m_\sigma = (2, 0)$. The monomial frame is $e_\sigma = s^2 s_D$, and division by it identifies

$$H^0(U_\sigma, \mathcal{O}(D)) \xrightarrow{\widetilde{s} \mapsto \widetilde{s}/e_\sigma} \Bbbk[\sigma^\vee \cap M] = \Bbbk[s^{-1}, t/s].$$

The translated lattice points $(P_D \cap M) - m_\sigma$ record the normalized *global* sections; the whole cone records all local regular monomials. For example, s^{-3} is regular on this chart, but

$$s^{-3} e_\sigma = s^{-1} s_D$$

does not extend to a global regular section, since $(-1, 0) \notin P_D$. Thus trivializing the bundle and extending a section globally are different operations.

5.4.4 When the vertex coordinates give an embedding

The semigroup check also makes the distinction between ample and very ample precise. Let $P \subseteq M_\mathbb{R}$ be a full-dimensional lattice polytope, and let D_P be its ample Cartier divisor on X_{Σ_P}. At a vertex v, write

$$S_v = \mathbb{N}\{m - v : m \in P \cap M\}, \qquad \sigma_v^\vee = \mathrm{Cone}(P - v).$$

The normalized projective coordinates generate $\Bbbk[S_v]$, whereas the coordinate ring of the prescribed normal affine chart is $\Bbbk[\sigma_v^\vee \cap M]$. Consequently,

$$\boxed{D_P \text{ is very ample} \quad \Longleftrightarrow \quad S_v = \sigma_v^\vee \cap M \text{ at every vertex } v.}$$

Indeed, the nonvanishing locus of the vertex section is U_{σ_v}. On this open set, a closed embedding is equivalent to surjectivity of the coordinate-ring map, which is exactly the displayed semigroup equality. These vertex opens cover the variety and the corresponding affine target opens cover its image. That image is closed because X_{Σ_P} is projective. Equality of the *real cones* alone does not ensure equality of their integral semigroups.

Every lattice polygon satisfies this condition: it admits a triangulation into unimodular lattice triangles, and dilating those triangles shows that every lattice point of kP is a sum of k lattice points of P. Applied near a vertex, this also gives the displayed tangent-semigroup equality. In dimension three the equality can fail, even for an ample divisor.

Example 5.32 (An ample divisor that needs to be doubled)**.** For this example work over $\mathbb{C}$, as in the lecture, and take

$$M = \mathbb{Z}^3, \qquad w = (1, 1, 2), \qquad P = \mathrm{Conv}\{0, e_1, e_2, w\}.$$

This is the height-two Reeve tetrahedron. Its only lattice points are its four vertices, so its complete linear system restricts on the torus to

$$\phi_{D_P}(s, t, u) = [1 : s : t : stu^2].$$

The points (s,t,u) and $(s,t,-u)$ have the same image. Moreover, every $[1:a:b:c]$ with $abc \neq 0$ is obtained by choosing $u^2 = c/(ab)$. Hence the projective image is all of $\mathbb{P}^3$. The exponent differences generate

$$M' = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}w = \{(a,b,c) \in M : c \text{ is even}\},$$

an index-two sublattice. Thus the map $X_{\Sigma_P} \to \mathbb{P}^3$ is finite of degree two, and is not an embedding. This is the lattice issue in Observation 5.30, rather than a normalization of $\mathbb{P}^3$.

At the vertex 0, the missing lattice point

$$h = (1,1,1) = \tfrac{1}{2}(e_1 + e_2 + w)$$

lies in the tangent cone but not in S_0. Its character is stu. The cone semigroup is generated by e_1, e_2, w, h, so the normal-fan chart is

$$\operatorname{Spec} \mathbb{C}[s,t,stu^2,stu] \cong \operatorname{Spec} \mathbb{C}[a,b,c,d]/(d^2 - abc).$$

Its origin is singular, whereas the image chart $\operatorname{Spec} \mathbb{C}[s,t,stu^2]$ is an affine three-space. In particular, smoothness of the image does not imply smoothness of the prescribed normal-fan variety.

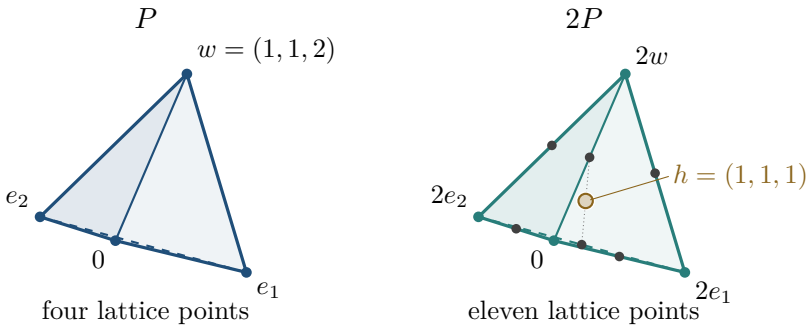


Figure 5.6: The height-two Reeve tetrahedron and its dilation, drawn at matching display sizes. The only lattice points of P are its four vertices. The eleven lattice points of $2P$ consist of four vertices, six edge midpoints, and the highlighted interior point h, which supplies the missing character $\chi^h = stu$. Dashed edges are hidden.

Now replace P by $2P$, keeping the same normal fan. If V denotes the set of four vertices of P, then

$$2P \cap M = \{v + v' : v, v' \in V\} \cup \{h\}.$$

There are eleven points: four vertices, six edge midpoints, and the interior point h. They give the map to $\mathbb{P}^{10}$ with coordinates

$$[1 : s : t : st : stu : stu^2 : s^2 : t^2 : s^2tu^2 : st^2u^2 : s^2t^2u^4].$$

The new coordinates recover $u = (stu)/(st)$ on the torus. If the coordinates corresponding to st, stu, stu^2 are X_3, X_4, X_5, then $X_4^2 = X_3X_5$; this is one relation, rather than a complete set of defining equations.

To check the boundary as well, let $v \in V$, and let r_1, r_2, r_3 be its differences to the other three vertices. Their determinant has absolute value two. The half-open fundamental parallelepiped contains precisely 0 and

$$h_v = h - 2v = \tfrac{1}{2}(r_1 + r_2 + r_3).$$

Subtracting integer parts of the cone coefficients gives

$$\mathrm{Cone}(P - v) \cap M = \mathbb{N}r_1 + \mathbb{N}r_2 + \mathbb{N}r_3 + \mathbb{N}h_v.$$

All four generators occur in $(2P \cap M) - 2v$: use the points $v + v_i$ and h. The vertex criterion therefore proves that $2D_P$ is very ample. Doubling supplies the missing local generators; it changes neither the fan nor the singularities.

5.4.5 What changes the normal fan?

The normal fan records directions, not lengths or absolute position. Consequently:

- translating P by a vector does not change Σ_P;
- multiplying P by a positive scalar does not change Σ_P;
- changing an edge length without changing the directions of the facets often preserves the same normal fan.

For example, every axis-parallel rectangle has the normal fan of

$$\mathbb{P}^1 \times \mathbb{P}^1,$$

and every positive dilation of the standard triangle has the normal fan of $\mathbb{P}^2$.

The normal fan changes when a facet disappears, a new facet becomes active, or the polytope loses dimension. These are exactly the phenomena visible in Figure 5.4.

5.4.6 The normal fan of a divisor polytope

Let D be a basepoint-free torus-invariant Cartier divisor on a complete toric variety X_Σ. Its Cartier data

$$\{m_\sigma\}_{\sigma \in \Sigma(n)}$$

lie in P_D, and the distinct values among them are precisely the vertices of P_D.

Theorem 5.33 (The fan remembered by a divisor)**.** *Let D be basepoint free. For every vertex v of P_D, the corresponding maximal cone of the normal fan is*

$$\sigma_v = \bigcup_{\substack{\sigma \in \Sigma(n) \\ m_\sigma = v}} \sigma.$$

Consequently,

$$\boxed{\Sigma \text{ refines } \Sigma_{P_D}.}$$

In words: the normal fan of P_D is obtained by gluing together all maximal cones that carry the same Cartier datum.

Reference. This is Cox–Little–Schenck, Proposition 6.2.5.

Conceptual proof sketch. For $u \in \sigma$, convexity gives

$$\psi_D(u) = \langle m_\sigma, u \rangle = \min_{m \in P_D} \langle m, u \rangle.$$

Thus every direction in σ sees the Cartier datum m_σ as a minimizing vertex of P_D. If several adjacent cones have the same datum, their union belongs to one normal cone. If the data differ, the minimizing vertex changes, so one has crossed a wall of the normal fan. □

Combining this theorem with strict convexity gives

$$\boxed{\begin{array}{c} D \text{ basepoint free} \Longrightarrow \Sigma \text{ refines } \Sigma_{P_D}, \\ D \text{ ample} \Longleftrightarrow \Sigma = \Sigma_{P_D}. \end{array}}$$

Strict convexity means that neighboring maximal cones never share the same Cartier datum, so no wall is glued away.

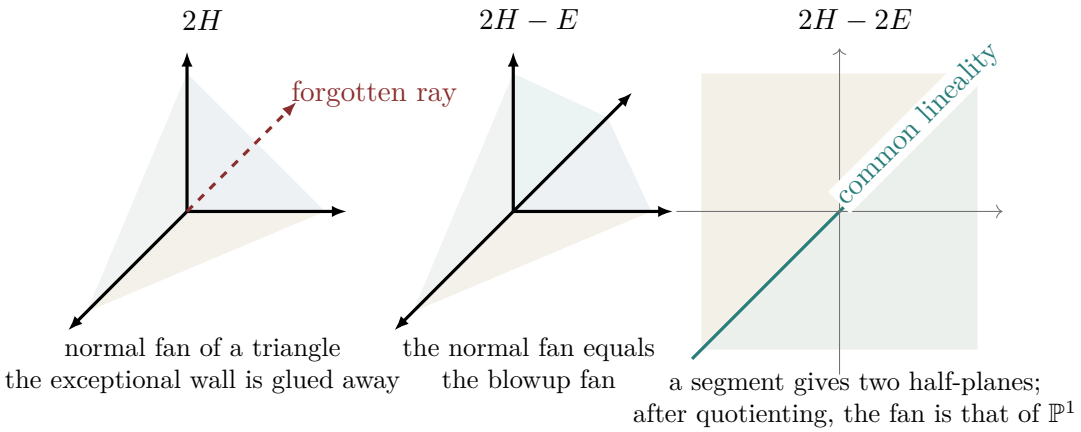


Figure 5.7: The normal fans associated to the three polytopes in Figure 5.4. A redundant inequality glues away the exceptional wall; an active facet remembers it; a one-dimensional polytope produces a degenerate normal fan with lineality.

5.4.7 The morphism encoded by the normal fan

Because Σ refines Σ_{P_D}, the identity on the lattice induces a proper toric morphism

$$\pi_D : X_\Sigma \longrightarrow X_{\Sigma_{P_D}}.$$

When P_D is lower dimensional, one first quotients by the common lineality space of its normal cones. The resulting toric variety has dimension $\dim P_D$.

Theorem 5.34 (A basepoint-free divisor comes from an ample one)**.** *Let D be basepoint free on the complete toric variety X_Σ. Then D is linearly equivalent to the pullback of an ample divisor on the projective toric variety associated to P_D:*

$$D \sim \pi_D^* A_{P_D}.$$

Reference. This is Cox–Little–Schenck, Theorem 6.2.8.

Conceptual proof sketch. The support function of D is

$$\psi_D(u) = \min_{m \in P_D} \langle m, u \rangle.$$

It is linear precisely on the cones of the normal fan of P_D, so it descends from Σ to Σ_{P_D}. On the normal fan it is strictly convex, hence defines an ample divisor A_{P_D}. Pulling that support function back along the refinement recovers ψ_D, and therefore recovers D up to linear equivalence. □

The three blowup examples now have a geometric interpretation:

D	normal fan of P_D	behavior of the projective map
$2H$	fan of $\mathbb{P}^2$	contracts the exceptional curve
$2H - E$	fan of $\mathrm{Bl}_p(\mathbb{P}^2)$	forgets no toric wall
$2H - 2E$	fan of $\mathbb{P}^1$ after quotient	records the pencil of lines through p.

Surface laboratory

Use Figure 5.7 to explain the following maps.

(a) Why does the map defined by $2H$ factor through the blowdown

$$\mathrm{Bl}_p(\mathbb{P}^2) \longrightarrow \mathbb{P}^2?$$

(b) Why does $2H - E$ retain the exceptional wall?

(c) Why does $2H - 2E$ have one-dimensional image, even though its complete linear system has three monomial sections?

A full-dimensional polytope and its normal fan determine each other only up to changes of support numbers: many different polytopes can have the same normal fan. On a toric surface, varying these support numbers changes edge lengths while preserving the directions of the boundary divisors.

Guiding question

Given the rays

$$u_1, \dots, u_r$$

of a complete toric surface, which polygons have these vectors as primitive inward facet normals? How do their edge lengths and lattice points encode sections and intersection geometry?

5.5 Toric surfaces through their polygons

In dimension two, the fan–polytope dictionary is especially complete. A polygon can be drawn, measured, and modified by hand, while still remembering the projective geometry of its toric surface.

The key dimension-two fact is the following.

Theorem 5.35 (Every complete toric surface is projective)**.** *Every complete toric surface is projective. Equivalently, every complete fan in $N_{\mathbb{R}} \simeq \mathbb{R}^2$ is the normal fan of a lattice polygon.*

Reference. This is Cox–Little–Schenck, Proposition 6.3.25.

Conceptual proof sketch. Let

$$u_1, \dots, u_s$$

be primitive generators of the rays, listed cyclically. Because the fan is complete, the origin lies in the interior of their convex hull. Hence there is a relation

$$\lambda_1 u_1 + \cdots + \lambda_s u_s = 0$$

with every $\lambda_i > 0$. The relation can be chosen rational and then scaled so that all λ_i are positive integers.

Rotate each vector u_i by ninety degrees, choosing the direction compatible with the cyclic ordering, and use the resulting vector of lattice length λ_i as the i-th edge of a polygon. The relation above says that the edge path closes. Its cyclic ordering makes the polygon convex, and its primitive inward facet normals are exactly the u_i. Therefore its normal fan is the original fan. □

Thus every complete toric surface can be studied through many polygons: changing the support numbers changes the divisor while preserving the surface.

5.5.1 A four-polygon catalogue

The same familiar surfaces now appear as familiar shapes.

For the Hirzebruch surface, use the ray generators

$$u_1 = e_1, \qquad u_2 = e_2, \qquad u_3 = -e_1 + re_2, \qquad u_4 = -e_2.$$

The divisor

$$D = aD_3 + bD_4$$

has polygon

$$P_D = \{(x, y) : 0 \le y \le b, \quad 0 \le x \le a + ry\},$$

with vertices

$$(0,0), \quad (a,0), \quad (a+rb, b), \quad (0,b).$$

The parameter r is visible as the slope of the nonvertical side.

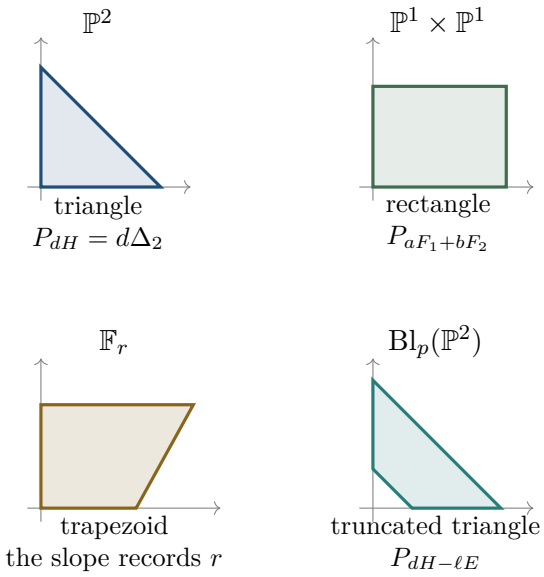


Figure 5.8: Four standard projective toric surfaces and characteristic shapes of their divisor polygons. The normal directions determine the surface; the edge positions and lengths record the chosen divisor.

Pause and think

Starting only from this trapezoid, recover its four primitive inward facet normals. In what order do they appear in the fan of $\mathbb{F}_r$?

5.5.2 Smoothness is visible at the corners

Let P be a lattice polygon. At each vertex, two facets meet, with primitive inward normals $u, v \in N$.

Proposition 5.36 (Smooth polygon criterion)**.** *The projective toric surface X_{Σ_P} is smooth if and only if, at every vertex of P, the two primitive inward facet normals form a basis of N:*

$$|\det(u, v)| = 1.$$

Equivalently, the primitive directions of the two edges leaving each vertex form a basis of M.

Proof. A vertex of P corresponds to a maximal cone of Σ_P, and that cone is generated by the primitive inward normals of the two incident facets. Apply the smoothness criterion for affine toric charts. □

A lattice polygon satisfying this condition is often called a *smooth* or *Delzant* polygon.

Observation 5.37 (Use primitive edge directions). At $(2, 0)$ in the polygon of Example 5.29, the primitive outgoing directions are $(-1, 0)$ and $(-1, 1)$, with determinant -1. The entire bottom-edge vector is $(-2, 0)$; using it would incorrectly multiply the determinant by two. The smoothness test records directions in the lattice, independently of edge lengths.

Example 5.38 (A singular triangle)**.** The triangle

$$P = \mathrm{Conv}\,(0,0),(n,0),(0,1)$$

has inward normals

$$e_1, \qquad e_2, \qquad -e_1 - ne_2.$$

At one vertex, the determinant of the two incident normals has absolute value n. Its normal fan therefore gives $\mathbb{P}(1,1,n)$, which is singular for $n > 1$.

5.5.3 Edge lengths are intersection numbers

Algebraic geometry cheatsheet

Let X be a smooth complete surface, let D be a Cartier divisor, and let $C \subseteq X$ be an irreducible complete curve. The intersection number

$$D \cdot C$$

is the degree of the restricted line bundle

$$\mathcal{O}_X(D)|_C.$$

If D gives a projective embedding, then $D \cdot C$ is the degree of the image of C with respect to a hyperplane. The self-intersection D^2 is the degree of the embedded surface.

Observation 5.39 (Count zeros after restricting the bundle). Suppose C is smooth and a nonzero regular section $\widetilde{s}$ of $L = \mathcal{O}_X(D)$ has zero divisor A. If C is not a component of A, then

$$D \cdot C = \deg(L|_C) = \sum_{p \in C} \mathrm{ord}_p(\widetilde{s}|_C).$$

Geometrically, intersect the section graph with the zero section, then restrict to C. Transverse intersections contribute one; all intersections are counted with multiplicity.

If C is a component of A, the restricted section is identically zero, although the restricted line bundle still exists. Choose another section when the linear system permits it; basepoint freeness guarantees a section not vanishing identically on C. Any two nonzero rational sections differ by a rational function:

$$\widetilde{s}' = g\widetilde{s}, \qquad \mathrm{div}(\widetilde{s}') = A + \mathrm{div}(g).$$

This is the section interpretation of changing the divisor by a principal divisor. The replacement is regular exactly when its section divisor is effective. Without such a regular replacement, one can use a rational section avoiding C at its generic point; its restriction computes degree as zeros *minus* poles. Thus moving a divisor does not require pretending every line bundle is basepoint free.

For an edge F of a lattice polygon, define its *lattice length* by

$$\mathrm{length}_M(F) = |F \cap M| - 1.$$

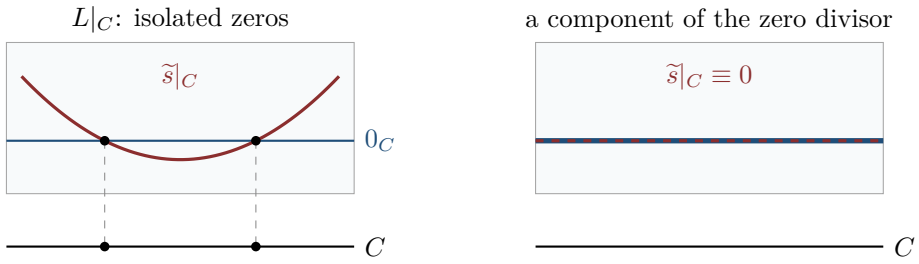


Figure 5.9: Schematic slices of the restricted line bundle. On the left, the section gives a finite zero divisor on the curve. On the right, it coincides with the zero section, so a different rational or regular representative is needed to compute degree by points.

Equivalently, if its endpoints differ by ℓw, where w is primitive, then $\text{length}_M(F) = \ell$.

Theorem 5.40 (The edge–intersection dictionary)**.** *Let X_Σ be a smooth projective toric surface, and let D be an ample torus-invariant Cartier divisor. If F_ρ is the edge of P_D with primitive inward normal u_ρ, then*

$$\boxed{D \cdot D_\rho = \text{length}_M(F_\rho).}$$

Reference. This is the surface interpretation of Cox–Little–Schenck, Proposition 6.3.8.

Conceptual proof sketch. The invariant curve D_ρ is a copy of $\mathbb{P}^1$. The two vertices of F_ρ are the Cartier data of the two maximal cones adjacent to ρ. Their difference lies in the rank-one lattice

$$M \cap u_\rho^\perp$$

and is ℓ times its primitive generator, where $\ell = \text{length}_M(F_\rho)$. Proposition 6.3.8 computes the degree of $\mathcal{O}(D)|_{D_\rho}$ from exactly this difference, giving $D \cdot D_\rho = \ell$. □

For

$$D = dH - \ell E \quad \text{on} \quad \text{Bl}_p(\mathbb{P}^2),$$

the four edge lengths of

$$P_D = \{(a, b) : a, b \geq 0,\ \ell \leq a + b \leq d\}$$

are

$$d, \qquad d - \ell, \qquad d - \ell, \qquad \ell.$$

They recover the intersections

$$D \cdot H = d, \qquad D \cdot (H - E) = d - \ell, \qquad D \cdot E = \ell.$$

Thus positivity on every invariant curve can literally be read as positivity of every edge length.

Surface laboratory

For the Hirzebruch trapezoid

$$\text{Conv}\,(0,0), (a,0), (a + rb, b), (0, b),$$

compute the four lattice edge lengths. Match them with the four intersection numbers $D \cdot D_i$.
Which lengths vanish when $a = 0$ or $b = 0$, and which toric curves should the corresponding projective map contract?

5.5.4 Supplementary example: the classroom trapezoid

Return to the polygon from Example 5.29, in the same lattice coordinates:

$$P = \mathrm{Conv}\{(0,0),(2,0),(1,1),(0,1)\}.$$

Its inward normal rays are $e_1, e_2, -e_1-e_2, -e_2$. Their fan is the fan of $X = \mathbb{F}_1 \cong \mathrm{Bl}_p(\mathbb{P}^2)$. Name the four invariant curves

$$F = D_{e_1}, \qquad B = D_{e_2}, \qquad F' = D_{-e_1-e_2}, \qquad E = D_{-e_2}.$$

The notation D_u here means the divisor of the ray generated by u. The coefficients of the divisor defining P are

$$D = 2F' + E, \qquad P_D = \{(r,q) : r,q \geq 0,\ q \leq 1,\ r+q \leq 2\}.$$

These coordinates differ from the earlier Hirzebruch catalogue: the affine lattice transformation $(r,q) \mapsto (r, 1-q)$ sends this polygon to $\mathrm{Conv}\{(0,0),(1,0),(2,1),(0,1)\}$. Its linear part reflects the normal fan. In particular, the names of the upper and lower section curves must be matched through that reflection.

Example 5.41 (Recover the intersections from the class group)**.** Write $s = \chi^{(1,0)}$ and $t = \chi^{(0,1)}$. The character-divisor formula gives

$$\mathrm{div}(s) = F - F', \qquad \mathrm{div}(t) = B - F' - E.$$

Thus $F' \sim F$ and $B \sim E + F$. The divisor-class sequence from Chapter 4 gives

$$\mathrm{Cl}(X) = \mathrm{Pic}(X) = \mathbb{Z}[F] \oplus \mathbb{Z}[E].$$

Here F, F' are distinct fibers of the ruling, E is its negative section, and B is a disjoint section. Adjacent invariant curves meet transversely once, while nonadjacent ones are disjoint. It follows that

$$F^2 = F \cdot F' = 0, \qquad E \cdot F = 1, \qquad 0 = E \cdot B = E \cdot (E+F),$$

so $E^2 = -1$ and $B^2 = (E+F)^2 = 1$. The full intersection table is

$\cdot$	F	B	F'	E
F	0	1	0	1
B	1	1	1	0
F'	0	1	0	1
E	1	0	1	-1

Taking $H = B$, the pullback of a line on $\mathbb{P}^2$, recovers

$$D \sim 2F + E = 2H - E.$$

Thus the class-group calculation gives exactly the polygon lengths:

curve	dual edge of P_D	$D \cdot$ curve
F	$r = 0$	1
B	$q = 0$	2
F'	$r + q = 2$	1
E	$q = 1$	1

For instance, $F \cdot (2F + E) = 1$, whereas $B \cdot (2F + E) = 2$. The four numbers come from the same divisor class, even though its chosen invariant representative visibly contains only F' and E.

Example 5.42 (Restrict using the star fan)**.** The invariant curve $C = V(\rho)$ is the *closure* of the orbit $O(\rho) \cong \mathbb{G}_m$, and here $C \cong \mathbb{P}^1$. By Proposition 3.18, its fan is formed by retaining only cones containing ρ, then taking their images in

$$N_\rho = N/\mathbb{Z}u_\rho.$$

One must restrict to this star before taking the quotient; projecting every ray can include a divisor that does not meet C.

First let $C = F$. Its coefficient in $D = 2F' + E$ is zero, so the canonical section does not vanish identically on it. The neighboring rays are e_2 and $-e_2$, and the quotient map $q(a, b) = b$ sends them to $+1$ and -1. Their coefficients are 0 and 1. Hence

$$D|_F = [E \cap F], \qquad \mathcal{O}_X(D)|_F \cong \mathcal{O}_{\mathbb{P}^1}(1).$$

Although $-e_1 - e_2$ also maps to -1, its divisor F' is disjoint from F and contributes nothing.

Next take $C = F'$. The canonical section now vanishes identically on C, since its coefficient in D is two. The characters st and s^2 give two regular replacements:

$$A_1 = D + \operatorname{div}(st) = F + B,$$
$$A_2 = D + \operatorname{div}(s^2) = 2F + E.$$

Both are effective and have coefficient zero on F'. This is the board's instruction to add a linear function. In the chapter's support-function convention the precise formula is

$$\psi_{D+\operatorname{div}(\chi^m)}(u) = \psi_D(u) - \langle m, u\rangle.$$

Equivalently, the opposite function $\eta_D = -\psi_D$ from Observation 5.20 gains $x + y$ or $2x$. Both have value -2 at $(-1, -1)$, cancelling the coefficient two there. The new linear pieces on the star annihilate that ray and descend to the quotient.

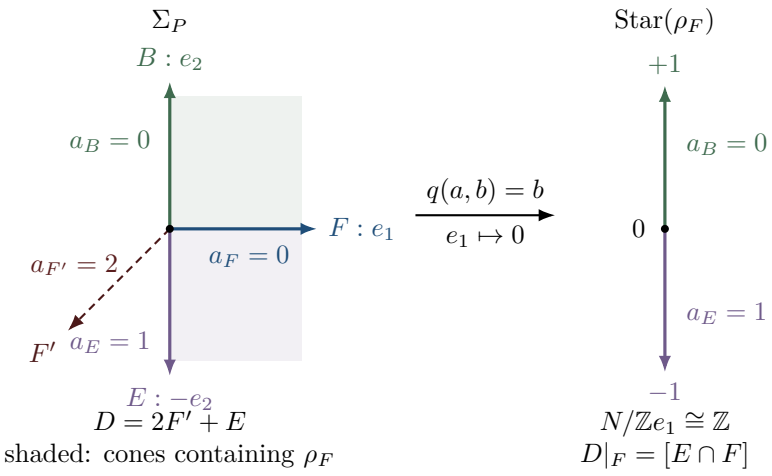


Figure 5.10: Restrict first to the cones containing ρ_F, then quotient by $\mathbb{Z}e_1$. The resulting fan describes the orbit closure $F = V(\rho_F) \cong \mathbb{P}^1$, whose dense orbit is $O(\rho_F) \cong \mathbb{G}_m$. Since $a_F = 0$, the adjacent coefficients $0, 1$ descend directly, giving a divisor of degree one. The dashed ray of the disjoint fiber F' is outside this star and contributes nothing.

For F', use $q'(a,b) = b - a$. The neighboring rays of B and E map to $+1$ and -1; the other fiber F is disjoint. With $p_B = B \cap F'$, $p_E = E \cap F'$, we obtain

$$A_1|_{F'} = p_B, \qquad A_2|_{F'} = p_E.$$

The coefficient pairs $(1,0)$ and $(0,1)$ both represent $\mathcal{O}_{\mathbb{P}^1}(1)$. They differ by the principal divisor of the quotient character $z = (t/s)|_{F'}$:

$$\operatorname{div}_{F'}(z) = p_B - p_E.$$

This explains why different choices of moving section give the same intersection number.

On the polygon side, the edge dual to F' is $G = [(2,0),(1,1)]$. Translating by the two character exponents puts it into the character lattice of the curve:

$$M \cap (-1,-1)^{\perp} = \mathbb{Z}(-1,1).$$

In that lattice,

$$G - (1,1) = [(1,-1),(0,0)] \cong [-1,0],$$
$$G - (2,0) = [(0,0),(-1,1)] \cong [0,1].$$

The two intervals differ by a lattice translation and both have length one, recovering $D \cdot F' = 1$.

Observation 5.43 (Why a negative self-intersection needs a pole). Apply the same procedure to $\mathcal{O}_X(E)|_E$. Since

$$E + \operatorname{div}(t) = B - F',$$

a rational replacement has coefficient zero on E. Restricting gives $-[E \cap F']$, because B is disjoint from E. There is one pole and no zero, so the degree is -1. This is $E^2 = -1$; no nonzero regular section of the restricted line bundle could have that degree.

5.5.5 Area is self-intersection

Normalize area so that a fundamental lattice parallelogram in M has area 1.

Theorem 5.44 (Area and self-intersection)**.** *Let D be a nef torus-invariant Cartier divisor on a smooth complete toric surface. Then*

$$\boxed{D^2 = 2\,\mathrm{Area}(P_D).}$$

Moreover,

$$-D \cdot K_X = |\partial P_D \cap M|.$$

Reference. This is Cox–Little–Schenck, Proposition 10.5.6; see also Theorem 13.4.1.

Conceptual proof sketch. The number of sections of kD is

$$|kP_D \cap M|.$$

As k grows, the leading term of this lattice-point count is

$$\mathrm{Area}(P_D)k^2.$$

On the geometric side, the leading term of the Hilbert polynomial is

$$\frac{D^2}{2}k^2.$$

Comparing the leading coefficients gives the area formula. Comparing the linear coefficients gives the boundary formula. □

For the classroom trapezoid in Section 5.5.4, the area is $3/2$. Thus

$$D^2 = 2\,\mathrm{Area}(P_D) = 3,$$

also obtained from $(2F + E)^2 = 4(F \cdot E) + E^2 = 3$. The embedding by its five sections therefore realizes $\mathbb{F}_1$ as a degree-three surface in $\mathbb{P}^4$.

When P_D is two-dimensional, combining these identities with Pick's theorem,

$$\mathrm{Area}(P) = |\,\mathrm{Int}(P) \cap M| + \frac{1}{2}|\partial P \cap M| - 1,$$

produces the surface Riemann–Roch count

$$|P_D \cap M| = \frac{D^2 - D \cdot K_X}{2} + 1.$$

In toric geometry, the lattice-point formula and the intersection formula are two views of the same polynomial.

Observation 5.45. For an ample divisor, the interior lattice points also have a direct divisor meaning. Since

$$K_X = -\sum_{\rho} D_\rho,$$

we have

$$P_{D+K_X} \cap M = \mathrm{Int}(P_D) \cap M.$$

Thus interior lattice points index the monomial sections of $\mathcal{O}(D + K_X)$.

5.5.6 Blowing up cuts a corner

Suppose P is smooth at a vertex v, and the two incident facets have primitive inward normals u and v'. In suitable integral coordinates near that corner, the polygon is locally

$$x \geq 0, \qquad y \geq 0.$$

Cutting off a corner of lattice size ℓ adds the inequality

$$x + y \geq \ell.$$

The new facet has primitive inward normal

$$u + v'.$$

This is precisely the star subdivision that performs the toric blowup of the corresponding fixed point.

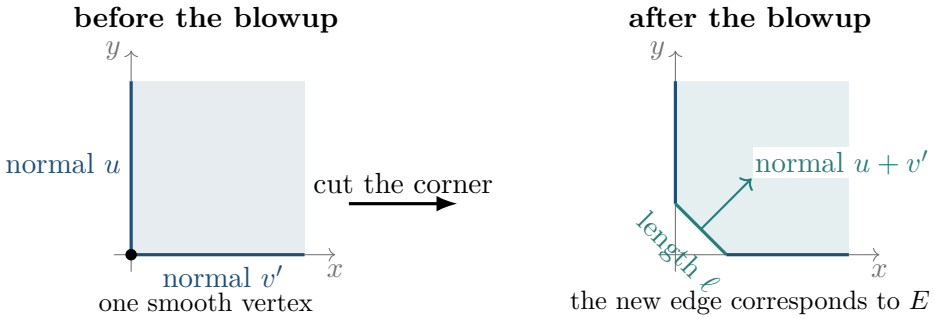


Figure 5.11: At a smooth polygon corner, a toric blowup cuts off a triangle. The new edge has inward normal equal to the sum of the two old normals and corresponds to the exceptional divisor.

Conversely, when an edge shrinks to length zero, the corresponding ray disappears from the normal fan and the invariant curve is contracted. This is the polygon version of a toric blowdown.

Pause and think

Begin with the triangle $d\Delta_2$. Cut all three corners by lattice sizes ℓ_0, ℓ_1, ℓ_2.
Write the six inequalities of the resulting polygon. What numerical conditions must the ℓ_i satisfy for all six edges to have positive length?

5.5.7 A surface-recognition workflow

For a lattice polygon P, one can now read the associated surface and divisor in a consistent order:

Look at	Combinatorial information	Geometric meaning
facet normals	normal fan	the toric surface
corners	determinants of adjacent normals	smooth or quotient-singular fixed points
edge lengths	lattice lengths	intersection with invariant curves
area	normalized two-dimensional volume	self-intersection of the divisor
lattice points	monomial exponents	global sections and projective coordinates
corner cuts	new facets and new normals	toric blowups and exceptional curves

A polygon is therefore not merely a picture of a toric surface. It is a simultaneous record of its fan, an ample divisor, its sections, its projective map, and several intersection numbers.

The next chapter will replace the single projective coordinate system coming from one divisor by a universal coordinate system with one variable for every ray.

Guiding question

Can one build one polynomial ring

$$\Bbbk[x_\rho : \rho \in \Sigma(1)]$$

whose variables record all invariant boundary divisors at once, and then recover X_Σ as a quotient of an open subset of affine space?

What this chapter has accomplished

A torus-invariant divisor

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

produces the polyhedron

$$P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho\}.$$

Its lattice points are precisely the exponents of its monomial global sections:

$$H^0(X_\Sigma, \mathcal{O}(D)) = \bigoplus_{m \in P_D \cap M} \Bbbk \chi^m.$$

Those sections define projective maps. For a finite set

$$A = \{m_0, \ldots, m_r\} \subseteq M,$$

the monomial map

$$t \longmapsto [\chi^{m_0}(t) : \cdots : \chi^{m_r}(t)]$$

produces a projective toric variety by taking the orbit closure.

The support function then detects how much projective geometry the sections see:

$$\boxed{\begin{aligned} \psi_D \text{ convex} &\iff D \text{ is basepoint free,} \\ \psi_D \text{ strictly convex} &\iff D \text{ is ample.} \end{aligned}}$$

Equivalently, convexity makes the complete linear system define a morphism, while strict convexity makes a sufficiently large multiple give an embedding.

A polytope remembers these bends through its normal fan. For a basepoint-free divisor,

$$\Sigma \text{ refines } \Sigma_{P_D},$$

and strict convexity is exactly the condition

$$\Sigma = \Sigma_{P_D}.$$

Thus an ample toric divisor is the same data as a lattice polytope whose normal fan is the fan of the variety.

For surfaces, the dictionary becomes especially visible:

$$\begin{array}{ccc} \text{edge lengths of } P_D & \longleftrightarrow & D \cdot D_\rho, \\ 2\,\mathrm{Area}(P_D) & \longleftrightarrow & D^2, \\ P_D \cap M & \longleftrightarrow & \text{global sections,} \\ \text{cutting a corner} & \longleftrightarrow & \text{a toric blowup.} \end{array}$$

Every complete toric surface is projective. The challenge exercise shows that this phenomenon stops in dimension three.

Comprehension exercises

5.1. Let

$$D = aF_1 + bF_2$$

on $\mathbb{P}^1 \times \mathbb{P}^1$, with $a, b \geq 0$.

(a) Show that P_D is a rectangle with side lengths a and b.

(b) Count its lattice points and recover

$$h^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(a,b)) = (a+1)(b+1).$$

(c) Write the monomial map associated to the complete linear system.

(d) Determine when D is very ample and describe what the map does when one of a, b is zero.

5.2. Let D, E be torus-invariant divisors and let $m_0 \in M$. Prove the following statements directly from the defining inequalities:

(a)

$$P_{kD} = kP_D \qquad (k \in \mathbb{Z}_{>0}).$$

(b)

$$P_{D+\mathrm{div}(\chi^{m_0})} = P_D - m_0.$$

(c)

$$P_D + P_E \subseteq P_{D+E}.$$

Interpret each statement using sections and projective maps.

5.3. Let

$$A = \{m_0, \ldots, m_r\} \subseteq M.$$

(a) Show that translating every element of A by the same lattice vector does not change the monomial map to $\mathbb{P}^r$.

(b) Show that the kernel of

$$\phi_A : T \longrightarrow \mathbb{P}^r$$

is determined by the subgroup generated by the differences

$$m_i - m_0.$$

(c) Give a criterion for $\phi_A|_T$ to be injective.

5.4. On $X = \mathrm{Bl}_p(\mathbb{P}^2)$, consider

$$D = dH - rE, \qquad 0 \leq r \leq d.$$

(a) Count the lattice points in

$$P_D = \{(a,b) : a, b \geq 0,\ r \leq a + b \leq d\}.$$

(b) Recover the dimension of the space of degree-d plane curves vanishing to order at least r at p.

(c) Determine when the complete linear system is basepoint free and when D is ample.

(d) Describe the maps arising in the boundary cases $r = 0$ and $r = d$.

5.5. Let

$$P = \mathrm{Conv}\{(0,0),(b,0),(b+ra,a),(0,a)\}, \qquad a,b>0,$$

where $r \geq 0$.

(a) Compute the primitive inward normal to each edge.

(b) Show that the normal fan is the fan of the Hirzebruch surface $\mathbb{F}_r$.

(c) Determine the lattice lengths of the four edges.

(d) Translate those lengths into intersection numbers of the corresponding ample divisor with the invariant curves.

5.6. Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional lattice polygon.

(a) Show that P is smooth exactly when the primitive inward normals to the two edges meeting at each vertex form a lattice basis of N.

(b) Apply the test to the standard triangle, a rectangle, and the trapezoid from the previous exercise.

(c) Construct a lattice triangle whose normal fan gives $\mathbb{P}(1,1,n)$, and identify its singular corner.

5.7. For $D = 2H - E$ on $\mathrm{Bl}_p(\mathbb{P}^2)$:

(a) List the five lattice points of P_D.

(b) Write the associated monomial map on the dense torus.

(c) Compute the lattice lengths of the four edges and recover the intersection numbers of D with the four invariant curves.

(d) Compute D^2 from the area of P_D.

5.8. Let P be a smooth lattice polygon, and let P' be obtained by cutting off a corner by a segment of lattice length r.

(a) Show that the new inward normal is the sum of the two old inward normals.

(b) Interpret the operation as a toric blowup.

(c) Determine how the area changes.

(d) Compare the change in area with

$$(\pi^* D - rE)^2 = D^2 - r^2.$$

5.9. Use Pick's theorem to show that, for an ample divisor D on a smooth projective toric surface,

$$h^0(X,\mathcal{O}(D)) = \mathrm{Area}(P_D) + \frac{1}{2}\,|\partial P_D \cap M| + 1.$$

Then rewrite the right-hand side using D^2, $D \cdot (-K_X)$, and $\chi(\mathcal{O}_X) = 1$.

Challenge exercise

Complete but not projective: Fulton's toric threefold.

For toric surfaces, completeness forces projectivity. We now explore a smooth complete toric threefold for which no ample divisor exists. The obstruction will be a cycle of strict-convexity inequalities that cannot all hold simultaneously.

Let

$$N = \mathbb{Z}^3$$

with standard basis e_1, e_2, e_3, and define

$$v_1 = -e_1, \quad v_2 = -e_2, \quad v_3 = -e_3, \quad v_4 = e_1 + e_2 + e_3,$$
$$v_5 = v_3 + v_4 = e_1 + e_2, \quad v_6 = v_1 + v_4 = e_2 + e_3, \quad v_7 = v_2 + v_4 = e_1 + e_3.$$

Write

$$\sigma_{ijk} = \mathrm{Cone}(v_i, v_j, v_k).$$

Consider the fan Σ whose maximal cones are

$$\begin{aligned} &\sigma_{123}, \\ &\sigma_{126},\ \sigma_{267},\ \sigma_{467}, \\ &\sigma_{135},\ \sigma_{156},\ \sigma_{456}, \\ &\sigma_{237},\ \sigma_{357},\ \sigma_{457}. \end{aligned}$$

Let

$$X = X_\Sigma.$$

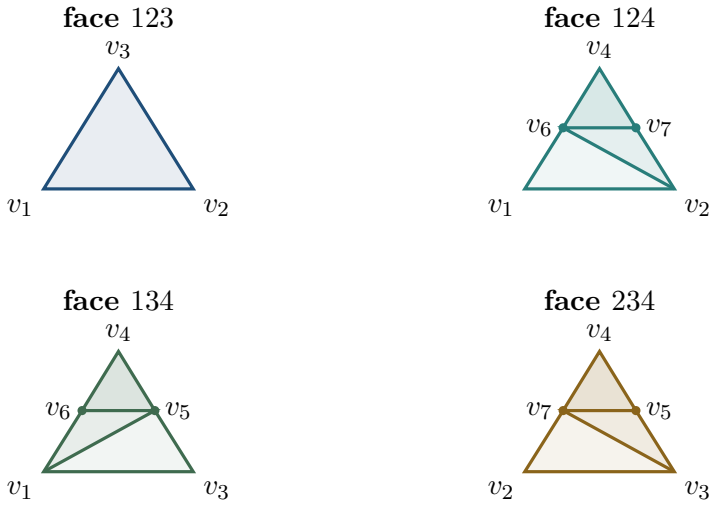


Figure 5.12: The four three-dimensional cones of the fan of $\mathbb{P}^3$, shown through their triangular cross-sections. Three faces are subdivided by the new rays v_5, v_6, v_7. The small triangles are exactly the maximal cones listed above.

Hint 1. Begin with the fan of $\mathbb{P}^3$.
Show that

$$v_1 + v_2 + v_3 + v_4 = 0$$

and that the cones generated by any three of v_1, v_2, v_3, v_4 form the fan of $\mathbb{P}^3$.
Verify that

$$v_5 \in \mathrm{Cone}(v_3, v_4), \qquad v_6 \in \mathrm{Cone}(v_1, v_4), \qquad v_7 \in \mathrm{Cone}(v_2, v_4).$$

Use Figure 5.12 to check that the listed cones subdivide the four maximal cones of the $\mathbb{P}^3$ fan. Conclude that

$$|\Sigma| = N_{\mathbb{R}},$$

so X is complete.

Hint 2. The example is smooth.
For every maximal cone σ_{ijk}, compute

$$\det(v_i, v_j, v_k).$$

Show that each determinant is ± 1. Hence every maximal cone is smooth and X is a smooth complete toric threefold.

Hint 3. Translate projectivity into a support-function problem.
Assume, for contradiction, that X is projective. Explain why there must exist an integral strictly convex support function

$$\psi : N_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

Set

$$h_i = \psi(v_i).$$

If τ is a maximal cone with linear piece m_τ, recall that strict convexity gives

$$\langle m_\tau, u \rangle > \psi(u) \qquad \text{whenever } u \notin \tau.$$

Hint 4. The first incompatible inequality.
Check the vector relation

$$v_1 + v_7 = e_3 = v_2 + v_6.$$

The vector e_3 lies in the common face

$$\mathrm{Cone}(v_2, v_6) = \sigma_{126} \cap \sigma_{267}.$$

Use the linear piece on σ_{267} to show

$$\psi(e_3) > h_1 + h_7,$$

and use the linear piece on σ_{126} to show

$$\psi(e_3) = h_2 + h_6.$$

Deduce

$$\boxed{h_2 + h_6 > h_1 + h_7.}$$

Hint 5. Complete the cycle.
Use the relations

$$v_1 + v_5 = e_2 = v_3 + v_6$$

and

$$v_3 + v_7 = e_1 = v_2 + v_5$$

together with the cones σ_{156} and σ_{357} to obtain

$$\boxed{h_1 + h_5 > h_3 + h_6}$$

and

$$\boxed{h_3 + h_7 > h_2 + h_5.}$$

Hint 6. Add the inequalities.
Add the three boxed inequalities. Explain why the left- and right-hand sides contain exactly the same six terms. Conclude that no strictly convex support function can exist on Σ.
Therefore X has no ample divisor and is not projective.

Hint 7. Complete does not mean devoid of projective maps.
The fan Σ refines the fan of $\mathbb{P}^3$. Explain why this gives a proper birational toric morphism

$$\pi : X \longrightarrow \mathbb{P}^3.$$

The pullback

$$\pi^* \mathcal{O}_{\mathbb{P}^3}(1)$$

is basepoint free. Describe its support function on Σ, and explain why it is convex but not strictly convex.
Thus X has a nonconstant projective morphism even though X itself is not projective.

Hint 8. The missing polytope.
Suppose there were a lattice polytope $P \subseteq M_{\mathbb{R}}$ with

$$\Sigma_P = \Sigma.$$

What support function would P produce? Use the contradiction above to show that no such polytope exists.
Compare this with the surface theorem that every complete toric surface fan is the normal fan of a lattice polygon. What genuinely new phenomenon appears in dimension three?

The lesson of the example

Completeness asks only that the cones cover $N_{\mathbb{R}}$. Projectivity asks for a global convex shape whose normal cones are exactly those cones. In dimension three, a complete fan may contain a cyclic pattern of walls whose required bends are mutually incompatible.

Reference. This is the classical example in Fulton, *Introduction to Toric Varieties*, §3.4, p. 71. The strict-convexity contradiction is also written out in Nicolas Perrin's lecture notes, Proposition 4.8.2.

Notes and references

The global-section formula and divisor polyhedra are developed in Cox–Little–Schenck, §4.3. Convexity, basepoint freeness, ampleness, and normal fans are treated in Chapter 6. Projective toric varieties from finite sets of lattice points are developed in Chapter 7. Intersection numbers, volumes, and the surface polygon dictionary are treated in Chapters 10 and 13.

The challenge exercise is Fulton's classical smooth complete nonprojective toric threefold. It marks the precise point at which the surface story stops generalizing: complete toric surfaces are projective, but a complete three-dimensional fan need not admit any strictly convex integral support function.

The next chapter introduces the Cox ring

$$S = \Bbbk[x_\rho : \rho \in \Sigma(1)],$$

which places one homogeneous coordinate on every invariant boundary divisor and recovers the toric variety as a quotient of an open subset of affine space.

Chapter 6

Cox Coordinates and Quotient Constructions

Chapter overview

Projective space is built from homogeneous coordinates by deleting the origin and quotienting by one scalar action:

$$\mathbb{P}^n = (\mathbb{A}^{n+1} \setminus \{0\})/\mathbb{G}_m.$$

A toric variety admits a striking generalization of this construction. Every ray $\rho \in \Sigma(1)$ contributes a coordinate x_ρ, and the polynomial ring

$$S_\Sigma = \Bbbk[x_\rho : \rho \in \Sigma(1)]$$

is graded by the divisor class group:

$$\deg(x_\rho) = [D_\rho] \in \mathrm{Cl}(X_\Sigma).$$

The grading records the independent rescalings of the coordinates. After deleting a fan-determined exceptional set and quotienting by the corresponding diagonal group, we recover X_Σ.
The chapter builds toward

$$\boxed{X_\Sigma \simeq (\mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma))/G, \qquad G = \mathrm{Hom}(\mathrm{Cl}(X_\Sigma), \mathbb{G}_m).}$$

The quotient must be interpreted with care when the fan is not simplicial; we will make the distinction precise when it arises.

The three dictionaries

Cox coordinates package the fan, the boundary, and a multigraded polynomial ring into one construction.

Algebra	Geometry	Combinatorics
variable x_ρ	invariant divisor D_ρ	ray ρ
$\mathrm{Cl}(X_\Sigma)$-grading	independent coordinate rescalings	relations among ray generators
irrelevant ideal	exceptional set to delete	collections of rays missing from cones
homogeneous quotient	toric variety X_Σ	the fan Σ

Surface laboratory

The surface examples will keep the construction concrete:

$$\mathbb{P}^2, \qquad \mathbb{P}^1 \times \mathbb{P}^1, \qquad \mathbb{F}_r, \qquad \mathrm{Bl}_p(\mathbb{P}^2).$$

Their Cox rings are polynomial rings with three or four variables, but the gradings and deleted coordinate loci differ. Those differences encode the divisor classes and the fan.

6.1 Why homogeneous coordinates need a grading

The last chapter ended with a polynomial ring waiting in the wings:

$$S = \Bbbk[x_\rho : \rho \in \Sigma(1)].$$

There is one variable for every ray, hence one variable for every irreducible component D_ρ of the toric boundary. At first sight, this looks almost too easy. A polynomial ring is familiar; why should it contain the geometry of an arbitrary toric variety?

The subtlety is not the ring itself. The subtlety is the word *homogeneous*. For projective space, all coordinates may be rescaled by one common scalar, and ordinary degree records that one scaling direction. A general toric variety can have several independent scaling directions. Its homogeneous coordinates therefore need a grading by a group larger than $\mathbb{Z}$.

Algebraic geometry cheatsheet

A point of projective space is an equivalence class

$$[z_0 : \cdots : z_n] = [\lambda z_0 : \cdots : \lambda z_n], \qquad \lambda \in \Bbbk^\times.$$

Equivalently,

$$\mathbb{P}^n = (\mathbb{A}^{n+1} \setminus \{0\})/\mathbb{G}_m,$$

where $\mathbb{G}_m$ acts by simultaneous scalar multiplication.
A polynomial equation

$$f(z_0, \ldots, z_n) = 0$$

is well defined on projective points only when replacing $(z_0, \ldots, z_n)$ by $(\lambda z_0, \ldots, \lambda z_n)$ multiplies f by one common power of λ. This is exactly what happens when every monomial of f has the same total degree:

$$f(\lambda z_0, \ldots, \lambda z_n) = \lambda^d f(z_0, \ldots, z_n).$$

Thus ordinary homogeneity is the algebraic record of one scalar symmetry.

6.1.1 One scaling direction: projective space

For $\mathbb{P}^2$, write homogeneous coordinates as

$$[x_0 : x_1 : x_2].$$

The triples

$$(x_0, x_1, x_2) \qquad \text{and} \qquad (\lambda x_0, \lambda x_1, \lambda x_2)$$

represent the same point. Therefore the three variables have the same weight:

$$\deg(x_0) = \deg(x_1) = \deg(x_2) = 1.$$

A polynomial such as

$$x_0 x_1 + x_2^2$$

is homogeneous of degree 2, since both terms acquire the factor λ^2. By contrast,

$$x_0 + x_1 x_2$$

is not homogeneous: its two terms acquire different powers of λ, so its zero set is not preserved by rescaling.

This familiar story can be read in divisor language. The three coordinate hyperplanes define linearly equivalent divisors:

$$[D_0] = [D_1] = [D_2] = H \in \mathrm{Cl}(\mathbb{P}^2).$$

Thus the assignment

$$\deg(x_i) = [D_i]$$

recovers the ordinary grading after identifying $\mathrm{Cl}(\mathbb{P}^2) \simeq \mathbb{Z}$ with generator H.

Pause and think

Explain directly from the fan of $\mathbb{P}^2$ why the three invariant boundary divisors have the same class. Which principal divisors give the two independent relations?

6.1.2 Two scaling directions: $\mathbb{P}^1 \times \mathbb{P}^1$

Now consider

$$\mathbb{P}^1 \times \mathbb{P}^1.$$

A point is written

$$([x_0 : x_1], [y_0 : y_1]).$$

There are two independent rescalings:

$$(x_0, x_1; y_0, y_1) \sim (\lambda x_0, \lambda x_1; \mu y_0, \mu y_1), \qquad (\lambda, \mu) \in (\Bbbk^\times)^2.$$

The x-coordinates respond only to λ, while the y-coordinates respond only to μ. One integer can no longer record both behaviors. The natural grading is the bidegree

$$\deg(x_0) = \deg(x_1) = (1, 0), \qquad \deg(y_0) = \deg(y_1) = (0, 1).$$

The polynomial

$$x_0 y_0 + x_1 y_1$$

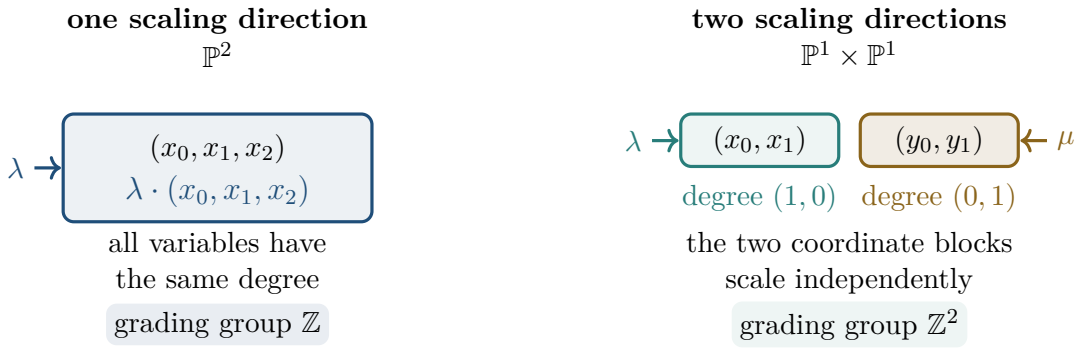


Figure 6.1: A grading records how coordinate blocks transform under rescaling. Projective space has one common scaling direction; $\mathbb{P}^1 \times \mathbb{P}^1$ has two independent directions.

has bidegree $(1, 1)$. Under rescaling, both terms acquire the same factor:

$$(\lambda x_0)(\mu y_0) + (\lambda x_1)(\mu y_1) = \lambda\mu(x_0y_0 + x_1y_1).$$

Its zero set is therefore well defined on $\mathbb{P}^1 \times \mathbb{P}^1$.

By contrast, an ordinary total-degree grading would declare

$$x_0 + y_0$$

to be homogeneous of degree 1. But

$$x_0 + y_0 \longmapsto \lambda x_0 + \mu y_0,$$

and there is no common factor unless $\lambda = \mu$. The equation $x_0 + y_0 = 0$ is not well defined on the product. Ordinary degree has forgotten which scaling acts on which coordinate.

Pause and think

Determine which of the following polynomials are bihomogeneous, and record their bidegrees when they are:

$$x_0^2y_0 + x_0x_1y_1, \qquad x_0y_0 + x_1y_1^2, \qquad x_0x_1 + y_0y_1.$$

For each failure, identify the two monomials whose scaling behaviors do not match.

The divisor class group explains the bidegree. If F_1 and F_2 are the two ruling classes, then

$$\mathrm{Cl}(\mathbb{P}^1 \times \mathbb{P}^1) \simeq \mathbb{Z}F_1 \oplus \mathbb{Z}F_2,$$

and the four invariant boundary divisors satisfy

$$[D_{x_0}] = [D_{x_1}] = F_1, \qquad [D_{y_0}] = [D_{y_1}] = F_2.$$

Hence the bidegree is again obtained by the rule

$$\deg(x_\rho) = [D_\rho].$$

6.1.3 One variable for every boundary divisor

The examples suggest the general construction. Let X_Σ be a toric variety whose rays span $N_\mathbb{R}$; in particular, this holds for every complete toric variety. For each ray

$$\rho \in \Sigma(1),$$

introduce a variable x_ρ. We want the equation

$$x_\rho = 0$$

to remember the invariant prime divisor D_ρ. More generally, a monomial

$$x^a = \prod_{\rho\in\Sigma(1)} x_\rho^{a_\rho}$$

records the effective invariant divisor

$$D_a = \sum_{\rho\in\Sigma(1)} a_\rho D_\rho.$$

This makes the correct notion of degree unavoidable:

$$\boxed{\deg(x^a) = [D_a] = \left[\sum_\rho a_\rho D_\rho\right] \in \mathrm{Cl}(X_\Sigma).}$$

In particular,

$$\boxed{\deg(x_\rho) = [D_\rho].}$$

The class group is not being added as decoration. It is precisely the group that remembers when two invariant divisors differ by a principal divisor. By theorem 4.14, we have the exact sequence

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{\Sigma(1)} \xrightarrow{\deg} \mathrm{Cl}(X_\Sigma) \longrightarrow 0,$$

where

$$m \longmapsto \big(\langle m, u_\rho\rangle\big)_\rho$$

and

$$(a_\rho)_\rho \longmapsto \left[\sum_\rho a_\rho D_\rho\right].$$

The first map records the principal-divisor relation

$$\mathrm{div}(\chi^m) = \sum_\rho \langle m, u_\rho\rangle D_\rho.$$

Thus two monomials x^a and x^b have the same degree exactly when

$$\sum_\rho (a_\rho - b_\rho) D_\rho$$

is principal.

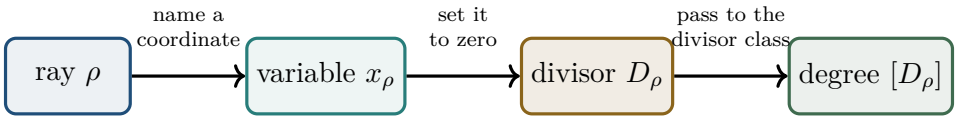


Figure 6.2: The basic chain behind Cox coordinates. Each ray gives both a boundary divisor and a variable; the divisor class becomes the variable's degree.

Definition 6.1 (The class-group grading)**.** Let

$$S_\Sigma = \Bbbk[x_\rho : \rho \in \Sigma(1)].$$

Give S_Σ the $\mathrm{Cl}(X_\Sigma)$-grading determined by

$$\deg(x_\rho) = [D_\rho].$$

A polynomial is *homogeneous of degree* $\beta \in \mathrm{Cl}(X_\Sigma)$ if every monomial appearing in it has degree β.

Reference. This is the grading introduced in Cox–Little–Schenck, §5.2, "The Grading." Their exact sequence (5.1.1) is the same class-group sequence used above.

Observation 6.2. There is optional representation-theoretic language for this picture. If an abelian group A grades a ring

$$S = \bigoplus_{\alpha \in A} S_\alpha,$$

then the group

$$\mathrm{Hom}(A, \mathbb{G}_m)$$

acts diagonally on $\mathrm{Spec}(S)$, and the spaces S_α are its weight spaces. In our setting,

$$A = \mathrm{Cl}(X_\Sigma).$$

Thus the class-group grading is the algebraic shadow of the rescalings that will eventually be quotiented out. No familiarity with weight spaces is needed: the concrete content is simply that all monomials of a homogeneous polynomial acquire the same common factor.

6.1.4 Surface laboratory: the blowup grading

Return to the toric blowup

$$X = \mathrm{Bl}_p(\mathbb{P}^2)$$

from Chapter 5. Its rays are generated by

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2.$$

Let the corresponding variables be

$$x_0, x_1, x_E, x_2.$$

The principal-divisor relations are

$$-D_0 + D_1 + E \sim 0, \qquad -D_0 + E + D_2 \sim 0.$$

Writing $H = [D_0]$, we obtain

$$[D_1] = [D_2] = H - E.$$

Therefore the polynomial ring

$$S_X = \Bbbk[x_0, x_1, x_E, x_2]$$

has the $\mathrm{Cl}(X) \simeq \mathbb{Z}H \oplus \mathbb{Z}E$ grading

$$\deg(x_0) = H, \qquad \deg(x_1) = \deg(x_2) = H - E, \qquad \deg(x_E) = E.$$

Now consider the divisor class $2H - E$. The five monomials

$$x_0x_1, \qquad x_0x_2, \qquad x_1^2x_E, \qquad x_1x_2x_E, \qquad x_2^2x_E$$

all have degree $2H - E$. They are the Cox-coordinate counterparts of the five lattice points in the truncated polygon P_{2H-E} from Chapter 5.

This is our first glimpse of what the Cox ring will accomplish: one multigraded polynomial ring simultaneously contains the monomial sections of many divisor classes.

Prediction

For the divisor class $dH - rE$, predict the exponents of a general monomial

$$x_0^{a_0}x_1^{a_1}x_E^{a_E}x_2^{a_2}$$

of degree $dH - rE$. How should the resulting integer conditions be related to the inequalities

$$a, b \geq 0, \qquad r \leq a + b \leq d$$

that define the section polytope P_{dH-rE}?

The three dictionaries

The need for a class-group grading can now be read in all three languages.

Algebra	Geometry	Combinatorics
variable x_ρ	boundary divisor D_ρ	ray ρ
monomial x^a	effective invariant divisor $\sum a_\rho D_\rho$	nonnegative exponent vector $a \in \mathbb{Z}^{\Sigma(1)}$
degree in $\mathrm{Cl}(X_\Sigma)$	linear-equivalence class	exponent vectors modulo principal-divisor relations
homogeneous polynomial	a scaling-invariant equation	monomials lying in one fiber of the degree map

The lesson is compact:

one scaling direction gives ordinary degree;
many scaling directions give the class-group grading.

Guiding question

For a fixed divisor class $\beta \in \mathrm{Cl}(X_\Sigma)$, what geometric object is represented by the homogeneous piece

$$(S_\Sigma)_\beta?$$

Can it be identified with the global sections of a divisor in the class β, and how does multiplication in the polynomial ring encode multiplication of sections?

6.2 The Cox ring and its homogeneous pieces

Section 6.1 explained why the variables of a toric homogeneous coordinate ring must be graded by $\mathrm{Cl}(X_\Sigma)$. We now take the next step: instead of looking at one degree at a time, we place all divisor classes and all their monomial sections inside a single polynomial ring.

Throughout this section, the rays of Σ span $N_\mathbb{R}$. This holds in particular when X_Σ is complete. Equivalently, X_Σ has no torus factor, and the map

$$M \longrightarrow \mathbb{Z}^{\Sigma(1)}, \qquad m \longmapsto \big(\langle m, u_\rho\rangle\big)_\rho$$

is injective.

6.2.1 One polynomial ring for all divisor classes

Definition 6.3 (The Cox ring)**.** The *Cox ring*, or *total coordinate ring*, of X_Σ is the polynomial ring

$$\mathrm{Cox}(X_\Sigma) = S_\Sigma = \Bbbk[x_\rho : \rho \in \Sigma(1)]$$

with the $\mathrm{Cl}(X_\Sigma)$-grading determined by

$$\deg(x_\rho) = [D_\rho].$$

For $\alpha \in \mathrm{Cl}(X_\Sigma)$, the homogeneous piece of degree α is denoted

$$(S_\Sigma)_\alpha.$$

Thus

$$S_\Sigma = \bigoplus_{\alpha \in \mathrm{Cl}(X_\Sigma)} (S_\Sigma)_\alpha, \qquad (S_\Sigma)_\alpha (S_\Sigma)_\beta \subseteq (S_\Sigma)_{\alpha+\beta}.$$

Reference. This is the total coordinate ring and grading of Cox–Little–Schenck, §5.2.

The ring is polynomial, but its grading is geometric. A monomial

$$x^a = \prod_\rho x_\rho^{a_\rho}$$

represents the effective invariant divisor

$$E_a = \sum_\rho a_\rho D_\rho,$$

and its degree is precisely the divisor class

$$\deg(x^a) = [E_a].$$

Consequently,

$$(S_\Sigma)_\alpha$$

is spanned by the monomials whose effective invariant divisors lie in the class α.

Example 6.4 (The familiar graded pieces of $\mathbb{P}^2$)**.** For $\mathbb{P}^2$,

$$S = \Bbbk[x_0, x_1, x_2], \qquad \deg(x_i) = H.$$

After identifying $\mathrm{Cl}(\mathbb{P}^2) \simeq \mathbb{Z}H$, the homogeneous piece S_{dH} is the usual vector space of degree-d homogeneous polynomials. For example,

$$S_{2H} = \mathrm{span}_\Bbbk\{x_0^2, x_0x_1, x_0x_2, x_1^2, x_1x_2, x_2^2\}.$$

Thus ordinary homogeneous coordinates are the rank-one instance of the Cox construction.

Example 6.5 (Bihomogeneous pieces on $\mathbb{P}^1 \times \mathbb{P}^1$)**.** For

$$S = \Bbbk[x_0, x_1, y_0, y_1]$$

with

$$\deg(x_i) = (1, 0), \qquad \deg(y_j) = (0, 1),$$

the piece $S_{(a,b)}$ is spanned by the bihomogeneous monomials

$$x_0^i x_1^{a-i} y_0^j y_1^{b-j}, \qquad 0 \le i \le a, \quad 0 \le j \le b.$$

Hence

$$\dim_\Bbbk S_{(a,b)} = (a+1)(b+1)$$

when $a, b \ge 0$, while the piece is zero if either coordinate is negative.

These examples suggest that a homogeneous piece should be more than an abstract collection of polynomials: it should be the space of sections of the corresponding divisor class.

6.2.2 A degree fiber is a section polytope

Choose a torus-invariant Weil divisor

$$D = \sum_{\rho \in \Sigma(1)} d_\rho D_\rho, \qquad \alpha = [D] \in \mathrm{Cl}(X_\Sigma).$$

Recall from Chapter 5 that

$$P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle + d_\rho \geq 0 \text{ for every } \rho\}.$$

For a lattice point $m \in P_D \cap M$, define the monomial

$$x^{\langle m,D\rangle} := \prod_{\rho \in \Sigma(1)} x_\rho^{\langle m,u_\rho\rangle + d_\rho}.$$

The inequalities defining P_D guarantee that every exponent is nonnegative.

Lemma 6.6 (Homogenizing a character)**.** *For every $m \in P_D \cap M$, the monomial $x^{\langle m,D\rangle}$ has degree $[D]$.*

Proof. Its degree is

$$\left[\sum_\rho (\langle m, u_\rho\rangle + d_\rho) D_\rho\right] = [D + \operatorname{div}(\chi^m)] = [D].$$

The principal-divisor formula is the entire reason the degree does not change when the character χ^m is homogenized. □

The exponent vector of this monomial is

$$\big(\langle m, u_\rho\rangle + d_\rho\big)_\rho \in \mathbb{Z}_{\geq 0}^{\Sigma(1)}.$$

Thus the lattice points of P_D are converted into the nonnegative integer points lying over $[D]$ under the degree map.

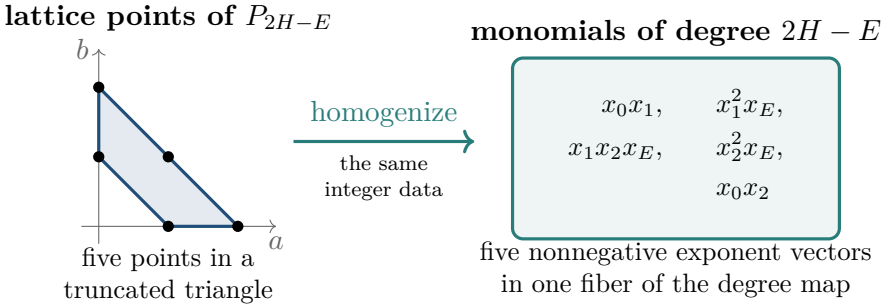


Figure 6.3: For $D = 2H - E$ on $\mathrm{Bl}_p(\mathbb{P}^2)$, homogenization turns the five lattice points of P_D into the five monomials in the homogeneous piece S_{2H-E}.

6.2.3 Homogeneous pieces are spaces of sections

Theorem 6.7 (Sections inside the Cox ring)**.** *Let D be a torus-invariant Weil divisor on a toric variety without torus factors, and set $\alpha = [D]$. Then the assignment*

$$\chi^m \longmapsto x^{\langle m,D\rangle}$$

induces an isomorphism

$$\boxed{H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) \simeq (S_\Sigma)_\alpha.}$$

Under this isomorphism, the character basis indexed by $P_D \cap M$ becomes the monomial basis of the homogeneous piece.

Reference. This is Cox–Little–Schenck, Proposition 5.4.1.

Proof. By theorem 5.4, the left-hand side has basis

$$\{\chi^m : m \in P_D \cap M\}.$$

By theorem 6.6, each basis element maps to a monomial of degree α.

Conversely, let

$$x^a = \prod_\rho x_\rho^{a_\rho}$$

be any monomial of degree α. Then

$$\left[\sum_\rho a_\rho D_\rho\right] = [D],$$

so the invariant divisor

$$\sum_\rho a_\rho D_\rho - D$$

is principal. Therefore it equals $\operatorname{div}(\chi^m)$ for some $m \in M$, and hence

$$a_\rho = \langle m, u_\rho \rangle + d_\rho \qquad \text{for every } \rho.$$

Because all $a_\rho \geq 0$, we have $m \in P_D \cap M$. The assumption that the rays span $N_\mathbb{R}$ makes m unique. Thus the two monomial bases are in bijection. □

This theorem answers the guiding question from the previous section:

$$\boxed{(S_\Sigma)_{[D]} \text{ is a canonical polynomial model for } H^0(X_\Sigma, \mathcal{O}(D)).}$$

The explicit isomorphism depends on the chosen representative D, but the homogeneous piece depends only on its class.

Warning

The notation $S_{[D]}$ does not mean that every class has a nonzero homogeneous piece. It is possible that

$$(S_\Sigma)_\alpha = 0.$$

Geometrically, this means that the class α contains no effective torus-invariant divisor, equivalently that the corresponding section polyhedron has no lattice points.

6.2.4 Multiplication of sections becomes multiplication of monomials

Let D and E be torus-invariant divisors with classes

$$\alpha = [D], \qquad \beta = [E].$$

There is a natural multiplication map

$$H^0(X_\Sigma, \mathcal{O}(D)) \otimes H^0(X_\Sigma, \mathcal{O}(E)) \longrightarrow H^0(X_\Sigma, \mathcal{O}(D+E)).$$

On character sections,

$$\chi^m \cdot \chi^{m'} = \chi^{m+m'}.$$

The corresponding Cox monomials satisfy

$$x^{\langle m,D\rangle} x^{\langle m',E\rangle} = x^{\langle m+m',D+E\rangle}.$$

Therefore the diagram

$$\begin{array}{ccc} H^0(\mathcal{O}(D)) \otimes H^0(\mathcal{O}(E)) & \longrightarrow & H^0(\mathcal{O}(D+E)) \\ \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} \\ S_\alpha \otimes S_\beta & \longrightarrow & S_{\alpha+\beta} \end{array}$$

commutes, where the bottom arrow is ordinary polynomial multiplication.

This is why the direct sum over all divisor classes is so useful: the Cox ring does not merely list section spaces. It remembers how their sections multiply.

Observation 6.8. For a general normal variety, a Cox ring is often introduced as a direct sum of section spaces over divisor classes. In the toric case, theorem 6.7 identifies that direct sum with the exceptionally concrete polynomial ring

$$\Bbbk[x_\rho : \rho \in \Sigma(1)].$$

The absence of algebraic relations among the variables is a special feature of toric geometry: each variable independently records one boundary component.

6.2.5 Surface laboratory: solving the blowup prediction

For

$$X = \mathrm{Bl}_p(\mathbb{P}^2),$$

Section 6.1 found

$$\deg(x_0) = H, \qquad \deg(x_1) = \deg(x_2) = H - E, \qquad \deg(x_E) = E.$$

Consider a monomial

$$x_0^{a_0} x_1^{a_1} x_E^{a_E} x_2^{a_2}.$$

Its degree is

$$(a_0 + a_1 + a_2)H + (a_E - a_1 - a_2)E.$$

It has degree $dH - rE$ precisely when

$$a_0 + a_1 + a_2 = d, \qquad a_E - a_1 - a_2 = -r.$$

Set

$$a = a_1, \qquad b = a_2.$$

Then

$$a_0 = d - a - b, \qquad a_E = a + b - r.$$

The requirement that all four exponents be nonnegative becomes

$$a, b \geq 0, \qquad r \leq a + b \leq d.$$

These are exactly the inequalities defining P_{dH-rE}. Thus the lattice point (a, b) corresponds to the Cox monomial

$$\boxed{x_0^{d-a-b} x_1^a x_E^{a+b-r} x_2^b.}$$

The class-group grading and the section polytope are therefore two ways of organizing the same integer solutions:

$$\boxed{P_{dH-rE} \cap M \longleftrightarrow \{\text{monomials of Cox degree } dH - rE\}.}$$

Pause and think

For $D = 3H - 2E$, list the lattice points of P_D and the corresponding Cox monomials. Verify the degree of each monomial directly from the four variable degrees.

The three dictionaries

A homogeneous piece of the Cox ring is the same object viewed in three languages.

Algebra	Geometry	Combinatorics
homogeneous piece $S_{[D]}$	sections of $\mathcal{O}(D)$	lattice points of P_D
monomial $x^{\langle m,D\rangle}$	character section χ^m	exponent vector over the degree fiber
product in the Cox ring	product of sections	addition of lattice points and exponent vectors

The Cox ring has now collected every section space into one multigraded polynomial ring. But this ring is the coordinate ring of the full affine space

$$\mathbb{A}^{\Sigma(1)} = \operatorname{Spec}(S_\Sigma),$$

and that affine space is too large. Some coordinate patterns do not represent any point of X_Σ.

Guiding question

For projective space, the only forbidden coordinate pattern is $(0, \ldots, 0)$. For a general fan, which collections of Cox coordinates are forbidden to vanish simultaneously, and how can the cones of Σ encode them as the zero set of an ideal?

6.3 The irrelevant ideal and the exceptional set

The Cox ring remembers the rays of the fan and the linear-equivalence relations among their divisors. It does *not* yet remember which collections of rays span cones. That missing incidence data is exactly what tells us which Cox coordinates are allowed to vanish together.

For projective space, the answer is familiar: all homogeneous coordinates may vanish individually, but they may not vanish simultaneously. Thus one deletes the origin before quotienting. For a general toric variety, the forbidden locus can be a union of several coordinate subspaces. The fan determines precisely which ones.

Throughout this section, write

$$S_\Sigma = \Bbbk[x_\rho : \rho \in \Sigma(1)]$$

and regard

$$\mathbb{A}^{\Sigma(1)} = \operatorname{Spec}(S_\Sigma)$$

as the affine space of Cox-coordinate tuples

$$z = (z_\rho)_{\rho \in \Sigma(1)}.$$

6.3.1 The monomial attached to a cone

Let $\sigma \in \Sigma$. Define

$$\boxed{x^{\widehat{\sigma}} = \prod_{\rho \notin \sigma(1)} x_\rho.}$$

The hat means that we multiply the variables whose rays are *omitted* from σ.

The principal open set

$$D(x^{\widehat{\sigma}}) = \{z : x^{\widehat{\sigma}}(z) \neq 0\}$$

consists of those coordinate tuples for which

$$z_\rho \neq 0 \qquad \text{whenever } \rho \notin \sigma(1).$$

The coordinates indexed by rays inside σ may vanish; all the others must be invertible.

Pause and think

Suppose σ has rays ρ_1, ρ_2 and the fan has five rays in total. Write $x^{\widehat{\sigma}}$. Which two coordinates may vanish on $D(x^{\widehat{\sigma}})$, and which three are forced to be nonzero?

If $\tau \preceq \sigma$, then $\tau(1) \subseteq \sigma(1)$, so

$$x^{\widehat{\tau}}$$

is divisible by $x^{\widehat{\sigma}}$. Consequently, the monomials coming from maximal cones already generate all the others.

Definition 6.9 (The irrelevant ideal)**.** The *irrelevant ideal* of the fan Σ is

$$\boxed{B(\Sigma) = \left\langle x^{\widehat{\sigma}} : \sigma \in \Sigma \right\rangle = \left\langle x^{\widehat{\sigma}} : \sigma \in \Sigma_{\max} \right\rangle \subseteq S_\Sigma.}$$

Its zero set

$$\boxed{Z(\Sigma) = V(B(\Sigma)) \subseteq \mathbb{A}^{\Sigma(1)}}$$

is called the *exceptional set*.

Reference. This is the construction in Cox–Little–Schenck, §5.1, under "The Exceptional Set."

The name "irrelevant" comes from projective algebraic geometry. It does not mean that the ideal is unimportant. Quite the opposite: $B(\Sigma)$ is the part of the Cox construction that remembers how the rays are assembled into cones.

Since the complement of the zero set of an ideal generated by $f_1, \dots, f_s$ is the union of the principal opens $D(f_i)$, we obtain

$$\boxed{\mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma) = \bigcup_{\sigma \in \Sigma_{\max}} D(x^{\widehat{\sigma}}).}$$

Thus every maximal cone contributes one affine chart upstairs in Cox coordinate space. In the next section, quotienting this chart will recover the toric affine chart U_σ.

6.3.2 Which coordinates are allowed to vanish together?

For a point $z = (z_\rho) \in \mathbb{A}^{\Sigma(1)}$, define its *zero-coordinate set*

$$I(z) = \{\rho \in \Sigma(1) : z_\rho = 0\}.$$

The fan gives a remarkably simple test for whether z survives the deletion of $Z(\Sigma)$.

Proposition 6.10 (The cone test for Cox coordinates)**.** *A point $z \in \mathbb{A}^{\Sigma(1)}$ lies outside $Z(\Sigma)$ if and only if there is a cone $\sigma \in \Sigma$ such that*

$$\boxed{I(z) \subseteq \sigma(1).}$$

In words: the coordinates that vanish must be indexed by rays that all fit inside one cone of the fan.

Proof. The point z lies outside $Z(\Sigma)$ exactly when

$$x^{\widehat{\sigma}}(z) \neq 0$$

for at least one maximal cone σ. This nonvanishing says that $z_\rho \neq 0$ for every $\rho \notin \sigma(1)$. Equivalently, every ray whose coordinate vanishes belongs to $\sigma(1)$, which is the condition

$$I(z) \subseteq \sigma(1).$$

□

This criterion makes the geometry of the deletion visible:

a collection of coordinates may vanish together $\iff$ their rays lie together in a cone.

Warning

The grading alone does not determine this condition. The grading knows the ray generators and their divisor-class relations. The irrelevant ideal additionally knows *which subsets of rays span cones*. Thus two fans built from the same ray data can have the same Cox ring and grading but different irrelevant ideals.

6.3.3 Minimal forbidden collections

It is often inefficient to list every forbidden set of rays. Instead, we record the minimal ones.

Definition 6.11 (Primitive collection)**.** A subset $C \subseteq \Sigma(1)$ is a *primitive collection* if

(a) the rays in C are not all contained in any cone of Σ;
(b) every proper subset of C is contained in the rays of some cone of Σ.

Thus a primitive collection is a minimal forbidden simultaneous vanishing pattern.

Proposition 6.12 (The exceptional set from primitive collections)**.** *The irreducible coordinate-subspace components of $Z(\Sigma)$ are indexed by the primitive collections:*

$$\boxed{Z(\Sigma) = \bigcup_{C \text{ primitive}} V(x_\rho : \rho \in C).}$$

Reference. This is Cox–Little–Schenck, Definition 5.1.5 and Proposition 5.1.6.

Conceptual proof sketch. If all coordinates indexed by a primitive collection C vanish, then their zero-coordinate set cannot lie in one cone. By theorem 6.10, every such point belongs to $Z(\Sigma)$.

Conversely, take a forbidden zero-coordinate set. Since the fan is finite, it contains a minimal subset that still fails to lie in one cone. That minimal subset is a primitive collection. Hence every point of the exceptional set lies in one of the displayed coordinate subspaces. □

6.3.4 Projective space: delete only the origin

For $\mathbb{P}^n$, the fan has rays

$$\rho_0, \dots, \rho_n,$$

and each maximal cone contains all but one of them. If σ_i is the maximal cone omitting ρ_i, then

$$x^{\widehat{\sigma_i}} = x_i.$$

Therefore

$$B(\Sigma) = \langle x_0, \dots, x_n \rangle, \qquad Z(\Sigma) = \{0\}.$$

The Cox open set is exactly

$$\mathbb{A}^{n+1} \setminus \{0\},$$

recovering the familiar starting point for homogeneous coordinates on projective space.

The unique primitive collection is the full set

$$\{\rho_0, \dots, \rho_n\}.$$

Any proper subset spans a cone, but all $n+1$ rays do not.

6.3.5 Two independent projective factors

For $\mathbb{P}^1 \times \mathbb{P}^1$, use Cox coordinates

$$(x_0, x_1; y_0, y_1),$$

where x_0, x_1 belong to the first factor and y_0, y_1 to the second. A maximal cone contains one ray from each opposite pair. The four complementary monomials are

$$x_0y_0, \qquad x_0y_1, \qquad x_1y_0, \qquad x_1y_1.$$

Hence

$$\begin{aligned} B(\Sigma) &= \langle x_0y_0, x_0y_1, x_1y_0, x_1y_1 \rangle \\ &= \langle x_0, x_1 \rangle \cap \langle y_0, y_1 \rangle, \end{aligned}$$

and therefore

$$\boxed{Z(\Sigma) = V(x_0, x_1) \cup V(y_0, y_1).}$$

This is exactly the expected coordinate rule:

$$(x_0, x_1) \neq (0, 0) \qquad \text{and} \qquad (y_0, y_1) \neq (0, 0).$$

The two primitive collections are the two opposite pairs of rays.

Pause and think

Why is the origin not the only point that must be removed from $\mathbb{A}^4$ in this example? Give a point other than the origin that cannot represent an element of $\mathbb{P}^1 \times \mathbb{P}^1$, and identify the primitive collection responsible for excluding it.

6.3.6 Surface laboratory: the blowup of $\mathbb{P}^2$

For

$$X = \mathrm{Bl}_p(\mathbb{P}^2),$$

recall the cyclically ordered rays

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2,$$

with Cox variables

$$x_0, x_1, x_E, x_2.$$

The maximal cones and their complementary monomials are

maximal cone	$x^{\widehat{\sigma}}$
$\mathrm{Cone}(u_0, u_1)$	x_Ex_2
$\mathrm{Cone}(u_1, u_E)$	x_0x_2
$\mathrm{Cone}(u_E, u_2)$	x_0x_1
$\mathrm{Cone}(u_2, u_0)$	x_1x_E

Consequently,

$$\begin{aligned} B(\Sigma) &= \langle x_E x_2, x_0 x_2, x_0 x_1, x_1 x_E \rangle \\ &= \langle x_0, x_E \rangle \cap \langle x_1, x_2 \rangle, \end{aligned}$$

so

$$\boxed{Z(\Sigma) = V(x_0, x_E) \cup V(x_1, x_2).}$$

The primitive collections are

$$\{\rho_0, \rho_E\} \qquad \text{and} \qquad \{\rho_1, \rho_2\}.$$

These are exactly the two pairs of rays that do not lie together in a cone.

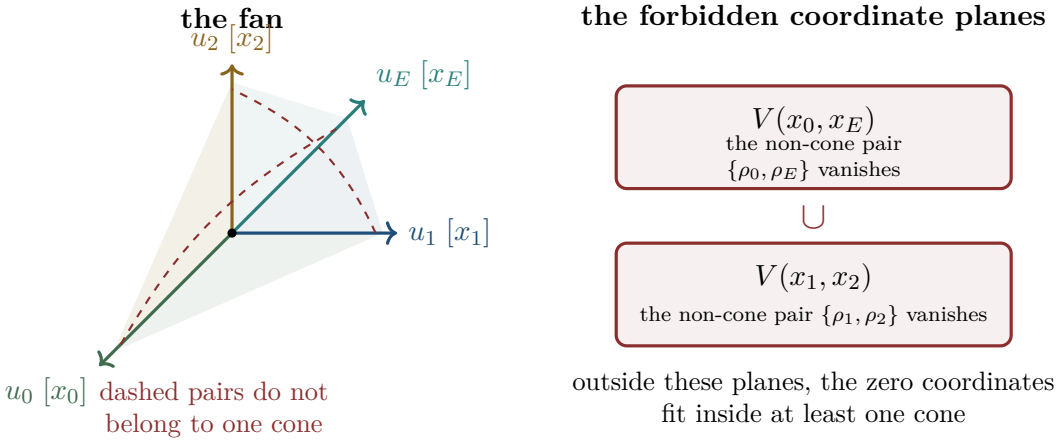


Figure 6.4: For $\mathrm{Bl}_p(\mathbb{P}^2)$, the fan has two primitive collections. Each gives one forbidden coordinate plane in Cox space.

Notice how the blowup changes more than the grading. Inserting the ray ρ_E changes which rays are adjacent, hence changes which coordinate pairs may vanish simultaneously. The fan is now visible in the open subset

$$\mathbb{A}^4 \setminus Z(\Sigma).$$

Prediction

The open set $D(\widehat{x^\sigma})$ has coordinate ring

$$(S_\Sigma)_{\widehat{x^\sigma}},$$

where every variable outside σ is invertible. Which monomials in this localization should be unchanged by all class-group rescalings? Predict why the degree-zero subring should be

$$\Bbbk[\sigma^\vee \cap M],$$

the coordinate ring of the affine toric chart U_σ.

Guiding question

We now have two ingredients:

$$\mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma) \qquad \text{and} \qquad G = \operatorname{Hom}(\operatorname{Cl}(X_\Sigma), \mathbb{G}_m).$$

Why does quotienting each open set $D(\widehat{x^\sigma})$ by the class-group rescalings recover U_σ, and when are the fibers of the resulting map exactly the G-orbits?

6.4 The quotient construction

The previous two sections produced the ingredients of a homogeneous coordinate construction:

$$S_\Sigma = \Bbbk[x_\rho : \rho \in \Sigma(1)], \qquad \deg(x_\rho) = [D_\rho], \qquad \widehat{X}_\Sigma = \mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma).$$

The open set $\widehat{X}_\Sigma$ remembers which coordinate patterns are allowed by the fan. What remains is to identify points that differ only by the rescalings encoded by the class-group grading.

Throughout this section, the rays of Σ span $N_\mathbb{R}$, so X_Σ has no torus factor.

6.4.1 The rescaling group

The grading group itself determines the group that acts on Cox space.

Definition 6.13 (The Cox rescaling group)**.** Set

$$G = \operatorname{Hom}_\mathbb{Z}(\operatorname{Cl}(X_\Sigma), \mathbb{G}_m).$$

For $g \in G$, the value $g(\alpha) \in \mathbb{G}_m$ is defined for every class $\alpha \in \operatorname{Cl}(X_\Sigma)$. The action on Cox coordinates is

$$g \cdot (z_\rho)_\rho = \big(g([D_\rho])z_\rho\big)_\rho.$$

Thus a coordinate is rescaled according to its degree. More generally, if $f \in (S_\Sigma)_\alpha$ is homogeneous, then

$$f(g \cdot z) = g(\alpha) f(z).$$

In particular,

the G-invariant elements of S_Σ are precisely the elements of degree 0.

Observation 6.14. If

$$\operatorname{Cl}(X_\Sigma) \simeq \mathbb{Z}^r \oplus A_{\text{tors}},$$

then G is a product of $(\mathbb{G}_m)^r$ with a finite diagonalizable group. The technical name is a *quasitorus*. When the class group is free, as in all of our smooth surface examples, G is an ordinary algebraic torus.

Dualizing the class-group sequence

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{\Sigma(1)} \longrightarrow \operatorname{Cl}(X_\Sigma) \longrightarrow 0$$

gives an exact sequence of diagonalizable groups

$$\boxed{1 \longrightarrow G \longrightarrow (\mathbb{G}_m)^{\Sigma(1)} \longrightarrow T_N \longrightarrow 1.}$$

The middle group rescales every Cox coordinate independently. Its subgroup G consists exactly of the rescalings that disappear when we pass to the dense torus T_N of X_Σ.
Reference. This is Cox–Little–Schenck, exact sequence (5.1.2) and Lemma 5.1.1.

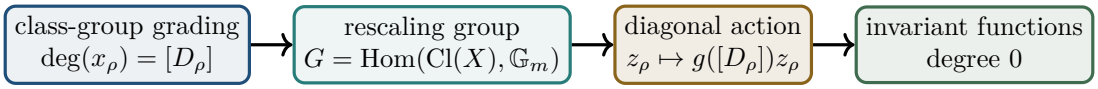


Figure 6.5: A grading and a diagonal group action are two descriptions of the same symmetry. Quotient functions are the homogeneous expressions whose total class-group degree is zero.

Algebraic geometry cheatsheet

Suppose an algebraic group G acts on an affine variety $Y = \mathrm{Spec}(A)$. The invariant ring

$$A^G = \{f \in A : g \cdot f = f \text{ for every } g \in G\}$$

defines the affine quotient

$$Y /\!/ G = \mathrm{Spec}(A^G).$$

The double slash reminds us that this construction is governed by invariant functions.
A *geometric quotient* has the stronger property that its fibers are exactly the G-orbits. A categorical or good quotient may put several orbits in one fiber; in that case the invariant functions cannot distinguish those orbits.
For Cox coordinates, the local quotient rings will therefore be the degree-zero parts of suitable localizations of S_Σ.

6.4.2 Recovering one affine chart

Fix a cone $\sigma \in \Sigma$. Recall the monomial

$$x^{\widehat{\sigma}} = \prod_{\rho \notin \sigma(1)} x_\rho.$$

The corresponding principal open subset is

$$\widehat{U}_\sigma = D(x^{\widehat{\sigma}}) \subseteq \widehat{X}_\Sigma.$$

On this open set, every coordinate x_ρ with $\rho \notin \sigma(1)$ is invertible. Its coordinate ring is

$$\Bbbk[\widehat{U}_\sigma] = (S_\Sigma)_{x^{\widehat{\sigma}}}.$$

The prediction at the end of the previous section is now the local heart of the quotient construction.

Theorem 6.15 (The degree-zero localization is the affine toric chart)**.** *For every $\sigma \in \Sigma$, there is a natural isomorphism*

$$\boxed{\left((S_\Sigma)_{\widehat{x^\sigma}}\right)_0 \simeq \Bbbk[\sigma^\vee \cap M].}$$

Equivalently,

$$\widehat{U}_\sigma \,/\!/\, G \simeq U_\sigma.$$

Reference. This is the local invariant-ring computation (5.1.4) in the proof of Cox–Little–Schenck, Theorem 5.1.11.

Proof. For $m \in \sigma^\vee \cap M$, define

$$\Phi_\sigma(\chi^m) = \prod_{\rho \in \Sigma(1)} x_\rho^{\langle m, u_\rho \rangle}.$$

When $\rho \in \sigma(1)$, the exponent is nonnegative because $m \in \sigma^\vee$. When $\rho \notin \sigma(1)$, the exponent may be negative, but x_ρ is invertible in $(S_\Sigma)_{\widehat{x^\sigma}}$. Thus the expression is well defined.

Its degree is zero, since its exponent vector is the principal-divisor vector

$$\big(\langle m, u_\rho \rangle\big)_\rho \in \operatorname{im}(M \to \mathbb{Z}^{\Sigma(1)}).$$

Hence Φ_σ maps into the degree-zero subring.

Conversely, consider a degree-zero Laurent monomial

$$x^a = \prod_\rho x_\rho^{a_\rho} \in (S_\Sigma)_{\widehat{x^\sigma}}.$$

We have $a_\rho \geq 0$ for $\rho \in \sigma(1)$, since those variables were not inverted. Degree zero means that the exponent vector a lies in the kernel of

$$\mathbb{Z}^{\Sigma(1)} \longrightarrow \operatorname{Cl}(X_\Sigma).$$

By exactness, there is a unique $m \in M$ such that

$$a_\rho = \langle m, u_\rho \rangle \qquad \text{for every } \rho.$$

The inequalities on the coordinates belonging to σ say precisely that $m \in \sigma^\vee$. Therefore every degree-zero monomial is in the image of Φ_σ, and the same follows for linear combinations of such monomials. □

The formula is worth reading in both directions:

$$\boxed{\chi^m \quad \longleftrightarrow \quad \prod_\rho x_\rho^{\langle m, u_\rho \rangle}.}$$

A character on the toric chart becomes a degree-zero Laurent monomial in Cox coordinates. Negative exponents are allowed only on the coordinates that were declared nonzero on that chart.

Example 6.16 (The usual affine chart of projective space)**.** For $\mathbb{P}^2$, let σ be the cone whose complementary ray corresponds to x_0. Then

$$\widehat{U}_\sigma = D(x_0), \qquad (S_{x_0})_0 = \Bbbk\left[\frac{x_1}{x_0}, \frac{x_2}{x_0}\right].$$

Thus

$$D(x_0) /\!/ \mathbb{G}_m \simeq \mathbb{A}^2,$$

which is the familiar projective chart $x_0 \neq 0$.

6.4.3 Gluing the local quotients

If $\tau \preceq \sigma$, then

$$\widehat{U}_\tau \subseteq \widehat{U}_\sigma.$$

Algebraically, passing from σ to τ inverts additional Cox variables. Under theorem 6.15, this is exactly the usual localization

$$\Bbbk[\sigma^\vee \cap M] \longrightarrow \Bbbk[\tau^\vee \cap M]$$

that defines the open immersion $U_\tau \hookrightarrow U_\sigma$. The local quotients therefore glue in precisely the same pattern as the affine toric charts.

Theorem 6.17 (The Cox quotient)**.** *Let X_Σ be a toric variety without torus factors. There is a surjective toric morphism*

$$\pi : \widehat{X}_\Sigma = \mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma) \longrightarrow X_\Sigma$$

that is constant on G-orbits, and

$$\boxed{X_\Sigma \simeq \widehat{X}_\Sigma /\!/ G.}$$

Moreover, the quotient is geometric—its fibers are exactly the G-orbits—if and only if Σ is simplicial.

Reference. The construction of π is Cox–Little–Schenck, Proposition 5.1.9, and the quotient statement is their Theorem 5.1.11.

Conceptual proof sketch. On every G-stable open set $\widehat{U}_\sigma$, invariant functions are the degree-zero localized functions. By theorem 6.15, their spectrum is U_σ. These affine quotients agree on overlaps, so they glue to a global morphism

$$\pi : \widehat{X}_\Sigma \to X_\Sigma.$$

The remaining issue is orbit structure. If σ is simplicial, its ray generators are linearly independent over $\mathbb{R}$; this prevents a nontrivial one-parameter subgroup of G from moving an orbit toward a different orbit inside the same local fiber. Hence each fiber is one orbit. For a nonsimplicial cone, a linear dependence among its rays produces distinct orbits that invariant functions cannot separate. □

Warning

The notation matters:

$$X_\Sigma = \widehat{X}_\Sigma \,//\, G$$

is valid in general. Writing a single slash

$$X_\Sigma = \widehat{X}_\Sigma / G$$

as a literal orbit space is justified when Σ is simplicial. For a smooth fan the action is even free, so the quotient behaves like the most familiar projective-space construction.

6.4.4 Surface laboratory: one chart of the blowup

For

$$X = \mathrm{Bl}_p(\mathbb{P}^2),$$

use the basis

$$L = H - E, \qquad E$$

of $\mathrm{Cl}(X)$. Then

$$\deg(x_0) = L + E, \qquad \deg(x_1) = \deg(x_2) = L, \qquad \deg(x_E) = E.$$

Consequently,

$$G \simeq (\mathbb{G}_m)^2$$

acts by

$$(\lambda, \mu) \cdot (x_0, x_1, x_E, x_2) = (\lambda\mu x_0, \lambda x_1, \mu x_E, \lambda x_2).$$

Consider the cone

$$\sigma = \mathrm{Cone}(u_1, u_E).$$

The complementary variables are x_0, x_2, so

$$\widehat{U}_\sigma = D(x_0 x_2).$$

Two degree-zero functions are

$$s = \frac{x_1}{x_2}, \qquad t = \frac{x_E x_2}{x_0}.$$

They are invariant under both λ and μ, and the local theorem gives

$$((S_X)_{x_0x_2})_0 = \Bbbk[s, t].$$

Hence

$$\widehat{U}_\sigma \,//\, (\mathbb{G}_m)^2 \simeq \mathbb{A}^2.$$

The same calculation on the other three maximal cones gives four copies of $\mathbb{A}^2$. Their degree-zero localizations glue according to the fan, recovering the blowup.

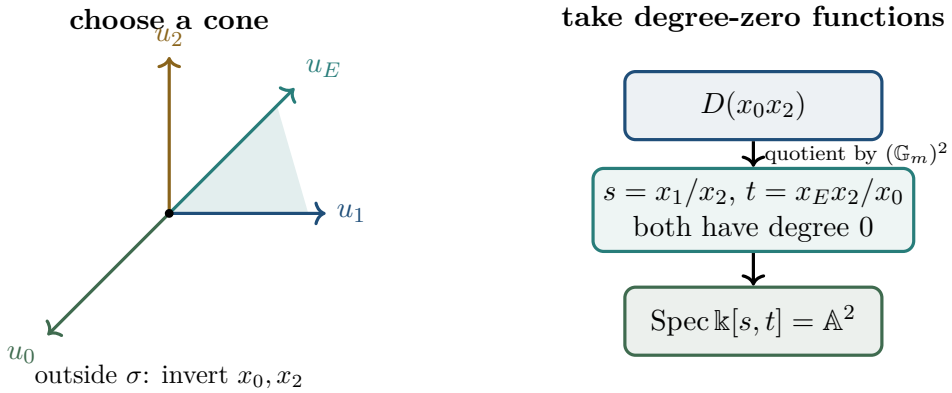


Figure 6.6: One Cox chart of the blowup. The variables outside the chosen cone are inverted, and the degree-zero Laurent monomials become the ordinary coordinates on the affine toric chart.

Pause and think

For $\mathbb{P}^1 \times \mathbb{P}^1$, use Cox coordinates

$$(x_0, x_1, y_0, y_1)$$

with bidegrees

$$\deg(x_i) = (1,0), \qquad \deg(y_j) = (0,1).$$

On the open set $D(x_0y_0)$, find two degree-zero generators and show that the local quotient is $\mathbb{A}^2$. How do your generators compare with the usual affine coordinates on the two factors?

Guiding question

The quotient construction is now formal. What does it look like in specific examples when we write down, side by side,

$$\text{the grading,} \qquad Z(\Sigma), \qquad G, \qquad \text{and the resulting quotient?}$$

How can two polynomial rings with the same number of variables produce different toric varieties because their gradings or exceptional sets differ?

6.5 Four quotient constructions

The quotient theorem becomes most useful when its ingredients are written next to one another:

$$\text{Cox ring,} \qquad \text{grading,} \qquad \text{exceptional set,} \qquad \text{group action.}$$

The examples in this section all begin with a polynomial ring, but small changes in the grading and the deleted coordinate locus produce different varieties.

6.5.1 Weight matrices

Suppose for the moment that

$$\mathrm{Cl}(X_\Sigma) \simeq \mathbb{Z}^k$$

and choose a basis. If the rays are $\rho_1, \dots, \rho_s$, write the degree of each Cox variable as a column vector. The resulting matrix

$$Q = \begin{pmatrix} | & & | \\ \deg(x_{\rho_1}) & \cdots & \deg(x_{\rho_s}) \\ | & & | \end{pmatrix} \in \mathrm{Mat}_{k\times s}(\mathbb{Z})$$

is called a *weight matrix* for the chosen basis.

If

$$g = (\lambda_1, \dots, \lambda_k) \in (\mathbb{G}_m)^k,$$

and the j-th column of Q is $(q_{1j}, \dots, q_{kj})^{\mathsf{T}}$, then

$$g \cdot x_{\rho_j} = \lambda_1^{q_{1j}} \cdots \lambda_k^{q_{kj}} x_{\rho_j}.$$

Thus the weight matrix turns the grading immediately into the diagonal action.

Warning

A weight matrix depends on the chosen basis of $\mathrm{Cl}(X_\Sigma)$. Changing the basis changes the displayed matrix and the parameters used for G, but not the quotient variety.

6.5.2 Projective space

For $\mathbb{P}^n$, the Cox ring is

$$S = \Bbbk[x_0, \dots, x_n], \qquad \mathrm{Cl}(\mathbb{P}^n) = \mathbb{Z}H,$$

and every variable has degree H. Thus

$$Q_{\mathbb{P}^n} = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}.$$

The irrelevant ideal is

$$B(\Sigma) = \langle x_0, \dots, x_n \rangle,$$

so the exceptional set is only the origin. The group

$$G = \mathbb{G}_m$$

acts by simultaneous scaling:

$$\lambda \cdot (x_0, \dots, x_n) = (\lambda x_0, \dots, \lambda x_n).$$

Therefore the Cox quotient is exactly the classical construction

$$\boxed{\mathbb{P}^n = (\mathbb{A}^{n+1} \setminus \{0\})/\mathbb{G}_m.}$$

Reference. This is Cox–Little–Schenck, Example 5.1.12.

On the chart $x_i \neq 0$, the invariant functions are

$$\frac{x_0}{x_i}, \dots, \widehat{\frac{x_i}{x_i}}, \dots, \frac{x_n}{x_i},$$

recovering the usual affine chart $\mathbb{A}^n$.

6.5.3 A product: $\mathbb{P}^1 \times \mathbb{P}^1$

Write the Cox coordinates as

$$(u_0, u_1, v_0, v_1).$$

The class group and grading are

$$\mathrm{Cl}(\mathbb{P}^1 \times \mathbb{P}^1) \simeq \mathbb{Z}F_1 \oplus \mathbb{Z}F_2,$$

$$\deg(u_0) = \deg(u_1) = F_1, \qquad \deg(v_0) = \deg(v_1) = F_2.$$

Hence

$$Q_{\mathbb{P}^1 \times \mathbb{P}^1} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

The opposite pairs of rays are the primitive collections, so

$$B(\Sigma) = \langle u_0, u_1 \rangle \cap \langle v_0, v_1 \rangle$$

and

$$Z(\Sigma) = V(u_0, u_1) \cup V(v_0, v_1).$$

Thus an allowed point satisfies

$$(u_0, u_1) \neq (0, 0), \qquad (v_0, v_1) \neq (0, 0).$$

The group $(\mathbb{G}_m)^2$ acts by

$$(\lambda, \mu) \cdot (u_0, u_1, v_0, v_1) = (\lambda u_0, \lambda u_1, \mu v_0, \mu v_1).$$

The two blocks are independently projectivized, giving

$$\boxed{\mathbb{P}^1 \times \mathbb{P}^1 = \frac{\mathbb{A}^4 \setminus \big(V(u_0, u_1) \cup V(v_0, v_1)\big)}{(\mathbb{G}_m)^2}.}$$

Reference. This is Cox–Little–Schenck, Example 5.1.13.

Pause and think

On the open set $D(u_0 v_0)$, check that

$$s = \frac{u_1}{u_0}, \qquad t = \frac{v_1}{v_0}$$

have degree zero. Explain why they give the usual affine coordinates on one product chart.

6.5.4 A twist: the Hirzebruch surface $\mathbb{F}_r$

Recall the fan of $\mathbb{F}_r$, with rays in cyclic order

$$w_1 = (1,0), \qquad w_2 = (0,1), \qquad w_3 = (-1,r), \qquad w_4 = (0,-1).$$

Use Cox coordinates

$$(u_0, v_0, u_1, v_1)$$

in the same cyclic order. The principal-divisor relations are

$$D_1 - D_3 \sim 0, \qquad D_2 + rD_3 - D_4 \sim 0.$$

Set

$$F = [D_1] = [D_3], \qquad S = [D_2].$$

Then

$$[D_4] = S + rF,$$

so the Cox grading is

$$\deg(u_0) = \deg(u_1) = F, \qquad \deg(v_0) = S, \qquad \deg(v_1) = S + rF.$$

With respect to the ordered basis (F, S),

$$Q_{\mathbb{F}_r} = \begin{pmatrix} 1 & 0 & 1 & r \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

Consequently,

$$(\lambda, \mu) \cdot (u_0, v_0, u_1, v_1) = (\lambda u_0, \mu v_0, \lambda u_1, \lambda^r \mu v_1).$$

The fan is still a four-cycle, so its primitive collections are the same opposite pairs as for $\mathbb{P}^1 \times \mathbb{P}^1$. Hence

$$Z(\Sigma_r) = V(u_0, u_1) \cup V(v_0, v_1).$$

We therefore obtain

$$\boxed{\mathbb{F}_r = \frac{\mathbb{A}^4 \setminus \big(V(u_0, u_1) \cup V(v_0, v_1)\big)}{(\mathbb{G}_m)^2},}$$

but with the twisted action displayed above.

The quotient map to the base is visible immediately:

$$\pi : \mathbb{F}_r \longrightarrow \mathbb{P}^1, \qquad [u_0, v_0, u_1, v_1] \longmapsto [u_0 : u_1].$$

On the chart $u_0 \neq 0$, use λ to set $u_0 = 1$. The remaining pair of fiber coordinates is

$$[v_0 : u_0^{-r} v_1].$$

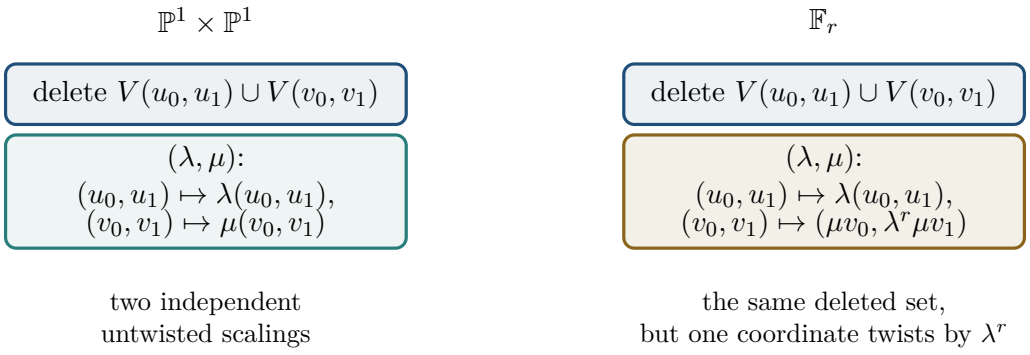


Figure 6.7: The product and the Hirzebruch surface use the same ambient space and the same exceptional set. The difference lies entirely in the grading, hence in the diagonal action.

On the chart $u_1 \neq 0$, it is

$$[v_0 : u_1^{-r} v_1].$$

On the overlap, these differ by the factor

$$\left(\frac{u_1}{u_0}\right)^{-r}.$$

The exponent r in the Cox action is therefore exactly the twisting seen when the two local product charts are glued.

When $r = 0$, the twist disappears and

$$\mathbb{F}_0 \simeq \mathbb{P}^1 \times \mathbb{P}^1.$$

Prediction

The exceptional set does not depend on r, but the quotient variety does. Which part of the quotient data must therefore remember the self-intersection of the distinguished section of $\mathbb{F}_r$?

6.5.5 The blowup of $\mathbb{P}^2$

For

$$X = \mathrm{Bl}_p(\mathbb{P}^2),$$

use the basis

$$L = H - E, \qquad E$$

of $\mathrm{Cl}(X)$. The degrees from Section 6.1 become

$$\deg(x_0) = L + E, \qquad \deg(x_1) = \deg(x_2) = L, \qquad \deg(x_E) = E.$$

In the coordinate order

$$(x_0, x_1, x_E, x_2),$$

the weight matrix is

$$Q_X = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

Thus

$$(\lambda, \mu) \cdot (x_0, x_1, x_E, x_2) = (\lambda\mu x_0, \lambda x_1, \mu x_E, \lambda x_2).$$

From Section 6.3,

$$Z(\Sigma) = V(x_0, x_E) \cup V(x_1, x_2).$$

Hence

$$\boxed{\mathrm{Bl}_p(\mathbb{P}^2) = \frac{\mathbb{A}^4 \setminus (V(x_0, x_E) \cup V(x_1, x_2))}{(\mathbb{G}_m)^2}.}$$

This is the $r = 1$ Hirzebruch quotient in different names. Indeed, under the relabeling

$\mathbb{F}_1$	u_0	u_1	v_0	v_1
$\mathrm{Bl}_p(\mathbb{P}^2)$	x_1	x_2	x_E	x_0,

and the identification

$$F = L, \qquad S = E,$$

both the degree matrix and the exceptional set agree. Thus Cox coordinates make the isomorphism

$$\boxed{\mathbb{F}_1 \simeq \mathrm{Bl}_p(\mathbb{P}^2)}$$

visible as a simple relabeling of variables and divisor classes.

Surface laboratory

The four examples can be compared at a glance:

Variety	Cox group	Deleted locus	Decisive feature
$\mathbb{P}^n$	$\mathbb{G}_m$	origin	one common scaling
$\mathbb{P}^1 \times \mathbb{P}^1$	$(\mathbb{G}_m)^2$	$V(u_0, u_1) \cup V(v_0, v_1)$	two untwisted blocks
$\mathbb{F}_r$	$(\mathbb{G}_m)^2$	the same two planes	weight r twists one coordinate
$\mathrm{Bl}_p(\mathbb{P}^2)$	$(\mathbb{G}_m)^2$	$V(x_0, x_E) \cup V(x_1, x_2)$	new ray gives a new coordinate and class

The examples support a useful slogan:

> The fan determines what must be deleted
> and how the remaining coordinates must be rescaled.

The irrelevant ideal remembers which rays fit into cones, while the class-group grading remembers the linear relations among those rays. Neither piece alone is enough.

Guiding question

All four fans in this section are smooth, so the Cox group acts freely on the Cox open set and the quotient is geometric. What changes when a cone is simplicial but not smooth, or when it is not simplicial at all?
Should one expect finite stabilizers, positive-dimensional stabilizers, or fibers containing more than one orbit?

6.6 Singular and non-simplicial fans

The examples in Section 6.5 were smooth. Their Cox groups acted freely on the Cox open set, so passing to the quotient simply identified points related by rescaling.

For a singular toric variety, two new phenomena can occur:

a point may have a nontrivial stabilizer,

and

one quotient fiber may contain several distinct orbits.

The fan determines exactly which phenomenon occurs. The decisive question is whether a cone is smooth, merely simplicial, or not simplicial at all.

6.6.1 Which rescalings fix a Cox point?

Let

$$z = (z_\rho)_\rho \in \widehat{X}_\Sigma$$

be a point of the Cox open set, and let

$$J(z) = \{\rho \in \Sigma(1) : z_\rho \neq 0\}.$$

The coordinates indexed by $J(z)$ are the ones that constrain a rescaling: if $z_\rho \neq 0$, then a group element fixing z must act trivially on that coordinate.

Set

$$L(z) = \langle [D_\rho] : \rho \in J(z) \rangle \subseteq \mathrm{Cl}(X_\Sigma).$$

Proposition 6.18 (Stabilizers from surviving degrees)**.** *The stabilizer of z under the Cox group*

$$G = \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Cl}(X_\Sigma), \mathbb{G}_m)$$

is

$$\boxed{G_z \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Cl}(X_\Sigma)/L(z), \mathbb{G}_m).}$$

Proof. An element $g \in G$ fixes z precisely when

$$g([D_\rho])z_\rho = z_\rho$$

for every ray ρ. When $z_\rho = 0$, this imposes no condition. When $z_\rho \neq 0$, it is equivalent to

$$g([D_\rho]) = 1.$$

Thus g fixes z exactly when it is trivial on the subgroup $L(z)$. Such homomorphisms are precisely those that factor through $\mathrm{Cl}(X_\Sigma)/L(z)$. □

The formula becomes especially geometric at one distinguished point for each cone. For $\sigma \in \Sigma$, define

$$z_\sigma = (z_\rho)_\rho, \qquad z_\rho = \begin{cases} 0, & \rho \in \sigma(1), \\ 1, & \rho \notin \sigma(1). \end{cases}$$

The zero coordinates fit inside σ, so $z_\sigma \in \widehat{X}_\Sigma$. Its image in X_Σ lies in the orbit $O(\sigma)$.

The boundary divisors D_ρ with $\rho \notin \sigma(1)$ do not meet the affine chart U_σ. Restriction therefore gives

$$\mathrm{Cl}(U_\sigma) \simeq \frac{\mathrm{Cl}(X_\Sigma)}{\langle [D_\rho] : \rho \notin \sigma(1) \rangle}.$$

This is the toric instance of the divisor-class restriction sequence of Cox–Little–Schenck, Theorem 4.0.20. Equivalently, one can apply theorem 4.14 directly to the affine fan of U_σ.

Corollary 6.19 (The local class group is the stabilizer group)**.** *For every cone $\sigma \in \Sigma$,*

$$\boxed{G_{z_\sigma} \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Cl}(U_\sigma), \mathbb{G}_m).}$$

This gives a useful interpretation of the local class group:

$$\boxed{\mathrm{Cl}(U_\sigma) \text{ measures the rescaling symmetry left over above } O(\sigma).}$$

6.6.2 The smooth–simplicial–nonsimplicial trichotomy

Let $\sigma(1) = \{\rho_1, \dots, \rho_s\}$, with primitive ray generators $u_1, \dots, u_s$. The affine class-group sequence is

$$M \longrightarrow \mathbb{Z}^s \longrightarrow \mathrm{Cl}(U_\sigma) \longrightarrow 0, \qquad m \longmapsto (\langle m, u_1 \rangle, \dots, \langle m, u_s \rangle).$$

The image has rank $\dim \sigma$. Hence

$$\operatorname{rank} \mathrm{Cl}(U_\sigma) = |\sigma(1)| - \dim \sigma.$$

Theorem 6.20 (Three quotient regimes)**.** *For a cone $\sigma \in \Sigma$, the following three cases occur.*

(1) If σ is smooth, then

$$\mathrm{Cl}(U_\sigma) = 0, \qquad G_{z_\sigma} = \{1\}.$$

The Cox action is free near the corresponding orbit.

(2) If σ is simplicial but not smooth, then

$$\mathrm{Cl}(U_\sigma)$$

is a nonzero finite group, and G_{z_σ} is finite. The quotient is geometric, but it has finite quotient singularities.

(3) If σ is not simplicial, then

$$\operatorname{rank} \mathrm{Cl}(U_\sigma) > 0,$$

so G_{z_σ} contains a positive-dimensional torus. The Cox quotient is not geometric: invariant functions may identify several orbits in one fiber.

Reference. The geometric-quotient criterion is Cox–Little–Schenck, Theorem 5.1.11. The equivalence between simplicial fans and finite quotient singularities is their Theorem 3.1.19. The divisor-theoretic counterpart is theorem 4.29.

Proof. The rank formula shows that $\mathrm{Cl}(U_\sigma)$ is finite exactly when

$$|\sigma(1)| = \dim \sigma,$$

which is precisely the simplicial condition. If the cone is not simplicial, the local class group has positive rank, so its dual contains a positive-dimensional torus.

When σ is simplicial, the group is trivial precisely when the map

$$M \longrightarrow \mathbb{Z}^{\sigma(1)}$$

is surjective. Dually, this says that the primitive ray generators form part of a lattice basis, which is exactly the smoothness condition. Now apply theorem 6.19 and the quotient theorem theorem 6.17. □

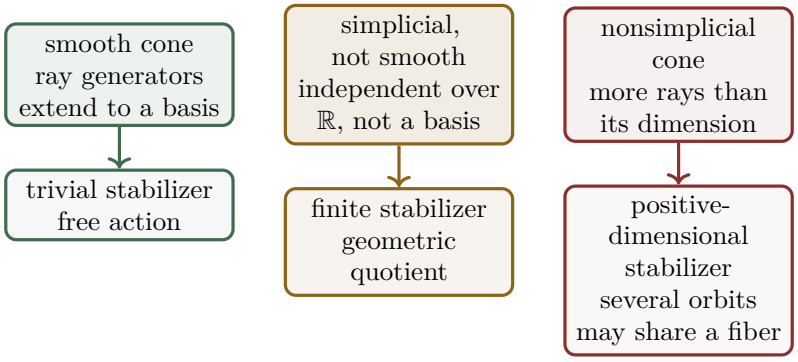


Figure 6.8: The local geometry of a toric chart is reflected in the stabilizer of its distinguished Cox point.

6.6.3 Finite stabilizers: a cyclic quotient surface

Work over $\Bbbk = \mathbb{C}$, and let

$$\sigma_d = \mathrm{Cone}((d,-1),(0,1)) \subseteq \mathbb{R}^2, \qquad d \geq 2.$$

The cone has two rays in dimension two, so it is simplicial. The determinant of its primitive generators has absolute value d, so it is not smooth.

Let the Cox variables be x_1, x_2. Since the fan has a single maximal cone containing both rays,

$$B(\Sigma) = \langle 1 \rangle, \qquad Z(\Sigma) = \varnothing.$$

The class-group computation gives

$$\mathrm{Cl}(U_{\sigma_d}) \simeq \mathbb{Z}/d\mathbb{Z}, \qquad \deg(x_1) = \deg(x_2) = 1 \pmod{d}.$$

Consequently,

$$G = \mathrm{Hom}(\mathbb{Z}/d\mathbb{Z}, \mathbb{G}_m) = \mu_d,$$

and a primitive d-th root of unity acts by

$$\zeta \cdot (x_1, x_2) = (\zeta x_1, \zeta x_2).$$

Thus

$$\boxed{U_{\sigma_d} \simeq \mathbb{A}^2/\mu_d.}$$

The invariant monomials are those whose total degree is divisible by d. In particular, the invariant ring is generated by

$$x_1^d, \quad x_1^{d-1}x_2, \quad \ldots, \quad x_1x_2^{d-1}, \quad x_2^d.$$

This is the affine cone over the degree-d rational normal curve. For $d = 2$, setting

$$A = x_1^2, \qquad B = x_1x_2, \qquad C = x_2^2$$

gives

$$\mathbb{C}[x_1, x_2]^{\mu_2} \simeq \mathbb{C}[A, B, C]/(AC - B^2).$$

Reference. The class-group calculation is Cox–Little–Schenck, Example 4.1.4.

Pause and think

Away from the origin, determine which points of $\mathbb{A}^2$ have a nontrivial stabilizer under the scalar μ_d-action. Why can the quotient still be geometric even though the action is not free?

The quotient map remembers each finite orbit as one point. The singularity comes from the fact that the orbit of the origin has a nontrivial finite stabilizer, not from several different orbits being collapsed together.

6.6.4 Positive-dimensional stabilizers: the cone over a square

Now let $N = \mathbb{Z}^3$, and consider the cone generated by

$$u_1 = (0, 0, 1), \quad u_2 = (1, 0, 1),$$
$$u_3 = (1, 1, 1), \quad u_4 = (0, 1, 1).$$

These four rays are the cone from the origin over the vertices of a unit square in the plane $z = 1$. The cone is three-dimensional but has four rays, so it is not simplicial. The relation

$$u_1 - u_2 + u_3 - u_4 = 0$$

shows that

$$\mathrm{Cl}(U_\sigma) \simeq \mathbb{Z},$$

with Cox degrees

$$\deg(x_1) = \deg(x_3) = 1, \qquad \deg(x_2) = \deg(x_4) = -1.$$

Hence $G = \mathbb{G}_m$ acts by

$$\lambda \cdot (x_1, x_2, x_3, x_4) = (\lambda x_1, \lambda^{-1} x_2, \lambda x_3, \lambda^{-1} x_4).$$

Again the fan is affine with one maximal cone, so no locus is deleted:

$$\widehat{X}_\Sigma = \mathbb{A}^4.$$

The degree-zero invariant ring is generated by

$$A = x_1 x_2, \qquad B = x_1 x_4, \qquad C = x_3 x_2, \qquad D = x_3 x_4.$$

These generators satisfy one relation,

$$AD = BC.$$

Therefore

$$\boxed{U_\sigma \simeq \mathbb{A}^4 /\!/ \mathbb{G}_m \simeq \operatorname{Spec} \frac{\Bbbk[A, B, C, D]}{(AD - BC)}.}$$

This is the three-dimensional affine quadric cone.

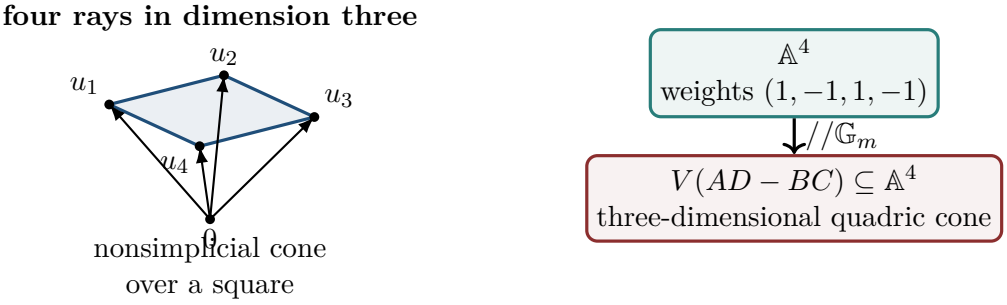


Figure 6.9: A non-simplicial cone produces a positive-dimensional stabilizer. The Cox quotient of $\mathbb{A}^4$ is the affine quadric cone, and the fiber over its vertex is larger than one orbit.

The failure of a geometric quotient can be seen directly. The vertex of the quadric cone is defined by

$$A = B = C = D = 0.$$

Its inverse image is

$$V(x_1, x_3) \cup V(x_2, x_4).$$

For example, every point

$$(x_1, 0, x_3, 0) \neq 0$$

maps to the vertex, but two such points need not lie in the same $\mathbb{G}_m$-orbit: the ratio $[x_1 : x_3] \in \mathbb{P}^1$ is unchanged by the action. Thus one quotient fiber contains a whole family of distinct orbits, together with the origin whose stabilizer is all of $\mathbb{G}_m$.

This is exactly what the double slash warns us about:

$$\mathbb{A}^4 /\!/ \mathbb{G}_m$$

is governed by invariant functions, and those functions cannot separate the different orbits in the vertex fiber.

Warning

Two uses of the word "quotient" must be kept separate.
The *Cox quotient*

$$\widehat{X}_\Sigma /\!/ G$$

is a global presentation of every toric variety. The group G may have positive dimension even when the toric variety is smooth; for example, $\mathbb{P}^n$ is a quotient by $\mathbb{G}_m$.
A *finite quotient singularity* is a local statement: near a singular point, the variety is locally a quotient of a smooth variety by a finite group. For toric varieties, this happens precisely for simplicial fans. What must be finite are the relevant point stabilizers, not necessarily the entire Cox group.

The three dictionaries

The singular quotient picture can now be read in three languages.

Algebra	Geometry	Combinatorics
$\mathrm{Cl}(U_\sigma) = 0$	trivial stabilizer; smooth chart	σ smooth
$\mathrm{Cl}(U_\sigma)$ finite	finite stabilizer; quotient singularity	σ simplicial
$\operatorname{rank} \mathrm{Cl}(U_\sigma) > 0$	positive-dimensional stabilizer; several orbits may share a fiber	σ nonsimplicial

Guiding question

The quotient construction began with a polynomial ring, a grading, and an irrelevant ideal. How much of the original fan can be recovered from those data alone?
More specifically, if one changes the deleted locus while keeping the same graded polynomial ring and the same group action, what kinds of different toric quotients might appear?

What this chapter has accomplished

A fan Σ with rays spanning $N_{\mathbb{R}}$ produces the polynomial ring

$$S_\Sigma = \Bbbk[x_\rho : \rho \in \Sigma(1)].$$

The ring itself remembers only the rays. The class-group grading

$$\deg(x_\rho) = [D_\rho] \in \mathrm{Cl}(X_\Sigma)$$

records the relations among their primitive generators and turns ordinary homogeneity into a multigraded notion.

For a torus-invariant divisor D, the homogeneous piece of degree $[D]$ recovers its global sections:

$$\boxed{(S_\Sigma)_{[D]} \simeq H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)).}$$

The homogenization formula

$$\chi^m \longmapsto \prod_\rho x_\rho^{\langle m, u_\rho\rangle + a_\rho}, \qquad D = \sum_\rho a_\rho D_\rho,$$

identifies a lattice point of P_D with a Cox monomial of degree $[D]$.

The fan determines which coordinate patterns are allowed. Its irrelevant ideal is

$$B(\Sigma) = \left\langle x^{\widehat{\sigma}} = \prod_{\rho \notin \sigma(1)} x_\rho : \sigma \in \Sigma_{\max} \right\rangle,$$

and the exceptional set

$$Z(\Sigma) = V(B(\Sigma))$$

removes precisely the points whose zero coordinates do not fit inside one cone.

The grading determines the rescaling group

$$G = \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Cl}(X_\Sigma), \mathbb{G}_m),$$

and the toric variety is recovered as

$$\boxed{X_\Sigma \simeq \left(\mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma)\right) /\!/ G.}$$

Locally, the construction is the degree-zero identity

$$\left((S_\Sigma)_{x^{\widehat{\sigma}}}\right)_0 \simeq \Bbbk[\sigma^\vee \cap M].$$

Thus Cox coordinates do not replace the affine toric charts; they place all of them inside one multigraded affine space.

Finally, the stabilizers of the Cox action detect singularities. At the distinguished point associated to a cone σ,

$$G_{z_\sigma} \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Cl}(U_\sigma), \mathbb{G}_m).$$

Consequently,

$$\begin{array}{ccc} \sigma \text{ smooth} & \iff & G_{z_\sigma} \text{ trivial}, \\ \sigma \text{ simplicial} & \iff & G_{z_\sigma} \text{ finite}, \\ \sigma \text{ nonsimplicial} & \iff & G_{z_\sigma} \text{ positive-dimensional}. \end{array}$$

The challenge exercise uses the same quotient language to construct line bundles by adding one more coordinate.

Comprehension exercises

6.1. For $\mathbb{P}^n$, recover from the fan the sequence

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{n+1} \longrightarrow \mathbb{Z} \longrightarrow 0.$$

Show that every Cox variable has degree 1, that

$$B(\Sigma) = \langle x_0, \dots, x_n \rangle,$$

and that the Cox quotient is the usual homogeneous-coordinate construction of projective space. Identify S_d with $H^0(\mathbb{P}^n, \mathcal{O}(d))$.

6.2. For $X = \mathbb{P}^1 \times \mathbb{P}^1$, let

$$S = \Bbbk[x_0, x_1, y_0, y_1]$$

with the standard bidegree.

(a) Determine all monomials of bidegree $(2, 1)$.

(b) Recover

$$h^0(X, \mathcal{O}(2, 1)) = 6.$$

(c) Compute the irrelevant ideal and exceptional set.

(d) Verify on the chart $x_0 y_0 \neq 0$ that the degree-zero localized ring is a polynomial ring in two variables.

6.3. Let $X = \mathrm{Bl}_p(\mathbb{P}^2)$, with Cox degrees

$$\deg(x_0) = H, \qquad \deg(x_1) = \deg(x_2) = H - E, \qquad \deg(x_E) = E.$$

(a) Write the weight matrix for the action of $(\mathbb{G}_m)^2$.

(b) Compute all monomials of degree $2H - E$.

(c) Recompute

$$B(\Sigma) = \langle x_E x_2, x_0 x_2, x_0 x_1, x_1 x_E \rangle.$$

(d) On the open set $D(x_0 x_2)$, find two algebraically independent degree-zero Laurent monomials and recover an affine-plane chart.

6.4. For the Hirzebruch surface $\mathbb{F}_r$, use the quotient presentation from Section 6.5.

(a) Check directly that the two pairs of forbidden simultaneous vanishings are independent of r.

(b) Show that the exponent r appears in the transition function between the two local product charts.

(c) Explain how the grading distinguishes $\mathbb{F}_r$ from $\mathbb{P}^1 \times \mathbb{P}^1$, even though the ambient affine space and deleted locus are the same.

6.5. Let Σ be a fan and let

$$C = \{\rho_1, \dots, \rho_s\} \subseteq \Sigma(1).$$

Prove directly from theorem 6.10 that

$$V(x_{\rho_1}, \dots, x_{\rho_s}) \subseteq Z(\Sigma)$$

if and only if C is contained in no cone of Σ. Deduce the description of $Z(\Sigma)$ as the union of the coordinate subspaces associated to primitive collections.

6.6. Consider the simplicial cone

$$\sigma_d = \mathrm{Cone}(de_1 - e_2, e_2) \subseteq \mathbb{R}^2.$$

(a) Compute $\mathrm{Cl}(U_{\sigma_d})$.

(b) Determine the corresponding finite Cox group.

(c) Compute its action on $\mathbb{A}^2$.

(d) Explain why the quotient is geometric even though the origin has a nontrivial stabilizer.

6.7. For the action

$$\lambda \cdot (x_1, x_2, x_3, x_4) = (\lambda x_1, \lambda^{-1} x_2, \lambda x_3, \lambda^{-1} x_4),$$

show that the invariant ring is generated by

$$A = x_1 x_2, \quad B = x_1 x_4, \quad C = x_3 x_2, \quad D = x_3 x_4$$

with the single relation $AD - BC = 0$. Find two points in the fiber over the vertex that are not in the same orbit, and explain why this cannot happen for a geometric quotient.

6.8. Let $z \in \widehat{X}_\Sigma$, and let

$$I(z) = \{\rho : z_\rho \neq 0\}.$$

Show that the stabilizer of z is the subgroup of G on which all characters $[D_\rho]$, $\rho \in I(z)$, are trivial. Apply this to a distinguished point z_σ and recover

$$G_{z_\sigma} \simeq \mathrm{Hom}(\mathrm{Cl}(U_\sigma), \mathbb{G}_m).$$

6.9. In $N = \mathbb{Z}^3$, let

$$u_1 = (1, 0, 1), \quad u_2 = (0, 1, 1), \quad u_3 = (-1, 0, 1), \quad u_4 = (0, -1, 1).$$

The cone over the square with these vertices has two simplicial subdivisions:

$$\Sigma_+ = \{\mathrm{Cone}(u_1, u_2, u_3), \mathrm{Cone}(u_1, u_3, u_4)\}$$

and

$$\Sigma_- = \{\mathrm{Cone}(u_1, u_2, u_4), \mathrm{Cone}(u_2, u_3, u_4)\},$$

together with all faces. Compare their Cox constructions.

(a) Show that the polynomial ring, class-group grading, and group G are the same for both fans.

(b) Compute the two irrelevant ideals and show that they differ.

(c) Explain why the irrelevant ideal is essential for recovering the whole fan rather than only its rays.

Challenge exercise

Line bundles as an extra Cox coordinate.

Let $X = X_\Sigma$ be a smooth toric variety without torus factors. Write

$$\widehat{X} = \mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma), \qquad G = \operatorname{Hom}_{\mathbb{Z}}(\operatorname{Cl}(X), \mathbb{G}_m),$$

and let

$$\pi : \widehat{X} \longrightarrow X$$

be the Cox quotient. Since X is smooth, the action of G on $\widehat{X}$ is free.

Every class $\alpha \in \operatorname{Cl}(X)$ determines a character

$$\chi_\alpha : G \longrightarrow \mathbb{G}_m, \qquad \chi_\alpha(g) = g(\alpha).$$

Use this character to let G act on one extra affine coordinate:

$$g \cdot (z, t) = \big(g \cdot z, \chi_\alpha(g)t\big), \qquad (z, t) \in \widehat{X} \times \mathbb{A}^1.$$

The goal is to understand the quotient

$$L_\alpha = (\widehat{X} \times \mathbb{A}^1)/G$$

and to show that it is the total space of the line bundle represented by α.

$$\begin{array}{ccc} \widehat{X} \times \mathbb{A}^1 & \xrightarrow{/G} & L_\alpha \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow \\ \widehat{X} & \xrightarrow[\pi]{} & X \end{array}$$

Figure 6.10: A character of the Cox group tells the extra coordinate how to rescale. Quotienting the trivial line over $\widehat{X}$ then produces a line bundle over X.

Hint 1. The quotient still lies over X.
Show that the projection

$$\mathrm{pr}_1 : \widehat{X} \times \mathbb{A}^1 \longrightarrow \widehat{X}$$

is G-equivariant. Deduce a map

$$p_\alpha : L_\alpha \longrightarrow X.$$

Explain why the class of $(z, 0)$ gives a zero section. Why should the fiber over $\pi(z)$ be a one-dimensional vector space rather than a copy of $\mathbb{A}^1$ with no preferred linear structure?

Hint 2. Calibrate the sign on $\mathbb{P}^1$.
For $X = \mathbb{P}^1$, we have

$$\widehat{X} = \mathbb{A}^2 \setminus \{0\}, \qquad G = \mathbb{G}_m.$$

For $d \in \mathbb{Z}$, consider

$$\lambda \cdot (x_0, x_1, t) = (\lambda x_0, \lambda x_1, \lambda^d t).$$

On the chart $x_i \neq 0$, show that

$$\tau_i = \frac{t}{x_i^d}$$

is invariant. On the overlap, compute

$$\tau_1 = \left(\frac{x_0}{x_1}\right)^d \tau_0.$$

Recognize the quotient as the total space of

$$\mathcal{O}_{\mathbb{P}^1}(d).$$

Hint 3. Build a local homogeneous unit.
Choose a torus-invariant divisor

$$D = \sum_\rho a_\rho D_\rho$$

with $[D] = \alpha$. Since X is smooth, D is Cartier. For a maximal cone σ, let $m_\sigma \in M$ be its Cartier datum, so

$$\langle m_\sigma, u_\rho \rangle = -a_\rho \qquad (\rho \in \sigma(1)).$$

On

$$\widehat{U}_\sigma = D(x^{\widehat{\sigma}}),$$

define the Laurent monomial

$$q_\sigma = \prod_\rho x_\rho^{a_\rho + \langle m_\sigma, u_\rho \rangle}.$$

Show that:

(a) q_σ has Cox degree α;
(b) its exponent is zero for every $\rho \in \sigma(1)$;
(c) it is therefore an invertible function on $\widehat{U}_\sigma$;
(d) it transforms by

$$q_\sigma(g \cdot z) = \chi_\alpha(g) q_\sigma(z).$$

Hint 4. Trivialize the quotient locally.
On $\widehat{U}_\sigma \times \mathbb{A}^1$, set

$$\tau_\sigma = \frac{t}{q_\sigma}.$$

Show that τ_σ is G-invariant. Use it to construct an isomorphism

$$p_\alpha^{-1}(U_\sigma) \simeq U_\sigma \times \mathbb{A}^1.$$

Conclude that $p_\alpha : L_\alpha \to X$ is a line bundle.

Hint 5. Read the transition functions.
On $U_\sigma \cap U_\tau$, show that

$$\tau_\tau = \frac{q_\sigma}{q_\tau} \tau_\sigma.$$

Compute

$$\frac{q_\sigma}{q_\tau} = \prod_\rho x_\rho^{\langle m_\sigma - m_\tau, u_\rho \rangle}.$$

Use theorem 6.15 to identify this degree-zero Laurent monomial with the torus character

$$\chi^{m_\sigma - m_\tau}.$$

Compare with the transition functions of $\mathcal{O}_X(D)$, and deduce

$$\boxed{L_\alpha \simeq \mathcal{O}_X(D).}$$

Hint 6. Why only the grading matters.
Let

$$a = (a_\rho)_\rho, \qquad b = (b_\rho)_\rho$$

be two exponent vectors in $\mathbb{Z}^{\Sigma(1)}$ with the same class-group degree. Show that

$$a - b = \big(\langle m, u_\rho \rangle\big)_\rho$$

for some $m \in M$.
The vectors a, b define characters of the large coordinate torus:

$$\widetilde{\chi}_a((t_\rho)_\rho) = \prod_\rho t_\rho^{a_\rho}, \qquad \widetilde{\chi}_b((t_\rho)_\rho) = \prod_\rho t_\rho^{b_\rho}.$$

Show that their ratio factors through

$$(\mathbb{G}_m)^{\Sigma(1)} \longrightarrow T_N.$$

Since G is the kernel of this map, conclude that

$$\widetilde{\chi}_a|_G = \widetilde{\chi}_b|_G.$$

Thus two choices with the same Cox grading give the same action on the extra direction, hence isomorphic line bundles.
Interpret this as the familiar statement

$$D \sim D' \quad \Longrightarrow \quad \mathcal{O}_X(D) \simeq \mathcal{O}_X(D').$$

Hint 7. Tensor products add weights.
Construct natural isomorphisms

$$L_\alpha \otimes L_\beta \simeq L_{\alpha+\beta}, \qquad L_\alpha^\vee \simeq L_{-\alpha}.$$

Explain why this is already visible from the action on the extra coordinate.

Hint 8. Homogeneous polynomials become sections.
Let $f \in (S_\Sigma)_\alpha$. Then

$$f(g \cdot z) = \chi_\alpha(g) f(z).$$

Show that

$$\pi(z) \longmapsto [z, f(z)]$$

defines a global section of L_α. Use theorem 6.7 and the identification from Hint 5 to recover

$$(S_\Sigma)_\alpha \simeq H^0(X, L_\alpha).$$

Hint 9. The smooth conclusion: $\mathrm{Cl} = \mathrm{Pic}$**.**
The construction gives a homomorphism

$$\Phi : \mathrm{Cl}(X) \longrightarrow \mathrm{Pic}(X), \qquad \alpha \longmapsto [L_\alpha].$$

Use the preceding hints to show that Φ sends $[D]$ to $[\mathcal{O}_X(D)]$. Explain why smoothness implies that every Weil divisor is Cartier, and conclude that

$$\boxed{\mathrm{Cl}(X) \simeq \mathrm{Pic}(X).}$$

Where would this argument fail for a singular toric variety?

Hint 10. Return to the running blowup.
For

$$X = \mathrm{Bl}_p(\mathbb{P}^2), \qquad \mathrm{Cl}(X) = \mathbb{Z}H \oplus \mathbb{Z}E,$$

write $g = (\lambda, \mu) \in (\mathbb{G}_m)^2$. The class

$$\alpha = dH - rE$$

gives the character

$$\chi_\alpha(\lambda, \mu) = \lambda^d \mu^{-r}.$$

Write the full action on

$$\widehat{X} \times \mathbb{A}^1$$

and describe the quotient line bundle. Explain how a Cox monomial of degree $dH - rE$ becomes a section of this bundle. Compare your answer with the truncated polygon P_{dH-rE} from Chapter 5.

The lesson of the construction

The Cox quotient does more than recover the base variety. Once $\widehat{X} \to X$ is known, every character of the rescaling group tells one additional coordinate how to transform. The resulting associated line bundle depends only on the character—equivalently, only on the divisor class. For a smooth toric variety, all divisor classes are Cartier, so this construction realizes the entire Picard group:

$$\mathrm{Cl}(X) = \mathrm{Pic}(X) = \mathrm{Hom}(G, \mathbb{G}_m).$$

Reference. See Cox–Little–Schenck, Lemma 5.1.1, Proposition 4.2.6, and §14.1.

Notes and references

The Cox quotient construction is developed in Cox–Little–Schenck, §5.1. The class-group grading and homogeneous coordinate ring are treated in §5.2, homogeneous ideals in §5.3, and graded modules and sheaves in §5.4. The local identity

$$\left((S_\Sigma)_{\widehat{x^\sigma}}\right)_0 \simeq \Bbbk[\sigma^\vee \cap M]$$

is the algebraic core of the quotient construction.

The relation between smoothness, factoriality, and $\mathrm{Pic}(X_\Sigma) = \mathrm{Cl}(X_\Sigma)$ is developed in Chapter 4. The challenge exercise recasts that equality through associated bundles: for a free Cox quotient, characters of G produce line bundles on the quotient. This is also the basic linearization idea used in toric geometric invariant theory; see Cox–Little–Schenck, §14.1.

The phrase *universal torsor* is often used for the free Cox quotient

$$\widehat{X}_\Sigma \longrightarrow X_\Sigma$$

of a smooth toric variety. No torsor formalism was needed here: the concrete content is the familiar associated-bundle construction

$$(\widehat{X}_\Sigma \times \mathbb{A}^1)/G.$$

Later chapters can use Cox coordinates not only to describe the ambient toric variety, but also to write and study subvarieties inside it by means of homogeneous equations.

Part II

(More) Advanced Topics

Chapter 7

Canonical Geometry and Reflexive Polytopes

Chapter overview

Chapter 4 gave the striking formula

$$K_{X_\Sigma} = -\sum_{\rho\in\Sigma(1)} D_\rho.$$

At the time, we used it mainly as a divisor calculation. We now ask what this formula means geometrically and what it can do.
The starting point is the invariant differential

$$\frac{d\chi^m}{\chi^m}.$$

It is regular on the dense torus, but it acquires logarithmic poles when the torus is compactified by its boundary divisors. The residues of those poles are exactly the pairings

$$\langle m, u_\rho\rangle.$$

Thus the character lattice, the toric boundary, and differential forms fit into one dictionary. The rest of the chapter will turn to the anticanonical divisor

$$-K_{X_\Sigma} = \sum_\rho D_\rho,$$

its positivity, and the lattice polytope

$$P_{-K_X} = \{m \in M_\mathbb{R} : \langle m, u_\rho\rangle \geq -1 \text{ for every } \rho\}.$$

This will lead to Fano toric varieties, polar duality, and reflexive polytopes.

The three dictionaries

The canonical geometry of a toric variety begins with one simple operation: logarithmically differentiating a character.

Algebra	Geometry	Combinatorics
character χ^m	invariant logarithmic form $d\chi^m/\chi^m$	lattice vector $m \in M$
pole along D_ρ	residue on a boundary component	pairing $\langle m, u_\rho \rangle$
top exterior power	canonical sheaf	negative of the toric boundary
anticanonical sections	Fano geometry	lattice points and polar polytopes

Surface laboratory

For a smooth complete toric surface, the boundary is a cycle of smooth rational curves. The identity

$$K_X + \partial X = 0$$

says that the pair $(X, \partial X)$ carries a nowhere-vanishing logarithmic two-form.
Later, the polygon of $-K_X$ will let us see whether the surface is Fano, compute $(-K_X)^2$, and recognize how toric blowups change the anticanonical geometry.

7.1 Logarithmic forms and the toric boundary

In Section 4.6, we computed the canonical divisor by examining one invariant top form. We now widen the lens. Instead of studying only the top exterior power, we will understand all invariant one-forms and how they interact with every boundary component.

Throughout the main discussion, X_Σ is smooth of dimension n, and

$$\partial X = \sum_{\rho \in \Sigma(1)} D_\rho$$

is its reduced toric boundary. The singular case will be recorded at the end as an optional extension.

Algebraic geometry cheatsheet

Let X be smooth and let

$$D = D_1 + \cdots + D_r$$

be a divisor with simple normal crossings. Near a point of X, one can choose local coordinates

$$x_1, \ldots, x_n$$

so that

$$D = V(x_1 \cdots x_s)$$

for some $s \leq n$.
Ordinary regular one-forms are locally generated by

$$dx_1, \ldots, dx_n.$$

A *logarithmic one-form along D* is allowed to have a simple pole in each boundary direction. Such forms are locally generated by

$$\frac{dx_1}{x_1}, \dots, \frac{dx_s}{x_s}, \quad dx_{s+1}, \dots, dx_n.$$

They form a locally free sheaf denoted

$$\Omega^1_X(\log D).$$

The coefficient of dx_i/x_i, restricted to D_i, is the *residue* along D_i. Thus

$$\mathrm{Res}_{D_i}\left(\frac{dx_i}{x_i}\right) = 1.$$

A logarithmic pole is therefore controlled: it remembers one regular function on the divisor where the pole occurs.

7.1.1 Logarithmically differentiating a character

Let $m \in M$. On the dense torus T_N, the character χ^m is an invertible regular function, so the expression

$$\mathrm{dlog}(\chi^m) := \frac{d\chi^m}{\chi^m}$$

is a regular one-form on T_N.

The notation is justified by the familiar rule

$$\mathrm{dlog}(fg) = \mathrm{dlog}(f) + \mathrm{dlog}(g).$$

Since

$$\chi^{m+m'} = \chi^m \chi^{m'},$$

we obtain

$$\mathrm{dlog}(\chi^{m+m'}) = \mathrm{dlog}(\chi^m) + \mathrm{dlog}(\chi^{m'}).$$

Thus logarithmic differentiation turns multiplication of characters into addition of differential forms.

Choose a basis $m_1, \dots, m_n$ of M, and write

$$t_i = \chi^{m_i}.$$

If

$$m = a_1 m_1 + \cdots + a_n m_n,$$

then

$$\chi^m = t_1^{a_1} \cdots t_n^{a_n},$$

and hence

$$\boxed{\operatorname{dlog}(\chi^m) = \sum_{i=1}^{n} a_i \frac{dt_i}{t_i}.}$$

Proposition 7.1 (Invariant one-forms on the torus)**.** *The assignment*

$$m \longmapsto \operatorname{dlog}(\chi^m)$$

induces an isomorphism

$$M \otimes_{\mathbb{Z}} \Bbbk \simeq H^0(T_N, \Omega^1_{T_N})^{T_N}$$

from the character lattice tensored with $\Bbbk$ *to the vector space of torus-invariant regular one-forms.*

Proof. For a basis $m_1, \ldots, m_n$ of M, the torus is

$$T_N \simeq \mathbb{G}_m^n$$

with coordinates $t_1, \ldots, t_n$. Its regular one-forms form the free $\Bbbk[M]$-module generated by

$$\frac{dt_1}{t_1}, \ldots, \frac{dt_n}{t_n}.$$

These generators are invariant under translation by the torus. An invariant form cannot have a nonconstant Laurent polynomial as a coefficient, so the invariant forms are precisely their constant linear combinations. The displayed formula identifies those combinations with $M \otimes_{\mathbb{Z}} \Bbbk$. □

Reference. This is the torus calculation at the beginning of Cox–Little–Schenck, §8.1, especially equation (8.1.1).

Observation 7.2. The representation-theoretic language is optional here. The torus acts on itself by translation, hence on differential forms by pullback. The forms

$$\frac{dt_i}{t_i}$$

are fixed by this action. Thus the lattice M gives a preferred basis of invariant cotangent directions everywhere on the torus.

7.1.2 What happens at the boundary

A logarithmic form is designed to survive when the torus is completed by boundary divisors. The local toric chart makes this completely visible.

Let $\sigma \in \Sigma$ be a smooth cone of dimension r. Choose a basis

$$u_1, \ldots, u_n$$

of N such that

$$\sigma = \operatorname{Cone}(u_1, \ldots, u_r).$$

Let $m_1, \dots, m_n$ be the dual basis, and set

$$x_i = \chi^{m_i}.$$

Then

$$U_\sigma \simeq \mathbb{A}^r \times \mathbb{G}_m^{n-r},$$

and the toric boundary on this chart is

$$\partial X \cap U_\sigma = V(x_1 \cdots x_r).$$

The divisor associated to the ray $\rho_i = \mathbb{R}_{\geq 0} u_i$ is

$$D_{\rho_i} \cap U_\sigma = V(x_i).$$

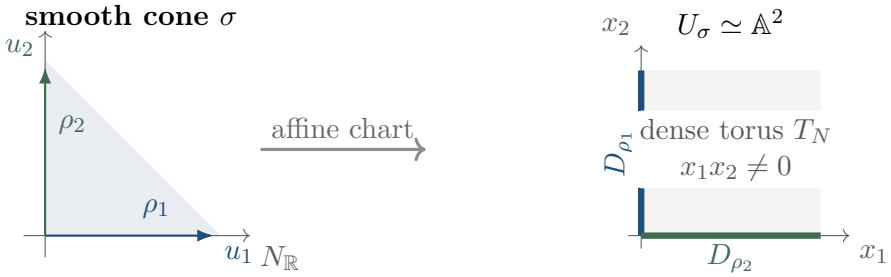


Figure 7.1: On a smooth toric chart, rays become coordinate boundary components. The form $\frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2}$ has a simple pole along each colored boundary axis.

For a general lattice vector

$$m = a_1 m_1 + \cdots + a_n m_n,$$

we have

$$\operatorname{dlog}(\chi^m) = \sum_{i=1}^{n} a_i \frac{dx_i}{x_i}.$$

Along $D_{\rho_i} = V(x_i)$, the residue is therefore a_i. But

$$a_i = \langle m, u_i \rangle.$$

This gives the local formula

$$\boxed{\operatorname{Res}_{D_\rho}\left(\frac{d\chi^m}{\chi^m}\right) = \langle m, u_\rho \rangle.}$$

Pause and think

Why is the primitive generator u_ρ, rather than an arbitrary nonzero lattice point on the ray, the correct vector in the residue formula?

Compare this with the order formula

$$\operatorname{ord}_{D_\rho}(\chi^m) = \langle m, u_\rho \rangle.$$

What is the common local calculation behind the two statements?

7.1.3 The logarithmic cotangent bundle is trivial

The local forms

$$\frac{dx_1}{x_1}, \dots, \frac{dx_n}{x_n}$$

are not an accidental coordinate basis. They are the forms attached to the dual lattice basis $m_1, \dots, m_n$. Changing the lattice basis changes these forms by the same integral matrix. Consequently, they patch canonically over all toric charts.

Theorem 7.3 (Logarithmic forms on a smooth toric variety)**.** *Let X_Σ be a smooth toric variety with toric boundary ∂X. Then*

$$\boxed{\Omega^1_{X_\Sigma}(\log \partial X) \simeq M \otimes_{\mathbb{Z}} \mathcal{O}_X.}$$

Under this isomorphism,

$$m \otimes 1 \longleftrightarrow \frac{d\chi^m}{\chi^m}.$$

In particular, the logarithmic cotangent bundle is trivial.

Reference. This is Cox–Little–Schenck, equation (8.1.5), together with the local logarithmic-form construction preceding it.

Proof. On a smooth chart

$$U_\sigma \simeq \mathbb{A}^r \times \mathbb{G}_m^{n-r},$$

the boundary has equation $x_1 \cdots x_r = 0$. Hence $\Omega^1(\log \partial X)$ is freely generated by

$$\frac{dx_1}{x_1}, \dots, \frac{dx_r}{x_r}, \frac{dx_{r+1}}{x_{r+1}}, \dots, \frac{dx_n}{x_n}.$$

For $i > r$, the coordinate x_i is a unit, so dx_i/x_i is an ordinary regular form. Sending m_i to dx_i/x_i gives an isomorphism

$$M \otimes_{\mathbb{Z}} \mathcal{O}_{U_\sigma} \simeq \Omega^1_{U_\sigma}(\log \partial X).$$

The construction is expressed entirely through characters, so the local isomorphisms agree on overlaps and patch globally. □

The residues assemble into an exact sequence.

Theorem 7.4 (The toric residue sequence)**.** *For a smooth toric variety,*

$$0 \longrightarrow \Omega^1_{X_\Sigma} \longrightarrow M \otimes_{\mathbb{Z}} \mathcal{O}_X \xrightarrow{\operatorname{Res}} \bigoplus_{\rho \in \Sigma(1)} \mathcal{O}_{D_\rho} \longrightarrow 0$$

is exact, where

$$\operatorname{Res}(m \otimes f) = \big(\langle m, u_\rho\rangle f|_{D_\rho}\big)_\rho.$$

Reference. This is Cox–Little–Schenck, Theorem 8.1.1, interpreted through logarithmic forms and the Poincaré residue sequence (8.1.6).

Conceptual proof sketch. On a smooth affine chart, the boundary is a union of coordinate hyperplanes. A logarithmic form lies in the ordinary cotangent sheaf exactly when all of its logarithmic residues vanish. Conversely, one may prescribe an arbitrary regular residue along each coordinate boundary component by choosing the corresponding coefficient of dx_i/x_i. This proves exactness locally, hence globally. □

The middle term is simple, but the three terms encode different parts of the geometry:

ordinary differentials	lattice-valued logarithmic differentials	residues on the boundary
Ω^1_X	$M \otimes \mathcal{O}_X$	$\bigoplus_\rho \mathcal{O}_{D_\rho}$

$$\Omega^1_X \longrightarrow M \otimes \mathcal{O}_X \xrightarrow{\text{Res}} \bigoplus_\rho \mathcal{O}_{D_\rho}$$

7.1.4 Taking the top exterior power

Choose a basis $m_1, \dots, m_n$ of M. The wedge

$$\Omega_{\log} = \frac{d\chi^{m_1}}{\chi^{m_1}} \wedge \cdots \wedge \frac{d\chi^{m_n}}{\chi^{m_n}}$$

is a nowhere-vanishing global section of

$$\Omega^n_{X_\Sigma}(\log \partial X).$$

A different lattice basis changes this form only by determinant ± 1. Thus the logarithmic canonical bundle is canonically trivial up to the choice of orientation of M.

For a simple-normal-crossing divisor D on a smooth variety,

$$\Omega^n_X(\log D) \simeq \omega_X(D).$$

Applying this to the toric boundary gives

$$\omega_{X_\Sigma}(\partial X) \simeq \mathcal{O}_{X_\Sigma}.$$

Equivalently,

$$\boxed{K_{X_\Sigma} + \partial X \sim 0.}$$

In fact, using the particular rational form $\Omega_{\log}$, we recover the representative already proved in theorem 4.26:

$$\operatorname{div}(\Omega_{\log}) = -\sum_{\rho \in \Sigma(1)} D_\rho.$$

Observation 7.5 (The toric pair is logarithmically Calabi–Yau). A smooth pair (X, D) is often called *log Calabi–Yau* when

$$K_X + D \sim 0.$$

Thus every smooth toric variety, together with its full toric boundary, is log Calabi–Yau.

This does *not* mean that $K_X \sim 0$. The boundary is essential: it is exactly what compensates for the poles of the invariant volume form.

7.1.5 Two compactifications of a torus

Example 7.6 (The two ends of $\mathbb{P}^1$)**.** The dense torus of $\mathbb{P}^1$ has coordinate t. The invariant form

$$\frac{dt}{t}$$

has a simple pole at $t = 0$. Near infinity, use $s = 1/t$. Then

$$\frac{dt}{t} = -\frac{ds}{s},$$

so there is also a simple pole at $s = 0$, which is the point at infinity. Therefore

$$\operatorname{div}\left(\frac{dt}{t}\right) = -[0] - [\infty].$$

The two boundary points are the two ends added when $\mathbb{G}_m$ is compactified to $\mathbb{P}^1$.

Example 7.7 (The three boundary lines of $\mathbb{P}^2$)**.** Let

$$D_i = V(X_i) \subseteq \mathbb{P}^2.$$

On the torus chart $X_0X_1X_2 \neq 0$, set

$$x = \frac{X_1}{X_0}, \qquad y = \frac{X_2}{X_0}.$$

The invariant volume form is

$$\Omega_{\log} = \frac{dx}{x} \wedge \frac{dy}{y}.$$

Its simple poles along D_1 and D_2 are visible from $x = 0$ and $y = 0$.

To inspect D_0, work on $X_1 \neq 0$ with

$$u = \frac{X_0}{X_1} = \frac{1}{x}, \qquad v = \frac{X_2}{X_1} = \frac{y}{x}.$$

Then

$$\frac{dx}{x} = -\frac{du}{u}, \qquad \frac{dy}{y} = \frac{dv}{v} - \frac{du}{u},$$

and hence

$$\Omega_{\log} = -\frac{du}{u} \wedge \frac{dv}{v}.$$

Thus $u = 0$, which is D_0, also contributes a simple pole. We recover

$$K_{\mathbb{P}^2} = -(D_0 + D_1 + D_2) \sim -3H.$$

Pause and think

On $\mathbb{P}^1 \times \mathbb{P}^1$, choose torus coordinates s, t and analyze

$$\frac{ds}{s} \wedge \frac{dt}{t}.$$

Locate its four poles and recover

$$K_{\mathbb{P}^1 \times \mathbb{P}^1} \sim -2F_1 - 2F_2.$$

Observation 7.8 (What changes in the singular case). For a normal toric variety, the ordinary cotangent sheaf need not be locally free. One replaces it by the reflexive sheaf of Zariski one-forms. Cox–Little–Schenck, Theorem 8.1.4, gives the corresponding residue sequence, and Theorem 8.2.3 gives

$$\omega_{X_\Sigma} \simeq \mathcal{O}_{X_\Sigma}\left(-\sum_\rho D_\rho\right).$$

No singular differential-form technology is needed for the next sections. The codimension-one formula from Chapter 4 remains the working tool.

The boundary has now appeared in two complementary roles:

as poles of invariant forms:	$K_X = -\partial X,$
as a positive divisor:	$-K_X = \partial X.$

The next section studies when this positive-looking divisor is actually ample.

Guiding question

The anticanonical divisor of a toric variety is the sum of all boundary components:

$$-K_{X_\Sigma} = \sum_\rho D_\rho.$$

When does this divisor define an embedding into projective space? What should its support function and its polytope look like when that happens?

7.2 The anticanonical divisor and Fano varieties

The previous section explained why the toric boundary is anticanonical:

$$-K_{X_\Sigma} = \sum_{\rho \in \Sigma(1)} D_\rho.$$

The coefficients are all positive, but that alone does not make the divisor positive in the geometric sense. The boundary components may interact in a way that makes their sum ample, merely basepoint free, or not even nef.

This section turns that question into a picture in $M_{\mathbb{R}}$.

Algebraic geometry cheatsheet

Let X be a smooth projective variety. The variety is called *Fano* if its anticanonical divisor is ample:

$$-K_X \text{ is ample.}$$

Equivalently, some positive tensor power of

$$\omega_X^{-1} = \mathcal{O}_X(-K_X)$$

gives a closed embedding into projective space.
For a normal variety, one must first ask whether K_X is Cartier. A normal variety is called *Gorenstein Fano* if

$$-K_X \text{ is Cartier and ample.}$$

On a smooth variety every Weil divisor is Cartier, so the two expressions describe the same condition.

Reference. This is Cox–Little–Schenck, Definition 8.3.1.

7.2.1 The anticanonical polytope

Since every coefficient of $-K_{X_\Sigma}$ is equal to 1, its divisor polytope has the especially uniform form

$$\boxed{P_{-K_X} = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -1 \text{ for every } \rho \in \Sigma(1)\}\,.}$$

Each ray contributes a half-space whose boundary is one lattice unit from the origin.

The origin satisfies every inequality strictly:

$$\langle 0, u_\rho \rangle = 0 > -1.$$

Thus 0 lies in the interior of P_{-K_X}. When Σ is complete, the polytope is bounded by theorem 5.7.

Proposition 7.9 (The distinguished interior lattice point)**.** *Let X_Σ be complete. Then the origin is the unique interior lattice point of P_{-K_X}:*

$$\operatorname{Int}(P_{-K_X}) \cap M = \{0\}.$$

Proof. Suppose $m \in M$ lies in the interior. Then

$$\langle m, u_\rho \rangle > -1$$

for every ray. The pairing is integral, so

$$\langle m, u_\rho \rangle \geq 0 \qquad \text{for every } \rho.$$

Since the cones of a complete fan cover $N_{\mathbb{R}}$, the linear functional m is nonnegative in every direction. Applying this to both u and $-u$ shows that $m = 0$. □

This conclusion does not yet say that $-K_X$ is ample. The inequalities may be redundant, the vertices may fail to be lattice points, or the normal fan of the polytope may be coarser than the fan we started with.

7.2.2 The toric Fano criterion

Assume now that X_Σ is smooth and complete. Then $-K_X$ is Cartier, and its support function satisfies

$$\psi_{-K_X}(u_\rho) = -1 \qquad \text{for every ray } \rho.$$

Theorem 7.10 (The toric Fano criterion)**.** *For a smooth complete toric variety X_Σ, the following are equivalent:*

(a) X_Σ is Fano.

(b) The support function ψ_{-K_X} is strictly convex.

(c) The anticanonical polytope P_{-K_X} is a lattice polytope whose normal fan is exactly Σ.

(d) For every maximal cone σ, its anticanonical Cartier datum m_σ is a vertex of P_{-K_X}, and the normal cone at that vertex is σ.

Proof. The equivalence of (a) and (b) is theorem 5.25 applied to $-K_X$.

When a Cartier divisor is ample, its support function is strictly convex, its Cartier data are the vertices of its polytope, and the polytope has normal fan Σ. Conversely, if the normal fan of P_{-K_X} is Σ, then the support function has a genuinely different linear piece on every maximal cone, hence is strictly convex. This gives the remaining equivalences. □

Corollary 7.11 (Facets at lattice distance one)**.** *If X_Σ is smooth and Fano, then every ray inequality*

$$\langle m, u_\rho \rangle \geq -1$$

cuts out a facet of P_{-K_X}. Its primitive inward normal is u_ρ, so every facet is exactly one lattice unit from the origin.

The three dictionaries

For a smooth complete toric variety, Fano geometry is the same condition viewed in three languages.

Algebra	Geometry	Combinatorics
anticanonical ring $\bigoplus_{k\geq 0} H^0(X, -kK_X)$	anticanonical embeddings for $k \gg 0$	lattice points of kP_{-K_X}
$-K_X = \sum_\rho D_\rho$	X is Fano	ψ_{-K_X} is strictly convex
Cartier data m_σ	distinct local anticanonical frames	vertices of a polytope with normal fan Σ

7.2.3 The first anticanonical polygons

Example 7.12 (The anticanonical triangle of $\mathbb{P}^2$)**.** For the rays

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_2 = e_2,$$

the anticanonical polygon is

$$P_{-K_{\mathbb{P}^2}} = \{(a,b) : a \geq -1,\ b \geq -1,\ a+b \leq 1\}\,.$$

Hence

$$P_{-K_{\mathbb{P}^2}} = \mathrm{Conv}\{(-1,-1),(2,-1),(-1,2)\}.$$

All three inequalities are active, and its normal fan is the fan of $\mathbb{P}^2$. Thus $-K_{\mathbb{P}^2} \sim 3H$ is ample.

Example 7.13 (The anticanonical square of $\mathbb{P}^1 \times \mathbb{P}^1$)**.** The four ray generators are $\pm e_1, \pm e_2$, so

$$P_{-K_{\mathbb{P}^1\times\mathbb{P}^1}} = [-1,1] \times [-1,1].$$

Its normal fan is the product fan, and

$$-K_{\mathbb{P}^1\times\mathbb{P}^1} \sim 2F_1 + 2F_2$$

is ample.

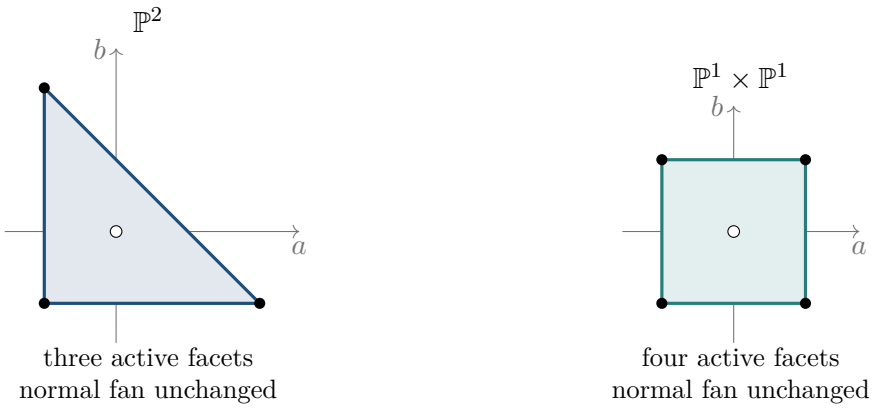


Figure 7.2: The anticanonical polygons of $\mathbb{P}^2$ and $\mathbb{P}^1 \times \mathbb{P}^1$. The open circle is the unique interior lattice point. Every facet is at lattice distance one from the origin.

7.2.4 The Hirzebruch surfaces: ample, borderline, and negative

Recall the fan of the Hirzebruch surface $\mathbb{F}_r$, with rays in cyclic order

$$w_1 = (1,0), \qquad w_2 = (0,1), \qquad w_3 = (-1,r), \qquad w_4 = (0,-1).$$

Its anticanonical polygon is

$$\boxed{P_r = \{(x,y) : x \geq -1, \quad -1 \leq y \leq 1, \quad x \leq 1 + ry\}\,.}$$

The Cartier data on the four maximal cones are

$$m_{12} = (-1, -1), \quad m_{23} = (1 - r, -1),$$
$$m_{34} = (r + 1, 1), \quad m_{41} = (-1, 1).$$

These four points tell the whole positivity story.

Proposition 7.14 (Anticanonical positivity on $\mathbb{F}_r$)**.** *For $r \geq 0$:*

$$\mathbb{F}_r \text{ is Fano} \iff r = 0 \text{ or } 1,$$
$$-K_{\mathbb{F}_2} \text{ is basepoint free but not ample,}$$
$$-K_{\mathbb{F}_r} \text{ is not basepoint free} \iff r \geq 3.$$

Proof. For $r = 0$ and $r = 1$, all four Cartier data are distinct vertices of P_r, so the normal fan of the polygon is the original fan. Thus $-K$ is ample.

For $r = 2$,

$$m_{12} = m_{23} = (-1, -1).$$

All Cartier data still lie in P_2, so $-K$ is basepoint free, but one wall is invisible and the support function is not strictly convex. Hence $-K$ is not ample.

For $r \geq 3$,

$$m_{23} = (1 - r, -1)$$

fails the inequality $x \geq -1$. Thus one local anticanonical frame does not extend to a global section, and $-K$ is not basepoint free. □

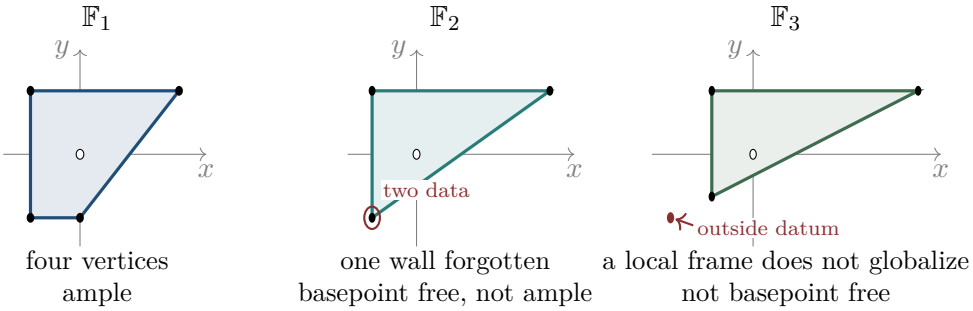


Figure 7.3: The anticanonical polygons of $\mathbb{F}_1$, $\mathbb{F}_2$, and $\mathbb{F}_3$. At $r = 2$, two neighboring Cartier data coincide. At $r = 3$, the red Cartier datum $(-2, -1)$ lies outside the polytope.

There is also a curve-theoretic way to read the same transition. Let S be the torus-invariant section with

$$S^2 = -r$$

and let F be a fiber. Then

$$-K_{\mathbb{F}_r} = 2S + (r + 2)F,$$

so

$$(-K_{\mathbb{F}_r}) \cdot F = 2, \qquad (-K_{\mathbb{F}_r}) \cdot S = 2 - r.$$

The anticanonical divisor is positive on S for $r = 0, 1$, zero on S for $r = 2$, and negative on S for $r \geq 3$. The polygon and the invariant curve tell the same story.

Surface laboratory

The identification

$$\mathbb{F}_1 \simeq \mathrm{Bl}_p(\mathbb{P}^2)$$

turns the first panel of Figure 7.3 into the familiar corner-cut picture. On the blowup,

$$-K_X = 3H - E,$$

and the representative used in Chapter 5 has polytope

$$\{(a, b) : a, b \geq 0, \quad 1 \leq a + b \leq 3\}.$$

Translating by $(-1, -1)$ gives the centered anticanonical polygon. Translation changes neither the normal fan nor the associated projective map.

Pause and think

The fan of $\mathbb{P}^2$ has three rays. Insert the ray $e_1 + e_2$ to blow up one fixed point, and then insert a ray in a second maximal cone to blow up another fixed point.
Write the new anticanonical inequalities. Does the second corner cut still leave every inequality active? What does your answer predict about the Fano property?

The Fano condition has produced a very rigid kind of lattice polytope:

0 is the unique interior lattice point, and every facet is at lattice distance 1 from 0.

The next section studies this condition intrinsically, without first mentioning a toric variety.

Prediction

For each facet F of a Fano polytope $P \subseteq M_{\mathbb{R}}$, record its primitive inward normal $u_F \in N$, and form

$$\mathrm{Conv}(u_F : F \text{ a facet of } P) \subseteq N_{\mathbb{R}}.$$

Compute this new polygon for the anticanonical triangle of $\mathbb{P}^2$ and the square of $\mathbb{P}^1 \times \mathbb{P}^1$. What relation do you expect between faces of the original polygon and faces of the new one? What might happen if the construction is applied twice?

7.3 Reflexive polytopes and polar duality

The anticanonical polygons in the previous section all had the form

$$P = \{m \in M_{\mathbb{R}} : \langle m, u_F \rangle \geq -1 \text{ for every facet } F\}.$$

The primitive inward normals u_F were not merely auxiliary data: they assembled into a second polytope in the dual vector space $N_{\mathbb{R}}$. We now study that construction intrinsically.

7.3.1 The polar remembers supporting hyperplanes

Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional polytope with the origin in its interior. Its *polar polytope*, in the sign convention used for toric divisors, is

$$\boxed{P^\circ = \{u \in N_{\mathbb{R}} : \langle m, u\rangle \geq -1 \text{ for every } m \in P\}.}$$

The condition $0 \in \operatorname{int}(P)$ matters. It ensures that P° is bounded and full dimensional. It also puts the origin in the interior of P°.

Observation 7.15. Some texts define the polar using

$$\langle m, u\rangle \leq 1.$$

Replacing u by $-u$ converts that convention into ours. The present sign is convenient because toric divisor polytopes are already written using inequalities

$$\langle m, u_\rho\rangle \geq -a_\rho.$$

Suppose the facets of P have primitive inward normals $u_F \in N$ and facet presentation

$$P = \{m \in M_{\mathbb{R}} : \langle m, u_F\rangle \geq -a_F \text{ for every facet } F\}, \qquad a_F > 0.$$

Because u_F is primitive, the positive integer a_F is the *lattice distance* from the origin to the supporting hyperplane of F.

Proposition 7.16 (Facets become vertices of the polar)**.** *With the notation above,*

$$\boxed{P^\circ = \operatorname{Conv}\left(\frac{u_F}{a_F} : F \text{ a facet of } P\right).}$$

In particular, the exact position of each facet determines a vertex of the polar.

Reference. This is the calculation in Cox–Little–Schenck, Exercise 2.2.1.

Proof. Set

$$Q = \operatorname{Conv}\left(\frac{u_F}{a_F} : F \text{ a facet of } P\right).$$

Then

$$\begin{aligned} Q^\circ &= \left\{m \in M_{\mathbb{R}} : \left\langle m, \frac{u_F}{a_F}\right\rangle \geq -1 \text{ for every } F\right\} \\ &= \{m \in M_{\mathbb{R}} : \langle m, u_F\rangle \geq -a_F \text{ for every } F\} \\ &= P. \end{aligned}$$

Polarity is involutive for full-dimensional polytopes containing the origin in their interiors, so

$$P^\circ = (Q^\circ)^\circ = Q.$$

□

The formula makes a crucial contrast visible:

> The normal fan remembers the directions u_F,
> whereas the polar remembers the scaled vectors u_F/a_F.

Translations and positive dilations do not change a normal fan, but they do change the polar.

Pause and think

Let $P = [-1, 2] \times [-1, 1]$. Compute its polar from the four facet inequalities. Which vertex of P° fails to be a lattice point?
Now compare P with $[-1, 1]^2$. The two rectangles have the same normal fan. Why does polarity distinguish them?

7.3.2 The anticanonical triangle and its polar

For $X = \mathbb{P}^2$, the anticanonical polygon is

$$P = \{(x, y) : x \geq -1, \quad y \geq -1, \quad x + y \leq 1\}.$$

Its three primitive inward facet normals are

$$u_1 = (1, 0), \qquad u_2 = (0, 1), \qquad u_0 = (-1, -1).$$

Every facet has lattice distance one from the origin, so

$$P^\circ = \mathrm{Conv}(u_0, u_1, u_2).$$

Thus the polar is exactly the convex hull of the ray generators of the fan of $\mathbb{P}^2$.



Figure 7.4: For the anticanonical triangle of $\mathbb{P}^2$, each colored facet of P becomes the matching vertex of P°. The polar is the convex hull of the primitive ray generators of the fan.

7.3.3 Reflexive polytopes

The previous example has a special integrality property: both P and P° are lattice polytopes.

Definition 7.17 (Reflexive polytope)**.** A full-dimensional lattice polytope $P \subseteq M_{\mathbb{R}}$ is *reflexive* if

$$0 \in \operatorname{int}(P)$$

and its polar $P^\circ \subseteq N_{\mathbb{R}}$ is also a lattice polytope.

Theorem 7.18 (The distance-one criterion)**.** *Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional lattice polytope with $0 \in \operatorname{int}(P)$. The following are equivalent:*

(a) P is reflexive.

(b) Every facet of P is at lattice distance one from the origin.

(c) The facet presentation of P is

$$P = \{m \in M_{\mathbb{R}} : \langle m, u_F \rangle \geq -1 \text{ for every facet } F\},$$

where every u_F is the primitive inward normal.

When these conditions hold,

$$\boxed{P^\circ = \operatorname{Conv}(u_F : F \text{ a facet of } P).}$$

Reference. This is Cox–Little–Schenck, Definition 2.3.12 together with Exercise 2.2.1.

Proof. Write the facet presentation as

$$P = \{m : \langle m, u_F \rangle \geq -a_F \text{ for every } F\},$$

with u_F primitive and $a_F \in \mathbb{Z}_{>0}$. By theorem 7.16, the vertices of the polar are the vectors

$$\frac{u_F}{a_F}.$$

Since u_F is primitive, this vector lies in N if and only if $a_F = 1$. Thus P° is a lattice polytope exactly when every facet has lattice distance one. □

Corollary 7.19 (Reflexivity is symmetric)**.** *If P is reflexive, then P° is reflexive and*

$$(P^\circ)^\circ = P.$$

Moreover, the origin is the unique interior lattice point of both P and P°.

Proof. Both P and P° are lattice polytopes containing the origin in their interiors, and polarity is involutive. Hence the definition is symmetric.

The unique-interior-point assertion follows from the distance-one facet presentation exactly as in theorem 7.9. □

Warning

Having the origin in the interior is not enough. For example,

$$[-1, 2] \times [-1, 1]$$

is a lattice polytope containing the origin, but the facet $x = 2$ is at lattice distance two. Its polar has the nonintegral vertex

$$\left(-\frac{1}{2}, 0\right),$$

so the rectangle is not reflexive.

7.3.4 Faces reverse dimension

Polarity exchanges faces of complementary dimensions. If $Q \subsetneq P$ is a nonempty face, define its *dual face* by

$$Q^* = \{u \in P^\circ : \langle m, u\rangle = -1 \text{ for every } m \in Q\}.$$

Proposition 7.20 (The polar face dictionary)**.** *Let P be reflexive of dimension n. Then:*

(a) Q^* is a face of P°.

(b) The correspondence $Q \mapsto Q^*$ reverses inclusions.

(c)

$$\dim Q + \dim Q^* = n - 1.$$

(d) Applying the construction twice returns the original face:

$$(Q^*)^* = Q.$$

Conceptual proof sketch. A face Q is cut out by the facets of P that contain it. The primitive normals of precisely those facets are vertices of P°, and their convex hull is Q^*. Adding equations on one side enlarges the dual face on the other, which reverses inclusions. The dimension formula and involutivity are the usual dual face statements for a polytope and its polar. □

For a polygon, the dictionary is especially visible:

$$\boxed{\text{edges of } P \longleftrightarrow \text{vertices of } P^\circ, \qquad \text{vertices of } P \longleftrightarrow \text{edges of } P^\circ.}$$

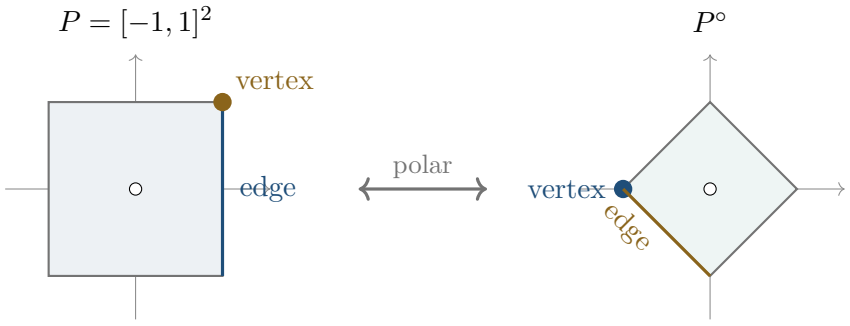


Figure 7.5: For the square and diamond, the blue edge $x = 1$ becomes the blue polar vertex $(-1, 0)$. The gold vertex $(1, 1)$ becomes the gold edge joining $(-1, 0)$ to $(0, -1)$.

7.3.5 The normal fan is already inside the polar

The polar does more than record the normals one at a time. Its whole face structure reconstructs the normal fan.

Theorem 7.21 (Normal cones from polar faces)**.** *Let $P \subseteq M_{\mathbb{R}}$ be reflexive. For every face $Q \subseteq P$, the normal cone of Q is*

$$\boxed{\sigma_Q = \mathrm{Cone}(Q^*) \subseteq N_{\mathbb{R}}.}$$

Consequently, the normal fan Σ_P consists of the cones over the proper faces of P°.

Reference. This is Cox–Little–Schenck, Exercise 2.3.4.

Conceptual proof sketch. A vector $u \in N_{\mathbb{R}}$ lies in σ_Q precisely when the linear function $\langle -, u\rangle$ reaches its minimum on P along a face containing Q. After rescaling a nonzero u until that minimum is -1, the resulting point lies in the dual face Q^*. Undoing the rescaling shows that the whole normal cone is the cone over Q^*. □

For the anticanonical polygon of a smooth toric Fano variety, this says

$$\Sigma = \Sigma_{P_{-K_X}} = \{\text{cones over proper faces of } P^\circ_{-K_X}\}.$$

In particular,

$$P^\circ_{-K_X} = \mathrm{Conv}(u_\rho : \rho \in \Sigma(1)).$$

The fan and the anticanonical polytope are therefore two sides of one polar pair.

The three dictionaries

For a reflexive polytope, the three dictionaries become symmetric.

Polytope P	**Polar** P°	**Toric variety** X_{Σ_P}
facet F	vertex u_F	ray ρ_F and divisor D_{ρ_F}
vertex v	facet v^*	maximal cone and torus fixed point
face inclusion	reverse face inclusion	reverse orbit–cone correspondence
lattice distance one	lattice vertex	integral anticanonical local data

Prediction

Suppose X_Σ is singular and its anticanonical inequalities cut out a bounded polytope with normal fan Σ. Its vertices need not lie in M.
On each maximal cone σ, what condition would produce a lattice vector $m_\sigma \in M$ satisfying

$$\langle m_\sigma, u_\rho\rangle = -1 \qquad (\rho \in \sigma(1))?$$

How is this the Cartier condition for $-K_{X_\Sigma}$?

7.4 Gorenstein Fano toric varieties

The previous two sections isolated two different features of the anticanonical divisor:

- ampleness is detected by the strict convexity of the anticanonical support function;
- Cartierness is detected by integral height-one data on the maximal cones.

A Gorenstein Fano toric variety is what we obtain when both conditions hold at once. On the polytope side, the same combination becomes reflexivity.

7.4.1 Three levels of anticanonical positivity

Let X be a complete normal variety.

Definition 7.22. The variety X is

(a) $\mathbb{Q}$-*Gorenstein* if K_X is $\mathbb{Q}$-Cartier;
(b) $\mathbb{Q}$-*Fano* if K_X is $\mathbb{Q}$-Cartier and $-K_X$ is ample;
(c) *Gorenstein Fano* if $-K_X$ is Cartier and ample.

When X is smooth and $-K_X$ is ample, we simply say that X is Fano.

Reference. The Gorenstein Fano terminology is Cox–Little–Schenck, Definition 8.3.1.

Thus

$$\text{smooth Fano} \Longrightarrow \text{Gorenstein Fano} \Longrightarrow \mathbb{Q}\text{-Fano}.$$

Neither converse holds. The point of this section is to see the two failures directly in a fan and its anticanonical polytope.

7.4.2 Local height-one data become polytope vertices

Let X_Σ be complete and $\mathbb{Q}$-Gorenstein. For each maximal cone σ, the height-one criterion from theorem 4.32 gives a rational vector

$$h_\sigma \in M_{\mathbb{Q}}$$

such that

$$\langle h_\sigma, u_\rho \rangle = 1 \qquad (\rho \in \sigma(1)).$$

Equivalently, the local Cartier datum for $-K_X$ is

$$m_\sigma = -h_\sigma, \qquad \langle m_\sigma, u_\rho \rangle = -1 \qquad (\rho \in \sigma(1)).$$

If $-K_X$ is ample, then m_σ is not merely a local linear function: it is the vertex of P_{-K_X} whose normal cone is σ. Cartierness asks whether these vertices lie in the lattice M.

Proposition 7.23 (Integrality of anticanonical vertices)**.** *Let X_Σ be a $\mathbb{Q}$-Fano toric variety. The following are equivalent:*

(a) X_Σ is Gorenstein Fano.
(b) Every anticanonical Cartier datum m_σ belongs to M.
(c) Every vertex of P_{-K_X} is a lattice point.

(d) For every maximal cone σ, the primitive ray generators of σ lie on an integral height-one hyperplane.

Proof. The equivalence of (a), (b), and (d) is exactly the height-one criterion for the Cartier divisor $-K_X$. Since $-K_X$ is ample, strict convexity identifies the local data m_σ with the vertices of P_{-K_X}. This gives the equivalence with (c). □

This is the local-to-global mechanism behind reflexivity:

the equations $\langle m, u_\rho \rangle = -1$ define rational vertices,
and Gorensteinness says that all of those vertices are integral.

7.4.3 Gorenstein Fano varieties and reflexive polytopes

We can now state the central theorem of the chapter.

Theorem 7.24 (Gorenstein Fano–reflexive correspondence)**.** *The following constructions are inverse to one another, up to lattice equivalence.*

(a) If X_Σ is a projective Gorenstein Fano toric variety, then its anticanonical polytope

$$P_{-K_X} = \{m \in M_\mathbb{R} : \langle m, u_\rho \rangle \geq -1 \text{ for every } \rho\}$$

is reflexive and has normal fan Σ.
(b) If $P \subseteq M_\mathbb{R}$ is reflexive, then the projective toric variety X_{Σ_P} is Gorenstein Fano, and

$$P = P_{-K_{X_{\Sigma_P}}}.$$

Reference. This is Cox–Little–Schenck, Theorem 8.3.4.

Proof. Suppose first that X_Σ is Gorenstein Fano. Since $-K_X$ is Cartier, its polytope is a lattice polytope. Since it is ample, every ray inequality is active and the normal fan of the polytope is exactly Σ. Its facet presentation is

$$P_{-K_X} = \{m \in M_\mathbb{R} : \langle m, u_\rho \rangle \geq -1 \text{ for every } \rho\},$$

with each u_ρ primitive. Every facet is therefore at lattice distance one from the origin. By theorem 7.18, the polytope is reflexive.

Conversely, let P be reflexive. Its facet presentation is

$$P = \{m \in M_\mathbb{R} : \langle m, u_F \rangle \geq -1 \text{ for every facet } F\},$$

where $u_F \in N$ is the primitive inward normal. The divisor attached to this polytope on X_{Σ_P} is therefore

$$D_P = \sum_F D_F = -K_{X_{\Sigma_P}}.$$

Because P is a lattice polytope, this divisor is Cartier; because Σ_P is its normal fan, it is ample. Hence X_{Σ_P} is Gorenstein Fano. □

For a Gorenstein Fano toric variety, polarity takes the particularly concrete form

$$\boxed{P^\circ_{-K_X} = \mathrm{Conv}(u_\rho : \rho \in \Sigma(1)).}$$

Moreover,

$$\Sigma = \{\text{cones over the proper faces of } P^\circ_{-K_X}\}.$$

Thus the polar simultaneously records the ray generators and how they assemble into cones.

Observation 7.25. Since the polar of a reflexive polytope is again reflexive, a reflexive pair

$$P \subseteq M_{\mathbb{R}}, \qquad P^\circ \subseteq N_{\mathbb{R}}$$

produces two Gorenstein Fano toric varieties. They need not be isomorphic. This symmetry is one reason reflexive polytopes play a central role in toric mirror symmetry; no familiarity with mirror symmetry is needed here.

The three dictionaries

The Gorenstein Fano condition is the same phenomenon in three languages.

Algebra	Geometry	Combinatorics
$-K_X$ is Cartier	X is Gorenstein	local height-one data are integral
$-K_X$ is ample	anticanonical powers embed X	its support function is strictly convex
Cartier and ample together	X is Gorenstein Fano	P_{-K_X} is reflexive
local anticanonical frames	torus fixed points	vertices of P_{-K_X}

7.4.4 Weighted projective spaces

Weighted projective spaces show that ampleness and Cartierness are independent questions. Let

$$X = \mathbb{P}(q_0, \dots, q_n), \qquad \gcd(q_0, \dots, q_n) = 1.$$

Its Cox ring is

$$S = \Bbbk[x_0, \dots, x_n], \qquad \deg(x_i) = q_i \in \mathrm{Cl}(X) \simeq \mathbb{Z}.$$

Since the coordinate divisor D_i has class q_i,

$$[-K_X] = \left[\sum_i D_i\right] = (q_0 + \cdots + q_n) \in \mathrm{Cl}(X).$$

This class is positive, so it is ample as a rational divisor. The remaining question is whether it is Cartier.

Proposition 7.26 (The weighted-projective Gorenstein criterion)**.** *The weighted projective space* $\mathbb{P}(q_0, \dots, q_n)$ *is Gorenstein Fano if and only if*

$$q_i \mid q_0 + \cdots + q_n \qquad \textit{for every } i.$$

Every weighted projective space is $\mathbb{Q}$-Fano.

Reference. This is Cox–Little–Schenck, Example 8.3.3 and Exercise 8.3.2.

Conceptual proof sketch. The anticanonical class is the positive class

$$q_0 + \cdots + q_n,$$

so it is ample. A class of degree d defines a line bundle on the weighted projective space precisely when every weight divides d, or equivalently when

$$\operatorname{lcm}(q_0, \ldots, q_n) \mid d.$$

Applying this to $d = q_0 + \cdots + q_n$ gives the criterion. □

7.4.5 A singular comparison: $\mathbb{P}(1,1,2)$ and $\mathbb{P}(1,1,3)$

For $X_n = \mathbb{P}(1,1,n)$, choose the ray generators

$$u_0 = (1,0), \qquad u_1 = (-1,n), \qquad u_2 = (0,-1).$$

They satisfy

$$u_0 + u_1 + nu_2 = 0.$$

The anticanonical region is

$$P_{-K_{X_n}} = \{(x,y) : x \geq -1, \quad -x + ny \geq -1, \quad y \leq 1\}.$$

Its vertices are

$$(-1,1), \qquad (n+1,1), \qquad \left(-1, -\frac{2}{n}\right).$$

Thus

$$P_{-K_{X_n}} \text{ is a lattice polygon} \quad \Longleftrightarrow \quad n \mid 2.$$

This recovers the conclusion from Chapter 4:

$$\mathbb{P}(1,1,2) \text{ is singular Gorenstein Fano,}$$

whereas

$$\mathbb{P}(1,1,3) \text{ is } \mathbb{Q}\text{-Fano but not Gorenstein.}$$

Pause and think

The singular cone of $\mathbb{P}(1,1,n)$ is generated by

$$(1,0) \qquad \text{and} \qquad (-1,n).$$

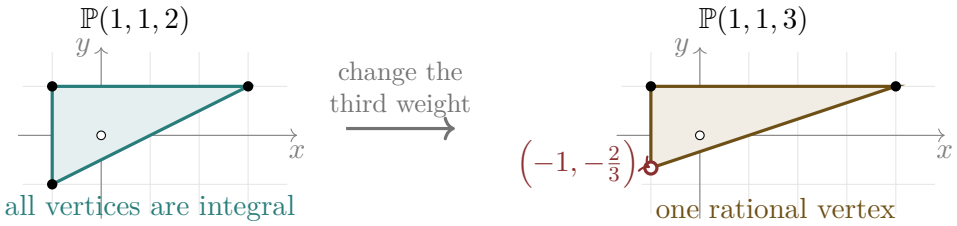


Figure 7.6: The anticanonical polygons of two singular weighted projective planes. The polygon for $\mathbb{P}(1,1,2)$ is reflexive. For $\mathbb{P}(1,1,3)$, ampleness survives but one anticanonical vertex leaves the lattice, recording the failure of Gorensteinness.

Solve the height-one equations on this cone and compare the resulting functional with the nonintegral vertex

$$\left(-1,-\frac{2}{n}\right)$$

of the anticanonical polytope. Why do the two calculations contain the same divisibility condition?

7.4.6 Reflexive does not mean smooth

The polygon for $\mathbb{P}(1,1,2)$ is reflexive, but the corresponding surface is singular. Reflexivity guarantees that $-K_X$ is Cartier and ample; it does not say that the cones of the fan are smooth.

Let $P \subseteq M_{\mathbb{R}}$ be reflexive. Since

$$\Sigma_P = \{\text{cones over faces of } P^\circ\},$$

the toric variety X_{Σ_P} is smooth precisely when every maximal cone over a facet of P° is smooth. Equivalently,

$$\boxed{X_{\Sigma_P} \text{ is smooth} \iff \text{every facet of } P^\circ \text{ is a unimodular simplex.}}$$

For a polygon, this becomes especially concrete. If

$$u_1, \dots, u_s$$

are the vertices of P° in cyclic order, then

$$X_{\Sigma_P} \text{ is smooth} \iff |\det(u_i, u_{i+1})| = 1 \qquad \text{for every } i.$$

Thus a smooth toric Fano surface corresponds not merely to a reflexive polygon, but to a reflexive polygon whose polar has unimodular edges. That is the starting point for the final surface laboratory.

Prediction

Begin with the anticanonical triangle of $\mathbb{P}^2$. A toric blowup cuts off one corner and inserts the sum of the two neighboring ray generators.
Which torus-fixed corner cuts preserve both of the following properties?

every facet remains at lattice distance one from the origin,

every consecutive pair of polar vertices remains a lattice basis.

How many times can this be done before the anticanonical divisor stops being ample?

7.5 Toric del Pezzo surfaces through their polygons

A *del Pezzo surface* is a smooth projective surface whose anticanonical divisor is ample. For a smooth toric surface, the previous sections turn this definition into a problem about a lattice polygon:

$$X_\Sigma \text{ is del Pezzo} \iff P_{-K_X} \text{ is reflexive and } \Sigma_{P_{-K_X}} = \Sigma.$$

Smoothness asks that consecutive primitive ray generators form lattice bases. Ampleness asks that every wall of the fan remain visible to the anticanonical polygon. In dimension two, these two restrictions leave only five possibilities.

7.5.1 The five smooth toric del Pezzo surfaces

Theorem 7.27 (Classification of smooth toric del Pezzo surfaces)**.** *Up to isomorphism, the smooth toric del Pezzo surfaces are*

$$\boxed{\mathbb{P}^2, \qquad \mathbb{P}^1 \times \mathbb{P}^1, \qquad \mathrm{Bl}_p(\mathbb{P}^2), \qquad \mathrm{Bl}_{p,q}(\mathbb{P}^2), \qquad \mathrm{Bl}_{p,q,r}(\mathbb{P}^2),}$$

where p, q, r are distinct torus-fixed points of $\mathbb{P}^2$, with as many of them blown up as the notation indicates. Their anticanonical degrees are respectively

$$9, \qquad 8, \qquad 8, \qquad 7, \qquad 6.$$

Reference. The polytope classification is Cox–Little–Schenck, Exercises 8.3.6–8.3.8. In particular, Exercise 8.3.7 isolates the five smooth cases among the sixteen reflexive polygons, and Exercise 8.3.8 identifies the corresponding toric surfaces.

Conceptual proof sketch. A smooth toric Fano surface corresponds, by theorem 7.24, to a reflexive polygon whose polar has unimodular edges. The classification of reflexive polygons leaves exactly five such lattice-equivalence classes.

There is also a geometric way to organize the same list. Every smooth complete toric surface is obtained from $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or a Hirzebruch surface by toric blowups; this is Cox–Little–Schenck, Theorem 10.4.3. By theorem 7.14, the only Fano Hirzebruch surfaces are

$$\mathbb{F}_0 \simeq \mathbb{P}^1 \times \mathbb{P}^1 \qquad \text{and} \qquad \mathbb{F}_1 \simeq \mathrm{Bl}_p(\mathbb{P}^2).$$

Starting from $\mathbb{P}^2$, at most three successive torus-fixed blowups preserve ampleness of the anticanonical divisor. These produce the remaining three surfaces in the list. □

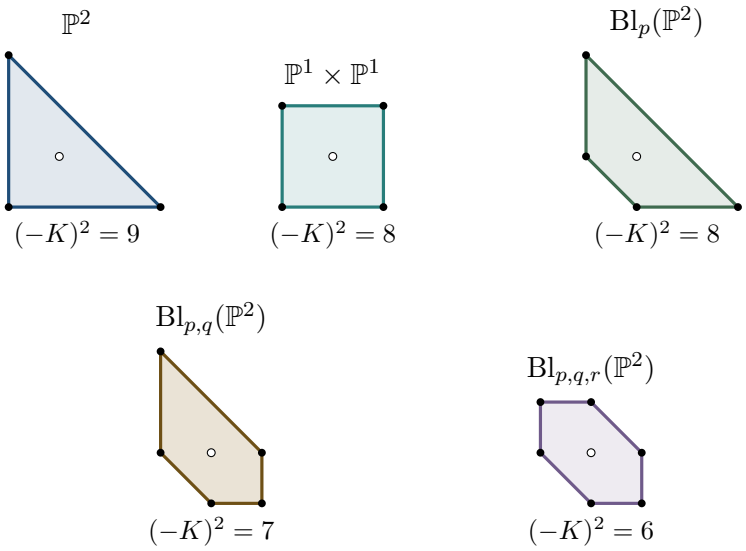


Figure 7.7: The five anticanonical polygons of smooth toric del Pezzo surfaces, up to lattice equivalence. The open circle is the origin, the unique interior lattice point.

The two degree-eight surfaces are genuinely different. The polygon of $\mathbb{P}^1 \times \mathbb{P}^1$ is centrally symmetric, while the polygon of $\mathrm{Bl}_p(\mathbb{P}^2) \simeq \mathbb{F}_1$ is not. Equivalently, the first surface has no curve of self-intersection -1, while the second contains the exceptional curve.

7.5.2 Degree, area, and boundary lattice points

For any ample divisor on a projective toric surface, theorem 5.44 gives

$$D^2 = 2\,\mathrm{Area}(P_D).$$

For an anticanonical reflexive polygon, the origin is the unique interior lattice point. Pick's theorem therefore gives

$$\mathrm{Area}(P_{-K_X}) = 1 + \frac{1}{2}|\partial P_{-K_X} \cap M| - 1 = \frac{1}{2}|\partial P_{-K_X} \cap M|.$$

Hence

$$\boxed{(-K_X)^2 = |\partial P_{-K_X} \cap M|.}$$

The degree of a smooth toric del Pezzo surface is literally the number of lattice points around the boundary of its anticanonical polygon. Since the only interior lattice point is the origin,

$$h^0(X, -K_X) = |P_{-K_X} \cap M| = (-K_X)^2 + 1.$$

Pause and think

Check the degree formula directly on the five polygons in Figure 7.7. Remember that a lattice edge of length ℓ contributes ℓ to the number of boundary lattice points when the edge lengths are summed cyclically.

There is also a curve-by-curve version. By theorem 5.40, if the facet F_ρ of P_{-K_X} corresponds to the invariant curve D_ρ, then

$$(-K_X) \cdot D_\rho = \mathrm{length}_M(F_\rho).$$

Thus ampleness of $-K_X$ is visible in the polygon as the positivity of every edge length.

7.5.3 Toric blowups are anticanonical corner cuts

Suppose a fixed point corresponds to a smooth cone

$$\sigma = \mathrm{Cone}(u, v), \qquad \det(u, v) = 1.$$

The toric blowup inserts the ray generated by $u + v$. For the anticanonical polygon, this adds the inequality

$$\langle m, u + v \rangle \geq -1.$$

At the old corner, both

$$\langle m, u \rangle = -1 \qquad \text{and} \qquad \langle m, v \rangle = -1,$$

so the new inequality cuts that corner off. In the smooth case, the removed triangle is unimodular and has area $1/2$. Consequently,

$$(-K_{\widetilde{X}})^2 = (-K_X)^2 - 1,$$

as expected from the blowup formula

$$K_{\widetilde{X}} = \pi^* K_X + E.$$

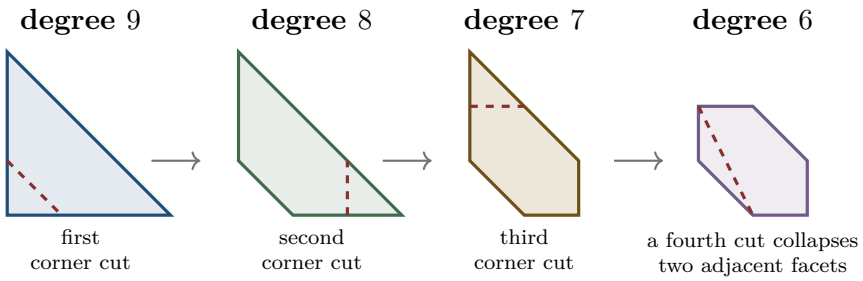


Figure 7.8: Successive torus-fixed blowups of $\mathbb{P}^2$ cut unimodular corners from the anticanonical polygon and lower the degree by one. At degree six, every edge has lattice length one; a further blowup makes two neighboring old facets collapse, so the new anticanonical divisor is no longer ample.

Why does the process stop after the third blowup? On $\mathrm{Bl}_{p,q,r}(\mathbb{P}^2)$, every edge of the anticanonical hexagon has lattice length one. Blow up the fixed point where two such invariant curves meet. Their strict transforms satisfy

$$(-K_{\widetilde{X}}) \cdot D_i' = (-K_X) \cdot D_i - 1 = 0.$$

Therefore $-K_{\widetilde{X}}$ cannot be ample. In the polygon, the same failure appears when two old facets shrink to length zero. The new support function remains convex, but it is no longer strictly convex along those walls.

Surface laboratory

For a smooth toric del Pezzo surface, one polygon simultaneously shows:

facet normals	the fan,
edge lengths	$(-K_X) \cdot D_\rho$,
twice the area	$(-K_X)^2$,
lattice points	$H^0(X, -K_X)$,
corner cuts	toric blowups.

This is the surface form of all three dictionaries at once.

Prediction

For each of the five polygons $P = P_{-K_X}$, draw the polar polygon P° and count boundary lattice points on both sides. The five examples suggest the identity

$$|\partial P \cap M| + |\partial P^\circ \cap N| = 12.$$

Why should the number 12 appear? We will return to this question when Riemann–Roch turns lattice-point counts into intersection numbers.

Guiding question

The anticanonical polygon controls global sections and positivity, but what can it tell us about the topology of the underlying toric variety—its fixed points, Betti numbers, and cohomology ring?

What this chapter has accomplished

The toric boundary is not merely a collection of invariant divisors. It is the divisor along which invariant differential forms acquire logarithmic poles. For every character χ^m,

$$\frac{d\chi^m}{\chi^m}$$

extends as a logarithmic one-form, and its residue along D_ρ is

$$\operatorname{Res}_{D_\rho}\left(\frac{d\chi^m}{\chi^m}\right) = \langle m, u_\rho \rangle.$$

On a smooth toric variety, these invariant forms trivialize the logarithmic cotangent bundle:

$$\Omega^1_{X_\Sigma}(\log \partial X) \simeq M \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma}.$$

Taking the top exterior power gives the conceptual form of the canonical-divisor formula:

$$\boxed{K_{X_\Sigma} + \partial X \sim 0.}$$

The anticanonical divisor therefore has the particularly uniform polytope

$$P_{-K_X} = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -1 \text{ for every } \rho\}.$$

Its positivity is read from its support function:

$$X_\Sigma \text{ Fano} \quad \Longleftrightarrow \quad -K_{X_\Sigma} \text{ ample} \quad \Longleftrightarrow \quad \Sigma_{P_{-K_X}} = \Sigma.$$

The origin is the unique interior lattice point of this polytope.

Polarity remembers more than the normal fan. If

$$P = \{m : \langle m, u_F \rangle \geq -a_F\},$$

then

$$P^\circ = \operatorname{Conv}\left(\frac{u_F}{a_F}\right).$$

Thus the normal fan remembers only the directions of the facet normals, while the polar remembers their lattice distances from the origin. A lattice polytope is reflexive precisely when every facet is at lattice distance one from the origin.

This gives the central correspondence of the chapter:

$$\boxed{\begin{array}{c} P \text{ reflexive} \Longleftrightarrow X_{\Sigma_P} \text{ Gorenstein Fano,} \\ P^\circ \text{ lattice} \Longleftrightarrow K_{X_{\Sigma_P}} \text{ Cartier.} \end{array}}$$

Smoothness is stronger: the relevant cones must be unimodular, not merely Gorenstein.

In dimension two, the dictionary becomes completely visible. The five smooth toric del Pezzo surfaces are

$$\mathbb{P}^2, \qquad \mathbb{P}^1 \times \mathbb{P}^1, \qquad \mathrm{Bl}_p(\mathbb{P}^2), \qquad \mathrm{Bl}_{p,q}(\mathbb{P}^2), \qquad \mathrm{Bl}_{p,q,r}(\mathbb{P}^2),$$

with anticanonical degrees

$$9, \qquad 8, \qquad 8, \qquad 7, \qquad 6.$$

Their anticanonical polygons record the fan, the invariant intersection numbers, the global sections, and the toric blowups all at once.

Comprehension exercises

7.1. On the torus $T = (\mathbb{G}_m)^2$, let

$$\chi^{(a,b)} = x^a y^b.$$

Compute

$$\frac{d\chi^{(a,b)}}{\chi^{(a,b)}}.$$

Compactify T to $\mathbb{P}^1 \times \mathbb{P}^1$, and compute the residue of this logarithmic form along each of the four boundary divisors. Explain how the four residues recover the pairings with the four ray generators.

7.2. Let X_Σ be a smooth complete toric variety of dimension n. Choose a basis $m_1, \dots, m_n$ of M, and set

$$\omega = \frac{d\chi^{m_1}}{\chi^{m_1}} \wedge \cdots \wedge \frac{d\chi^{m_n}}{\chi^{m_n}}.$$

Show on a smooth affine chart that ω has a simple pole along each boundary component and no other zero or pole. Recover

$$K_{X_\Sigma} = -\sum_\rho D_\rho.$$

7.3. For $X = \mathbb{P}^2$, compute P_{-K_X} directly from the three ray generators. Then:

(a) list all lattice points of P_{-K_X};

(b) verify that the origin is its unique interior lattice point;

(c) recover $(-K_X)^2$ from its area;

(d) recover $h^0(X, -K_X)$ from its lattice points.

7.4. For the Hirzebruch surface $\mathbb{F}_r$, use the rays

$$(1,0), \quad (0,1), \quad (-1,r), \quad (0,-1)$$

to rederive

$$P_{-K_{\mathbb{F}_r}} = \{(x,y) : x \geq -1,\ -1 \leq y \leq 1,\ x \leq 1 + ry\}.$$

Draw the cases $r = 0, 1, 2, 3$. Determine directly from the polygons when $-K$ is ample, basepoint free but not ample, or not basepoint free. Compare with

$$(-K_{\mathbb{F}_r}) \cdot S = 2 - r.$$

7.5. Let

$$P = \{m \in M_{\mathbb{R}} : \langle m, u_i \rangle \geq -a_i,\ i = 1, \ldots, s\}$$

be full dimensional with $0 \in \operatorname{int}(P)$, where the inequalities are irredundant and the u_i are primitive. Prove

$$P^\circ = \operatorname{Conv}\left(\frac{u_1}{a_1}, \ldots, \frac{u_s}{a_s}\right).$$

Deduce the lattice-distance-one criterion for reflexivity.

7.6. Let $P \subseteq M_{\mathbb{R}}$ be reflexive. For a face $Q \preceq P$, define

$$Q^* = \{u \in P^\circ : \langle m, u \rangle = -1 \text{ for every } m \in Q\}.$$

Show in dimension two that vertices of P correspond to edges of P°, and edges of P correspond to vertices of P°. Verify

$$\dim Q + \dim Q^* = 1.$$

Work this out explicitly for the square and diamond.

7.7. Consider the weighted projective plane $\mathbb{P}(1, 1, n)$.

(a) Show that it is $\mathbb{Q}$-Fano for every $n \geq 1$.

(b) Use the divisibility criterion to determine for which n it is Gorenstein Fano.

(c) Compute the anticanonical polygon for $n = 2$ and $n = 3$, and locate the nonintegral vertex in the latter case.

7.8. A reflexive polygon need not define a smooth toric surface. For each of the following polygons, compute the determinants of consecutive primitive ray generators of its normal fan:

$$\operatorname{Conv}\{(1,0),(0,1),(-1,-1)\}, \qquad \operatorname{Conv}\{(1,0),(0,1),(-1,-2)\}.$$

Determine which associated toric surfaces are smooth. Explain why both polygons can nevertheless be reflexive.

7.9. Let X be a smooth toric del Pezzo surface and let $\widetilde{X} \to X$ be the blowup of a torus fixed point. Show directly from the corner-cut picture that

$$(-K_{\widetilde{X}})^2 = (-K_X)^2 - 1.$$

Then explain why blowing up a fixed point of $\mathrm{Bl}_{p,q,r}(\mathbb{P}^2)$ produces a surface whose anticanonical divisor is not ample.

7.10. For each of the five smooth toric del Pezzo polygons in Figure 7.7, compute the polar polygon and verify experimentally that

$$|\partial P \cap M| + |\partial P^\circ \cap N| = 12.$$

Record how the two summands change under each toric blowup. Do not try to prove the identity in general yet; formulate instead what a proof would need to explain.

Challenge exercise

A crepant resolution hidden in a reflexive triangle.

Let $N = \mathbb{Z}^2$, and consider the complete fan Σ with rays

$$v_0 = (1, 0), \qquad v_1 = (-1, 2), \qquad v_2 = (0, -1),$$

and maximal cones generated by each consecutive pair. The positive relation

$$v_0 + v_1 + 2v_2 = 0$$

identifies the corresponding toric surface with

$$Y = X_\Sigma \simeq \mathbb{P}(1, 1, 2).$$

The cone $\mathrm{Cone}(v_0, v_1)$ is singular. Insert the primitive ray

$$w = (0, 1)$$

and let $\widetilde{\Sigma}$ be the resulting subdivision. Write

$$\pi : X_{\widetilde{\Sigma}} \longrightarrow Y$$

for the induced proper birational toric morphism.

The goal is to prove that π is a *crepant resolution*:

$$X_{\widetilde{\Sigma}} \text{ is smooth}, \qquad K_{X_{\widetilde{\Sigma}}} = \pi^* K_Y.$$

You will also see why resolving the singularity changes the anticanonical divisor from ample to merely basepoint free.

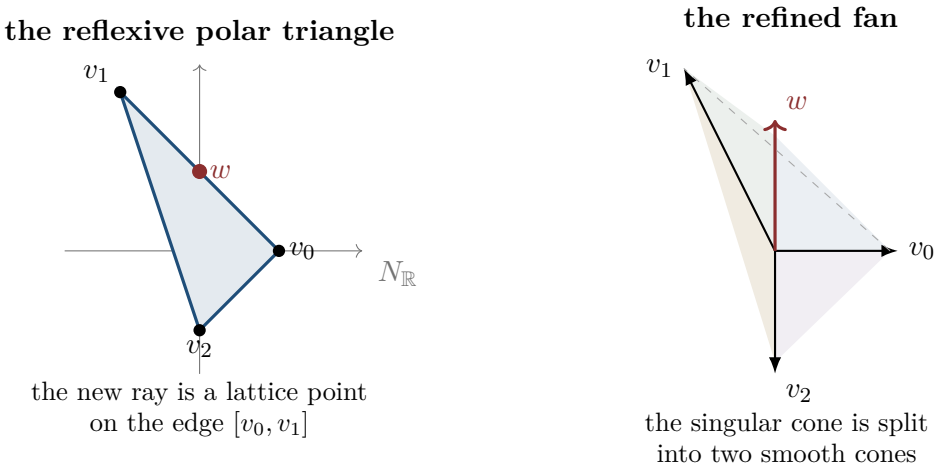


Figure 7.9: The midpoint w of the polar edge $[v_0, v_1]$ lies on the same height-one hyperplane as its endpoints. Adding its ray resolves the singular cone without changing the anticanonical polytope.

Hint 1. Locate the singularity.
Compute the determinants of the three maximal cones of Σ. Show that two cones are smooth and that

$$|\det(v_0, v_1)| = 2.$$

Conclude that Y has one cyclic quotient singularity of order two. Why is the fan nevertheless complete?

Hint 2. Recover the reflexive anticanonical triangle.
Write $m = (a, b) \in M_{\mathbb{R}}$. Show that

$$P_{-K_Y} = \{(a, b) : a \geq -1,\ b \leq 1,\ a - 2b \leq 1\}.$$

Find its three vertices and verify that it is a lattice polygon with unique interior lattice point 0. Compute its polar and show that

$$P^{\circ}_{-K_Y} = \mathrm{Conv}\{v_0, v_1, v_2\}.$$

Hint 3. Check the Gorenstein condition locally.
On the singular cone $\tau = \mathrm{Cone}(v_0, v_1)$, find an element $m_\tau \in M$ satisfying

$$\langle m_\tau, v_0\rangle = \langle m_\tau, v_1\rangle = 1.$$

Show that one may take

$$m_\tau = (1, 1).$$

Conclude from the height-one criterion that K_Y is Cartier near the singular point. Explain why this is the local reason that Y is Gorenstein rather than merely $\mathbb{Q}$-Gorenstein.

Hint 4. Resolve the cone.
Observe that

$$w = \frac{v_0 + v_1}{2} = (0, 1).$$

Compute

$$\det(v_0, w), \qquad \det(w, v_1),$$

and show that the subdivided fan $\widetilde{\Sigma}$ is smooth. Compare its four rays with the standard fan of $\mathbb{F}_2$, and deduce

$$X_{\widetilde{\Sigma}} \simeq \mathbb{F}_2.$$

Hint 5. Why the resolution is crepant.
Check that the new ray lies on the same height-one hyperplane:

$$\langle m_\tau, w\rangle = 1.$$

Use the Cartier data for K_Y to compute the coefficient of the new divisor $E = D_w$ in $\pi^* K_Y$. Compare with the toric formula

$$K_{X_{\widetilde{\Sigma}}} = -D_{v_0} - D_{v_1} - D_{v_2} - E$$

and prove

$$\boxed{K_{X_{\widetilde{\Sigma}}} = \pi^* K_Y.}$$

This equality is the definition of a crepant resolution.

Hint 6. The anticanonical polygon does not change.
The new ray contributes the additional inequality

$$\langle (a,b), w \rangle = b \geq -1.$$

Show that this inequality is already implied by

$$a \geq -1, \qquad a - 2b \leq 1.$$

Conclude that

$$P_{-K_{X_{\widetilde{\Sigma}}}} = P_{-K_Y}.$$

Interpret this equality as

$$-K_{X_{\widetilde{\Sigma}}} = \pi^*(-K_Y).$$

Hint 7. Ample downstairs, only nef upstairs.
Since P_{-K_Y} has normal fan Σ, the divisor $-K_Y$ is ample. Explain why the refined fan $\widetilde{\Sigma}$ is strictly finer than the normal fan of the same polygon. Deduce that

$$-K_{X_{\widetilde{\Sigma}}}$$

is basepoint free but not ample.
Equivalently, show that the support function is linear across the new wall generated by w. Which curve is contracted by the associated anticanonical morphism?

Hint 8. See the exceptional curve numerically.
The neighboring rays of w satisfy

$$v_0 + v_1 = 2w.$$

Use the smooth toric surface intersection formula to show that

$$E^2 = -2.$$

Then compute

$$(-K_{X_{\widetilde{\Sigma}}}) \cdot E = 0.$$

Explain why this confirms both that $-K$ is not ample and that the anticanonical morphism contracts exactly the (-2)-curve E.

Hint 9. The general pattern.
Summarize the example in the following three languages:

Fan	**Divisors**	**Polytope**
insert a height-one ray	$K_{\widetilde{X}} = \pi^* K_X$	add a redundant distance-one inequality
refine one cone	$-K_{\widetilde{X}}$ loses ampleness	P_{-K} stays unchanged

Why is this phenomenon special to rays lying on the height-one slice? What discrepancy would you expect if the new ray lay above or below that slice?

The lesson of the resolution

A crepant toric resolution does not create a new anticanonical polytope. It refines the fan by inserting lattice points on the height-one boundary of the polar polytope. The canonical class is preserved, but strict convexity may be lost:

$$\boxed{\text{Gorenstein Fano downstairs} \quad \longrightarrow \quad \text{basepoint-free } -K \text{ upstairs.}}$$

For $\mathbb{P}(1,1,2)$, the resolution is $\mathbb{F}_2$, and the new ray produces the (-2)-curve contracted by the anticanonical map.

Notes and references

Invariant differential forms and the toric residue sequence are developed in Cox–Little–Schenck, §8.1. The canonical divisor and the Cartier and Gorenstein criteria are treated in §8.2. Fano toric varieties, weighted projective examples, and the correspondence with reflexive polytopes appear in §8.3.

Our polar convention is

$$P^\circ = \{u : \langle m, u\rangle \geq -1 \text{ for every } m \in P\}.$$

This is adapted to the toric divisor inequalities used throughout the notes. With the opposite sign convention, some sources write the polar using ≤ 1; the geometry is the same after replacing one polytope by its negative.

The classification of smooth complete toric surfaces is developed in Cox–Little–Schenck, Chapter 10. Combining that classification with the anticanonical positivity criterion gives the five smooth toric del Pezzo surfaces displayed in theorem 7.27. The sixteen reflexive polygons include singular Gorenstein del Pezzo surfaces as well.

The challenge exercise illustrates a basic principle of toric birational geometry. Subdividing a Gorenstein cone by rays lying on its height-one slice is crepant. In the example

$$\mathbb{F}_2 \longrightarrow \mathbb{P}(1, 1, 2),$$

the exceptional curve has self-intersection -2, the canonical class pulls back unchanged, and the anticanonical polygon remains the same. This picture will reappear when resolutions and birational transformations are studied more systematically.

The boundary identity

$$|\partial P \cap M| + |\partial P^\circ \cap N| = 12$$

for reflexive polygons is traditionally called the twelve-point theorem. We have used it only as an experimentally visible prediction. A later Riemann–Roch discussion will explain why the number twelve is natural from the geometry of smooth toric surfaces.

Chapter 8

Moment Polytopes and Topology

Chapter overview

Until now, a lattice polytope has played several algebraic and combinatorial roles. Its lattice points indexed global sections, its normal fan constructed a projective toric variety, and its edges and area encoded intersection numbers on surfaces.
Over $\mathbb{C}$, the same polytope acquires a new meaning: it is the image of a continuous map from the toric variety itself. The mechanism begins with the compact torus

$$K_N \simeq (S^1)^n \subseteq T_N \simeq (\mathbb{C}^*)^n.$$

The compact torus changes the phases of monomial coordinates but not their absolute values. Normalized coordinate magnitudes therefore give convex coefficients, and the lattice weights of the coordinates combine to produce a point of the polytope.
This chapter develops the progression

$$\boxed{\begin{array}{c}\text{moment map} \longrightarrow \text{polytope as orbit space} \longrightarrow \text{cells and Betti numbers} \\ \longrightarrow \text{the cohomology ring from the fan.}\end{array}}$$

Throughout the chapter, we set $\Bbbk = \mathbb{C}$ and use the classical topology.

The three dictionaries

The topology of a projective toric variety is governed by the same weights that defined its projective coordinates.

Algebra	Geometry and topology	Combinatorics
monomial coordinates χ^{m_i}	projective toric embedding	lattice weights $m_i \in M$
coordinate phases	compact-torus orbits	directions that disappear under μ
normalized values $\lvert z_i\rvert^2$	moment map	convex coefficients
torus-invariant divisors	degree-two cohomology classes	rays and Stanley–Reisner relations

Surface laboratory

For a projective toric surface, the moment image is a polygon. Over an interior point sits a two-dimensional compact torus; over an edge, one circle direction has collapsed; over a vertex, the entire compact torus has collapsed to a fixed point.

A toric blowup cuts a corner from the moment polygon. Later in the chapter, the same corner cut will produce one new two-dimensional cohomology class and change the intersection form by adjoining an exceptional class of square -1.

8.1 Compact torus actions and the moment map

Throughout this chapter, we work over

$$\Bbbk = \mathbb{C}$$

and use the classical topology on complex varieties. This change of field matters: an element of $\mathbb{C}^*$ has both a positive size and a phase, and the phase directions form a compact torus.

8.1.1 The compact torus inside the algebraic torus

Recall that

$$T_N = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^*).$$

Let

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

The *compact torus* associated to N is

$$K_N := \mathrm{Hom}_{\mathbb{Z}}(M, S^1) \subseteq T_N.$$

After choosing a basis of M, these groups become

$$T_N \simeq (\mathbb{C}^*)^n, \qquad K_N \simeq (S^1)^n.$$

Every nonzero complex number has a unique polar decomposition

$$z = |z|\frac{z}{|z|}.$$

Taking the logarithm of its positive factor gives

$$\mathbb{C}^* \simeq \mathbb{R} \times S^1, \qquad z \longmapsto \left(\log|z|, \frac{z}{|z|}\right).$$

Applying $\mathrm{Hom}_{\mathbb{Z}}(M, -)$ yields the intrinsic toric version.

Proposition 8.1 (Radial and angular directions)**.** *There is an isomorphism of real Lie groups*

$$\boxed{T_N \simeq N_{\mathbb{R}} \times K_N.}$$

The first factor records absolute values of characters, while the second records their phases.

Proof. For $t \in T_N$, define

$$\nu_t : M \longrightarrow \mathbb{R}, \qquad m \longmapsto \log|\chi^m(t)|.$$

Because absolute value turns products into products and logarithms turn products into sums, ν_t is a homomorphism. Hence

$$\nu_t \in \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{R}) = N_{\mathbb{R}}.$$

The phase

$$m \longmapsto \frac{\chi^m(t)}{|\chi^m(t)|}$$

is an element of K_N. These two assignments are inverse to the map obtained by multiplying radial and angular characters. □

Observation 8.2. No background in Lie groups is needed here. The proposition is simply the coordinatewise statement

$$(\mathbb{C}^*)^n \simeq (\mathbb{R}_{>0})^n \times (S^1)^n \simeq \mathbb{R}^n \times (S^1)^n.$$

The noncompact factor $N_{\mathbb{R}}$ moves toward and away from the toric boundary. The compact factor K_N rotates without changing any absolute value.

Pause and think

Let $k \in K_N$. Why does

$$|\chi^m(k)| = 1$$

for every $m \in M$? Explain why a formula involving only absolute values of projective coordinates should be constant on K_N-orbits.

8.1.2 A projective toric variety with weighted coordinates

Let

$$A = \{m_0, \dots, m_r\} \subseteq M$$

be a finite set of lattice points. As in Section 5.2, it defines a monomial map

$$\varphi_A : T_N \longrightarrow \mathbb{P}^r, \qquad t \longmapsto [\chi^{m_0}(t) : \cdots : \chi^{m_r}(t)].$$

Write

$$X_A := \overline{\varphi_A(T_N)} \subseteq \mathbb{P}^r.$$

The coordinates carry the weights $m_0, \dots, m_r$. Accordingly, K_N acts on $\mathbb{P}^r$ by

$$k \cdot [z_0 : \cdots : z_r] = [\chi^{m_0}(k) z_0 : \cdots : \chi^{m_r}(k) z_r].$$

Since every factor $\chi^{m_i}(k)$ has absolute value one, the action changes phases but not coordinate magnitudes.

This suggests discarding the phases and retaining only the relative sizes

$$|z_0|^2, \dots, |z_r|^2.$$

Once normalized by their sum, these become coefficients of a convex combination of the weights m_i.

Definition 8.3 (Projective moment map)**.** The *moment map* associated to A is

$$\mu_A : X_A \longrightarrow M_{\mathbb{R}}, \qquad \boxed{\mu_A([z_0 : \cdots : z_r]) = \frac{\sum_{i=0}^r |z_i|^2 m_i}{\sum_{i=0}^r |z_i|^2}.}$$

Proposition 8.4 (The first properties of the moment map)**.** *The map μ_A is well defined, continuous, and invariant under the action of K_N. Moreover,*

$$\mu_A(X_A) \subseteq \mathrm{Conv}(A).$$

Proof. Multiplying all homogeneous coordinates by $c \in \mathbb{C}^*$ multiplies both the numerator and denominator by $|c|^2$, so the quotient is unchanged. Thus the formula is well defined on projective space.

For $k \in K_N$,

$$|\chi^{m_i}(k) z_i|^2 = |z_i|^2,$$

which proves invariance. Continuity is immediate. Finally, the numbers

$$p_i = \frac{|z_i|^2}{\sum_j |z_j|^2}$$

are nonnegative and satisfy $\sum_i p_i = 1$. Hence

$$\mu_A([z]) = \sum_i p_i m_i$$

is a convex combination of the lattice points in A. □

The important surprise is that the inclusion in theorem 8.4 is an equality. Although X_A occupies only a small part of $\mathbb{P}^r$, it contains exactly the right points to fill the entire convex hull of its weights.

For a projective toric variety X_P, its *positive real part* $(X_P)_{\geq 0}$ is the closure of the points of the dense torus on which every character takes a positive real value. It is a concrete slice transverse to the compact-torus orbits.

Theorem 8.5 (The moment image is the polytope)**.** *Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional lattice polytope, and assume the lattice points*

$$P \cap M = \{m_0, \ldots, m_r\}$$

define a projective embedding

$$X_P \hookrightarrow \mathbb{P}^r.$$

Then the associated moment map satisfies

$$\boxed{\mu_P(X_P) = P.}$$

More precisely, μ_P maps the positive real part of X_P homeomorphically onto P.

Reference. Cox–Little–Schenck, Theorem 12.2.5 proves the homeomorphism for the algebraic moment map. Formula (12.2.4) gives the squared-norm moment map used here, and Exercise 12.2.8 establishes the same conclusion for this version.

Conceptual proof sketch. On the positive real part of the dense torus, write a point as $\exp(u)$ with $u \in N_{\mathbb{R}}$. Then

$$|\chi^{m_i}(\exp(u))|^2 = \exp(2\langle m_i, u\rangle),$$

so

$$\mu_P(u) = \frac{\sum_i e^{2\langle m_i,u\rangle} m_i}{\sum_i e^{2\langle m_i,u\rangle}}.$$

This is the gradient, up to a factor of two, of the strictly convex function

$$F(u) = \log\left(\sum_i e^{2\langle m_i,u\rangle}\right).$$

Strict convexity makes the gradient injective. Convex duality shows that its image is the interior of P.

As u tends to infinity in a chosen direction, the exponential weights concentrate on the face where the corresponding linear functional is largest. Thus boundary directions in the torus land on faces of P, and taking the closure fills all of P. □

The moment map converts *relative coordinate magnitudes* into *barycentric coordinates in a lattice polytope.*

8.1.3 The interval inside $\mathbb{P}^1$

Example 8.6 (The moment map of $\mathbb{P}^1$)**.** Take

$$A = \{0, 1\} \subseteq \mathbb{Z}.$$

Then $X_A = \mathbb{P}^1$, and

$$\mu([z_0 : z_1]) = \frac{|z_1|^2}{|z_0|^2 + |z_1|^2} \in [0, 1].$$

Every $s \in [0, 1]$ occurs: one may take

$$[z_0 : z_1] = [\sqrt{1-s} : \sqrt{s}].$$

The two endpoints come from the fixed points

$$[1 : 0] \qquad \text{and} \qquad [0 : 1].$$

All other values retain one circle of phase information.

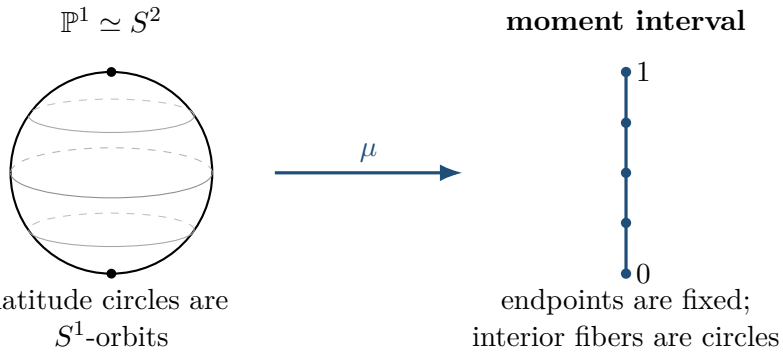


Figure 8.1: The moment map collapses each circle orbit in $\mathbb{P}^1$ to one point of the interval. Section 8.2 will show that this behavior is the model for every projective toric variety.

8.1.4 Triangles, rectangles, and corner cuts

Example 8.7 (The triangle of $\mathbb{P}^2$)**.** For

$$A = \{0, e_1, e_2\},$$

the moment map is

$$\mu([z_0 : z_1 : z_2]) = \frac{|z_1|^2}{S} e_1 + \frac{|z_2|^2}{S} e_2, \qquad S = |z_0|^2 + |z_1|^2 + |z_2|^2.$$

Thus its two coordinates are nonnegative and have sum at most one. The image is the standard triangle

$$\Delta_2 = \mathrm{Conv}(0, e_1, e_2).$$

Conversely, a point $(a, b) \in \Delta_2$ is the image of

$$[\sqrt{1-a-b} : \sqrt{a} : \sqrt{b}].$$

So here the theorem is visible directly from barycentric coordinates.

Example 8.8 (The rectangle of $\mathbb{P}^1 \times \mathbb{P}^1$)**.** For the line bundle $\mathcal{O}(1, 1)$, the moment map can be written without using Segre coordinates:

$$\mu([x_0 : x_1], [y_0 : y_1]) = \left(\frac{|x_1|^2}{|x_0|^2 + |x_1|^2}, \frac{|y_1|^2}{|y_0|^2 + |y_1|^2} \right).$$

Its image is the square $[0, 1]^2$. More generally, $\mathcal{O}(a, b)$ gives a rectangle with side lengths a and b.

Example 8.9 (A blowup becomes a corner cut)**.** On

$$X = \mathrm{Bl}_p(\mathbb{P}^2),$$

the divisor $2H - E$ has section polytope

$$P_{2H-E} = \{(a, b) \in \mathbb{R}^2 : a, b \geq 0,\ 1 \leq a + b \leq 2\}.$$

Its five lattice points give an embedding into $\mathbb{P}^4$, and theorem 8.5 says that the moment image is exactly this truncated triangle. The exceptional divisor is visible as the new edge created by cutting off the corner of the triangle for $2H$.

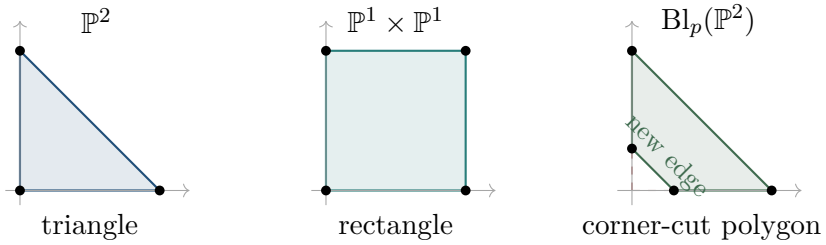


Figure 8.2: Moment images of three projective toric surfaces. The same polygons previously indexed sections and determined normal fans; over $\mathbb{C}$, they are also the actual images of continuous maps from the varieties.

Observation 8.10 (Dependence on the chosen line bundle). The moment polytope belongs to a polarized toric variety, not merely to the underlying variety. Translating every weight by $m_0 \in M$ changes the moment map by

$$\mu_{A+m_0} = \mu_A + m_0,$$

while replacing a line bundle by its d-th tensor power dilates the polytope:

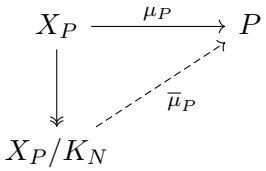
$$P \longmapsto dP.$$

Thus the precise position and scale of the moment image depend on the chosen projective embedding. Its normal fan, and hence the underlying toric variety when the divisor is ample, does not change.

Observation 8.11 (Why symplectic geometers call this a moment map). The Fubini–Study form on projective space makes the compact torus action Hamiltonian, and the formula in theorem 8.3 is its symplectic moment map. We will not need the general language of symplectic forms or Hamiltonian vector fields. For us, the important facts are the explicit weighted-average formula, its invariance under K_N, and its polytope image.

8.1.5 What information has the map forgotten?

Because μ_P is constant on compact-torus orbits, it factors through the quotient map:



The map $\overline{\mu}_P$ is surjective by theorem 8.5. What remains unclear is whether it is injective. The examples suggest that the only information forgotten by μ_P is phase information, but the number of surviving phases changes along the boundary of P.

Prediction

The map μ_P is constant on every K_N-orbit. In the examples, its fibers seem to change dimension as the image point moves from the interior of P to an edge and then to a vertex. For a point in the relative interior of a codimension-k face, which k circle directions should collapse? Formulate a prediction for the dimension of the compact-torus orbit in that fiber.

Guiding question

Does the moment map identify *exactly* the points lying in the same compact-torus orbit? Equivalently, is the quotient space

$$X_P/K_N$$

actually homeomorphic to the polytope P?

8.2 The polytope as an orbit space

The answer to the question at the end of the previous section is yes. The moment map forgets the angular coordinates, but it makes no additional identifications: two points have the same moment image precisely when they lie in the same compact-torus orbit.

The shape of the fiber is not constant. Over the interior of the polytope it is the full compact torus K_N. Each time the image reaches a new boundary direction, one or more circle directions become stabilizers and disappear from the orbit. Our first task is to match those disappearing directions with faces of the polytope.

8.2.1 Faces and torus strata

Let $P \subseteq M_{\mathbb{R}}$ be a full-dimensional lattice polytope whose lattice points define a projective toric embedding. For a face

$$Q \preceq P,$$

write σ_Q for its normal cone. Recall from Section 5.4 that

$$\dim Q + \dim \sigma_Q = n.$$

Since the fan of X_P is the normal fan Σ_P, the orbit–cone correspondence attaches an algebraic torus orbit

$$O_Q := O(\sigma_Q) \subseteq X_P$$

to the face Q. Its dimension is

$$\dim O_Q = n - \dim \sigma_Q = \dim Q.$$

Thus the face–orbit correspondence preserves dimension, even though the intermediate passage from faces to normal cones reverses it.

In the projective coordinates indexed by $P \cap M$, the orbit O_Q has a particularly useful description:

$$x = [z_m]_{m \in P \cap M} \in O_Q \quad \Longrightarrow \quad z_m \neq 0 \iff m \in Q \cap M.$$

Coordinates whose weights lie outside Q vanish, while the coordinates indexed by lattice points of Q remain nonzero. The closure

$$X_Q := \overline{O_Q}$$

is itself the projective toric variety associated to the face Q.

Proposition 8.12 (The moment map respects face strata)**.** *For every face $Q \preceq P$,*

$$\boxed{\mu_P(X_Q) = Q, \qquad \mu_P(O_Q) = \operatorname{Relint}(Q).}$$

Equivalently,

$$\mu_P^{-1}\big(\operatorname{Relint}(Q)\big) = O_Q.$$

Reference. The face-by-face argument in Cox–Little–Schenck, Theorem 12.2.5 proves the statement for the algebraic moment map; Exercise 12.2.8 gives the same result for the squared-norm moment map used here.

Conceptual proof sketch. On X_Q, precisely the coordinates with weights in Q can be nonzero, so the moment-map formula restricts to the moment map for Q. Thus X_Q maps onto Q. On the dense orbit O_Q, every one of these coordinates is nonzero, so the resulting convex combination cannot lie in a proper face of Q: equality in a supporting inequality would force all nonzero weights into that face. Hence O_Q maps onto $\operatorname{Relint}(Q)$, while its smaller orbit strata map to the boundary faces. □

The proposition gives the old orbit–face dictionary a new topological column. It also explains why the moment map knows exactly which algebraic torus orbit contains a point: one first finds the unique face whose relative interior contains its image.

8.2.2 The compact stabilizer of a face

The algebraic orbit O_Q is a complex torus of dimension $\dim Q$. Its maximal compact subgroup will be the fiber of the moment map over $\operatorname{Relint}(Q)$.

To describe it intrinsically, set

$$N_{\sigma_Q} = N \cap \operatorname{span}_{\mathbb{R}}(\sigma_Q), \qquad N(\sigma_Q) = N/N_{\sigma_Q}.$$

The character lattice of O_Q is

$$\sigma_Q^\perp \cap M.$$

Restriction of characters gives a surjective map of compact tori

$$K_N \longrightarrow K_{N(\sigma_Q)} = \mathrm{Hom}_{\mathbb{Z}}(\sigma_Q^\perp \cap M, S^1).$$

Definition 8.13 (Compact stabilizer of a face)**.** The *compact stabilizer* associated to Q is

$$K_Q := \left\{ k \in K_N : \chi^m(k) = 1 \text{ for every } m \in \sigma_Q^\perp \cap M \right\}.$$

Equivalently, K_Q is the kernel in the exact sequence

$$1 \longrightarrow K_Q \longrightarrow K_N \longrightarrow K_{N(\sigma_Q)} \longrightarrow 1.$$

The lattice $\sigma_Q^\perp \cap M$ has rank $\dim Q$. Therefore

$$\boxed{K_Q \simeq (S^1)^{\operatorname{codim} Q}, \qquad K_N/K_Q \simeq (S^1)^{\dim Q}.}$$

Every point of O_Q has stabilizer K_Q under the K_N-action. Indeed, the algebraic torus is abelian, so all points in one algebraic orbit have the same stabilizer, and the action on O_Q factors through its compact torus $K_{N(\sigma_Q)}$.

Reference. This is the compact-torus form of Cox–Little–Schenck, Lemma 3.2.5 and Proposition 12.2.3(b).

For a facet F, the normal cone is the ray generated by its primitive inward normal u_F. The subgroup K_F is the circle

$$S^1 \longrightarrow K_N, \qquad e^{i\theta} \longmapsto \exp(i\theta u_F).$$

This is the precise meaning of the statement that the circle normal to a facet collapses along that facet.

Pause and think

Let P be a polygon and let v be a vertex where the two edges F_1 and F_2 meet. The stabilizers over the interiors of the edges are the circles generated by u_{F_1} and u_{F_2}.
What is the stabilizer over v? What dimension should the moment-map fiber over v have?

8.2.3 Every fiber is one compact-torus orbit

We can now state the central result. Its key input is the positive real part introduced in Section 8.1. It gives one canonical point over each point of P; polar decomposition supplies all remaining points in the fiber.

Theorem 8.14 (Fibers of the moment map)**.** *Let $p \in P$, and let $Q \preceq P$ be the unique face satisfying*

$$p \in \mathrm{Relint}(Q).$$

Then the fiber of the moment map is a single compact-torus orbit:

$$\boxed{\mu_P^{-1}(p) = K_N \cdot x_p \simeq K_N/K_Q \simeq (S^1)^{\dim Q},}$$

where x_p is the unique point of the positive real part mapping to p.

Reference. This is Cox–Little–Schenck, Proposition 12.2.3(b), combined with Theorem 12.2.5 and Exercise 12.2.8.

Conceptual proof sketch. The positive-part homeomorphism supplies the unique point x_p over p with no angular coordinates. Polar decomposition of $O_Q \simeq (\mathbb{C}^*)^{\dim Q}$ says that every other point over p is obtained by restoring those angular coordinates, hence lies in $K_N \cdot x_p$. The stabilizer is K_Q, so the fiber is $K_N/K_Q \simeq (S^1)^{\dim Q}$. □

Corollary 8.15 (The polytope is the orbit space)**.** *The moment map induces a homeomorphism*

$$\boxed{\overline{\mu}_P : X_P/K_N \xrightarrow{\sim} P.}$$

Reference. Cox–Little–Schenck, Proposition 12.2.3(a) identifies the compact-torus quotient with the nonnegative part; Theorem 12.2.5 and Exercise 12.2.8 identify that part with P via the moment map.

Conceptual proof sketch. The preceding theorem says exactly that the fibers of μ_P are the K_N-orbits. Thus μ_P induces a continuous bijection $X_P/K_N \to P$, which is a homeomorphism because the source is compact and the target is Hausdorff. □

The three dictionaries

For a face $Q \preceq P$ of codimension k, let $p \in \mathrm{Relint}(Q)$. The three versions of the orbit–face dictionary are:

Polytope	**Algebraic geometry**	**Compact-torus topology**
$\mathrm{Int}(P)$	dense orbit T_N	free orbit $K_N \simeq (S^1)^n$
facet F	dense orbit in the divisor D_F	one circle stabilizer; fiber $(S^1)^{n-1}$
codimension-k face Q	codimension-k orbit closure X_Q	k-torus stabilizer; fiber $(S^1)^{n-k}$
vertex v	torus-fixed point x_v	stabilizer K_N; fiber is one point

8.2.4 A toric surface over its polygon

When $\dim X_P = 2$, the theorem can be seen at a glance. The compact torus is

$$K_N \simeq (S^1)^2.$$

An interior point of the polygon has trivial stabilizer, so its fiber is a two-dimensional torus. Along the relative interior of an edge, one circle is a stabilizer and the fiber is the other circle. At a vertex, both circle directions stabilize the point, so the fiber has collapsed completely.

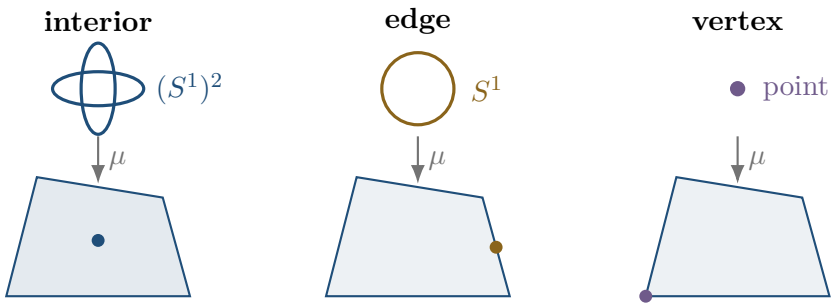


Figure 8.3: The moment fibers of a toric surface. Two circle directions survive over the polygon interior, one survives over an edge, and none survive over a vertex. Thus the boundary records exactly where compact torus directions collapse.

Example 8.16 (Which circles collapse on a rectangle?)**.** For $X = \mathbb{P}^1 \times \mathbb{P}^1$, write

$$K_N = (S^1)_x \times (S^1)_y$$

and use the square moment polytope $[0,1]^2$. The product formula from the rectangle example in Section 8.1 gives

$$\mu^{-1}(s,t) = \mu_{\mathbb{P}^1}^{-1}(s) \times \mu_{\mathbb{P}^1}^{-1}(t).$$

Consequently:

position of (s,t)	**stabilizer**	**fiber**
$0 < s < 1,\ 0 < t < 1$	$\{1\}$	$(S^1)_x \times (S^1)_y$
$s \in \{0,1\},\ 0 < t < 1$	$(S^1)_x$	$(S^1)_y$
$0 < s < 1,\ t \in \{0,1\}$	$(S^1)_y$	$(S^1)_x$
$s,t \in \{0,1\}$	$(S^1)_x \times (S^1)_y$	point

The two vertical sides record the collapse of the x-circle, and the two horizontal sides record the collapse of the y-circle. At a corner both collapses occur simultaneously.

8.2.5 Reconstructing the space from collapsing tori

Over the interior, the moment map is a genuine product. Indeed, the positive real slice and polar decomposition give

$$\mu_P^{-1}(\operatorname{Int} P) \simeq \operatorname{Int}(P) \times K_N.$$

The boundary prevents this product description from extending unchanged to all of P: the subgroup K_Q must be collapsed over $\operatorname{Relint}(Q)$.

This can be packaged as a global topological model. Let

$$s : P \longrightarrow (X_P)_{\geq 0}$$

be the inverse of the positive-real-part homeomorphism. For $p \in \operatorname{Relint}(Q)$, impose the relation

$$(p,k) \sim (p,k') \quad \Longleftrightarrow \quad k^{-1}k' \in K_Q.$$

Proposition 8.17 (The collapsing-torus model)**.** *The map*

$$P \times K_N \longrightarrow X_P, \qquad (p,k) \longmapsto k \cdot s(p),$$

induces a homeomorphism

$$\boxed{X_P \simeq (P \times K_N)/\!\sim.}$$

Over $\mathrm{Relint}(Q)$*, the relation collapses exactly the subtorus* K_Q*.*

Reference. This is the topological model in Cox–Little–Schenck, equation (12.2.7); its construction is Exercise 12.2.9.

Conceptual proof sketch. The section s chooses the positive representative of each orbit, and polar decomposition restores an arbitrary angular part $k \in K_N$. Over $p \in \mathrm{Relint}(Q)$, two angular parts give the same point precisely when they differ by the stabilizer K_Q. This is exactly the relation $\sim$; compactness then turns the resulting continuous bijection into a homeomorphism. □

This model makes the slogan "the polytope is the orbit space" more informative. The polytope specifies where the orbits live, while its face normals specify which circle subgroups collapse. The total space is obtained by starting with a full compact torus above the interior and performing the prescribed collapses along the boundary.

Warning

The homeomorphism

$$X_P/K_N \simeq P$$

does *not* say that X_P is homeomorphic to P. The left-hand side has already collapsed every compact-torus orbit. Nor does the underlying topological shape of P alone remember which circle subgroup collapses along each facet; that information comes from the lattice normals.

8.2.6 What changes in the singular case?

Neither Theorem 8.14 nor Corollary 8.15 requires X_P to be smooth. For a singular projective toric variety, the orbit space is still the polytope, and the fiber over $\mathrm{Relint}(Q)$ is still the compact torus K_N/K_Q.

What can fail is the local topology of the total space near a collapsed fiber. For example, let P be a lattice polygon and let v be a vertex. Suppose the normal cone at v is generated by primitive rays

$$\sigma_v = \mathrm{Cone}(u_1, u_2).$$

Then

$$d = |\det(u_1, u_2)|$$

measures the local lattice index. If $d = 1$, the affine chart at the fixed point is smooth and isomorphic to $\mathbb{C}^2$. If $d > 1$, the fixed point is a cyclic quotient singularity. In both cases the moment fiber over v is one point, and a neighborhood of v in the orbit space still looks like a corner of a polygon.

Thus the orbit-space theorem records the stratification correctly but does not, by itself, guarantee smooth local models. Smoothness requires the familiar lattice-basis condition on every maximal cone.

Observation 8.18 (A triangle need not mean $\mathbb{P}^2$). The moment polytope of a polarized weighted projective plane is also a triangle. Over its interior sit two-dimensional compact tori, over its edges sit circles, and over its vertices sit fixed points, exactly as for $\mathbb{P}^2$. A non-unimodular normal cone at one vertex nevertheless produces a quotient singularity there. The integral geometry of the triangle, not its shape as an unlabelled disk, detects the difference.

Prediction

Choose a generic vector $u \in N_{\mathbb{R}}$ and use it to define the height function

$$h_u(m) = \langle m, u \rangle$$

on P. Orient every edge in the direction of increasing height.
At a vertex v, the fiber has collapsed to a fixed point x_v. How might the one-parameter subgroup determined by u use the oriented edges at v to select a whole region of X_P flowing toward x_v? What should the dimension of that region be?

Guiding question

Can a generic height function decompose a smooth projective toric variety into affine cells indexed by the vertices of P, so that the Betti numbers can be read directly from the oriented polytope?

8.3 One-parameter flows, cells, and Betti numbers

The orbit-space picture tells us where the compact-torus directions collapse. We now let the noncompact part of the algebraic torus move points toward those collapsed fibers. A generic one-parameter subgroup will divide the whole variety into attracting regions, one for each torus fixed point. When the variety is smooth, these regions are affine spaces. Thus a height function on the polytope becomes a tool for computing topology.

Throughout this section, $P \subset M_{\mathbb{R}}$ is a full-dimensional lattice polytope and X_P is smooth and projective of dimension n.

8.3.1 One vector, two kinds of motion

Choose a lattice vector $u \in N$. It defines a one-parameter subgroup

$$\lambda_u : \mathbb{C}^* \longrightarrow T_N, \qquad t \longmapsto \lambda_u(t),$$

characterized by

$$\chi^m(\lambda_u(t)) = t^{\langle m,u \rangle} \qquad (m \in M).$$

The same vector defines a linear height function

$$h_u : P \longrightarrow \mathbb{R}, \qquad h_u(m) = \langle m, u \rangle.$$

We call u *generic for* P if it is not perpendicular to any edge of P. Equivalently, h_u takes distinct values at the two endpoints of every edge. For such a vector, orient each edge from lower height to higher height.

Observation 8.19 (The flow is visible on the polytope)**.** On the dense torus, write a point as $k\exp(-w)$, with $k \in K_N$ and $w \in N_{\mathbb{R}}$. Multiplication by $\lambda_u(t)$, for positive real t, translates the real coordinate in the direction $-\log(t)u$. Under the moment map, the resulting path moves monotonically in the direction of decreasing h_u as $t \to 0$. Its limiting vertex is the minimum of h_u on the appropriate face.

The convention in this observation is worth remembering: the limit $t \to 0$ corresponds to movement toward *smaller* height. Replacing u by $-u$, or taking $t \to \infty$, reverses every arrow and interchanges attracting and repelling sets.

8.3.2 Attracting sets at the fixed points

The vertices of P correspond to the torus fixed points of X_P. Write x_v for the fixed point corresponding to a vertex v. Its attracting set is

$$C_v = \left\{ x \in X_P : \lim_{t \to 0} \lambda_u(t) \cdot x = x_v \right\}.$$

To read its dimension, look only at the oriented edges incident to v. Let

$$d_u(v) = \#\{\text{edges directed away from } v\}.$$

Thus $d_u(v)$ is the number of edge directions along which height increases from v.

Theorem 8.20 (Attracting cells from oriented edges)**.** *Let X_P be a smooth projective toric variety and let $u \in N$ be generic for P. Then*

$$X_P = \coprod_{v \in \mathrm{Vert}(P)} C_v, \qquad C_v \simeq \mathbb{A}_{\mathbb{C}}^{d_u(v)}.$$

Moreover, the cells can be ordered so that successive unions are closed. Consequently they give a CW decomposition with one real $2d_u(v)$-dimensional cell for each vertex v.

Reference. This is the toric form of the Białynicki–Birula decomposition. Cox–Little–Schenck do not formulate the decomposition in this language; their orbit filtration in §12.3 gives the corresponding Betti-number count in Theorem 12.3.12. The local weight calculation uses the smooth affine charts described in §3.3.

Conceptual proof sketch. At a fixed point x_v, smoothness identifies the invariant affine chart with $\mathbb{A}^n$, one coordinate for each edge leaving v. If the corresponding primitive edge direction is $e \in M$, then $\lambda_u(t)$ acts on that coordinate with weight $\langle e, u \rangle$. A coordinate may be arbitrary when this weight is positive, since multiplication by $t^{\langle e,u \rangle}$ sends it to zero as $t \to 0$; it must vanish when the weight is negative. Hence the attracting set is an affine coordinate subspace of dimension $d_u(v)$. Projectivity orders these strata by height and makes their successive unions closed, producing the asserted cell decomposition. □

Warning

The word *generic* is essential. If u is perpendicular to an edge, the fixed locus of λ_u can contain a positive-dimensional toric subvariety rather than isolated fixed points. The resulting Białynicki–Birula strata are then affine bundles over fixed components, not necessarily affine spaces indexed by vertices.

8.3.3 Reading the cells from a polygon

For a toric surface, each vertex has two incident edges. After orienting them, its attracting cell has dimension 0, 1, or 2, according to whether zero, one, or two arrows leave the vertex.

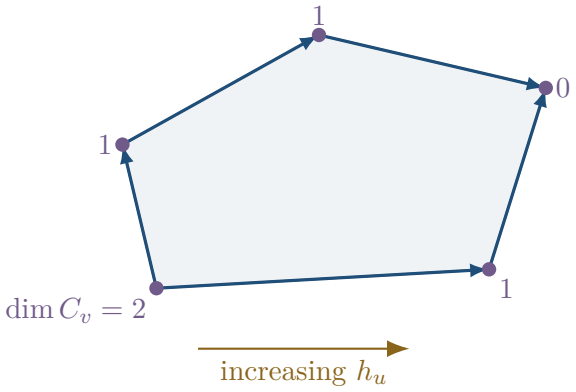


Figure 8.4: A generic height orients the edges. The label at a vertex is the number of arrows leaving it, hence the complex dimension of its attracting cell. This pentagon gives one 0-cell, three 1-cells, and one 2-cell.

Example 8.21 (The projective plane)**.** For the standard triangle, a generic height orders the three vertices. At the minimum, both edges point outward; at the middle vertex, one points in and one points out; at the maximum, both point inward. Thus

$$\mathbb{P}^2 = \mathbb{A}^2 \sqcup \mathbb{A}^1 \sqcup \mathbb{A}^0.$$

The three cells have real dimensions $4, 2, 0$, exactly as in the usual CW decomposition of projective space.

Example 8.22 (The product of two projective lines)**.** Choose a generic diagonal height on the square. The numbers of outgoing edges at its four vertices are

$$2, \quad 1, \quad 1, \quad 0.$$

Therefore

$$\mathbb{P}^1 \times \mathbb{P}^1 = \mathbb{A}^2 \sqcup \mathbb{A}^1 \sqcup \mathbb{A}^1 \sqcup \mathbb{A}^0.$$

The two one-dimensional cells already foreshadow the two independent degree-two classes coming from the two projections.

Example 8.23 (Cutting off a corner)**.** The moment polygon of $\mathrm{Bl}_p(\mathbb{P}^2)$ is obtained by cutting one corner off a triangle. A generic orientation of the resulting quadrilateral has outgoing-edge counts

$$2, \quad 1, \quad 1, \quad 0.$$

Thus the blowup has the same number of cells in each dimension as $\mathbb{P}^1 \times \mathbb{P}^1$, even though the two toric surfaces have different intersection forms. Additive topology sees the extra degree-two class; the ring structure will distinguish how that class intersects itself.

8.3.4 Odd cohomology vanishes

Because every cell has even real dimension, the cellular cochain complex has no groups in odd degree. There are also no differentials between even-dimensional cells: a cellular differential changes degree by one.

Corollary 8.24 (Betti numbers from the oriented polytope)**.** *For a smooth projective toric variety,*

$$H^{2k+1}(X_P;\mathbb{Z}) = 0 \qquad \text{and} \qquad b_{2k}(X_P) = \#\{v \in \mathrm{Vert}(P) : d_u(v) = k\}.$$

In particular, the integral cohomology groups are free abelian and

$$\chi(X_P) = |\,\mathrm{Vert}(P)| = |\Sigma(n)|.$$

Reference. The vanishing of odd cohomology and the combinatorial formula for the even Betti numbers are Cox–Little–Schenck, Theorems 12.3.11 and 12.3.12. The Euler-characteristic formula is Theorem 12.3.9.

Conceptual proof sketch. Theorem 8.20 supplies one even-dimensional cell for each vertex. Cellular cohomology therefore has no odd groups and no nonzero differentials. A cell $C_v \simeq \mathbb{C}^k$ contributes one generator in degree $2k$. Summing the ranks gives one contribution per vertex, so the Euler characteristic is the number of vertices. Under the orbit–cone dictionary, vertices of P correspond to maximal cones of Σ. □

Observation 8.25 (Why the answer cannot depend on u)*.* Changing the generic height may reverse some arrows and move cells from one fixed point to another, but Corollary 8.24 shows that the number of vertices with k outgoing edges is invariant. The individual cells depend on u; their number in each dimension is a topological invariant of X_P.

8.3.5 The h-vector packages the count

The preceding examples suggest attaching a polynomial to a simple polytope by counting vertices according to the number of outgoing edges. This is one of the standard descriptions of the h-vector.

Definition 8.26 (The h-vector of a simple polytope)**.** Let P be a simple n-polytope and choose a generic height function. Define

$$h_k(P) = \#\{v \in \mathrm{Vert}(P) : d_u(v) = k\}, \qquad 0 \le k \le n.$$

The vector

$$h(P) = (h_0, h_1, \dots, h_n)$$

is independent of the chosen generic height and is called the *h-vector* of P.

For readers who prefer a definition using only face numbers, let f_i denote the number of i-dimensional faces of P, with $f_{-1} = 1$. The same numbers are characterized by

$$\sum_{k=0}^{n} h_k t^{n-k} = \sum_{i=0}^{n} f_{i-1}(t-1)^{n-i}.$$

The oriented-vertex description explains geometrically why these apparently combinatorial coefficients must be nonnegative.

Theorem 8.27 (The h-vector is the Betti vector)**.** *If P is a smooth lattice polytope, then*

$$\boxed{b_{2k}(X_P) = h_k(P), \qquad b_{2k+1}(X_P) = 0.}$$

Equivalently, the Poincaré polynomial is

$$\mathrm{Poin}_{X_P}(q) = \sum_{k=0}^{n} h_k(P) q^{2k}.$$

Reference. This is the smooth projective case of Cox–Little–Schenck, Theorem 12.3.12.

Conceptual proof sketch. By definition, $h_k(P)$ counts precisely the vertices whose attracting cells have complex dimension k. The result is therefore a repackaging of Corollary 8.24. □

Poincaré duality now translates into the symmetry

$$h_k(P) = h_{n-k}(P).$$

Thus the topology of X_P gives a geometric explanation for the Dehn–Sommerville symmetry of the h-vector in the smooth case.

Example 8.28 (Three familiar h-vectors)**.** The cell counts above give

P	$h(P)$	$\mathrm{Poin}_{X_P}(q)$
triangle	$(1,1,1)$	$1 + q^2 + q^4$
square	$(1,2,1)$	$1 + 2q^2 + q^4$
corner-cut triangle	$(1,2,1)$	$1 + 2q^2 + q^4$

The last two rows again show that Betti numbers are coarser than the toric or algebraic geometry.

8.3.6 Toric surfaces and blowups

Every convex r-gon has one vertex with no outgoing edge, one vertex with two outgoing edges, and $r-2$ vertices with one outgoing edge. Indeed, the first two are the maximum and minimum of a generic height; every other vertex has one neighbor above it and one below it. Therefore its h-vector is

$$(1, r-2, 1).$$

Corollary 8.29 (Betti numbers of a smooth complete toric surface)**.** *Let X_Σ be a smooth complete toric surface and let $r = |\Sigma(1)|$. Then*

$$b_0 = b_4 = 1, \qquad b_2 = r - 2, \qquad b_1 = b_3 = 0.$$

Each toric blowup at a fixed point adds one ray to the fan and therefore increases b_2 by one.

Conceptual proof sketch. A complete two-dimensional fan has as many rays as maximal cones, so its moment polygon has r vertices. The oriented polygon count gives one cell in complex dimensions 0 and 2, and $r - 2$ cells in complex dimension 1. A toric blowup star-subdivides one maximal cone, adding one ray and replacing one polygon vertex by an edge. □

Example 8.30 (The first blowup of $\mathbb{P}^2$)**.** The fan of $\mathbb{P}^2$ has three rays, so $b_2 = 1$. Blowing up a torus fixed point adds the exceptional ray and gives

$$b_2(\mathrm{Bl}_p \mathbb{P}^2) = 4 - 2 = 2.$$

One degree-two class is inherited from a line in $\mathbb{P}^2$, while the other is represented by the exceptional divisor. The cell decomposition detects that there are two classes, but it does not yet compute $H^2 = 1$, $E^2 = -1$, or $H \cdot E = 0$. For those products we need the cohomology ring.

Pause and think

Choose a generic height on each polygon below and verify the stated Betti numbers directly from the oriented edges.

1. A triangle gives $(b_0, b_2, b_4) = (1, 1, 1)$.
2. A quadrilateral gives $(1, 2, 1)$.
3. A hexagon gives $(1, 4, 1)$.

Why can no choice of generic height change these counts?

The three dictionaries

	Polytope	One-parameter flow	Topology
Indexing object	vertex v	fixed point x_v	cell C_v
Local number	outgoing edges $d_u(v)$	positive weights at x_v	$\dim_{\mathbb{C}} C_v$
Global count	$h_k(P)$	fixed points with k positive weights	$b_{2k}(X_P)$
Parity	—	affine attracting spaces	$H^{\mathrm{odd}} = 0$

Prediction

The cell decomposition determines the additive groups $H^*(X_P; \mathbb{Z})$, but not their multiplication. The codimension-one cells suggest that degree-two cohomology should be generated by invariant divisors. Which two families of relations among those divisors should come from empty intersections and principal divisors?

Guiding question

Can the fan recover not only the Betti numbers but the entire integral cohomology ring of a smooth complete toric variety?

8.4 The cohomology ring from the fan

The cell decomposition from the previous section determines the additive groups

$$H^k(X_\Sigma;\mathbb{Z}).$$

It does not yet describe the cup product. Two smooth toric surfaces can have the same number of cells in every dimension while their degree-two classes multiply differently.

The fan contains exactly the extra information we need. Its rays index the invariant divisors, its cones record which of those divisors meet, and its lattice vectors record the linear equivalences among them. Those two kinds of data become the two kinds of relations in the cohomology ring.

8.4.1 Invariant divisors as degree-two classes

Let X_Σ be a smooth complete toric variety. For every ray $\rho \in \Sigma(1)$, let

$$D_\rho = V(\rho)$$

be the corresponding invariant prime divisor. Its cohomology class is

$$x_\rho := c_1(\mathcal{O}_{X_\Sigma}(D_\rho)) = \mathrm{PD}[D_\rho] \in H^2(X_\Sigma;\mathbb{Z}).$$

Here PD denotes Poincaré duality. Thus the symbol $[D_\rho]$, when used in cohomology, means the degree-two class Poincaré dual to the divisor rather than merely its class in $\mathrm{Pic}(X_\Sigma)$.

Cup product turns intersections into multiplication. If distinct rays $\rho_1,\dots,\rho_k$ span a cone σ, smoothness implies that the corresponding boundary divisors meet transversely and

$$x_{\rho_1}\cdots x_{\rho_k} = \mathrm{PD}[V(\sigma)].$$

If the rays do not span a cone, the common intersection is empty and the product vanishes. This gives the first family of relations.

Definition 8.31 (The two fan ideals)**.** Let

$$S_\Sigma^{\mathrm{top}} = \mathbb{Z}[x_\rho : \rho \in \Sigma(1)], \qquad \deg x_\rho = 2.$$

The *Stanley–Reisner ideal* is

$$I_\Sigma = \langle x_{\rho_1}\cdots x_{\rho_k} : \rho_1,\dots,\rho_k \text{ do not span a cone of } \Sigma\rangle,$$

where the rays in each generator are distinct.

For $m \in M$, set

$$\theta_m = \sum_{\rho\in\Sigma(1)} \langle m, u_\rho\rangle x_\rho.$$

The *linear-relation ideal* is

$$J_\Sigma = \langle \theta_m : m \in M \rangle.$$

It is enough to use the elements $\theta_{m_1}, \dots, \theta_{m_n}$ coming from a basis of M.

The second ideal is familiar from Chapter 4. Every character has principal divisor

$$\operatorname{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho.$$

A principal divisor defines the trivial line bundle, so its first Chern class vanishes. Therefore

$$\theta_m = \sum_\rho \langle m, u_\rho \rangle x_\rho = 0 \qquad \text{in } H^2(X_\Sigma; \mathbb{Z}).$$

We have now found an evident graded ring homomorphism

$$\Phi : \frac{\mathbb{Z}[x_\rho : \rho \in \Sigma(1)]}{I_\Sigma + J_\Sigma} \longrightarrow H^*(X_\Sigma; \mathbb{Z}), \qquad x_\rho \longmapsto \mathrm{PD}[D_\rho].$$

The remarkable statement is that there are no additional generators or relations.

Theorem 8.32 (Jurkiewicz–Danilov presentation)**.** *Let X_Σ be a smooth complete toric variety. Then*

$$\boxed{H^*(X_\Sigma; \mathbb{Z}) \simeq \frac{\mathbb{Z}[x_\rho : \rho \in \Sigma(1)]}{I_\Sigma + J_\Sigma}, \qquad \deg x_\rho = 2.}$$

Under this isomorphism, x_ρ is the cohomology class of the invariant divisor D_ρ.

Reference. This is Cox–Little–Schenck, Theorem 12.4.4, where it is called the Jurkiewicz–Danilov theorem. The rational presentation for complete simplicial toric varieties is their Theorem 12.4.1.

Conceptual proof sketch. The divisor classes give a map from the polynomial ring to cohomology. Boundary incidence puts I_Σ in its kernel, and principal divisors put J_Σ in its kernel.

The conceptual reason these are all the relations comes from equivariant cohomology. Before forgetting the compact-torus action, the equivariant cohomology ring is the Stanley–Reisner ring

$$H^*_{K_N}(X_\Sigma; \mathbb{Z}) \simeq \frac{\mathbb{Z}[x_\rho]}{I_\Sigma}.$$

The coefficient ring $H^*(BK_N; \mathbb{Z}) \simeq \mathrm{Sym}_{\mathbb{Z}}(M)$ acts, with our principal-divisor convention, by sending $m \in M$ to the linear form $-\theta_m$. Passing from equivariant to ordinary cohomology sets all positive-degree classes from BK_N equal to zero, which imposes exactly the ideal J_Σ.

Smoothness and completeness ensure that this passage introduces no hidden integral torsion or extra relations. In the projective case, the even-dimensional cell decomposition from Section 8.3 is the geometric shadow of this freeness. □

Warning

Both hypotheses matter. For a complete simplicial fan, the analogous presentation holds with rational coefficients, but integral cohomology can be subtler. For a nonsimplicial

singular toric variety, ordinary cohomology need not be generated by divisor classes at all. The integral formula in theorem 8.32 is the smooth complete statement.

8.4.2 Cox variables and cohomology variables

The polynomial ring in theorem 8.32 looks deliberately similar to the Cox ring from Chapter 6. The variables have the same indexing set, but they play different roles.

Dictionary summary

	Cox construction	Cohomology ring
Coefficient ring	$\mathbb{C}$	$\mathbb{Z}$
Variable	coordinate z_ρ cutting out D_ρ	class $x_\rho = \mathrm{PD}[D_\rho]$
Degree	$\deg z_\rho = [D_\rho] \in \mathrm{Cl}(X_\Sigma)$	$\deg x_\rho = 2$
Fan nonfaces	the irrelevant ideal records excluded coordinate strata	I_Σ kills empty divisor intersections
Principal divisors	determine the class-group grading	generate the linear ideal J_Σ
Final operation	remove $V(B(\Sigma))$, then take a quotient	quotient the polynomial ring by $I_\Sigma + J_\Sigma$

Thus the two rings are not the same ring in different notation. The Cox ring uses variables as coordinates before taking a geometric quotient. The cohomology ring uses variables as divisor classes after the variety has been constructed.

8.4.3 A four-step computation

For a smooth complete fan, the presentation can be calculated by a short procedure.

Step 1. List the rays $\rho \in \Sigma(1)$ and introduce one variable x_ρ for each.

Step 2. Find the minimal collections of rays that do not span cones. Their squarefree monomials generate I_Σ.

Step 3. Choose a basis of M. Pair its elements with the primitive ray generators to obtain generators of J_Σ.

Step 4. Use the linear relations to eliminate variables. The remaining monomial relations reveal the cup products.

We now carry out all four steps for the three running examples.

8.4.4 The projective plane

Example 8.33 (The cohomology ring of $\mathbb{P}^2$)**.** Use the primitive ray generators

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_2 = e_2.$$

Every pair spans a cone, but the three rays together do not. Hence

$$I_\Sigma = \langle x_0 x_1 x_2 \rangle.$$

For the dual basis m_1, m_2 of M, the two principal-divisor relations are

$$-x_0 + x_1 = 0, \qquad -x_0 + x_2 = 0.$$

Therefore

$$J_\Sigma = \langle x_1 - x_0,\ x_2 - x_0 \rangle.$$

If H denotes the common class of the three coordinate lines, then

$$x_0 = x_1 = x_2 = H.$$

The Stanley–Reisner relation becomes $H^3 = 0$, and the theorem gives

$$\boxed{H^*(\mathbb{P}^2; \mathbb{Z}) \simeq \frac{\mathbb{Z}[H]}{\langle H^3 \rangle}, \qquad \deg H = 2.}$$

As an abelian group, the ring has basis

$$1, \quad H, \quad H^2,$$

in degrees $0, 2, 4$. The product H^2 is the class of the intersection point of two distinct coordinate lines.

8.4.5 The product of two projective lines

Example 8.34 (The cohomology ring of $\mathbb{P}^1 \times \mathbb{P}^1$)**.** Order the rays cyclically as

$$u_1 = e_1, \qquad u_2 = e_2, \qquad u_3 = -e_1, \qquad u_4 = -e_2.$$

The opposite pairs do not span cones, while every adjacent pair does. Thus

$$I_\Sigma = \langle x_1 x_3,\ x_2 x_4 \rangle.$$

The character relations are

$$x_1 - x_3 = 0, \qquad x_2 - x_4 = 0,$$

so

$$J_\Sigma = \langle x_1 - x_3,\ x_2 - x_4 \rangle.$$

Set

$$A = x_1 = x_3, \qquad B = x_2 = x_4.$$

Geometrically, A and B are the classes of fibers of the two projections. The two nonface relations become

$$A^2 = 0, \qquad B^2 = 0.$$

Consequently,

$$\boxed{H^*(\mathbb{P}^1 \times \mathbb{P}^1; \mathbb{Z}) \simeq \frac{\mathbb{Z}[A, B]}{\langle A^2, B^2 \rangle}, \qquad \deg A = \deg B = 2.}$$

An additive basis is

$$1, \quad A, \quad B, \quad AB.$$

The top class AB is represented by the fixed point where one fiber from each ruling meets. The equations $A^2 = B^2 = 0$ say that two distinct fibers from the same ruling are disjoint.

8.4.6 The blowup of the projective plane

Example 8.35 (The cohomology ring of $\mathrm{Bl}_p(\mathbb{P}^2)$)**.** Blow up the fixed point corresponding to $\mathrm{Cone}(e_1, e_2)$. As in Chapters 4 and 5, order the rays cyclically as

$$u_0 = -e_1 - e_2, \qquad u_1 = e_1, \qquad u_E = e_1 + e_2, \qquad u_2 = e_2.$$

The two nonadjacent pairs are

$$\{\rho_0, \rho_E\} \qquad \text{and} \qquad \{\rho_1, \rho_2\}.$$

Therefore

$$I_\Sigma = \langle x_0 x_E,\ x_1 x_2 \rangle.$$

Pairing the ray generators with the dual basis of M gives

$$-x_0 + x_1 + x_E = 0, \qquad -x_0 + x_E + x_2 = 0.$$

Hence

$$J_\Sigma = \langle -x_0 + x_1 + x_E,\ -x_0 + x_E + x_2 \rangle.$$

Before eliminating variables, the theorem reads

$$H^*(\mathrm{Bl}_p(\mathbb{P}^2); \mathbb{Z}) \simeq \frac{\mathbb{Z}[x_0, x_1, x_E, x_2]}{\langle x_0 x_E,\ x_1 x_2,\ -x_0 + x_1 + x_E,\ -x_0 + x_E + x_2 \rangle}.$$

Now put

$$H = x_0, \qquad E = x_E.$$

The linear relations say

$$x_1 = x_2 = H - E.$$

The two Stanley–Reisner relations become

$$HE = 0, \qquad (H - E)^2 = 0.$$

Since $HE = 0$, the second equation is equivalent to

$$H^2 + E^2 = 0.$$

We obtain the two useful forms of the answer

$$\boxed{H^*(\mathrm{Bl}_p(\mathbb{P}^2); \mathbb{Z}) \simeq \frac{\mathbb{Z}[H, E]}{\langle HE, (H - E)^2 \rangle} \simeq \frac{\mathbb{Z}[H, E]}{\langle HE, H^2 + E^2 \rangle}, \quad \deg H = \deg E = 2.}$$

The classes have their familiar meanings. The divisor D_0 misses the blown-up point, so H is the pullback of the line class. The new divisor is the exceptional curve E, while

$$x_1 = x_2 = H - E$$

are the classes of the strict transforms of the two coordinate lines through the blown-up point.

An additive basis is

$$1, \quad H, \quad E, \quad H^2.$$

Indeed, $E^2 = -H^2$, and every product of three positive-degree generators vanishes. The class $H^2 = x_0x_1$ is represented by a torus fixed point.

Surface laboratory

The square and the corner-cut triangle gave the same Betti numbers:

$$b_0 = b_4 = 1, \qquad b_2 = 2.$$

Their cohomology rings are nevertheless different.
On $\mathbb{P}^1 \times \mathbb{P}^1$, every degree-two class has the form $aA + bB$, and

$$(aA + bB)^2 = 2ab\, AB.$$

Thus every square is an even multiple of the top-degree generator. On $\mathrm{Bl}_p(\mathbb{P}^2)$, however,

$$E^2 = -H^2,$$

an odd multiple of a top-degree generator. Hence the two integral cohomology rings cannot be isomorphic as graded rings.
Additive topology counts the two degree-two classes. Multiplication remembers the parity of their intersection form.

8.4.7 From ring products to intersection numbers

If X_Σ has complex dimension n, then

$$H^{2n}(X_\Sigma; \mathbb{Z}) \simeq \mathbb{Z}.$$

After multiplying n divisor classes, evaluation on the fundamental class produces an integer:

$$\int_{X_\Sigma} x_{\rho_1} \cdots x_{\rho_n} \in \mathbb{Z}.$$

This is the intersection number

$$D_{\rho_1} \cdots D_{\rho_n}.$$

For a smooth maximal cone $\sigma = \mathrm{Cone}(u_{\rho_1}, \ldots, u_{\rho_n})$, the corresponding divisors meet transversely at one fixed point, so

$$\int_{X_\Sigma} x_{\rho_1} \cdots x_{\rho_n} = 1.$$

In the surface examples, this normalizes

$$\int_{\mathbb{P}^2} H^2 = 1, \qquad \int_{\mathbb{P}^1\times\mathbb{P}^1} AB = 1, \qquad \int_{\mathrm{Bl}_p(\mathbb{P}^2)} H^2 = 1.$$

The three dictionaries

Fan	Divisor geometry	Cohomology ring
rays not contained in a cone	empty common intersection	monomial in I_Σ
character pairing with ray generators	principal divisor	linear form in J_Σ
smooth maximal cone	transverse fixed point	top product equal to the point class

Prediction

Let the rays of a smooth complete toric surface be ordered cyclically, and let $x_i = [D_i]$. The Stanley–Reisner relations immediately give

$$x_i x_j = 0$$

for nonadjacent rays, while adjacent rays give the point class. The only products still to determine are the self-intersections

$$x_i^2.$$

How should the lattice relation among u_{i-1}, u_i, u_{i+1} determine this square?

What this chapter has accomplished

Over $\mathbb{C}$, the algebraic torus contains its compact part

$$K_N \simeq (S^1)^n \subseteq T_N \simeq (\mathbb{C}^*)^n.$$

For a projective toric embedding with lattice weights $A = \{m_0, \dots, m_r\}$, normalized squared coordinate magnitudes produce the moment map

$$\mu_A([z_0 : \cdots : z_r]) = \frac{\sum_i |z_i|^2 m_i}{\sum_i |z_i|^2}.$$

Its coefficients form a convex combination, and the positive real part of the toric variety shows that every point of the convex hull is attained:

$$\boxed{\mu_A(X_A) = \operatorname{Conv}(A).}$$

The moment map does more than record an image. Its fibers are exactly the compact-torus orbits. If $p \in \operatorname{Relint}(Q)$ for a face $Q \preceq P$, then

$$\mu_P^{-1}(p) \simeq K_N / K_Q \simeq (S^1)^{\dim Q},$$

where the stabilizer K_Q has dimension $\operatorname{codim} Q$. Consequently,

$$\boxed{X_P / K_N \simeq P.}$$

The polytope is therefore a stratified orbit space: full compact tori sit over its interior, circle directions collapse along facets, and the whole compact torus collapses over a vertex.

A generic vector $u \in N_{\mathbb{R}}$ turns this orbit picture into a cell decomposition. The height function

$$h_u(m) = \langle m, u \rangle$$

orients every edge of P, and the fixed point over a vertex v has an attracting cell

$$C_v \simeq \mathbb{C}^{d_u(v)},$$

where $d_u(v)$ is the number of edges leaving v. All cells have even real dimension, so

$$H^{2k+1}(X_P; \mathbb{Z}) = 0, \qquad b_{2k}(X_P) = h_k(P).$$

In particular,

$$\chi(X_P) = |\operatorname{Vert}(P)| = |\Sigma(n)|.$$

For a smooth complete toric surface with s rays this specializes to

$$b_0 = b_4 = 1, \qquad b_2 = s - 2, \qquad b_1 = b_3 = 0.$$

Finally, the fan recovers not only the additive groups but their multiplication. The invariant divisors generate integral cohomology, and the two familiar families of toric relations give the complete presentation

$$\boxed{H^*(X_\Sigma;\mathbb{Z}) \simeq \frac{\mathbb{Z}[x_\rho : \rho \in \Sigma(1)]}{I_\Sigma + J_\Sigma}, \qquad \deg x_\rho = 2.}$$

The Stanley–Reisner ideal I_Σ records empty intersections, while J_Σ records principal divisors. Thus the same fan data now govern three layers of topology:

faces and compact-torus fibers, cells and Betti numbers, divisor intersections and the cohomology ring.

The challenge exercise completes the surface story by extracting the entire intersection form from a cyclic fan and asking how much of the Hirzebruch parameter survives after the toric structure is forgotten.

Comprehension exercises

8.1. For $\mathbb{P}^1$, use the weights $0, 1 \in \mathbb{Z}$ and show that

$$\mu([z_0 : z_1]) = \frac{|z_1|^2}{|z_0|^2 + |z_1|^2} \in [0, 1].$$

Describe the fiber over an interior point and the fibers over the two endpoints. Write down an explicit representative in the nonnegative real part over each $t \in [0, 1]$.

8.2. For $\mathbb{P}^2$, take the weights

$$(0, 0), \qquad (1, 0), \qquad (0, 1).$$

Write the moment map in homogeneous coordinates. For points in the interior of the triangle, in the relative interior of an edge, and at a vertex, determine:

(a) which homogeneous coordinates are nonzero;

(b) the compact stabilizer;

(c) the dimension of the moment-map fiber.

8.3. Regard $\mathbb{P}^1 \times \mathbb{P}^1$ as the toric variety of the square $[0, 1]^2$. Prove directly from the product formula that

$$\mu_{\mathbb{P}^1 \times \mathbb{P}^1} = (\mu_{\mathbb{P}^1}, \mu_{\mathbb{P}^1}).$$

Choose a path from the square's interior to a vertex that first meets an edge. Describe, in order, which circle directions collapse along the path.

8.4. Let $Q \preceq P$ have codimension k. Using the exact sequence defining K_Q, show that

$$\dim K_Q = k, \qquad \dim(K_N/K_Q) = n - k = \dim Q.$$

Explain why the dimensions of the moment fibers decrease precisely when the image crosses into a proper face.

8.5. Compute the h-vector in each of the following two ways: first by orienting edges with a convenient generic height, and then by using the face-number identity from Section 8.3.

(a) the n-simplex;

(b) the n-cube.

Deduce the Betti numbers of $\mathbb{P}^n$ and $(\mathbb{P}^1)^n$.

8.6. Let P be a smooth lattice hexagon. Choose a generic height and prove that its outgoing-edge counts are

$$2, \quad 1, \quad 1, \quad 1, \quad 1, \quad 0$$

in some order. Determine the integral cohomology groups of X_P as abelian groups and compute its Euler characteristic. How do these answers change after one toric blowup at a fixed point?

8.7. Use the standard fan of $\mathbb{P}^n$ to compute I_Σ and J_Σ. Recover

$$H^*(\mathbb{P}^n;\mathbb{Z}) \simeq \frac{\mathbb{Z}[H]}{\langle H^{n+1}\rangle}, \qquad \deg H = 2.$$

Identify geometrically the class H^k.

8.8. Compute the cohomology ring of $\mathbb{P}^1 \times \mathbb{P}^2$ directly from the product fan. Show that

$$H^*(\mathbb{P}^1 \times \mathbb{P}^2;\mathbb{Z}) \simeq \frac{\mathbb{Z}[A,B]}{\langle A^2, B^3\rangle},$$

and list an additive basis. Verify that its ranks agree with the h-vector of the triangular prism.

8.9. The square and a corner-cut triangle both have h-vector $(1,2,1)$. Explain why their associated smooth toric surfaces have isomorphic cohomology groups as graded abelian groups but nonisomorphic integral cohomology rings. Formulate the extra information supplied by the cup product in terms of self-intersection numbers.

Challenge exercise

Toric surfaces and how much topology remembers.

Let X_Σ be a smooth complete toric surface. Order the rays of its fan counterclockwise,

$$\rho_1, \dots, \rho_s,$$

write $u_i \in N$ for the primitive generator of ρ_i, and let

$$D_i = V(\rho_i), \qquad x_i = [D_i] \in H^2(X_\Sigma; \mathbb{Z}).$$

Indices are taken cyclically. Smoothness implies that each consecutive pair (u_i, u_{i+1}) is an oriented lattice basis. Consequently, for every i there is a unique integer a_i such that

$$\boxed{u_{i-1} + u_{i+1} = a_i u_i.}$$

The first goal is to recover the complete intersection matrix of the invariant curves from these relations. The second is to apply the result to the Hirzebruch surfaces and determine exactly how much of the parameter r survives in integral cohomology.

Hint 1. Distinct invariant curves.
Use the orbit–cone correspondence to show that

$$D_i \cdot D_j = \begin{cases} 1, & \rho_i \text{ and } \rho_j \text{ are adjacent,} \\ 0, & \rho_i \text{ and } \rho_j \text{ are not adjacent.} \end{cases}$$

For adjacent rays, explain why smoothness makes the intersection at the corresponding fixed point transverse. Translate both cases into products of the classes x_i in the Jurkiewicz–Danilov presentation.

Hint 2. Self-intersection from three rays.
Choose $m_i \in M$ dual to the second vector in the ordered basis (u_{i-1}, u_i), so that

$$\langle m_i, u_{i-1} \rangle = 0, \qquad \langle m_i, u_i \rangle = 1.$$

Pair the relation

$$u_{i-1} + u_{i+1} = a_i u_i$$

with m_i, and then multiply the principal-divisor relation

$$\sum_j \langle m_i, u_j \rangle x_j = 0$$

by x_i. Use the nonadjacent Stanley–Reisner relations to make all but two terms vanish. Prove

$$\boxed{D_i^2 = x_i^2 = -a_i.}$$

Conclude that the cyclic list of primitive ray generators determines the entire intersection matrix of the invariant curves.

Hint 3. Four surface checks.
Apply the formula to the following cyclic fans.

(a) For $\mathbb{P}^2$, use

$$(1,0), \quad (0,1), \quad (-1,-1)$$

and show that each invariant line has square 1.

(b) For $\mathbb{P}^1 \times \mathbb{P}^1$, use

$$(1,0), \quad (0,1), \quad (-1,0), \quad (0,-1)$$

and show that all four invariant fibers have square 0.

(c) For $\mathrm{Bl}_p(\mathbb{P}^2)$, use

$$(-1,-1), \quad (1,0), \quad (1,1), \quad (0,1).$$

Identify $H = D_{(-1,-1)}$ and $E = D_{(1,1)}$, and recover

$$H^2 = 1, \qquad E^2 = -1, \qquad H \cdot E = 0.$$

(d) For the Hirzebruch surface $\mathbb{F}_r$, $r \geq 0$, use

$$u_1 = (1,0), \quad u_2 = (0,1), \quad u_3 = (-1,r), \quad u_4 = (0,-1).$$

Show that

$$D_1^2 = D_3^2 = 0, \qquad D_2^2 = -r, \qquad D_4^2 = r.$$

Hint 4. The cohomology ring of $\mathbb{F}_r$.
Introduce variables $x_1, \ldots, x_4$ for the four rays. Show that

$$I_\Sigma = \langle x_1 x_3, x_2 x_4 \rangle$$

and that the character relations are

$$x_1 - x_3 = 0, \qquad x_2 + r x_3 - x_4 = 0.$$

Set

$$F = x_1 = x_3, \qquad S = x_2, \qquad x_4 = S + rF.$$

Deduce

$$\boxed{H^*(\mathbb{F}_r;\mathbb{Z}) \simeq \frac{\mathbb{Z}[S,F]}{\langle F^2, S^2 + rSF \rangle}, \qquad \deg S = \deg F = 2.}$$

Normalize the point class by $S \cdot F = 1$, and reconcile the ring relations with

$$F^2 = 0, \qquad S^2 = -r.$$

Hint 5. Changing basis: the even case.
Suppose $r = 2k$ and define

$$S' = S + kF.$$

Compute $(S')^2$ and $S'F$. Show that the basis (S', F) has the same intersection form as the two ruling classes on $\mathbb{F}_0 \simeq \mathbb{P}^1 \times \mathbb{P}^1$. Construct an explicit graded ring isomorphism

$$H^*(\mathbb{F}_{2k}; \mathbb{Z}) \simeq H^*(\mathbb{F}_0; \mathbb{Z}).$$

Hint 6. Changing basis: the odd case.
Suppose $r = 2k + 1$ and again set $S' = S + kF$. Show that

$$(S')^2 = -S'F, \qquad F^2 = 0.$$

Construct an explicit graded ring isomorphism

$$H^*(\mathbb{F}_{2k+1}; \mathbb{Z}) \simeq H^*(\mathbb{F}_1; \mathbb{Z}).$$

Hint 7. Why parity cannot disappear.
For $\alpha = aS + bF \in H^2(\mathbb{F}_r; \mathbb{Z})$, prove

$$\alpha^2 = (-ra^2 + 2ab)[\mathrm{pt}].$$

Show that when r is even, every degree-two integral class has even self-intersection, whereas when r is odd, some integral class has odd self-intersection. Deduce the complete classification

$$\boxed{H^*(\mathbb{F}_r; \mathbb{Z}) \simeq H^*(\mathbb{F}_t; \mathbb{Z}) \quad \Longleftrightarrow \quad r \equiv t \pmod 2.}$$

Hint 8. Fan, ring, and polygon.
On $\mathbb{P}^1 \times \mathbb{P}^1$, let A and B be the two ruling classes. Compute $A \cdot B$ in three ways:

(a) from adjacency of the corresponding rays;

(b) from multiplication in $\mathbb{Z}[A, B]/\langle A^2, B^2 \rangle$;

(c) from polygon geometry.

For the last method, take P_A and P_B to be unit lattice segments in the two coordinate directions. Use

$$P_{A+B} = P_A + P_B$$

and polarize the identity $D^2 = 2\,\mathrm{Area}(P_D)$ to obtain

$$D \cdot E = \mathrm{Area}(P_D + P_E) - \mathrm{Area}(P_D) - \mathrm{Area}(P_E)$$

for nef divisors in this setting. Verify that the unit square gives $A \cdot B = 1$.
Finally, explain why the area formula cannot be applied directly to the exceptional curve E on $\mathrm{Bl}_p(\mathbb{P}^2)$, even though the fan and cohomology-ring calculations still give $E^2 = -1$.

The three dictionaries

Fan	**Cohomology ring**	**Polygon geometry**
adjacent rays	product is the point class	transverse boundary meeting
three-ray relation	self-intersection $D_i^2 = -a_i$	change of neighboring edge directions
principal divisors	integral change of basis	translate supporting inequalities
Minkowski sum	polarized cup product	mixed area

The fans distinguish every $\mathbb{F}_r$, but their integral cohomology rings distinguish only even from odd. Which information in the parameter r belongs to the chosen toric structure, which is visible to algebraic geometry, and which survives in the topology of the underlying manifold?

Notes and references

The projective moment map, the nonnegative part of a toric variety, and the compact-torus quotient are developed in Cox–Little–Schenck, §12.2. In particular, Proposition 12.2.3, Theorem 12.2.5, and Exercises 12.2.8–12.2.9 give the ingredients used in Sections 8.1 and 8.2.

The orbit filtration, Euler characteristic, vanishing of odd cohomology, and the h-vector formula appear in Cox–Little–Schenck, §12.3; see especially Theorems 12.3.9, 12.3.11, and 12.3.12. The attracting-cell language used here is the toric form of the Białynicki–Birula decomposition.

Cox–Little–Schenck, Theorem 12.4.4, gives the integral cohomology-ring presentation for smooth complete toric varieties; this is the Jurkiewicz–Danilov theorem. The rational version for complete simplicial toric varieties is Theorem 12.4.1. Our treatment emphasizes the parallel between the Stanley–Reisner and principal-divisor relations in cohomology and their earlier roles in the Cox construction.

The surface calculation in the challenge is a direct application of that presentation. The identity

$$u_{i-1} + u_{i+1} = a_i u_i \quad \Longrightarrow \quad D_i^2 = -a_i$$

turns a cyclic fan into its intersection matrix. For Hirzebruch surfaces, an integral change of basis removes all information except the parity of r from the cohomology ring. The fan retains the full integer because it remembers the distinguished torus action and its invariant boundary, not merely the topology of the underlying four-manifold.

Chapter 9

Localization and Toric Riemann–Roch

Chapter overview

Chapter 8 showed that the fan of a smooth complete toric variety determines its cohomology ring. We now ask what we can *do* with that ring. The answer will be unexpectedly arithmetic: characteristic classes in cohomology compute Euler characteristics of line bundles, and on a projective toric variety those Euler characteristics become counts of lattice points.

There is a second idea running through the chapter. Ordinary cohomology forgets the torus action, even though that action is the source of most of the geometry. Equivariant cohomology remembers it. For a smooth toric variety it has a strikingly combinatorial model as piecewise polynomial functions on the fan, and localization reduces global integrals to contributions from torus fixed points.

The chapter follows the progression

Chern character and Todd class $\longrightarrow$ piecewise polynomial equivariant classes $\longrightarrow$ fixed-point localization $\longrightarrow$ toric Riemann–Roch and lattice points.

Throughout, varieties are defined over $\mathbb{C}$. Unless stated otherwise, the toric varieties in this chapter are smooth and complete.

The three dictionaries

The same toric data will now appear in three new roles.

Fan and polytope	Equivariant topology	Index theory
ray ρ and divisor D_ρ	degree-two class	factor in $\mathrm{Td}(X)$
polynomial on each cone	equivariant cohomology class	fixed-point restriction
maximal cone σ	torus fixed point x_σ	localization contribution
lattice points of P_D	weights of global sections	Euler characteristic of $\mathcal{O}(D)$

Surface laboratory

On a smooth complete toric surface, the theory developed in this chapter will collapse to an especially concrete formula. For a divisor D,

$$\chi(X, \mathcal{O}_X(D)) = 1 + \frac{1}{2} D \cdot (D - K_X).$$

Thus the intersection form recovered from the fan in the challenge at the end of Chapter 8 already contains enough information to compute Euler characteristics of every line bundle. The first section will derive this formula; the rest of the chapter will explain why the same phenomenon in every dimension can be localized at the fixed points and, for ample divisors, read as lattice-point counting.

9.1 Chern characters, Todd classes, and Riemann–Roch

The cohomology ring from Chapter 8 records intersections of divisor classes. Riemann–Roch asks us to package those classes in a slightly different way. Given a line bundle L, we want to recover the alternating sum

$$\chi(X, L) := \sum_{i=0}^{\dim X} (-1)^i \dim_{\mathbb{C}} H^i(X, L)$$

from cohomological data on X.

Only three ingredients will be needed: Chern classes, the Chern character, and the Todd class. We introduce only the parts of their general theory that enter the toric calculation.

9.1.1 The tangent bundle from the boundary divisors

Let E be a complex vector bundle of rank r. Its Chern classes are classes

$$c_i(E) \in H^{2i}(X; \mathbb{Z}), \qquad c(E) = 1 + c_1(E) + \cdots + c_r(E).$$

For us, the two essential rules are

$$c(L) = 1 + c_1(L)$$

for a line bundle L, and the Whitney product formula

$$c(E) = c(E')c(E'')$$

for a short exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0.$$

If D is a Cartier divisor, then

$$c_1(\mathcal{O}_X(D)) = [D].$$

For a smooth toric variety, there is a generalized Euler sequence that plays the role of the familiar Euler sequence on projective space. If $r = |\Sigma(1)|$ and $n = \dim X_\Sigma$, its tangent-bundle form is

$$0 \longrightarrow \mathcal{O}_{X_\Sigma}^{r-n} \longrightarrow \bigoplus_{\rho \in \Sigma(1)} \mathcal{O}_{X_\Sigma}(D_\rho) \longrightarrow T_{X_\Sigma} \longrightarrow 0.$$

It is the dual of Cox–Little–Schenck, Theorem 8.1.6. We need only this consequence of the sequence here.

The trivial bundle on the left has total Chern class 1, so the Whitney formula immediately makes the Chern classes combinatorial.

Proposition 9.1 (Chern class of a smooth toric variety)**.** *Let X_Σ be smooth and complete. Then*

$$\boxed{c(T_{X_\Sigma}) = \prod_{\rho \in \Sigma(1)} (1 + [D_\rho]).}$$

In particular,

$$c_1(T_{X_\Sigma}) = \sum_\rho [D_\rho] = [-K_{X_\Sigma}],$$

and, if $n = \dim X_\Sigma$,

$$c_n(T_{X_\Sigma}) = |\Sigma(n)|[\mathrm{pt}].$$

Reference. This is Cox–Little–Schenck, Proposition 13.1.2. Their Example 13.1.1 derives the same formula from the cotangent form of the generalized Euler sequence.

Conceptual proof sketch. Apply the Whitney product formula to the tangent Euler sequence. Each summand $\mathcal{O}(D_\rho)$ contributes $1 + [D_\rho]$, while the trivial bundle contributes nothing. The degree-two part of the product is $\sum_\rho [D_\rho] = [-K_X]$.

For the top class, expand the product in the cohomology ring from Chapter 8. A product of n distinct invariant-divisor classes is the class of a point precisely when the corresponding rays form a maximal cone, and is zero otherwise. Thus every maximal cone contributes one copy of [pt]. □

The last statement reconnects immediately with Chapter 8:

$$\int_X c_n(T_X) = |\Sigma(n)| = \chi_{\mathrm{top}}(X).$$

The topological Euler characteristic counted fixed points. The holomorphic Euler characteristics we want next require a different combination of the same Chern data.

9.1.2 The Chern character is an exponential

For a line bundle there is almost nothing to remember.

Definition 9.2 (Chern character of a line bundle)**.** Let L be a line bundle on a smooth complete variety. Its *Chern character* is

$$\mathrm{ch}(L) := e^{c_1(L)} = 1 + c_1(L) + \frac{c_1(L)^2}{2!} + \frac{c_1(L)^3}{3!} + \cdots \in H^*(X; \mathbb{Q}).$$

The apparent infinite series is actually finite on an n-dimensional variety: every term above cohomological degree $2n$ vanishes. Thus, for $L = \mathcal{O}_X(D)$,

$$\boxed{\mathrm{ch}(\mathcal{O}_X(D)) = e^{[D]}.}$$

The exponential is exactly the right operation because tensor product of line bundles becomes multiplication:

$$\mathrm{ch}(L_1 \otimes L_2) = \mathrm{ch}(L_1)\,\mathrm{ch}(L_2).$$

Remark 9.3. The denominators are the first reason rational cohomology naturally appears in Riemann–Roch. Nothing is being completed analytically: all power series in this chapter are truncated by dimension.

9.1.3 The Todd class is the correction factor

Suppose formally that the Chern classes of T_X factor as

$$c(T_X) = \prod_{i=1}^{n} (1 + \xi_i).$$

The symbols ξ_i are called the *Chern roots*. The splitting principle allows us to calculate with them as though the tangent bundle were a sum of line bundles. Define

$$Q(x) := \frac{x}{1 - e^{-x}} = 1 + \frac{x}{2} + \frac{x^2}{12} - \frac{x^4}{720} + \cdots .$$

Definition 9.4 (Todd class)**.** For a smooth complete n-fold X, the *Todd class* is

$$\mathrm{Td}(X) := \prod_{i=1}^{n} Q(\xi_i) = \prod_{i=1}^{n} \frac{\xi_i}{1 - e^{-\xi_i}} \in H^*(X; \mathbb{Q}).$$

The first terms, written without Chern roots, are

$$\mathrm{Td}(X) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1 c_2 + \cdots .$$

As always, only terms through degree $2 \dim X$ matter.

For a toric variety the formal Chern roots can disappear entirely.

Theorem 9.5 (The toric Todd class)**.** *Let X_Σ be a smooth complete toric variety. Then*

$$\boxed{\mathrm{Td}(X_\Sigma) = \prod_{\rho \in \Sigma(1)} \frac{[D_\rho]}{1 - e^{-[D_\rho]}}.}$$

The right-hand side is interpreted through the power series $Q(x)$, so every factor has constant term 1*.*

Reference. This is Cox–Little–Schenck, Theorem 13.1.6.

Conceptual proof sketch. Characteristic classes defined by a power series are multiplicative for exact sequences. Apply this to the tangent Euler sequence. The trivial summands have Todd class 1, while $\mathcal{O}(D_\rho)$ contributes

$$Q([D_\rho]) = \frac{[D_\rho]}{1 - e^{-[D_\rho]}}.$$

Multiplying over the rays gives the formula. Equivalently, theorem 9.1 says that the divisor classes behave, for symmetric characteristic-class calculations, like a redundant list of Chern roots. □

This is the first major payoff of the toric boundary. A characteristic class of the tangent bundle has become a product indexed by the rays of the fan, and every factor lives in the cohomology ring we already know how to compute.

9.1.4 Hirzebruch–Riemann–Roch

We can now state the theorem that motivates the chapter. If $\alpha \in H^*(X;\mathbb{Q})$ and $\dim X = n$, write

$$\int_X \alpha$$

for evaluation of the degree-$2n$ part of α on the fundamental class of X. In particular,

$$\int_X [\mathrm{pt}] = 1.$$

Theorem 9.6 (Hirzebruch–Riemann–Roch for toric line bundles)**.** *Let $X = X_\Sigma$ be a smooth complete toric variety and let L be a line bundle on X. Then*

$$\boxed{\chi(X,L) = \int_X \mathrm{ch}(L)\,\mathrm{Td}(X).}$$

For $L = \mathcal{O}_X(D)$, this becomes

$$\boxed{\chi(X,\mathcal{O}_X(D)) = \int_X e^{[D]} \prod_{\rho\in\Sigma(1)} \frac{[D_\rho]}{1-e^{-[D_\rho]}}.}$$

Reference. The general HRR formula is equation (13.0.1) in Cox–Little–Schenck. The line-bundle statement for smooth complete toric varieties is their Theorem 13.3.8, obtained from equivariant Riemann–Roch by taking the nonequivariant limit.

Conceptual proof sketch. We will not reproduce the general proof of Riemann–Roch. Its message is that the Chern character converts the algebra of a line bundle into cohomology, while the Todd class corrects for the geometry of the tangent bundle. Their product has a top-degree component, and integration of that component recovers the alternating sum of sheaf cohomology dimensions.

For toric varieties there is a more concrete mechanism behind this global statement. In Sections 9.2 and 9.3 we will retain the torus action in equivariant cohomology and then use localization to reduce an integral to fixed-point contributions. Taking the nonequivariant limit of that calculation is precisely the route used in CLS to prove the toric statement above. □

A four-step computation. For a smooth complete toric variety, HRR has become a calculation in the ring from Section 8.4:

1. write $[D]$ and the invariant classes $[D_\rho]$ in $H^*(X;\mathbb{Q})$;
2. expand $e^{[D]}$ only through degree $2\dim X$;
3. expand each Todd factor $1 + \frac{1}{2}[D_\rho] + \frac{1}{12}[D_\rho]^2 + \cdots$;
4. use the fan relations and keep only the top-degree coefficient.

That coefficient is $\chi(X, \mathcal{O}_X(D))$.

9.1.5 The first calculation: line bundles on $\mathbb{P}^1$

Let $H \in H^2(\mathbb{P}^1; \mathbb{Z})$ be the class of a point divisor. From Chapter 8,

$$H^*(\mathbb{P}^1; \mathbb{Z}) \simeq \frac{\mathbb{Z}[H]}{\langle H^2 \rangle}, \qquad \int_{\mathbb{P}^1} H = 1.$$

There are two invariant divisors, both of class H. Since $H^2 = 0$, the Todd product truncates immediately:

$$\mathrm{Td}(\mathbb{P}^1) = \left(1 + \frac{1}{2}H\right)^2 = 1 + H.$$

For $L = \mathcal{O}_{\mathbb{P}^1}(d)$,

$$\mathrm{ch}(L) = e^{dH} = 1 + dH.$$

Therefore

$$\begin{aligned} \chi\big(\mathbb{P}^1, \mathcal{O}(d)\big) &= \int_{\mathbb{P}^1} (1 + dH)(1 + H) \\ &= \int_{\mathbb{P}^1} (d + 1)H = d + 1. \end{aligned}$$

Notice that HRR computes the *Euler characteristic*, not merely the number of global sections. When $d \geq 0$, higher cohomology vanishes and this says

$$h^0\big(\mathbb{P}^1, \mathcal{O}(d)\big) = d + 1.$$

For $d = -2$, however, $\chi(\mathcal{O}(-2)) = -1$: the negative sign records a nonzero H^1, not a negative-dimensional space of sections.

9.1.6 Surfaces: Riemann–Roch becomes intersection theory

Let X be a smooth complete surface. Since $c_1(T_X) = -K_X$,

$$\mathrm{Td}(X) = 1 - \frac{1}{2}K_X + \frac{1}{12}(K_X^2 + c_2(X)),$$

while

$$\mathrm{ch}\big(\mathcal{O}_X(D)\big) = 1 + D + \frac{1}{2}D^2.$$

Only cohomological degree four survives integration. HRR therefore gives

$$\chi(X, \mathcal{O}_X(D)) = \frac{1}{2}D \cdot (D - K_X) + \frac{1}{12}\int_X (K_X^2 + c_2(X)).$$

Applying the same formula to $D = 0$ identifies the last term with $\chi(X, \mathcal{O}_X)$. Hence

$$\boxed{\chi(X, \mathcal{O}_X(D)) = \chi(X, \mathcal{O}_X) + \frac{1}{2}D \cdot (D - K_X).}$$

This is the familiar surface Riemann–Roch formula. For a smooth complete toric variety,

$$H^i(X, \mathcal{O}_X) = 0 \qquad (i > 0),$$

so $\chi(X, \mathcal{O}_X) = 1$. We recover the formula announced at the beginning of the chapter:

$$\boxed{\chi(X, \mathcal{O}_X(D)) = 1 + \frac{1}{2} D \cdot (D - K_X).}$$

Reference. The surface calculation is Cox–Little–Schenck, Example 13.1.7 and Exercise 13.1.2; see their equation (13.1.7).

9.1.7 A triangle already knows Riemann–Roch

Take $X = \mathbb{P}^2$ and let H be the hyperplane class. We know

$$H^*(\mathbb{P}^2; \mathbb{Z}) \simeq \frac{\mathbb{Z}[H]}{\langle H^3 \rangle}, \qquad \int_{\mathbb{P}^2} H^2 = 1.$$

The three invariant divisors all represent H, so

$$\begin{aligned} \mathrm{Td}(\mathbb{P}^2) &= \left(1 + \frac{1}{2}H + \frac{1}{12}H^2\right)^3 \\ &= 1 + \frac{3}{2}H + H^2. \end{aligned}$$

For $L = \mathcal{O}_{\mathbb{P}^2}(d)$,

$$\mathrm{ch}(L) = 1 + dH + \frac{d^2}{2}H^2.$$

Multiplying and taking the coefficient of H^2,

$$\begin{aligned} \chi\big(\mathbb{P}^2, \mathcal{O}(d)\big) &= 1 + \frac{3}{2}d + \frac{1}{2}d^2 \\ &= \frac{(d+1)(d+2)}{2}. \end{aligned}$$

For $d \geq 0$, higher cohomology vanishes, and the last number is also

$$\#\big(d\Delta_2 \cap \mathbb{Z}^2\big),$$

the number of lattice points in the degree-d triangle. At the moment this is a familiar coincidence: monomials of degree d already explain the same count. By the end of the chapter, it will be the model for a general toric Riemann–Roch principle.

Dictionary summary

Section 8.4 turned fan data into a cohomology ring. The present section has added two universal operations inside that ring:

$$[D] \longmapsto e^{[D]}, \qquad \{[D_\rho]\}_\rho \longmapsto \prod_\rho \frac{[D_\rho]}{1 - e^{-[D_\rho]}}.$$

Their top-degree product is an integer: the Euler characteristic of a line bundle. What is still hidden is the torus action itself. The next section keeps that information and turns cohomology classes into polynomials living directly on the fan.

Guiding question

Ordinary cohomology has forgotten the torus action. Is there a version of cohomology in which a class can be read cone by cone from the fan, and whose restrictions to maximal cones remember the torus fixed points?

9.2 Equivariant cohomology and piecewise polynomials

There is a small paradox in the cohomology ring from Chapter 8. The torus action organizes nearly everything we do with a toric variety, yet the ordinary cohomology ring does not remember the action itself. Equivariant cohomology repairs this by keeping track of a space *together with* its torus action.

For the rest of this section we write

$$K_N = \mathrm{Hom}_{\mathbb{Z}}(M, S^1) \simeq (S^1)^n$$

for the compact torus from Section 8.1. The algebraic torus $T_N \simeq (\mathbb{C}^*)^n$ deformation retracts onto K_N, so this is the same topological equivariant theory that Cox–Little–Schenck denote by $H^*_T(-)$. We use rational coefficients in the main statements; for a smooth fan the same combinatorics also has an integral version.

9.2.1 The Borel construction: remembering the action

If a compact group K acts freely on a space X, the quotient X/K is often the right space to study. A torus action on a toric variety is very far from free: fixed points are central to the story. The Borel construction replaces the action by an auxiliary free one.

Choose a contractible space EK on which K acts freely, and put

$$BK := EK/K.$$

The diagonal action on $EK \times X$ is free, so we may form

$$EK \times_K X := (EK \times X)/K.$$

Definition 9.7 (Equivariant cohomology)**.** The *K-equivariant cohomology* of a K-space X is

$$H_K^*(X;\mathbb{Q}) := H^*(EK \times_K X;\mathbb{Q}).$$

Algebraic geometry cheatsheet

For an algebraic geometer, the easiest way to remember equivariant cohomology is as a cohomology theory for a space *together with an action*. Write

$$X_K := EK \times_K X$$

for the Borel construction, sometimes called the *homotopy quotient*. It is designed to satisfy the following basic tests. An equivariant map $f : X \to Y$ induces a map $X_K \to Y_K$, so $H_K^*(-)$ is a contravariant functor on K-spaces, just as ordinary cohomology is a contravariant functor on spaces.

1. **Free actions give ordinary quotients.** If K acts freely on X, then

$$X_K \simeq X/K, \qquad H_K^*(X) \simeq H^*(X/K).$$

Thus equivariant cohomology really does extend the familiar operation of taking the cohomology of a geometric quotient.

2. **A trivial action gives a tensor product.** If K acts trivially on X, then

$$X_K = BK \times X,$$

so, over $\mathbb{Q}$, the Künneth theorem gives

$$H_K^*(X;\mathbb{Q}) \simeq H^*(X;\mathbb{Q}) \otimes_{\mathbb{Q}} H_K^*(\mathrm{pt};\mathbb{Q}).$$

The second factor is the extra cohomology contributed by the group.

3. **Equivariant cohomology is a module over the cohomology of a point.** The projection $X_K \to BK$ makes

$$H_K^*(X;\mathbb{Q})$$

an algebra, and in particular a module, over

$$H_K^*(\mathrm{pt};\mathbb{Q}) = H^*(BK;\mathbb{Q}).$$

For a torus this coefficient ring is a polynomial ring. Its variables are the *equivariant parameters* that keep track of the action.

4. **There is a nonequivariant limit.** The fibration

$$X \longrightarrow X_K \longrightarrow BK$$

gives a forgetful map $H^*_K(X) \to H^*(X)$. Algebraically, forgetting the action means sending every positive-degree class in $H^*(BK)$ to zero. When X is *equivariantly formal*—in particular, when $H^{\mathrm{odd}}(X;\mathbb{Q}) = 0$—this gives

$$H^*(X;\mathbb{Q}) \simeq H^*_K(X;\mathbb{Q}) \otimes_{H^*(BK;\mathbb{Q})} \mathbb{Q},$$

where $H^*(BK;\mathbb{Q}) \to \mathbb{Q}$ kills all positive-degree elements.
Smooth complete toric varieties are equivariantly formal. In that case, as a module over $H^*(BK;\mathbb{Q})$, one may think additively of

$$H^*_K(X;\mathbb{Q}) \simeq H^*(X;\mathbb{Q}) \otimes_{\mathbb{Q}} H^*(BK;\mathbb{Q}).$$

This module isomorphism is not canonical in general, and the tensor product formula should *not* be taken as a property of every group action.

These four tests are usually enough intuition for the calculations in this chapter: equivariant cohomology interpolates between the cohomology of a quotient and the ordinary cohomology of X, while remembering the group action through the coefficient ring $H^*(BK)$.

Reference. This is Cox–Little–Schenck, Definition 12.4.6. Their equations (12.4.7)–(12.4.9) compute the coefficient ring for a torus.

Example 9.8 (The coefficient ring for S^1)**.** Let S^1 act on the infinite-dimensional unit sphere $S^\infty \subset \mathbb{C}^\infty$ by scalar multiplication. This action is free, and S^∞ is contractible, so one may take

$$ES^1 = S^\infty, \qquad BS^1 = \mathbb{P}^\infty,$$

where $\mathbb{P}^\infty$ is the direct limit of the finite-dimensional projective spaces. Since

$$H^*(BS^1;\mathbb{Z}) \simeq \mathbb{Z}[t], \qquad t = c_1\big(\mathcal{O}_{\mathbb{P}^\infty}(1)\big), \qquad \deg t = 2,$$

we obtain

$$\boxed{H^*_{S^1}(\mathrm{pt};\mathbb{Z}) \simeq \mathbb{Z}[t].}$$

The finite-dimensional approximations are already familiar to an algebraic geometer:

$$S^{2r+1}/S^1 = \mathbb{P}^r.$$

As r grows, their cohomology stabilizes in every fixed degree and produces the polynomial ring above.

Example 9.9 (The algebraic torus)**.** Now let

$$T_N = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^*) \simeq (\mathbb{C}^*)^n$$

be the algebraic torus. Over $\mathbb{C}$, it deformation retracts onto its maximal compact torus

$$K_N = \mathrm{Hom}_{\mathbb{Z}}(M, S^1) \simeq (S^1)^n.$$

Consequently their classifying spaces have the same homotopy type, and taking products of the preceding example gives

$$BT_N \simeq BK_N \simeq (\mathbb{P}^\infty)^n.$$

More intrinsically, every character $m \in M$ gives a homomorphism $T_N \to \mathbb{C}^*$, hence a map of classifying spaces to $B\mathbb{C}^* \simeq \mathbb{P}^\infty$. Pulling back the universal degree-two class identifies the character lattice with the degree-two part of the coefficient ring. Thus

$$\boxed{H^*_{T_N}(\mathrm{pt};\mathbb{Z}) \simeq H^*_{K_N}(\mathrm{pt};\mathbb{Z}) \simeq \mathrm{Sym}_{\mathbb{Z}}(M),}$$

where elements of M have cohomological degree 2. After tensoring with $\mathbb{Q}$, we will write

$$S := \mathrm{Sym}_{\mathbb{Q}}(M_{\mathbb{Q}}).$$

If a basis $m_1, \dots, m_n$ of M is chosen, this simply reads

$$S \simeq \mathbb{Q}[t_1, \dots, t_n], \qquad \deg t_i = 2.$$

The same approximation can be written entirely in algebraic geometry. After choosing coordinates on T_N, let it act factor by factor on

$$U_r := \left(\mathbb{A}^{r+1} \setminus \{0\}\right)^n.$$

The action is free and

$$U_r/T_N \simeq (\mathbb{P}^r)^n.$$

As $r \to \infty$, these quotients approximate BT_N in increasing cohomological degrees. Thus the topological classifying space above is already visible through a familiar sequence of algebraic quotients.

Thus equivariant cohomology is not merely a graded $\mathbb{Q}$-algebra: it is naturally an S-algebra. The extra variables in S are the parameters that remember the torus action.

Pause and think

Test the cheatsheet on S^1 itself. If S^1 acts on itself by rotation, the action is free, so what is $H^*_{S^1}(S^1;\mathbb{Q})$? What does it become if the action is trivial instead? The underlying space has not changed; only the action has. Where does equivariant cohomology record the difference?

9.2.2 What a toric equivariant class looks like

Now take a smooth complete fan $\Sigma \subseteq N_{\mathbb{R}}$. Every maximal cone $\sigma \in \Sigma(n)$ corresponds to a torus fixed point x_σ. Restriction to the fixed points gives a map

$$H^*_{K_N}(X_\Sigma;\mathbb{Q}) \longrightarrow \bigoplus_{\sigma\in\Sigma(n)} H^*_{K_N}(x_\sigma;\mathbb{Q}) \simeq \bigoplus_{\sigma\in\Sigma(n)} S.$$

So an equivariant class determines one polynomial $f_\sigma \in S$ at each maximal cone. Those polynomials cannot be chosen independently: neighboring fixed points are joined by one-dimensional orbits, and this forces their polynomials to agree on the common face.

This is exactly the gluing rule for piecewise polynomial functions.

Definition 9.10 (Piecewise polynomial functions on a fan)**.** Let $PP^*(\Sigma;\mathbb{Q})$ be the ring of functions

$$f : |\Sigma| \longrightarrow \mathbb{R}$$

whose restriction to every cone is given by a polynomial with rational coefficients. Equivalently, for a complete fan it is the ring of collections

$$\{f_\sigma \in S : \sigma \in \Sigma(n)\}$$

such that

$$f_\sigma|_{\sigma\cap\sigma'} = f_{\sigma'}|_{\sigma\cap\sigma'}$$

for every pair of maximal cones. A polynomial of ordinary degree k is placed in cohomological degree $2k$.

For instance, degree-two equivariant classes correspond to rational functions that are linear on each cone. The integral classes recover precisely the integral support functions that first appeared in Chapter 4.

Theorem 9.11 (Equivariant cohomology is piecewise polynomial)**.** *Let X_Σ be a smooth complete toric variety. Then restriction to the torus fixed points induces an isomorphism of S-algebras*

$$\boxed{H^*_{K_N}(X_\Sigma;\mathbb{Q}) \simeq PP^*(\Sigma;\mathbb{Q}).}$$

The same ring also has the Stanley–Reisner presentation

$$\boxed{H^*_{K_N}(X_\Sigma;\mathbb{Q}) \simeq \frac{\mathbb{Q}[x_\rho : \rho \in \Sigma(1)]}{I_\Sigma}, \qquad \deg x_\rho = 2.}$$

Here x_ρ is the equivariant class of D_ρ.

Reference. The Stanley–Reisner presentation is Cox–Little–Schenck, Theorem 12.4.14. The piecewise-polynomial interpretation is the discussion following their exact sequence (12.4.18); see especially Example 12.4.18 for $\mathbb{P}^2$. They explain immediately before that discussion that for a smooth fan the same argument may be carried out integrally.

Conceptual proof sketch. On the affine chart U_σ of a maximal cone, the torus action equivariantly retracts onto the fixed point x_σ. Thus an equivariant class restricts there to an element of $H^*_{K_N}(x_\sigma) \simeq S$, giving the polynomial f_σ. On an overlap, the two restrictions come from the same global class, so they agree on the common face.

The nontrivial direction is that these compatibility conditions are enough. The Chang–Skjelbred exact sequence used in CLS says, in this toric setting, that equivariant cohomology is determined by the fixed points and the one-dimensional orbits joining them. Hence every compatible collection glues to an equivariant class. Describing the same gluing with one generator for each invariant divisor gives the Stanley–Reisner presentation: nonfaces give exactly the monomial relations in I_Σ. □

This theorem explains something that was only foreshadowed in Section 8.4. Ordinary cohomology required *two* ideals, $I_\Sigma + J_\Sigma$. Equivariant cohomology requires only I_Σ. The linear relations have not vanished; they have become the coefficient-ring variables that remember the action.

There is one sign convention worth recording. With our convention

$$\operatorname{div}(\chi^m) = \sum_\rho \langle m, u_\rho \rangle D_\rho,$$

the S-algebra structure on the Stanley–Reisner presentation sends

$$m \longmapsto -\theta_m = -\sum_\rho \langle m, u_\rho \rangle x_\rho.$$

This is Cox–Little–Schenck's convention and is compatible with the support-function convention from Chapter 4. The sign will disappear when we pass to ordinary cohomology.

9.2.3 The first picture: $\mathbb{P}^1$

Let $N = M = \mathbb{Z}$, and write t for the degree-two generator of

$$S = \mathbb{Q}[t].$$

The fan of $\mathbb{P}^1$ has two maximal cones

$$\sigma_+ = \mathbb{R}_{\geq 0}, \qquad \sigma_- = \mathbb{R}_{\leq 0},$$

meeting at the origin. A piecewise polynomial is therefore a pair $(f_+, f_-) \in S^2$ whose values agree at 0. Algebraically,

$$f_+ - f_- \in (t).$$

Hence

$$\boxed{H^*_{K_N}(\mathbb{P}^1; \mathbb{Q}) \simeq \{(f_+, f_-) \in \mathbb{Q}[t]^2 : f_+ - f_- \in (t)\}.}$$

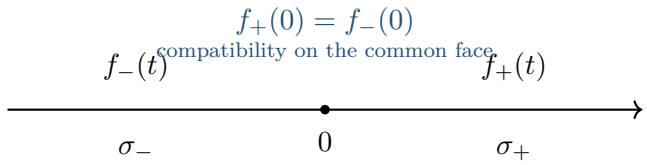


Figure 9.1: An equivariant class on $\mathbb{P}^1$ is a polynomial on each ray of the fan, with the two polynomials agreeing where the rays meet.

This description can be rewritten as a familiar quotient ring. Let

$$\eta := (0, t).$$

Every compatible pair is uniquely $a(t)(1,1) + b(t)\eta$, and

$$\eta^2 = t\eta.$$

Therefore

$$H^*_{K_N}(\mathbb{P}^1;\mathbb{Q}) \simeq \frac{\mathbb{Q}[t,\eta]}{\langle \eta(\eta - t)\rangle}.$$

The relation has not said that η is nilpotent: the parameter t records the torus weight. Nilpotence appears only after the torus parameters are forgotten.

9.2.4 Degree two: our old support functions return

The piecewise-polynomial theorem has a particularly satisfying degree- two interpretation. Let

$$D = \sum_\rho a_\rho D_\rho$$

be an invariant Cartier divisor with Cartier data $\{m_\sigma\}$. Recall from Section 4.2 that its support function is

$$\psi_D|_\sigma = \langle m_\sigma, -\rangle, \qquad \psi_D(u_\rho) = -a_\rho.$$

The divisor also has an equivariant first Chern class

$$[D]_{K_N} \in H^2_{K_N}(X_\Sigma;\mathbb{Q}).$$

At the fixed point x_σ, its restriction is precisely the linear form m_σ. Thus under theorem 9.11,

$$\boxed{[D]_{K_N} \longleftrightarrow \psi_D.}$$

Reference. The equivariant divisor class and its restriction properties are Cox–Little–Schenck, Proposition 12.4.13. Combining that proposition with the Cartier data from their Theorem 4.2.8 gives the identification above.

This is not a new analogy: it is the same object seen in two subjects. The linear piece m_σ that gives a local equation for the divisor is also the weight seen by equivariant cohomology at the fixed point x_σ.

The three dictionaries

For an invariant Cartier divisor D, the following data are three descriptions of the same local information.

Algebraic geometry	Fan	Equivariant topology
local equation χ^{-m_σ}	linear piece $\psi_D\|_\sigma = m_\sigma$	restriction $[D]_{K_N}\|_{x_\sigma} = m_\sigma$
Cartier compatibility	pieces agree on faces	fixed-point restrictions glue
principal character	global linear function	class from $H^*(BK_N)$

9.2.5 Forgetting the action

The inclusion of a fiber

$$X_\Sigma \hookrightarrow EK_N \times_{K_N} X_\Sigma$$

induces the *nonequivariant limit*

$$H^*_{K_N}(X_\Sigma;\mathbb{Q}) \longrightarrow H^*(X_\Sigma;\mathbb{Q}).$$

On the coefficient ring this is the augmentation

$$S \longrightarrow \mathbb{Q}, \qquad m \longmapsto 0 \quad (m \in M).$$

For a smooth complete toric variety,

$$\boxed{H^*(X_\Sigma;\mathbb{Q}) \simeq H^*_{K_N}(X_\Sigma;\mathbb{Q}) \otimes_S \mathbb{Q} \simeq \frac{PP^*(\Sigma;\mathbb{Q})}{S_+\, PP^*(\Sigma;\mathbb{Q})},}$$

where S_+ is the ideal of positive-degree global polynomials.

Reference. This is Cox–Little–Schenck, Proposition 12.4.7 and Corollary 12.4.8. Conceptually, the relevant Serre spectral sequence degenerates because a complete simplicial toric variety has no odd rational cohomology.

In the Stanley–Reisner presentation, setting every $m \in M$ to zero imposes

$$\theta_m = \sum_\rho \langle m, u_\rho\rangle x_\rho = 0.$$

Thus the ideal J_Σ from Chapter 8 reappears automatically:

$$\frac{\mathbb{Q}[x_\rho]}{I_\Sigma} \quad\longrightarrow\quad \frac{\mathbb{Q}[x_\rho]}{I_\Sigma + J_\Sigma}.$$

In degree two, the same statement says

$$\boxed{\text{piecewise linear functions/global linear functions } \simeq\ H^2(X_\Sigma;\mathbb{Q}),}$$

which is the support-function description of divisor classes in yet another guise.

For $\mathbb{P}^1$, the calculation above makes this visible at once:

$$\frac{\mathbb{Q}[t,\eta]}{\langle \eta(\eta - t)\rangle} \otimes_{\mathbb{Q}[t]} \mathbb{Q} \simeq \frac{\mathbb{Q}[\eta]}{\langle \eta^2\rangle} = H^*(\mathbb{P}^1;\mathbb{Q}).$$

Dictionary summary

The difference between the two cohomology rings can now be summarized without any new generators.

	Equivariant	**Ordinary**
coefficient ring	$S = \mathrm{Sym}_{\mathbb{Q}}(M_{\mathbb{Q}})$	$\mathbb{Q}$
fan presentation	$\mathbb{Q}[x_\rho]/I_\Sigma$	$\mathbb{Q}[x_\rho]/(I_\Sigma + J_\Sigma)$
fan functions	piecewise polynomials	quotient by global positive-degree polynomials
degree two	piecewise linear functions	piecewise linear modulo global linear

9.2.6 A full two-dimensional example: $\mathbb{P}^2$

Let the rays of the fan of $\mathbb{P}^2$ be generated by

$$u_0 = (-1,-1), \qquad u_1 = (1,0), \qquad u_2 = (0,1),$$

and let

$$\sigma_0 = \mathrm{Cone}(u_1, u_2), \qquad \sigma_1 = \mathrm{Cone}(u_2, u_0), \qquad \sigma_2 = \mathrm{Cone}(u_0, u_1).$$

Write $S = \mathbb{Q}[x, y]$, where x, y are the dual coordinates on $N_{\mathbb{R}}$.

An equivariant class is a triple

$$(f_0, f_1, f_2) \in S^3.$$

The three common rays impose three congruences. Since x vanishes on ρ_2, y on ρ_1, and $y - x$ on ρ_0, we obtain

$$\boxed{\begin{aligned} &f_1 - f_0 \in (x),\\ &f_2 - f_0 \in (y),\\ &f_2 - f_1 \in (y - x). \end{aligned}}$$

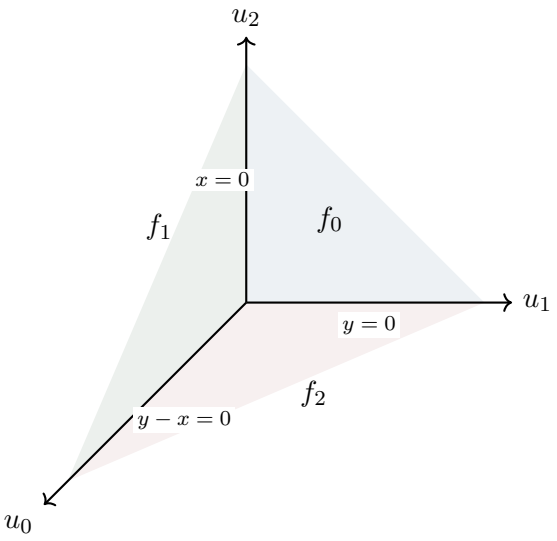


Figure 9.2: Piecewise polynomial data on the fan of $\mathbb{P}^2$. Across each common ray, the difference of the adjacent polynomials is divisible by the linear form vanishing on that ray.

Let us solve the compatibility conditions rather than merely state them. The first two give

$$f_1 = f_0 + xg, \qquad f_2 = f_0 + yh$$

for some $g, h \in S$. The third condition says

$$yh - xg \in (y - x).$$

Setting $y = x$ shows $x(h - g)|_{y=x} = 0$, hence

$$h - g \in (y - x).$$

Write $h = g + (y - x)k$. Then

$$\boxed{(f_0, f_1, f_2) = f_0(1, 1, 1) + g(0, x, y) + k(0, 0, y(y - x)).}$$

Thus $H^*_{K_N}(\mathbb{P}^2; \mathbb{Q})$ is a free S-module with one generator in cohomological degrees $0, 2, 4$. This is the equivariant shadow of the three ordinary cohomology classes

$$1, \ H, \ H^2$$

that we used in Section 9.1.

Reference. This calculation is Cox–Little–Schenck, Example 12.4.18. We have spelled out the divisibility step because it is the simplest place to see fixed-point compatibility become an actual computation.

Warning

Taking the nonequivariant limit is a quotient of the *piecewise-polynomial ring* by global positive-degree polynomials. It is not the operation of evaluating each fixed-point polynomial separately at $x = y = 0$. For example, the class $(0, x, y)$ above survives as the hyperplane class after the nonequivariant quotient even though all three displayed polynomials have zero constant term.

We have now reached the form of equivariant cohomology needed for localization. A class is a compatible polynomial at every maximal cone, hence at every torus fixed point. The next question is whether a *global integral* can also be reconstructed from these local restrictions.

Guiding question

If an equivariant class is determined by its values at the finitely many fixed points x_σ, can integration over all of X_Σ also be expressed as a sum of fixed-point contributions? What must appear in the denominator to compensate for replacing an n-dimensional variety by isolated points?

9.3 Fixed points and Atiyah–Bott localization

Section 9.2 turned an equivariant cohomology class into a collection of polynomials

$$\alpha \longleftrightarrow \left(\alpha|_{x_\sigma}\right)_{\sigma\in\Sigma(n)}$$

indexed by the torus fixed points. Localization says something much stronger: after allowing ourselves to divide by the torus weights, *integration itself* can be carried out on this finite set.

The denominator has a simple geometric meaning. A point has lost all of the tangent directions of X, so its contribution must be divided by the equivariant Euler class of those missing directions. For a toric variety, those tangent directions are read directly from the maximal cone.

9.3.1 The tangent weights at a fixed point

Let $\sigma \in \Sigma(n)$ be a maximal cone of a smooth fan and write

$$\sigma = \mathrm{Cone}(u_{\sigma,1}, \dots, u_{\sigma,n}).$$

Smoothness says that the primitive ray generators $u_{\sigma,1}, \dots, u_{\sigma,n}$ form a basis of N. Let

$$m_{\sigma,1}, \dots, m_{\sigma,n} \in M$$

be the dual basis. Equivalently, these are the primitive generators of the dual cone $\sigma^\vee$.

The affine chart U_σ is isomorphic to $\mathbb{C}^n$, the point x_σ is its origin, and the characters determined by the $m_{\sigma,i}$ describe the n coordinate directions. Thus the fan does not merely tell us where the fixed points are: it tells us the torus representation on the tangent space at each one.

Proposition 9.12 (The equivariant Euler class at a toric fixed point)**.** *Let X_Σ be smooth and complete and let $x_\sigma \in X_\Sigma^{K_N}$ correspond to a maximal cone σ. With the sign convention for equivariant divisor classes used in Section 9.2,*

$$\boxed{e_{K_N}(T_{x_\sigma}X_\Sigma) = (-1)^n \prod_{i=1}^{n} m_{\sigma,i} \in S = \mathrm{Sym}_{\mathbb{Q}}(M_{\mathbb{Q}}).}$$

In particular, this class becomes invertible after we invert the nonzero linear forms in S.

Reference. This is the fixed-point calculation behind Cox–Little–Schenck, Lemma 13.3.5 and Proposition 13.3.6. The interpretation of the denominator as an equivariant Euler class is made explicit immediately after their Theorem 13.3.7 and in Exercise 13.3.2.

Conceptual proof sketch. The invariant divisors corresponding to the rays of σ meet transversely at x_σ. Their equivariant normal directions therefore give the n tangent directions at the fixed point. The restriction calculation from Section 9.2 gives

$$[D_{\rho_i}]_{K_N}|_{x_\sigma} = -m_{\sigma,i}.$$

The equivariant Euler class is the product of these degree-two classes, which gives the displayed formula. This is exactly where the factor $(-1)^n$ in our localization formula will come from. □

Pause and think

There are two pieces of fan data at x_σ: the value $\alpha|_{x_\sigma}$, which Section 9.2 records as the polynomial on σ, and the dual basis $m_{\sigma,1}, \dots, m_{\sigma,n}$, which records the tangent directions. Localization will use exactly these two pieces and no others.

9.3.2 Equivariant integration

Ordinary integration over a smooth complete n-fold is the pushforward to a point,

$$\int_X : H^k(X;\mathbb{Q}) \longrightarrow H^{k-2n}(\mathrm{pt};\mathbb{Q}).$$

There is an equivariant version of the same pushforward,

$$\int_X^{K_N} : H^k_{K_N}(X;\mathbb{Q}) \longrightarrow H^{k-2n}_{K_N}(\mathrm{pt};\mathbb{Q}) = S^{k-2n}.$$

It remembers the equivariant parameters rather than setting them equal to zero. In top degree the answer lies in $S^0 = \mathbb{Q}$, so for an equivariant lift of an ordinary top-degree class it agrees with the ordinary integral after forgetting the action.

To state localization, let

$$S_{\mathrm{loc}} := S\big[\ell^{-1} : 0 \neq \ell \in M_{\mathbb{Q}}\big] \subseteq \mathrm{Frac}(S).$$

Thus we are allowed to divide by nonzero torus weights.

Theorem 9.13 (Atiyah–Bott localization)**.** *Let X_Σ be a smooth complete n-dimensional toric variety. For every $\alpha \in H^*_{K_N}(X_\Sigma;\mathbb{Q})$,*

$$\boxed{\int_{X_\Sigma}^{K_N} \alpha = \sum_{\sigma\in\Sigma(n)} \frac{\alpha|_{x_\sigma}}{e_{K_N}(T_{x_\sigma}X_\Sigma)} = (-1)^n \sum_{\sigma\in\Sigma(n)} \frac{\alpha|_{x_\sigma}}{\prod_{i=1}^n m_{\sigma,i}}.}$$

The equality is computed in S_{loc}, but the sum belongs to S: all apparent poles cancel.

Reference. This is Cox–Little–Schenck, Theorem 13.3.7, which is the toric form of the Atiyah–Bott/Berline–Vergne localization formula. The localization of the fixed-point restriction map used in the proof is their equation (13.3.3), following the discussion of the localization theorem on p. 643.

Conceptual proof sketch. Restriction to the fixed points gives

$$H^*_{K_N}(X_\Sigma) \longrightarrow \bigoplus_{\sigma\in\Sigma(n)} S.$$

Section 9.2 explained why its image consists of compatible polynomial data. After the nonzero weights are inverted, the localization theorem says that this restriction map becomes an

isomorphism. In other words, the fixed points become a complete basis for equivariant cohomology once fractions in the weights are allowed.

The class supported at x_σ has restriction $e_{K_N}(T_{x_\sigma}X)$ at that fixed point and zero at the others. Dividing by the Euler class therefore normalizes it to have value one at x_σ. Decompose α into these normalized fixed-point classes and push each one to a point. The result is exactly the sum in the theorem. □

At first sight the formula looks perverse. The left side is a polynomial, while the individual terms on the right are usually rational functions. The cancellation of the denominators is not an accident: it is the algebraic shadow of the fact that the fixed-point data came from one global equivariant class.

There is also a useful degree check. If $\deg \alpha < 2n$, then the equivariant pushforward would have negative degree and hence must vanish. Localization therefore manufactures nontrivial identities among rational functions of the torus weights.

9.3.3 The first localization calculation: $\mathbb{P}^1$

Return to the notation of Section 9.2. We have

$$S = \mathbb{Q}[t], \qquad \sigma_+ = \mathbb{R}_{\geq 0}, \qquad \sigma_- = \mathbb{R}_{\leq 0},$$

and an equivariant class is a pair

$$(f_+, f_-), \qquad f_+ - f_- \in (t).$$

The dual primitive generators are

$$m_+ = t, \qquad m_- = -t.$$

Since $n = 1$, Proposition 9.12 gives

$$e_{K_N}(T_{x_+}\mathbb{P}^1) = -t, \qquad e_{K_N}(T_{x_-}\mathbb{P}^1) = t.$$

Therefore localization is simply

$$\boxed{\int_{\mathbb{P}^1}^{K_N} (f_+, f_-) = -\frac{f_+}{t} + \frac{f_-}{t} = \frac{f_- - f_+}{t}.}$$

The gluing condition $f_+ - f_- \in (t)$ is precisely what makes the answer a polynomial.

For the unit class,

$$\int_{\mathbb{P}^1}^{K_N} 1 = -\frac{1}{t} + \frac{1}{t} = 0,$$

as it must by degree. More interestingly, recall the equivariant divisor class

$$\eta = (0, t), \qquad H^*_{K_N}(\mathbb{P}^1; \mathbb{Q}) \simeq \frac{\mathbb{Q}[t, \eta]}{\langle \eta(\eta - t) \rangle}.$$

Then

$$\boxed{\int_{\mathbb{P}^1}^{K_N} \eta = 0 + \frac{t}{t} = 1.}$$

After forgetting the equivariant parameter, η becomes the point class H, and we recover the familiar identity

$$\int_{\mathbb{P}^1} H = 1.$$

This example is worth remembering as a miniature of the entire theory: two local rational expressions, each meaningless at $t = 0$ on its own, combine to give a perfectly regular global answer.

9.3.4 A two-dimensional calculation: $\mathbb{P}^2$

Now use exactly the fan notation from the end of Section 9.2:

$$u_0 = (-1, -1), \qquad u_1 = (1, 0), \qquad u_2 = (0, 1),$$

with maximal cones $\sigma_0, \sigma_1, \sigma_2$ and $S = \mathbb{Q}[x, y]$. Taking dual bases in the three cones gives the following fixed-point data.

Cone	**Dual generators**	**Euler class**
$\sigma_0 = \mathrm{Cone}(u_1, u_2)$	$x,\ y$	xy
$\sigma_1 = \mathrm{Cone}(u_2, u_0)$	$y - x,\ -x$	$x(x - y)$
$\sigma_2 = \mathrm{Cone}(u_0, u_1)$	$-y,\ x - y$	$y(y - x)$

Already the unit class produces an identity:

$$\begin{aligned}\int_{\mathbb{P}^2}^{K_N} 1 &= \frac{1}{xy} + \frac{1}{x(x-y)} + \frac{1}{y(y-x)} \\ &= 0.\end{aligned}$$

Again this is forced by degree, but localization turns that simple topological fact into a cancellation among three rational functions.

Now recall the degree-two class from the piecewise-polynomial calculation of Section 9.2,

$$h_T = (0, x, y).$$

Its nonequivariant limit is the hyperplane class H. Since $h_T^2 = (0, x^2, y^2)$, localization gives

$$\begin{aligned}\int_{\mathbb{P}^2}^{K_N} h_T^2 &= \frac{0}{xy} + \frac{x^2}{x(x-y)} + \frac{y^2}{y(y-x)} \\ &= \frac{x}{x-y} + \frac{y}{y-x} = 1.\end{aligned}$$

Therefore, after taking the nonequivariant limit,

$$\boxed{\int_{\mathbb{P}^2} H^2 = 1.}$$

Notice what we did *not* do: we never multiplied divisor classes in the ordinary cohomology ring. The same intersection number was recovered solely from the three fixed-point restrictions of an equivariant class and the tangent weights encoded by the fan.

Dictionary summary

Localization turns the fan description of Section 9.2 into an integration algorithm.

	Global object	**Fixed-point data**
equivariant class	$\alpha \in H^*_{K_N}(X_\Sigma)$	polynomial $\alpha\|_{x_\sigma}$ on each maximal cone
tangent geometry	TX_Σ	$e_{K_N}(T_{x_\sigma}X) = (-1)^n \prod_i m_{\sigma,i}$
integration	$\displaystyle\int_X^{K_N} \alpha$	$\displaystyle\sum_\sigma \frac{\alpha\|_{x_\sigma}}{e_{K_N}(T_{x_\sigma}X)}$

9.3.5 Why this is exactly the tool Riemann–Roch needs

We can already see the shape of the next section. Let

$$D = \sum_\rho a_\rho D_\rho$$

be an invariant Cartier divisor with Cartier data $\{m_\sigma\}_{\sigma\in\Sigma(n)}$. Section 9.2 identified the restriction of its equivariant divisor class as

$$[D]_{K_N}|_{x_\sigma} = m_\sigma.$$

Hence the equivariant Chern character restricts to

$$\mathrm{ch}_{K_N}(\mathcal{O}(D))|_{x_\sigma} = e^{m_\sigma}.$$

At the same fixed point, the equivariant Chern roots of the tangent space are, in our convention,

$$-m_{\sigma,1}, \dots, -m_{\sigma,n}.$$

Consequently

$$\frac{\mathrm{Td}_{K_N}(X)|_{x_\sigma}}{e_{K_N}(T_{x_\sigma}X)} = \prod_{i=1}^n \frac{1}{1-e^{m_{\sigma,i}}}.$$

Putting the two restrictions into localization gives the formal fixed-point expression

$$\boxed{\int_X^{K_N} \mathrm{ch}_{K_N}(\mathcal{O}(D))\,\mathrm{Td}_{K_N}(X) = \sum_{\sigma\in\Sigma(n)} \frac{e^{m_\sigma}}{\prod_{i=1}^n (1-e^{m_{\sigma,i}})}.}$$

Here the exponentials are interpreted in the completed equivariant cohomology ring; no analytic convergence is involved.

Reference. This is the local calculation used in Cox–Little–Schenck, Theorems 13.3.4 and 13.3.7, in their proof of equivariant Riemann–Roch. We have isolated it here because it shows exactly why localization is the correct bridge between the fan and the Riemann–Roch integral.

The remarkable feature is the denominator. The Euler-class factor from localization cancels the numerator in the Todd factor, leaving $1 - e^{m_{\sigma,i}}$. These are precisely the geometric-series denominators one expects when counting lattice points in a cone. We will not develop Brion's theorem separately; in the next section we will use just this observation to reach the chapter's intended punchline.

Guiding question

The left side of the last formula is an equivariant characteristic class integral, while the right side resembles a sum of generating functions for lattice points near the vertices of a polytope. Why should its nonequivariant limit be the Euler characteristic of $\mathcal{O}(D)$, and when does that Euler characteristic equal $\#(P_D \cap M)$?

9.4 Toric Riemann–Roch and lattice points

We can now answer the guiding question with almost no new machinery. There are three quantities that have appeared at different moments in the notes:

$$\chi(X, \mathcal{O}_X(D)), \qquad \int_X e^{[D]} \operatorname{Td}(X), \qquad \#(P_D \cap M).$$

The first is sheaf cohomology, the second is topology, and the third is combinatorics. For a nef toric divisor on a smooth complete toric variety, all three are the same number.

The equality between the first two is Hirzebruch–Riemann–Roch. To identify the first with the third, we need only recall two facts that we have already encountered in toric geometry: lattice points give a basis of global sections, and a nef toric divisor has no higher cohomology.

9.4.1 The equivariant Euler characteristic remembers the weights

Before forgetting the torus action, it is useful to keep track not only of the dimensions of the cohomology groups but also of their characters. If a torus T_N acts on a line bundle L, define formally

$$\chi_{T_N}(X, L) = \sum_{i \geq 0} (-1)^i \operatorname{ch}_{T_N} H^i(X, L).$$

Thus the ordinary Euler characteristic remembers only the total dimension, while the equivariant Euler characteristic remembers the weight of every torus eigenspace.

For an invariant divisor D, the monomial basis of global sections is

$$H^0(X_\Sigma, \mathcal{O}(D)) = \bigoplus_{m \in P_D \cap M} \mathbb{C}\, \chi^m.$$

If D is nef and X_Σ is complete, Demazure vanishing gives

$$H^i(X_\Sigma, \mathcal{O}(D)) = 0 \qquad (i > 0).$$

Consequently the equivariant Euler characteristic becomes the finite character sum

$$\boxed{\chi_{T_N}\big(X_\Sigma, \mathcal{O}(D)\big) = \sum_{m \in P_D \cap M} e^m.}$$

Here e^m is notation for the character of weight m, written in the exponential notation used by the equivariant Chern character.

References. The monomial basis of global sections is Cox–Little–Schenck, Proposition 4.3.3. The higher-cohomology statement is their Theorem 9.2.3 (Demazure vanishing). Completeness makes $|\Sigma| = N_{\mathbb{R}}$ convex, so the hypotheses of that theorem are automatic once D is nef.

Now compare this finite sum with the expression at the end of Section 9.3. Equivariant Riemann–Roch and localization give

$$\chi_{T_N}\big(X_\Sigma, \mathcal{O}(D)\big) = \sum_{\sigma \in \Sigma(n)} \frac{e^{m_\sigma}}{\prod_{i=1}^{n} (1 - e^{m_{\sigma,i}})}.$$

Combining the two descriptions gives the striking identity

$$\boxed{\sum_{m \in P_D \cap M} e^m = \sum_{\sigma \in \Sigma(n)} \frac{e^{m_\sigma}}{\prod_{i=1}^{n} (1 - e^{m_{\sigma,i}})}.} \tag{9.1}$$

When D is ample, each term on the right has a direct polyhedral meaning. The Cartier datum m_σ is the vertex of P_D corresponding to the maximal cone σ, and the vectors $m_{\sigma,1}, \dots, m_{\sigma,n}$ generate the tangent cone of P_D at that vertex. Formally expanding geometric series gives

$$\frac{e^{m_\sigma}}{\prod_i (1 - e^{m_{\sigma,i}})} = e^{m_\sigma} \prod_i (1 + e^{m_{\sigma,i}} + e^{2m_{\sigma,i}} + \cdots).$$

Thus a fixed-point contribution is the generating function of lattice points in the tangent cone at one vertex. The infinite tails cancel when the vertex contributions are added, leaving the finite generating function of the polytope itself.

Pause and think

This is the precise bridge promised at the end of Section 9.3:

$$\boxed{\begin{array}{c}\text{torus fixed point} \longleftrightarrow \text{vertex of } P_D, \\ \text{tangent weights} \longleftrightarrow \text{edge directions at that vertex.}\end{array}}$$

We are seeing one of Brion's identities, but we do not need a separate theory of rational generating functions here. For us its role is to explain geometrically why the localization denominators look exactly like geometric-series denominators.

9.4.2 The one-dimensional identity in full

Let us watch the cancellation happen on $\mathbb{P}^1$. Write the rays of the fan as

$$\sigma_+ = \mathbb{R}_{\geq 0}, \qquad \sigma_- = \mathbb{R}_{\leq 0},$$

and take

$$D = dD_-, \qquad d \geq 0,$$

where D_- is the invariant divisor belonging to the ray generated by -1. The associated polytope is

$$P_D = [0, d],$$

and its Cartier data are

$$m_+ = 0, \qquad m_- = dt.$$

The primitive generators of the two dual cones are t and $-t$, respectively. Therefore the right side of (9.1) is

$$\frac{1}{1-e^t} + \frac{e^{dt}}{1-e^{-t}}.$$

Put $q = e^t$. Then

$$\begin{aligned} \frac{1}{1-q} + \frac{q^d}{1-q^{-1}} &= \frac{1}{1-q} - \frac{q^{d+1}}{1-q} \\ &= \frac{1-q^{d+1}}{1-q} \\ &= 1 + q + \cdots + q^d. \end{aligned}$$

In other words,

$$\boxed{\frac{1}{1-e^t} + \frac{e^{dt}}{1-e^{-t}} = 1 + e^t + \cdots + e^{dt}.}$$

The left side is a sum of two infinite-cone contributions, one from each fixed point. The right side contains exactly the $d+1$ lattice points of the interval $[0, d]$. Finally forgetting the torus action means sending every character to its dimension, or informally taking $e^t \to 1$. Although the two fixed-point fractions individually have poles there, their sum has the perfectly regular limit

$$\chi(\mathbb{P}^1, \mathcal{O}(d)) = d + 1.$$

Reference. This is the one-dimensional instance of the calculation in Cox–Little–Schenck, Example 13.3.9. Their Theorem 13.3.4 gives the general fixed-point character decomposition.

9.4.3 The toric Riemann–Roch counting formula

We can now forget the equivariant parameters and state the numerical payoff of the chapter.

Theorem 9.14 (Toric Riemann–Roch as lattice-point counting)**.** *Let X_Σ be a smooth complete toric variety and let D be a nef invariant Cartier divisor. Then*

$$\boxed{\#(P_D \cap M) = \chi(X_\Sigma, \mathcal{O}(D)) = \int_{X_\Sigma} e^{[D]} \prod_{\rho \in \Sigma(1)} \frac{[D_\rho]}{1 - e^{-[D_\rho]}}.}$$

In particular, the formula holds when D is ample.

References. This is the direct combination of Cox–Little–Schenck, Proposition 4.3.3, Theorem 9.2.3, Theorem 13.1.6, and Theorem 13.3.8. The last result is their toric Hirzebruch–Riemann–Roch theorem.

Conceptual proof. Hirzebruch–Riemann–Roch gives

$$\chi(X_\Sigma, \mathcal{O}(D)) = \int_{X_\Sigma} e^{[D]} \operatorname{Td}(X_\Sigma).$$

The toric Euler sequence turns the Todd class into the displayed product over invariant divisors. Since D is nef, Demazure vanishing kills every higher cohomology group, so

$$\chi(X_\Sigma, \mathcal{O}(D)) = h^0(X_\Sigma, \mathcal{O}(D)).$$

Finally, the characters χ^m, for $m \in P_D \cap M$, form a basis of global sections. Hence

$$h^0(X_\Sigma, \mathcal{O}(D)) = \#(P_D \cap M),$$

as claimed. □

There is a useful distinction in the theorem. Riemann–Roch itself always computes the Euler characteristic of a line bundle on our smooth complete toric variety. The positivity hypothesis is used only in the second step, when we identify that alternating sum with h^0 and hence with an honest lattice-point count.

Warning

Do not silently replace $\chi(X, \mathcal{O}(D))$ by $h^0(X, \mathcal{O}(D))$ for an arbitrary divisor. For negative divisors the higher cohomology can be nonzero. The example $\chi(\mathbb{P}^1, \mathcal{O}(-2)) = -1$ from Section 9.1 is the simplest warning: an Euler characteristic can be negative, while a number of lattice points cannot.

9.4.4 A corner cut: two computations on $\mathrm{Bl}_p(\mathbb{P}^2)$

Let

$$X = \mathrm{Bl}_p(\mathbb{P}^2)$$

at a torus fixed point, with the notation from Section 8.4. Thus

$$H^*(X; \mathbb{Z}) \simeq \frac{\mathbb{Z}[H, E]}{\langle HE, H^2 + E^2 \rangle}, \qquad \int_X H^2 = 1,$$

where H is the pullback of a line and E is the exceptional curve. The canonical class is

$$K_X = -3H + E.$$

Take

$$D = dH - mE, \qquad d \geq m \geq 0.$$

This divisor is nef. In the ray notation from Section 8.4 it can be represented by

$$D = dD_0 - mD_E,$$

so its polytope is cut out by

$$x \geq 0, \qquad y \geq 0, \qquad m \leq x + y \leq d.$$

It is the degree-d triangle with the corner $x + y < m$ removed.

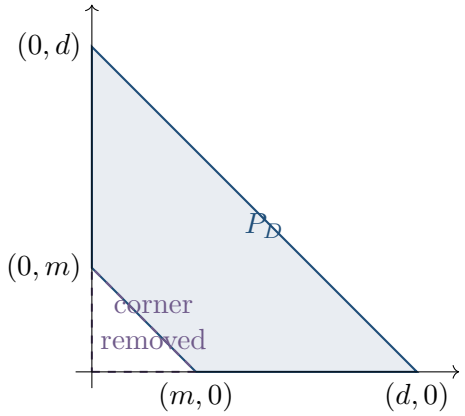


Figure 9.3: The polytope of $D = dH - mE$ is obtained from the degree-d triangle by cutting off the lattice corner of size m.

First compute by Riemann–Roch. From the intersection relations,

$$D^2 = d^2 - m^2,$$

and

$$\begin{aligned} D \cdot (D - K_X) &= (dH - mE)\big((d+3)H - (m+1)E\big) \\ &= d(d+3) - m(m+1). \end{aligned}$$

The surface formula from Section 9.1 therefore gives

$$\begin{aligned} \chi(X, \mathcal{O}_X(dH - mE)) &= 1 + \frac{1}{2}\big[d(d+3) - m(m+1)\big] \\ &= \binom{d+2}{2} - \binom{m+1}{2}. \end{aligned}$$

Now compute from the polygon. The full degree-d triangle has

$$\binom{d+2}{2}$$

lattice points. The discarded lattice points satisfy $x, y \geq 0$ and $x + y < m$. There are

$$1 + 2 + \cdots + m = \binom{m+1}{2}$$

of them. Hence

$$\boxed{\#(P_D \cap M) = \binom{d+2}{2} - \binom{m+1}{2} = \chi(X, \mathcal{O}_X(dH - mE)).}$$

The calculation has a familiar algebraic interpretation as well: sections of $\mathcal{O}_{\mathbb{P}^2}(d)$ are degree-d monomials, and passing to $dH - mE$ imposes vanishing of order at least m at the blown-up point. The monomials removed by that vanishing condition are exactly the lattice points in the cut-off corner.

The three dictionaries

The corner-cut calculation is the chapter in miniature.

Algebraic geometry	Topology	Polytope
$\mathcal{O}_X(dH - mE)$	$dH - mE \in H^2(X)$	$m \leq x + y \leq d$
vanish to order m	exceptional class $E^2 = -1$	cut off a corner of size m
Euler characteristic	HRR integral	number of lattice points

9.4.5 Dilating the polytope

There is one final consequence worth recording. If D is ample and $P = P_D$, then kD corresponds to the dilated polytope kP. Applying Theorem 9.14 to every integer $k \geq 0$ gives

$$\boxed{\#(kP \cap M) = \int_X e^{k[D]} \operatorname{Td}(X).}$$

Because cohomology vanishes above degree $2n$, the exponential on the right truncates. The right side is therefore a polynomial in k of degree at most n. Thus, for the smooth lattice polytopes arising here, Riemann–Roch gives a geometric explanation for the polynomial behavior of lattice-point counts under dilation.

We will not develop the general theory of Ehrhart polynomials or the volume-polynomial and Khovanskii–Pukhlikov refinements here. The important lesson for these notes is already visible: characteristic classes convert the geometry of a toric variety into arithmetic data of its polytope.

The punchline

For a smooth projective toric variety, the same integer can now be computed in three genuinely different ways:

$$\begin{array}{rl} \text{sheaf theory:} & \chi(X, \mathcal{O}(D)), \\ \text{cohomology:} & \displaystyle\int_X e^{[D]} \operatorname{Td}(X), \\ \text{combinatorics:} & \#(P_D \cap M). \end{array}$$

Equivariant cohomology and localization explain why the bridge between the last two is controlled one fixed point—equivalently, one vertex of the polytope—at a time.

What this chapter has accomplished

Chapter 8 gave us the cohomology ring of a smooth complete toric variety. The first lesson of this chapter was that this ring can be used to compute an algebro-geometric invariant. For a line bundle L on a smooth complete variety, Hirzebruch–Riemann–Roch says

$$\boxed{\chi(X, L) = \int_X \operatorname{ch}(L) \operatorname{Td}(X).}$$

For $L = \mathcal{O}_X(D)$, the Chern character is $e^{[D]}$, while the generalized toric Euler sequence turns the Todd class into the explicit product

$$\boxed{\operatorname{Td}(X_\Sigma) = \prod_{\rho \in \Sigma(1)} \frac{[D_\rho]}{1 - e^{-[D_\rho]}}.}$$

On a surface this recovers the particularly concrete formula

$$\chi(X, \mathcal{O}_X(D)) = 1 + \frac{1}{2} D \cdot (D - K_X).$$

Equivariant cohomology then put the torus action back into the story. For the compact torus $K_N \subset T_N$, its coefficient ring is

$$H^*_{K_N}(\mathrm{pt}; \mathbb{Q}) \simeq \operatorname{Sym}_{\mathbb{Q}}(M_{\mathbb{Q}}) =: S,$$

and a smooth complete toric variety is equivariantly formal. Most importantly, its equivariant cohomology has a fan-theoretic model:

$$\boxed{H^*_{K_N}(X_\Sigma; \mathbb{Q}) \simeq PP^*(\Sigma; \mathbb{Q}).}$$

Thus an equivariant class is a polynomial on each cone, with the polynomials agreeing on common faces. In degree two these are the piecewise linear functions already familiar from toric divisors. Forgetting the action amounts to setting the positive-degree elements of S equal to zero; this recovers ordinary cohomology.

Localization showed why the piecewise-polynomial description is so powerful. A maximal cone σ determines both a fixed point x_σ and the tangent weights $m_{\sigma,1}, \ldots, m_{\sigma,n}$. After those weights are made invertible, integration becomes a finite sum:

$$\boxed{\int_{X_\Sigma}^{K_N} \alpha = (-1)^n \sum_{\sigma \in \Sigma(n)} \frac{\alpha|_{x_\sigma}}{\prod_i m_{\sigma,i}}.}$$

The individual summands are rational functions, but the apparent poles cancel because the local data come from one global equivariant class.

Finally, Riemann–Roch, positivity, and the monomial basis of toric sections brought the entire chapter back to the polytope. If D is nef and Cartier, then Demazure vanishing gives

$$\boxed{\chi(X_\Sigma, \mathcal{O}(D)) = \int_{X_\Sigma} e^{[D]} \operatorname{Td}(X_\Sigma) = \#(P_D \cap M).}$$

Before forgetting the action, the same statement remembers each lattice point as a character, and localization decomposes the answer into one contribution from each vertex of the polytope. The fan, equivariant topology, and lattice-point combinatorics are therefore not three parallel descriptions: they are three stages of one calculation.

The challenge exercise reverses an argument we used earlier in the notes. Chapter 7 invoked Pick's theorem to extract geometry from lattice polygons. We now have enough machinery to derive Pick's theorem itself from toric Riemann–Roch.

Comprehension exercises

9.1. Let A and B be the two ruling classes on $X = \mathbb{P}^1 \times \mathbb{P}^1$, so that

$$A^2 = B^2 = 0, \qquad A \cdot B = 1, \qquad K_X = -2A - 2B.$$

For $D = aA + bB$, use surface Riemann–Roch to prove

$$\chi\big(X, \mathcal{O}_X(a,b)\big) = (a+1)(b+1).$$

When $a, b \geq 0$, recover the same number by counting lattice points in the rectangle $[0,a] \times [0,b]$.

9.2. Let $S, F \in H^2(\mathbb{F}_r; \mathbb{Z})$ be the section and fiber classes from the Chapter 8 challenge, with

$$S^2 = -r, \qquad F^2 = 0, \qquad S \cdot F = 1.$$

Use the four invariant boundary divisors to show that

$$K_{\mathbb{F}_r} = -2S - (r+2)F.$$

Then compute $\chi(\mathbb{F}_r, \mathcal{O}(aS + bF))$ directly from surface Riemann–Roch. Which part of your answer remembers the integer r, even though Chapter 8 showed that the abstract cohomology ring remembers only its parity?

9.3. Let S^1 act on $S^3 \subset \mathbb{C}^2$ by scalar multiplication. The action is free. Use the Borel construction to compute

$$H^*_{S^1}(S^3; \mathbb{Q}).$$

Now give the same underlying space the trivial S^1-action and compute its equivariant cohomology again. Explain precisely where the two answers record the difference between the actions.

9.4. Let $T = (\mathbb{C}^*)^2$ act in the standard way on $X = \mathbb{P}^1 \times \mathbb{P}^1$. Without finding a presentation of the whole ring, use equivariant formality to determine the rank of $H^*_T(X; \mathbb{Q})$ as a module over

$$H^*_T(\mathrm{pt}; \mathbb{Q}) = \mathbb{Q}[t_1, t_2].$$

Then explain algebraically how setting $t_1 = t_2 = 0$ recovers a four-dimensional ordinary cohomology ring. Why should this rank also equal the number of torus fixed points?

9.5. Consider the fan of $\mathbb{P}^1 \times \mathbb{P}^1$ with rays generated by

$$(1,0), \quad (0,1), \quad (-1,0), \quad (0,-1).$$

Describe an element of $PP^*(\Sigma;\mathbb{Q})$ as four polynomials, one on each quadrant, and write the compatibility condition across each ray. Specialize your description to degree two and explain how the resulting piecewise linear functions recover torus-invariant Cartier divisor classes. What changes after quotienting by global linear functions?

9.6. On each factor of $\mathbb{P}^1 \times \mathbb{P}^1$, use the equivariant class $\eta = (0,t)$ from Section 9.3. Let η_1, η_2 be its pullbacks from the two factors, with equivariant parameters t_1, t_2. Use the four fixed points and localization to prove

$$\int_{\mathbb{P}^1\times\mathbb{P}^1}^{T} \eta_1\eta_2 = 1.$$

Then use localization again and watch the two contributions in the other factor cancel to show

$$\int_X^T \eta_1^2 = \int_X^T \eta_2^2 = 0.$$

Recover the ordinary intersection number $(A+B)^2 = 2$.

9.7. For $d \geq 0$, apply Riemann–Roch on $\mathbb{P}^1$ to $\mathcal{O}_{\mathbb{P}^1}(-d-2)$. Show that

$$\chi(\mathbb{P}^1, \mathcal{O}(-d-2)) = -d-1.$$

Since this line bundle has no nonzero global sections, determine $h^1(\mathbb{P}^1, \mathcal{O}(-d-2))$. Explain why this example shows that the nef hypothesis in the lattice-point version of toric Riemann–Roch cannot simply be discarded.

9.8. Let a nef divisor on $\mathbb{F}_r$ have polygon

$$P = \operatorname{Conv}(0,0), (a,0), (a+rb,b), (0,b), \qquad a,b,r \geq 0.$$

Count its lattice points directly by slicing at integer heights $y = 0, 1, \ldots, b$, and prove

$$\#(P \cap M) = (a+1)(b+1) + \frac{r}{2}b(b+1).$$

Now compute the area and the four lattice edge lengths of P. Use the area/intersection and edge-length/intersection dictionaries from Chapter 5 together with surface Riemann–Roch to recover exactly the same answer.

Challenge exercise

Riemann–Roch proves Pick's theorem.

Let $P \subset M_{\mathbb{R}} \simeq \mathbb{R}^2$ be a two-dimensional lattice polygon. Write

$$A(P) = \mathrm{Area}(P), \qquad B(P) = |\partial P \cap M|, \qquad I(P) = |\,\mathrm{Int}(P) \cap M|.$$

Pick's theorem is the identity

$$\boxed{A(P) = I(P) + \frac{1}{2}B(P) - 1.}$$

We used this formula in Chapter 7. The goal of this challenge is to reverse the logic: derive it from the toric Riemann–Roch theorem of this chapter. Begin with a smooth polygon, where every piece of the dictionary is already available, and then use a toric resolution to remove the smoothness assumption.

Hint 1. Turn the polygon into a line bundle.
First assume that P is smooth, let $X = X_P$ be its smooth projective toric surface, and choose the ample torus-invariant divisor D with

$$P_D = P.$$

Explain why

$$P_{kD} = kP \qquad (k \geq 0)$$

and why Theorem 9.14 gives

$$L_P(k) := \#(kP \cap M) = \chi(X, \mathcal{O}_X(kD)).$$

Hint 2. Write the surface Riemann–Roch polynomial.
Apply the surface formula from Section 9.1 to kD. Prove

$$\boxed{L_P(k) = 1 + \frac{1}{2}k^2 D^2 - \frac{1}{2}k\, D \cdot K_X.}$$

Why is the constant term equal to 1? Give both interpretations: $L_P(0) = 1$ on the polytope side, and $\chi(X, \mathcal{O}_X) = 1$ on the toric side.

Hint 3. Recognize the quadratic coefficient as area.
Use the area/self-intersection theorem from Chapter 5 to replace the self-intersection by

$$D^2 = 2A(P).$$

The preceding polynomial becomes

$$L_P(k) = A(P)k^2 - \frac{1}{2}(D \cdot K_X)k + 1.$$

Notice that no interior lattice-point count has entered the argument.

Hint 4. Recognize the linear coefficient as boundary length.
Write the toric boundary as

$$-K_X = \sum_{\rho \in \Sigma(1)} D_\rho.$$

If F_ρ is the edge of P corresponding to ρ, use the edge-length/intersection dictionary from Chapter 5 to show

$$-D \cdot K_X = \sum_{\rho} D \cdot D_\rho = \sum_{\rho} \text{length}_M(F_\rho).$$

Now prove the elementary boundary identity

$$\boxed{\sum_{\substack{F \prec P \\ \dim F = 1}} \text{length}_M(F) = B(P).}$$

Be careful about the vertices: summing $|F \cap M|$ would count every vertex twice, whereas summing $|F \cap M| - 1$ counts the cyclic boundary exactly once.
Conclude that

$$\boxed{L_P(k) = A(P)k^2 + \frac{1}{2}B(P)k + 1.}$$

This is the Ehrhart polynomial of a smooth lattice polygon, obtained here from Riemann–Roch rather than from Ehrhart theory.

Hint 5. Set $k = 1$.
Since every lattice point of P is either interior or on the boundary,

$$L_P(1) = I(P) + B(P).$$

Substitute $k = 1$ into the Riemann–Roch polynomial and show

$$I(P) + B(P) = A(P) + \frac{1}{2}B(P) + 1.$$

Rearrange. You have proved

$$\boxed{A(P) = I(P) + \frac{1}{2}B(P) - 1,}$$

Pick's theorem for smooth lattice polygons.

Hint 6. Recover the Chapter 7 anticanonical calculation.
Suppose $P = P_{-K_X}$ is a smooth reflexive polygon. Recall that the origin is its unique interior lattice point, so $I(P) = 1$. Use the formula you have just proved to recover

$$2A(P) = B(P).$$

Then use $(-K_X)^2 = 2A(P)$ to obtain

$$\boxed{(-K_X)^2 = B(P),}$$

the degree formula used in Section 7.5. Explain why the logical direction is now the reverse of the argument in Chapter 7.

Hint 7. Remove the smoothness assumption.
Now let P be an arbitrary two-dimensional lattice polygon. Its normal fan Σ_P need not be smooth. Choose a smooth complete refinement $\widetilde{\Sigma}$ and let

$$\pi : X_{\widetilde{\Sigma}} \longrightarrow X_{\Sigma_P}$$

be the corresponding toric resolution. If A_P is the ample Cartier divisor determined by P on X_{Σ_P}, set

$$D = \pi^* A_P.$$

Use the pullback description of basepoint-free toric divisors from Chapter 5 to explain why D is nef and

$$P_D = P.$$

Thus Theorem 9.14 still applies on the *smooth* surface $X_{\widetilde{\Sigma}}$.
The new rays of $\widetilde{\Sigma}$ correspond to curves contracted by π, so D has intersection zero with those curves. The uncontracted boundary curves correspond to the original edges of P, and their intersections with D are the lattice lengths of those edges. Deduce again that

$$D^2 = 2A(P), \qquad -D \cdot K_{X_{\widetilde{\Sigma}}} = B(P).$$

Repeat Hints 2–5 on the resolution and conclude that Pick's theorem holds for every lattice polygon.

Hint 8. One theorem, three languages.
Summarize the proof by completing the following dictionary in your own words:

$$\boxed{D^2 \longleftrightarrow 2A(P), \qquad -D \cdot K_X \longleftrightarrow B(P), \qquad \chi(\mathcal{O}_X(D)) \longleftrightarrow I(P) + B(P).}$$

Which input is topological, which is intersection-theoretic, and which is combinatorial? Where, exactly, does positivity enter the proof? Where, exactly, is smoothness used, and how did the resolution remove it?

The three dictionaries

The proof of Pick's theorem is the chapter's three-way dictionary in its most compact form.

Intersection theory	Riemann–Roch	Lattice polygon
D^2	quadratic coefficient	$2\,\mathrm{Area}(P)$
$-D \cdot K_X$	linear coefficient	$\lvert\partial P \cap M\rvert$
$\chi(\mathcal{O}_X)$	constant coefficient	1
$\chi(\mathcal{O}_X(D))$	value at $k = 1$	$\lvert P \cap M\rvert$

The proof explains why Pick's formula has exactly three terms. On a surface, Riemann–Roch has a quadratic intersection term, a linear canonical term, and the constant $\chi(\mathcal{O}_X) = 1$; the toric dictionary turns them into area, boundary length, and 1.

Notes and references

The Chern-class calculation for a smooth toric variety follows from the generalized Euler sequence; see Cox–Little–Schenck, Proposition 13.1.2. Their Theorem 13.1.6 gives the product formula for the Todd class, and Example 13.1.7 gives the surface Riemann–Roch calculation used repeatedly in this chapter. Hirzebruch–Riemann– Roch for line bundles on smooth complete toric varieties is stated in their Theorem 13.3.8.

Our equivariant cohomology discussion follows Cox–Little–Schenck, §12.4. Definition 12.4.6 introduces the Borel construction, Proposition 12.4.7 and Corollary 12.4.8 describe the passage back to ordinary cohomology, and Theorem 12.4.14 gives the equivariant Stanley–Reisner presentation. The piecewise-polynomial viewpoint is developed in the discussion surrounding their exact sequence (12.4.18). In degree two it recovers the support functions and Cartier data already used throughout these notes.

The fixed-point formula in Section 9.3 is the toric form of the Atiyah–Bott/Berline–Vergne localization theorem. In Cox–Little–Schenck, Lemma 13.3.5 and Proposition 13.3.6 compute the toric fixed-point weights, while Theorem 13.3.7 gives the localization formula itself. Their Theorem 13.3.4 and Example 13.3.9 explain the equivariant Riemann–Roch character calculation that underlies our one-dimensional example.

The final lattice-point formula combines three earlier results: global sections are indexed by $P_D \cap M$ by Cox–Little–Schenck, Proposition 4.3.3; higher cohomology vanishes for nef toric divisors by their Theorem 9.2.3; and toric Hirzebruch–Riemann–Roch is Theorem 13.3.8. The area and edge-length intersection formulas used in the challenge were proved earlier in Chapter 5 and correspond to Cox–Little–Schenck, Proposition 10.5.6 and Proposition 6.3.8, respectively.

We have intentionally not developed the volume-polynomial, Khovanskii–Pukhlikov, or general Ehrhart-theoretic refinements of Chapter 13 of Cox–Little–Schenck. The point here was narrower: to see equivariant cohomology and localization turn Riemann–Roch into a concrete toric computation, and then to watch that computation become lattice-point arithmetic. The next chapter changes direction from invariants of a fixed toric variety to birational geometry: varying a Cox linearization will move us through the secondary fan and make contractions, flips, and the toric minimal model program visible.

Chapter 10

VGIT and Toric Birational Geometry

Chapter overview

The preceding chapters asked what a fan can tell us about a fixed toric variety. We now change the question: how can one toric variety change birationally into another?

The Cox construction already contains the answer. Fix an affine space with a diagonal action of a torus G. In Chapter 6 we deleted a fan-determined exceptional set and took a quotient. Geometric invariant theory gives a more flexible version of the same construction: a character of G determines which points are allowed before we take the quotient. As the character moves, the allowed set changes only when we cross certain walls. The resulting chamber decomposition is the *secondary fan*.

Our goal in this final chapter is to make the progression

$$\boxed{\begin{array}{c}\text{Cox action and a character} \longrightarrow \text{GIT quotient} \longrightarrow \text{secondary fan} \\ \longrightarrow \text{wall crossing and contractions} \longrightarrow \text{toric MMP}\end{array}}$$

as concrete as possible. We work over $\mathbb{C}$, and most examples begin with a smooth projective toric variety so that $\mathrm{Cl}(X) = \mathrm{Pic}(X)$ is free and the Cox group is itself a torus.

The three dictionaries

The same data will be read in three languages throughout the chapter.

Cox coordinates	GIT	Birational geometry
class $\theta \in \mathrm{Cl}(X)$	character / linearization	choice of model and polarization
homogeneous monomial	semi-invariant section	witness for semistability
region of characters	constant semistable locus	one toric birational model
wall	limiting quotient	contraction or flip

Surface laboratory

Our running two-dimensional example will be

$$\mathbb{F}_1 \simeq \mathrm{Bl}_p(\mathbb{P}^2).$$

Chapter 6 constructed it from $\mathbb{A}^4$ by a $(\mathbb{G}_m)^2$-action. In this chapter we will *keep that affine space and that action fixed* and vary only the character used to form the quotient. One

region of characters gives $\mathbb{F}_1$; another gives $\mathbb{P}^2$; on the wall between them we will see the exceptional curve of the blowup contract.
This is the surface model for the entire chapter. In dimension three the same kind of wall crossing can also produce small contractions and flips, which have no surface analogue.

10.1 One Cox quotient, many linearizations

Recall the Cox construction for a smooth projective toric variety X_Σ. If the rays are $\Sigma(1) = \{\rho_1, \dots, \rho_r\}$, then

$$S = \mathbb{C}[x_1, \dots, x_r]$$

is graded by $\mathrm{Cl}(X_\Sigma)$, with

$$\deg(x_i) = [D_{\rho_i}].$$

The Cox group

$$G = \mathrm{Hom}\big(\mathrm{Cl}(X_\Sigma), \mathbb{G}_m\big)$$

acts diagonally on $\mathbb{A}^r$. Chapter 6 then formed

$$X_\Sigma \simeq (\mathbb{A}^r \setminus Z(\Sigma))/G.$$

There are three ingredients in this display:

the affine space	+	the action of G	+	the open set we quotient.

The first two are encoded by the Cox grading. GIT will tell us how to produce the third from one additional choice: a character of G.

Guiding question

What changes if we keep $\mathbb{A}^r$ and the action of G fixed, but change the character used to form the quotient?

10.1.1 Why the affine quotient is not enough

If a reductive group G acts on an affine variety V, the affine categorical quotient is

$$V /\!/ G = \mathrm{Spec}(\mathbb{C}[V]^G).$$

For Cox constructions this often forgets far too much. The simplest example is already projective space: scalar multiplication by $\mathbb{G}_m$ on $\mathbb{A}^{n+1}$ has only constant polynomial invariants, and hence

$$\mathbb{A}^{n+1} /\!/ \mathbb{G}_m = \{\mathrm{pt}\}.$$

But deleting the origin and remembering how functions *transform*, rather than insisting that they be invariant, recovers $\mathbb{P}^n$.

This is the elementary idea behind the GIT linearization. Let

$$\widehat{G} = \mathrm{Hom}(G, \mathbb{G}_m)$$

be the character group. A character $\chi \in \widehat{G}$ lifts the action on $\mathbb{A}^r$ to the trivial line bundle by

$$g \cdot (z, t) = (g \cdot z, \chi(g)t).$$

Although the underlying line bundle is still trivial, its G-action is not. The choice of χ is called a *linearization*.

For the Cox group there is a canonical identification

$$\boxed{\widehat{G} \simeq \mathrm{Cl}(X_\Sigma).}$$

Thus a divisor class can be read as a character. If $\theta \in \mathrm{Cl}(X_\Sigma)$ and χ_θ is the corresponding character, then a Cox polynomial F of degree θ satisfies

$$F(g \cdot z) = \chi_\theta(g) F(z).$$

Such an F is a *semi-invariant* of weight χ_θ. This is precisely the relation between the Cox grading and GIT used in Cox–Little–Schenck, Example 14.1.4.

Algebraic geometry cheatsheet

For the diagonal torus actions used in this chapter, it is enough to remember the following translation.

$$\begin{aligned} F \in S_{d\theta} \quad &\Longleftrightarrow \quad F(gz) = \chi_\theta(g)^d F(z), \\ F(z) \neq 0 \quad &\Longrightarrow \quad z \text{ is allowed by the linearization } \theta. \end{aligned}$$

A point is semistable when *some* positive multiple $d\theta$ has such a homogeneous polynomial not vanishing there. So in the Cox setting, semistability is a question about which homogeneous monomials can be nonzero at a point.

10.1.2 Semistable points and the GIT quotient

We now make that rule precise.

Definition 10.1 (Semistability for a Cox linearization)**.** Fix $\theta \in \mathrm{Cl}(X_\Sigma)$. A point $z \in \mathbb{A}^r$ is *semistable with respect to* θ if there exist an integer $d > 0$ and a polynomial

$$F \in S_{d\theta}$$

such that $F(z) \neq 0$. The semistable locus is denoted $(\mathbb{A}^r)^{ss}_\theta$.

This is the toric form of Cox–Little–Schenck, Definition 14.1.6. Their definition is phrased in terms of invariant sections of the linearized bundle; the Cox grading identifies those sections with the homogeneous pieces $S_{d\theta}$.

The quotient is built from all of these pieces at once.

Definition 10.2 (The toric GIT quotient)**.** Set

$$R_\theta = \bigoplus_{d\geq 0} S_{d\theta}.$$

The GIT quotient of $\mathbb{A}^r$ by G with linearization θ is

$$\boxed{\mathbb{A}^r /\!/_\theta G := \operatorname{Proj}(R_\theta).}$$

Equivalently,

$$\mathbb{A}^r /\!/_\theta G \simeq (\mathbb{A}^r)^{ss}_\theta /\!/ G.$$

Reference. This is Cox–Little–Schenck, Definition 14.1.11 and Proposition 14.1.12.

The point of the construction is that G and its action have not changed. Only θ has changed, but θ changes the graded ring R_θ and the open set $(\mathbb{A}^r)^{ss}_\theta$. The quotient can therefore change as well.

For a diagonal torus action there is a remarkably efficient test for semistability. Write the weight of the coordinate x_i as

$$\beta_i = \deg(x_i) \in \widehat{G},$$

and tensor the character lattice with $\mathbb{R}$.

Proposition 10.3 (The weight-cone test)**.** *Let $G \simeq (\mathbb{G}_m)^k$ act diagonally on $\mathbb{A}^r$ with weights $\beta_1, \dots, \beta_r$, and let $\theta \in \widehat{G}$. For $z = (z_1, \dots, z_r)$, set*

$$I(z) = \{i : z_i \neq 0\}.$$

Then

$$\boxed{z \in (\mathbb{A}^r)^{ss}_\theta \quad \Longleftrightarrow \quad \theta \in \operatorname{Cone}(\beta_i : i \in I(z)) \subseteq \widehat{G}_{\mathbb{R}}.}$$

Proof. Suppose first that z is semistable. Some $F \in S_{d\theta}$ does not vanish at z. At least one monomial

$$x^a = x_1^{a_1} \cdots x_r^{a_r}$$

appearing in F is nonzero at z. Hence $a_i = 0$ whenever $z_i = 0$, and homogeneity gives

$$d\theta = \sum_{i \in I(z)} a_i \beta_i.$$

After dividing by d, the character θ lies in the stated cone.

Conversely, if θ lies in this rational polyhedral cone, then after clearing denominators there are $d > 0$ and integers $a_i \geq 0$, supported on $I(z)$, such that

$$d\theta = \sum_i a_i \beta_i.$$

The monomial $x^a \in S_{d\theta}$ does not vanish at z, so z is semistable. □

This test is the computational engine for variation of toric GIT. The semistable locus can change only when θ moves across one of the cones spanned by subsets of the weights.

Example 10.4 (Scalar multiplication revisited)**.** Let $G = \mathbb{G}_m$ act on $\mathbb{A}^{n+1}$ with every coordinate of weight 1, and let $\theta = d \in \widehat{G} \simeq \mathbb{Z}$. The weight-cone test gives

d	$(\mathbb{A}^{n+1})^{ss}_d$	$\mathbb{A}^{n+1} /\!/_d \mathbb{G}_m$
$d > 0$	$\mathbb{A}^{n+1} \setminus \{0\}$	$\mathbb{P}^n$
$d = 0$	$\mathbb{A}^{n+1}$	$\{\mathrm{pt}\}$
$d < 0$	$\varnothing$	$\varnothing$.

Thus even the usual construction of projective space already exhibits the basic VGIT phenomenon: the quotient depends on the character, not only on the action.

10.1.3 An ample class recovers the Cox quotient

The original Cox construction has not disappeared inside this larger family. It is recovered by choosing a positive character.

Proposition 10.5 (Cox quotient as a GIT quotient)**.** *Let X_Σ be a smooth projective toric variety and let $\theta \in \mathrm{Cl}(X_\Sigma)$ be ample. Then*

$$\boxed{(\mathbb{A}^{\Sigma(1)})^{ss}_\theta = \mathbb{A}^{\Sigma(1)} \setminus Z(\Sigma),}$$

and

$$\mathbb{A}^{\Sigma(1)} /\!/_\theta G \simeq X_\Sigma.$$

In this smooth case the semistable points are stable and the quotient is geometric.

Reference. The equality of the semistable locus with the Cox open set is Cox–Little–Schenck, Proposition 14.1.9. In their more general projective setting, the stable locus is obtained from the subfan of simplicial cones; for a smooth fan this is all of Σ.

This proposition changes the way we should think about Chapter 6. There we started from Σ, built $Z(\Sigma)$, and then took a quotient. Now we can start from the diagonal Cox action and an ample class θ; semistability reconstructs the deleted set for us. Better still, θ is free to move.

10.1.4 Surface laboratory: the same action gives $\mathbb{F}_1$ and $\mathbb{P}^2$

Return to the Cox action for

$$X = \mathbb{F}_1 \simeq \mathrm{Bl}_p(\mathbb{P}^2)$$

from Chapter 6. With Cox coordinates

$$(u_0, v_0, u_1, v_1)$$

and the basis (F, S) of $\mathrm{Cl}(X) \simeq \mathbb{Z}^2$, the weights are

$$\deg(u_0) = \deg(u_1) = F, \qquad \deg(v_0) = S, \qquad \deg(v_1) = F + S.$$

Equivalently,

$$(\lambda, \mu) \cdot (u_0, v_0, u_1, v_1) = (\lambda u_0, \mu v_0, \lambda u_1, \lambda\mu v_1).$$

Now keep this action fixed and vary

$$\theta = aF + bS \in \mathrm{Cl}(X)_{\mathbb{R}}.$$

The three distinct weight rays divide the positive quadrant into two regions.

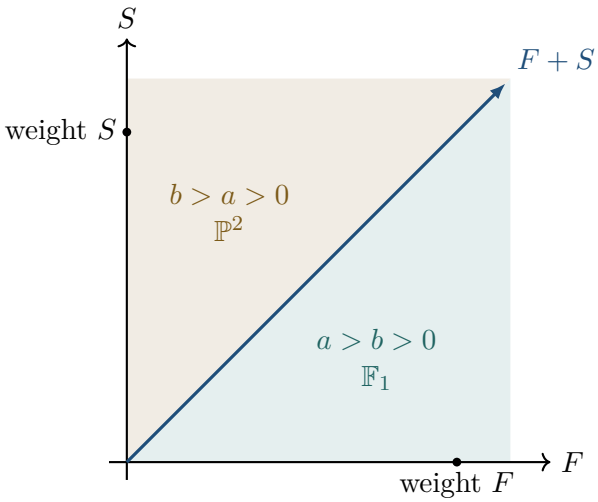


Figure 10.1: The character space for the $\mathbb{F}_1$ Cox action. The diagonal ray already predicts the wall between two generic GIT quotients. Section 10.2 will turn this picture into the secondary fan.

First suppose

$$a > b > 0.$$

The character lies between the rays F and $F + S$. By the weight-cone test, a semistable point must have at least one of u_0, u_1 nonzero and at least one of v_0, v_1 nonzero. Conversely those two conditions are sufficient. Hence

$$(\mathbb{A}^4)^{ss}_{\theta} = \mathbb{A}^4 \setminus \big(V(u_0, u_1) \cup V(v_0, v_1)\big).$$

This is exactly the Cox open set of $\mathbb{F}_1$, so

$$\boxed{\mathbb{A}^4 /\!/_{\theta} (\mathbb{G}_m)^2 \simeq \mathbb{F}_1 \qquad (a > b > 0).}$$

In the blowup notation of Chapter 6, $F = H - E$ and $S = E$, so

$$aF + bS = aH - (a - b)E.$$

The inequalities $a > b > 0$ are exactly the familiar ample inequalities for the blowup.

Now move θ across the diagonal and suppose

$$b > a > 0.$$

The weight-cone test gives a different open set:

$$(\mathbb{A}^4)^{ss}_\theta = \{v_0 \neq 0\} \cap \{(u_0, u_1, v_1) \neq (0,0,0)\}.$$

Indeed, the weight S is now indispensable: the cone generated by F and $F+S$ lies on or below the diagonal. Once $v_0 \neq 0$, semistability requires one additional nonzero coordinate of positive F-weight.

Use the μ-action to set $v_0 = 1$. The residual λ-action on the remaining coordinates is

$$\lambda \cdot (u_0, u_1, v_1) = (\lambda u_0, \lambda u_1, \lambda v_1).$$

Therefore

$$\boxed{\mathbb{A}^4 /\!/_\theta (\mathbb{G}_m)^2 \simeq \mathbb{P}^2 \qquad (b > a > 0).}$$

Nothing about $\mathbb{A}^4$ changed. Nothing about the $(\mathbb{G}_m)^2$-action changed. The character crossed one ray, the semistable locus changed, and the quotient changed from the blowup to the projective plane.

10.1.5 What lives on the wall?

The separating ray is

$$\theta = F + S = H.$$

Its degree-one semi-invariants are

$$u_0 v_0, \qquad u_1 v_0, \qquad v_1.$$

In fact every monomial of degree $d(F+S)$ is a product of these three, so

$$R_{F+S} \simeq \mathbb{C}[u_0 v_0, u_1 v_0, v_1]$$

with all three generators in degree one. Consequently the wall quotient is again

$$\mathbb{A}^4 /\!/_{F+S} (\mathbb{G}_m)^2 \simeq \mathbb{P}^2.$$

From the $\mathbb{F}_1$-side, the quotient map is visibly given by

$$\mathbb{F}_1 \longrightarrow \mathbb{P}^2, \qquad [u_0, v_0, u_1, v_1] \longmapsto [u_0 v_0 : u_1 v_0 : v_1].$$

On the divisor $v_0 = 0$, every point is sent to $[0:0:1]$. Under the Chapter 6 identification $S = E$, this is the exceptional curve of $\mathrm{Bl}_p(\mathbb{P}^2)$. We have therefore rediscovered the blowdown

$$\boxed{\mathbb{F}_1 \simeq \mathrm{Bl}_p(\mathbb{P}^2) \longrightarrow \mathbb{P}^2}$$

by moving a character to a wall.

Pause and think

The equation $a = b$ did not come from the fan of $\mathbb{F}_1$ by inspection. It came from the arrangement of the *Cox weights* in $\mathrm{Cl}(X)_{\mathbb{R}}$. Why should all characters with $a > b > 0$ give the same semistable locus? Why should all characters with $b > a > 0$ do the same? What finite collection of cones in character space organizes this behavior in general?

The answer is the secondary fan. Its chambers are precisely the regions on which the toric GIT quotient is constant in the appropriate sense. The example above has already drawn the relevant part of that fan; the next section will construct it systematically and explain its relation with regular subdivisions and the secondary polytope.

10.2 The secondary fan and the secondary polytope

Section 10.1 found a wall by hand. The weights for the Cox action on $\mathbb{F}_1$ were

$$F, \qquad S, \qquad F, \qquad F + S,$$

and the ray through $F + S$ separated two regions with different quotients. Nothing in that computation, however, explained why there should be only finitely many such regions or why they should fit together as a fan.

The answer is remarkably rigid. For a diagonal torus action, the weights themselves cut character space into a rational polyhedral fan. This is the *secondary fan*, also called the *GKZ decomposition*. Its chambers record the generic GIT quotients, while its faces record the quotients that appear when the character becomes special.

Guiding question

How can we read *all* toric GIT quotients of a fixed diagonal action from one finite combinatorial object?

10.2.1 From weight cones to chambers

Let

$$G \simeq (\mathbb{G}_m)^k$$

act diagonally on $\mathbb{A}^r$ with weights

$$\beta_1, \dots, \beta_r \in \widehat{G},$$

and set

$$C_\beta = \mathrm{Cone}(\beta_1, \dots, \beta_r) \subseteq \widehat{G}_{\mathbb{R}}.$$

The weight-cone test of Proposition 10.3 immediately shows that characters outside C_β have empty semistable locus. Thus C_β is the region in character space where nonempty GIT quotients can occur.

For a point $z = (z_1, \dots, z_r)$, recall

$$I(z) = \{i : z_i \neq 0\}.$$

It is useful to package exactly the weights visible at z.

Definition 10.6 (Weight cone of a point)**.** The *weight cone* of $z \in \mathbb{A}^r$ is

$$\omega(z) = \mathrm{Cone}(\beta_i : i \in I(z)) \subseteq \widehat{G}_{\mathbb{R}}.$$

Thus

$$z \in (\mathbb{A}^r)^{ss}_\theta \quad \Longleftrightarrow \quad \theta \in \omega(z).$$

There are only finitely many possible cones $\omega(z)$, since each is spanned by a subset of the finite set $\{\beta_1, \dots, \beta_r\}$. For $\theta \in C_\beta$, define

$$\Gamma(\theta) = \bigcap_{\theta \in \omega(z)} \omega(z).$$

In words: intersect every weight cone in which θ lies. The result is the smallest region that the semistability test can detect around θ.

Theorem 10.7 (The secondary fan)**.** *The cones $\Gamma(\theta)$, together with their faces, form a finite rational polyhedral fan*

$$\Sigma_{\mathrm{GKZ}} \quad \textit{in} \quad \widehat{G}_{\mathbb{R}}$$

whose support is C_β. Moreover, if θ and θ' lie in the relative interior of the same cone, then

$$(\mathbb{A}^r)^{ss}_\theta = (\mathbb{A}^r)^{ss}_{\theta'}$$

and hence

$$\mathbb{A}^r /\!/_\theta G \simeq \mathbb{A}^r /\!/_{\theta'} G.$$

The maximal cones of Σ_{GKZ} are called chambers*; their codimension-one faces are called* walls*.*

Reference. This is the diagonal-torus form of Cox–Little–Schenck, Theorem 14.4.7. Their GKZ cones keep track simultaneously of the fan of the quotient and of coordinates that never vanish; the description above is the same chamber structure viewed directly through semistability.

The theorem is substantial, but its basic mechanism is already visible from the weight-cone test. For every point z, the statement "z is semistable" is the polyhedral condition

$$\theta \in \omega(z).$$

Since only finitely many $\omega(z)$ occur, only finitely many semistability patterns can occur. Passing from this finite collection of cones to their common chamber decomposition gives Σ_{GKZ}. The content of the theorem is that these regions fit together with the face-to-face behavior required of a fan.

Algebraic geometry cheatsheet

For computations, one can think of the secondary fan as a *master map of linearizations*:

$\theta \notin C_\beta$	$\Longrightarrow$	empty semistable locus,
θ in a chamber	$\Longrightarrow$	one generic toric quotient,
θ on a wall	$\Longrightarrow$	a limiting quotient,
θ on a lower-dimensional face	$\Longrightarrow$	a still more special quotient.

The important point is not merely that the quotient is constant inside a chamber. The *way the chambers meet* records which models can be connected by varying the linearization.

10.2.2 The $\mathbb{F}_1$ picture really is a fan

Return to the action from the previous section. The distinct weight rays are

$$\mathbb{R}_{\geq 0}F, \qquad \mathbb{R}_{\geq 0}(F+S), \qquad \mathbb{R}_{\geq 0}S,$$

and

$$C_\beta = \mathrm{Cone}(F, S).$$

There are two two-dimensional GKZ chambers:

$$\Gamma_{\mathbb{F}_1} = \mathrm{Cone}(F, F+S), \qquad \Gamma_{\mathbb{P}^2} = \mathrm{Cone}(F+S, S).$$

Indeed, if $\theta = aF + bS$, then

location of θ	semistable locus	quotient
$a > b > 0$	$\mathbb{A}^4 \setminus (V(u_0, u_1) \cup V(v_0, v_1))$	$\mathbb{F}_1$
$a = b > 0$	wall semistable locus	$\mathbb{P}^2$
$b > a > 0$	$\{v_0 \neq 0\} \cap \{(u_0, u_1, v_1) \neq (0,0,0)\}$	$\mathbb{P}^2$.

Thus Figure 10.1 was not just a helpful picture: it was the two-dimensional part of Σ_{GKZ}. The common face

$$\mathbb{R}_{\geq 0}(F+S)$$

is the wall across which the exceptional curve of $\mathbb{F}_1$ is contracted.

There is a small subtlety worth remembering. The chamber $\Gamma_{\mathbb{F}_1}$ is literally

$$\mathrm{Nef}(\mathbb{F}_1) = \mathrm{Cone}(H - E, H) = \mathrm{Cone}(F, F+S).$$

On the other side, however, the quotient is $\mathbb{P}^2$, whose Picard group has rank one even though $\Gamma_{\mathbb{P}^2}$ is two-dimensional. The reason is visible in the semistable locus: v_0 is forced to be nonzero, so one Cox coordinate has disappeared from the quotient fan. The secondary fan remembers this extra GIT information. We will return to the relation with nef cones in Section 10.4.

10.2.3 Where triangulations enter: Gale duality

So far the secondary fan lives in character space. Its second life is as a space of *regular triangulations*. The bridge is Gale duality, the same linear algebra that relates Cox weights to ray generators.

For the cleanest geometric picture, suppose that

$$\nu_1, \dots, \nu_r \in N$$

are distinct lattice points lying in an affine hyperplane $H \subset N_{\mathbb{R}}$ with $0 \notin H$, and that

$$Q_\nu = \mathrm{Conv}(\nu_1, \dots, \nu_r)$$

has full dimension in H. The map

$$M \longrightarrow \mathbb{Z}^r, \qquad m \longmapsto \big(\langle m, \nu_1\rangle, \dots, \langle m, \nu_r\rangle\big)$$

has a free cokernel after making the usual lattice hypotheses. Write that cokernel as $\widehat{G}$. After tensoring with $\mathbb{R}$, we have

$$0 \longrightarrow M_{\mathbb{R}} \longrightarrow \mathbb{R}^r \longrightarrow \widehat{G}_{\mathbb{R}} \longrightarrow 0.$$

The images β_i of the standard basis vectors are precisely the Gale-dual weights.

Now assign a height $\omega_i \geq 0$ to every ν_i and form the lifted cone

$$C_{\nu,\omega} = \mathrm{Cone}((\nu_1, \omega_1), \dots, (\nu_r, \omega_r)) \subseteq N_{\mathbb{R}} \times \mathbb{R}.$$

Project the lower faces back to $N_{\mathbb{R}}$. Their cones give a subdivision Σ_ω of C_ν. Intersecting with the hyperplane H gives a subdivision of Q_ν.

Definition 10.8 (Regular triangulation)**.** A triangulation $\mathcal{T}$ of Q_ν, using vertices among the ν_i, is *regular* (or *coherent*) if it is obtained by this lifting-and-lower-hull construction for some choice of heights $\omega = (\omega_1, \dots, \omega_r)$.

This is Cox–Little–Schenck, Definition 15.2.8. The construction explains why the height vector naturally lives in the same space as a GIT character. If we change ω by

$$\omega_i \longmapsto \omega_i + \langle m, \nu_i\rangle,$$

we have merely tilted the lifting by a linear function. The regular subdivision does not change. Hence the meaningful parameter is the class

$$[\omega] \in \mathbb{R}^r / M_{\mathbb{R}} \simeq \widehat{G}_{\mathbb{R}}.$$

That quotient is exactly the character space in which the secondary fan lives.

Theorem 10.9 (Chambers and regular triangulations)**.** *In the affine-hyperplane situation above, the chambers of Σ_{GKZ} are in bijection with the regular triangulations of ν. A character in a chamber determines the corresponding triangulation by the lower-hull construction.*

Reference. This is Cox–Little–Schenck, Proposition 15.2.9. We will use the theorem as a dictionary rather than reproduce its full proof: Gale duality identifies height vectors modulo

linear functions with GIT characters, while strict convexity of the resulting piecewise linear function is exactly the condition that the projected lower hull be a triangulation.

Thus, in this setting, the same chamber can be read in three ways:

$$\boxed{\begin{array}{c}\text{GIT chamber in } \widehat{G}_{\mathbb{R}} \quad \longleftrightarrow \quad \text{regular triangulation of } Q_\nu \\ \longleftrightarrow \text{ simplicial fan of the corresponding toric quotient.}\end{array}}$$

10.2.4 The secondary polytope

There is now a beautiful dual way to package all the chambers at once. For a triangulation $\mathcal{T}$, define its *GKZ vector*

$$\phi_{\mathcal{T}} = (\phi_1, \dots, \phi_r) \in \mathbb{R}^r,$$

where

$$\phi_i = \sum_{\substack{\Delta \in \mathcal{T} \\ \nu_i \in \Delta}} \mathrm{Vol}(\Delta).$$

Here Δ runs over the maximal simplices and Vol denotes normalized lattice volume in H. The classical GKZ secondary polytope is the convex hull of these vectors; equivalently, after passing to its natural affine span, it is a polytope dual to the secondary fan.

Theorem 10.10 (The secondary polytope)**.** *Under the affine-hyperplane hypotheses above, there is a full dimensional lattice polytope*

$$P_{\mathrm{GKZ}} \subseteq \widehat{G}^*_{\mathbb{R}}$$

whose normal fan is Σ_{GKZ}. Its vertices are in bijection with regular triangulations of ν.

Reference. See Cox–Little–Schenck, Proposition 15.2.11, for the existence and normal-fan statement. The vertex–triangulation correspondence is Corollary 15.2.12. The GKZ-vector construction above gives the classical distinguished realization of the secondary polytope. Our normal-fan convention may be replaced by the opposite convention if one uses upper rather than lower hulls; none of the chamber combinatorics changes.

The word *secondary* is now rather apt. The original polytope Q_ν lives in the lattice space and admits many triangulations. The secondary polytope lives in the dual parameter space and organizes *the triangulations themselves.*

Secondary polytope	**Secondary fan / VGIT**
vertex	chamber / regular triangulation
edge	wall between adjacent chambers
higher-dimensional face	coarser regular subdivision
normal cone to a face	characters producing that subdivision

The reversal of dimensions is exactly what one expects from a normal fan: vertices of a polytope have full-dimensional normal cones, while edges correspond to codimension-one cones. Thus a path from one vertex of P_{GKZ} to a neighboring vertex is dual to crossing a wall in the VGIT fan.

10.2.5 A square: two triangulations and one wall

The smallest nontrivial secondary polytope can be computed completely by hand. Take the four lattice points

$$\nu_1 = (0,0,1), \quad \nu_2 = (1,0,1), \quad \nu_3 = (1,1,1), \quad \nu_4 = (0,1,1)$$

in the affine plane $z = 1$. Their convex hull is a square. There are exactly two triangulations:

$$\mathcal{T}_{13} = \{123, 134\}, \qquad \mathcal{T}_{24} = \{124, 234\},$$

according to which diagonal is chosen.

The unique affine dependence is

$$\nu_1 - \nu_2 + \nu_3 - \nu_4 = 0.$$

Consequently the one-dimensional quotient of height space is measured by

$$\delta(\omega) = \omega_1 + \omega_3 - \omega_2 - \omega_4.$$

If $\delta(\omega) < 0$, the points ν_1, ν_3 lie relatively lower and the lower hull produces $\mathcal{T}_{13}$. If $\delta(\omega) > 0$, it produces $\mathcal{T}_{24}$. When $\delta(\omega) = 0$, the four lifted points are coplanar and the projected lower face is the entire square: the diagonal disappears.

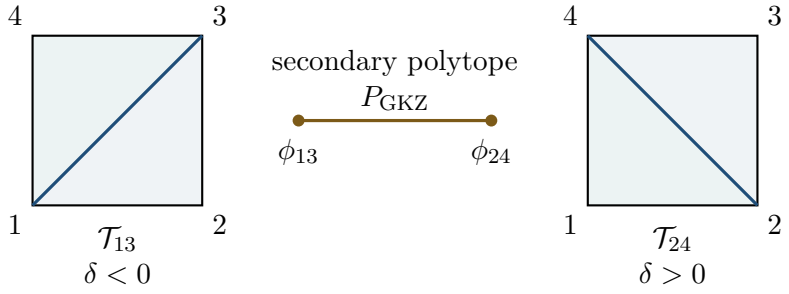


Figure 10.2: The square has two regular triangulations, hence its secondary polytope is a segment. The wall $\delta = 0$ is the coarser subdivision consisting of the unsplit square.

Because each triangle has normalized area one, the two GKZ vectors are

$$\phi_{13} = (2,1,2,1), \qquad \phi_{24} = (1,2,1,2).$$

Thus

$$P_{\mathrm{GKZ}} = \mathrm{Conv}(\phi_{13}, \phi_{24})$$

is a line segment. Its normal fan is one-dimensional: it consists of two rays separated by the origin. Those two chambers are exactly

$$\delta < 0 \qquad \text{and} \qquad \delta > 0,$$

with the wall $\delta = 0$ between them. If one defines normal cones using maximization rather than the lower-hull/minimization convention, the two ray labels are exchanged; the fan itself is unchanged.

This tiny example contains the entire VGIT pattern. A generic parameter chooses one triangulation. Moving to the wall erases a diagonal and produces a coarser subdivision. Crossing the wall inserts the other diagonal. In higher dimension this elementary replacement is called a *bistellar flip*; it is the combinatorial shadow of the birational transformations we will study next.

Pause and think

The secondary fan and the secondary polytope contain the same incidence data, but with dimensions reversed. A chamber of the fan is dual to a vertex of the polytope, and a wall between two chambers is dual to an edge between two vertices. Thus there are two equivalent ways to imagine varying a linearization: move *through* the secondary fan, or move *along edges* of the secondary polytope. Which picture makes a wall crossing easier to see?

We are now ready to put geometry back into the picture. A wall of the secondary fan tells us that two GIT chambers are adjacent; a flip of regular triangulations tells us how their combinatorics differ. The next question is what the induced map between the corresponding toric varieties actually does. In Section 10.3 we will see that, depending on the wall, it can contract a divisor, produce a small contraction and a flip, or lead to a fiber-type morphism.

10.3 Crossing a wall: contractions, flips, and fibrations

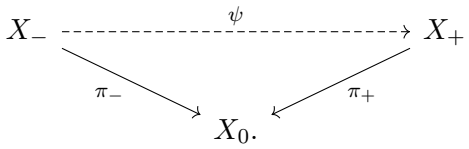
The secondary fan tells us *when* the quotient changes. We now ask what the change means geometrically. Let C_- and C_+ be adjacent chambers, choose characters

$$\theta_- \in \operatorname{Relint}(C_-), \qquad \theta_+ \in \operatorname{Relint}(C_+),$$

and let θ_0 lie in the relative interior of their common wall. Write

$$X_- = \mathbb{A}^r /\!/_{\theta_-} G, \qquad X_0 = \mathbb{A}^r /\!/_{\theta_0} G, \qquad X_+ = \mathbb{A}^r /\!/_{\theta_+} G.$$

Under the usual semiprojective hypotheses, variation of GIT produces projective toric morphisms



The wall quotient X_0 forgets just enough information that the two nearby models can map to it. The dashed arrow is the induced birational map when both morphisms are birational.

Guiding question

Which subvarieties become invisible when a character reaches a wall, and how does the answer determine the birational transformation on the other side?

10.3.1 The three geometric possibilities

The geometry is controlled by the exceptional locus. Recall that the exceptional locus of a birational morphism is the set on which it is not an isomorphism.

Definition 10.11 (Three types of contraction)**.** Let $\pi : X \to Y$ be a projective toric contraction with connected fibers.

1. It is of *fiber type* if $\dim Y < \dim X$.
2. It is *divisorial* if it is birational and its exceptional locus contains a divisor.
3. It is *small* if it is birational and its exceptional locus has codimension at least two.

If two small contractions

$$X_- \longrightarrow X_0 \longleftarrow X_+$$

arise from the two sides of a wall, the birational map $X_- \dashrightarrow X_+$ is an *elementary flip*. It is a *flop* when the canonical divisor has degree zero on the curves contracted on either side.

The adjective *small* refers to the exceptional locus, not to the size of the fibers. A small contraction may collapse curves, but it collapses no divisor. Consequently it does not delete a ray from the fan: the fans on the two sides have the same rays and differ only in how those rays are assembled into cones.

Proposition 10.12 (Wall-crossing trichotomy)**.** *Let X_Σ be a simplicial semiprojective toric variety, and let a codimension-one face of its nef chamber determine the wall morphism*

$$\pi : X_\Sigma \longrightarrow X_{\Sigma_0}.$$

Exactly one of the following occurs:

type	***geometry of π***	***fan behavior***
fiber type	$\dim X_{\Sigma_0} < \dim X_\Sigma$	Σ_0 *is degenerate*
divisorial	π *contracts an invariant divisor*	*one ray is removed*
small	$\operatorname{codim} \operatorname{Exc}(\pi) \geq 2$	$\Sigma(1) = \Sigma_0(1)$.

In the small case, a chamber on the other side supplies a second simplicial refinement Σ_+ of Σ_0 and hence an elementary flip

$$X_\Sigma \dashrightarrow X_{\Sigma_+}.$$

Reference. The classification by exceptional locus is Cox–Little–Schenck, Proposition 15.4.5. The wall diagram and the description of the exceptional loci for a flipping wall are given in Theorem 15.3.13; see also Lemma 15.3.7. We postpone the interpretation in terms of extremal rays of the Mori cone until Section 10.4.

Here is the practical picture to retain.

a whole direction survives in every fiber	$\Longrightarrow$	fiber type,
one Cox ray disappears	$\Longrightarrow$	divisorial contraction,
the rays stay but the cones change	$\Longrightarrow$	flip or flop.

10.3.2 Surface laboratory: the two boundary rays of $\mathrm{Nef}(\mathbb{F}_1)$

The chamber for $\mathbb{F}_1$ in Section 10.1 was

$$\mathrm{Cone}(F, H), \qquad H = F + S,$$

with interior classes

$$\theta = aF + bS, \qquad a > b > 0.$$

Its two boundary rays have completely different meanings. The ray through H gives the divisorial blowdown already seen in Section 10.1; the ray through F gives the ruling $\mathbb{F}_1 \to \mathbb{P}^1$.

Example 10.13 (The wall H: a divisorial contraction)**.** At $\theta = H = F + S$, the degree-one semi-invariants are

$$u_0v_0, \qquad u_1v_0, \qquad v_1.$$

They generate the section ring, so the wall quotient is

$$\mathrm{Proj}(R_H) \simeq \mathbb{P}^2.$$

The induced morphism is

$$\pi_H : \mathbb{F}_1 \longrightarrow \mathbb{P}^2, \qquad [u_0, v_0, u_1, v_1] \longmapsto [u_0v_0 : u_1v_0 : v_1].$$

On the invariant divisor

$$E = V(v_0),$$

the first two coordinates vanish, so all of $E \simeq \mathbb{P}^1$ maps to $[0 : 0 : 1]$. Away from E, the map is an isomorphism onto $\mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$. Thus

$$\boxed{\pi_H : \mathrm{Bl}_p(\mathbb{P}^2) \longrightarrow \mathbb{P}^2 \text{ contracts the divisor } E.}$$

Combinatorially, crossing this wall removes the ray introduced by the star subdivision that created the blowup.

Example 10.14 (The wall F: a fiber-type contraction)**.** Now move to the other boundary ray, $\theta = F$. The only monomials of degree dF are homogeneous polynomials in u_0, u_1. Therefore

$$R_F = \bigoplus_{d \geq 0} \mathbb{C}[u_0, u_1]_d, \qquad \mathrm{Proj}(R_F) \simeq \mathbb{P}^1.$$

The morphism defined by this ring is

$$\pi_F : \mathbb{F}_1 \longrightarrow \mathbb{P}^1, \qquad [u_0, v_0, u_1, v_1] \longmapsto [u_0 : u_1].$$

Fixing the point $[u_0 : u_1]$ leaves the pair (v_0, v_1), not both zero, modulo one scalar action. Hence every fiber is a copy of $\mathbb{P}^1$. We recover the Hirzebruch-surface ruling

$$\boxed{\mathbb{P}^1 \longrightarrow \mathbb{F}_1 \xrightarrow{\pi_F} \mathbb{P}^1.}$$

Nothing exceptional is confined to a proper subset of $\mathbb{F}_1$: the dimension of the quotient itself drops from two to one. This is the fiber-type case.

The intersection pairing gives a second way to recognize the two maps. Since

$$S^2 = -1, \qquad F^2 = 0, \qquad S \cdot F = 1,$$

we have

$$H \cdot S = (F + S) \cdot S = 0, \qquad F \cdot F = 0.$$

The class H therefore forgets the exceptional curve $S = E$, while the class F has degree zero along every fiber of the ruling. The wall in divisor space and the curves contracted by its linear system are dual pieces of information. Section 10.4 will turn this observation into the nef-cone/Mori-cone dictionary.

wall class	**target**	**contracted curves**	**type**
$H = F + S$	$\mathbb{P}^2$	the single curve $S = E$	divisorial
F	$\mathbb{P}^1$	all fibers of the ruling	fiber type

10.3.3 Why a genuine flip first appears in dimension three

A proper birational morphism of normal surfaces cannot be small: every nontrivial exceptional component is a curve, and a curve on a surface has codimension one. Thus a birational wall move between surfaces must contract a divisor. To see a small contraction, and therefore a flip, we need dimension at least three.

The square from Section 10.2 gives the smallest possible example. Its ray generators satisfy the circuit relation

$$\nu_1 - \nu_2 + \nu_3 - \nu_4 = 0.$$

The Gale-dual action is therefore the $\mathbb{G}_m$-action on $\mathbb{A}^4$ with weights

$$(1, -1, 1, -1): \qquad \lambda \cdot (x_1, x_2, x_3, x_4) = (\lambda x_1, \lambda^{-1} x_2, \lambda x_3, \lambda^{-1} x_4).$$

There are two chambers, $\theta > 0$ and $\theta < 0$, separated by the wall $\theta = 0$.

Example 10.15 (The two chamber quotients)**.** For $\theta > 0$, the weight-cone test says

$$(\mathbb{A}^4)^{ss}_{+} = \{(x_1, x_3) \neq (0, 0)\}.$$

The positive-weight coordinates determine a point $[x_1 : x_3] \in \mathbb{P}^1$, while x_2, x_4 have weight -1. Hence

$$X_+ = \mathbb{A}^4 /\!/_+ \mathbb{G}_m \simeq \mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}).$$

Similarly,

$$(\mathbb{A}^4)^{ss}_{-} = \{(x_2, x_4) \neq (0, 0)\}, \qquad X_- \simeq \mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}),$$

where the base is now $[x_2 : x_4]$.

Both threefolds are abstractly total spaces of the same vector bundle, but their maps to the wall quotient will remember different $\mathbb{P}^1$'s. Under the sign convention of Section 10.2, X_+ corresponds to the triangulation $\mathcal{T}_{24}$, while X_- corresponds to $\mathcal{T}_{13}$.

Example 10.16 (The wall quotient is the quadric cone)**.** At $\theta = 0$, all four coordinates are semistable and we take the affine quotient. The invariant ring is generated by

$$a = x_1x_2, \qquad b = x_1x_4, \qquad c = x_3x_2, \qquad d = x_3x_4.$$

These generators satisfy one relation,

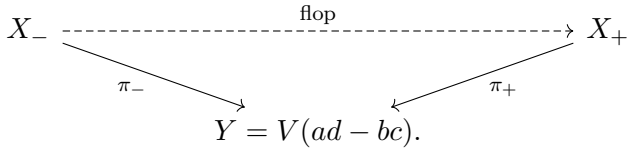
$$ad - bc = 0.$$

Thus

$$\boxed{X_0 = \mathbb{A}^4 /\!/_0 \mathbb{G}_m \simeq Y := V(ad - bc) \subseteq \mathbb{A}^4_{a,b,c,d}.}$$

The hypersurface Y is three-dimensional and singular only at the origin. It is the affine cone over the quadric surface $\mathbb{P}^1 \times \mathbb{P}^1 \subseteq \mathbb{P}^3$.

The four invariants define morphisms



It is worth checking the fibers directly. Regard a point of Y as the rank-at-most-one matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} x_1 \\ x_3 \end{pmatrix} \begin{pmatrix} x_2 & x_4 \end{pmatrix}.$$

Away from the zero matrix, both its column line and its row line are uniquely determined. Therefore π_+ and π_- are isomorphisms over $Y \setminus \{0\}$.

Over the origin, the answer changes. In the positive chamber, (x_1, x_3) cannot vanish, so the equations

$$x_1x_2 = x_1x_4 = x_3x_2 = x_3x_4 = 0$$

force $x_2 = x_4 = 0$. The remaining quotient is

$$\pi_+^{-1}(0) = \{(x_1, x_3) \neq (0, 0)\}/\mathbb{G}_m \simeq \mathbb{P}^1.$$

Similarly,

$$\pi_-^{-1}(0) \simeq \{(x_2, x_4) \neq (0, 0)\}/\mathbb{G}_m \simeq \mathbb{P}^1.$$

Each exceptional locus is a curve inside a threefold, hence has codimension two. Both contractions are small: no divisor disappears. The birational map replaces the column-line $\mathbb{P}^1$ by the row-line $\mathbb{P}^1$.

This is also exactly what happened to the square. On one side we used the diagonal 13; on the other we used the diagonal 24; on the wall the diagonal disappeared. The correspondence is

$\mathcal{T}_{13}$	$\longleftrightarrow$	X_-
unsplit square	$\longleftrightarrow$	Y
$\mathcal{T}_{24}$	$\longleftrightarrow$	X_+.

Finally, the exceptional curve in either resolution is the zero section of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, so its normal bundle is

$$N_{\mathbb{P}^1/X_\pm} \simeq \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1).$$

Adjunction gives

$$K_{X_\pm}|_{\mathbb{P}^1} \simeq K_{\mathbb{P}^1} \otimes \det(N_{\mathbb{P}^1/X_\pm})^{-1} \simeq \mathcal{O}_{\mathbb{P}^1}.$$

Thus $K_{X_\pm} \cdot \mathbb{P}^1 = 0$. The elementary flip is crepant, so it is in fact the standard threefold *flop*.

Algebraic geometry cheatsheet

The terms *flip* and *flop* describe the same wall diagram with different canonical behavior.

wall move	canonical degree on contracted curves
flip, with a chosen direction	$K \cdot C < 0$ before and $K \cdot C' > 0$ after
flop	$K \cdot C = K \cdot C' = 0$.

In the toric setting, both are produced by replacing one simplicial subdivision of a nonsimplicial cone by the subdivision on the other side of its circuit relation. The square relation has coefficient sum zero, which is the combinatorial shadow of the crepant condition.

10.3.4 Reading a wall crossing in four steps

The examples suggest a workflow that applies well beyond them.

1. **Find the wall.** Locate a codimension-one face of a GIT chamber in character space.
2. **Compute the wall ring.** Determine the semi-invariants of degree $d\theta_0$. Form $\operatorname{Proj}(R_{\theta_0})$, or take the invariant ring when the wall is $\theta_0 = 0$.
3. **Find the exceptional locus.** Compare the chamber semistable locus with the wall semistable locus and determine which orbit closures acquire the same image.
4. **Classify the map.** A dimension drop is fiber type; a contracted divisor is divisorial; a codimension-at-least-two exceptional locus is small, and the neighboring chamber supplies the flip.

Pause and think

In the $\mathbb{F}_1 \to \mathbb{P}^2$ wall move, a ray disappears and the exceptional curve is a divisor. In the resolved-conifold wall move, all four rays remain and only the triangulation changes. Why do equal ray sets but different maximal cones force the induced birational map to be an isomorphism in codimension one? Section 10.4 will identify the contracted curves canonically by dualizing this nef wall to an extremal ray of the Mori cone.

10.4 Extremal rays and contractions

Section 10.3 classified a wall morphism by looking at its exceptional locus. We now approach the same morphism from the opposite direction. Instead of starting with the map and asking what it contracts, we start with the numerical classes of curves and ask which map they determine.

The bridge is intersection theory. A divisor on the boundary of the nef cone has degree zero on some curves. Those curve classes lie on a face of the Mori cone. In the toric setting, every nef divisor is basepoint free, so its sections produce a morphism that contracts exactly those curves. Thus dual polyhedral cones turn a numerical vanishing condition into an honest geometric contraction.

Guiding question

How does a facet of the nef cone identify the curves contracted by a wall morphism, and how can those curve classes be computed directly from the fan?

10.4.1 Divisors and curves modulo numerical equivalence

Let X be a smooth projective variety. Two Cartier divisors are *numerically equivalent* if they have the same intersection number with every irreducible curve. After tensoring with $\mathbb{R}$, the space of numerical divisor classes is

$$N^1(X)_{\mathbb{R}} = \bigl(\mathrm{Pic}(X)/\equiv_{\mathrm{num}}\bigr) \otimes_{\mathbb{Z}} \mathbb{R}.$$

Similarly, two real one-cycles are numerically equivalent if every Cartier divisor has the same degree on them. The resulting space is

$$N_1(X)_{\mathbb{R}} = Z_1(X)_{\mathbb{R}}/\equiv_{\mathrm{num}}.$$

The definitions give a perfect pairing of finite-dimensional real vector spaces

$$\boxed{N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \qquad ([D],[C]) \longmapsto D \cdot C.}$$

Their common dimension is the *Picard number*

$$\rho(X) = \dim N^1(X)_{\mathbb{R}} = \dim N_1(X)_{\mathbb{R}}.$$

Inside the curve space, form the cone generated by effective curves.

Definition 10.17 (The Mori cone)**.** The *Mori cone*, or *cone of curves*, is

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{Cone}([C] : C \subseteq X \text{ an irreducible curve})} \subseteq N_1(X)_{\mathbb{R}}.$$

A one-dimensional face of $\overline{\mathrm{NE}}(X)$ is an *extremal ray*.

The closure in this definition matters for general varieties. For a projective toric variety the cone will be rational polyhedral and hence already closed.

A divisor class is nef precisely when it has nonnegative degree on every curve. Therefore

$$\boxed{\mathrm{Nef}(X) = \overline{\mathrm{NE}}(X)^{\vee} = \{\alpha \in N^1(X)_{\mathbb{R}} : \alpha \cdot \gamma \geq 0 \text{ for every } \gamma \in \overline{\mathrm{NE}}(X)\}.}$$

This equality is not an analogy: the nef and Mori cones are literal dual cones under the intersection pairing. In particular, a facet of one is perpendicular to an extremal ray of the other.

10.4.2 The toric cone theorem

Let X_Σ be an n-dimensional smooth projective toric variety. A codimension-one cone

$$\tau \in \Sigma(n-1)$$

is traditionally called a *wall of the fan*. The orbit–cone correspondence gives

$$C_\tau := V(\tau), \qquad \dim C_\tau = n - (n-1) = 1,$$

so every fan wall produces a complete torus-invariant curve.

These finitely many curves generate every numerical curve class that is effective.

Theorem 10.18 (The toric cone theorem)**.** *For a smooth projective toric variety X_Σ,*

$$\boxed{\overline{\mathrm{NE}}(X_\Sigma) = \sum_{\tau \in \Sigma(n-1)} \mathbb{R}_{\geq 0}[V(\tau)].}$$

Consequently $\overline{\mathrm{NE}}(X_\Sigma)$ and $\mathrm{Nef}(X_\Sigma)$ are dual strongly convex rational polyhedral cones of full dimension. Every extremal ray is generated by the class of a torus-invariant curve.

Reference. The generation statement is Cox–Little–Schenck, Theorem 6.3.20, and the projective duality statement is Theorem 6.3.22. Those results hold under weaker hypotheses than the smooth projective case used here.

The theorem is the curve-side counterpart of a fact already used throughout the notes: a toric divisor is nef if and only if its support function is convex. Convexity can be tested across the codimension-one walls of the fan, and crossing such a wall measures the intersection number with $V(\tau)$. Thus

$$D \cdot V(\tau) \geq 0 \quad \text{for every } \tau \in \Sigma(n-1) \quad \Longleftrightarrow \quad D \text{ is nef.}$$

Dualizing these finitely many inequalities gives the displayed description of the Mori cone.

Two cautions are useful. First, every extremal ray is generated by an invariant curve, but an arbitrary invariant curve need not generate an extremal ray. Several invariant curves may also be numerically equivalent and hence determine the same ray. Second, the word "wall" now occurs in three closely related places.

Algebraic geometry cheatsheet

Keep the three walls distinct.

wall	where it lives	what it determines
$\tau \in \Sigma(n-1)$	the fan in $N_\mathbb{R}$	the invariant curve $V(\tau)$
facet of $\mathrm{Nef}(X)$	$N^1(X)_\mathbb{R}$	an extremal ray of $\overline{\mathrm{NE}}(X)$
wall of Σ_{GKZ}	character space	a limiting GIT quotient.

For a Cox chamber with all rays present, the last two walls are the same wall under the identification of that chamber with the nef cone. The first wall supplies an invariant curve representing the dual extremal ray.

10.4.3 Wall relations compute curve classes

The class $[V(\tau)]$ can be read from one local relation among ray generators. Let

$$\tau = \mathrm{Cone}(u_2, \dots, u_n)$$

be a wall of a smooth complete fan, and write the two adjacent maximal cones as

$$\sigma = \mathrm{Cone}(u_1, u_2, \dots, u_n), \qquad \sigma' = \mathrm{Cone}(u_2, \dots, u_n, u_{n+1}).$$

Smoothness gives a unique integral relation of the form

$$\boxed{u_1 + b_2 u_2 + \cdots + b_n u_n + u_{n+1} = 0,} \qquad b_i \in \mathbb{Z}.$$

This is the *wall relation* of τ.

Proposition 10.19 (The wall relation as an intersection vector)**.** *Let $C_\tau = V(\tau)$, and let D_i be the invariant divisor corresponding to the ray through u_i. Then*

$$D_1 \cdot C_\tau = D_{n+1} \cdot C_\tau = 1, \qquad D_i \cdot C_\tau = b_i \quad (2 \le i \le n),$$

and every other invariant divisor has intersection zero with C_τ. Hence, for $D = \sum_\rho a_\rho D_\rho$,

$$\boxed{D \cdot C_\tau = a_1 + b_2 a_2 + \cdots + b_n a_n + a_{n+1}.}$$

Why this formula is natural. Intersecting a principal divisor with C_τ gives zero. Thus for every $m \in M$,

$$0 = \mathrm{div}(\chi^m) \cdot C_\tau = \sum_\rho \langle m, u_\rho \rangle (D_\rho \cdot C_\tau).$$

Equivalently,

$$\sum_\rho (D_\rho \cdot C_\tau) u_\rho = 0$$

in N. The orbit–cone correspondence shows that only the rays in $\sigma \cup \sigma'$ can contribute. Moreover, D_1 and D_{n+1} meet C_τ transversely at its two torus-fixed points, so both intersections are one. The unique wall relation then forces all the remaining coefficients. □

Reference. This is the smooth case of Cox–Little–Schenck, Proposition 6.4.4; their formula includes the cone multiplicities needed in the simplicial singular case. The interpretation of a numerical curve class as a relation among ray generators is Equation (15.1.4).

For a surface, this recovers a familiar calculation. If the rays are cyclically ordered and

$$u_{i-1} + u_{i+1} = a_i u_i,$$

then the wall relation for the invariant curve D_i is

$$u_{i-1} - a_i u_i + u_{i+1} = 0.$$

Proposition 10.19 therefore gives

$$D_i^2 = -a_i.$$

The surface self-intersection formula from Chapter 8 is thus the two-dimensional instance of the general wall-relation calculation.

10.4.4 A nef facet produces an extremal contraction

Let $R \subseteq \overline{\mathrm{NE}}(X)$ be an extremal ray. Its dual facet is

$$F_R = \{\alpha \in \mathrm{Nef}(X) : \alpha \cdot \gamma = 0 \text{ for every } \gamma \in R\}.$$

Choose a rational divisor class $[D]$ in the relative interior of F_R, and replace it by a positive integral multiple. Since D is nef and toric, it is basepoint free. Its section ring therefore defines a morphism

$$\varphi_D : X \longrightarrow X_D := \mathrm{Proj}\left(\bigoplus_{k\geq 0} H^0(X, \mathcal{O}_X(kD))\right).$$

Theorem 10.20 (Toric extremal contraction)**.** *Let X_Σ be a smooth projective toric variety and let $R \subseteq \overline{\mathrm{NE}}(X_\Sigma)$ be an extremal ray. There is a projective toric morphism with connected fibers*

$$\varphi_R : X_\Sigma \longrightarrow X_0$$

such that, for every irreducible curve $C \subseteq X_\Sigma$,

$$\boxed{\varphi_R(C) \text{ is a point} \quad \Longleftrightarrow \quad [C] \in R.}$$

The morphism may be obtained from any integral nef class in the relative interior of the dual facet F_R.

Reference. This is the projective smooth case of Cox–Little–Schenck, Proposition 15.4.1, which treats simplicial semiprojective toric varieties. The equivalence "nef = basepoint free" is Theorem 6.3.12.

Here is the mechanism. The morphism associated to a basepoint-free toric divisor factors D, up to linear equivalence, as

$$D \sim \varphi_D^* A$$

for an ample divisor A on the image. Therefore

$$D \cdot C = 0 \quad \Longleftrightarrow \quad \varphi_D(C) \text{ is a point.}$$

Because $[D]$ lies in the relative interior of F_R, the only effective classes on which it vanishes are those in R. This proves the numerical characterization of the contracted curves.

The same statement can be seen directly in the fan and polytope. The normal fan of P_D is obtained by joining adjacent maximal cones whose Cartier data agree. Across a fan wall $\tau = \sigma \cap \sigma'$,

$$m_\sigma = m_{\sigma'} \quad \Longleftrightarrow \quad D \cdot V(\tau) = 0.$$

Joining those cones makes $V(\tau)$ invisible to the resulting toric morphism. This is exactly the fan map constructed by the extremal contraction theorem.

More generally, if $[D]$ lies in the relative interior of a lower-dimensional face $F \subseteq \mathrm{Nef}(X)$, then it contracts the entire dual face

$$F^{\perp} \cap \overline{\mathrm{NE}}(X).$$

An *elementary* contraction is the special case in which F is a facet and the dual face is one extremal ray.

10.4.5 Surface laboratory: the dual cones of $\mathbb{F}_1$

We can now recover both morphisms from Section 10.3 by one two-by-two intersection calculation. Write

$$\mathbb{F}_1 = \mathrm{Bl}_p(\mathbb{P}^2), \qquad E^2 = -1, \qquad H^2 = 1, \qquad H \cdot E = 0,$$

and let

$$F = H - E$$

be the class of a fiber of the ruling. To keep the two numerical spaces visually distinct, write

$$e = [E] \in N_1(\mathbb{F}_1)_{\mathbb{R}}, \qquad f = [F] \in N_1(\mathbb{F}_1)_{\mathbb{R}}$$

for the curve classes, while $H, F \in N^1(\mathbb{F}_1)_{\mathbb{R}}$ denote divisor classes. Their pairing is

	$e = [E]$	$f = [F]$
H	0	1
$F = H - E$	1	0.

Every effective curve class is a nonnegative combination of e and f, so

$$\boxed{\overline{\mathrm{NE}}(\mathbb{F}_1) = \mathrm{Cone}(e, f).}$$

Indeed, if $D = \alpha H + \beta F$, then the table gives

$$D \cdot E = \beta, \qquad D \cdot F = \alpha.$$

Hence

$$\boxed{\mathrm{Nef}(\mathbb{F}_1) = \mathrm{Cone}(H, F),}$$

and D is ample exactly when $\alpha, \beta > 0$.

The two boundary rays of the nef cone are perpendicular to the two extremal rays of the Mori cone:

nef facet	**vanishing**	**dual extremal ray**	**contraction**
$\mathbb{R}_{\geq 0}H$	$H \cdot E = 0$	$\mathbb{R}_{\geq 0}e$	$\mathbb{F}_1 \to \mathbb{P}^2$
$\mathbb{R}_{\geq 0}F$	$F \cdot F = 0$	$\mathbb{R}_{\geq 0}f$	$\mathbb{F}_1 \to \mathbb{P}^1$.

Thus the blowdown and the ruling are not two unrelated constructions. They are the extremal contractions belonging to the two edges of one nef cone.

This also clarifies their different geometric types. The ray $\mathbb{R}_{\geq 0}e$ contains the class of a single irreducible curve, the exceptional section E; contracting it is divisorial. The ray $\mathbb{R}_{\geq 0}f$ contains

the class of every fiber in a moving family covering $\mathbb{F}_1$; contracting all of them lowers dimension and is of fiber type.

Surface laboratory

The same computation works for every Hirzebruch surface. If S is the section with $S^2 = -r$ and F is a fiber, then

$$\overline{\mathrm{NE}}(\mathbb{F}_r) = \mathrm{Cone}([S], [F]), \qquad \mathrm{Nef}(\mathbb{F}_r) = \mathrm{Cone}(F, S + rF).$$

Indeed, for $D = aS + bF$,

$$D \cdot F = a, \qquad D \cdot S = b - ar.$$

The two inequalities $a \geq 0$ and $b - ar \geq 0$ cut out the nef cone. When $r = 1$, the class $S + F = H$ gives the blowdown. When $r > 1$, the same nef boundary contracts the negative section to a singular torus-fixed point rather than to a smooth point.

10.4.6 The conifold relation revisited

The square example from Section 10.3 shows the same dictionary in dimension three. On the side using the diagonal 24, the two maximal cones meet along

$$\tau = \mathrm{Cone}(\nu_2, \nu_4),$$

and the wall relation is

$$\nu_1 - \nu_2 + \nu_3 - \nu_4 = 0.$$

For the exceptional curve $C = V(\tau) \simeq \mathbb{P}^1$, Proposition 10.19 gives

$$(D_1 \cdot C, D_2 \cdot C, D_3 \cdot C, D_4 \cdot C) = (1, -1, 1, -1).$$

The curve space is one-dimensional, so

$$\overline{\mathrm{NE}}(X_+) = \mathbb{R}_{\geq 0}[C].$$

Moving the linearization to the origin reaches the unique nef wall and contracts precisely this ray, producing the quadric cone $Y = V(ad - bc)$.

The wall relation also detects crepancy without using the normal bundle. Since

$$K_{X_+} = -D_1 - D_2 - D_3 - D_4,$$

we obtain

$$K_{X_+} \cdot C = -(1 - 1 + 1 - 1) = 0.$$

On the other resolution the relation, and hence the curve class, is reversed. The canonical degree is still zero. This is the numerical reason that the wall crossing is a flop.

10.4.7 Nef chambers inside the secondary fan

We can now close the circle begun in Section 10.2. For a simplicial semiprojective toric quotient in which every Cox coordinate remains present, the corresponding GKZ cone is naturally identified with the nef cone of the quotient:

$$\boxed{\Gamma_{\Sigma,\varnothing} \simeq \operatorname{Nef}(X_\Sigma).}$$

This is Cox–Little–Schenck, Proposition 15.1.3. Under this identification, the three constructions from this chapter line up:

secondary fan	divisors and curves	geometry
chamber for X_Σ	$\operatorname{Nef}(X_\Sigma)$	one polarized toric model
facet of the chamber	$R^\perp \cap \operatorname{Nef}(X_\Sigma)$	extremal contraction of R
neighboring chamber	nef cone of the next model	flip or another birational model.

This also explains the subtlety noticed for $\mathbb{F}_1$ in Section 10.2. Its chamber $\operatorname{Cone}(F, H)$ is exactly $\operatorname{Nef}(\mathbb{F}_1)$. Across the H-wall, the coordinate v_0 is forced to be nonzero and its ray disappears. The adjacent two-dimensional GKZ chamber therefore contains an extra GIT direction; after removing that redundant direction, the quotient $\mathbb{P}^2$ has the expected one-dimensional nef cone.

Pause and think

Suppose K_X is not nef. By duality there is an extremal ray $R \subseteq \overline{\operatorname{NE}}(X)$ with

$$K_X \cdot R < 0.$$

Theorem 10.20 now supplies a canonical next geometric move: contract R, and flip if the contraction is small. Repeating this choice is the minimal model program. What does this algorithm look like as a path through the secondary fan, and where can it end for a projective toric variety? Section 10.5 will assemble the final picture.

10.5 The toric minimal model program

We now have all of the ingredients promised at the beginning of the chapter. A Cox character chooses a GIT quotient, the secondary fan organizes those quotients into chambers, a chamber is the nef cone of one model, and a facet of that nef cone is dual to an extremal ray of curves. The minimal model program adds one final instruction: *among all extremal rays, move only in a direction on which the canonical divisor is negative.*

We begin with a smooth projective toric variety, but we must immediately allow a slightly larger class of spaces. A divisorial contraction or a flip can create singularities even when the starting variety is smooth. The natural setting is therefore a projective $\mathbb{Q}$-factorial toric variety with terminal singularities. For a toric variety, $\mathbb{Q}$-factoriality is equivalent to the fan being simplicial.

Guiding question

How does the sign of $K_X \cdot R$ orient the wall crossings in the secondary fan, why must the resulting process stop, and what geometric object is left at the end?

10.5.1 The algorithm: contract, flip, or stop

Let X be a projective $\mathbb{Q}$-factorial toric variety with terminal singularities. If K_X is not nef, the toric cone theorem supplies an extremal ray

$$R \subseteq \overline{\mathrm{NE}}(X) \qquad \text{with} \qquad K_X \cdot R < 0.$$

The contraction theorem then gives a projective toric morphism with connected fibers

$$\varphi_R : X \longrightarrow Y$$

that contracts exactly the curves whose classes lie in R. There are three possibilities.

1. If $\dim Y < \dim X$, then φ_R is of *fiber type*. The process stops.
2. If φ_R is birational and contracts a divisor, replace X by Y and continue. This is a *divisorial step*.
3. If φ_R is birational but contracts no divisor, then it is small. Replace X by the other side of the toric flip

$$\begin{array}{ccccc} X & & \xrightarrow{\quad\psi\quad} & & X^+ \\ & {\scriptstyle\varphi_R}\searrow & & \swarrow{\scriptstyle\varphi_R^+} & \\ & & Y. & & \end{array}$$

   and continue on X^+.

In the last case the contracted curve class changes sign. More precisely, if R^+ is the extremal ray of X^+ contracted by φ_R^+, then

$$K_X \cdot R < 0 \qquad \Longrightarrow \qquad K_{X^+} \cdot R^+ > 0.$$

Thus a flip does not merely replace one triangulation by another: it turns a K-negative direction into a K-positive direction.

Definition 10.21 (Mori fiber space)**.** A *Mori fiber space* is a contraction

$$f : X \longrightarrow Z$$

with connected fibers such that

$$\dim Z < \dim X, \qquad \rho(X) - \rho(Z) = 1, \qquad -K_X \text{ is } f\text{-ample}.$$

For a K_X-negative extremal contraction of fiber type, these conditions are automatic.

If instead the process reaches a model X' for which $K_{X'}$ is nef, then X' is called a *minimal model*. The general MMP has these two possible outcomes: a minimal model or a Mori fiber space. For projective toric varieties, however, there is a stronger conclusion.

Theorem 10.22 (The projective toric K-MMP)**.** *Starting from a positive-dimensional projective $\mathbb{Q}$-factorial toric variety with terminal singularities, any sequence of K-negative divisorial contractions and flips terminates after finitely many steps in a Mori fiber space.*

Why the two assertions hold. First, a positive-dimensional projective toric variety cannot have nef canonical divisor. If A is ample and $K_X = -\sum_\rho D_\rho$, then

$$K_X \cdot A^{n-1} = -\sum_\rho D_\rho \cdot A^{n-1} < 0,$$

because every invariant prime divisor has positive intersection with A^{n-1}. Hence the algorithm cannot end with K_X nef; when it stops, it must stop at a fiber-type contraction.

It remains to know that it stops. Every divisorial step decreases the Picard number by one, so only finitely many divisorial steps can occur. Between them, flips preserve the set of rays of the fan, and there are only finitely many fans built from a fixed finite ray set. Finiteness alone does not exclude a cycle, but K-negativity does. On a common toric refinement of a K-negative flip, the pullbacks of the two canonical divisors differ by a nonzero effective exceptional divisor. Going around a cycle would therefore say that one fixed pullback of K equals itself plus a nonzero effective divisor, which is impossible. Thus there is no infinite sequence of flips. □

Reference. The toric procedure is Cox–Little–Schenck, Procedure 15.5.5. Proposition 15.5.3 shows that K-negative divisorial contractions and flips preserve terminal singularities, and Theorem 15.5.10 proves termination of toric flips. Their results are stated in the more general semiprojective and relative setting.

10.5.2 The MMP as a directed path through the secondary fan

The algorithm has a particularly concrete GIT interpretation. Let A be an ample divisor on the current model and consider the segment in divisor-class space

$$\theta(t) = (1-t)[A] + t[K_X], \qquad 0 \le t \le 1.$$

It begins in the interior of $\mathrm{Nef}(X)$. Suppose it first reaches the boundary at $t = t_0$, on the facet dual to an extremal ray R. Then

$$0 = \theta(t_0) \cdot R = (1-t_0)A \cdot R + t_0 K_X \cdot R,$$

and therefore

$$\boxed{K_X \cdot R = -\frac{1-t_0}{t_0}\, A \cdot R < 0.}$$

So moving from an ample class toward the canonical class automatically selects a K-negative wall.

What happens at that wall is exactly the trichotomy from Section 10.3.

Algebraic geometry cheatsheet

The toric MMP in the secondary fan.

wall reached	fan / GIT change	MMP instruction
flipping wall	replace one circuit triangulation by the other	cross to the adjacent nef chamber and continue
divisorial wall	delete the ray of the exceptional divisor	lower the Picard number, then rebuild the chamber picture
boundary of the effective region	the quotient becomes degenerate in one direction	take the fiber-type contraction and stop.

The word *directed* is essential. VGIT allows us to cross a wall in either direction. The K-MMP allows only the direction for which the curves being removed have negative canonical degree. It is this numerical sign that turns the secondary fan from a catalogue of models into an algorithm.

10.5.3 A complete run on $\mathbb{F}_1$

Return one last time to

$$\mathbb{F}_1 = \mathrm{Bl}_p(\mathbb{P}^2).$$

Use the curve classes

$$e = [E], \qquad f = [F], \qquad \overline{\mathrm{NE}}(\mathbb{F}_1) = \mathrm{Cone}(e, f)$$

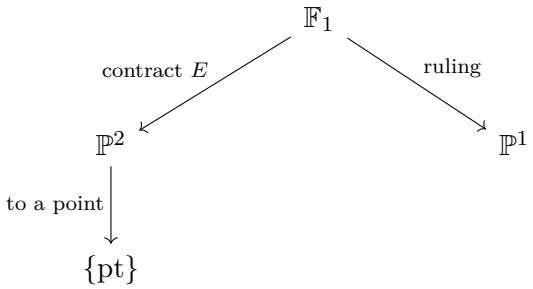
from Section 10.4. Since

$$K_{\mathbb{F}_1} = -3H + E = -2E - 3F,$$

we obtain

$$K_{\mathbb{F}_1} \cdot E = -1, \qquad K_{\mathbb{F}_1} \cdot F = -2.$$

Both extremal rays are K-negative, so the first MMP step is not unique.



On the left branch, contracting e is the divisorial blowdown

$$\mathbb{F}_1 \longrightarrow \mathbb{P}^2.$$

The Mori cone of $\mathbb{P}^2$ has one ray, generated by the class ℓ of a line, and

$$K_{\mathbb{P}^2} \cdot \ell = -3.$$

Its extremal contraction is $\mathbb{P}^2 \to \{\mathrm{pt}\}$, a Mori fiber space.

On the right branch, choosing f gives the ruling immediately:

$$\mathbb{F}_1 \longrightarrow \mathbb{P}^1.$$

This is already a Mori fiber space, since its relative Picard number is one and

$$-K_{\mathbb{F}_1} \cdot F = 2 > 0.$$

Thus one starting variety can have more than one valid MMP. The two runs even end over bases of different dimensions. The algorithm is finite and tightly constrained, but it is not canonical until the extremal ray is chosen.

Surface laboratory

The same calculation for $\mathbb{F}_r$ exposes an important warning. If $S^2 = -r$ and F is a fiber, then

$$K_{\mathbb{F}_r} = -2S - (r+2)F,$$

so

$$K_{\mathbb{F}_r} \cdot S = r - 2, \qquad K_{\mathbb{F}_r} \cdot F = -2.$$

Hence

r	$K \cdot S$	$K \cdot F$	allowed first K-MMP step
0	-2	-2	either ruling
1	-1	-2	blow down S, or take the ruling
2	0	-2	the ruling only
$r \geq 3$	$r-2>0$	-2	the ruling only.

For $r \geq 2$, the negative section still spans an extremal ray of the Mori cone, but that ray is not K-negative. In particular, on $\mathbb{F}_2$ its contraction is K-trivial, not a step of the ordinary K-MMP. A curve having negative self-intersection is not the same thing as its ray being K-negative.

10.5.4 Flips are directed; flops are not MMP steps

The sign test can also be read directly from a wall relation. Suppose an invariant curve C is represented by

$$\sum_i b_i u_i = 0$$

and the coefficients b_i are its intersections with the corresponding invariant divisors. Since $K_X = -\sum_i D_i$,

$$\boxed{K_X \cdot C = -\sum_i b_i.}$$

Thus the circuit itself tells us whether the associated wall crossing is permitted by the K-MMP:

$$\sum_i b_i > 0 \quad \Longleftrightarrow \quad K_X \cdot C < 0.$$

If the contraction is small, crossing the wall reverses the circuit class. The new exceptional curves consequently have positive canonical degree, which records the direction of the flip.

Now compare the conifold wall from Sections 10.3 and 10.4. Its relation is

$$\nu_1 - \nu_2 + \nu_3 - \nu_4 = 0.$$

The coefficient sum is zero, and hence

$$K_X \cdot C = -(1 - 1 + 1 - 1) = 0.$$

The two small resolutions are related by a *flop*, not by a K-negative flip. VGIT sees the two neighboring chambers perfectly well, but the ordinary K-MMP does not prefer either direction. Flops belong to the comparison of minimal models, where the canonical class is numerically unchanged; they are adjacent to the MMP story, but they are not descent steps in the ordinary K-MMP.

10.5.5 The chapter's final dictionary

We can now read one geometric move in every language developed in the notes. Begin with a toric variety X_Σ and a K-negative extremal ray R. Then

fan:	a wall relation represents R,
intersection theory:	$K_X \cdot R < 0$,
nef–Mori duality:	$R^\perp \cap \mathrm{Nef}(X)$ is a facet,
Cox/GIT:	the corresponding character reaches a wall,
secondary fan:	cross, delete a ray, or reach the boundary,
birational geometry:	flip, contract a divisor, or form a Mori fiber space.

This is the promised finale. The Cox quotient from Chapter 6 did not merely construct one toric variety. Once its character is allowed to move, it organizes a finite birational world:

Cox quotient $\longrightarrow$ secondary fan $\longrightarrow$ wall crossing $\longrightarrow$ toric MMP.

Pause and think

The two runs from $\mathbb{F}_1$ show exactly where choice enters the MMP: both rays e and f are K-negative, and the canonical class does not choose between them. What the theory guarantees is not a unique path, but a legal move at every stage and finite termination.
To test whether you now own the full dictionary, start with a wall relation, compute its canonical degree, identify the dual nef facet, and predict whether the corresponding GIT wall gives a divisorial contraction, a flip, a flop, or a Mori fiber space. The concluding

exercises will ask you to carry out precisely this chain without being told which language to begin in.

What this chapter has accomplished

The chapter began by returning to the Cox construction and changing only one ingredient. We fixed an affine space with a diagonal torus action, but allowed the character

$$\theta \in \mathrm{Hom}(G, \mathbb{G}_m)_{\mathbb{R}}$$

to move. The basic calculation was the weight-cone criterion

$$\boxed{z \text{ is } \theta\text{-semistable} \quad \Longleftrightarrow \quad \theta \in \mathrm{Cone}(w_i : z_i \neq 0).}$$

It turns semistability into convex geometry: a point survives exactly when the weights of its nonzero coordinates can support the chosen character. For an ample Cox character this recovers the usual Cox open set, while other characters may produce different toric quotients from the same polynomial ring and the same group action.

The regions on which the semistable locus remains constant assemble into the secondary, or GKZ, fan. Its support is the effective weight cone, its maximal cones are GIT chambers, and its faces record limiting quotients. Gale duality translates this chamber structure into regular subdivisions of a ray or point configuration. In the affine-hyperplane setting, the same incidence data are encoded dually by the secondary polytope:

$$\boxed{\begin{array}{c}\text{GIT chamber} \longleftrightarrow \text{regular triangulation} \longleftrightarrow \text{vertex of } P_{\mathrm{GKZ}},\\ \text{GIT wall} \longleftrightarrow \text{coarser regular subdivision} \longleftrightarrow \text{edge or higher face of } P_{\mathrm{GKZ}}.\end{array}}$$

Crossing a wall put geometry back into this combinatorics. A fiber-type wall lowers the dimension of the quotient; a divisorial wall removes a ray and contracts an invariant divisor; a small wall keeps the rays and changes how they are assembled into cones. In the last case the two neighboring models are related by a flip, or by a flop when the contracted curves have canonical degree zero. The two boundary rays of $\mathrm{Nef}(\mathbb{F}_1)$ displayed the fiber-type and divisorial possibilities, while the two resolutions of the conifold displayed the first genuinely three-dimensional flop.

The numerical meaning of a wall came from nef–Mori duality:

$$\boxed{\mathrm{Nef}(X) = \overline{\mathrm{NE}}(X)^{\vee}.}$$

For a projective toric variety, invariant curves generate the Mori cone. A fan wall gives such a curve, its wall relation gives the intersection vector of that curve, and the perpendicular nef facet gives the morphism that contracts it. Thus the three different walls used in the chapter—a codimension-one cone of the fan, a facet of the nef cone, and a wall of the GKZ fan—are three appearances of one numerical contraction.

Finally, the canonical class oriented the chamber picture. If a wall relation represents an invariant curve C, then

$$K_X \cdot C = -\sum_i (D_i \cdot C).$$

The toric K-MMP repeatedly chooses a ray on which this number is negative and then contracts it, flips it, or stops at a Mori fiber space. Divisorial steps decrease the Picard number, flips

cannot cycle, and the process terminates. Moreover, a positive-dimensional projective toric variety cannot have nef canonical divisor, so the absolute toric K-MMP ends with a Mori fiber space. In the language of this chapter, it is a directed path through finitely many chambers.

The exercises first test each part of this dictionary separately. The challenge then asks you to reconstruct an entire toric flip from a single one-dimensional weight matrix.

Comprehension exercises

10.1. Let $\mathbb{G}_m$ act on $\mathbb{A}^3$ with coordinates (x_0, x_1, y) and weights

$$1, \qquad 1, \qquad -1.$$

Use the weight-cone criterion to determine the semistable loci for characters $\theta > 0$ and $\theta < 0$. Show that the corresponding quotients are

$$\operatorname{Tot}\big(\mathcal{O}_{\mathbb{P}^1}(-1)\big) \qquad \text{and} \qquad \mathbb{A}^2,$$

respectively. Compute the invariant ring at $\theta = 0$, identify the wall quotient with $\mathbb{A}^2$, and explain why the wall morphism from the positive chamber contracts the zero section of $\mathcal{O}_{\mathbb{P}^1}(-1)$.

10.2. Return to the Cox weights of $\mathbb{F}_1$,

$$F, \qquad S, \qquad F, \qquad F + S.$$

Starting only from these four weights, use the weight-cone criterion to recover the two chambers

$$a > b > 0, \qquad b > a > 0$$

for $\theta = aF + bS$. Determine which coordinate patterns are unstable in each chamber. Then explain why the first quotient is $\mathbb{F}_1$, the second is $\mathbb{P}^2$, and the common wall gives the blowdown of the exceptional curve.

10.3. For the square configuration

$$(0,0), \quad (1,0), \quad (1,1), \quad (0,1),$$

write down its two triangulations. Compute the GKZ vector of each triangulation using normalized lattice area, and verify that their convex hull is a segment. Show directly that the normal fan of this segment has two chambers separated by one wall. Which subdivision of the square corresponds to the wall itself?

10.4. Consider the two circuit relations

$$u_1 + u_3 = u_2 + u_4 \qquad \text{and} \qquad u_1 + u_2 + u_3 = u_4 + u_5.$$

Regard their coefficients as intersection vectors of contracted curves. Compute $K_X \cdot C$ in each case. Explain why the first relation is compatible with a flop, whereas the second has a preferred K-MMP direction. What additional geometric information is needed to decide whether the second contraction is divisorial or small?

10.5. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, with ruling curve classes f_1, f_2 and divisor classes H_1, H_2. Use

$$H_i \cdot f_j = \delta_{ij}$$

to prove

$$\overline{\mathrm{NE}}(X) = \mathrm{Cone}(f_1, f_2), \qquad \mathrm{Nef}(X) = \mathrm{Cone}(H_1, H_2).$$

Identify the extremal contraction determined by each nef boundary ray. Since $K_X = -2H_1 - 2H_2$, show that either contraction is a valid first and final step of the K-MMP.

10.6. Use the standard fan of $\mathbb{P}^3$, whose rays are generated by

$$e_1, \quad e_2, \quad e_3, \quad -e_1 - e_2 - e_3.$$

Take the fan wall $\tau = \mathrm{Cone}(e_1, e_2)$. Write the wall relation using the two adjacent rays, and use it to compute the intersections of the invariant curve C_τ with all four invariant divisors. Deduce

$$-K_{\mathbb{P}^3} \cdot C_\tau = 4.$$

Identify C_τ geometrically and reconcile the answer with $-K_{\mathbb{P}^3} = 4H$.

10.7. On the Hirzebruch surface $\mathbb{F}_r$, let $S^2 = -r$ and let F be a ruling fiber. Starting from

$$K_{\mathbb{F}_r} = -2S - (r+2)F,$$

determine which extremal rays are K-negative for every $r \geq 0$. Run every possible ordinary K-MMP beginning at $\mathbb{F}_0$, $\mathbb{F}_1$, and $\mathbb{F}_2$. Explain why the contraction of the negative section on $\mathbb{F}_2$ belongs to the VGIT picture but not to the ordinary K-MMP.

10.8. Let X be a positive-dimensional projective toric variety and let A be ample. Use

$$K_X = -\sum_\rho D_\rho$$

to show

$$K_X \cdot A^{n-1} < 0.$$

Why does this rule out K_X being nef? Explain why an absolute toric K-MMP can therefore terminate only with a fiber-type contraction. Finally, explain which part of the argument would have to change for a relative MMP over a nontrivial base.

Challenge exercise

Build a toric flip from one line of weights.

Let $G = \mathbb{G}_m$ act on

$$\mathbb{A}^5 = \operatorname{Spec} \mathbb{C}[x_0, x_1, x_2, y_0, y_1]$$

by

$$\lambda \cdot (x_0, x_1, x_2, y_0, y_1) = (\lambda x_0, \lambda x_1, \lambda x_2, \lambda^{-1} y_0, \lambda^{-1} y_1).$$

Thus the weight matrix is the single row

$$Q = \begin{pmatrix} 1 & 1 & 1 & -1 & -1 \end{pmatrix}.$$

There are two nonzero chambers in $\operatorname{Hom}(G, \mathbb{G}_m)_{\mathbb{R}} \simeq \mathbb{R}$. The goal is to reconstruct the two quotients, their common wall contraction, and the direction of the resulting flip using only this row of weights.

Hint 1. Find the two semistable loci.
Apply the weight-cone criterion. Prove that for $\theta > 0$,

$$U_+ = \{(x, y) : (x_0, x_1, x_2) \neq (0, 0, 0)\},$$

whereas for $\theta < 0$,

$$U_- = \{(x, y) : (y_0, y_1) \neq (0, 0)\}.$$

Draw the one-dimensional GKZ fan: two rays separated by the wall $\theta = 0$. Which coordinate subspace is removed in each chamber?

Hint 2. Identify the chamber quotients.
For $\theta > 0$, the nonzero vector $x = (x_0, x_1, x_2)$ determines a point of $\mathbb{P}^2$. Check from the G-action that each y_j is a fiber coordinate of $\mathcal{O}_{\mathbb{P}^2}(-1)$. Deduce

$$\boxed{X_+ := U_+/G \simeq \operatorname{Tot}\left(\mathcal{O}_{\mathbb{P}^2}(-1)^{\oplus 2}\right).}$$

Reverse the roles of x and y to prove

$$\boxed{X_- := U_-/G \simeq \operatorname{Tot}\left(\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 3}\right).}$$

In particular, both quotients are smooth fourfolds.

Hint 3. Compute the wall quotient.
At $\theta = 0$, the quotient is affine:

$$X_0 = \operatorname{Spec} \mathbb{C}[x_0, x_1, x_2, y_0, y_1]^G.$$

Show that the invariant ring is generated by the six quadratic monomials

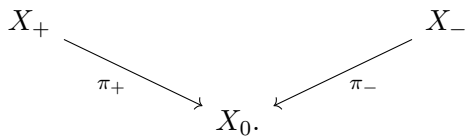
$$z_{ij} = x_i y_j, \qquad 0 \leq i \leq 2, \quad 0 \leq j \leq 1.$$

Arrange them in a 3×2 matrix $Z = (z_{ij})$. Prove that the relations are the 2×2 minors of Z, and conclude

$$\boxed{X_0 = \{Z \in \mathrm{Mat}_{3\times 2}(\mathbb{C}) : \mathrm{rank}(Z) \leq 1\}.}$$

Equivalently, X_0 is the affine cone over the Segre embedding $\mathbb{P}^2 \times \mathbb{P}^1 \subset \mathbb{P}^5$.

Hint 4. Find the exceptional loci.
The invariant functions define morphisms



Show that π_+ contracts the zero section $E_+ \simeq \mathbb{P}^2$ to the vertex of the cone and that π_- contracts the zero section $E_- \simeq \mathbb{P}^1$. Prove that each map is an isomorphism away from its zero section. Since

$$\mathrm{codim}_{X_+}(E_+) = 2, \qquad \mathrm{codim}_{X_-}(E_-) = 3,$$

deduce that both contractions are small. Thus the induced map

$$X_+ \dashrightarrow X_-$$

is an elementary toric flip in one of its two orientations.

Hint 5. Let the canonical class choose the orientation.
For a vector bundle $p : E \to B$, use the tangent sequence of its total space to prove

$$K_{\mathrm{Tot}(E)} \simeq p^*\Big(K_B \otimes (\det E)^{-1}\Big).$$

Apply this formula to the two chamber quotients. On X_+, show

$$K_{X_+}|_{E_+} \simeq \mathcal{O}_{\mathbb{P}^2}(-1),$$

so $K_{X_+} \cdot L = -1$ for a line $L \subset E_+$. On X_-, show

$$K_{X_-}|_{E_-} \simeq \mathcal{O}_{\mathbb{P}^1}(1),$$

so $K_{X_-} \cdot E_- = 1$. Conclude that

$$\boxed{X_+ \dashrightarrow X_- \text{ is directed from the } K\text{-negative side to the } K\text{-positive side.}}$$

Hint 6. Recover the sign from the circuit.
Let $u_0, \ldots, u_4$ be the images of the standard basis of $\mathbb{Z}^5$ in the quotient lattice determined by Q. Show that the unique circuit relation is

$$u_0 + u_1 + u_2 - u_3 - u_4 = 0.$$

Its coefficient sum is 1. Use the wall-relation formula from Section 10.4 to recover

$$K_{X_+} \cdot L = -1.$$

Explain how the two subdivisions of this circuit keep the same rays but assemble them into different maximal cones, exactly as required for a small wall crossing.

Hint 7. Compare a flip with a flop.
Replace the weights by

$$(1, 1, -1, -1).$$

Repeat the canonical-class calculation, or simply use the coefficient sum of the circuit relation, to show that the two chamber quotients are K-trivial over their common affine quotient. Recover the resolved-conifold flop from Section 10.3. Which single numerical feature of the weight row distinguishes the flop from the flip above?

Hint 8. Read the whole construction in six languages.
Summarize the example by completing the following chain without looking back at the preceding hints:

weight row $\longrightarrow$ semistable loci $\longrightarrow$ two GKZ chambers
$\longrightarrow$ two circuit subdivisions $\longrightarrow$ small contractions $\longrightarrow$ K-directed flip.

At which arrow is convex geometry used? Where does invariant theory enter? Where is the orbit–cone dictionary hiding? Which calculation turns an undirected VGIT wall crossing into an MMP step?

The three dictionaries

The entire flip can be recovered from the row $Q = (1, 1, 1, -1, -1)$.

Weights / fan	GIT quotient	Birational geometry
positive chamber	$\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^2}(-1)^{\oplus 2})$	small K-negative contraction
wall	cone over $\mathbb{P}^2 \times \mathbb{P}^1$	common contraction target
negative chamber	$\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 3})$	flipped, K-positive model
coefficient sum 1	unbalanced circuit	flip rather than flop

This example contains the whole chapter in miniature. A grading produces a torus action; a character selects an open set; the chambers organize the possible quotients; a circuit describes the wall; and one canonical intersection number directs the birational move.

A final perspective

These notes began with a strongly convex rational polyhedral cone. From its lattice points we built a semigroup algebra and an affine toric variety. Gluing cones into a fan produced global varieties; support functions and polytopes produced divisors, line bundles, and projective embeddings; Cox coordinates replaced gluing by one multigraded quotient. The compact torus then turned the polytope into an orbit space, the fan became a cohomology ring, and localization and Riemann–Roch turned characteristic classes into lattice-point counts. In this final chapter, varying the Cox character made the same data move through an entire birational world.

The progression can be compressed into one last diagram:

$$\boxed{\begin{array}{c}\text{cone and semigroup} \longrightarrow \text{affine toric variety},\\ \text{fan and support function} \longrightarrow \text{global variety and line bundle},\\ \text{polytope and Cox grading} \longrightarrow \text{sections, topology, and quotients},\\ \text{moving character} \longrightarrow \text{secondary fan and toric MMP}.\end{array}}$$

What makes toric geometry unusually coherent is not merely that its objects admit pictures. It is that the same finite combinatorial data can be read, without loss, in several mathematical languages. A ray may be a divisor, a Cox variable, or a facet normal. A wall may be an invariant curve, a nef equation, or a change of quotient. A lattice point may be a character, a monomial section, or a contribution to an index. Moving between these descriptions is the central skill the book has tried to develop.

Toric varieties are therefore both a testing ground and a source of models. They make singularities, resolutions, positivity, cohomology, localization, and the minimal model program explicit enough to compute, while retaining the structure of the general theories they illuminate. The combinatorics does not replace the geometry. It reveals how the geometry is organized.

Notes and references

The GIT construction used in Section 10.1 is developed in Cox–Little–Schenck, §14.1. Proposition 14.1.9 identifies the semistable locus of an ample Cox character with the usual Cox open set. Definition 14.1.11 and Proposition 14.1.12 give the section-ring construction of the GIT quotient; the weight-cone test used throughout the chapter is its direct monomial specialization, proved in Section 10.1.

The GKZ decomposition of the effective cone is treated in Cox–Little–Schenck, §14.4; Theorem 14.4.7 states that the GKZ cones form a fan and that the quotient is constant on the relative interior of each cone. Regular triangulations and the secondary polytope are developed in their §15.2. We used Definition 15.2.8 and Proposition 15.2.9 for the passage between chambers and regular triangulations, and Proposition 15.2.11 together with Corollary 15.2.12 for the secondary polytope and its vertices.

Our wall-crossing discussion follows Cox–Little–Schenck, §15.3 and §15.4. Theorem 15.3.13 describes the birational maps attached to a flipping wall, Proposition 15.4.5 gives the fiber-type, divisorial, and small trichotomy, and Proposition 15.4.1 constructs the contraction dual to a nef facet. The resolved conifold is the basic balanced circuit: its two triangulations give small crepant resolutions, so the wall crossing is a flop.

The numerical input comes from the toric intersection theory of Cox–Little–Schenck, Chapter 6. Theorems 6.3.20 and 6.3.22 give the generation of the Mori cone by invariant curves and its duality with the nef cone. Proposition 6.4.4 computes the intersection vector of an invariant curve from its wall relation, and Theorem 6.3.12 supplies the crucial toric fact that nef divisors are basepoint free.

The toric MMP appears in Cox–Little–Schenck, §15.5. Procedure 15.5.5 gives the algorithm, Proposition 15.5.3 records preservation of terminal singularities under the relevant divisorial contractions and flips, and Theorem 15.5.10 proves termination of toric flips. Our treatment concentrated on the absolute projective case; the same combinatorics also works in relative and semiprojective settings.

The final challenge is the standard unbalanced circuit with weight row $(1, 1, 1, -1, -1)$. Its wall quotient is the determinantal cone of rank-at-most-one 3×2 matrices. The two small resolutions have exceptional loci $\mathbb{P}^2$ and $\mathbb{P}^1$, and the unequal numbers of positive and negative weights make the canonical degree nonzero. The balanced row $(1, 1, -1, -1)$ instead gives the conifold flop.

The notes have emphasized a single habit of thought: whenever a toric calculation appears in one language, look for its counterparts in the lattice, fan, polytope, divisor, Cox, topological, and birational languages. That dictionary is more durable than any isolated formula, and it is the real conclusion of the book.

Index

affine chart
    toric, 154
affine plane, 3
affine semigroup, 108
affine toric variety, 103
    classification, 143
algebraic torus, 47, 51
ample divisor, 248
ampleness criterion, 253
anticanonical divisor, 341
anticanonical polytope, 341
Atiyah–Bott localization, 422
attracting cell, 384

basepoint-free divisor, 253
Betti number, 384
binomial equation, 108
blowup
    Cox quotient, 310
    of the projective plane, 33
    star subdivision, 180
Borel construction, 412
boundary incidence, 194

canonical divisor
    MMP direction, 466
    toric formula, 224
Cartier data, 212
Cartier divisor, 212
cell decomposition
    toric variety, 384
change of lattice basis, 79
character, 56
    extending to an affine variety, 103
character lattice, 60
Chern character, 406
Chern class, 406
class-group grading, 288
class-group sequence, 205
classification
    normal affine toric varieties, 143
cocharacter lattice, 65
cohomology ring
    toric presentation, 390
compact torus, 372
complete fan, 158
completeness
    fan criterion, 175
cone
    rational polyhedral, 116
contraction
    divisorial, 454
    fiber type, 454
    from a normal fan, 261
    small, 454
coordinate strata, 3
coordinate triangle, 8
corner cut
    anticanonical polygon, 357
    blowup, 242
    toric blowup, 270
Cox coordinate, 288
    vanishing test, 300
Cox quotient, 305
Cox ring, 294
cyclic quotient singularity, 316

degree
    del Pezzo surface, 357
degree-zero localization, 305
del Pezzo surface
    toric, 357
dense torus, 103
distance-one criterion, 346
divisor class group, 205
divisor of a character, 200
divisor polyhedron, 242
dual cone, 125
dual lattice, 73

Ehrhart polynomial, 427
equivariant cohomology, 412

equivariant Euler characteristic, 427
equivariant Euler class, 422
equivariant integration, 422
exceptional divisor, 33
exceptional set
    Cox construction, 300
extremal contraction, 459
extremal ray, 459

face duality, 346
face localization, 132
face stratum, 378
factorial toric variety, 224
fan
    definition, 158
    of the projective plane, 27
    support, 158
Fano criterion, 341
Fano toric variety, 341
fixed point
    localization, 422
    of an affine toric variety, 132
    torus-fixed, 22
flip
    toric, 454
flop
    toric, 454

Gale duality, 448
general linear group over the integers, 79
geometric invariant theory, 442
GKZ fan, *see* secondary fan
global section, 16
    inside the Cox ring, 294
    monomial basis, 242
gluing
    affine toric charts, 154
    affine toric varieties, 163
Gordan's lemma, 116
Gorenstein cone, 224
Gorenstein Fano toric variety, 352
group algebra, 51, 60

h-vector, 384
height-one criterion, 224
Hilbert basis, 116
Hirzebruch surface, 180
    Cox quotient, 310
Hirzebruch–Riemann–Roch, 406
homogeneous coordinates, 8
    toric, 288
homogeneous piece
    of the Cox ring, 294

intersection number
    edge length, 270
    from cohomology, 390
invariant prime divisor, 194
invariant vector, 86
irrelevant ideal, 300

Jurkiewicz–Danilov presentation, 390

lattice
    character, 60
lattice pairing, 73
lattice point
    and monomial sections, 16
    as a section, 242
    Riemann–Roch count, 427
lattice polygon
    area and degree, 357
Laurent polynomial, 51
linear equivalence, 200
    global linear function, 218
linearization, 442
local equation
    of a divisor, 212
logarithmic cotangent bundle, 334
logarithmic differential form, 334

minimal model program
    toric, 466
moment map, 372
    fibers, 378
moment polytope, 372
monomial, 16
    as character, 56
monomial automorphism, 79
monomial map, 108
Mori cone, 459
Mori fiber space, 466

nef cone, 459
nonsimplicial fan, 316
normal cone, 261
normal fan, 27, 261
normality, 116
normalization, 116

one-parameter flow, 384
one-parameter subgroup, 65
    limit criterion, 125
    limits, 27
one-parameter subgroup lattice, 65
orbit closure, 22, 169
orbit space
    moment polytope, 378
orbit–cone correspondence, 169
order of vanishing, 73, 200

Picard group, 212
piecewise polynomial, 412
piecewise-linear function, 218
polar duality, 346
polytope
    faces, 261
    triangle of monomials, 16
primitive collection, 300
primitive lattice vector, 65
principal divisor, 200
product fan, 175
projective map
    from lattice points, 248
projective plane, 8
    dense torus, 12
    fan, 180
projective space
    Cox quotient, 310
projective toric variety, 248

Q-factorial toric variety, 224
Q-Gorenstein cone, 224
quotient construction
    Cox, 305

ray
    invariant divisor, 194
reflexive polytope, 346
    Gorenstein Fano correspondence, 352
regular triangulation, 448
representation
    of a torus, 86
    one-dimensional, 56
rescaling group, 305
residue sequence, 334
Riemann–Roch
    toric, 427

saturation, 116
secondary fan, 448
secondary polytope, 448
self-intersection, 33
    polytope area, 270
semigroup algebra, 108
semigroup homomorphism, 132
semistable point, 442
separation lemma, 163
simplicial fan, 316
smooth polytope, 270
smoothness criterion
    affine cone, 132
    fan, 175
stabilizer
    Cox action, 316
    of a polytope face, 378
Stanley–Reisner ideal, 390
star of a cone, 169
star subdivision, 33
strongly convex cone, 125
subfan, 158
support function, 218
    as an equivariant class, 412
    convexity, 253
    strict convexity, 253

tangent weight, 422
termination
    toric flips, 466
Todd class, 406
toric boundary, 194, 334
toric ideal, 108
toric morphism, 132
    fan criterion, 175
toric surface

projectivity, 270
toric variety
from a fan, 163
torsion
in the class group, 205
torus
dense torus, 12
topological and algebraic, 47
torus action
on the projective plane, 12
torus factor, 132
torus orbit, 22, 169

variation of GIT quotient, 442
very ample divisor, 248

wall crossing, 454
wall relation, 459
weight cone, 442
weight matrix, 310
weight space, 86
weight-space decomposition, 86
weighted projective plane, 180
weighted projective space
Gorenstein criterion, 352
Weil divisor, 205